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Structure and Function of Membrane Proteins

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Structure and Function of Membrane Proteins

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Preface

The volume *Structure and Function of Membrane Proteins* is aimed at biologists, biochemists, and biophysicists. The volume outlines detailed experimental and computational approaches for the analysis of many aspects important to the understanding of membrane protein structure and function. Readers will receive guidance on the selection and use of methods for overexpression and purification, tools to characterize membrane proteins within different phospholipid bilayers, direction on functional studies, and approaches to determine the structures of membrane proteins. Detailed experimental steps for specific membrane proteins with critical notes allow the protocols to be modified for different systems.

The book is grouped into chapters starting with overexpression and purification with the example of an enzyme (Chapter 1). The chapter emphasizes steps to obtain large quantities of purified protein relevant to the use of structural and functional studies, which can be adapted to other membrane proteins. Chapter 2 covers the preparation of phospholipids for reconstitution and the reconstitution of membrane proteins into different sizes and types of phospholipid bilayers. Chapter 3 focuses on studies of membrane protein complex formation, binding properties, and complex dissociation of G protein-coupled receptors (GPCR) and G proteins. The protocols apply affinity purification and gel filtration. Thus, the approach is easily modified and equally valuable for large-scale purification for the range of structural approaches discussed in later chapters. Chapter 4 covers comprehensive techniques in electrophysiology applied to characterize channel activity and pharmacology to study a channel in situ, heterologously expressed, and purified as well as partially purified and reconstituted in planar lipid bilayers. Chapter 5 describes isothermal titration calorimetry (ITC) applied to protein-protein interactions of membrane proteins in phospholipid bicelles to provide critical thermodynamic insights on these interactions. In addition, the role of phospholipids in transmembrane helix-helix association can be elucidated by ITC. The protocols use the association of integrin α IIb and β 3 transmembrane domains. Atomic force microscopy to study conformational dynamics over different timescales is described in Chapter 6. The chapter highlights the importance of biochemical activity in the measurement and its maintenance. In addition, it provides critical guidance on objective AFM image processing techniques.

Structure determination by X-ray crystallography (Chapter 7) has successfully been applied to the structure of proteins for decades. Despite critical breakthroughs for membrane proteins, the method still faces challenges in overexpression, purification, and structure determination. Chapter 7 covers a detailed general pipeline from cloning to structure determination of membrane proteins. The protocol is specific to BamA, and strategies are discussed on its application to other membrane proteins as well as general considerations to achieve high quantities of purified protein. These approaches for membrane protein expression and purification are valuable for studies by X-ray crystallography and other structural methods. Microcrystal electron diffraction (MicroED) has rapidly evolved as a method to provide a tremendous number of structures of many proteins that appeared intractable to structural analysis. Chapter 8 describes two approaches that are key advances for MicroED sample preparation: application of a crystal slurry directly to EM grids and FIB-SEM. The chapter also covers the use of an energy filter to decrease inelastic scattering during data

collection and hence noise in the resulting data. Chapter 9 details single-particle cryo-EM with significant guidance on preparing membrane proteins, their initial analysis by negative stain, overcoming specific challenges for single-particle cryo-EM grid preparation, data collection, and image processing. Cryo-EM of helical crystals (Chapter 10) allows for rapid data collection as the helical nature of the sample provides data in multiple orientations in one image. Chapter 10 covers the helical crystallization of a sarco(endo)plasmic reticulum Ca^{2+} -ATPase, cryo-EM grid preparation, data collection and analysis. Large parts of these protocols can be modified for the study of other membrane proteins and are also equally applicable to soluble proteins that form filaments. Chapter 11 targets membrane protein preparation for solution and magic angle spinning NMR studies in detergent micelles and phospholipid bilayers. Protocols use the example of the recombinant production of the stable-isotope labeled channel hVDAC1. The signal in small-angle neutron scattering (SANS) experiments can be significantly enhanced by deuteration. Chapter 12 covers the detailed steps from transformation of plasmid into *E. coli*, overproduction of deuterated membrane protein to preparation for SANS experiments.

The last third of the book switches focus to methods and tools for setting up, running, and analyzing molecular dynamics (MD) simulations of membrane proteins. Chapter 13 describes the application of a commonly used webserver, CHARMM-GUI, to build an accurate, detailed membrane for a bacterial outer membrane protein. Chapter 14 also covers setting up a membrane protein simulation system, but instead of an atomistic approach, it uses a coarse-grained one. Novel methods are the focus of the next two chapters, including constant pH MD simulations (Chapter 15) and biased simulations for determining energetics and kinetics (Chapter 16). Chapter 17 describes an analysis method for uncovering the precise origins of allosteric communication between distinct sites in a protein. Finally, Chapter 18 illustrates the past, present, and future of MD simulations of membrane proteins, with a focus on the chromatophore, a photosynthetic organelle containing dozens of membrane proteins.

We are tremendously grateful to our authors for their time and expertise in preparing this book. Our series editor John M. Walker, Professor Emeritus in the School of Life Sciences at the University of Hertfordshire, has generously shared his experience in all aspects of editing.

Atlanta, GA, USA

*Ingeborg Schmidt-Krey
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Chapter 1

Expression and Purification of Human Mitochondrial Intramembrane Protease PARL

Elena Arutyunova, Laine Lysyk, Melissa Morrison, Cory Brooks, and M. Joanne Lemieux

Abstract

Rhomboid proteases are a ubiquitous superfamily of serine intramembrane peptidases that play a role in a wide variety of cellular processes. The mammalian mitochondrial rhomboid protease, Presenilin-Associated Rhomboid Like (PARL), is a critical regulator of mitochondrial homeostasis through the cleavage of its substrates, which have roles in mitochondrial quality control and apoptosis. However, neither structural nor functional information for this important protease is available, because the expression of eukaryotic membrane proteins to sufficient levels in an active form often represents a major bottleneck for in vitro studies. Here we present an optimized protocol for expression and purification of the human PARL protease using the eukaryotic expression host *Pichia pastoris*. The PARL gene construct was generated in tandem with green fluorescent protein (GFP), which allowed for the selection of high expressing clones and monitoring during the large-scale expression and purification steps. We discuss the production protocol with precise details for each step. The protocol yields 1 mg of pure PARL per liter of yeast culture.

Key words PARL, PINK1, Rhomboid, Intramembrane serine proteases, *Pichia pastoris*, Membrane protein expression, Polytopic membrane protein, Detergent

1 Introduction

The mammalian mitochondrial Presenilin-Associated Rhomboid Like (PARL) belongs to the rhomboid family of serine intramembrane proteases that cleaves transmembrane substrates, which regulate diverse cell signaling events [1]. PARL has been shown to have pleiotropic roles in mitochondria, including mitophagy and apoptosis [2–5]. The identified substrates highlight PARL's involvement in various pathways and its relevance for disease. Several substrates have been identified that include mitochondrial kinase PINK1 [2, 6], mitochondrial phosphatase PGAM5 [3], pro-apoptotic protein Smac, lipid transfer protein STARD7, the complex III surveillance factor TTC19, and putative mitochondrial

chaperone protein CLBP [7]. To date, the cleavage of these substrates was assessed using cellular lysates or in vitro translated PARL [8]. Despite PARL's importance in cellular processes, neither structural information nor functional characterization in vitro is available and recombinant protein is required for further studies.

The main challenge associated with eukaryotic membrane protein expression and purification is the selection of a suitable expression system and obtaining sufficient amounts of homogenous protein for structural and functional studies [9, 10]. Membrane proteins are sensitive to the lipid environment [11], protein processing, folding, and post-translational modifications, which makes heterologous recombinant expression a significant task. Protein aggregation, misfolding, and low yields of functional protein often occur. Therefore, finding the optimal expression system is extremely valuable.

Here we describe the recombinant expression of the human PARL protease in *Pichia pastoris*. The yeast expression system *P. pastoris* has been successfully used by our group and others to produce many eukaryotic membrane proteins for structural studies including: PEMT [12], ENT7 [13], G-protein coupled receptors [14, 15], ion channels [16, 17], aquaporins [18, 19], and ABC transporters [20]. This system offers several advantages for heterologous protein expression including a low cost of the media and inducing agent, making large scale expression more feasible, and the ability of yeast to grow to a high cell density to allow for more fruitful yield. In addition, a strong promoter, AOX1, used in the vectors for *P. pastoris* is powerful and tightly regulated, which aids for high yields as well. Finally, *P. pastoris* is capable of generating posttranslational modifications such as N- and O-linked glycosylation, which can be essential for proper folding and function, and resemble those of higher eukaryotic organisms [21].

After successful expression, solubilization of a membrane protein and its subsequent purification, while preserving its activity and stability, are other important steps of obtaining a functional enzyme, which will be discussed in this chapter. Our optimized protocol yields 1 mg of pure PARL per liter of yeast culture.

2 Materials

2.1 Protein Expression and Purification Materials, Solutions, and Media

Commercial pPICZA vector (Invitrogen, Canada), Red shifted variant of enhanced GFP containing the mutations F64L and S65T (eGFP) from pEGFP-N1 (Takara, USA), *Pichia pastoris* yeast GS115 strain (ThermoFisher, Canada), PmeI enzyme (New England BioLabs, Canada), Gene Pulser Cuvette (Bio-Rad, Canada), DNase (Sigma, Canada), 100 mM PMSF (dissolved in anhydrous EtOH) (Sigma, Canada), 1 M TCEP (dissolved in anhydrous EtOH) (Sigma, Canada), cOmplete™ EDTA-free protease

inhibitor cocktail tablets (PIC) (Roche, Canada), Constant Systems cell disruptor (Constant Systems Ltd., UK), Fluorescent Imager “Image Quant LAS4000” (GE Healthcare Canada), FPLC column Sephadex 200 10/300 (GE Healthcare, USA), HisPur™ Cobalt Resin (ThermoFisher, Canada), Ni-NTA agarose (Invitrogen, Canada). BCA protein assay kit (ThermoFisher, Canada), Amicon® Ultra Centrifugal Filter Units, 10,000 NMWL (Merck Millipore Ltd, Canada).

2.2 Media

2.2.1 Growth Media Recipes

Low salt LB (growth media for pPICZ-transformed *E. coli*): 5 g yeast extract, 10 g tryptone, 5 g NaCl. Dissolve in 900 mL ddH₂O, adjust pH to 7.5 with 5 M NaOH. Top up to 1 L with ddH₂O. Autoclave.

YPD (growth media for wild type GS115 yeast cells): 10 g yeast extract, 20 g peptone. Dissolve in 800 mL of ddH₂O. For plates, add 20 g agar. Autoclave. Add 100 mL of dextrose (20%)*.

YPDS (growth media for pPICZ-transformed yeast cells): 10 g of yeast extract, 20 g of peptone, 182.17 g of sorbitol. Dissolve in 600 mL ddH₂O. For plates, add 20 g agar. Autoclave. Add 100 mL of dextrose (20%)*.

*Sterile 20% (w/v) dextrose is prepared by filter sterilizing the solution into an autoclaved bottle using a 0.22µm filter and added to YPD and YPDS media after autoclaving to a final concentration of 2% (w/v).

The required antibiotic is added once autoclaved media is cooled; ampicillin is added to low salt LB and YPD media to a final concentration of 100µg/mL and Zeocin is added to YPDS media to a final concentration of 25µg/mL.

Plates and media containing Zeocin were stored in the dark as Zeocin is light sensitive.

2.2.2 Expression Media Recipes

BMGY or BMMY (growth media and induction media respectively): 10 g yeast extract, 20 g peptone. Dissolve in 780 mL ddH₂O if making BMGY or 790 mL ddH₂O if making BMMY. For plates, add 20 g agar. Autoclave.

500× Biotin: 0.02% (w/v) biotin dissolved in 0.05 M NaOH. Filter-sterilize. Store at 4 °C.

10× Yeast Nutrient Broth (YNB): 34 g yeast nitrogen base without amino acids and without ammonium sulfate, 10 g ammonium sulfate. Dissolve in 800 mL ddH₂O while heating. Top up to 1 L with ddH₂O. Filter-sterilize into autoclaved 1 L bottle. Store at 4 °C.

10× Potassium Phosphate buffer: 23 g K₂HPO₄, 118.13 g KH₂PO₄. Dissolve in 800 mL ddH₂O. pH to 6.0 with KOH. Filter sterilize into autoclaved 1 L bottle. Store at 4 °C.

If making plates, the components of BMGY or BMMY media should be added in a specific order at a specific temperature.

When at ~70 °C add 100 mL of 10× Potassium Phosphate buffer.
*.

When at ~60 °C add 100 mL of 10× YNB.†.

* These two solutions are stored at 4 °C so they will decrease the media temperature by a lot if not warmed to room temperature; therefore, prepare media quickly before agar starts to solidify.

† If YNB is added when the solution is too hot it will denature and the solution will turn cloudy.

When at ~50 °C add.:

1 mL 1000× (100 mg/mL) Ampicillin.

2 mL of 500× Biotin.

For BMGY add 20 mL of 50% sterile glycerol (autoclaved).

For BMMY add 10 mL 100% methanol (filter sterilized).

BMGY and BMMY broth is prepared in 4 L baffled flasks as these allow for greater aeration than standard 4 L culture flasks.

2.3 Buffers and Solutions

Phosphate-buffered Saline (PBS): 10 mM Na₂HPO₄, 1.8 mM KH₂PO₄, 137 mM NaCl, 2.7 mM KCl, pH 7.4.

Resuspension buffer for detergent optimization: 50 mM KPO₄, 10% glycerol, 300 mM NaCl, 10 mM βME, 1 PIC tablet.

Detergent solutions (w/v): 20% neopentyl glycol (NG), 10% n-decyl-β-D-maltopyranoside (DM), 10% n-dodecyl-β-D-maltopyranoside (DDM), 10% dodecyl octaethylene glycol ether (C12E8), 10% lauryldimethylamine-N-oxide (LDAO), 10% Triton X-100, 10% Fos-choline-12 (FC-12).

Gel filtration buffer: 50 mM K₂PO₄, 5% glycerol, 1 mM NaCl, 10 mM βME and 0.1% of the corresponding detergent.

Tris buffered saline (TBS): 50 mM Tris-HCl, pH 8.0, 150 mM NaCl.

Lysis buffer: 50 mM Tris-HCl pH 8.0, 200 mM NaCl, 5% glycerol.

Solubilization buffer: 50 mM Tris-HCl, pH 8.0, 200 mM NaCl, 20% (v/v) glycerol, 20 mM imidazole.

Buffer A: 50 mM Tris-HCl, pH 8.0, 300 mM NaCl, 20% (v/v) glycerol, 0.1% (w/v) DDM.

Buffer B: 50 mM Tris-HCl, pH 8.0, 300 mM NaCl, 20% (v/v) glycerol, 0.1% (w/v) DDM, 500 mM imidazole.

Dialysis buffer: 50 mM Tris-HCl, pH 8.0, 300 mM NaCl, 20% (v/v) glycerol.

All solutions were filter sterilized into autoclaved bottles using a 0.22μm filter before use and stored at 4 °C.

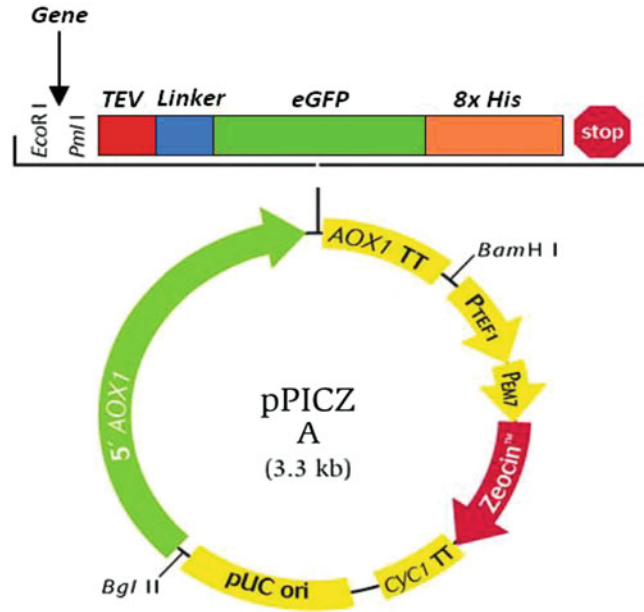


Fig. 1 pPICZ-GFP plasmid map. The modified pPICZA vector includes a TEV protease cleavage site (ENLYFQ*S, where * denotes the cleavage site) and C-terminal GFP-fusion protein with a hexahistidine tag

3 Methods

3.1 Construction of GFP Fusion Vector and Cloning of PARL

The commercial pPICZA vector was modified by cloning in a Tobacco Etch Virus protease (TEV) cut site, red shifted variant of enhanced GFP containing the mutations F64L and S65T (eGFP) from pEGFP-N1 and 6× His-tag (pPICZA-GFP vector) (*see Note 1*).

The human PARL gene was cloned into the pPICZA-GFP vector having a C-terminal hexahistidine tag and contained a TEV protease cleavage site within the linker between PARL and GFP (*see Note 2*) (Fig. 1).

3.2 Transformation of *Pichia pastoris*

The advantageous feature of *P. pastoris* is genetic integration of the gene of interest into the host genome. This genetic integration allows for increased chances of highly expressing a eukaryotic membrane protein of interest as opposed to plasmid-based bacterial expression systems [22].

3.2.1 Linearizing Plasmid DNA

1. Digest 50μg DNA vector in 100μL total volume with PmeI enzyme (*see Note 3*).
2. Add 1/10th volume of 3 M NaOAc to the sample and mix well.

3. Add $3\times$ volume of 95% EtOH and mix well—the DNA should precipitate immediately.
4. Spin at max speed for 5 min at RT.
5. Wash the DNA with 200 μ L 70% EtOH and repellet at max speed for 3 min.
6. Remove supernatant and let residual EtOH dry for <3 min.
7. Resuspend DNA in sterile water (*see* **Note 4**).

3.2.2 Preparing

Electrocompetent

P. pastoris GS115 Cells

1. Inoculate 5 mL of YPD + Ampicillin with a colony (1–2 mm diameter) of *P. pastoris* GS115 and grow overnight at 28 °C, 280 rpm in a tilted 50 mL conical tube. Typical polystyrene tubes held in an upright position used for *E. coli* cultures are not good as the denser yeast settles out because the shaking is not vigorous enough.
2. Use 100–500 μ L of overnight culture to subinoculate 100 mL of YPD + Ampicillin and continue to grow overnight (this amount of cells will be enough for 20 transformations; adjust volumes of the following washes accordingly).
3. Pellet yeast at $500 \times g$ for 5 min at 4 °C.
4. Rinse with 100 mL cold sterile water and repellet.
5. Rinse with 50 mL cold sterile water and repellet.
6. Rinse with 20 mL cold 1 M sorbitol and repellet.
7. Resuspend cells in 2 mL cold 1 M sorbitol and keep on ice.

3.2.3 Transformation and Plate Screening

The typical strategy when expressing proteins in *P. pastoris* is to choose several clones (typically 5–10) to screen for expression, owing to variable expression of clones, ranging from essentially background to highly expressing clones. After testing several membrane proteins, our lab demonstrated that only a small fraction of the clones express at relatively high levels, while most clones give essentially no protein expression. Thus, it may be necessary to screen 50–100 clones using conventional liquid culture based screening approaches to find the best expressing clone, or if insufficient clones are chosen for screening, no high-expressing clone may be found. Therefore, the induction plate based screening method is a highly efficient way of screening a very large number of clones for expression [12].

1. Add 5 μ g of linearized plasmid DNA to 95 μ L of electrocompetent yeast in sorbitol, mix well and keep on ice.
2. Electroporate at 2.5 kV in 2 mm cuvettes and immediately add 1 mL of cold 1 M sorbitol.
3. Incubate at 28 °C with shaking for at least 1 h to let yeast recover and express zeocin resistance gene.

4. Plate 5 μ L, 15 μ L, 50 μ L, and 200 μ L of your transformation on YPDS + Zeocin plates (prepare duplicate plates to screen many colonies) and incubate at 28 °C. Allow 2–3 days for the colonies to appear (*see Note 5*).
5. Transfer individual colonies in grid pattern to YPDS + Zeocin plates to prepare for screening on BMMY plates; the YPDS plate serves as a master plate for future small/large scale cultures (*see Note 6*).
6. Let colonies grow on YPDS for 1–2 days and transfer to BMMY + Ampicillin plates in the same grid pattern, having GS115 yeast as negative control in position 1–2 on the plate.
7. Visualize BMMY plates every 24 h under blue light using a Fluorescent Imager to identify high expressing colonies and visualize the progression of expression of each transformant (*see Note 7*). The representative screening plate for colonies expressing PARL Δ 77-GFP is depicted in Fig. 2.
8. Pick the colony with the highest fluorescence (the fluorescence can be judged by densitometry) to start large scale expression.

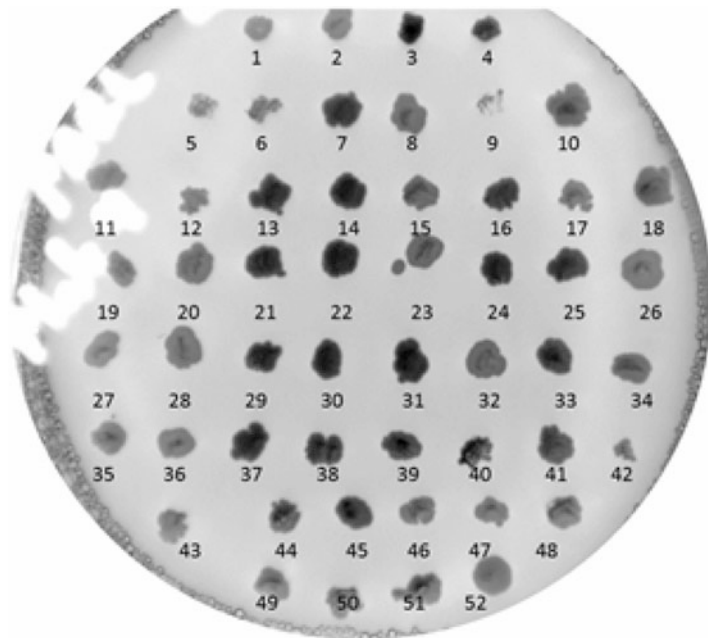


Fig. 2 BMMY induction plate from the expression screening of PARL Δ 77-GFP after 48 h of induction. PARL Δ 77-GFP colonies were plated onto BMMY plates in a grid pattern and incubated at 28 °C. Plates were visualized under blue light every 24 h to monitor fluorescence. Two negative controls of untransformed GS115 yeast cells were included in colony positions 1 and 2. Colony 14 was selected for the culture growth as an overexpressing colony

Table 1

Relative fluorescence of PARL Δ 77-GFP liquid cultures. Five milliliters of liquid culture was centrifuged and resuspended in 200 μ L of PBS buffer for each flask. Resuspended cells were placed in a multiwell plate and measured for relative fluorescence units in a fluorimeter

PARL Δ 77-GFP liquid culture flask #	Relative fluorescence units		
	24 h post induction	48 h post induction	72 h post induction
1	3717	7611	22,341
2	8657	12,903	16,917
3	14,928	39,202	43,929
4	14,496	29,668	45,629
5	4507	28,873	28,105

3.3 Large Scale Expression of PARL-GFP

1. At the end of the day, inoculate 10 mL of BMGY + Ampicillin media in a 50 mL conical with a high expressing colony (loosen cap for aeration) and shake at 250–300 rpm at 28 °C overnight.
2. In the next morning check the OD₆₀₀ of a 1/5 and 1/10 dilution.
3. Inoculate the large starter culture (250–500 mL) of BMGY + Ampicillin such that the OD₆₀₀ is <0.05 and grow overnight at 28 °C with shaking.
4. Inoculate 6 $\times$ 1 L BMGY + Ampicillin with 1/6th of the large starter culture and let grow at 28 °C, 250 rpm until the OD₆₀₀ is ~4.0, which takes roughly 24 h.
5. Pellet cells in autoclaved 1 L centrifuge bottles at 500 $\times$ g for 10 min at 4 °C (*see Note 8*).
6. Resuspend cells in 6 $\times$ 1 L BMMY + Ampicillin and continue shaking at 24 °C, 250 rpm.
7. Add 20 mL MeOH per 1 L every 24 h to continue expression.
8. The expression level can be monitored by pelleting 5 mL of cell culture and resuspending the cells in 200 μ L of PBS. The example of fluorescence measured in each flask every 24 h is represented in Table 1. The optimal time for PARL Δ 77-GFP expression was found to be between 48 and 72 h.
9. Harvest the cells by centrifugation at 500 $\times$ g for 10 min at 4 °C. The cell pellet can be frozen and stored at –20 °C if not being used immediately.

3.4 Cell Lysis and Membrane Fraction Isolation

1. Resuspend cell pellet in 300 mL of lysis buffer, containing 1 μ g/mL DNase and Protease Inhibitor Cocktail (PIC) tablet.
2. Stir until all cells are fully resuspended.

3. Following the instructions for the Constant Systems cell disruptor, lyse the cells at 40 kPSI, performing six sequential passages.
4. Perform low-speed centrifugation of the lysate at $3000 \times g$ for 20 min to pellet the cell debris. If the supernatant is very cloudy, repeat this step.
5. Collect the supernatant from the previous centrifugation and perform high-speed ultracentrifugation at $100,000 \times g$ for 2 h to isolate the membrane fraction. Decant the supernatant. The membranes can be stored at $-80\text{ }^{\circ}\text{C}$ until further purification steps are undertaken.

3.5 Detergent Optimization for PARL-GFP

A key parameter in the purification of membrane proteins is the choice of detergent [23, 24]. Optimization of membrane protein solubility in detergents can be a laborious and time-consuming endeavor, however in case of PARL-GFP, detergent optimization experiments are facilitated by the presence of the fluorescent GFP tag. Here the detergent was optimized using PARL-FL-GFP (*see Note 2*).

1. Homogenize 3 g of crude membranes containing PARL-GFP with a glass homogenizer in 30 mL of resuspension buffer.
2. Separate homogenized membranes into 7×2.7 mL aliquots and add 0.3 mL of the following detergents to each: 20% NG, 10% DM, 10% DDM, 10% C₁₂E₈, 10% LDAO, 10% Triton X-100, 10% FC-12 (*see Note 9*). Include a blank sample with water.
3. Incubate each aliquot of homogenized membrane with its detergent on a nutator at $4\text{ }^{\circ}\text{C}$ for 30 min.
4. Solubility may be assessed via three approaches:
 - (a) *SDS-PAGE gel combined with in-gel fluorescence* for assessment of irreversible protein aggregation and proteolysis: Mix each sample (7.5 μL) with SDS loading dye and electrophorese on 14% SDS-PAGE gels. PARL-FL-GFP fusion was found at the correct molecular weight with some proteolysis and free GFP (Fig. 3).
 - (b) *Relative fluorescence units measured before and after solubilization* in order to obtain a quantitative measure of solubilization:
 - Use aliquots (200 μL) of homogenized and detergent solubilized membranes for relative fluorescence comparison.
 - Measure the fluorescence using a fluorescence plate reader, at an excitation wavelength of 488 nm and emission wavelength of 509 nm.

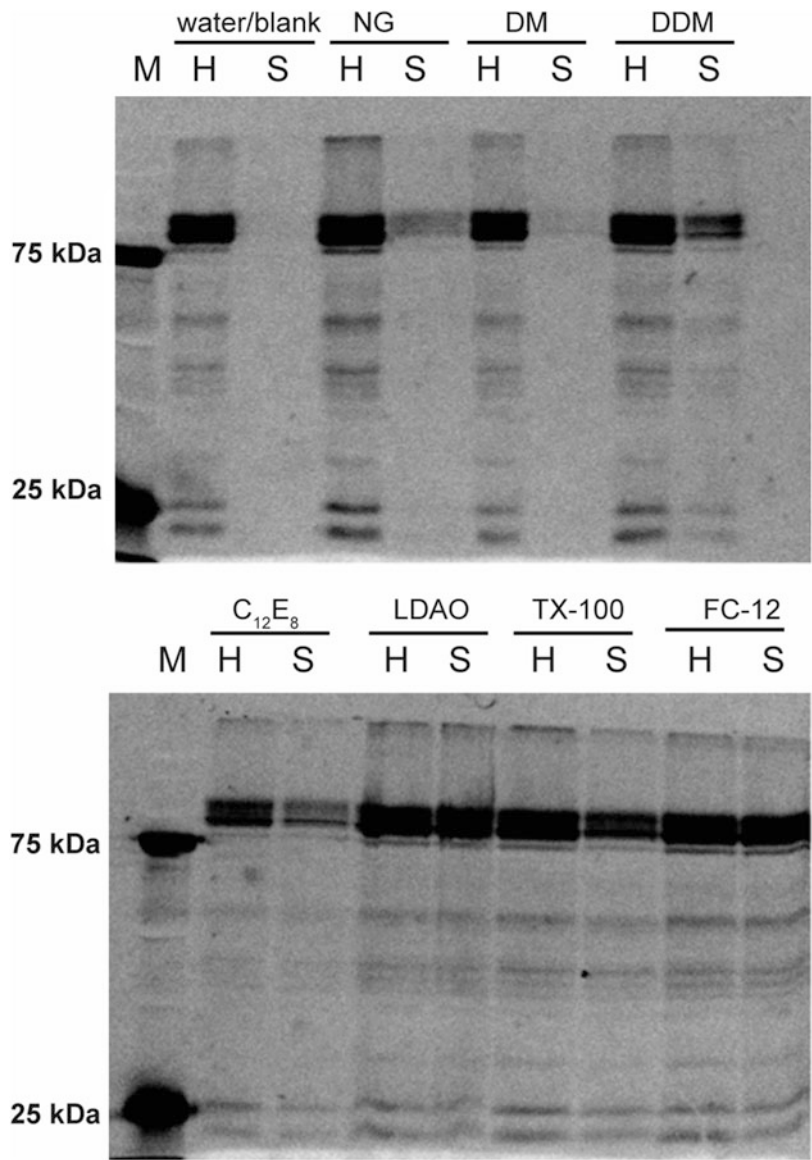


Fig. 3 PARL-FL-GFP in-gel fluorescence solubility test. Yeast membranes with PARL-FL-GFP were homogenized and solubilized in different detergents (*NG* neopentyl glycol, *DM* n-decyl- β -D-maltopyranoside, *DDM* n-dodecyl- β -D-maltopyranoside, *C12E8* dodecyl octaethylene glycol ether, *LDAO* lauryldimethylamine-N-Oxide, *Tx-100* Triton X-100, *FC12* fos-choline-12). 14% SDS-PAGE gels were loaded, alternating homogenized fractions (H) and detergent solubilized homogenized fractions (S). 7.5 μ L of each sample was loaded with SDS loading dye. Gels were imaged under blue light with the ImageQuant Imager

Based on the percent solubilization (Table 2) and the homogeneity of the protein in an SDS-PAGE gel (Fig. 3), LDAO and FC-12 appeared to be the most efficient detergents for the solubilization of PARL-GFP. The commonly used detergent DDM was also selected for further

Table 2

Solubility of PARL-GFP in various detergents. Samples of PARL-GFP were analyzed for solubility in several detergents. Aliquots of each detergent sample were taken once homogenized and solubilized. Relative fluorescence was measured in a fluorimeter to assess the percent solubility in the following detergents. (*NG* neopentyl glycol, *DM* n-decyl- β -D-maltopyranoside, *DDM* n-dodecyl- β -D-maltopyranoside, *C12E8* dodecyl octaethylene glycol ether, *LDAO* lauryldimethylamine-N-oxide, *Tx-100* Triton X-100, *FC12* fos-choline-12)

Relative fluorescence	Water/blank	NG	DM	DDM	C ₁₂ E ₈	LDAO	Tx-100	FC12
Homogenized	9576	9398	9636	8405	9570	7449	6896	8614
Solubilized	117	1112	169	4718	4780	7944	4973	7297
% Solubilized	1.2	11.8	1.8	56.1	49.9	106.6	72.1	84.7

analysis due to its successful history in solubilizing membrane proteins and its compatibility for crystallization [10, 25].

- (c) *Fluorescence size exclusion chromatography (FSEC)*: Three promising detergents from the first two means were selected to determine the monodispersity of the protein in detergent (Fig. 4) [26].
- Centrifuge the samples of crude membranes with PARL-GFP from the solubilization step in the corresponding detergent (200 μ L) using membrane filters with a benchtop centrifuge at 4 °C until completely filtered.
 - Equilibrate an analytical Size Exclusion Chromatography column (Sephadex 200 10/300 was used by our lab) with 2 column volumes of gel-filtration buffer, containing 0.1% of the corresponding detergent.
 - Load the fusion protein on the column and run at 0.3 mL/min.
 - Collect fractions every 300 μ L and measure aliquots of each fraction (200 μ L) in a 96 well plate for relative fluorescence using a fluorescence plate reader, at an excitation wavelength of 488 nm and emission wavelength of 509 nm.

Analytical FSEC in 0.1% FC-12 and 0.1% DDM revealed PARL-FL-GFP to be 53% and 43% homogenous, respectively, making these detergents suitable for solubilization. FSEC in LDAO resulted in a large fluorescent peak at void volume, suggesting that PARL-FL-GFP aggregated in LDAO. DDM was chosen for PARL purification not only due to feasibility but also because it has a low critical micellar concentration (CMC) and is considered to be one of the gentler detergents for maintaining functional properties of membrane proteins. DDM is a nonionic detergent

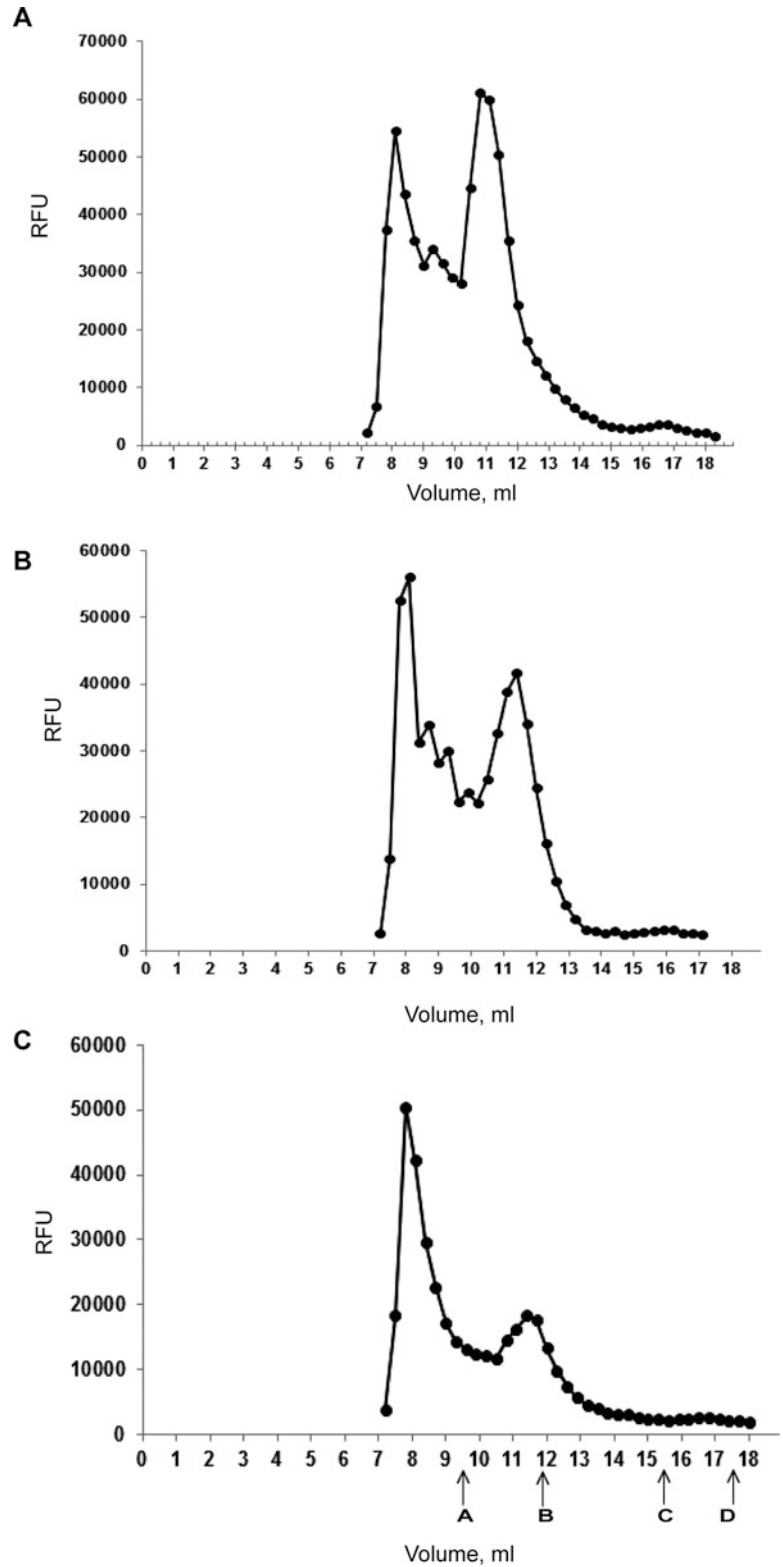


Fig. 4 Fluorescence size exclusion chromatography of PARL-FL-GFP. FSEC in fos-choline-12 (a), n-dodecyl β -D-maltopyranoside (b) and lauryldimethylamine-N-oxide (c). Analytical fluorescent size exclusion chromatography was used to

with a maltose head group and C₁₂ aliphatic chain; because of its long alkyl chain, DDM is able to fully mask the transmembrane regions of a protein, which is important as even partial exposure of these regions to the aqueous phase will lead to hydrophobic mismatch and protein aggregation. Therefore, DDM is a commonly used detergent for initial solubilization of membrane proteins [25].

3.6 IMAC Purification of PARL

The choice of construct with a hexahistidine tag at the C-terminus allows for purification of PARL-GFP using immobilized metal affinity chromatography (IMAC). This method is known to provide a very good yield and high purification factor. Moreover, IMAC is compatible with various detergents and buffer compositions, permitting optimization in accordance with research goals. Here we describe an example of PARL Δ 77 purification.

1. Homogenize membranes in Solubilization buffer in a 10:1 buffer volume to pellet weight ratio using a Dounce homogenizer until fully resuspended.
2. Transfer the solution into a beaker and add solid DDM (or a 20% (w/v) stock solution) to a final concentration of 1% (w/v) with gentle stirring at 4 °C for 1–2 h.
3. Remove insoluble material by high-speed centrifugation at 100,000 $\times g$, 4 °C for 30 min.
4. Equilibrate 2 mL of settled Co²⁺ IMAC resin with 2 column volumes of Buffer A.

Both Ni²⁺ and Co²⁺ have six metal coordination sites and both can be used for purification of His-tagged proteins. However, Ni²⁺ has a higher affinity for polyhistidine tags, especially when the tags are not fully exposed, leading to binding of more contaminating proteins. The major contaminating protein of yeast expression system from IMAC purification appears to be alcohol dehydrogenase (ADH). The MW of it is 36 kDa, which is very close to the MW of pure PARL Δ 77. On SDS-PAGE gels, ADH masks PARL and complicates purification. Interestingly, Co²⁺ beads do not bind ADH (Fig. 5) and therefore a



Fig. 4 (continued) analyze the optimal detergent solubilization. Filter-centrifuged crude membranes of PARL-FL-GFP (200 μ L) solubilized in selected detergent was loaded on a Superdex 200 10/300 in 50 mM KPO₄, 5% glycerol, 0.1 M NaCl, 10 mM β ME, 0.1% detergent. Fractions were collected every 300 μ L. Relative fluorescence was measured separately in a fluorimeter. The column void volume was 8.32 mL. The arrows represent molecular weight standards: A—Thyroglobulin (MW 670 kDa; Stokes radius 85 Å); B— γ -Globulin (MW 158 kDa; Stokes radius 52.9 Å); C—Ovalbumin (MW 44 kDa; Stokes radius 30.5 Å); D—Myoglobin (MW 17 kDa; Stokes radius 20.7 Å)

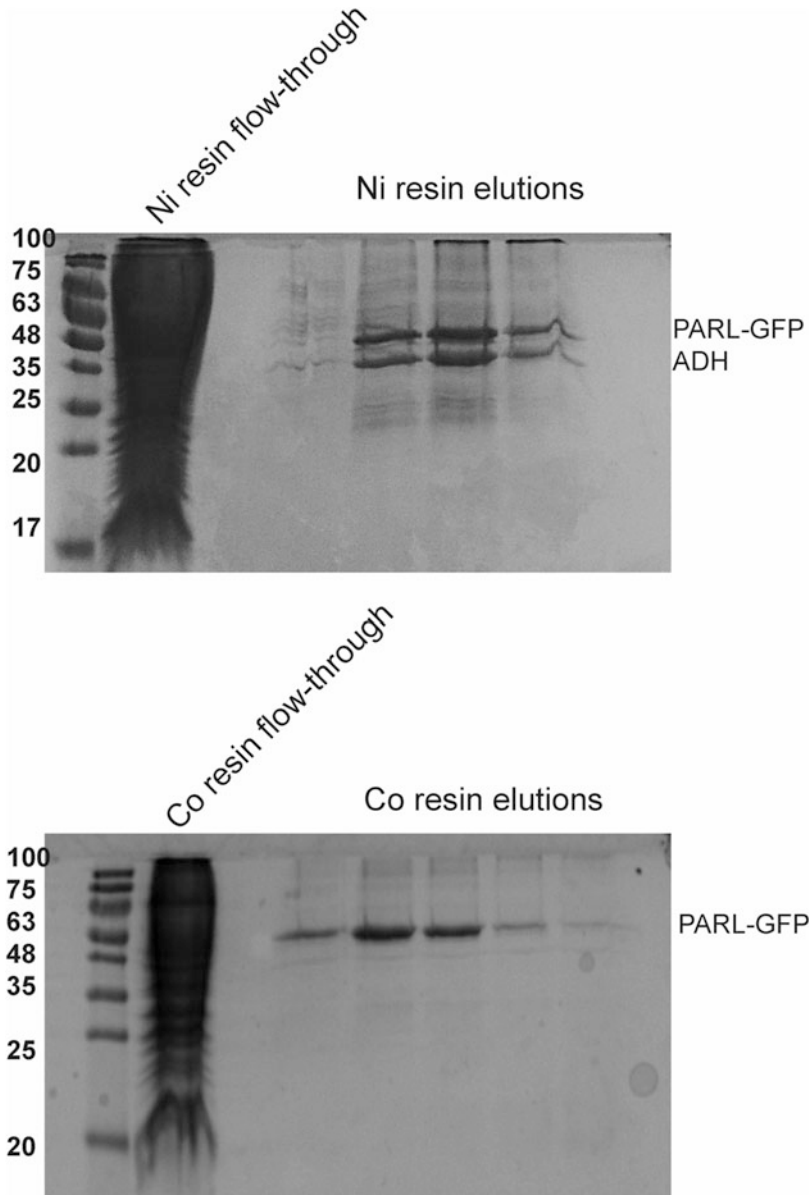


Fig. 5 SDS-PAGE of PARL Δ 77-GFP purifications. Purification of PARL Δ 77-GFP using either Ni²⁺-NTA (a) or HisPurTM Co²⁺ resin (b). Ni²⁺-NTA or HisPurTM Co²⁺ resin was incubated with DDM- solubilized membrane fractions containing PARL Δ 77-GFP. The resin was then washed and protein was eluted with 0.5 M imidazole. Fluorescence of the gels was visualized under blue light with an ImageQuant Imager to identify GFP and then gels were stained with Coomassie blue and destained in 15% acetic acid

Co²⁺-affinity column was used for PARL Δ 77-GFP purification.

5. Pass the supernatant from the high-speed centrifugation step through 2 mL of settled Co²⁺ resin by gravity flow (two times) to allow binding of PARL Δ 77-GFP-His to the resin (*see Note 10*).

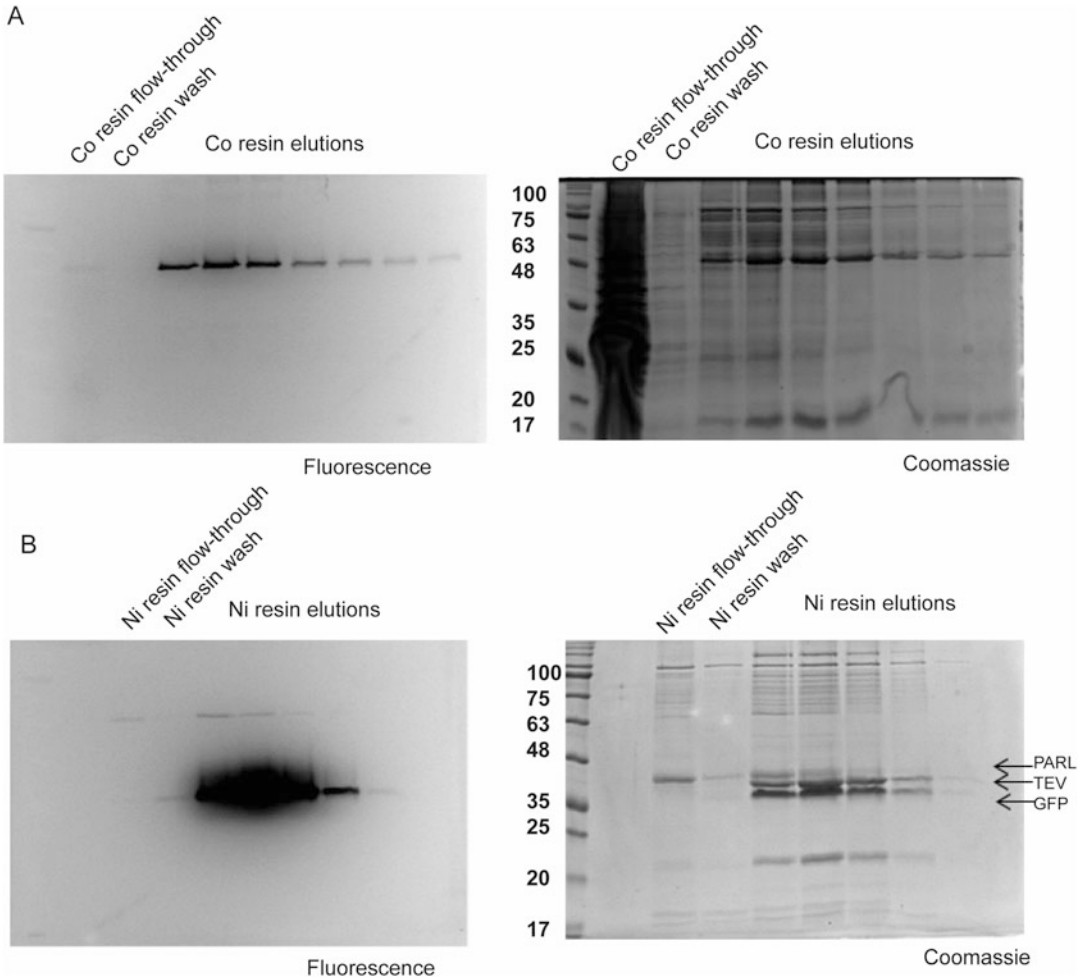


Fig. 6 SDS-PAGE of PARL Δ 77 IMAC purification steps. Purification fractions from HisPurTM Co²⁺ resin (**a**) and Ni²⁺-NTA resin (**b**) were loaded on 14% SDS-PAGE gels and run at 170 V. Fluorescence of the gels was visualized under blue light with an ImageQuant Imager to identify GFP and then gels were stained with Coomassie blue and destained in 15% acetic acid. Purified PARL Δ 77 is found in the Ni²⁺-resin column flow-through (FT)

6. Wash the protein-bound resin with 20 mM imidazole in Buffer A to disrupt any non-specific binding.
7. Elute the protein of interest with Buffer B in 1 mL fractions (PARL-GFP has a green colour and elution can easily be monitored by eye) until eluted fractions are clear (*see Note 11*).
8. Analyze cobalt column fractions by SDS-PAGE and pool elution fractions containing PARL-GFP (Fig. 6a).
9. Digest the purified PARL-GFP fusion protein by incubating the protein sample with TEV protease and 1 mM TCEP overnight at 4 °C (*see Note 12*).

10. Dialyze the protein sample for 2 h with stirring at 4 °C to remove imidazole and TCEP in 1 L of dialysis buffer.
11. Purify PARL from free GFP and TEV using Ni-NTA agarose resin. Apply the dialyzed protein sample on the Ni²⁺ resin and collect flow-through. Analyze flow-through by SDS-PAGE to assess PARL purity (Fig. 6b). Both free GFP and TEV have C-terminal His- tags and will bind to the resin. PARL will be found in the flow-through. If the flow-through still has a green shade, perform a second negative Ni²⁺ column the same way.
12. Determine the concentration of PARL via a BCA protein assay kit.
13. If the determined concentration of PARL is lower than 0.7 mg/mL, concentrate PARL using a 10,000 MWCO concentrator. PARL can be concentrated up to 1–1.5 mg/mL without significant aggregation.
14. Add *E. coli* polar lipids to purified PARL to a final lipid concentration of 0.1 mg/mL. Lipids are usually dissolved in organic solvents. Thus, aliquot necessary amount of lipids into a glass tube using a Hamilton syringe, dry gently with air stream until the lipid forms a thin layer and then add purified PARL. Incubate on ice for 30 min (*see* **Note 13**).
15. Aliquot protein-lipid sample, flash freeze in liquid nitrogen, and store at –80 °C.

4 Notes

1. The advantage of having a C-terminal GFP fusion protein is the ability to monitor and visualize expression of our protein as induction is occurring by measuring the fluorescence. In addition, because GFP is at the C-terminus of the protein of interest, the presence of fluorescence indicates proper folding of the fusion protein [27].
2. PARL is 379 amino acid protein; several truncated forms of PARL have been identified in vivo. Upon removal of the MTS, the resulting PARL Δ 53 truncation is considered the mature form of the protease and is the predominant species present in lung, brain, heart, and muscle tissues [28]. A second processing event occurs at Ser77, termed β -cleavage, resulting in the PARL Δ 77 truncation of the enzyme, which is observed in spleen, lung, brain, and kidney tissues [28]. This β -cleavage event is proposed to be autocatalytic as mutation of Ser277, the catalytic serine, results in reduced formation of PARL Δ 77 [28–30]. We started our initial screening with full length protein; however, this form of enzyme resulted in a very low expression yield. Generating PARL Δ 53 and PARL Δ 77 significantly

increased the expression level. Therefore, the detergent optimization experiments were performed using PARL-FL-GFP and the later purification steps are described using PARL $\Delta 77$.

3. PmeI was chosen because it cuts the pPICZ-PARL-GFP plasmid in only one site, providing efficient linearization.
4. At the final resuspension step, water must be used for efficient transformation. The presence of salts can lower electroporation efficiency and cause arching.
5. If transformation is efficient, plating 15 μ L yields about 50–100 isolated colonies and plating any higher volumes results in too high of a colony density. The 200 μ L plate is recommended, just in case the transformation was very inefficient. Plating $2 \times 5 \mu$ L, $2 \times 15 \mu$ L, $1 \times 50 \mu$ L, and $1 \times 200 \mu$ L is optimal for maximum screening potential.
6. It is not necessary to add Zeocin on the plate after the initial screen because transformants are stable, but it will prevent contamination from non-transformants.
7. The time required to see expression is variable—it takes between 24 to 48 h to see black colonies. This variability can arise from the difference in cell amount deposited to each spot on the grid, as well as transfer of any agar chunks to the BMMY plate because the yeast will continue to grow off the dextrose before using methanol.
8. If your culture has bacterial contamination, it will not pellet at $500 \times g$ and the supernatant will be cloudy.
9. The selection of detergents was critical to determine which class was most preferable for the protein. OG represents glucosides, DDM and DM are maltosides, FC12 is a lipid-like detergent, LDAO is zwitterionic, and C12E8 and Triton X-100 are polyoxyethylenes (Anatrace). A key factor for successful membrane protein extraction is the critical solubilization concentration, which is the minimal concentration of detergent that is required to transfer a membrane system into micellar dispersion. The critical solubilization concentration of a detergent is highly dependent on initial lipid concentration and is usually individually calculated based on the CMC of the detergent.
10. The amount of Co^{2+} resin to be used in the preparation is determined empirically. The binding capacity of Co^{2+} -NTA (or HisPur™ Cobalt resin) is protein dependent, but it is significantly lower for membrane proteins than for soluble proteins. For PARL-GFP, we determined that 2 mL of settled resin is an optimal amount to capture the recombinant protein expressed from 6 L of yeast culture, without binding a lot of contaminating proteins and being able to tolerate DDM.

11. In the case of PARL purification, we avoid a gradient or step elution. Co^{2+} resin is more specific than Ni^{2+} resin and, if the amount of resin is optimized (*see Note 10*), the eluted sample is not highly contaminated. A more important point to consider in this case is that the subsequent steps of purification require concentrating the protein sample, which can lead to aggregation. Moreover, DDM is a detergent with a low CMC and high micelle molecular weight (MW). Therefore, it is concentrated along with the protein, which leads to an increasing concentration of detergent in the protein sample, and again may cause aggregation. Therefore, it is crucial to obtain the eluted protein in a minimal volume.
12. Tobacco etch virus (TEV) protease was chosen to remove the His-tag for several reasons. TEV protease is very specific [31] and leaves only a serine or glycine residue after digestion [32] of its recognition sequence. Additionally, it is easy to recombinantly express and purify in large quantities [33] with a His-tag for purification; hence, there is no problem removing it from the digested protein sample using Ni-NTA resin.
13. The functionality of integral membrane proteins is often affected by lipid molecules [11, 34] that directly surround the membrane proteins in the membrane. PARL is very sensitive to lipid presence and loses activity much faster if left in a DDM-only environment.

References

1. Freeman M (2016) Rhomboids, signalling and cell biology. *Biochem Soc Trans* 44 (3):945–950. <https://doi.org/10.1042/BST20160035>
2. Jin SM, Lazarou M, Wang C, Kane LA, Narendra DP, Youle RJ (2010) Mitochondrial membrane potential regulates PINK1 import and proteolytic destabilization by PARL. *J Cell Biol* 191(5):933–942. <https://doi.org/10.1083/jcb.201008084>
3. Sekine S, Kanamaru Y, Koike M, Nishihara A, Okada M, Kinoshita H, Kamiyama M, Maruyama J, Uchiyama Y, Ishihara N, Takeda K, Ichijo H (2012) Rhomboid protease PARL mediates the mitochondrial membrane potential loss-induced cleavage of PGAM5. *J Biol Chem* 287(41):34635–34645. <https://doi.org/10.1074/jbc.M112.357509>
4. Dusterhoft S, Kunzel U, Freeman M (2017) Rhomboid proteases in human disease: mechanisms and future prospects. *Biochim Biophys Acta* 1864(11 Pt B):2200–2209. <https://doi.org/10.1016/j.bbamcr.2017.04.016>
5. Chen G, Han Z, Feng D, Chen Y, Chen L, Wu H, Huang L, Zhou C, Cai X, Fu C, Duan L, Wang X, Liu L, Liu X, Shen Y, Zhu Y, Chen Q (2014) A regulatory signaling loop comprising the PGAM5 phosphatase and CK2 controls receptor-mediated mitophagy. *Mol Cell* 54(3):362–377. <https://doi.org/10.1016/j.molcel.2014.02.034>
6. Yamano K, Youle RJ (2013) PINK1 is degraded through the N-end rule pathway. *Autophagy* 9(11):1758–1769. <https://doi.org/10.4161/auto.24633>
7. Saita S, Nolte H, Fiedler KU, Kashkar H, Venne AS, Zahedi RP, Kruger M, Langer T (2017) PARL mediates Smac proteolytic maturation in mitochondria to promote apoptosis. *Nat Cell Biol* 19(4):318–328. <https://doi.org/10.1038/ncb3488>
8. Saita S, Tatsuta T, Lampe PA, Konig T, Ohba Y, Langer T (2018) PARL partitions the lipid transfer protein STARD7 between the cytosol and mitochondria. *EMBO J* 37(4). <https://doi.org/10.15252/embj.201797909>

9. Koth CM, Payandeh J (2009) Strategies for the cloning and expression of membrane proteins. *Adv Protein Chem Struct Biol* 76:43–86. [https://doi.org/10.1016/S1876-1623\(08\)76002-4](https://doi.org/10.1016/S1876-1623(08)76002-4)
10. Bill RM, Henderson PJ, Iwata S, Kunji ER, Michel H, Neutze R, Newstead S, Poolman B, Tate CG, Vogel H (2011) Overcoming barriers to membrane protein structure determination. *Nat Biotechnol* 29 (4):335–340. <https://doi.org/10.1038/nbt.1833>
11. Lee AG (2004) How lipids affect the activities of integral membrane proteins. *Biochim Biophys Acta* 1666(1–2):62–87. <https://doi.org/10.1016/j.bbame.2004.05.012>
12. Brooks CL, Morrison M, Joanne Lemieux M (2013) Rapid expression screening of eukaryotic membrane proteins in *Pichia pastoris*. *Protein Sci* 22(4):425–433. <https://doi.org/10.1002/pro.2223>
13. Girke C, Arutyunova E, Syed M, Traub M, Mohlmann T, Lemieux MJ (2015) High yield expression and purification of equilibrative nucleoside transporter 7 (ENT7) from *Arabidopsis thaliana*. *Biochim Biophys Acta* 1850 (9):1921–1929. <https://doi.org/10.1016/j.bbagen.2015.06.003>
14. Hino T, Arakawa T, Iwanari H, Yurugi-Kobayashi T, Ikeda-Suno C, Nakada-Nakura Y, Kusano-Arai O, Weyand S, Shimamura T, Nomura N, Cameron AD, Kobayashi T, Hamakubo T, Iwata S, Murata T (2012) G-protein-coupled receptor inactivation by an allosteric inverse-agonist antibody. *Nature* 482(7384):237–240. <https://doi.org/10.1038/nature10750>
15. Shimamura T, Shiroishi M, Weyand S, Tsujimoto H, Winter G, Katritch V, Abagyan R, Cherezov V, Liu W, Han GW, Kobayashi T, Stevens RC, Iwata S (2011) Structure of the human histamine H1 receptor complex with doxepin. *Nature* 475 (7354):65–70. <https://doi.org/10.1038/nature10236>
16. Miller AN, Long SB (2012) Crystal structure of the human two-pore domain potassium channel K2P1. *Science* 335(6067):432–436. <https://doi.org/10.1126/science.1213274>
17. Long SB, Campbell EB, Mackinnon R (2005) Crystal structure of a mammalian voltage-dependent Shaker family K⁺ channel. *Science* 309(5736):897–903
18. Ho JD, Yeh R, Sandstrom A, Chorny I, Harries WE, Robbins RA, Miercke LJ, Stroud RM (2009) Crystal structure of human aquaporin 4 at 1.8 Å and its mechanism of conductance. *Proc Natl Acad Sci U S A* 106(18):7437–7442. <https://doi.org/10.1073/pnas.0902725106>
19. Tornroth-Horsefield S, Wang Y, Hedfalk K, Johanson U, Karlsson M, Tajkhorshid E, Neutze R, Kjellbom P (2006) Structural mechanism of plant aquaporin gating. *Nature* 439 (7077):688–694. <https://doi.org/10.1038/nature04316>
20. Aller SG, Yu J, Ward A, Weng Y, Chittaboina S, Zhuo R, Harrell PM, Trinh YT, Zhang Q, Urbatsch IL, Chang G (2009) Structure of P-glycoprotein reveals a molecular basis for poly-specific drug binding. *Science* 323 (5922):1718–1722. <https://doi.org/10.1126/science.1168750>
21. Norden K, Agemark M, Danielson JA, Alexandersson E, Kjellbom P, Johanson U (2011) Increasing gene dosage greatly enhances recombinant expression of aquaporins in *Pichia pastoris*. *BMC Biotechnol* 11:47. <https://doi.org/10.1186/1472-6750-11-47>
22. Cereghino JL, Cregg JM (2000) Heterologous protein expression in the methylotrophic yeast *Pichia pastoris*. *FEMS Microbiol Rev* 24 (1):45–66. <https://doi.org/10.1111/j.1574-6976.2000.tb00532.x>
23. Hjelmeland LM (1990) Solubilization of native membrane proteins. *Methods Enzymol* 182:253–264
24. Wagner S, Bader ML, Drew D, de Gier JW (2006) Rationalizing membrane protein overexpression. *Trends Biotechnol* 24(8):364–371. <https://doi.org/10.1016/j.tibtech.2006.06.008>
25. Carpenter EP, Beis K, Cameron AD, Iwata S (2008) Overcoming the challenges of membrane protein crystallography. *Curr Opin Struct Biol* 18(5):581–586. <https://doi.org/10.1016/j.sbi.2008.07.001>
26. Kawate T, Gouaux E (2006) Fluorescence-detection size-exclusion chromatography for precrystallization screening of integral membrane proteins. *Structure* 14(4):673–681. <https://doi.org/10.1016/j.str.2006.01.013>
27. Drew DE, von Heijne G, Nordlund P, de Gier JW (2001) Green fluorescent protein as an indicator to monitor membrane protein overexpression in *Escherichia coli*. *FEBS Lett* 507 (2):220–224. [https://doi.org/10.1016/S0014-5793\(01\)02980-5](https://doi.org/10.1016/S0014-5793(01)02980-5)
28. Jeyaraju DV, McBride HM, Hill RB, Pellegrini L (2011) Structural and mechanistic basis of Parl activity and regulation. *Cell Death Differ* 18(9):1531–1539. <https://doi.org/10.1038/cdd.2011.22>
29. Jeyaraju DV, Xu L, Letellier MC, Bandaru S, Zunino R, Berg EA, McBride HM, Pellegrini L (2006) Phosphorylation and cleavage of presenilin-associated rhomboid-like protein (PARL) promotes changes in mitochondrial morphology. *Proc Natl Acad Sci U S A* 103

- (49):18562–18567. <https://doi.org/10.1073/pnas.0604983103>. 0604983103 [pii]
30. Sik A, Passer BJ, Koonin EV, Pellegrini L (2004) Self-regulated cleavage of the mitochondrial intramembrane-cleaving protease PARL yields Pbeta, a nuclear-targeted peptide. *J Biol Chem* 279(15):15323–15329. <https://doi.org/10.1074/jbc.M313756200>. M313756200 [pii]
 31. Parks TD, Leuther KK, Howard ED, Johnston SA, Dougherty WG (1994) Release of proteins and peptides from fusion proteins using a recombinant plant virus proteinase. *Anal Biochem* 216(2):413–417. <https://doi.org/10.1006/abio.1994.1060>
 32. Kapust RB, Routzahn KM, Waugh DS (2002) Processive degradation of nascent polypeptides, triggered by tandem AGA codons, limits the accumulation of recombinant tobacco etch virus protease in *Escherichia coli* BL21(DE3). *Protein Expr Purif* 24(1):61–70. <https://doi.org/10.1006/prep.2001.1545>
 33. Tropea JE, Cherry S, Waugh DS (2009) Expression and purification of soluble His(6)-tagged TEV protease. *Methods Mol Biol* 498:297–307. https://doi.org/10.1007/978-1-59745-196-3_19
 34. Lemieux MJ, Huang Y, Wang DN (2004) The structural basis of substrate translocation by the *Escherichia coli* glycerol-3-phosphate transporter: a member of the major facilitator superfamily. *Curr Opin Struct Biol* 14(4):405–412



Chapter 2

Reconstitution of Detergent-Solubilized Membrane Proteins into Proteoliposomes and Nanodiscs for Functional and Structural Studies

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Abstract

Reconstitution of detergent-solubilized membrane proteins into phospholipid bilayers allows for functional and structural studies under close-to-native conditions that greatly support protein stability and function. Here we outline the detailed steps for membrane protein reconstitution to result in proteoliposomes and nanodiscs. Reconstitution can be achieved via a number of different strategies. The protocols for preparation of proteoliposomes use detergent removal via dialysis or via nonpolar polystyrene beads, or a mixture of the two methods. In this chapter, the protocols for nanodiscs apply polystyrene beads only. Proteoliposome preparation methods allow for substantial control of the lipid-to-protein ratio, from minimal amounts of phospholipid to high concentrations, type of phospholipid, and mixtures of phospholipids. In addition, dialysis affords a fairly large degree of control and variation of parameters such as rate of reconstitution, temperature, buffer conditions, and proteoliposome size. For the nanodisc approach, which is highly advantageous for ensuring equal access to both membrane sides of the protein as well as fast reconstitution of only a single membrane protein into a well-defined bilayer environment in each nanodisc, the protocols outline how a number of these parameters are more restricted in comparison to the proteoliposome protocols.

Key words Membrane protein, Reconstitution, Detergent, Phospholipids, Bilayer, Dialysis, Nonpolar polystyrene beads, Nanodiscs, Proteoliposomes

1 Introduction

Reconstitution of purified, detergent-solubilized membrane proteins into a phospholipid bilayer is a critical tool for functional and structural studies of membrane proteins. Reconstitution mimics the important biological environment of membrane proteins, where the specific membrane surroundings and components can

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be critical to protein function and/or necessary for experimental conditions. One example for which this was characterized in impressive detail is the TRPV1 ion channel [1]. In addition, the resulting phospholipid bilayer environment often greatly stabilizes the membrane protein in comparison to the detergent-solubilized protein [2, 3].

The reconstitution protocols in this chapter cover reconstitution by detergent-removal by dialysis as well as via nonpolar polystyrene beads to prepare proteoliposomes, reconstitution into nanodiscs to prepare small and larger volumes of membrane-protein nanodiscs, and the prerequisite preparation of phospholipid and polystyrene bead (Bio-Bead SM2™) stocks. Protocols are chosen based on the desired outcome in terms of number of reconstituted membrane proteins per bilayer, amount and type of phospholipid, size of membrane, rate of reconstitution, and accessibility of the membrane protein (Table 1). Reconstitution via detergent removal by dialysis allows for substantial variation in rate and often control of the resulting membrane size, from nanometers to microns [2–7]. The disadvantage of dialysis can be the longer timeframes of often days or more [2, 3]. Polystyrene beads [8, 9] have both the advantage and disadvantage of rapid detergent removal within hours or less, as the fast rate facilitates experiments, but the use of beads may limit membrane size. Proteoliposomes are typically closed, spherical membranes with membrane proteins in both membrane orientations, limiting the accessibility of the proteins, whereas nanodiscs make the membrane proteins accessible from both sides (Fig. 1). Furthermore, reconstitution into nanodiscs allows for the controlled reconstitution of individual membrane proteins or membrane protein complexes in a highly controlled environment within each nanodisc [10–12], allowing preparations for a large range of biochemical and biophysical experiments [13].

The protocols in this chapter focus on reconstitution. Thus, the use of native nanodiscs [14] is not discussed here. Native nanodiscs promise substantial advantages as these types of nanodiscs retain the native membrane environment without the need to remove the membrane protein from its immediate membrane surroundings of annular phospholipids via detergents [15, 16]. On the other hand, the presence of annular phospholipids in native nanodiscs does not offer the substantial degree of control over the phospholipid environment as reconstitution of detergent-solubilized membrane proteins with no or minimally copurified phospholipids does.

The protocols for phospholipid stock preparation and reconstitution into proteoliposomes build on our experience with reconstitution for two-dimensional (2D) crystallization of eukaryotic membrane proteins [2, 17–22]. Two-dimensional crystallization requires the membrane protein to be reconstituted under highly controlled conditions to induce 2D crystallization and at the same time ensure protein stability during and after long periods of

Table 1
Choice of reconstitution method

	Proteoliposomes	Proteoliposomes	MP nanodiscs
Detergent-removal method	Dialysis	Polystyrene beads	Polystyrene beads
Rate	Often >24 h, days, or weeks ^a	1–4 h	1–4 h
Number of MP reconstituted	Few to 100,000 s	Few to 100,000 s	1
Membrane size	Tens of nm to several μm^b	Tens of nm to several μm^b	nm-range
Control of membrane size	Substantial ^b	Limited	Uniform
Resulting volume	50 μL to several mL <50 μL^c	50 μL to several mL	50 μL to several mL

^aDependent on detergent type and concentration

^bThe proteoliposome size is determined by parameters such as the type and concentration of the solubilizing detergent and salt(s) as well as glycerol concentration. The size variation may be substantial, but can be controlled in a subsequent step via filtering

^cWith dialysis buttons

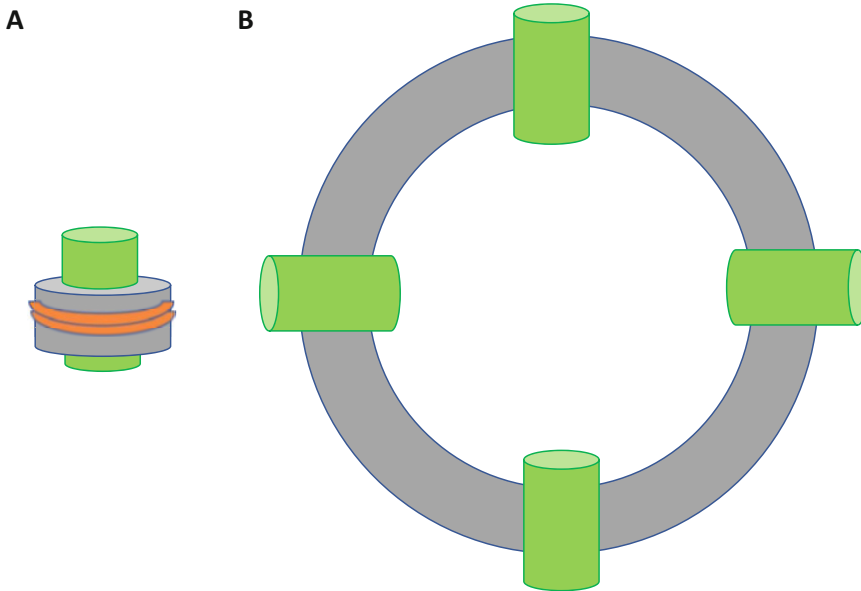


Fig. 1 Membrane protein reconstitution into (a) nanodiscs and (b) proteoliposomes. Nanodiscs have a defined diameter and are accessible from both membrane sides. Proteoliposomes are closed, spherical membranes with membrane proteins in both membrane orientations, limiting the accessibility of the membrane proteins

incubation. In addition, substantial modifications to the bilayer content and concentration are often required and thus considerable changes to buffers and incubation temperatures are necessary. The protocols for nanodisc reconstitution use our phospholipid preparation protocol and build largely on the pioneering work of Steven Sligar in developing the Membrane Scaffold Protein [10–13, 23], with several subsequent modifications for our membrane proteins of interest.

2 Materials

2.1 Preparation of Detergent-Solubilized Phospholipid Stock

1. 10 mg/mL phospholipid solubilized in chloroform (*see Note 1*).
2. 0.5% Na-deoxycholate in buffer or ultrapure water (*see Note 2*).
3. Chloroform for cleaning of glassware and Hamilton syringe.
4. Round-bottom flask with stopper.
5. Cork ring holder for round-bottom flask, or aluminum foil shaped as a stand.
6. Hamilton syringe.
7. Nitrogen gas.
8. Tygon tubing.
9. Water bath sonicator with basket or flask holder.
10. Microcentrifuge tubes (500 μ L or 1 mL).

2.2 Reconstitution into Proteoliposomes by Dialysis

1. ~1–3 mg/mL purified, detergent-solubilized membrane protein (*see Note 3*).
2. Detergent-solubilized phospholipid stock (*see Note 1*).
3. Detergent-free purification buffer.
4. Beakers (2 L, 1 L, 400 mL).
5. Dialysis tubing (1 cm flat width and ~10,000 MWCO).
6. Dialysis clips, weighed or unweighed, 40 mm length (*see Note 4*).
7. Blunt plastic forceps.
8. Razor blade or scalpel.
9. Heating and cooling incubator.

2.3 Polystyrene Bead Stock Preparation for Proteoliposome and Nanodisc Reconstitution

1. 5 g Bio-BeadsTM SM-2, stored in desiccator.
2. 100% methanol.
3. Ultrapure water.
4. Conical tubes (50 mL).
5. Storage bottle with wide opening (50 mL or 100 mL).
6. Rocking platform shaker.

2.4 Reconstitution into Proteoliposomes via Polystyrene Beads

1. ~1–3 mg/mL purified, detergent-solubilized membrane protein (*see Note 3*).
2. Detergent-solubilized phospholipid stock (*see Note 1*).
3. Purification buffer with detergent, if protein concentration will be decreased.

4. BioBead™ stock.
5. Fine-tipped forceps.
6. Heating and cooling incubator.

2.5 Small-Volume Reconstitution (100 μ L or Less) into Nanodiscs

1. 100 μ L of ~1 mg/mL purified, detergent-solubilized membrane protein (*see Note 3*).
2. Detergent- and glycerol-free buffer (based on purification buffer).
3. 1 mg/mL Membrane Scaffold Protein (Sigma M7074-5MG) (*see Note 5*).
4. 10 mg/mL detergent-solubilized phospholipid stock (*see Note 1*).
5. Washed polystyrene bead stock in ultrapure water.
6. Fine-tipped forceps.
7. 1.5 mL microcentrifuge tubes (small-volume nanodisc reconstitution).
8. Heating and cooling incubator.

2.6 Large-Volume Reconstitution (1–3 mL) into Nanodiscs

1. 300 μ L of 1 mg/mL purified, detergent-solubilized membrane protein (*see Note 3*).
2. 1 mg/mL membrane scaffold protein (Sigma M7074-5MG).
3. 10 mg/mL detergent-solubilized phospholipid stocks (*see Note 1*).
4. 700 μ L detergent-free, glycerol-free buffer purification buffer (“buffer A”).
5. 0.35 g/mL (wet mass) washed bead stock in ultrapure water.
6. 2 mL or 5 mL microcentrifuge tubes.
7. Heating and cooling incubator.

3 Methods

Clean gloves must be worn at all times to avoid contamination. Handling of liquid nitrogen requires training and safety precautions. The following abbreviations are used:

MP	Membrane protein
PL	Phospholipid
D	Detergent
LPR	Lipid-to-protein ratio
DMPC	1,2-dimyristoyl-sn-glycero-3-phosphocholine

(continued)

DOPC	1,2-dioleoyl-sn-glycero-3-phosphocholine
POPC	1-palmitoyl-2-oleoyl-sn-glycero-3-phosphocholine
DPPC	1,2-dipalmitoyl-sn-glycero-3-phosphocholine
MSP	Membrane Scaffold Protein

3.1 Preparation of Detergent-Solubilized Phospholipid Stock

The resulting phospholipid (PL) stocks can be used for all reconstitution protocols in this chapter.

1. Clean and dry a round-bottom flask (*see Note 6*) and place it on a cork holder inside the fume hood.
2. Attach a new pipette tip with Parafilm to the Tygon tubing connected to the Nitrogen gas.
3. Carefully test a stream of nitrogen gas against a piece of paper or a gloved hand to ensure that the gas is blowing at a pressure that is weak enough to not spatter the chloroform and PL in the round-bottom flask.
4. Use nitrogen gas to replace the air inside the round bottom flask by flowing nitrogen gas into the flask. Cap the flask.
5. Pipet the chloroform-solubilized PL with a Hamilton syringe into the round bottom flask.
6. Carefully insert the pipette tip into the opening of the round-bottom flask and evaporate the chloroform by applying nitrogen gas for ~10–15 min, ensuring that the PLs form a thin film at the bottom of the flask (*see Note 7*).
7. Pipet the detergent solution to achieve the desired concentration and volume (*see Note 8*).
8. Gently tap the round-bottom flask, while ensuring that the detergent solution stays in the bottom of the flask (*see Note 9*).
9. Place the round-bottom flask containing the PL–detergent mixture upright in the basket of the sonicator and sonicate for ~5–15 min. The solution will become clear within 1–5 min or less (*see Note 10*).
10. Pipet detergent-solubilized PL aliquots of 10–20 μ L into Eppendorf tubes and flash-freeze the tubes by dropping them into a small dewar of liquid nitrogen.
11. Store at either -20°C or -80°C .

3.2 Preparation of Polystyrene Beads

The protocol outlines steps for the preparation of polystyrene beads (BioBeads SM2™) that are used for either reconstitution into proteoliposomes or into nanodiscs. The beads need to be handled carefully in order to achieve the highest degree of accuracy and reproducibility of protocols as the cleaned beads are stored in water. Beads can be either weighed or selected by size and counted.

Weighing is often the preferred option for large numbers of beads needed for an experiment, while individual selection of beads is chosen when only a small number of beads is required. To ensure complete detergent removal, the critical micelle concentration (CMC) and final concentration of the specific solubilizing-detergent for the membrane protein of interest require careful consideration and testing to determine the optimal volume or number of beads.

1. Carefully weigh 5 g of beads (*see Note 11*).
2. Place the beads in a 50 mL conical tube and fill the tube with methanol to the 50 mL marking.
3. Agitate the tube at the lowest setting on a rocking platform shaker for 12–24 h at room temperature.
4. Place the tube upright. Once the beads have settled at the bottom of the tube, use a Pasteur pipette to slowly and carefully aspirate the methanol in order to avoid loss of beads.
5. Add ultrapure water to the 50 mL mark and repeat **steps 3 and 4** for 4–6 days with ultrapure water.
6. After the last day of rinsing, replace the ultrapure water and store the beads in a bottle at 4 °C (*see Note 12*).

3.3 Reconstitution into Proteoliposomes by Dialysis

While this protocol covers the reconstitution of volumes in the 75–100 μ L range with the use of dialysis tubing, the protocol can be easily modified for smaller volumes as low as 15 μ L with the use of dialysis buttons or larger volumes of several milliliters or more with the use of longer and/or wider dialysis tubing.

1. Calculate the desired final LPR (*see Notes 13 and 14*).
2. Clean beakers, stirring rods, and dialysis clips with laboratory glassware detergent and thoroughly rinse items with distilled water to ensure complete removal of detergent. Rinse with deionized water and air-dry.
3. Prepare 250 mL of detergent-free dialysis buffer for each experiment. Add 250 mL of the buffer to a 400 mL beaker and cover the beaker with aluminum foil (*see Note 15*).
4. Cut 8–10 cm dialysis tubing for each experiment. Thoroughly rinse and hydrate the tubing in deionized water or a buffer identical to the one used for dialysis.
5. If protein samples are frozen, thaw them on ice, or prepare 75–100 μ L aliquots of the protein (*see Note 16*).
6. Thaw the detergent-solubilized PL on ice and add it to the thawed MP sample. Remove any residual PL from the pipette by gently pipetting the mixture to avoid bubble formation (*see Note 17*). For faster detergent removal during the dialysis with

the help of polystyrene beads, follow protocol 3.2 and steps 2 and 3 in protocol 3.4. The beads can be placed in the dialysis tubing or directly in the beaker.

7. Close the tube and vortex the MP–detergent–PL solution for ~10 s.
8. Incubate the MP–detergent–PL mixture (dialysate) on ice for 10 min.
9. Place a regular or weighed dialysis clip ~1 cm from one end of the hydrated dialysis tubing and hold the tubing upright with the clip at the bottom. Pipet the dialysate into the lower end of the dialysis tubing.
10. Place an unweighted, regular dialysis clip ~1 cm from the top at a 90° angle, which will ensure that the dialysate remains under the air–water interface.
11. Place the clipped tubing containing the sample under the air–water interface of the buffer in the 400 mL beaker and seal with Parafilm or aluminum foil.
12. Place the beakers in an incubator at the desired temperature (*see Note 18*).
13. After the detergent removal is complete, remove the beaker from the incubator.
14. Use blunt forceps to remove the dialysis tube from the beaker, hold the tubing upright, and cut the dialysis tubing in half with a razor blade or scalpel.
15. Transfer the sample to a microcentrifuge tube and seal with Parafilm (*see Notes 19 and 20*).
16. Store dialysate at room temperature, at 4 °C, or flash-freeze in liquid nitrogen for storage at –80 °C (*see Note 21*).

3.4 Reconstitution into Proteoliposomes via Polystyrene Beads

The following protocol is optimized for volumes of 100 µL. For slightly larger volumes, the number of beads is increased and larger tubes are used for incubation, or the sample is divided into 100-µL aliquots. The smaller aliquots facilitate reproducibility and sample recovery.

1. Follow steps 5–8 in protocol 3.3 to thaw, mix, and incubate the MP and PL on ice. The PL may be thawed at room temperature, depending on the lipid phase transition temperature (*see Notes 18 and 20*).
2. Pour a small amount (approximately 50–100) of beads into a petri dish and use forceps to select eight medium-sized beads per 100 µL sample (*see Note 22*).
3. Blot the beads with a clean laboratory tissue to remove excess water and place the beads into the MP–PL–D mixture.

4. Incubate the tube containing the mixture and the beads at a temperature above the phase transition temperature of the PL (*see Note 18*) for 2 h (*see Note 23*).
5. Remove the solution containing the proteoliposomes with a micropipette, leaving only the beads in the tube (*see Note 20*).
6. Depending on the MP, the resulting proteoliposomes are stored at 4 °C or at room temperature.

3.5 Reconstitution into Nanodiscs

The protocols below separately cover optimal conditions for membrane protein reconstitution into nanodiscs of smaller volumes of 100 μ L or less and reconstitution volumes in the milliliter range. While very similar in approach, key differences are included to achieve the best results for either smaller or larger volumes.

3.5.1 Small-Volume Reconstitution (100 μ L or Less) into Nanodiscs

Volumes of 100 μ L or less are typically prepared due to experimental needs, first nanodisc reconstitution tests for a membrane protein, and/or limited availability of a particular membrane proteins. This protocol outlines the preparation of small volumes (100 μ L at 1 mg/mL) of a membrane protein of choice reconstituted in nanodiscs.

1. Prepare the detergent- and glycerol-free buffer and calculate the desired MP-MSP-PL ratio (*see Note 24*).
2. Thaw MSP and the detergent-solubilized MP (unless newly purified) on ice and the PL stock at room temperature.
3. Gently mix the tubes containing the MP and MSP by tapping. Tap or vortex the PL stock.
4. Add the desired PL volume to the MP, vortex, and incubate on ice for 10 min.
5. Add eight medium-sized beads to an Eppendorf tube and add a sufficient volume of detergent- and glycerol-free buffer to bring the final solution, once the MP-MSP-PL mixture is added (**step 7**), to 100 μ L (*see Note 25*).
6. Add the MSP to the tube with the MP-PL solution and mix by pipetting.
7. Immediately add the MP-MSP-PL solution to the tube containing the beads and buffer.
8. Seal the microcentrifuge tube with Parafilm and incubate at a temperature at least 1 °C above the phase transition temperature of the PL. The incubation time for DMPC and DPPC is ~2 h. For POPC the incubation time is >4 h (*see Note 26*).
9. Use the samples for subsequent experiments or store the reconstituted nanodiscs at 4 °C (*see Note 27*).

3.5.2 Large-Volume (1–3 mL) Reconstitution into Nanodiscs

Volumes of 1–3 mL of sample can be prepared in batch for applications requiring larger volumes of MP nanodiscs. The following protocol was optimized to prepare 1 mL of MP nanodiscs of an MP purified and solubilized in dodecyl maltoside. The example uses 300 μ L of MP and 0.35 g/mL [24].

1. Thaw MSP and the MP (unless newly purified) on ice and thaw PL at room temperature (*see Note 28*).
2. Carefully mix MSP as well as the MP by tapping and tap or vortex the PL.
3. Add the PL to the tube containing the MP. Incubate on ice for 10 min.
4. Add 0.35 g/mL beads to buffer A.
5. Pipet MSP into buffer A, carefully mix by pipetting, and add MP-PL mixture, followed by mixing.
6. Pipet the MP-PL-MSP solution into the tube containing the beads and buffer A (*see Note 29*).
7. Seal the microcentrifuge tube with Parafilm. On a rocking platform, incubate overnight at a temperature set to approximately 1 °C above the phase transition temperature of the PL (*see Note 30*).
8. Pipet the MP nanodiscs from the tube containing the beads, leaving the beads to be discarded, and use immediately or store at 4 °C (*see Note 24*).

4 Notes

1. Phospholipid in chloroform is preferable due to the hygroscopic nature of phospholipids in powder form. We typically purchase concentrations of 1 mg/mL, 10 mg/mL or 25 mg/mL phospholipid in chloroform, depending on the final use. Phospholipid stock concentrations of 1 mg/mL and 10 mg/mL are useful for many lipid-to-protein ratios. The phospholipids 1,2-dimyristoyl-sn-glycero-3-phosphocholine (DMPC), 1,2-dioleoyl-sn-glycero-3-phosphocholine (DOPC), and 1-palmitoyl-2-oleoyl-sn-glycero-3-phosphocholine (POPC) have been successfully used for reconstitution of a large number of membrane proteins, where DMPC proved particularly successful for two-dimensional crystallization [2, 3]. DMPC, DOPC, and/or 1,2-dipalmitoyl-sn-glycero-3-phosphocholine (DPPC) stocks are prepared for the nanodisc reconstitution protocol. Nanodiscs can be produced from many types as well as mixtures of phospholipids [25].

2. A solubilizing detergent of 0.5% Na-deoxycholate was initially used for reconstitution in two-dimensional crystallization trials [17–19] and has since been used successfully for a large number of different membrane protein reconstitutions in our laboratory.
3. No or minimal copurified phospholipid is present for subsequent experiments requiring reconstitution into specific phospholipid and/or specific LPR conditions, or for the reconstitution into nanodiscs.
4. Use dialysis buttons for volumes below 50 μ L.
5. The original MSP constructs are used widely and result in nanodiscs with diameters of either 9.7 nm or 11.9 nm [12]. Shorter or longer MSP can be selected to modify the nanodisc diameter [23, 26] in combination with optimization of the LPR to obtain smaller or larger nanodiscs, respectively [11, 23]. Covalently circularized nanodiscs allow for the formation of nanodiscs of $\sim$ 50 nm [27].
6. Wash all beakers and cylinders with laboratory glass ware detergents, chloroform, and dry in a dust free environment.
7. Rotary evaporators can be used for this task as well as to remove any remaining solvent.
8. Phospholipid concentrations of 1 mg/mL are recommended for low LPRs as pipetting of volumes smaller than 1 μ L decrease the accuracy of the final concentration.
9. Pipet any solution containing detergents very gently to avoid the formation of bubbles.
10. A cloudy appearance of the phospholipid after sonication for several minutes indicates that the phospholipid cannot be used for successful reconstitution. It is highly recommended to use a new batch of phospholipids.
11. Do not leave the bottle of polystyrene beads open for an unnecessary amount of time and return it to the desiccator immediately after use.
12. Carefully washed beads may be used for up to 1 year or more. A rinsing step with ultrapure water may be repeated before use.
13. The choice of phospholipid and LPR will depend on the purpose of the reconstitution. DMPC, DOPC, and POPC can be used for reconstitution of many membrane proteins, or a mixture purified from or mimicking native membranes may be required. The LPR is calculated by weight or as a molar LPR. We prefer molar LPRs as reconstitution experiments in our laboratory often require minimal LPRs. For membrane proteins of $\sim$ 50 kDa at concentrations of 1 mg/mL with no or minimal copurified PLs, LPRs as low as 2–5 have resulted in

successful reconstitution with a high MP concentration that was sufficient for closely packed 2D crystals [17, 19]. Significantly higher LPRs are required to decrease the number of MPs in each proteoliposome.

14. Depending on the purpose of the reconstitution, especially when the PL composition is critical, membrane proteins may require careful assessment for copurified PLs before reconstitution. The amount of copurified lipid can vary between purifications. A control experiment for a new purification batch can include an LPR of 0 to test, which will result in proteoliposomes when copurified PLs are present. In these cases, either copurified PLs are removed with additional washing steps, or the LPR is adjusted.
15. Keep buffer components identical, or as close as possible, to the buffer used for purification. The size of membranes can frequently be increased by testing types and concentrations of divalent cations and glycerol concentrations of 10–20% [17–19].
16. Generally, 100 μL of sample is used to study each parameter. However, if the sample is precious, a smaller sample volume may be used. Volumes smaller than 50 μL are best dialyzed in dialysis buttons.
17. The PL may be thawed at room temperature or on ice, depending on the lipid phase transition temperature and the temperature sensitivity of the MP. For PLs thawed on ice, however, we have observed successful reconstitutions by dialysis with the use of PLs with a wide range of phase transition temperatures and MPs. Thus, this is our standard approach for thawing PLs for reconstitution by dialysis, including thawing of DMPC.
18. Dialysis temperatures are typically chosen above the phase transition temperature of the PL. For temperature-sensitive MPs that do not reconstitute in PLs with suitable phase transition temperatures, consider temperatures below the transition phase of the PL. We have achieved successful reconstitution with DMPC and DOPC at 4 °C [17]. The presence of PLs, even before reconstitution, often has a stabilizing effect on the MP though. This may allow for higher temperatures than those used for experiments with only MP-D mixtures. The length of the dialysis, which can span from hours to weeks, is determined based on the CMC of the detergent, the dialysis temperature, and comparable MP reconstitutions.
19. The sample volume before and after dialysis can vary. We have not found an effect on reconstitution with an increase or a decrease in volume after dialysis. An increased volume may require simple concentration of the proteoliposomes by centrifugation for several minutes with a tabletop centrifuge.

20. Protein concentrations can vary from far below 1 mg/mL to several mg/mL for reconstitution via polystyrene beads. For MPs available only at low concentrations, the resulting proteoliposomes can be concentrated by centrifugation after reconstitution and removal from the beads. The solution with the proteoliposomes is centrifuged at ~3000–5000 rpm for several minutes, followed by removal of part of the supernatant and vortexing of the remaining microcentrifuge content.
21. Storage conditions will depend on time of use and sample stability. Further processing, such as preparation of uniformly sized vesicles may occur at this point.
22. The number of beads may require optimization, depending on the detergent and detergent concentration used to solubilize the PL and MP as well as the desired rate of reconstitution.
23. Two hours are usually more than sufficient for complete detergent removal. The incubation time can be increased by 1–2 h at lower temperatures, or if excess detergent is observed. Alternatively, additional beads can be used instead of a prolonged incubation time.
24. The MSP–PL ratio is critical for forming MP-loaded nanodiscs. Good starting points are ratios of MSP and PL (i.e., DMPC 1:80, DPPC 1:90, or POPC 1:65) combined with a 1:4 ratio of MP:MSP (Richie et al. [24] provide several examples of different membrane proteins). If a transmission electron microscope (TEM) is available, screening of the size exclusion peaks by negative stain TEM can aid in determining the optimal ratio. We have found this approach to be particularly fast and helpful for assessing the presence of excess lipids, which will result in liposomes rather than MP nanodiscs that can be identified by TEM within minutes. In order to optimize the MP–MSP–PL ratio for successful MP nanodisc formation, the PL concentration is decreased and the MP and MSP are kept at their original concentration.
25. We found that eight medium-sized beads are sufficient for forming membrane-protein-loaded nanodiscs for volumes of 100 μ L. Depending on the CMC and concentration of the detergent, the number of beads may need to be adjusted. Also note that it is critical to use glycerol-free buffer under these conditions.
26. We found that for small MP-MSP-DMPC volumes of 100 μ L, the incubation time can be decreased to 1 h. Depending on the MP and its solubilizing detergent type and concentration, the detergent-removal time may require optimization.
27. The integrity of nanodiscs upon storage can be assessed within minutes by transmission electron microscopy of negatively stained samples. Intact samples are then tested for activity.

28. For an MP with 12 transmembrane α -helices, molar ratios of MP, MSP, and PL of 1:11:30 have worked well as a starting point to determine volumes of MSP and PL.
29. Use a 2 mL microcentrifuge tube for 1 mL volumes (the volume of the solution, but not the bead volume) and a 5 mL tube for volumes of >1 mL.
30. Recommendations for incubation temperatures of commonly used phospholipids are 24 °C for DMPC, 37 °C for DPPC, and 4 °C for POPC.

References

1. Gao Y, Cao E, Julius D, Cheng Y (2016) TRPV1 structures in nanodiscs reveal mechanisms of ligand and lipid action. *Nature* 16:347–351
2. Schmidt-Krey I (2007) Electron crystallography of membrane proteins: two-dimensional crystallization and screening by electron microscopy. *Methods* 41:417–426
3. Kühlbrandt W (1992) Two-dimensional crystallization of membrane proteins. *Q Rev Biophys* 25:1–49
4. Jap BK, Zulauf M, Scheybani T, Hefti A, Baumeister W, Aepli U, Engel A (1992) 2D crystallization: from art to science. *Ultramicroscopy* 46:45–84
5. Engel A, Hoenger A, Hefti A, Henn C, Ford RC, Kistler J, Zulauf M (1992) Assembly of 2-D membrane protein crystals: dynamics, crystal order, and fidelity of structure analysis by electron microscopy. *J Struct Biol* 109:219–234
6. Vink M, Derr K, Love J, Stokes DL, Ubarretxena-Belandia I (2007) A high-throughput strategy to screen 2D crystallization trials of membrane proteins. *J Struct Biol* 160:295–304
7. Stahlberg H, Biyani N, Engel A (2015) 3D reconstruction of two-dimensional crystals. *Arch Biochem Biophys* 581:68–77
8. Rigaud J-L, Mosser G, Lacapère J-J, Olofsson A, Lévy D, Ranck J-L (1997) Bio-Beads: an efficient strategy for two-dimensional crystallization of membrane proteins. *J Struct Biol* 118:226–235
9. Rigaud J-L, Lévy D, Mosser G, Lambert O (1998) Detergent removal by non-polar polystyrene beads. *Eur Biophys J* 27:305–319
10. Bayburt TH, Carlson JW, Sligar SG (1998) Reconstitution and imaging of a membrane protein in a nanometer-size phospholipid bilayer. *J Struct Biol* 123:37–44
11. Bayburt TH, Grinkova YV, Sligar SG (2002) Self-assembly of discoidal phospholipid bilayer nanoparticles with membrane scaffold proteins. *Nano Lett* 2:853–856
12. Denisov IG, Grinkova YV, Lazarides AA, Sligar SG (2004) Directed self-assembly of monodisperse phospholipid bilayer nanodiscs with controlled size. *J Am Chem Soc* 126:3477–3487
13. Denisov IG, Sligar SG (2017) Nanodiscs in membrane biochemistry and biophysics. *Chem Rev* 117:4669–4713
14. Yeh V, Lee T-Y, Chen C-W, Kuo P-C, Shiue J, Chu L-K, Yu T-Y (2018) Highly efficient transfer of 7TM membrane protein from native membrane to covalently circularized nanodisc. *Sci Rep* 8:13501
15. Knowles TJ, Finka R, Smith C, Lin YP, Dafforn T, Overduin M (2009) Membrane proteins solubilized intact in lipid containing nanoparticles bounded by styrene maleic acid copolymer. *J Am Chem Soc* 131:7484–7485
16. Esmaili M, Acevedo-Morantes C, Wille H, Overduin M (2020) The effect of hydrophobic alkyl sidechains on size and solution behaviors of nanodiscs formed by alternating styrene maleamic copolymer. *Biochim Biophys Acta Biomembr* 1862:183360
17. Schmidt-Krey I, Lundqvist G, Morgenstern R, Hebert H (1998) Parameters for the two-dimensional crystallization of the membrane protein microsomal glutathione transferase. *J Struct Biol* 123:87–96
18. Schmidt-Krey I, Mutucumarana V, Haase W, Stafford DW, Kühlbrandt W (2007) Two-dimensional crystallization of human vitamin K-dependent γ -glutamyl carboxylase. *J Struct Biol* 157:437–442
19. Zhao G, Johnson MC, Schnell JR, Kanaoka Y, Haase W, Irikura D, Lam BK, Schmidt-Krey I (2010) Two-dimensional crystallization conditions of human leukotriene C₄ synthase

- requiring adjustment of a particularly large combination of specific parameters. *J Struct Biol* 169:450–454
20. Schmidt-Krey I, Rubinstein JL (2011) Electron cryomicroscopy of membrane proteins: specimen preparation for two-dimensional crystals and single particles. *Micron* 42:107–116
 21. Johnson MC, Schmidt-Krey I (2014) Towards a general protocol to form single-layered 2D crystal sheets of membrane proteins for electron crystallography. *Microsc Microanal* 20:1240–1241
 22. Uddin YM, Schmidt-Krey I (2015) Inducing two-dimensional crystallization of membrane proteins by dialysis for electron crystallography. *Methods Enzymol* 557:351–562
 23. Grinkova YV, Denisov IG, Sligar SG (2010) Engineering extended membrane scaffold proteins for self-assembly of soluble nanoscale lipid bilayers. *Protein Eng Des Sel* 23:843–848
 24. Ritchie TK, Grinkova YV, Bayburt TH, Denisov IG, Zolnerciks JK, Atkins WM, Sligar SG (2009) Chapter 11—reconstitution of membrane proteins in phospholipid bilayer nanodiscs. *Methods Enzymol* 464:211–231
 25. Rouck JE, Krapf JE, Roy J, Huff HC, Das A (2017) Recent advances in nanodisc technology for membrane protein studies (2012–2017). *FEBS Lett* 591:2057–2088
 26. Hagn F, Etzkorn M, Raschle T, Wagner G (2013) Optimized phospholipid bilayer nanodiscs facilitate high-resolution structure determination of membrane proteins. *J Am Chem Soc* 135:1919–1925
 27. Nasr ML, Baptista D, Strauss M, Sun Z-YJ, Grigoriu S, Huser S, Plückthun A, Hagn F, Walz T, Hogle JM, Wagner G (2017) Covalently circularized nanodiscs for studying membrane proteins and viral entry. *Nat Methods* 14:49–52



Chapter 3

Biochemical Characterization of GPCR–G Protein Complex Formation

Filip Pamula and Ching-Ju Tsai

Abstract

The complex of G protein-coupled receptors (GPCR) and G proteins is the core assembly in GPCR signaling in eukaryotes. With the recent development of cryo-electron microscopy, there has been a rapid growth in structures of GPCR–G protein complexes solved to near-atomic resolution, giving important insights into this signaling complex. Here we describe the biochemical protocol to study the interaction between GPCRs and G proteins before preparation of GPCR–G protein complexes for structural studies. We use gel filtration to analyze the binding properties between GPCR and G protein with the presence of agonist or antagonist, as well as the complex dissociation in the presence of GTP analogue. Methods used in the protocol are affinity purification and gel filtration, which are also commonly used in protein sample preparation for structural work. Therefore, the protocol can be easily adapted for large-scale sample preparation.

Key words GPCR, G protein, Ligand, Detergent, Affinity purification, Complex formation, Gel filtration

1 Introduction

G protein-coupled receptors (GPCRs) are a major membrane protein class in animals orchestrating signaling. GPCRs are divided to five major classes based on their sequence similarities: Rhodopsin-like (Class A); Secretin (Class B1) and Adhesion (Class B2); Glutamate (Class C); Frizzled (Class F) and Taste receptor 2 [1, 2]. To date, there are around 400 solved structures, and all of them possess the architecture of seven transmembrane helices. GPCRs bind the activating ligand, agonist, from the extracellular side, which triggers conformational changes in the cytoplasmic region to form a cleft for recruiting G proteins and arrestins [3]. The cleft is formed due to the outward movement of transmembrane helices 5 (TM5) and 6. Without binding of the signaling partner, the agonist-bound state is much less stable than the antagonist-bound

state due to the exposure of the residues at the binding interface. This also explains why the majority of the determined GPCR structures are in the antagonist-bound state.

Based on the current Protein Data Bank statistics (<https://www.rcsb.org/>), 90% of the GPCR structures were solved using crystallography. Since 2017, cryo-EM has contributed 7.7% of the GPCR structures to the total statistics due to the rapid development in EM imaging, data processing and implementation of EM facilities into research units worldwide [4, 5]. Those cryo-EM structures are either of dimeric receptors [6, 7], or of monomeric receptors coupled with G proteins [8–24] or arrestin [25, 26]. It shows the trend that cryo-EM has become a highly advantageous method to uncover the structures of GPCR signaling complexes to near-atomic resolution, because cryo-EM requires much less protein and also bypasses crystallization, where intrinsically flexible protein complexes are unlikely to form well-diffracting crystals.

Preparation of a GPCR–G protein complex is challenging due to the unstable behavior of an activated receptor as well as the complex formation. Here we develop a protocol using bovine rhodopsin mutant as the receptor template to study how antagonist and agonist affect the state of the receptor and the ability of G protein coupling (Fig. 1). First, opsin solubilized in detergent is purified by affinity purification chromatography (Fig. 2), followed by reconstitution with its antagonist 9-*cis* retinal or agonist all-*trans* retinal. The reconstitution of those light-sensitive ligands can be traced using ultraviolet–visible (UV-VIS) spectroscopy (Fig. 3). As rhodopsin can couple to the Gi/t/o family, the Gi protein heterotrimer is added to rhodopsin in order to form the rhodopsin–Gi complex. When the rhodopsin–Gi protein complex is formed, dissociation of the complex is triggered by adding GTP γ S, an analog of GTP. GTP binds to the G α subunit inducing a conformational change, which causes dissociation of the G protein from the receptor. Samples at different stages are evaluated using analytical gel filtration to identify the formation and dissociation of the rhodopsin–Gi complex (Fig. 4).

In humans, there are ~800 GPCRs but only 16 G α protein subtypes from four G protein families (Gs, Gi, Gq/11, G12/13) [27]. Most of the receptors can couple to more than one G protein subtype [28]. Therefore, cellular signaling has to be finely tuned, and any mis-signaling event can easily cause abnormal responses. Signaling via GPCR–G protein complexes needs three elements: agonist, receptor, and G protein. Solving novel structures of GPCR–G protein complexes with different ligands, receptors and G proteins will substantially increase insights into the coupling and selectivity between receptors and G proteins. This will shed light on subtle changes in the coupling mechanisms and guide us in understanding the big picture in the signaling network.

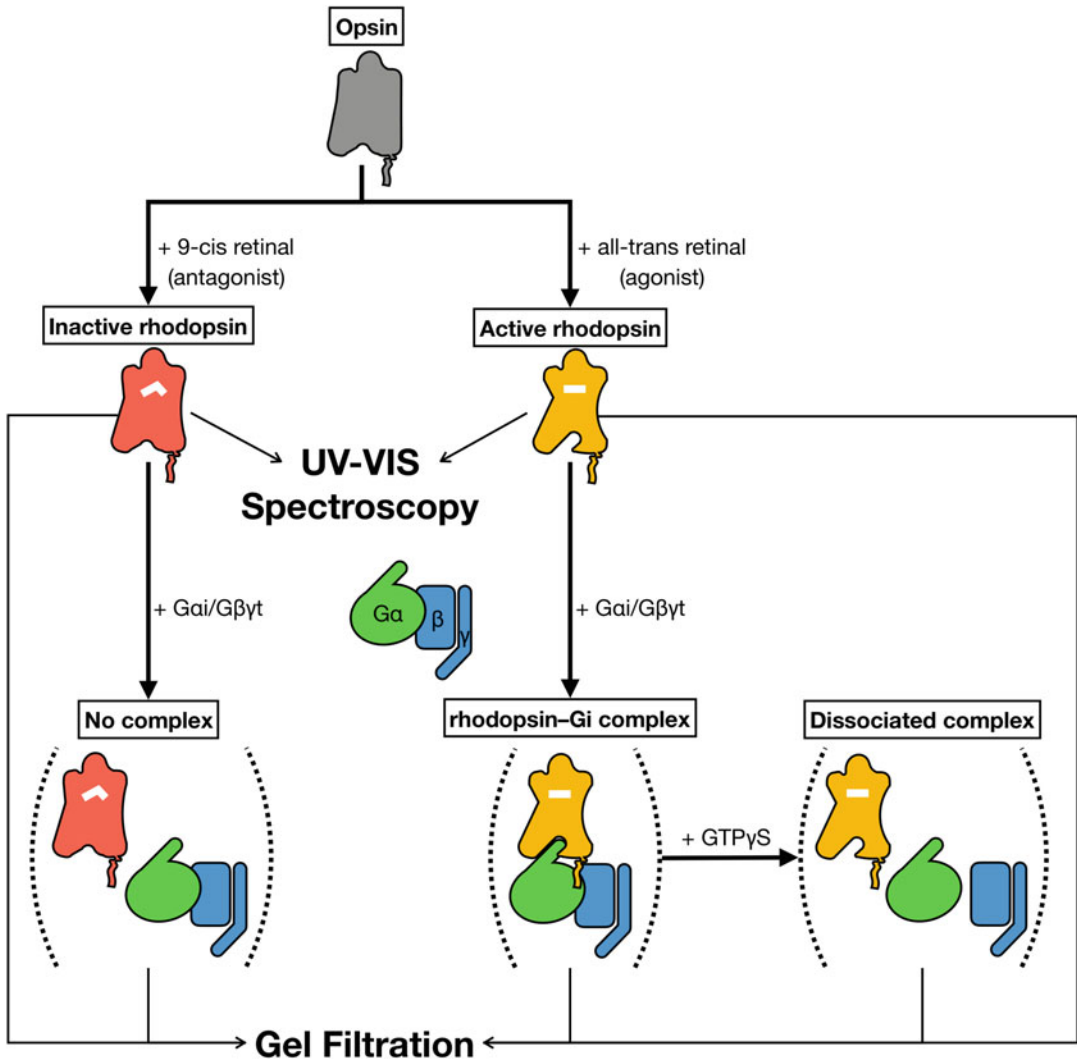


Fig. 1 Flowchart for characterization of opsin, rhodopsin, complex formation and dissociation. Opsin is firstly purified, followed by adding antagonist (9-cis retinal) or agonist (all-trans retinal) to prepare the inactive and active forms of rhodopsin. UV-VIS spectroscopy is used to monitor retinal binding in the rhodopsin. Forming the rhodopsin-G protein complex is by adding G protein heterotrimer to the rhodopsin samples. For the formed complex, GTP γ S is supplemented to dissociate the complex. All the samples are analyzed by gel filtration

2 Materials

Prepare all the buffers and solutions with ultrapure water.

2.1 Protein Samples and Chemicals

1. 30 g HEK 293 GnTI-deficient cell pellet expressing, constitutively active and stabilized mutant of bovine opsin N2C/M257Y/D282C [29].

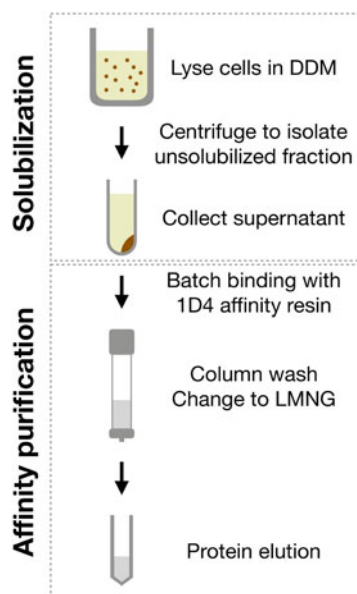


Fig. 2 Solubilization and affinity purification of opsin. To prepare purified opsin, HEK cells expressing opsin are solubilized in the detergent DDM. The unsolubilized cell debris is removed by centrifugation. Solubilized fraction is mixed with 1D4 affinity purification resin. Resin is collected in an open column and washed to remove protein contaminant as well as to change detergent from DDM to LMNG. Opsin is eluted using 1D4 peptide

- Human G α_i1 subunit, purified from *E. coli*. Adjust the concentration to 4 mg/mL [30].
- Bovine transducin $\beta_1\gamma_1$ subunit (G $\beta\gamma$ t), purified. Adjust the final concentration to 4 mg/mL [31].
- EDTA-free protease inhibitor cocktail tablet.
- 10 mM 9-cis retinal (9CR) dissolved in 100% ethanol (*see Note 1*).
- 10 mM all-trans retinal (ATR) dissolved in 100% ethanol (*see Note 1*).
- 10% dodecyl maltoside (DDM) dissolved in water (*see Note 2*)
- 5% lauryl maltose neopentyl glycol (LMNG) dissolved in water (*see Note 2*).
- 800 μ M 1D4 peptide (*see Note 3*).
- 10 mM GTP γ S dissolved in water.

2.2 Affinity Purification

- 10 $\times$ phosphate buffered saline (PBS): Weigh PBS powder and dissolve in water to 10 $\times$ concentration. Filter the buffer through a 0.22 μ m MCE membrane using a vacuum pump to maintain its sterility.

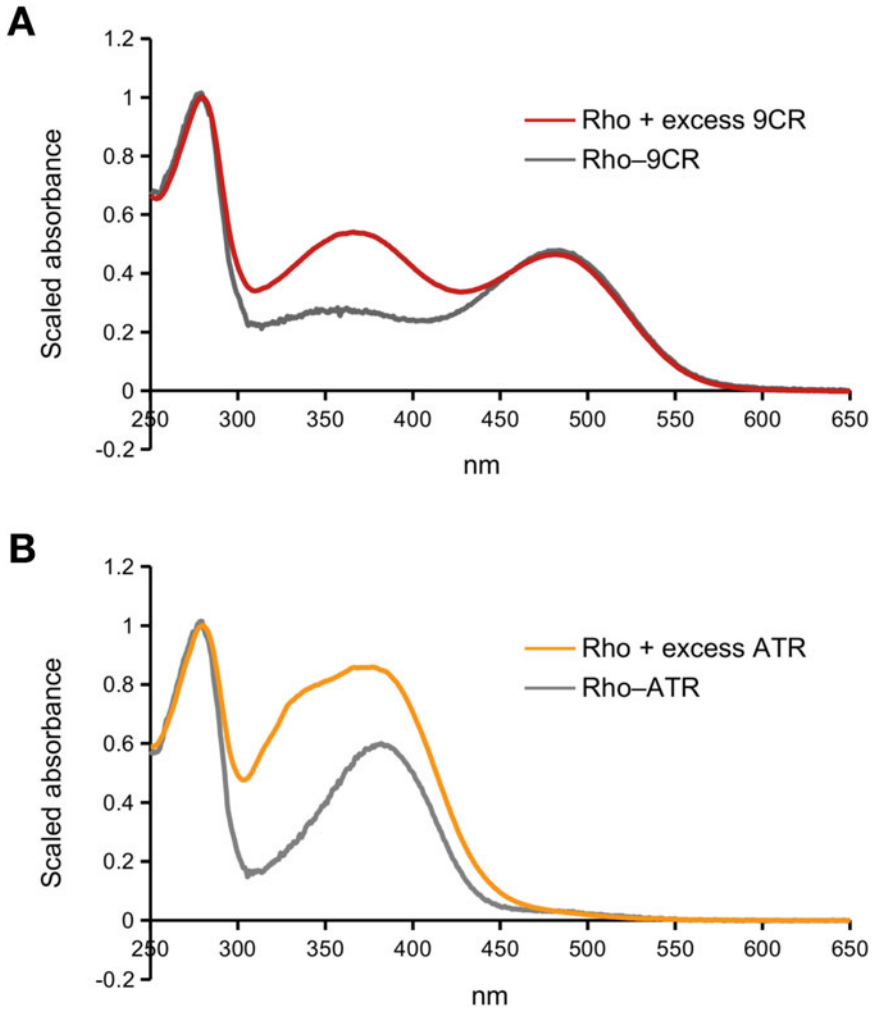


Fig. 3 UV-VIS spectroscopy of rhodopsin samples. (a) The UV-VIS spectrum of opsin supplemented with excess 9-cis retinal is shown in red. The gray curve is measured from 9CR-bound rhodopsin without free retinal in the sample. (b) The orange curve depicts the rhodopsin sample with excess all-trans retinal. The gray curve shows the spectrum of ATR-bound rhodopsin without free retinal in the sample

2. 1 M HEPES pH 7.5: weigh 119.15 g HEPES and add water to a volume of 450 mL. Adjust its pH value with 5 N NaOH to 7.5. Top up with water to a final volume of 500 mL. Filter the solution through a 0.22 μ m MCE membrane filter using a vacuum pump.
3. 5 M NaCl: weigh 146.1 g NaCl and dissolve in water to a final volume of 500 mL. Filter the solution through a 0.22 μ m MCE membrane filter using a vacuum pump.
4. 1 M $MgCl_2$: weigh 10.165 g $MgCl_2 \cdot 6H_2O$ (or 4.76 g anhydrous $MgCl_2$) and dissolve in water to a final volume of 50 mL. Filter the solution through a 0.22 μ m MCE membrane syringe filter with a syringe.

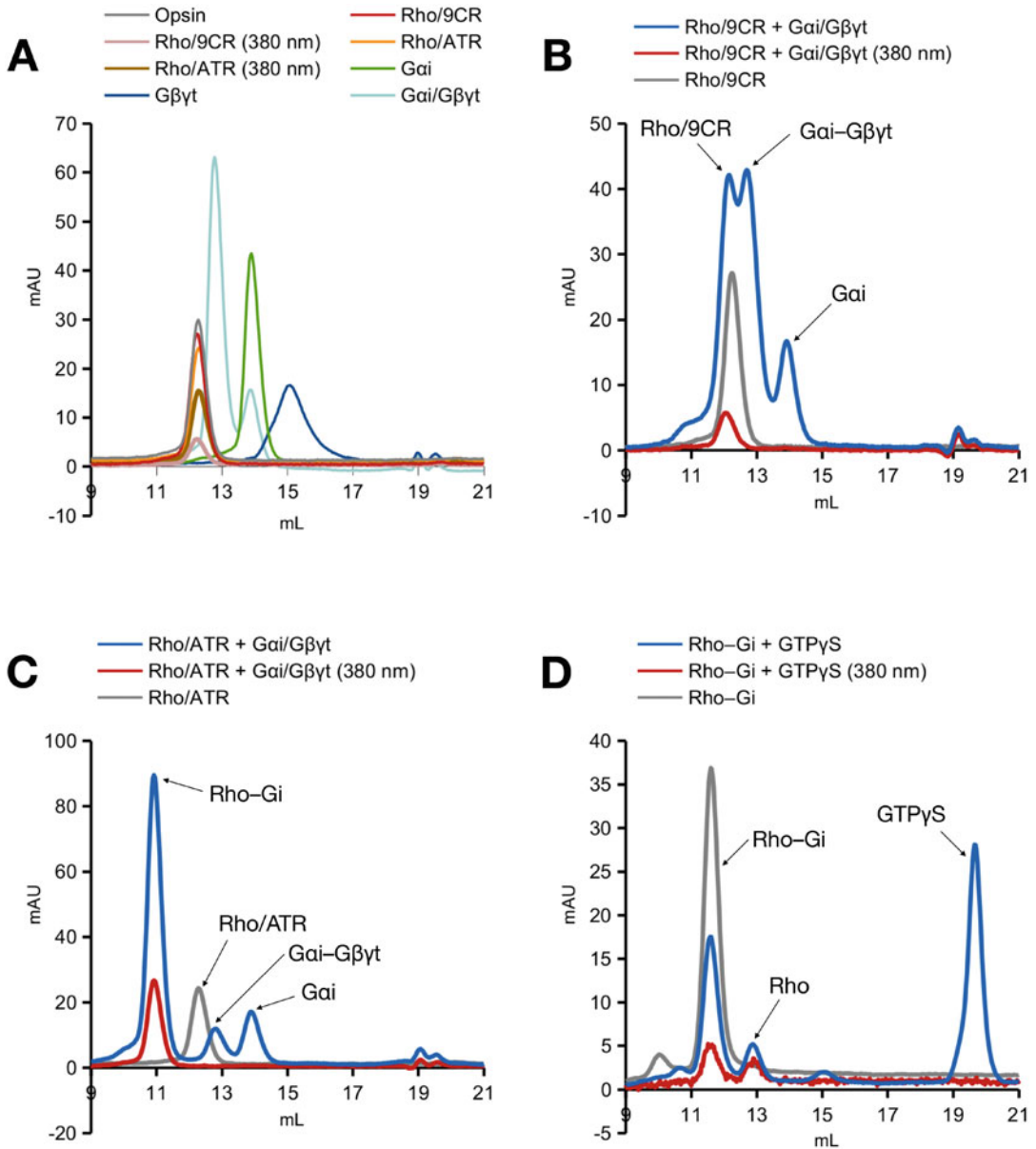


Fig. 4 Gel filtration analysis. The gel filtration curves if not specified are recorded using 280 nm absorbance. (a) Gel filtration profiles of opsin, rhodopsin supplemented with 9-cis retinal or all-trans retinal, G protein subunits, and G protein heterotrimer Gαi-Gβγt. (b) Gel filtration curves of inactive rhodopsin with and without G protein heterotrimer. (c) Gel filtration curves of active rhodopsin with and without G protein heterotrimer. (d) Gel filtration curves of rhodopsin-Gi complex before and after exposure to GTPγS

5. Buffer A: PBS, 0.04% DDM (*see Note 4*).
6. Buffer B: 20 mM pH 7.5 HEPES, 150 mM NaCl, 1 mM MgCl₂, 0.02% LMNG (*see Note 4*).
7. Buffer C: 20 mM pH 7.5 HEPES, 150 mM NaCl, 1 mM MgCl₂, 0.02% LMNG, 80 μM 1D4 peptide (*see Note 4*).

2.3 Gel Filtration

1. Superdex 200 Increase 10/300 GL column.
2. Buffer D: 20 mM pH 7.5 HEPES, 150 mM NaCl, 1 mM MgCl_2 , 0.01% LMNG (*see Note 5*).

3 Methods

All buffers are chilled to 4 °C before use and all steps are carried out at 4 °C or on ice.

3.1 Solubilization

1. Add 72 mL PBS to the HEK293 cell pellet and gently mix to resuspend the cell pellet (*see Note 6*).
2. Transfer the cell suspension to a Dounce homogenizer. Add two tablets of EDTA-free protease inhibitor cocktail and homogenize together with the cell suspension.
3. Under gentle stirring, slowly add 18 mL of 10% DDM stock to the cell suspension and adjust the final volume to 120 mL with PBS. This brings cell lysis to a final DDM concentration of 1.25%. Continue stirring for 1–2 h for solubilization of the cell membrane.
4. Spin down the lysate at $150,000 \times g$ for 45 min to remove the unsolubilized debris.
5. Collect the supernatant and transfer to a 500 mL bottle.

3.2 Affinity Purification of Opsin

1. Add 10 mL 50% slurry of 1D4 immunoaffinity resin to the solubilized cell lysate (*see Note 7*). Mix the suspension of resin and cell lysate by rolling the bottle gently for 4 h (*see Note 8*).
2. Collect resin in an open column and allow the cell lysate to flow out by gravity or using a peristaltic pump.
3. Wash the resin with 8 column volumes (CV) of Buffer A (*see Note 9*).
4. Wash the resin with 4 CV Buffer B.
5. Add 1.5 CV Buffer C. Close the column and gently rock the column overnight.
6. Collect the eluate from the column in a 50-mL tube (*see Note 10*).
7. Add 1.5 CV Buffer C. Close the column and gently roll the column for 2 h.
8. Collect the eluate from the column into the same tube.
9. Add 1.5 CV Buffer C. Close the column and gently roll the column for 2 h.
10. Collect the eluate from the column into the same tube.

11. Measure the concentration at 280 nm absorption using a spectrophotometer.
12. Concentrate the eluted protein using a spin concentrator with a molecular weight cut-off (MWCO) of 50 kDa.
13. Collect the concentrated fraction, and transfer to a 1.5 mL microcentrifuge tube. Centrifuge the concentrated protein at $20,000 \times g$ to remove protein aggregates.
14. Transfer the cleared supernatant to a clean 1.5-mL microcentrifuge tube. Measure the protein concentration and adjust to 4 mg/mL with Buffer B.

3.3 Reconstitute Opsin into Inactive and Active Rhodopsin

1. Under dim red-light conditions, pipet 60 μ L of opsin and 0.78 μ L of 9-cis retinal into a microcentrifuge tube. Mix the sample well, and incubate in the dark for 16 h (*see Note 11*). Measure the UV-VIS spectrum of the sample in the range of 250–650 nm (Fig. 3a).
2. Under dim red-light conditions, pipet 60 μ L of opsin and 0.78 μ L of all-trans retinal into a microcentrifuge tube. Mix the sample well, and incubate in the dark for 16 h (*see Note 11*). Measure the UV-VIS spectrum of the sample in the range of 250–650 nm (Fig. 3b).

3.4 Form G Protein Heterotrimer $G\alpha i/G\beta\gamma t$

1. Pipet 90 μ L of $G\alpha i$ (4 mg/mL), 90 μ L of $G\beta\gamma t$ (4 mg/mL), and 0.72 μ L of LMNG (5%) into a microcentrifuge tube (*see Note 12*). Mix well and incubate for 2 h.

3.5 Prepare Gel Filtration Control Samples

1. Prepare 110 μ L opsin at 1 mg/mL. Dilute with Buffer D (*see Note 13*).
2. Prepare 110 μ L opsin/9CR at 1 mg/mL in the dark. Dilute with Buffer D (*see Note 13*).
3. Prepare 110 μ L opsin/ATR at 1 mg/mL in the dark. Dilute with Buffer D (*see Note 13*).
4. Prepare 110 μ L $G\alpha i$ at 1 mg/mL. Dilute with Buffer D (*see Note 13*).
5. Prepare 110 μ L $G\beta\gamma t$ at 1 mg/mL. Dilute with Buffer D (*see Note 13*).
6. Prepare $G\alpha i/G\beta\gamma t$ at 2 mg/mL (*see Note 14*).

3.6 Prepare Rhodopsin–G Protein Complex Samples for Gel Filtration

1. Under dim red light, prepare 110 μ L containing 1 mg/mL of opsin/9CR and 2 mg/mL of the $G\alpha i/G\beta\gamma t$ mixture. Incubate in the dark for 1 h (*see Note 15*).
2. Under dim red light, prepare 110 μ L containing 1 mg/mL of opsin/ATR and 2 mg/mL of the $G\alpha i/G\beta\gamma t$ mixture. Incubate in the dark for 1 h (*see Note 16*).

3.7 Prepare Rhodopsin-G Protein Complex Dissociation Sample for Gel Filtration

1. Prepare 0.35 mg of gel-filtration purified rhodopsin-Gi complex and concentrate to higher than 1.5 mg/mL for the next two steps (*see Note 17*).
2. Prepare 110 μ L rhodopsin-Gi complex at 1.5 mg/mL. Dilute with Buffer D if necessary.
3. Prepare 110 μ L containing 1.5 mg/mL rhodopsin-Gi complex and 0.1 mM GTP γ S. Incubate in the dark for 1 h.

3.8 Gel Filtration

1. Connect a Superdex 200 Increase 10/300 gel filtration column on a liquid chromatography purifier equipped with multiple wavelength detection and autosampler (*see Note 18*).
2. Equilibrate the column with Buffer D by flushing 40 mL of Buffer D into the column at a flow rate of 0.3 mL/min.
3. Set up the autosampler to load 77 μ L into the column, followed by loading 24 mL of Buffer D at 0.3 mL/min flow rate (*see Note 19*). Record the UV signals at 280 nm, 380 nm and 500 nm.
4. Plot the UV curves versus elution volume in X-Y scatter chart style (Fig. 4).

4 Notes

1. Open the purchased bottle containing 9-cis and all-trans retinal only under dim-red light. Add 100% ethanol to dissolve retinal, which should happen immediately. Mix well and aliquot the dissolved retinal to microtubes. Store the aliquots at -80°C .
2. Detergent solids are usually stored at -20°C . Due to the hygroscopic property of detergent, the purchased bottle should be warmed up to room temperature before opening. For DDM, weigh 4 g DDM powder in a 50 mL falcon tube. Add water to ~ 35 mL marker line. Gently rock the tube until DDM is fully dissolved. Add water to adjust the final volume to 40 mL. Keep at -20°C for long-term storage. In contrast to DDM, LMNG solution at 10% is much more viscous. It is therefore recommended to prepare LMNG stock solutions at 5%. Weigh 0.5 g LMNG powder in a 15 mL falcon tube. Add water to ~ 9 mL and gently rock the tube until LMNG powder is fully dissolved. Adjust the final volume to 10 mL by adding water.
3. 1D4 peptide is a 9-residue peptide with the sequence TETSQ VAPA. It is synthesized via a commercial request. It is suggested to purchase the synthesized peptide in purity over 95%, otherwise low-yield protein purification may occur. Weigh powder and dissolve in water to 800 μ M. Aliquot into 2 mL microtubes and store at -20°C .

4. Dilute Buffer A, B and C from the stock solution.
5. Dilute Buffer D from the stock solution. Filter the solution through a 0.22 μm MCE membrane filter and degas under stirring using a vacuum pump.
6. HEK293 cell pellets are usually stored at -80°C and should be warmed up to room temperature and diluted with PBS buffer before homogenization.
7. The 1D4 immunoaffinity agarose consists of agarose beads linked with the monoclonal Rho1D4 antibody, which binds the last nine amino acids of bovine rhodopsin TETSQVAPA as an epitope. The 1D4 immunoaffinity agarose works as affinity purification material to capture proteins that contain a C-terminal 1D4 sequence. This purification material can be prepared [32] or purchased.
8. Magnetic stirrers should be avoided in suspending the resin. Resin beads may be shredded between the bottle and the stirrer. It can cause high back-pressure if the resin beads are used for another purification using a liquid chromatography purifier machine, even though the binding capacity would not be affected.
9. The column volume equals the volume of the settled resin bed. In this protocol, 1 CV equals to 5 mL. Therefore, 40 mL Buffer A is required for this step. The washing step can be sub-divided into four rounds of washing, each time using 10 mL to rinse the resin beads, which stick to the wall of the column, back to the resin bed.
10. To collect all the eluate solution trapped in the resin bed, gentle air pressure can be applied to the top of the column or the eluate can be aspirated from the bottom. When applying air pressure, use an electronic pipette controller or a 10 mL syringe attached with a 200 μL pipette tip to connect the opening of the lid. When using aspiration, connect a 5 mL syringe to the tap of the column through a short piece of tubing to draw the liquid, and then transfer the liquid to the tube used for eluate collection. This step should be performed quickly to avoid drying of the resin.
11. The molecular weight of opsin is 40 kDa. 60 μL of 4 mg/mL rhodopsin equals 6 nmole. 0.78 μL of 10 mM retinal equals 7.8 nmole. This results in mixing of opsin and retinal at a molar ratio of 1:1.3. This excess of retinal is to ensure the reconstitution of retinal into the ligand binding pocket in opsin.
12. The molecular weights of G α i and G β \gamma t are at 40.4 and 45.9 kDa, respectively. Therefore, this step is to mix G α i and G β \gamma t at a molar ratio of 1.14:1. LMNG is supplemented to a final concentration of 0.02%, which is also the same concentration in the eluted opsin sample.

13. Pipet 27.5 μL of each control protein (opsin, opsin/9CR, opsin/ATR, G α i, G β yt; protein stock solution at 4 mg/mL) and 82.5 μL of Buffer D in a microcentrifuge tube. Mix well.
14. Pipet 55 μL of G α i/G β yt mixture from **step 1** in Subheading 3.4 and 55 μL of Buffer D in a microcentrifuge tube. Mix well.
15. Under dim red light, pipet 27.5 μL of opsin/9CR from **step 1** in Subheading 3.3, 55 μL of G α i/G β yt mixture from **step 1** in Subheading 3.4 and 27.5 μL of Buffer D into a microcentrifuge tube. Mix well and incubate for 1 h.
16. Under dim red light, pipet 27.5 μL of opsin/ATR from **step 2** in Subheading 3.3, 55 μL of G α i/G β yt mixture from **step 1** in Subheading 3.4 and 27.5 μL of Buffer D into a microcentrifuge tube. Mix well and incubate for 1 h.
17. Rhodopsin–Gi complex is prepared in a similar manner as **step 2** in Subheading 3.6, mixing 2 mg rhodopsin/ATR and 4 mg G α i/G β yt mixtures in a microtube followed by incubation for 1 h. Concentrate the rhodopsin/ATR/G α i/G β yt mixture to 250 μL and inject into the LC purifier for gel filtration as described in **step 3** in Subheading 3.8. Collect the peak fraction of rhodopsin–Gi complex and adjust the final concentration to 1.5 mg/mL by dilution or by concentration. Preparation and purification are performed in the dark.
18. In the case of Superdex 200 Increase 10/300 column, the separation of globular proteins ranges from 10 to 600 kDa.
19. If samples can only be delivered using manual injection, then use a 200- μL sample loop.

References

1. Stevens RC, Cherezov V, Katritch V et al (2013) The GPCR network: a large-scale collaboration to determine human GPCR structure and function. *Nat Rev Drug Discov* 12:25–34. <https://doi.org/10.1038/nrd3859>
2. Pándy-Szekeres G, Munk C, Tsonkov TM et al (2018) GPCRdb in 2018: adding GPCR structure models and ligands. *Nucleic Acids Res* 46: D440–D446. <https://doi.org/10.1093/nar/gkx1109>
3. Hilger D, Masureel M, Kobilka BK (2018) Structure and dynamics of GPCR signaling complexes. *Nat Struct Mol Biol* 25:4–12. <https://doi.org/10.1038/s41594-017-0011-7>
4. Kühlbrandt W (2014) Biochemistry. The resolution revolution. *Science* 343:1443–1444. <https://doi.org/10.1126/science.1251652>
5. Bai X, McMullan G, Scheres SH (2015) How cryo-EM is revolutionizing structural biology. *Trends Biochem Sci* 40:49–57. <https://doi.org/10.1016/j.tibs.2014.10.005>
6. Zhao DY, Pöge M, Morizumi T et al (2019) Cryo-EM structure of the native rhodopsin dimer in nanodiscs. *J Biol Chem* 294:14215–14230. <https://doi.org/10.1074/jbc.RA119.010089>
7. Koehl A, Hu H, Feng D et al (2019) Structural insights into the activation of metabotropic glutamate receptors. *Nature* 566:79–84. <https://doi.org/10.1038/s41586-019-0881-4>
8. Liang Y-L, Khoshouei M, Radjainia M et al (2017) Phase-plate cryo-EM structure of a class B GPCR–G-protein complex. *Nature*:1–18. <https://doi.org/10.1038/nature22327>
9. Zhang Y, Sun B, Feng D et al (2017) Cryo-EM structure of the activated GLP-1 receptor in complex with a G protein. *Nature* 546:248–253. <https://doi.org/10.1038/nature22394>
10. Liang Y-L, Khoshouei M, Glukhova A et al (2018) Phase-plate cryo-EM structure of a biased agonist-bound human GLP-1 receptor-

- Gs complex. *Nature* 555:121–125. <https://doi.org/10.1038/nature25773>
11. García-Nafria J, Lee Y, Bai X et al (2018) Cryo-EM structure of the adenosine A2A receptor coupled to an engineered heterotrimeric G protein. *eLife* 7:e35946. <https://doi.org/10.7554/eLife.35946>
 12. Koehl A, Hu H, Maeda S et al (2018) Structure of the μ opioid receptor-Gi protein complex. *Nature* 558:1–23. <https://doi.org/10.1038/s41586-018-0219-7>
 13. Kang Y, Kuybeda O, de Waal PW et al (2018) Cryo-EM structure of human rhodopsin bound to an inhibitory G protein. *Nature* 558:553–558. <https://doi.org/10.1038/s41586-018-0215-y>
 14. Draper-Joyce CJ, Khoshouei M, Thal DM et al (2018) Structure of the adenosine-bound human adenosine A1 receptor-Gi complex. *Nature* 558:559–563. <https://doi.org/10.1038/s41586-018-0236-6>
 15. García-Nafria J, Nehmé R, Edwards PC, Tate CG (2018) Cryo-EM structure of the serotonin 5-HT1B receptor coupled to heterotrimeric Go. *Nature* 558:620–623. <https://doi.org/10.1038/s41586-018-0241-9>
 16. Liang Y-L, Khoshouei M, Deganutti G et al (2018) Cryo-EM structure of the active, Gs-protein complexed, human CGRP receptor. *Nature* 561:492–497. <https://doi.org/10.1038/s41586-018-0535-y>
 17. Krishna Kumar K, Shalev-Benami M, Robertson MJ et al (2019) Structure of a signaling cannabinoid receptor 1-G protein complex. *Cell*:1–11. <https://doi.org/10.1016/j.CELL.2018.11.040>
 18. Zhao L-H, Ma S, Sutkeviciute I et al (2019) Structure and dynamics of the active human parathyroid hormone receptor-1. *Science* 364:148–153. <https://doi.org/10.1126/science.aav7942>
 19. Maeda S, Qu Q, Robertson MJ et al (2019) Structures of the M1 and M2 muscarinic acetylcholine receptor/G-protein complexes. *Science* 364:552–557. <https://doi.org/10.1126/science.aaw5188>
 20. Qi X, Liu H, Thompson B et al (2019) Cryo-EM structure of oxysterol-bound human smoothened coupled to a heterotrimeric Gi. *Nature*. <https://doi.org/10.1038/s41586-019-1286-0>
 21. Kato HE, Zhang Y, Hu H et al (2019) Conformational transitions of a neurotensin receptor 1-Gil complex. *Nature* 572:80–85. <https://doi.org/10.1038/s41586-019-1337-6>
 22. Tsai C-J, Marino J, Adaixo R et al (2019) Cryo-EM structure of the rhodopsin-G α i- β γ complex reveals binding of the rhodopsin C-terminal tail to the g β subunit. *eLife* 8:547919. <https://doi.org/10.7554/eLife.46041>
 23. Gao Y, Hu H, Ramachandran S et al (2019) Structures of the rhodopsin-transducin complex: insights into G-protein activation. *Mol Cell* 75:781–790.e3. <https://doi.org/10.1016/j.molcel.2019.06.007>
 24. Zhao P, Liang Y-L, Belousoff MJ et al (2020) Activation of the GLP-1 receptor by a non-peptidic agonist. *Nature* 577:432–436. <https://doi.org/10.1038/s41586-019-1902-z>
 25. Yin W, Li Z, Jin M et al (2019) A complex structure of arrestin-2 bound to a G protein-coupled receptor. *Cell Res* 29:971–983. <https://doi.org/10.1038/s41422-019-0256-2>
 26. Nguyen AH, Thomsen ARB, Cahill TJ et al (2019) Structure of an endosomal signaling GPCR-G protein- β -arrestin megacomplex. *Nat Struct Mol Biol* 26:1123–1131. <https://doi.org/10.1038/s41594-019-0330-y>
 27. Simon MI, Strathmann MP, Gautam N (1991) Diversity of G proteins in signal transduction. *Science* 252:802–808. <https://doi.org/10.1126/science.1902986>
 28. Flock T, Hauser AS, Lund N et al (2017) Selectivity determinants of GPCR-G-protein binding. *Nature* 545:317–322. <https://doi.org/10.1038/nature22070>
 29. Deupi X, Edwards P, Singhal A et al (2012) Stabilized G protein binding site in the structure of constitutively active metarhodopsin-II. *Proc Natl Acad Sci* 109:119–124. <https://doi.org/10.1073/pnas.1114089108>
 30. Sun D, Flock T, Deupi X et al (2015) Probing G α i1 protein activation at single-amino acid resolution. *Nat Struct Mol Biol* 22:686–694. <https://doi.org/10.1038/nsmb.3070>
 31. Maeda S, Sun D, Singhal A et al (2014) Crystallization scale preparation of a stable GPCR signaling complex between constitutively active rhodopsin and G-protein. *PLoS One* 9:e98714. <https://doi.org/10.1371/journal.pone.0098714>
 32. Molday LL, Molday RS (2014) 1D4: a versatile epitope tag for the purification and characterization of expressed membrane and soluble proteins. *Methods Mol Biol* 1177:1–15. https://doi.org/10.1007/978-1-4939-1034-2_1



Chapter 4

Electrophysiological Approaches for the Study of Ion Channel Function

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Abstract

Ion channels play crucial roles in cell physiology, and are a major class of targets for clinically relevant pharmaceuticals. Because they carry ionic current, the function and pharmacology of ion channels can be studied using electrophysiological approaches that range in resolution from the single molecule to many millions of molecules. This chapter describes electrophysiological approaches for the study of one representative ion channel that is defective in a genetic disease, and that is the target of so-called highly effective modulator therapies now used in the clinic: the cystic fibrosis transmembrane conductance regulator (CFTR). Protocols are provided for studying CFTR expressed heterologously, for CFTR expressed *in situ* in airway epithelial cells, and for purified or partially purified CFTR protein reconstituted into planar lipid bilayers.

Key words Electrophysiology, CFTR, Patch clamp, Ussing chamber, Planar lipid bilayer

1 Introduction

Ion channels have been a favorite target of study for physiologists for decades, in part because methods are available that allow one to watch these molecular machines perform their function in real time. Using various forms of patch clamp electrophysiology, one may amplify the tiny amounts of current carried by the ions in flux, making it possible for channel function to be measured with exquisite resolution in both temporal and amplitude domains. With the recent advances in structural biology enabling high resolution analysis of integral membrane proteins, physiologists have been able to explore and understand the molecular bases for channel function and pharmacology. Indeed, the combination of structural approaches with electrophysiological approaches provides an incredibly satisfying opportunity to understand nature and evolution at the molecular level.

Of course, channels in living organisms never function in isolation, so it is also important to study their activity at higher levels of organization, such as in whole cells or in tissues. Channel proteins in the plasma membrane often are localized to complexes with other proteins, including systems that confer regulation by post-translational modification of the channel proteins, themselves. In fact, channels may be used as reporters for the activation of signaling pathways that impinge upon those channels. For example, we have used the activation of Ca^{2+} -dependent chloride channels to study the regulation of G protein-coupled receptors linked to $G_{\alpha q}$, and the activation of cAMP-stimulated chloride channels to study the regulation of G protein-coupled receptors linked to $G_{\alpha s}$ [1, 2]. All ion channels accomplish the passive diffusion of ions down an electrochemical gradient established by primary and secondary active transporters. In the case of channels used for electrical signaling in neurons, the channel's functional role in the cell does not actually require the movement of large amounts of ions. Instead, the opening of the channel in essence changes the electrical resistance in the membrane in order to alter membrane voltage, leading to further activation of other voltage-sensitive channels. This makes it straightforward to study the channel in isolation, when expressed heterologously. In contrast, in epithelial cells that accomplish the bulk movement of substrates (and often water) from one fluid-filled space to another, the functional role of the channel in one aspect of the plasma membrane (e.g., the apical membrane) is dependent upon the function of often several types of transporters, pumps, and channels in the other aspect (e.g., the basolateral membrane) [3]. Therefore, understanding the function and regulation of the channel under study often requires experiments in situ in whole epithelial cells.

In this chapter, we describe approaches to study ion channels at various levels of resolution, from one channel per experiment with single-molecule resolution, to many millions per experiment, in aggregate. All protocols provided here focus on the study of CFTR, the chloride channel protein that is defective in Cystic Fibrosis, a lethal genetic disease. Protocols are provided for the study of CFTR expressed heterologously in *Xenopus* oocytes, for the study of CFTR expressed endogenously in mammalian epithelial cells, and for the study of CFTR in proteoliposomes reconstituted into planar lipid bilayers.

2 Materials

2.1 Solutions

- *Ca^{2+} -free Barth solution*: 89 mM NaCl, 1.0 mM KCl, 2.4 mM NaHCO_3 , 10 mM HEPES, 0.82 mM MgSO_4 , pH 7.4.

- *Ca²⁺-containing Barth solution*: 88 mM NaCl, 1.0 mM KCl, 2.4 mM NaHCO₃, 10 mM HEPES, 0.82 mM MgSO₄, 0.33 mM Ca(NO₃)₂, 0.91 mM CaCl₂, pH 7.4.
- *Oocyte incubation media*: L-15 (Gibco/BRL, Gaithersburg, MD), 10 mM HEPES (pH 7.5), 50 I.U./mL penicillin, 50 µg/mL streptomycin.
- *Hypertonic stripping solution*: 200 mM monopotassium aspartate, 20 mM KCl, 1 mM MgCl₂, 10 mM EGTA, and 10 mM HEPES, pH 7.2 adjusted with KOH.
- *Chloride-containing pipette solution*: 150 mM NMDG-Cl, 5 mM MgCl₂, 10 mM TES, pH 7.5, filtered using 0.22–0.45 µm disposable filters before use.
- *Intracellular solution for oocyte patch*: 150 mM NMDG-Cl, 1.1 mM MgCl₂, 2 mM Tris-EGTA, 10 mM TES, pH 7.5, filtered using 0.22–0.45 µm disposable filters before use.
- *N96*: 96 mM NaCl, 2 mM KCl, 1 mM MgCl₂, and 5 mM HEPES, pH 7.5.
- *Normal-chloride Ringers*: 140 mM NaCl, 5 mM KCl, 0.36 mM K₂HPO₄, 0.44 mM KH₂PO₄, 1.3 mM CaCl₂, 0.5 mM MgCl₂, 4.2 mM NaHCO₃, 10 mM Na-HEPES, 10 mM glucose (for basolateral solution) or mannitol (for apical solution), pH 7.4.
- *Low-chloride Ringers*: 133.33 mM Na gluconate, 5 mM K gluconate, 2.5 mM NaCl, 0.36 mM K₂HPO₄, 0.44 mM KH₂PO₄, 5.7 mM CaCl₂, 0.5 mM MgCl₂, 4.2 mM NaHCO₃, 10 mM Na-HEPES, 10 mM mannitol, pH 7.4.

2.2 Reagents

- Recombinant catalytic subunit of bovine Protein Kinase A (PKA) is from Promega (Madison, WI).
- Quikchange site-directed mutagenesis kit is from Agilent (Santa Clara, CA).
- mMessage mMachine in vitro transcription kits are from Invitrogen (Carlsbad, CA).
- Contrad 70 detergent solution is from Decon Labs, Inc. (King of Prussia, PA).
- Lipids, either in powder form or in chloroform, are from Avanti Polar Lipids (Alabaster, AL).
- CFTRinh172 is from Calbiochem (Burlington, MA).
- All other reagents are from Sigma-Aldrich (St. Louis, MO).
- Compounds are prepared typically as a stock solution in DMSO then stored at –20 °C, unless otherwise noted. These compounds are diluted to the final concentration in the relevant recording solution immediately before use. DMSO has no effect on CFTR channel behavior when diluted at least 1:1000 in recording solution.

2.3 Hardware and Software

- Model P-2000 pipet puller is from Sutter Instrument Co. (Novato, CA).
- Axopatch 200B amplifier, Geneclamp 500B amplifier, Digidata 1322A analog to digital converter, and pClamp software are from Molecular Devices (Sunnyvale, CA).
- Four-pole low-pass Bessel filter is from Warner Instruments (Hamden, CT).
- Igor software is from WaveMetrics, Inc. (Lake Oswego, OR).
- Transwell filters are models 3470 or 3460 from Corning, Inc. (Corning, NY).
- Ussing Chamber equipment and Acquire & Analyze software are from Physiologic Instruments (San Diego, CA).
- Spin-2 bilayer stir plate and controller, stir bars, BCH-M13 bilayer chambers, and CD13A-200 cuvettes are from Warner Instruments (Hamden, CT).

3 Methods

3.1 Recording from Channels Heterologously Expressed in *Xenopus* Oocytes

For studies of CFTR structure and function, we mainly use *Xenopus laevis* oocytes as a system for heterologous expression of CFTR variants for the following reasons [4, 5]. First, *Xenopus* oocytes do not endogenously express CFTR protein or other chloride channels activated via the cAMP/PKA system [6]. Second, unlike mammalian cells cultured at 37 °C, *Xenopus* oocytes are usually maintained at ~17 °C which facilitates the trafficking of CFTR mutants to the oocyte plasma membrane when the same mutants fail to effectively traffic in mammalian cells maintained at 37 °C. Third, it is easier to control the expression level of CFTR protein by injecting a precise amount of complementary RNA (cRNA) into each cell. Finally, *Xenopus* oocytes provide a model for electrophysiological recordings at various levels of resolution, from millions of channels per record to a single channel per record, as described below [7–9]. We prefer to obtain oocytes from our own *Xenopus* colony. Some investigators are able to use oocytes available from commercial sources such as EcoCyte Bioscience (Austin, TX).

We understand that *Xenopus* oocytes represent a heterologous expression system that is quite dissimilar from human bronchial epithelial cells and might exhibit species specific behavior of CFTR. So far, our findings of functional features of the CFTR channel are consistent with data collected from studies of CFTR in mammalian cells. One major caveat that must be remembered is that oocyte experiments are performed at room temperature, rather than human body temperature. However, if one is interested in obtaining very high-resolution data on channel function, the advantages

of using the oocyte system greatly outweigh the disadvantages. Other electrophysiological approaches, such as recording short-circuit currents from cell monolayers studied with the Ussing chamber recording technique, also may be applied, as described below.

3.1.1 Preparation of *Xenopus* Oocytes and CFTR cRNAs

1. Ovarian lobes are surgically removed from *Xenopus* frogs and digested with 2.8–3.2 mg/mL collagenase I in Ca^{2+} -free Barth solution for two hours at room temperature. Methods of animal handling are in accordance with the NIH guidelines and the protocol was approved by the Institutional Animal Care and Use Committee of Emory University.
2. Enzymatic digestion is terminated by washing oocytes twice with Ca^{2+} -containing Barth solution.
3. Isolated single oocytes (stage V–VI) are selected and then incubated at 17 °C in oocyte media.
4. CFTR cRNAs are generated in vitro using a mMessage mMachine kit using linearized cDNA constructs encoding the appropriate CFTR variant: wild-type CFTR (WT-CFTR) or a mutant.
 - (a) *Note: All mutants of CFTR are generated using the Quik-change site-directed mutagenesis kit. The entire open reading frame of each construct is screened with full sequencing before use.*
5. Inject cRNAs into the apex of the animal pole of the oocyte, to standardize experiments. Different amounts of CFTR cRNA are injected into oocytes based on the experiment planned, as described below. For single channel recording, we aim to record one CFTR protein per patch, although often we find one to four CFTR channels per patch; we typically discard the patch if the maximum number of open channels is greater than four. For inside-out macropatch recording, we aim to record currents from a few hundred to over a thousand CFTR channels after they are activated by MgATP and PKA.
 - (a) *Note: CFTR protein expression varies from batch to batch of oocytes. The expression levels in the plasma membrane are not expected to be exactly the same across several oocytes from the same batch, even with the same amount of cRNA injected. Therefore, the amount of CFTR cRNA injected per oocyte requires adjustment depending on the purpose of recording.*
 - (b) *Note: The number of active channels per patch pulled from a given cell also can be modulated, somewhat, by choosing to patch on the cell at a particular distance from where the cRNA was injected, since channel expression is usually highest at the site of injection.*

6. Electrophysiology experiments are performed 24–96 h following injection of cRNAs.
7. Statistical analysis is performed using the t-test for unpaired or paired measurements in Sigmaplot 12.3, with $p < 0.05$ considered indicative of significance.

3.1.2 Inside-Out Single Channel Recording

1. For single-channel recordings, we typically inject 0.5–1 ng CFTR cRNA per oocyte.
2. Oocytes are prepared for study by shrinking in hypertonic stripping solution followed by manual removal of the vitelline membrane.
3. Pipettes are pulled from borosilicate glass with puller Model P-2000 and have resistances averaging $\sim 10\text{ M}\Omega$ for single channel recording when filled with chloride-containing pipette solution.
4. After forming the giga-ohm seal on the oocyte membrane (typical seal resistances are $200\text{ G}\Omega$ or greater), the pipette is pulled away from the oocyte abruptly, using the micromanipulator, to establish an inside-out patch for recording. In this case, the solution in the pipet represents the extracellular solution, and the solution in the bath represents the intracellular solution. An advantage of the *Xenopus* oocyte is that resistances can sometimes be so high that they cannot be measured, thus enabling extremely low-noise recordings.
5. CFTR channels are activated by addition of 1 mM MgATP with different concentrations of PKA to the intracellular solution in the chamber.
 - (a) *Note: Because of large variabilities in the activity per unit of PKA from commercial sources, we strictly use the product available from Promega.*
6. CFTR currents are measured with an Axopatch 200B amplifier, and are recorded at 10 kHz to DAT tape. Single channel amplitude of human WT-CFTR recorded at $V_M = -100\text{ mV}$ is about 0.72 pA, which is much lower than the amplitude of many channels studied. Consequently, it is crucial for successful single channel recording to remove all sources of electrical noise and maintain a perfectly “quiet” recording system. Our single channel recording systems over the past 25 years have relied upon the use of a modified digital tape recorder, so that no computer is used during recording in order to limit sources of noise. This has enabled the study of channels with unitary conductance amplitude even lower than that of WT-CFTR, and the analysis of events of submillisecond duration.
7. For subsequent analysis after the experiment, records are played back and filtered with a four-pole low-pass Bessel filter

at a corner frequency of usually 1 kHz and acquired using a Digidata 1322A and computer at typically 500 Hz using pClamp software.

8. For display, single channel records are typically filtered digitally in pClamp to 100 Hz.
9. pClamp is again used to analyze single channel current and to make all-points amplitude histograms.
10. Open duration histograms are made and fit with a single exponential function with Igor software (WaveMetrics, Inc., Portland, OR).
11. Apparent open probability (NP_o) is measured using Clampfit, within the pClamp suite (Fig. 1) [10].

3.1.3 Inside-Out Macropatch Recording

1. For macropatch recordings, we typically inject 5–10 ng CFTR cRNA per oocyte.
2. Oocytes are prepared for study by shrinking in hypertonic stripping solution followed by manual removal of the vitelline membrane.
3. Pipettes are pulled from borosilicate glass with puller Model P-2000 and have resistances averaging $\sim 1\text{--}2\text{ M}\Omega$ for macropatch recording when filled with chloride-containing pipette solution.
4. After forming the giga-ohm seal on the oocyte, the pipette is pulled away from the oocyte abruptly, using the micromanipulator, to establish an inside-out patch for recording.

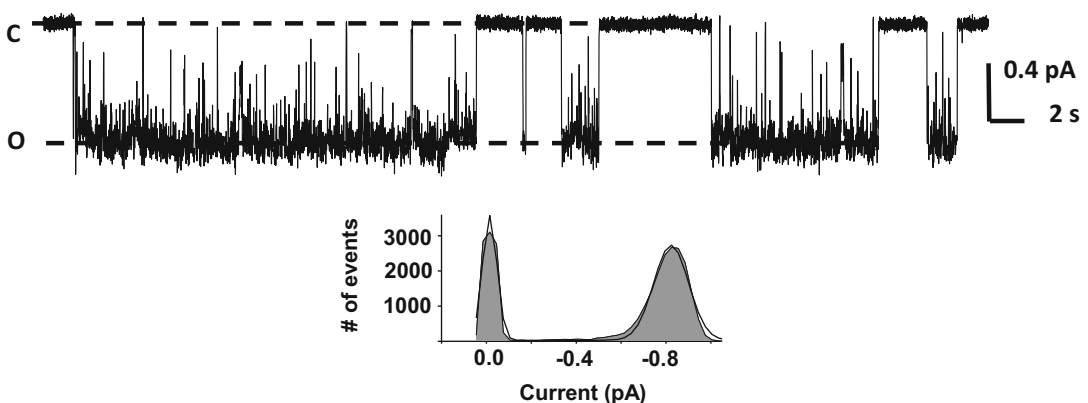


Fig. 1 Single channel behavior of CFTR. A representative single-channel current trace is shown for WT-CFTR in the presence of 1 mM MgATP and 127.6 U/mL PKA, from an inside-out membrane patch excised from a *Xenopus* oocyte, with symmetrical 150 mM Cl^- . The trace was recorded at $V_M = -100\text{ mV}$. C = closed state; O = full open state. An all-points amplitude histogram is shown under the trace, where the superimposed solid line is results from fit to a Gaussian function

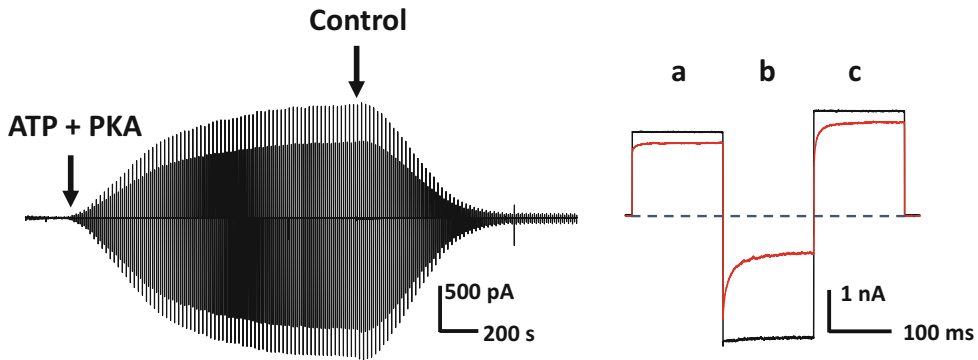


Fig. 2 Macroscopic channel behavior of CFTR from macropatch recording. (*Left*) A representative macropatch current trace is shown of WT-CFTR recorded in inside-out mode with symmetrical 150 mM Cl^- solution. Channels were fully activated in 1 mM MgATP + 127.6 U/mL PKA. A voltage-ramp protocol was applied every 5 s. (*Right*) Representative macropatch current of WT-CFTR recorded in inside-out mode with a step protocol in the absence (black line) and in the presence (red line) of 50 μM glibenclamide is also shown. Voltage steps from holding potential (0 mV) to +80 mV (a) then -120 mV (b) then +100 mV (c) and back to 0 mV. The blue dashed line indicates zero current level. The time-dependent nature of glibenclamide-mediated channel block (segment b) and relief from block (segments a, c) are evident in comparing the red and black traces

5. CFTR channels are activated by addition of 1 mM MgATP with different concentrations of PKA to the intracellular solution in the chamber.
 - (a) *Note: Rundown of currents from fully activated CFTR channels in the excised inside-out patch configuration is very minor after about 20 min in the continued presence of 1 mM MgATP + 127.6 U/mL PKA. However, rundown of CFTR currents is much more pronounced when PKA is removed from the cytoplasmic solution even in the continued presence of 1 mM MgATP; this likely reflects the activity of protein phosphatases endogenous to the oocyte membrane [11, 12].*
6. Current recordings are performed with an Axopatch 200B amplifier operated by pClamp 8.2 software via a Digidata 1322A.
7. Data are filtered at 100 Hz with a four-pole Bessel filter and acquired at 2 kHz.
8. Various voltage protocols are used, dependent on the purpose of the experiment.
 - (a) For example, a voltage ramp protocol is applied every 5 s to observe real time activation of CFTR channels by PKA (Fig. 2, left). The typical ramp protocol for this sort of experiment includes: holding at $V_M = 0$ mV, stepping up to +100 mV for 50 ms, ramping down to -100 mV over 300 ms, then stepping back to 0 mV [10].

- (b) A step protocol is applied to observe voltage-dependent block of CFTR by glibenclamide and other blockers [11]. The step protocol includes: holding at $V_M = 0$ mV, then stepping to +80, to -120, to +100 mV, then back to 0 mV. Each of the segments at a specific potential lasts 160 ms (Fig. 2, *right*). This specific protocol is used to enable the study of the interaction of these compounds with the CFTR channel, leading to block of current. Because these compounds are negatively charged (or zwitterionic) at physiological pH and they block CFTR from the cytoplasmic face, the pulse to $V_M = +80$ mV enables removal of blocker from the channel pore. The pulse to $V_M = -120$ mV enables one to observe the association of blocker to channel as current slowly diminishes at this potential. The pulse to $V_M = +100$ mV enables one to observe the dissociation of blocker from the channel as current slowly increases at this potential. Fractional block in the presence of blocker is measured from the steady-state currents at $V_M = -120$ mV.
- (c) *Note: We point out that much more complex voltage protocols can be generated using pClamp software for the study of voltage-activated ion channels. These protocols can include steps that enable subtraction of leak currents and capacitive artifacts.*

3.1.4 Two-Electrode Voltage-Clamp Recording

While patch clamp recording from excised membranes enables the study of a low number of channels, the oocyte also is amenable to the study of millions of channels per cell, by use of the two-electrode voltage clamp (TEVC) technique. While single-electrode techniques with amplifiers have been used for recording of channel currents from whole small cells that are intact (other than the damage induced by the electrode itself), the enormous surface area of the *Xenopus* oocyte means that amplifiers must have very high compliance (i.e., the ability to pass very large currents) in order to effectively clamp the potential across the whole cell membrane. This is best accomplished by use of the TEVC technique where, essentially, one electrode is used to measure membrane potential and the other is used to pass current, the magnitude of which is directly proportional to the current arising from movement of ions across the membrane. Fortunately, stage V–VI oocytes are large enough to accommodate two electrodes simultaneously, and sometimes even more (e.g., additional pipettes for rapid introduction of signaling molecules into the cell during voltage-clamp).

1. For TEVC recording, we typically inject 0.1–1 ng of CFTR cRNA per cell, sometimes along with 0.08 ng of cRNA encoding the human beta-2 adrenergic receptor (β_2AR). Use of β_2AR

to stimulate the cAMP/PKA signaling pathway—all other components of which are provided by the oocyte itself—enables CFTR to be activated with addition of only isoproterenol to the bath. This is much cheaper than using a cocktail of forskolin (an activator of adenylyl cyclase) and isobutylmethylxanthine (IBMX, an inhibitor of phosphodiesterase), perhaps with addition of a membrane-permeant form of cAMP such as dibutyryl cAMP [13].

2. We use ND96 as the common bath solution in TEVC. In experiments where activation of Ca^{2+} -dependent signaling also is studied, ND96 solution is supplemented with 1.8 mM CaCl_2 [14].
3. Electrodes are pulled as in macropatch experiments, except in this case to a sharp tip for cell impalement. Electrodes typically have resistances of 0.5–1.4 M Ω when filled with 3 M KCl and measured in ND96 bath solution.
4. We use a Geneclamp 500B amplifier equipped with a virtual ground, controlled by pClamp 10.2 software (e.g., refs. [9, 10, 15]).
5. Data are typically acquired at 1 kHz and software-filtered at 100 Hz.
6. Most voltage-clamp protocols use a holding potential in the range of -20 to -30 mV, because, in standard ND96 solution, the oocyte membrane potential is close to -30 mV.
 - (a) Voltage-clamp protocols for study of CFTR currents over minutes typically include a step to -60 mV followed by a ramp to 0 mV over 500 ms, with the protocol elicited at 0.1 Hz [16]; currents at -60 mV are reported (Fig. 3).

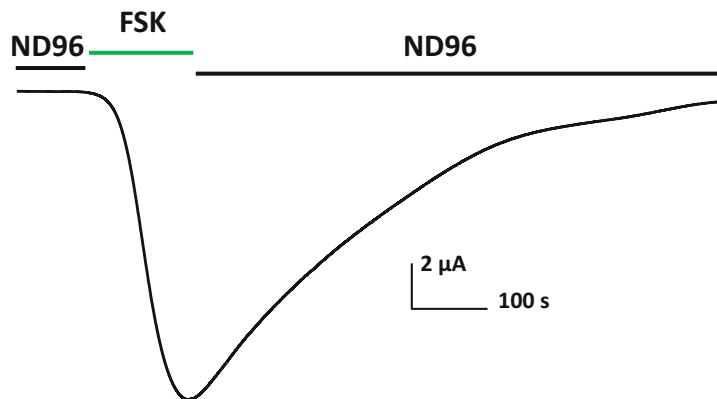


Fig. 3 Macroscopic channel behavior of CFTR from whole-oocyte recording. A representative current trace recorded by two-electrode voltage clamp in ND96 control bath solution is shown, with CFTR activated by exposure to 10 μM Forskolin (FSK). $V_M = -60$ mV

- (b) For experiments to study the effect of voltage-dependent blockers, we use either a step protocol (75 ms steps to potentials ranging from -140 to $+80$ mV, in 20 mV increments) or ramp protocols (typically from -100 to $+60$ mV) [17].

3.2 Recording Channel Currents in Epithelial Cell Monolayers Using the Ussing Chamber

As noted above, epithelial ion channels like CFTR only achieve their function in the apical membrane of airway epithelial cells via the coordinated activity of pumps, transporters, and other channels in the basolateral membrane. Hence, one can best understand the function of CFTR by studying it in situ in the epithelial cell, along with its functional partners. To accomplish this, epithelial cells can be cultured into polarized monolayers on permeable supports, and transepithelial short-circuit currents may be recorded from those cultures using the Ussing chamber technique [15]. This also gives one the opportunity to study CFTR in primary epithelial cells from the human airway (e.g., human bronchial epithelial cells, HBEs, or human tracheal epithelial cells, HTEs), where the protein is expressed at physiological densities and with the physiologically relevant interacting partners. Another model system highly used in CF research is Fischer Rat Thyroid (FRT) cells which have been altered to express either WT-CFTR or a number of disease-relevant mutants at the same level of abundance [18]. The following protocol is for the study of WT-CFTR currents in 16HBE airway epithelial cells grown on Transwell filters plated at 150,000–250,000 cells per well (Fig. 4).

Experiments in the Ussing Chamber can be conducted either in symmetric chloride conditions, with normal-chloride Ringers on either side of the monolayer, or under a chloride gradient in which the apical solution has been replaced with low-chloride Ringers [15]. Both have their advantages, in that symmetric chloride conditions allow the user to ascertain more information about the system through determining the conductance, while imposing a chloride gradient can increase the magnitude of the currents observed. In some cases, for example in 16HBEs, a gradient is necessary to be able to observe any cAMP-activated chloride currents (*unpublished observation*). Note that mannitol can be substituted for glucose in the apical buffer to maintain the osmolarity of the solution while preventing obtrusive currents from the sodium glucose cotransporter located on the apical side of bronchial epithelial cells.

A VCC-MC6 voltage/current clamp amplifier is used to control the electrodes, collect data, and output that data to the computer. Each chamber in the experiment is connected to a DM-MC6 input module and dummy membrane, which is connected to an individual amplifier and control panel on the VCC-MC6. The DM-MC6 serves as an integration point for the four P202-S electrodes, two voltage-sensing and two current-passing, that are

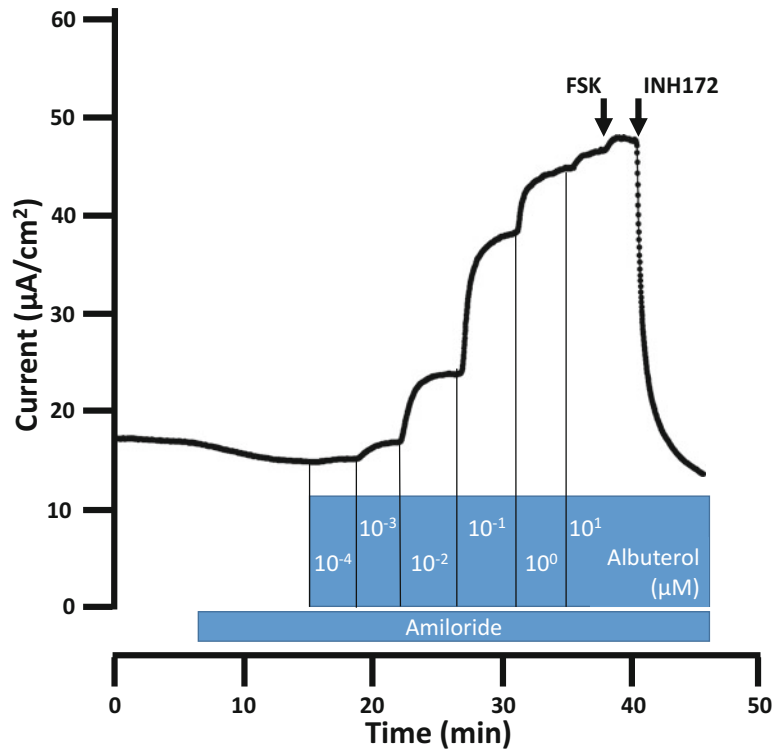


Fig. 4 Recording of CFTR activity by Ussing chamber technique. Short-circuit currents are measured from 16HBE epithelial cells grown on permeable supports under a chloride gradient. Additions to bathing solution are shown, including Amiloride (20 μM), Albuterol at various concentrations as noted, Forskolin (FSK, 10 μM), and channel inhibitor CFTR_{inh}172 (10 μM)

connected on either side of the P2300 chamber secured into an EM-RSYS-2 mount. The electrodes are secured into P2023 electrode tips that are filled with 3 M KCl 3% agar on the tip, and back-filled with 3 M KCl. The two sides of the chamber are separated by a P2302T slider with a Transwell filter inserted into the center.

3.2.1 Running an Ussing Chamber Experiment

1. Turn on the water bath to 42 °C to heat the chamber to 37 °C.
2. Place the recording solution(s) in a 37 °C water bath and the chambers in a 37 °C incubator for at least 1 h to warm.
3. Make electrode tips by melting 3% agar in 3 M KCl, pulling some of the agar solution into a small syringe, placing the tip of the electrode into the end of the syringe, injecting some of the agar into the tip, then immediately putting this tip into a beaker of 3 M KCl. Ensure that there are no bubbles in the agar once it is in the tip. Once the agar has solidified, use a syringe with a long needle to fill the rest of the tip with 3 M KCl. Insert electrodes into these tips, again making sure there are no bubbles in the tip's agar or fluid surrounding the electrode.

4. Assemble the chambers with a slider containing a Transwell filter with no cells on it, to be used for blanking the amplifier system.
5. Insert electrodes in tips into the chambers. Long, white-base current-passing electrodes go on the outside of the chamber. Short, black-base voltage-sensing electrodes go on the inside of the chamber. If the electrodes become white, they should be rechlorided by carefully submerging the silver-chloride part in bleach for 1 h.
6. Connect the electrodes to the lead, making sure to keep the pairs of the lead on the same sides of the Transwell with each other.
7. Fill each chamber with 4 mL of the appropriate Ringers solution. Make sure that there are no bubbles at the end of the electrode tips or inside the chambers. Bubbles at any point in the path of the electrodes will prevent the electrodes from being able to record.
8. Put the gas lines into each chamber and turn on the Blood Gas (95%/5% O₂/CO₂) tank. Adjust the nobs on the top of the chambers until the bubbling rate is appropriate for mixing the solution without being too vigorous.
9. Correct the electrode offset potential by setting “Meter” to “V,” pressing the “Function” button to read “Open,” and using the “Offset” button and dial to adjust until the meter reads “0.0”.
 - (a) *Note: The offset dial can only adjust ± 10 , so if the offset is greater than that, the electrodes may need to be switched out, or there may be a bubble somewhere in the circuit.*
10. Correct the fluid resistance compensation by pressing and holding the “Push To Adj” button while adjusting the “Fluid Resistance Comp” dial until the displayed voltage reads “0.0”.
 - (a) *Note: The FRC Range and Gain may need to be adjusted depending on the surface area of the slider being used.*
11. Wash the monolayer of cells in Ringers solution prior to clipping the plastic support off of the Transwell, if necessary, and mounting the tissues into the chamber.
 - (a) *Note: Fill the basolateral side of the chamber first, as it is detrimental for the basolateral side of epithelial cells to be exposed to air.*
12. Give the computer control of the amplifier by pressing the “Function” button to read “Clamp” and the “Rem” button to enable the computer to control the amplifier.
13. Reference tissues by selecting the “Acquire” tab, the “Reference” subtab, then the “Reference” button. Boxes should remain white to indicate no issues. Yellow indicates a small

offset issue. Red indicates a significant offset. Something is very wrong with the setup in this case and must be rectified immediately

- (a) *Note: If the problem cannot be determined, unselect the chamber in Acquire and Analyze, unselect “Rem,” take the chamber off of “Clamp” mode, and adjust the chamber to be a dummy membrane (flip both switches on the DM-MC6 box to “Test”).*

14. Begin data acquisition in the software.
15. After the experiment is done, wash the chambers and sliders by soaking overnight in Contrad 70 detergent solution diluted with tap water. Rinse three times with tap water, then rinse with distilled water.

3.3 Recording from Channels Reconstituted into Planar Lipid Bilayers

The reconstitution of ion channel proteins into planar lipid bilayers (PLB) enables the study of channel function in lipid membranes of known composition. These methods have been in use since Mueller and coworkers first described in 1962 the incorporation of proteoliposomes from the electric organ of *Electrophorus* into a lipid bilayer, and the resulting change in conductance across the bilayer [19]. This was followed by an explosion of studies, including from the de Robertis lab [20], that explored the ion selectivity of the reduction in resistance induced by introduction of proteolipid to the bilayer and the subsequent increase in conductance observed upon addition of acetylcholine. The resolution of this recording method was greatly improved upon the invention of the patch clamp amplifier [21], such that one may now investigate the activity of ion channels with even very low conductance and fast kinetics.

An advantage of this approach is that both the cytoplasmic and extracellular aspects of the channel—or other protein under study—are accessible from either the *cis* or *trans* chambers. This is particularly useful when studying a protein that is activated by a ligand added to either cytoplasmic or extracellular solution, in contrast to a voltage-stimulated channels that can be activated by the flip of a switch on the amplifier. A second major advantage is that the composition of the planar lipid bilayer is essentially at the discretion of the researcher. A wide variety of mixtures of phospholipids, sphingomyelin, cholesterol, or other amphiphiles can be included in the bilayer, and their effects on protein function studied at very high resolution.

Channels may be added to the PLB at various stages of purification, from proteoliposomes generated from mixed microsomes (membranes from both organellar and plasma membranes), proteoliposomes generated from only one membrane source (i.e., plasma membranes or mitochondrial membranes), or protein purified to monodispersity in detergent or exogenous lipid(s). The following protocol is for the study of CFTR protein added to the



Fig. 5 Recording of CFTR reconstituted into a planar lipid bilayer. Shown is a current trace measured at $V_M = -60$ mV with asymmetric chloride concentration resulting in 100 mV electrochemical driving force. C = closed current level. O = full open state. Currents were filtered at 0.1 kHz

PLB in the form of proteoliposomes generated from plasma membranes of baby hamster kidney (BHK) cells stably transfected to express human CFTR with a $6\times$ Histidine tag on the C-terminus (Fig. 5) [22].

3.3.1 Setting Up the Equipment for a Planar Lipid Bilayer Experiment

This protocol assumes the use of an Axopatch 200B amplifier, Digidata 1322A, and a Spin-2 stir plate and controller. The Spin-2 stir plate should be placed inside of a solid aluminum box (grounded), which is inside of a Faraday cage on a floating anti-vibration table (also grounded). A foam mat should be placed on top of the stir plate to partially buffer the planar lipid bilayer chamber from the stir plate. The stir plate should be connected to the Spin-2 controller, which also should be grounded. The head stage connected to the amplifier (grounded) should run through the Faraday cage and into the aluminum box. On the amplifier, “Pipette Offset” should remain around 5. The “Configuration” should be in “Patch $\beta = 1$ ” for the capacitive head stage, and should be in “Whole Cell $\beta = 1$ ” for the resistive head stage. “Mode” should be set to “V-clamp.”

3.3.2 Preparing for a Planar Lipid Bilayer Experiment

1. Prepare any glass rods needed to paint the lipids onto the aperture by using a Bunsen burner to melt the end of a capillary tube into a rounded end (with no hole) and give a slight curve to this knob.
2. Prepare any agar bridges needed to connect the electrodes to the recording solution by using a Bunsen burner to bend a capillary tube into a small arch that fits between the well for the electrode and the well for the recording solution. Trim the tube to be the correct length using a diamond-tipped pen.
3. Fill the bridge with agar by melting 3% agar in 1 M KCl, hooking the capillary arch on the rim of a small beaker, and pour the melted agar into the beaker so that capillary action draws the agar into the entire capillary tube. Make sure the bridges do not have any bubbles.
4. Store the agar bridge into a solution of 1 M KCl.

3.3.3 *Running a Planar Lipid Bilayer Experiment*

1. Prepare the lipids by first determining the concentration relevant to the planned experiment. Protocols use anywhere between 10 and 50 $\mu\text{g}/\text{mL}$ lipids in decane as the final concentration for painting a bilayer on the aperture. Lipids are water-sensitive, so before opening any vial, let the vial warm to room temperature to prevent condensation on the inside of the vial. Furthermore, once lipids are dissolved in chloroform and opened, lipids should be aliquoted with glass syringes into glass vials with appropriate chloroform-resistant lids and stored at -80°C . Note that glass syringes are important as chloroform dissolves plastic, so using standard plastic pipettes will contaminate the lipids. Furthermore, note that chloroform is highly volatile, carcinogenic, and potentially lethal if acutely exposed at high concentrations. Always work with chloroform under a chemical fume hood.
2. Using a glass syringe, cleaned three times with chloroform, collect the appropriate volume of lipids and dispense it into a glass Reacti-Vial.
3. Using N_2 gas, gently dry the lipids in the Reacti-Vial, making sure that all chloroform is gone.
4. Using a clean glass syringe, collect the appropriate amount of decane and dispense it onto the dried lipids in the Reacti-Vial.
5. Vortex the Reacti-Vial briefly to dissolve any lipids that dried to the walls of the vial.
6. On the outside of the removable cup of the bilayer chamber, use the glass rod to paint a little bit of lipids onto the aperture.
7. Use the pressure of your thumb to push air through the aperture, and allow the lipids to dry for one minute. Then, put the removable cup into the larger chamber.
8. Add an appropriate volume of recording solution into both chambers, keeping in mind the volumes of solutions that will need to be added later in the experiment and the capacity of the cups ($\sim 0.9\text{ mL}$ is normally appropriate).
9. Use the pressure of your thumb to force some solution through the aperture to remove any bubbles that have formed on either side of it.
10. Add 1 M KCl to the outside wells for the electrodes.
11. Add the stir bars to each side of the aperture, making sure that the magnets align with the chambers.
12. Connect the electrodes to the head stage and place the electrodes into the outside wells. Dental wax can be used to keep the electrode placement steady during the experiment.
13. Connect the electrode wells to the recording chambers using the agar bridges.

14. Turn on the amplifier, Digidata, and low-pass Bessel filter.
15. Switch the “ext. command” to “on” and run the protocol, using the “pipet offset” dial to 0 the setup as much as possible.
16. Use Clampex (part of the pClamp suite) to run a protocol that performs a voltage ramp from 0 mV to +200 mV to 0 mV over 1 s. This protocol is used to evaluate the capacitance of the bilayer when it is formed, using the equation

$$I_{\text{cap}} = C \frac{dv}{dt}$$

where I_{cap} is the capacitive current, C is the capacitance, and dv/dt is calculated from the protocol (0.2 V/1 s).

17. Continuing to run the protocol, use the glass bubble to paint on lipids (or smudge around the lipids that are already there) until a stable bilayer is formed, as determined by the emergence of a pseudo square-wave while the protocol is running.
18. Once a stable bilayer is formed, switch to a protocol that steps from 0 mV, to +100 mV, to −100 mV, back to 0 mV over 1.2 s.
19. Very carefully, add the protein of interest (and any other necessary components) to the *cis* side (the outer cup), adding an equal volume of control buffer to the *trans* side (the inner cup). For example, for CFTR recordings, add to the *cis* side:
 - (a) 10 μg of microsomes from BHKs overexpressing CFTR.
 - (b) 500 μM MgATP (to facilitate channel activity).
 - (c) 50 U/mL PKA (to activate CFTR).
 - (d) 300 mM KCl (up from the 50 mM base buffer) to establish an ion gradient and therefore a voltage potential.
 - *Note: Some people add an ion gradient from cis to trans at this point only to facilitate fusion. If they do this, they do not add an equal volume of control buffer to the trans side, but wait until incorporation occurs to add an equal volume of the ions to the trans side to abolish the gradient.*
20. Turn on the stir plate and wait for incorporation. An incorporation event can be detected as:
 - (a) An out-of-place capacitance spike.
 - (b) Unstable currents.
 - (c) An increase in the leak current between the +100 mV and −100 mV holding currents.
21. If no incorporation occurs after 20 min, break the bilayer by flicking the chamber and quickly repainting the bilayer over the aperture, preferably without adding new lipids. Sometimes, an ion channel will actually be painted into the bilayer, and recording can begin immediately.

22. Once incorporation has occurred, turn off the stir plate immediately, stop running the protocol, and begin recording.
23. Switch the “ext. command” to “off” and the holding command to “+” or “–,” depending on the channel being studied.
24. At this point, bilayers can last anywhere between 10 s and 45 min, depending on the stability of the membrane, which is based on lipid composition, membrane capacitance, applied voltage, and other factors.
25. After a while of painting bilayers, if more than five or six swipes of lipid have been added to the aperture, discard everything, force some solution through the aperture, and try again to form a fresh bilayer.

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References

1. Cohen SP, Haack KK, Halstead-Nussloch GE et al (2010) Identification of RL-TGR, a coreceptor involved in aversive chemical signaling. *Proc Natl Acad Sci U S A* 107:12339–12344
2. Haack KK, Tougas MR, Jones KT et al (2010) A novel bioassay for detecting GPCR heterodimerization: transactivation of beta 2 adrenergic receptor by bradykinin receptor. *J Biomol Screen* 15:251–260
3. Frizzell RA, Hanrahan JW (2012) Physiology of epithelial chloride and fluid secretion. *Cold Spring Harb Perspect Med* 2:a009563–a009563
4. Zhang ZR, Cui G, Zeltwanger S, McCarty NA (2004) Time-dependent interactions of glibenclamide with CFTR: kinetically complex block of macroscopic currents. *J Membr Biol* 201:139–155
5. Zhang ZR, Cui G, Liu X, Song B, Dawson DC, McCarty NA (2005) Determination of the functional unit of the cystic fibrosis transmembrane conductance regulator chloride channel. One polypeptide forms one pore. *J Biol Chem* 280:458–468
6. Machaca K, Qu Z, Kuruma A et al (2002) The endogenous calcium-activated Cl[−] channel in *Xenopus* oocytes: a physiologically and biophysically rich model system. In: Fuller C, Benos DJ (eds) *Current topics in membranes: chloride channels of excitable and non-excitabile cells*. Elsevier Science, San Francisco, CA
7. Cui G, Zhang ZR, O’Brien AR, Song B, McCarty NA (2008) Mutations at arginine 352 alter the pore architecture of CFTR. *J Membr Biol* 222:91–106
8. Cui G, Khazanov N, Stauffer BB et al (2016) Potentiators exert distinct effects on human, murine, and *Xenopus* CFTR. *Am J Physiol Lung Cell Mol Physiol* 311:L192–L207
9. Cui G, McCarty NA (2015) Murine and human CFTR exhibit different sensitivities to CFTR potentiators. *Am J Physiol Lung Cell Mol Physiol* 309:L687–L699
10. Cui G, Stauffer BB, Imhoff BR et al (2019) VX-770-mediated potentiation of numerous human CFTR disease mutants is influenced by phosphorylation level. *Sci Rep* 9:13460
11. Cui G, Song B, Turki HW et al (2012) Differential contribution of TM6 and TM12 to the pore of CFTR identified by three sulfonylurea-based blockers. *Pflügers Arch* 463:405–418
12. Cui G, Hong J, Chung-Davidson YW et al (2019) An Ancient CFTR ortholog informs molecular evolution in ABC Transporters. *Dev Cell* 51:421–430
13. McCarty NA, McDonough S, Cohen BN et al (1993) Voltage-dependent block of the cystic fibrosis transmembrane conductance regulator Cl[−] channel by two closely related arylamino-benzoates. *J Gen Physiol* 102:1–23

14. Cohen SA, Hatt H, Kubanek J et al (2008) Reconstitution of a chemical defense signaling pathway in a heterologous system. *J Exp Biol* 211:599–605
15. Stauffer BB, Cui G, Cottrill KA et al (2017) Bacterial Sphingomyelinase is a state-dependent inhibitor of the Cystic Fibrosis Transmembrane conductance Regulator (CFTR). *Sci Rep* 7:2931
16. Infield DT, Cui G, Kuang C et al (2016) Positioning of extracellular loop 1 affects pore gating of the cystic fibrosis transmembrane conductance regulator. *Am J Physiol Lung Cell Mol Physiol* 310:L403–L414
17. Zhang ZR, Zeltwanger S, McCarty NA (2000) Direct comparison of NPPB and DPC as probes of CFTR expressed in *Xenopus* oocytes. *J Membr Biol* 175:35–52
18. Rowe SM, Pyle LC, Jurkevante A et al (2010) DeltaF508 CFTR processing correction and activity in polarized airway and non-airway cell monolayers. *Pulm Pharmacol Ther* 23:268–278
19. Mueller P, Rudin DO, Tien HT et al (1962) Reconstitution of cell membrane structure *in vitro* and its transformation into an excitable system. *Nature* 194:979–980
20. Parisi M, Reader TA, De Robertis E (1972) Conductance properties of artificial lipidic membranes containing a proteolipid from *Electrophorus*: response to cholinergic agents. *J Gen Physiol* 60:454–470
21. Hamill OP, Marty A, Neher E et al (1981) Improved patch-clamp techniques for high-resolution current recording from cells and cell-free membrane patches. *Pflugers Arch* 391:85–100
22. Linsdell P, Hanrahan JW (1996) Disulphonic stilbene block of cystic fibrosis transmembrane conductance regulator Cl[−] channels expressed in a mammalian cell line and its regulation by a critical pore residue. *J Physiol* 496:687–693



Isothermal Titration Calorimetry of Membrane Proteins

Han N. Vu, Alan J. Situ, and Tobias S. Ulmer

Abstract

The ability to quantify protein–protein interactions without adding labels to protein has made isothermal titration calorimetry (ITC) a preferred technique to study proteins in aqueous solution. Here, we describe the application of ITC to the study of protein–protein interactions in membrane mimics using the association of integrin α IIb and β 3 transmembrane domains in phospholipid bicelles as an example. A higher conceptual and experimental effort compared to water-soluble proteins is required for membrane proteins and rewarded with rare thermodynamic insight into this central class of proteins.

Key words Bicelle, Isothermal titration calorimetry, Membrane protein, Protein–lipid interaction, Protein–protein interaction, Transmembrane helices

1 Introduction

Membrane proteins (MP) have evolved from controlling the exchange of metabolites and environmental sensing in colonies of prokaryotic cells to organizing organelles within eukaryotic cells and assembling such cells into multicellular organisms [1, 2]. These roles make their correct folding, appropriate stability, and balanced interactions central to cellular function and dysfunction. Most MP fold by assembling transmembrane (TM) α -helices into bundles [3, 4]. Accordingly, helix–helix interactions underlie folding, determine stability, and mediate interactions between protein (s) (segments) within the membrane. To understand these processes quantitatively, a thermodynamic description is necessary. Here, we protocol the thermodynamic characterization of the association of the heterodimeric integrin α IIb β 3 TM complex in phospholipid bicelles as a prototypical example for the application of isothermal titration calorimetry (ITC) to MP.

In ITC, one protein is placed in the sample cell, which together with the reference cell, is thermostated and surrounded by an adiabatic shield. The second protein (ligand) is loaded into a syringe with a stirring blade on the tip, and incrementally injected

into the sample cell at defined intervals and stirring speed until all binding sites on proteins are occupied by ligand [5]. An injection i , which results in the release or absorption of heat, leads to temperature differences between the reference and sample cells. The power needed to offset this difference (isothermal condition) yields a measurement of the underlying heat change, δH_i , which is quantified with respect to the number of injected ligand molecules (Fig. 1). In the most effective experimental design, protein and ligand concentrations are chosen where injected ligand fully binds to protein and no free ligand exits until all protein is occupied [5]. Nonlinear curve fitting of δH_i as a function of protein and ligand concentrations, P and L_i , respectively, then yields the binding (or association) constant $K_{PL} = [PL]/[P][L]$, binding enthalpy ΔH° and protein-to-ligand stoichiometry (N). Additionally, the binding free energy and entropy changes follow from $\Delta G^\circ = -RT \ln K_{PL} = \Delta H^\circ - T\Delta S^\circ$, where T denotes the absolute temperature and R the gas constant.

The application of ITC to MP mandates a number of additional considerations over water-soluble proteins. First, for protein and injected ligand to mix, a chosen membrane mimic must allow the exchange of immersed proteins. Micelles and bicelles but not vesicles and nanodiscs generally fulfill this requirement. Isotropic bicelles, which are made up of a long-chain bilayer disc that is stabilized by short-chain rim lipids at a ratio (q-factor) of long- to short-chain lipid of ≤ 1.5 (Fig. 2a), provide an environment that sustains the structural integrity of integrin $\alpha\text{IIb}\beta 3$ [6, 7]. Second, the preorientation of helix–helix alignments in the two-dimensional bicelle bilayer, as opposed to their random orientation in three-dimensions, increases with the q-factor (Fig. 2b–c) and leads to substantial entropy savings [8, 9]. Thermodynamic parameters in bicelles therefore need to be compared at the same q-factor. Third, free energy changes for lipid-associated proteins are expressed with respect to the mole fraction standard state, that is, with respect to the number of solvating lipid molecules and not the sample volume [10–12]. This approach permits the comparison of ΔG° between different membrane mimics, provided they form an ideal solvent [13]. In practice, this condition must be tested by determining ΔG° at different lipid-to-protein ratios. For an ideal-dilute solute–solvent system, ΔG° is invariant. Fourth, K_{PL} and N may not be obtainable in one experiment. For integrin $\alpha\text{IIb}\beta 3$, immediate binding of injected ligand was only achieved at nonideal solvent conditions [9], which allowed the determination of N but not K_{PL} (Fig. 1a). Nonetheless, with the knowledge of N , K_{PL} could be measured at higher lipid-to-protein ratios when binding saturation is obtainable only in excess of ligand (Fig. 1b) [9, 14]. In sum, a higher experimental effort to study MP by ITC is rewarded with rare thermodynamic insight into this class of proteins and additional information on the role of lipids in TM helix–helix association [9, 15].

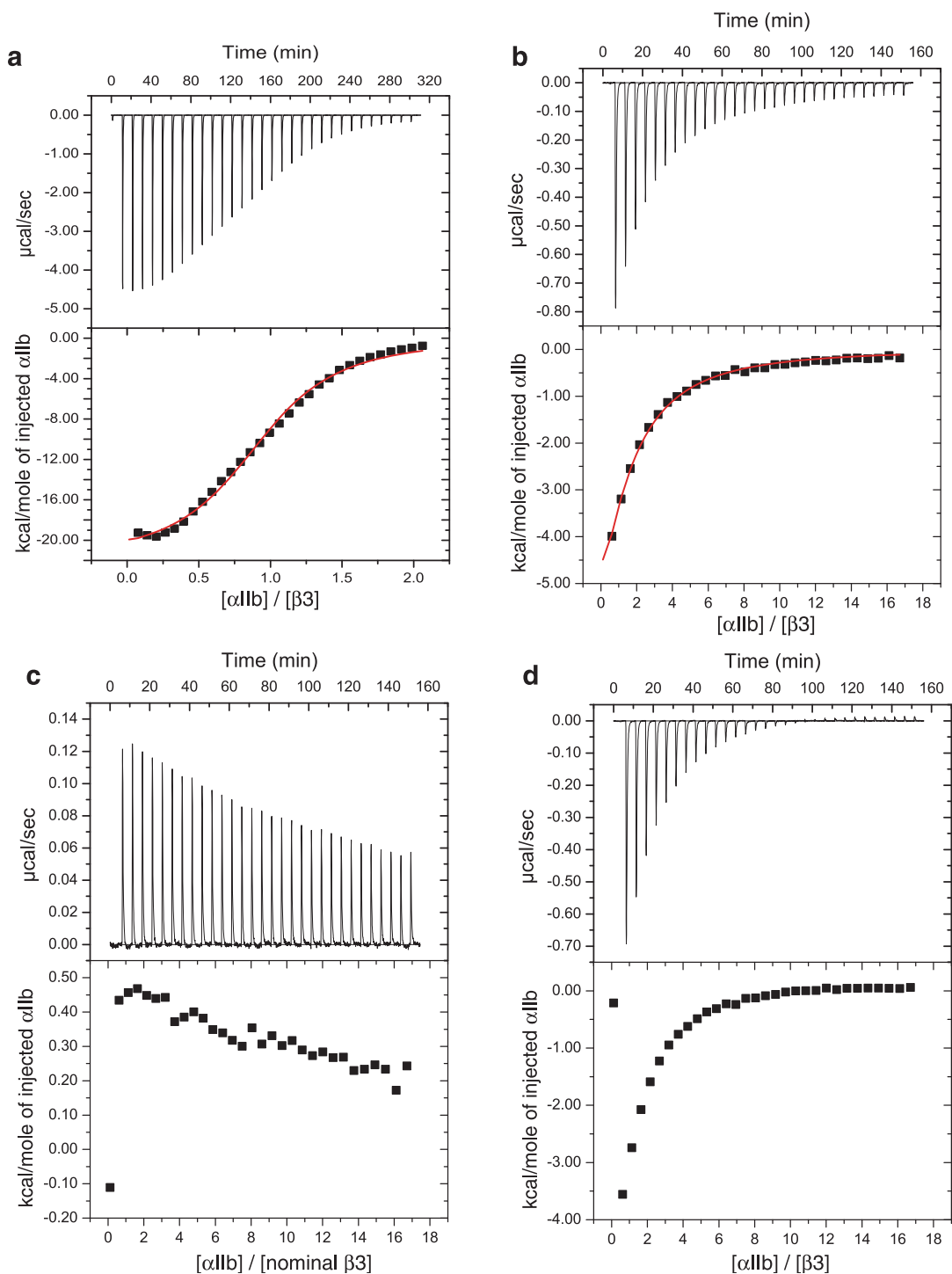


Fig. 1 Integrin $\beta 3$ titrations with αIIb peptide. Heat changes are depicted as a function of time and injected αIIb peptide. **(a)** At low lipid-to-peptide ratios, N could be evaluated. Moreover, $N = 1$ indicates the absence of homodimerization. Peptide and lipid concentrations of $81 \mu\text{M}$ $\beta 3$, 12 mM DHPC, and 18 mM DMPC (1,2-dimyristoyl-*sn*-glycero-3-phosphocholine) were used in 25 mM $\text{NaH}_2\text{PO}_4/\text{Na}_2\text{HPO}_4$, pH 7.4 solution at 28°C [9] (Figure reproduced from ref. [9] with permission from Elsevier). **(b)** At high lipid-to-peptide ratios, K_{PL} could be obtained. **(c, d)** From the titration of $\beta 3$ peptide with αIIb peptide (panel **d**), the heats of dilution of αIIb peptide alone (panel **c**) were manually subtracted to yield the final titration curve (panel **b**). Peptide and lipid concentrations of $10 \mu\text{M}$ $\beta 3$, 43 mM DHPC, and 17 mM POPC were used in 25 mM $\text{NaH}_2\text{PO}_4/\text{Na}_2\text{HPO}_4$, pH 7.4 solution at 28°C

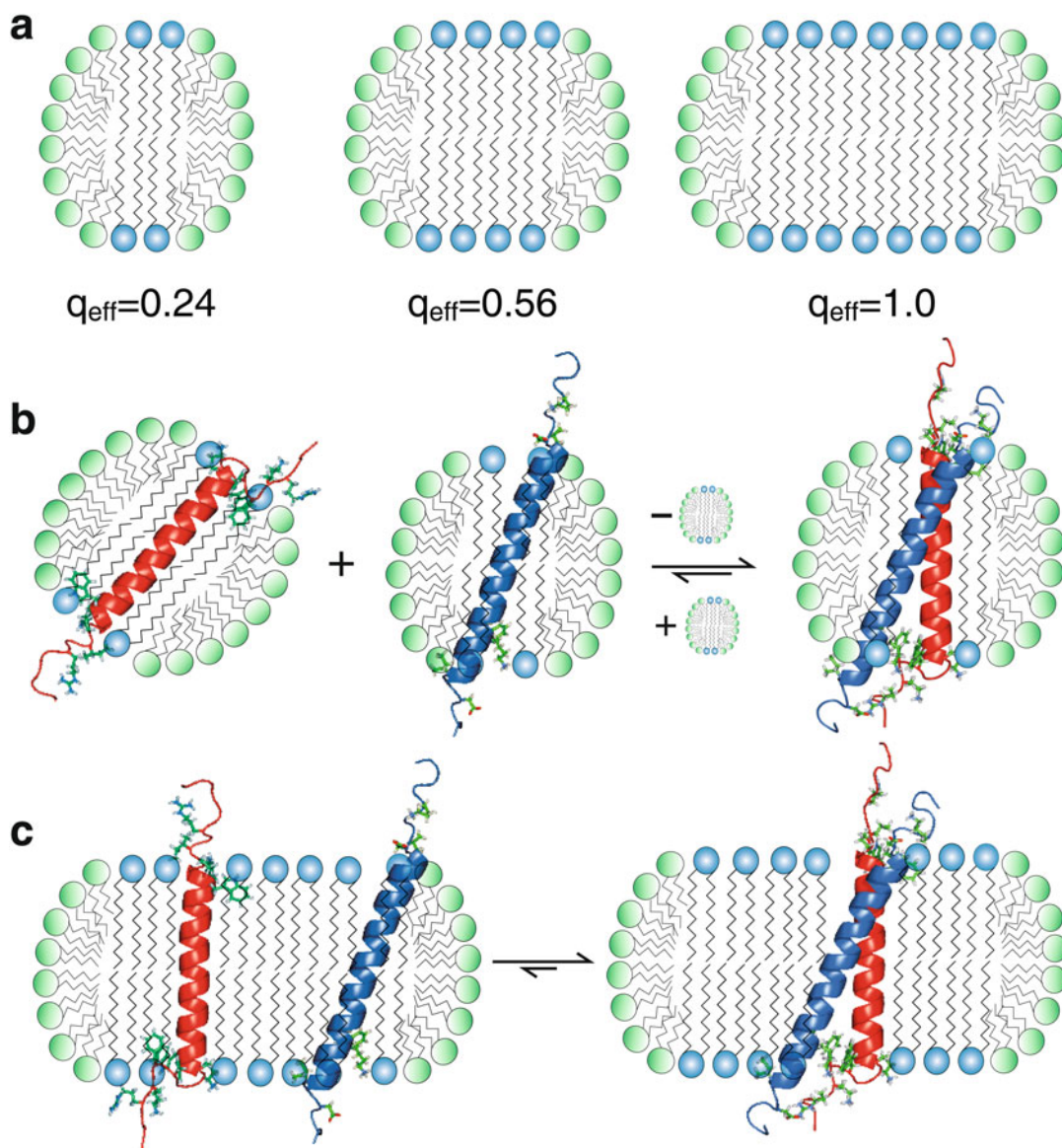


Fig. 2 Illustration of bicelle dimensions and helix–helix preorientation. Short- and long-chain lipids are shown with green and blue headgroups, respectively. **(a)** The ratio of long- to short-chain lipid (q -factor) determines bicelle dimensions. **(b, c)** In small bicelles, helices do not adopt a preferred orientation with respect to each other in contrast to their increasingly simultaneous presence in larger bicelles [9] (Reproduced from ref. [9] with permission from Elsevier)

2 Materials

A MicroCal VP-ITC system (Malvern Panalytical Ltd) including ThermoVac degasser is used in this protocol. All buffer solutions were prepared using MilliQ H₂O and filtered through a 0.2 μ M filter. Buffer is stored at room temperature.

2.1 Buffer Preparation

1. 0.5 M sodium phosphate, monobasic stock solution: Add 500 mL water to a 1 L graduated beaker. Weigh 59.99 g sodium phosphate, monobasic and transfer to the beaker. Add water to 1 L and mix with a magnetic stirring bar until dissolved.
2. 0.5 M sodium phosphate, dibasic stock solution: Add 500 mL water to a 1 L graduated beaker. Weigh 70.98 g sodium phosphate, dibasic and transfer to the beaker. Add water to 1 L and mix until dissolved.
3. 20 $\times$ sample buffer stock solution: 0.5 M NaH₂PO₄/Na₂HPO₄, pH 7.4. Add 800 mL 0.5 M sodium phosphate, dibasic solution to a beaker. Under the monitoring of a pH meter, slowly add 0.5 M sodium phosphate, monobasic to titrate the pH of the 0.5 M sodium phosphate, dibasic solution to a value of 7.4.
4. 1 $\times$ sample buffer solution: 25 mM NaH₂PO₄/Na₂HPO₄, pH 7.4. Dilute 50 mL of 20 $\times$ sample buffer stock solution with 950 mL of water to make 1 L of 25 mM NaH₂PO₄/Na₂HPO₄, pH 7.4 sample buffer solution. Filter the solution using a 0.2 μ M filter and store at room temperature.

2.2 Bicelle Preparation

1. Bicelle stock solution: Lipids may be purchased from Avanti Polar Lipids, Inc. The short-chain lipid 1,2-dihexanoly-*sn*-glycero-3-phosphocholine (DHPC), making up the bicelle rim (Fig. 2a), is best ordered in 200 mg aliquots (*see* **Note 1**). The long-chain lipid 1-palmitoyl-2-oleoyl-glycero-3-phosphocholine (POPC) may be ordered in any desired aliquot. A bicelle working solution of 43 mM DHPC and 17 mM POPC is desired for integrin α Ib β 3. This gives a q-factor of 0.40 and when subtracting the free, nonbicellar DHPC concentration, estimated at 9 mM [16], an effective q-factor of 0.50. At these lipid and employed α Ib β 3 peptide concentrations, ideal solvent conditions are encountered [15]. For uncharacterized systems, lipid-to-protein ratios have to be screened experimentally. Moreover, additional considerations apply to very weak and very strong interactions [17, 18]. To prepare the bicelle stock solution, add 882 μ L of 1 $\times$ sample buffer solution to a bottle containing 200 mg of DHPC (*see* **Note 1**). The concentration of DHPC and solution volume

may be quantified at this point (*see Note 2*). Then, carefully weigh 133 mg of POPC in a weighing boat and transfer to the DHPC bottle. Wait for the solution to become clear with intermittent gentle vortexing (*see Note 3*). The solution volume should be quantified at this point and the concentration of DHPC + POPC may be quantified (*see Note 2*). Typically, sample volumes of approximately 1200 μL are obtained and DHPC and POPC concentrations near 410 and 160 mM, respectively. If needed, additional POPC may be added to adjust the q-factor to 0.40 (*see Note 4*). Store dissolved lipids at -20°C .

2.3 Preparation of Integrin αIIb and $\beta 3\text{TM}$ Peptides

1. Bicelle working solution: Dilute the bicelle stock solution to final DHPC and POPC concentrations of 43 mM and 17 mM, respectively, using $1\times$ sample buffer solution.
2. Integrin αIIb will be loaded into the injections syringe whereas integrin $\beta 3$ is placed into the sample cell. Accordingly, prepare integrin αIIb and $\beta 3$ TM peptides at final concentrations of 0.8 mM and 10 μM in 500 μL and 2000 μL , respectively (*see Note 5*). Moreover, to determine the heat of ligand dilution, an additional 500 μL of 0.8 mM αIIb peptide is required.

3 Methods

3.1 Preparation of Calorimeter

1. Degas 2 mL water for 10 min using the MicroCal ThermoVac. Replace water in the reference cell with a long needle syringe (*see Note 6*).
2. Rinse the sample cell three times using $1\times$ sample buffer.

3.2 Loading of Sample Cell

1. Degas the sample solution for 5 min.
2. Remove all buffer in the sample cell, then load the integrin $\beta 3$ solution using a long-needle 2.5 mL glass syringe. Hold the syringe in the upright position and slowly lower the needle until it hits the bottom. Pull the needle up by about 1 mm, then gently release the sample solution.
3. Slightly overfill the sample cell with sample solution, then slowly pull up the needle and place the tip on the ledge where the cell stem meets the cell port and remove excess sample solution.

3.3 Loading of Injection Syringe

1. Transfer 500 μL integrin αIIb sample into a small glass test tube and put it in the holder of the instrument's pipette stand. Insert the injection syringe (auto-pipette) into the pipette stand until its tip touches the bottom of the glass test tube.

2. Attach the tube of the plastic filling syringe to the filling port of the injection syringe. Slowly pull the plunger of the plastic filling syringe to draw the integrin α IIB sample solution up until it exits through the top filling port.
3. Click the **Close Fill Port button** on the lower right corner of the ITC Controls window to lower the plunger of the injection syringe to close the port. Remove the hose of the plastic filling syringe from the filling port of the injection syringe.
4. Click on the **Purge** → **ReFill button** to dislodge any air bubbles from the walls of the injection syringe. Repeat this step two more times.
5. Carefully remove the injection syringe straight up from the stand and move above the center of the sample cell. Slowly insert the syringe into the sample cell.

3.4 Setting of ITC Run Parameters

1. In the ITC Controls window, select Thermostat/Calib, then, for the integrin titration, set the temperature to 28 °C, and click on Set Jacket Temp.
2. In the ITC controls tab, set total injections to 31, cell temperature to 28 °C, reference power to 10, initial delay to 60 s, molar syringe concentration to 0.8 mM (integrin α IIB), molar cell concentration to 0.01 mM (integrin β 3), and stirring speed to 307 RPM.
3. Select *high* for Feedback Mode/Gain, *Fast Equil* and *Auto* for ITC Equilibration Options.
4. Under the Injection Parameters, set the first injection volume to 2 μ L, duration to 4 s, and spacing to 300 s (*see Note 7*).
5. From the second to the 31st injections, set the injection volume to 9 μ L, duration to 10 s, and spacing to 300 s. Press start (the flag icon) at the top of the ITC Controls window to commence the ITC run.
6. Repeat steps starting from Subheading 3.1, **step 1**, with only bicelle working solution in the sample cell and 500 μ L of 0.8 mM α IIB peptide in the injection syringe to quantify the heat of ligand dilution (Fig. 1c).

3.5 Subtraction of Heat of Ligand Dilution and Data Fitting

1. To obtain the heats of α IIB + β 3 binding alone, the heats of ligand dilution need to be subtracted. In the ORIGIN analysis software, load the titration data and then the heats of ligand dilution (reference). Subtract the reference.
2. Under *Data Control*, click on *Remove Bad Data*, a square mouse pointer will appear. Move the pointer to the first injection data point and double-click to remove it (*see Note 7*).
3. Under *Model Fitting*, click on the *One Set of Sites* and a *Non-linear Curve Fitting Window* will pop-up.

4. Under *Parameter*, set the N value to 1 and uncheck the check mark on “Vary.” Click on the “100 Iter” button near the bottom until the displayed fitted values are constant. Press Done. A box with the determined N , ΔH° , and apparent K_{PL} , ΔG°_{app} , and ΔS°_{app} values will appear, which are based on the previously entered molar peptide concentrations.
5. Click on the *ITC tab*, then on *Final Figure*. In our version of MicroCal ORIGIN analysis software, the final figure is not displayed with the reference subtracted. We therefore subtract the reference manually. Nonetheless, the fitted values are not affected by the subtraction method.

3.6 Conversion of K_{PL} and ΔG° to the Mole Fraction Scale

1. With protein and ligand present only in bicelles, ΔH° expresses the heat absorbed or released to bind one mole of ligand (molar heat of ligand binding) in the hydrophobic phase. In contrast, the reported $K_{PL,app}$ and ΔG°_{app} need to be converted from the volume fraction to the mole fraction scale:

$$K_{PL} = K_{PL,app} \cdot [\text{lipid}],$$

where [lipid] denoted the sum of the bicellar DHPC + POPC concentrations in mol/L. As such, the concentration of free DHPC (9 mM [16]) must be subtracted, that is, $K_{PL} = K_{PL,app} \cdot [0.043 - 0.009 + 0.017] \text{ mol/L}$.

2. The mole fraction standard state free energy change, ΔG° , is then obtained from $\Delta G^\circ = -RT \ln K_{PL}$, where R is the gas constant (1.987 cal/K mol) and T is the absolute temperature.
3. Finally, the molar entropy change of ligand binding is updated: $\Delta S^\circ = (\Delta H^\circ - \Delta G^\circ)/T$.

4 Notes

1. Ordering freeze-dried DHPC in a bottle containing 200 mg avoids weighing this hygroscopic lipid. Allow the sealed bottle to warm to room temperature and, immediately after opening, add buffer to dissolve all lipid. Measure the volume by pipetting the solution into a new glass bottle.
2. Each lipid contains one phosphorus atom. Accordingly, the concentration of lipid can be determined by quantifying total phosphorus. The following protocol is taken from Avanti Polar Lipids, Inc.
 - (a) Prepare solutions of 8.9 N H_2SO_4 , 10% ascorbic acid, and 2.5% ammonium molybdate (VI) tetrahydrate.
 - (b) Prepare concentration standards. Pipet the following quantities from a 0.65 mM phosphorus standard solution

- (Sigma-Aldrich cat. no. P3869) into 12 × 75 mm test tubes: 0 μmol (0 μl) blank, 0.0325 μmol (50 μl), 0.065 μmol (100 μl), 0.114 μmol (175 μl), 0.163 μmol (250 μl), and 0.228 μmol (350 μl). Using a stream of nitrogen gas directed at the solution, gently evaporate all solvent.
- (c) Prepare the lipid sample. Pipet the equivalent of 0.1 μmol lipids into a 12 × 75 mm test tube. That is, take approximately 20 μl of the lipid sample, dilute with 1980 μl water and use 200 μl of this solution for quantification. Using a stream of nitrogen gas, gently evaporate all solvent. Analogously, prepare a sample of buffer alone.
 - (d) Add 0.45 ml H₂SO₄ to each tube of concentration standard, sample and buffer solution. Heat the tubes in an aluminum block in the chemical fume hood at 200 °C for 25 min.
 - (e) Let the solutions cool for 5 min, then add 150 μl of H₂O₂ to the bottom of each tubes.
 - (f) Put the tubes back in the heat block and incubate for another 30 min at 200 °C. The samples should be colorless at this point. If not, add 50 μl of H₂O₂ to all cooled tubes again and continue heating the tubes for 15 min. Let tubes cool to room temperature.
 - (g) Add 3.9 ml of water to all tubes, then add 0.5 ml of 2.5% ammonium molybdate (VI) tetrahydrate to each tube and vortex five times at moderate speed.
 - (h) Add 0.5 ml of 10% ascorbic acid to each tubes and vortex five times at moderate speed.
 - (i) Seal each tube with a cap to prevent evaporation. Then heat all tubes at 100 °C for 7 min. Cool the tubes to room temperature.
 - (j) Spectrophotometric analysis of samples: Zero the spectrophotometer using the 0 μmol standard. Determine the absorbance of the sample as well as of each standard at 820 nm. Create a linear calibration curve and then determine the amount of phosphorus in the lipid and buffer samples. Subtract buffer from sample values and use the determined volumes (*see Note 1*) to arrive at the lipid concentrations.
3. Allow the sealed bottle of POPC to warm to room temperature; freeze-dried POPC still absorbs moisture. Lipids dissolve at room temperature. However, up to three freeze-thaw cycles help to dissolve the lipids faster. Freezing also helps to eliminate bubbles formed during vortexing. To ensure a homogeneous

lipid solution, any liquid at the wall of the bottle should be spun down at $350 \times g$ for 5 min. Measure the volume by pipetting the solution into a new glass bottle.

4. For highest accuracy, especially when comparing for example mutant proteins, it is recommended to prepare all required lipid stock solutions simultaneously using the same batches of lipids and the same amount of POPC.
5. The concentration of purified integrin TM peptides [6, 7] solubilized in 56% acetonitrile, 24% n-propanol, and 0.01% trifluoroacetic acid is determined by UV spectroscopy at 280 nm. Subsequently, peptides are aliquoted and freeze-dried for solvation in bicelle working solution.
6. Water in the reference cell needs to be changed every 7 days.
7. The first injection data point will be deleted because of inaccuracy in its volume (an air pocket might form at the tip of the pipette needle during the transfer and insertion into the sample cell).

Acknowledgments

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References

1. Situ AJ, Ulmer TS (2019) Universal principles of membrane protein assembly, composition and evolution. *PLoS One* 14(8):e0221372
2. Markov AV, Kulikov AM (2005) Origin of Eukaryota: conclusions based on the analysis of protein homologies in the three superkingdoms. *Paleontol J* 39(4):345–357
3. Popot JL, Engelman DM (1990) Membrane-protein folding and oligomerization – the 2-stage model. *Biochemistry* 29(17):4031–4037
4. White SH, von Heijne G (2008) How translocons select transmembrane helices. *Annu Rev Biophys* 37:23–42
5. Wiseman T, Williston S, Brandts JF, Lin LN (1989) Rapid measurement of binding constants and heats of binding using a new titration calorimeter. *Anal Biochem* 179(1):131–137
6. Lau T-L, Partridge AP, Ginsberg MH, Ulmer TS (2008) Structure of the Integrin beta3 transmembrane segment in phospholipid bicelles and detergent micelles. *Biochemistry* 47(13):4008–4016
7. Lau T-L, Dua V, Ulmer TS (2008) Structure of the integrin alphaIIb transmembrane segment. *J Biol Chem* 283:16162–16168
8. Grasberger B, Minton AP, Delisi C, Metzger H (1986) Interaction between proteins localized in membranes. *Proc Natl Acad Sci U S A* 83(17):6258–6262
9. Situ AJ, Schmidt T, Mazumder P, Ulmer TS (2014) Characterization of membrane protein interactions by isothermal titration calorimetry. *J Mol Biol* 426:3670–3680
10. Carman GM, Deems RA, Dennis EA (1995) Lipid signaling enzymes and surface dilution kinetics. *J Biol Chem* 270(32):18711–18714
11. Fisher LE, Engelman DM, Sturgis JN (1999) Detergents modulate dimerization but not helicity, of the glycophorin A transmembrane domain. *J Mol Biol* 293(3):639–651
12. Kochendoerfer GG, Salom D, Lear JD, Wilk-Orescan R, Kent SBH, DeGrado WF (1999) Total chemical synthesis of the integral membrane protein influenza A virus M2: role of its C-terminal domain in tetramer assembly. *Biochemistry* 38(37):11905–11913

13. Fleming KG (2002) Standardizing the free energy change of transmembrane helix-helix interactions. *J Mol Biol* 323(3):563–571
14. Turnbull WB, Daranas AH (2003) On the value of c : can low affinity systems be studied by isothermal titration calorimetry? *J Am Chem Soc* 125(48):14859–14866
15. Schmidt T, Suk JE, Ye F, Situ AJ, Mazumder P, Ginsberg MH, Ulmer TS (2015) Annular anionic lipids stabilize the integrin α IIb β 3 transmembrane complex. *J Biol Chem* 290(13):8283–8293
16. Chou JJ, Baber JL, Bax A (2004) Characterization of phospholipid mixed micelles by translational diffusion. *J Biomol NMR* 29(3):299–308
17. Hong H, Blois TM, Cao Z, Bowie JU (2010) Method to measure strong protein-protein interactions in lipid bilayers using a steric trap. *Proc Natl Acad Sci U S A* 107(46):19802–19807
18. Mineev KS, Lesovoy DM, Usmanova DR, Goncharuk SA, Shulepko MA, Lyukmanova EN, Kirpichnikov MP, Bocharov EV, Arseniev AS (2014) NMR-based approach to measure the free energy of transmembrane helix-helix interactions. *Biochim Biophys Acta Biomembr* 1838(1):164–172



Chapter 6

Atomic Force Microscopy Reveals Membrane Protein Activity at the Single Molecule Level

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Abstract

Atomic force microscopy has emerged as a valuable complementary technique in membrane structural biology. The apparatus is capable of probing individual membrane proteins in fluid lipid bilayers at room temperature with spatial resolution at the molecular length scale. Protein conformational dynamics are accessible over a range of biologically relevant timescales. This chapter presents methodology our group uses to achieve robust AFM image data of the General Secretory system, the primary pathway of protein export from the cytoplasm to the periplasm of *E. coli*. Emphasis is given to measuring and maintaining biochemical activity and to objective AFM image processing methods. For example, the biochemical assays can be used to determine chemomechanical coupling efficiency of surface adsorbed translocases. The Hessian blob algorithm and its extension to nonlocalized linear features, the line detection algorithm, provide automated feature delineations. Many of the methods discussed here can be applied to other membrane protein systems of interest.

Key words AFM, Imaging, Active, Lipid, Bilayer, Peripheral, Single molecule

1 Introduction

Originally developed within the materials science community to study solid-state surfaces such as silicon [1], the atomic force microscope (AFM) has emerged as an increasingly important tool in membrane biology. In AFM, samples are gently scanned with a vanishingly sharp tip affixed to a miniature robotic arm. This direct single molecule imaging method does not require crystallization or cryogenic preservation. Though AFM images of proteins are generally inferior to crystallographic data in terms of resolution, the conformational dynamics data which AFM provides (typical resolution: ~ 10 Å lateral, ~ 1 Å vertical) complements traditional

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structural biology methods [2]. AFM images and kymographs (one dimensional line scans plotted versus time) can be obtained in physiological conditions and on large biomolecules. This opens the door for real-time observation of membrane protein complexes at work in fluid lipid bilayers. However, to take full advantage of the method, the biochemical activity of samples prepared for AFM, which generally requires surface adsorption, should be preserved [3]. An additional concern is image analysis which often requires significant user input that can bias AFM results [4].

The general secretory (Sec) system is a dynamic membrane transport complex responsible for exporting a large and diverse set of proteins from the cytosol of *E. coli* and has homologues across all life [5]. Our group has employed AFM to study central components of the Sec translocase in isolation [6, 7] and in complex [8] using both mica, the traditional supporting surface for biological AFM, as well as cleaned glass coverslips [9].

Does the requirement of surface adsorption alter protein activity? Questions of this type come up frequently in discussions of AFM results. Our group developed and employed two biochemical activity assays to characterize the biochemical activity of Sec system samples as prepared for AFM study [3, 6, 8]. These assays provide quantitative metrics of surface-adsorbed translocase activity including the chemomechanical coupling of the Sec translocase (ATP hydrolyzed per amino acid translocated) [3].

Analysis of AFM data largely consists of image processing. Membrane protein protrusions above the bilayer appear in AFM images as populations of “blobs,” which must be detected as different from the lipid background and characterized by parameters such as height, area, and volume. The precision of this procedure can be increased and human bias reduced by using a particle detection algorithm with minimal user input. Toward this end we developed the Hessian blob algorithm [4]. Briefly, given an input image, the first step is to create a stack of scale-space representations with increasing Gaussian smoothing characterized by a scale parameter. In the Hessian Blob feature extraction process, proteins are detected by applying and maximizing the determinant of the normalized Hessian matrix to the scale space representation of the image. While this approach works well for localized single-particle-like proteins, the method breaks down if the features of interest exhibit spatially extended or linear attributes. This is because Hessian blob detection only picks up endcaps of linear features. We developed the line detection algorithm to overcome this limitation. Here, the topological skeleton of an image is used as the starting point for feature extraction. Each pixel of the skeleton acts as the starting point for a blob. The line detection algorithm then scans outward until a change in the sign of the Gaussian curvature is detected. Overlapping blobs are stitched together, hence

delineating contiguous linear features, such as kymograph data of protein conformational dynamics.

This chapter describes methodology our group has developed for AFM imaging and quantitative analysis of the Sec system at work. We pay special attention to technical issues that can limit the impact of AFM results. For example, the choice of supporting surface species can result in differing biochemical activity levels of surface-adsorbed membrane proteins. Furthermore, conclusions drawn from AFM work are only as robust and objective as the underlying analysis. The Hessian blob and the line detection algorithms are presented. These methods provide rigorous particle and linear feature analysis, respectively. Overall, our methodology was developed in an effort to address longstanding questions in the protein export field. We anticipate that some of the approaches presented here will generalize to other membrane proteins of interest.

2 Materials

1. Atomic force microscope (Homebuilt or Commercial).
2. AFM Cantilevers (Olympus, Biolever mini).
3. Glass coverslips.
4. Glass microscope slides.
5. Ruler.
6. MQ water: 18.2 M-Ohm cm water from Milli-Q filtration system.
7. Metal specimen disks.
8. Teflon shim stock.
9. Hydrochloric acid, concentrated (37%).
10. Ultrasonic bath (Branson, model # 5200).
11. Anhydrous ethanol: absolute ethanol, 200 proof.
12. Stirrer plate.
13. Magnetic stir bar.
14. Aluminum foil.
15. Plastic wrap.
16. Parafilm.
17. Teflon coverslip basket.
18. Beakers, assorted sizes.
19. Punch set.
20. Mica.
21. Tape.

22. Superglue and superglue primer.
23. 5-min epoxy
24. Toothpicks.
25. Diamond scribe.
26. Glass petri dish.
27. Ultrapure Nitrogen gas (N_2).
28. Ultrapure Oxygen gas (O_2).
29. Ultrapure Argon gas (Ar).
30. Liquid Nitrogen (LN_2).
31. Oxygen plasma cleaner (Harrick, model PDC-001).
32. *E. coli* polar lipid extract.
33. Biobeads SM-2.
34. Needle, 21 G.
35. Culture tubes 15 × 85 mm.
36. Liposome extruder with filter supports (Avestin, Liposofast).
37. Membrane filter, 200 nm (Whatman, Nuclepore Track-Etch Membrane).
38. Aluminum dish.
39. Microfuge tubes.
40. Potassium hydroxide (KOH).
41. Potassium Acetate (KAc).
42. Magnesium Acetate ($Mg(OAc)_2$).
43. Dithiothreitol (DTT).
44. Dodecyl- β -maltoside (DBM).
45. Ethylenediaminetetraacetic acid (EDTA).
46. Potassium dihydrogen phosphate (KH_2PO_4).
47. 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES)
48. Ethylene glycol-bis(β -aminoethyl ether)-*N,N,N',N'*-tetraacetic acid (EGTA).
49. Urea.
50. Centrifuge.
51. Purified proteins (SecYEG, SecA, SecB).
52. Precursor proteins (e.g., pOmpA, pGBP) radiolabeled by ^{14}C -Leu (PerkinElmer).
53. Proteinase K.
54. Adenosine triphosphate (ATP).
55. Radiolabeled [γ - ^{32}P] ATP (PerkinElmer).
56. Thin layer chromatography (TLC) plates.

57. Laboratory oven.
58. Phosphor imaging plate (Fujifilm BAS-MS 2340).
59. Phosphor imager (GE healthcare Amersham Typhoon).
60. Analysis software (WaveMetrics, Igor Pro; GE Healthcare Life Sciences, ImageQuant).
61. Proteoliposome buffer: 10 mM HEPES, pH 7.6, 30 mM KAc, 1 mM Mg(OAc)₂.
62. Translocase buffer: 10 mM HEPES, pH 7.6, 300 mM KAc, 5 mM Mg(OAc)₂.
63. ATP hydrolysis reaction mixture: 3.3 mM radiolabeled ATP (0.908 nCi/ μ L), 10 mM HEPES, pH 7.6, 300 mM KAc, 5 mM Mg(OAc)₂.
64. Precursor buffer: 10 mM HEPES, pH 7.6, 300 mM KAc, 5 mM Mg(OAc)₂, 1 mM DTT, 1 mM EGTA.
65. Translocation reaction mixture: 1 μ M SecB, 1 μ M radiolabeled precursor (pOmpA or pGBP), 3.3 mM ATP in 10 mM HEPES, pH 7.6, 300 mM KAc, 5 mM Mg(OAc)₂, 1 mM DTT, and 1 mM EGTA.
66. SecYEG incubation buffer: 80 nM SecYEG, 10 mM HEPES, pH 7.6, 200 mM KAc, 5 mM Mg(OAc)₂.

3 Methods

3.1 Choosing the Supporting Surface

Mica is the traditional surface for biological AFM imaging. Freshly cleaved mica is atomically flat, clean, and hydrophilic, making it a popular supporting surface for imaging biomolecules. We have developed a protocol that optimizes glass coverslips as an alternative surface to mica. In recent work [3], glass was shown to be a better candidate for maintaining biochemical activity of the Sec translocase, possibly because the additional roughness of glass provides more space for translocating polypeptide chains to occupy. Despite the fact that glass itself is rougher than mica, once a lipid bilayer has been adsorbed, the difference in roughness between the upper bilayer leaflet on glass and on mica is minimal [9]. For these reasons glass is a viable supporting surface for membrane protein imaging. Here follows a protocol for the preparation of either a mica or glass surface, onto which one can prepare supported lipid bilayers containing membrane proteins.

3.1.1 Subsurface Preparation

1. Clean prefabricated metal specimen disks (*see* Fig. 1) by immersing disks in concentrated hydrochloric acid in a fume hood for 15 min.
2. Rinse with MQ water and blow dry with N₂.

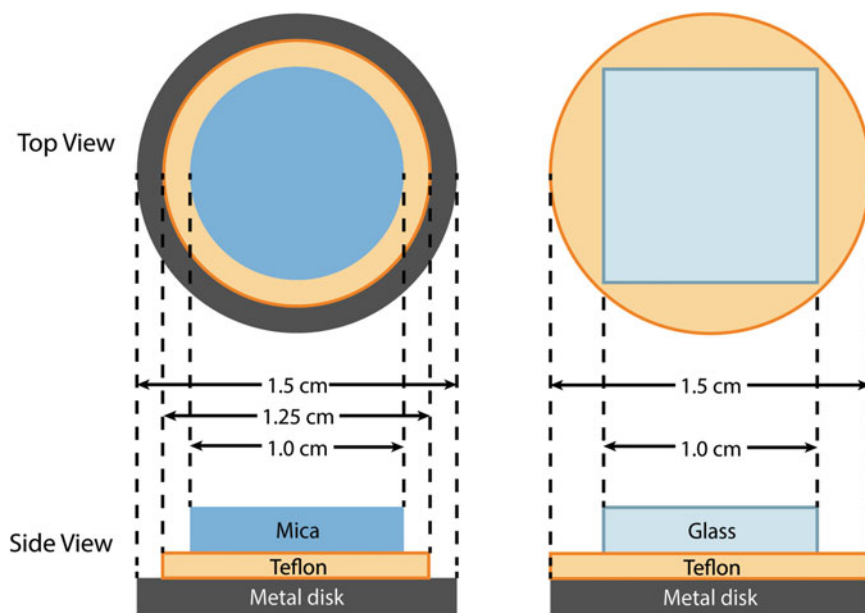


Fig. 1 Top and side views of supporting surface geometry for AFM imaging on either mica or glass. Dimensions shown are approximate

3. Using a punch set, punch out Teflon disks from the Teflon shim stock, 1.25–1.50 cm diameter. Be sure that the cut Teflon has a clean edge.
4. Wash the Teflon, alternating MQ water with ethanol in squirt bottles three times, then blow dry with N_2 .
5. Prime both the Teflon and the metal disk using primer (adhesion promoter). The metal disks dry slower, so coat these first and allow to dry. Teflon can be primed right before gluing onto the disk.
6. Dot superglue onto the primed metal disk and center Teflon on top. If there is a slight curl to the Teflon, lay it on the superglue dot with the concave side down (*see Note 1*).
7. Let dry for 10 min.

3.2 Mica Preparation

1. Punch out mica disks using a punch set. The diameter should be approximately 1 cm. Blow off any debris on the disks using N_2 .
2. Mix epoxy in an aluminum dish. Using a toothpick, dot the epoxy on the Teflon and center the mica on the Teflon. There should be a skirt of Teflon on all sides of the mica and the epoxy should not overflow. This step must be done quickly, as the epoxy begins to stiffen rapidly (about 1 min).
3. Cure for 1 h.

3.3 Glass Preparation

3.3.1 Wet Etching of the Glass

4. Immediately prior to use, cleave mica using scotch tape, making sure that the tape removes a full layer of mica.
1. To make the KOH-ethanol solution, put a large stir bar, 250 mL anhydrous ethanol, and 70 g potassium hydroxide pellets in a clean 1 L beaker. Cover the beaker with Aluminum foil to prevent splashing/spilling.
2. Set the stirrer to approximately 400 rpm. After about 1–2 h the solution should be saturated, indicated by a dark brown color. If this color is not achieved, add additional KOH pellets and continue stirring until the solution is the correct color (this may take 2–4 h). Note that the KOH-ethanol solution can be stored in an appropriate container at 4 °C.
3. Fill Branson 5200 ultrasonic bath $\frac{1}{4}$ full with water.
4. Fill two clean 1 L beakers with MQ water (enough to submerge coverslips, about $\frac{1}{2}$ full).
5. Arrange the beakers of MQ water and the beaker of KOH-ethanol solution in the bath. Turn the power “ON.” No heating is required.
6. Load coverslips into Teflon coverslip basket with the handle attached. A photograph of the custom made basket is shown in reference [10].
7. Submerge the coverslip basket and coverslips in the KOH-ethanol solution for 3 min. To enhance the cleaning process, move the basket up and down and side to side during this step.
8. Rinse coverslips with MQ water using squirt bottle.
9. Submerge coverslips into the first MQ water beaker for another 3 min.
10. Rinse coverslips with MQ water using squirt bottle again.
11. Submerge the coverslips into the second beaker of MQ water for another 3 min.
12. After the final submersion, rinse the coverslips over the sink with alternating MQ water and ethanol using squirt bottles, finishing with MQ water.
13. Blow dry using N₂. Visually inspect the glass. Check that there is no residue remaining.
14. Remove the basket handle and place in an auxiliary container. Glass should be stored in a dust-free, low-humidity environment.
15. Repeat **steps 7–14** for the rest of the coverslip baskets. Note, the KOH-ethanol solution can be stored and reused up to five times.

3.3.2 Mounting the Glass

1. Mark coverslips into quarters with a diamond-scribe and ruler, then gently break into four equal pieces.
2. Wash each glass square with ethanol and water squirt bottles three times, then blow dry with N₂.

3.3.3 Final Glass Cleaning

1. Place glass squares into a clean glass petri dish, with the preferred side facing up. Place the dish in the plasma-cleaner chamber uncovered.
2. Pump plasma-cleaner chamber down to <100 mTorr. Be sure all exterior-leading valves are closed and tubes have been back-pumped to remove any excess gas.
3. Open oxygen tank valve to allow flow, pressure should read ~20 psi.
4. Open the plasma-cleaner oxygen valve to allow O₂ to flow into the chamber. Adjust so that the pressure inside the chamber is ~250 mTorr.
5. Plasma clean on HIGH for 10 min. This should remove surface-bound organics and render the glass hydrophilic (*see Note 2*).
6. Turn off the plasma-cleaner. Vent the chamber and remove the dish of glass squares.
7. Mix epoxy and dot it onto the center of prepared Teflon/metal disks using a toothpick. Place a plasma-cleaned glass square on the dot of epoxy, being careful that the glass is flat to the eye and the epoxy does not overflow (*see Note 3*).
8. Deposit proteoliposome sample (*see Subheading 3.2* below) onto the glass. The epoxy will cure during the sample's incubation time.

3.4 Reconstitution of Purified Membrane Protein System

3.4.1 To Form Liposomes

1. Distribute *E. coli* polar lipid extract in chloroform to culture tubes such that there is approximately 13 mg lipid in each tube.
2. Evaporate chloroform using Ar gas under the hood, forming thin film of lipid on the wall of the tube. Keep rotating the tube while drying to evenly form film.
3. Place tube in the dry vacuum chamber and pump overnight (*see Note 4*).
4. For storage, backfill tubes with Ar gas, seal with parafilm, and store at -20 °C.
5. Take one tube of dry lipid prepared above (or out from the -20 °C freezer).
6. Add 550 µL of proteoliposome buffer (*see Subheading 2, item 61*).
7. Let sit at room temperature for 1 hour to swell the lipid.

8. Use pipet to rinse the wall of the tube and hydrate all lipids.
9. Wet a 200 nm membrane and filter supports with buffer.
10. Insert membrane into lipid extruder with filter supports on each side.
11. Label one side of the extruder with “IN.”
12. Load a 1000 μ L Hamilton syringe with lipid suspension.
13. Insert syringe into the extruder side labeled “IN.”
14. Insert an empty 1000 μ L Hamilton syringe to the other side of the extruder.
15. Pass the solution through the extruder 20–30 times (or until the solution becomes visually clear).
16. Remove solution from the side opposite “IN” into Eppendorf tubes.
17. Quantify lipid concentration using an appropriate assay.
18. Store at -80°C .

3.4.2 To Form Proteoliposomes

1. Use appropriate assays to purify and quantify desired membrane proteins such as the integral translocon SecYEG and the peripheral ATPase SecA.
2. Mix DBM, DTT and liposomes at the following molar ratio DBM:DTT:lipids = 10:4:10 in an Eppendorf tube.
3. Place the Eppendorf tube on a rotary mixer for 3 h at room temperature. This slowly inverts the tube with a period of about 1 s.
4. Add purified SecYEG to the tube at a 1:1000 molar ratio of SecYEG:lipid to the Eppendorf tube (*see* **Note 5**).
5. For coassembled reconstitution [11], add SecA dimer at a 1:1 molar ratio to SecYEG to the Eppendorf tube.
6. Continue incubation for 1 h at room temperature while rotary mixing.
7. Add Biobeads SM-2 (at an approximate ratio of 1:1 with the sample) to the tube to remove detergent.
8. Continue rotation for 90 min.
9. Use needle to poke a small hole at the bottom of the tube, place the Eppendorf tube into a larger centrifuge tube then spin at $5,000 \times g$ for 10 min at 10°C .
10. Load solution to another Eppendorf tube and centrifuge at $436,000 \times g$ for 20 min at 4°C .
11. Suspend the pellet in proteoliposome buffer (*see* Subheading 2, **item 61**).
12. Centrifuge the solution again at $436,000 \times g$ for 20 min at 4°C .

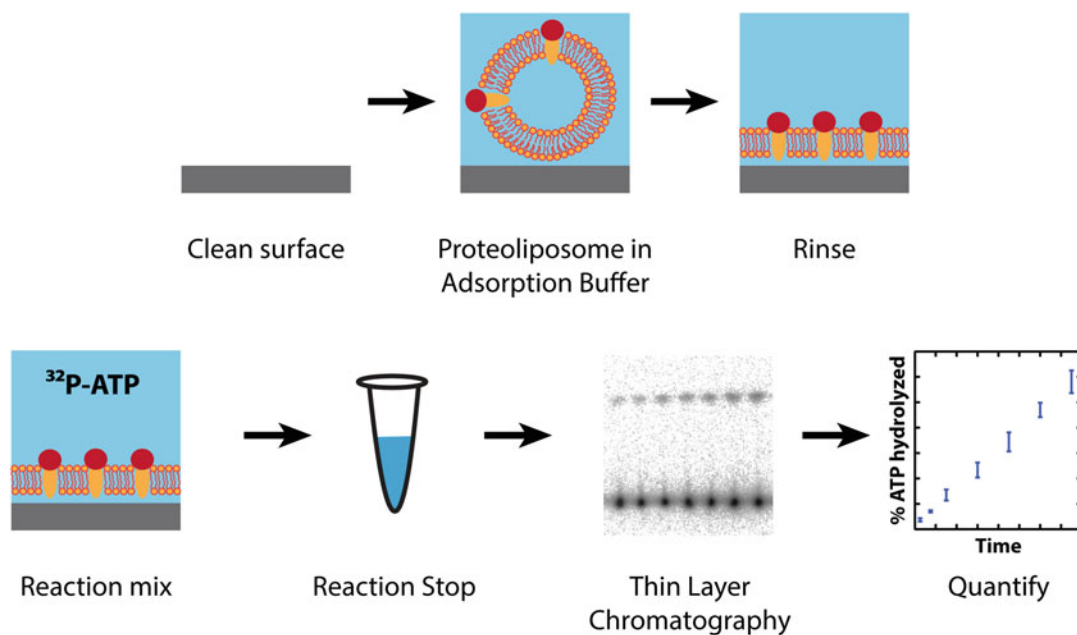


Fig. 2 Cartoon highlighting key steps of surface ATPase activity assay

13. Suspend the final pellet to give a concentration of ~10 mM lipid and 10 μ M SecYEG in proteoliposome buffer.
14. Aliquot into smaller Eppendorf tubes and flash freeze with LN₂ before storing at -80°C .

3.5 Biochemical Activity Assay of Surface-Adsorbed Translocase Complexes

3.5.1 ATP Hydrolysis Assay

1. To adsorb proteoliposomes onto surface, deposit 50 μ L of 80 nM proteoliposomes SecYEG coassembled with SecA (Y·A) on a freshly cleaned glass coverslip (or freshly cleaved mica surface) mounted on cleaned Teflon sheet adhered to a glass microscope slide.
2. Incubate the surface in a water saturated environment, on ice for 1 h.
3. Rinse the surface five times with 50 μ L of translocase buffer (*see* Subheading 2, item 62).
4. Add 50 μ L of ATP hydrolysis reaction mixture (Subheading 2, item 63) to the sample surface (Fig. 2 and *see* Note 6).
5. At desired times, remove 4 μ L of the surface solution and mix with 4 μ L of 50 mM EDTA.
6. Spot the solution on TLC plate and allow to dry.
7. Place the plate vertically in 125 mM KH₂PO₄ solution for ~1 h or until the level of the solution reaches ~1 in. below the top edge of the plate.
8. Dry the plate in oven at 70°C for 10 min.

9. Wrap the TLC plate in plastic wrap.
10. Expose the TLC plate to the imaging plate in a closed cassette for 15 min.
11. Develop the imaging plate.
12. Quantify activity with ImageQuant software.

3.5.2 Protein Translocation Activity Assay

1. Prepare one surface (using protocol described in Subheading 3.1) for each of the time points desired.
2. Deposit 50 μ L of 80 nM proteoliposomes Y-A on a freshly cleaned surface mounted on a cleaned Teflon sheet adhered to a glass microscope slide.
3. Incubate the surface in a water saturated environment, on ice for 1 h.
4. Rinse the surface five times with 50 μ L of translocase buffer (*see* Subheading 2, item 62).
5. Add 50 μ L of translocation reaction mixture (Subheading 2, item 65) to the surface.
6. At predetermined time points, vigorously rinse the surface with 50 μ L of 6 M Urea to stop the reaction (*see* Note 7).
7. Treat the surface with proteinase K by adding 70 μ L at 19 units/mL, incubate the surface for 15 min on ice.
8. Rinse the surface ten times with translocase buffer (*see* Subheading 2, item 62).
9. Pipet off the surface solution.
10. Dry the surfaces in the oven at 70 °C for 20 min.
11. Wrap the surfaces in plastic wrap.
12. Expose the surfaces to imaging plate in a closed cassette for 48 h.
13. Develop the imaging plate.
14. Quantify the translocation activity with ImageQuant software.
15. Compare with standard curve, made by spotting known quantities of radiolabeled precursor on surfaces which were developed and quantified in an identical manner to the translocation activity assay described here.

3.6 AFM Imaging of Membrane Proteins

3.6.1 Buffer Considerations

Buffers can be fine-tuned to optimize the biomolecular adsorption phase of an experiment as well as the AFM imaging phase. In order to minimize the probability of artifacts, buffers should have physiologically relevant salt and pH levels. Tip-sample interactions can be mediated by varying the ionic strength and valency of the imaging buffer which sets the Debye length, or the screening distance of the electric field within the fluid. For higher spatial resolution, the Debye length should be decreased, but not to the point that sample deformation occurs [12].

3.6.2 Preparing the Sample

1. Dilute aliquot of concentrated proteoliposomes in SecYEG incubation buffer (*see* Subheading 2, **item 66**, also *see* **Note 8**).
2. Deposit approximately 50 μL of dilute proteoliposome solution onto cleaned surface of your choice.
 - (a) Mica: The freshly cleaved layer (*see* Subheading 3.2) should be clean and unbroken when held to a light. Deposit sample to the newly exposed layer.
 - (b) Glass: *See* procedure described in Subheading 3.3 for preparing glass for AFM study.
3. Cover sample with a petri dish to minimize evaporation and prevent airborne particles from contaminating the sample during incubation.
4. Incubate sample on ice for 1 h (*see* **Note 9**).
5. Rinse the sample 4–6 times by exchanging the buffer on the surface with a pipette (removing previous volume of fluid and replacing with fresh buffer). The pipette may be depressed and released multiple times to remove any loosely bound aggregates (*see* **Note 10**).
6. Load an AFM tip into the tip holder.
7. To reduce probability of bubble formation, pipet a small volume ($\sim 5 \mu\text{L}$) of buffer near the AFM tip.
8. Load the tip holder into the AFM.
9. Initialize the course approach routine. This slowly raises the surface to the tip holder for imaging in fluid.
10. Image (*see* below).

3.7 Imaging

3.7.1 Force Control and Feedback Tuning

We apply tapping mode (also known as AC mode or intermittent contact mode) [13]. The cantilever is driven to oscillate at (or near) its resonance frequency in fluid with a free amplitude set while the cantilever is above (i.e., not interacting with) the surface. As the tip scans over a topographic feature, the amplitude of the cantilever motion changes. During imaging, the feedback loop rapidly adjusts the vertical position of the base of the cantilever so as to maintain a constant user defined amplitude (the imaging “set point”). When tuned properly, the error signal of the feedback loop reflects the local topography of the surface. To obtain robust results, the tip–sample force and feedback parameters need to be continuously monitored and carefully controlled (*see* **Note 11**).

1. With the tip disengaged from the surface, perform a cantilever tune operation. This sweeps the cantilever drive frequency over a range encompassing the fundamental resonance frequency. Select the drive frequency, phase, and drive amplitude for imaging. Record the resulting free amplitude.
2. Engage the tip with the surface.

3. If necessary, adjust the bandwidth and feedback loop gain settings to just under oscillation conditions (*see* **Note 12**).
4. Coarsely estimate the tip–sample force. Use the spring constant of the cantilever [nN/nm], the optical lever sensitivity [nm/V], and the difference between the free amplitude of the oscillation [V] and imaging set point [V].
5. Keep the force applied during imaging as low as possible, typically ≤ 100 pN. The challenge is to maintain as low a force as possible without losing intermittent contact with the surface.
6. Be aware of imaging artifacts that can emerge (*see* **Note 13**).
7. Periodically disengage the tip and reassess the drive frequency, phase, and drive amplitude. Update the resulting free amplitude.

3.7.2 Large Scale Imaging

To perform statistical analyses of the proteins under study:

1. Acquire images of representative areas of the sample (Fig. 3a).
2. Batch process the images using the Hessian blob algorithm (procedure described below) and the same parameters to minimize user bias (Fig. 3b–d).

3.7.3 Kymographs

An AFM kymograph typically comprises one-dimensional line scans plotted versus time. One can track conformational dynamics of individual proteins with enhanced temporal resolution using this technique. To acquire a kymograph:

1. Image following the guidelines above.
2. Identify proteins of interest, progressively alter scan location and scan size as desired.
3. Disable the slow scan axis, thus scan over the (nominally) identical surface features over time. As most AFM imaging software platforms do not transition seamlessly from space to temporal axes, note the pixel dwell time and/or scan rate for offline analysis.

3.8 Analysis of AFM Images of Membrane Proteins

To obtain the software, go to the following URL: <http://faculty.missouri.edu/~kinggm/research.htm>

To apply the Hessian blob algorithm:

3.8.1 Hessian Blob Algorithm

1. If necessary, mask any features above (or below) a certain height threshold that you wish to exclude from analysis, such as large aggregates (or small degradation products).
2. If necessary, perform background subtraction (e.g., second or third order flattening).

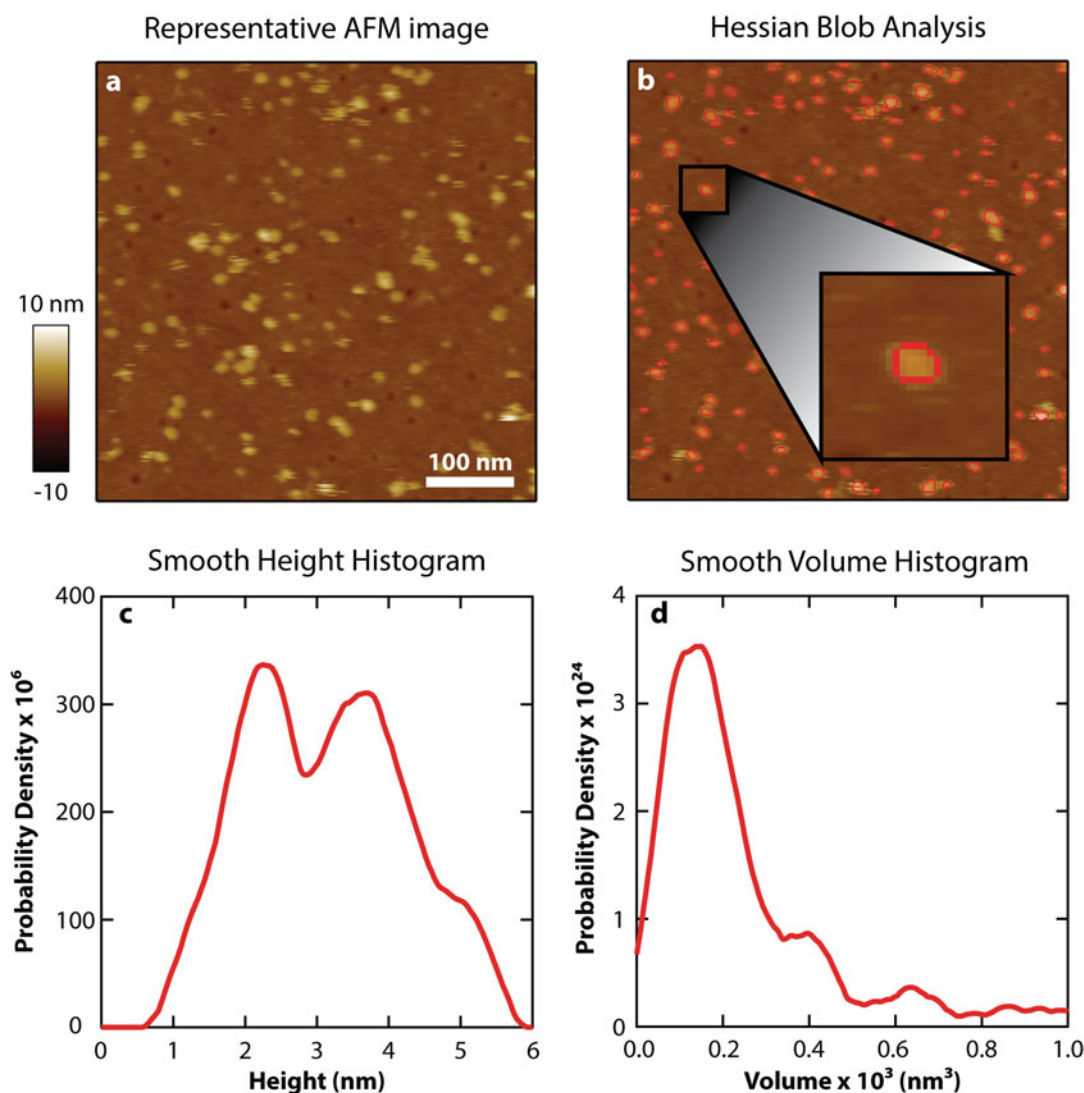


Fig. 3 AFM image data and analysis. (a) Image of Sec translocase (Y-A) adsorbed onto a glass coverslip obtained in aqueous buffer using tapping mode. (b) Hessian blob analysis applied to the image. The red perimeters highlight individual protrusions above the membrane surface. (c, d) Height and volume distributions of the highlighted features, respectively. Broad, multimodal distributions such as these are characteristic of dynamic membrane protein complexes imaged in near-native conditions

3. If necessary, remove any streaks that may appear in the image. This capability is included in many AFM software packages.
4. Perform Hessian blob detection, for detailed instructions see the tutorial at the following URL: [http://faculty.missouri.edu/~kinggm/HB_Tutorial\(8_21_18\).pdf](http://faculty.missouri.edu/~kinggm/HB_Tutorial(8_21_18).pdf)

This can be run on individual images or on a batch of images. The algorithm outputs the height, area, and volume of each detected particle.

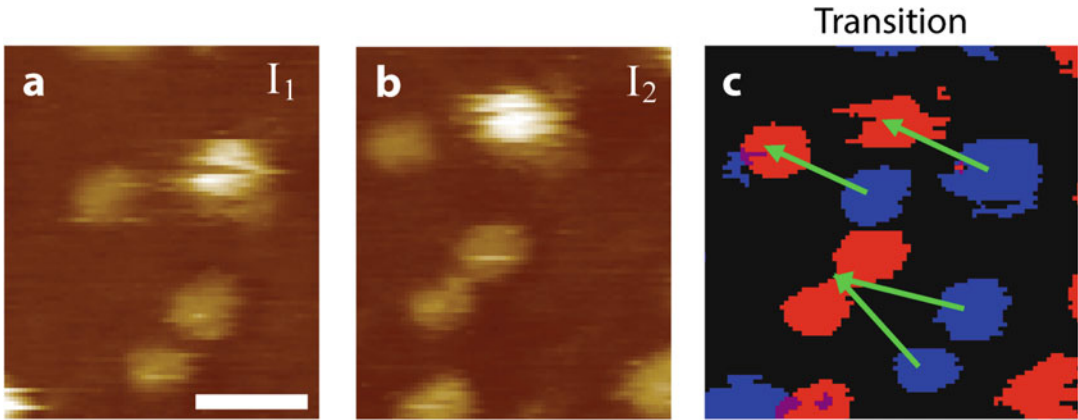


Fig. 4 Tracking individual proteins and interactions over time. **(a)** AFM image of Y-A. The scale bar represents 50 nm. **(b)** Image of nominally the same area at a later time. **(c)** The tracking algorithm identified and connected protrusions in both images (drawn in *blue* from I_1 , *red* from I_2). Green arrows indicate the direction and magnitude of lateral transitions of individual protrusions from I_1 to I_2 . Such analysis can be used to evaluate interactions between proteins facilitated through the membrane

- For each output data set, make a histogram, for example, by using Epanechnikov kernel density estimation.

3.8.2 Tracking Individual Proteins Over Time

Using the output of Hessian Blobs, one can track individual membrane proteins across a series of images $\{I_n, I_{n+1}, \dots, I_N\}$ taken at different times (Fig. 4). This can be achieved in an automated fashion through a set of drift vectors for each image transition pair, $I_n \rightarrow I_{n+1}$. Here, each protein location defined (by its center of mass) in image I_n is propagated to the nearest protein center of mass in image I_{n+1} .

To perform Particle tracking:

- Place desired sequence of AFM images in the same folder.
- Call *AnalyzeImages* function.
- The program will prompt a message asking for user input to perform particle tracking. Click “yes”.
- Review and accept the tracking. Note that **step 3** and **4** must be done for each consecutive image.

The program records the same information as the Hessian blob with additional data for each particle in each image separately. For example, it numbers every particle and tracks how they transition from one image to the next.

3.8.3 Precision Kymograph Analysis via the Line Detection Algorithm

To implement the line detection algorithm:

- If necessary, perform background subtraction (e.g., second or third order flattening).

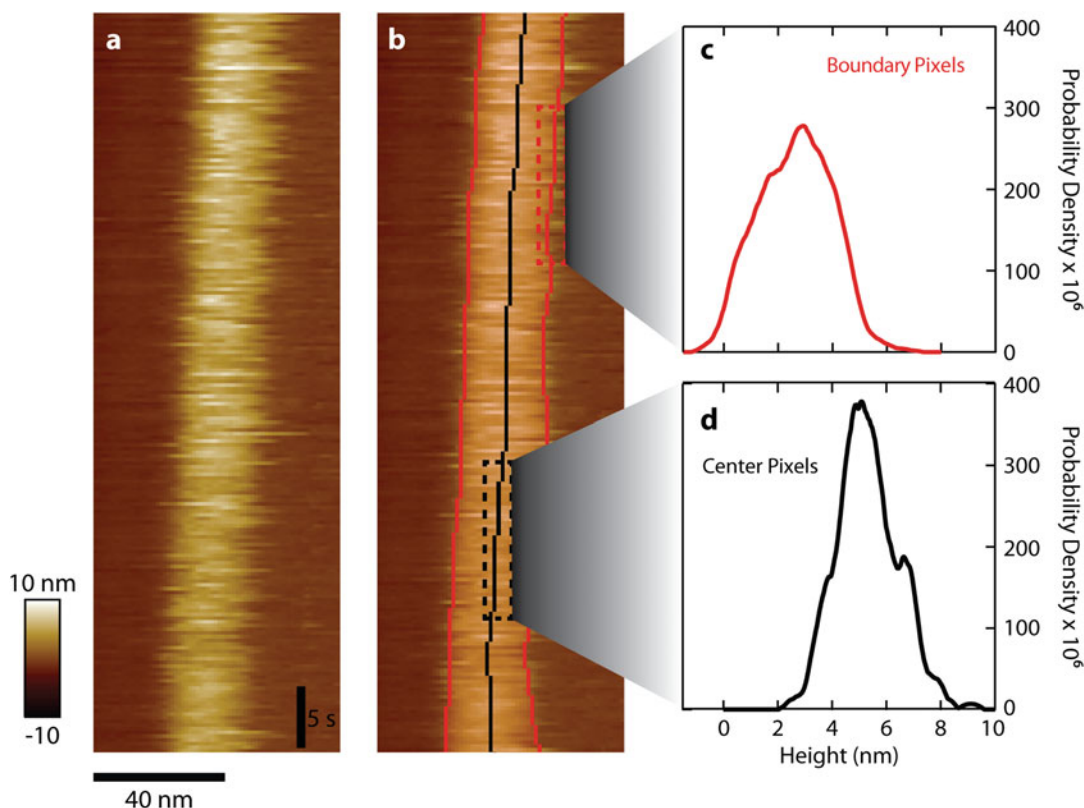


Fig. 5 Precise kymograph analysis allows separation between backbone and boundary dynamics. (a) AFM kymograph of Sec translocase (Y-A) on glass surface. (b) Line detection analysis applied to the image. The black line represents the backbone, or the kymograph center. The red lines indicate the boundary of the kymograph identified by the line detection algorithm. (c) Height distribution of the pixels along the kymograph boundary. (d) Height distribution of the backbone pixels of the kymograph [i.e., those pixels along the black line in (b)]. In principle, robust boundary and backbone delineations allow deconvolution between different modes of motion of a biochemically active protein under study

2. Perform line detection (Fig. 5). The algorithm outputs the same parameters as the Hessian blob: height, area, and volume of the linear feature.
3. After performing line detection, further analysis can be implemented. For example, in the case of a kymograph, one can deconvolve kymograph backbone dynamics from conformational changes at the kymograph boundaries.
4. Perform kymograph analysis. The algorithm will prompt the user to input optional padding on either side of the boundary line. For example, in Fig. 5c, the boundary pixels included the one-pixel-wide boundary itself along with three neighboring pixels on each side of the boundary. The user can also analyze manually selected segments of the kymograph, if required.

5. For each output data set, make a histogram, for example, by using Epanechnikov kernel density estimation as in Fig. 5c, d.

4 Notes

1. Be careful that the superglue does not overflow on top of the Teflon during this step and that the Teflon lies flat. If the Teflon is not flat, it can interfere with the AFM tip holder and cause instability during imaging.
2. In our experience the plasma cleaned glass maintains maximal hydrophilicity for <10 min, so a membrane protein sample should be ready to deposit as soon as possible.
3. This step is critical as overflowing epoxy will be unduly exposed to the sample, which can cause contamination.
4. A dry mechanical vacuum pump is used here to prevent back-streaming of oil, a potential contaminant.
5. This step should be accomplished when the liposomes are swelled, but not ruptured [11].
6. To assay translocation-associated ATP hydrolysis, add SecB and precursor proteins in precursor buffer (*see* Subheading 2, **item 64**) at this step.
7. For best results urea should be made fresh on the same day as the experiment.
8. Deposition concentration and buffer details will likely need to be fine-tuned for different membrane protein systems.
9. Typically, a lipid bilayer containing SecYEG needs 30–60 min to adsorb to the surface. For the Sec system, the sample is incubated on ice for 1 h. This preserves biochemical activity of the sample. Other samples may be incubated at room temperature either inside or outside the AFM. Be aware that the temperature inside an AFM can often increase above room temperature.
10. At this time, the absorption buffer may be exchanged for the imaging buffer.
11. Specific parameter values for feedback gains, bandwidth, as well as free amplitude vary widely.
12. The bandwidth of the feedback loop is often limited by the fundamental mechanical resonance frequency of the stage used to control the tip–sample separation. This frequency can vary depending on the load on the stage, but generally, the bandwidth does not have to be adjusted as frequently as the gain parameters.

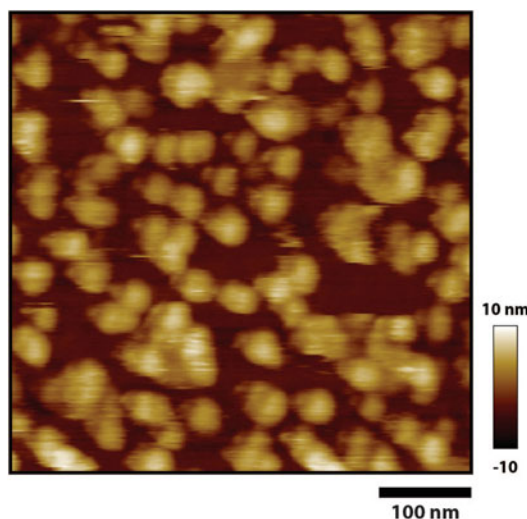


Fig. 6 AFM data showing a common artifact. Multiple proteins exhibiting the same topographic features in the same orientation is evidence of a double tip artifact. This can be fixed by cleaning or replacing the tip

13. Geometric and/or chemical properties of the tip apex can change during imaging, leading to artifacts. A common imaging artifact is shown in Fig. 6.

Acknowledgments

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References

1. Binnig G, Quate CF, Gerber C (1986) Atomic force microscope. *Phys Rev Lett* 56(9):930–933
2. Bippes C, Müller D (2011) High-resolution atomic force microscopy and spectroscopy of native membrane proteins. *Rep Prog Phys* 74:086601
3. Chattrakun K, Hoogerheide DP, Mao C et al (2019) Protein translocation activity in surface-supported lipid bilayers. *Langmuir* 35(37):12246–12256
4. Marsh BP, Chada N, Sanganna Gari RR et al (2018) The Hessian blob algorithm: precise particle detection in atomic force microscopy imagery. *Sci Rep* 8(1):978
5. Crane JM, Randall LL (2017) The sec system: protein export in *Escherichia coli*. *EcoSal Plus* 7(2). <https://doi.org/10.1128/ecosalplus.ESP-0002-2017>
6. Chada N, Chattrakun K, Marsh BP et al (2018) Single-molecule observation of nucleotide induced conformational changes in basal SecA-ATP hydrolysis. *Sci Adv* 4(10):eaat8797
7. Sanganna Gari RR, Frey NC, Mao C et al (2013) Dynamic structure of the translocon SecYEG in membrane: direct single molecule observations. *J Biol Chem* 288(23):16848–16854
8. Sanganna Gari RR, Chattrakun K, Marsh BP et al (2019) Direct visualization of the *E. coli* Sec translocase engaging precursor proteins in lipid bilayers. *Sci Adv* 5(6):eaav9404
9. Chada N, Sigdel KP, Sanganna Gari RR et al (2015) Glass is a viable substrate for precision force microscopy of membrane proteins. *Sci Rep* 5:12550

10. Sigdel KP, Pittman AE, Matin TR et al (2018) High-resolution AFM-based force spectroscopy. *Methods Mol Biol* 1814:49–62
11. Mao C, Cheadle CE, Hardy SJ et al (2013) Stoichiometry of SecYEG in the active translocase of *Escherichia coli* varies with precursor species. *Proc Natl Acad Sci U S A* 110(29):11815–11820
12. Muller DJ, Fotiadis D, Scheuring S et al (1999) Electrostatically balanced subnanometer imaging of biological specimens by atomic force microscope. *Biophys J* 76(2):1101–1111
13. Dufrene YF, Ando T, Garcia R et al (2017) Imaging modes of atomic force microscopy for application in molecular and cell biology. *Nat Nanotechnol* 12(4):295–307



Chapter 7

Structure Determination of Membrane Proteins Using X-Ray Crystallography

Evan Billings, Karl Lundquist, Claire Overly, Karthik Srinivasan, and Nicholas Noinaj

Abstract

Membrane proteins serve essential roles in all aspects of life and make up roughly one-third of all genomes from prokaryotes to eukaryotes. Their responsibilities include mediating cell signaling, nutrient import, waste export, cellular communication, trafficking, and immunity. For their critical role in many cellular processes, membrane proteins serve as targets for up to 50% of drugs currently on the market and remain primary targets in new therapeutics being developed. Despite their importance and abundance in nature, only ~1% of structures in the Protein Data Bank are of transmembrane proteins. This discrepancy can be directly attributed to the biochemical properties of membrane proteins and the difficulty in producing sufficient yields for structural studies or the difficulty in growing well-ordered crystals. Here, we present methods from our work that outline our general pipeline from cloning to structure determination of membrane proteins, with a focus on using X-ray crystallography, which still yields ~90% of all structures being deposited into the Protein Data Bank.

Key words Membrane proteins, Crystallization, Structure determination, Protein folding, BAM complex

1 Introduction

Transmembrane proteins can have either an α -helical or β -barrel membrane domain [1–7]. Both types serve similar roles in the cell; however, an overarching difference between the two types is that while α -helical membrane proteins can be found in all cellular membranes from prokaryotes to eukaryotes, β -barrel membrane proteins are only found in the outer membranes of gram-negative bacteria, mitochondria, and plastids such as chloroplasts [3, 5, 8]. This chapter will be mostly relevant to membrane proteins expressed in bacteria with a focus on the structure determination of β -barrel membrane proteins; however, the methods can be easily

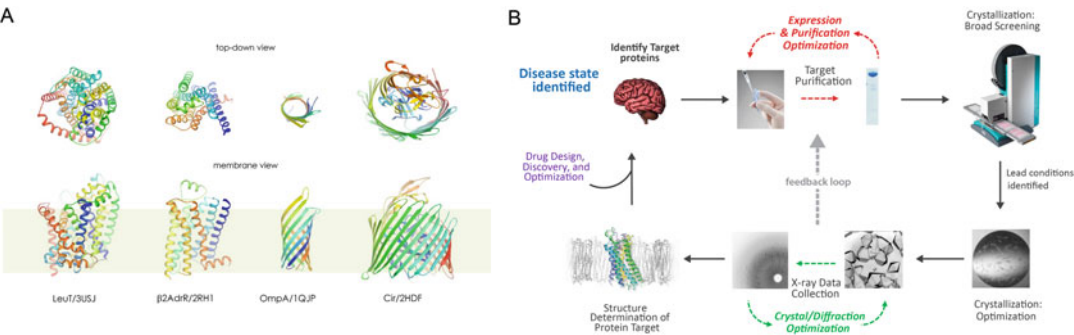


Fig. 1 The structure determination pipeline for membrane proteins using X-ray crystallography. (a) Examples of α -helical and β -barrel membrane proteins. (b) Highlighted are the two bottlenecks in the pipeline (1) expression/purification of the protein target and (2) growth of well diffracting crystals. A “feedback loop” would be necessary when crystallization fails to produce well-ordered crystals and a new construct must be explored

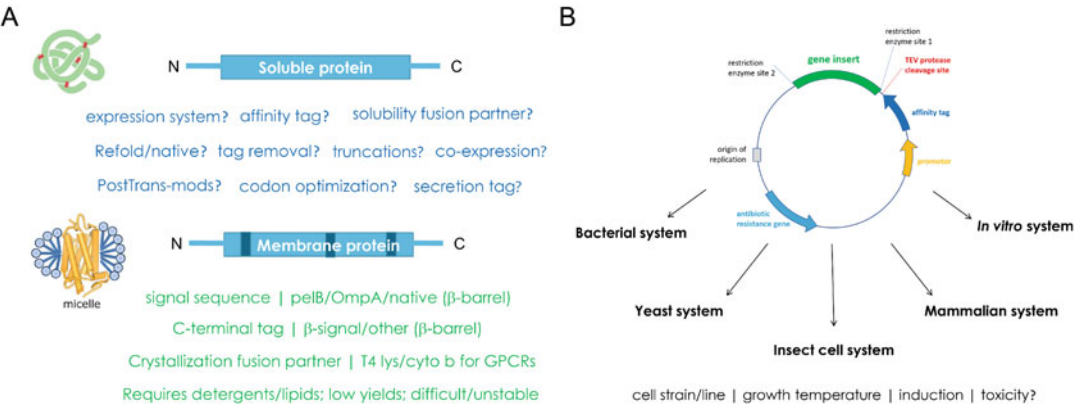


Fig. 2 Considerations and strategies for cloning membrane proteins for structure determination. (a) Like with soluble proteins (top), there are many factors to consider when studying membrane proteins. Additional factors (bottom) must also be considered that further complicate production of these families of proteins. (b) For cloning, one must carefully consider the expression system(s) that will be tested and an appropriate vector for delivering the gene and induction system. Each system has advantages and disadvantages that must be factored into the decision-making process

adapted to α -helical membrane proteins and for other expression systems (Fig. 1).

Many aspects of protein expression, purification, and crystallization may be addressed by considering them in the initial cloning of the protein (Fig. 2). Cloning choices that affect the expression of the protein include the construct size, the presence of tags or fusions, whether a native or denatured state is desired, and how the expression will be induced [9–11]. In instances where the entire protein cannot be successfully expressed or crystallized, truncations of the protein, such as those that omit a disordered portion of the

protein, may yield better results. Other truncations or fragments simply generate more manageable pieces of the protein specifically for structure determination.

The types of affinity tags, localization signal sequences, and protease sites encoded in a vector may affect both the expression and purification of the protein. Affinity tags may be moved from the N-terminus to the C-terminus if needed and can sometimes make a drastic difference in expression levels and stability. Commonly used affinity tags include 6 $\times$ - or 10 $\times$ -histidine, GST, and Strep. To aid its crystallization, a protein may be cloned as a fusion with another, easily crystallized protein fragment, such as lysozyme or maltose binding protein [10, 12]. Additionally, mutations can be engineered to reduce flexibility of the protein, increasing its thermostability [13, 14]. When working with membrane proteins, homologs and orthologs of the protein of interest are often screened for levels of expression, thermostability, mobility within a detergent or lipid, and initial crystallization screening to determine the most promising candidate(s).

In our chapter here, we will use an example protocol for the cloning to structure determination of BamA from *Neisseria gonorrhoeae* [15]; however, the protocols can be easily adapted to other membrane proteins of interest. Here, the pET20b vector was modified to include an N-terminal 10 $\times$ His-tag followed by a TEV protease site prior to the commercial NcoI site. The TEV protease site may be replaced with an enterokinase or thrombin site, if preferred [16]. A short list of common vectors commonly used for the production of membrane proteins include the pET family of vectors (specifically pET20b, pET-DUET, and pET26), pRSF-1b, pCDF-1b, pBAD, pTrc, pMAL, pGEX, pASG, and pHIS2-parallel. Primers may be designed to incorporate various restriction enzyme sequences onto the ends of the construct to accommodate the process of restriction enzyme digestion and ligation. Alternatively, other cloning strategies may be used including Golden Gate Assembly, Gibson Assembly, Gateway Cloning, and ligation-independent cloning.

Once an expression system has been chosen and a suitable clone or library of clones have been prepared, the next step is to transform the plasmids into our desired cell type which we will first use to verify expression of the gene in small-scale assays. Once expression is confirmed, we can proceed with large-scale expression. There are many factors affecting the conditions for optimal large-scale expression yields [9, 17, 18] (Fig. 3). Many of these include the cell type, medium being used, and expression temperature, all of which can be optimized during small-scale expression tests. There are other considerations that will be made in the interest of not only maximizing overall protein expression levels, but also maximizing the final yield of active, natively folded protein. With these interests in mind, large-scale expression of NgBamA was performed at low

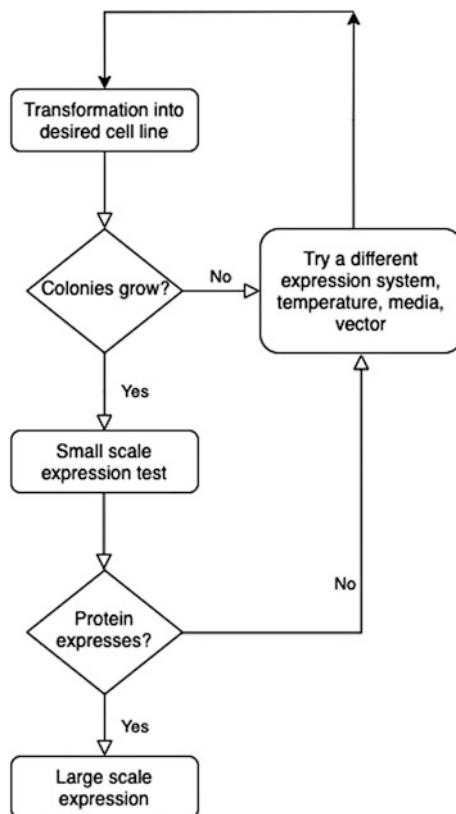


Fig. 3 Protein expression flow diagram. The diagram can be used to guide the decision-making process when expression membrane proteins, particularly in bacterial systems. Colony screening is sometimes necessary for difficult targets while easily expressed targets behave much like well-behaved soluble protein samples

temperature (20 °C) without IPTG induction [15, 19]. Cell lines such as BL21(DE3) often exhibit “leaky” expression of the recombinant gene. This means the T7 RNA polymerase is expressed at low amounts even in the absence of IPTG, yielding low levels of recombinant protein. Both the low temperature and the lack of induction have the effect of slowing the production of the protein of interest and in the case of membrane proteins, this can be advantageous for allowing sufficient time to complete transit and folding into the membrane. Alternatively, the use of autoinduction media is also commonly used in the expression of membrane proteins [20, 21]. Media can also be an important consideration for achieving high protein yields [17, 22]. For the expression of *NgBamA*, Terrific Broth (TB) was used since it can achieve very high cell densities ranging from an OD₆₀₀ of 10–20 and yielding 30–40 g of cells per liter of medium. Whereas less rich media like 2XYT will produce ~5 g of cells per liter of medium. In general, it

may be advantageous to grow in richer media to increase protein production, but it is also important to consider that richer media contains a greater variety of nutrients which may activate the expression of a greater variety of genes. This may interfere with the production of your protein or cause difficulty with protein purification due to the presence of additional impurities.

We will describe the following procedure using BL21(DE3) cells which utilize the T7 expression system allowing IPTG-mediated inducible expression [23]. This cell type also lacks proteases such as OmpT and is therefore sufficient for expressing nontoxic genes. However, in the event of low or no expression, we suggest trying alternate expression systems. Rosetta or Codon-Plus cells can be used to express genes with rare codons. C41 or C43 cells can be used to express toxic genes or many genes from other organisms [24, 25]. In the event that expression in several alternate cell lines fails, we suggest revisiting the vector selection.

The purification of membrane proteins is uniquely distinguished from the purification of soluble proteins and careful consideration must be applied through the process [11, 17, 19, 24]. In our example, *NgBamA* was cloned into the pET20b-HT vector which contains an N-terminal pelB signal sequence, routing it across the inner membrane where it is then folded and inserted into the outer membrane by the β -barrel assembly machinery [26]. Therefore, to purify membrane-embedded proteins such as *NgBamA*, the cellular membranes must be isolated, and the protein solubilized (released). The most obvious and important distinction from soluble protein purification is the requirement of a detergent or detergent-lipid mix to stabilize the hydrophobic, transmembrane regions of the protein. Most commonly, membranes are isolated by centrifugation and the membranes are then solubilized in a buffer supplemented with detergent (Fig. 4). Throughout the purification process, it is important to keep the detergent above the critical micelle concentration (CMC), so the protein does not aggregate or precipitate out of solution [27–29].

If the membrane protein is expressed as inclusion bodies, it may be refolded similar to methods used for soluble proteins [30–32]. However, in order to stabilize the protein during the refolding process, detergent must be present in the refolding buffers. Upon purification (and even during refolding), additives such as lipids, cholesterol, and other small molecules can often improve overall protein stability, which would assist in functional characterization and even structural studies.

Typical yields for the purification of natively folded membrane proteins are on average much lower when compared to soluble proteins; however, in some cases, yields can be greatly improved if the target protein is expressed as inclusion bodies and refolded. Producing sufficiently large yields of highly pure membrane proteins is most often the largest hurdle for successful structure

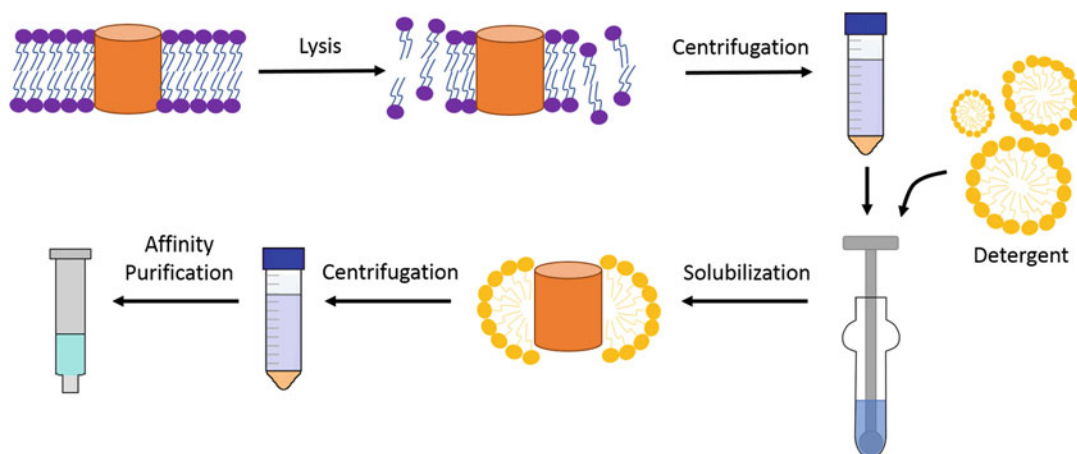


Fig. 4 Solubilization and purification of membrane proteins. The membrane is then disrupted when the cells are lysed. The insoluble cell membranes containing the protein of interest can then be isolated by centrifugation, separating them from the soluble portion of the lysate. In order to purify the protein, these isolated membranes must be solubilized in a detergent to put the protein into solution. To accomplish this, the collected membranes can be readily mixed with a buffer containing a solubilization detergent in a Dounce homogenizer. This solution can then be left to stir overnight. Afterward, the membrane protein will be solubilized in a detergent micelle. Thus, it can be separated by subsequent centrifugation to separate it from insoluble debris in the supernatant. The solubilized protein of interest is then ready for the proceeding steps of affinity purification and size exclusion chromatography

determination. Over the past few decades, several strategies have been developed to improve the purity of membrane proteins. For example, the target membrane protein can be expressed using higher affinity, more specific tags or even dual affinity tags, such as 6–10×HIS-tag on the N-terminus and a Strep-tag on the C-terminus. An ideal sample for crystallization would produce a single, symmetric, and monodisperse peak from SEC and would be >90% pure when visualized using SDS-PAGE. This is indicative of a stable and homogeneous protein sample, ideal for crystallization. One must also keep in mind that not all detergents are stabilizing to all membrane proteins, therefore, our suggestion is to try up to five different detergents (DDM, LDAO, OG, C₈E₄, LMNG) and consider even supplementing with cholesterol or lipids, if necessary [33]. And even once a suitable detergent is found, one must also carefully consider the increase in detergent concentration, as the protein is concentrated to 10 mg/mL for initial crystallization, and how this may affect crystallization screening [27].

Membrane proteins have large hydrophobic domains that mediate their embedment within the core of the membrane; therefore, once removed from the membrane bilayer, the exposed hydrophobic surfaces must be stabilized by detergents/lipids for *in vitro* studies. The presence of the highly dynamic detergent micelles occupying a large portion of the protein surface makes

crystallization even more challenging than normally expected. Crystallization of membrane proteins follows a parallel workflow to that of soluble proteins and variables like temperature, pH, type of buffer, precipitant concentration, salt concentration, and sample concentration should be optimized. However, there are additional considerations and methods specifically tailored for membrane proteins. For example, there are commercially available crystallization screens that are marketed specifically for membrane proteins, as well as, detergent additive screens and the use of hydrophobic crystallization sheets for hanging drop crystallization. Further, detergent screening, bicelle crystallization, and lipidic cubic phase (LCP) crystallization are all methods specifically for the crystallization of membrane proteins (Fig. 5). For the structure of *NgBamA*, only bicelle crystallization produced well-ordered crystals despite detergent screening yielding several lead conditions with very large, yet poorly diffraction, crystals.

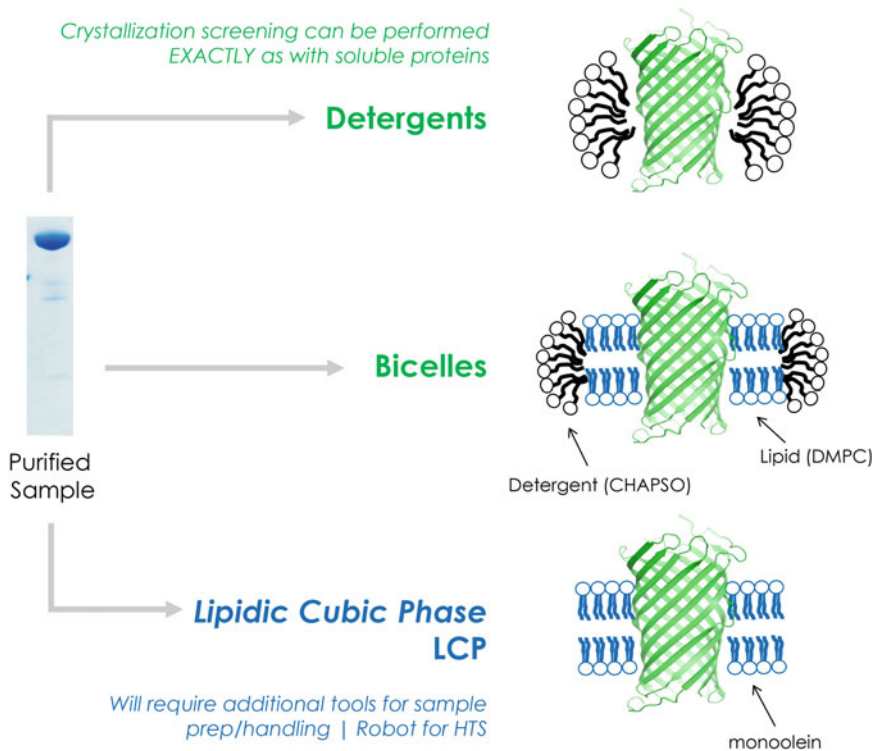


Fig. 5 Crystallization methods for membrane proteins. Outline of crystallization methods specifically for membrane proteins including detergent screening, bicelles crystallization, and lipidic cubic phase (LCP) crystallization. Note that detergent screening and bicelle crystallization can be performed with the same instrumentation as with soluble proteins; however, LCP crystallization will require additional tools and robotics for manual and automated methods

The presence of detergents in the sample buffer, however, does pose a challenge though since detergents significantly reduce the surface tension of the crystallization drops, making sitting drop crystallization and hanging drop crystallization with hydrophilic crystallization cover sheets nearly impossible for membrane proteins. Therefore, high-throughput screening requires the use of hydrophobic crystallization cover sheets to ensure a well-formed drop is maintained throughout the incubation period of the crystallization trials. One advantage here though is that most often, the hydrophobic cover sheets are UV transparent, making it convenient when identifying protein crystals from salt using UV imaging.

Protein crystals are fragile and harvesting them must be done with care. This is particularly true for membrane protein crystals which are grown in detergents and lipids, often making them even more fragile and more sensitive to even subtle temperature and buffer changes. As with all protein crystals, careful screening of cryo-protectants must be performed in order to find optimal conditions that preserve or enhance the diffraction quality of the crystals [34, 35]. There is often a reduced solvent content observed within membrane protein crystals due to the detergent, which can assist in cryoprotection; however, the LCP matrix, along with the precipitant, often serves as a sufficient cryoprotectant for these crystals [36].

Once crystals have been grown, structure determination of membrane proteins parallels that of soluble proteins with some additional considerations, which we will summarize here. But generally, the methods and instrumentation are shared, except for when LCP crystallization is being performed [33, 37, 38]. While crystals grown in detergent and bicelles [39, 40] are often similar in size to soluble proteins, LCP crystals are typically much smaller and more difficult to harvest, particularly from glass sandwich plates [36].

While earlier data collection on LCP crystals required multiple crystals for a complete dataset, advances in synchrotron instrumentation such as automated sample changing robotics, beam optics, and detectors can enable complete data collection on single LCP crystals. Microbeams ranging in size from 5 to 20 μm have enabled not only improved data collection from LCP harvested crystals but also in finding well diffracting regions that are more ordered compared to other regions on larger non-uniformly packed crystals [41, 42]. Both ultrafast goniometers and high-speed hybrid photon counting detectors with improved signal ratio have complemented the use of microbeams by allowing for superfast data collection to alleviate the effects of radiation damage without sacrificing data quality or resolution [43]. Rastering methods have also boosted data collection for membrane proteins by automating the search for “sweet spots” on larger crystals and the location of crystals within loops containing an LCP matrix [44, 45]. Lastly,

SONICC technology has improved the visualization of not only crystals within an LCP matrix, but also for the identification of nanocrystals which may be too small for UV imaging [46, 47]; however, its usefulness as a routine method for identifying protein crystals for X-ray or electron crystallography remains to be determined.

Crystals of membrane proteins often do not diffract to as high resolution as soluble protein crystals; a summary of all reported membrane protein structures can be found at the following website maintained by the White Laboratory at UC Irvine: <https://blanco.biomol.uci.edu/mpstruc/>. Therefore, data collection strategies and expectations must be adjusted accordingly, such that collecting a nice 3 Å complete dataset is the goal and anything higher resolution being a luxury, with anything lower resolution being the reality of working on membrane proteins. Of course, there are always some families of membrane proteins that are more stable and are more easily crystallized and routinely produce higher resolution diffraction, but this is not the norm. It is more likely that many crystals, sometime hundreds, will need to be screened in order to find a few that produce good enough diffraction quality for structure determination.

Once a dataset is collected and processed, the next step is structure determination using either molecular replacement or experimental phasing methods. Both phasing methods require knowing how many molecules are in the asymmetric unit to improve the likelihood of success. However, the need to stabilize membrane proteins with detergents or lipids leads to an increase in the average Matthew's coefficient (packing parameter, V_m) and an increase in the expected range of acceptable solvent contents; information used to estimate the number of molecules in the asymmetric unit [48]. A correct solution for a membrane protein might have a V_m of 3.9 and a solvent content of 70% for a resolution of 3 Å, while this would likely not be the case for a closely packed soluble protein (V_m ranges from 1.7 to 3.5) of similar size and resolution. Therefore, during initial phasing, if a correct number of molecules per asymmetric unit cannot be determined with high confidence, one should run jobs in parallel for each of the possible options determined from analysis of the cell contents.

2 Materials

2.1 Cloning

2.1.1 Polymerase Chain Reaction (PCR)

1. dH₂O (DNase and RNase-free).
2. Phusion HF buffer (5×, ThermoFisher).
3. Dimethyl sulfoxide (DMSO) (100%).
4. Deoxynucleotide triphosphate (dNTP) mix (10 mM of each nucleotide).

5. Phusion HF DNA polymerase (ThermoFisher).
6. *Ng*BamA genomic or optimized template DNA sequence (50–100 ng/mL concentration).
7. Forward and reverse primers (1000 ng/mL concentration each) for each construct.
8. Thermocycler (e.g., Eppendorf Mastercycler EP Gradient S).

2.1.2 Agarose Gel Electrophoresis

1. Agarose.
2. Tris-borate-EDTA (TBE, 1×) buffer: dissolve 10.8 g tris and 5.5 g boric acid in 900 mL of water, add 4 mL Na₂EDTA (0.5 M, pH 8.0) and adjust volume to 1 L.
3. Ethidium bromide (10 mg/mL).
4. DNA gel loading dye (e.g., Thermo Scientific DNA gel loading dye 6×).
5. DNA ladder (e.g., New England Biolabs 1-kb DNA ladder).
6. DNA gel casting system.
7. DNA gel electrophoresis apparatus (e.g., Bio-Rad Minisub Cell GT Cell + Powerpac).
8. UV light/imager (e.g., Bio-Rad UView Mini Transilluminator).

2.1.3 DNA Gel Extraction/ PCR Cleanup

1. DNA gel extraction/cleanup kit (e.g., IBI Scientific Gel/PCR DNA Fragments Extraction Kit).
2. *Ng*BamA PCR insert.
3. Benchtop centrifuge (e.g., Thermo Scientific Sorvall Legend Micro 17).
4. Microvolume spectrophotometer (e.g., ThermoFisher Nano-Drop, BioTek Epoch 2).

2.1.4 Restriction Enzyme Digestion

1. Purified *Ng*BamA PCR insert.
2. Purified plasmid expression vector (e.g., pET20bHT).
3. Relevant restriction enzymes (e.g., New England Biolabs NcoI and XhoI).
4. Cutsmart buffer (10×, New England Biolabs).
5. 37 °C incubator.

2.1.5 Ligation of Insert and Vector

1. Purified, digested *Ng*BamA PCR insert.
2. Purified, digested pET20bHT.
3. T4 DNA ligase (New England Biolabs).
4. T4 DNA ligase buffer (10×, New England Biolabs).

2.1.6 Transformation of Ligation Reaction into Chemically Competent DH5 α

1. DH5 α cells.
2. Ligation reaction.
3. 42 °C heat block or water bath.
4. LB agar plates with carbenicillin (50 μ g/mL) (LB/carb).
5. SOC outgrowth medium: 2% tryptone, 0.5% yeast extract, 10 mM NaCl, 2.5 mM KCl, 10 mM MgCl₂, 10 mM MgSO₄, 20 mM glucose; autoclave.

2.1.7 Overnight Cultures

1. Round-bottom 14 mL culture tubes.
2. Bacterial growth medium (e.g., 2XYT or LB).
3. Ampicillin: stock solution prepared at 50 mg/mL in water and stored in 500 μ L aliquots at –20 °C.
4. LB/carb agar plates with transformed DH5 α colonies.
5. 37 °C shaker.

2.1.8 DNA Plasmid Minipreps

1. Overnight cultures of transformed DH5 α .
2. DNA plasmid isolation kit (e.g., IBI High-speed Plasmid Mini Kit).

2.1.9 DNA Sequencing Analysis

1. Purified DNA plasmids.
2. Sequencing primers.
3. Computer.
4. Vector analysis software or BLAST tool (e.g., Vector NTI, SnapGene).

2.2 Expression

2.2.1 Transformation of Clone into BL21(DE3) Cells

1. LB agar plates with carbenicillin (50 μ g/mL) (LB/carb).
2. Incubator (37 °C).
3. BL21(DE3) chemically competent cells.
4. Sequence verified plasmid.
5. Heat block (42 °C).
6. LB medium: dissolve 5 g NaCl, 10 g peptone, and 5 g yeast extract into 800 mL of water. Adjust the volume to 1 L and autoclave.
7. Thermomixer (37 °C, 1000 RPM) (e.g., Eppendorf Thermomixer C).
8. Inoculation loop or L spreader.

2.2.2 Small-Scale Expression Test

1. Round-bottom 14 mL culture tubes.
2. LB medium: dissolve 5 g NaCl, 10 g peptone, and 5 g yeast extract into 800 mL of water. Adjust the volume to 1 L and autoclave.
3. Ampicillin: stock solution prepared at 50 mg/mL in water and stored in 500 μ L aliquots at –20 °C.

4. Inoculation loop.
5. Temperature controlled shaker (e.g., New Brunswick Innova 44R, ThermoFisher MaxQ 8000).
6. UV-vis spectrometer (e.g., Eppendorf Biophotometer Plus 6132).
7. Isopropyl β -d-1-thiogalactopyranoside (IPTG): stock solution prepared at 200 mM in water and stored in 500 μ L aliquots at -20°C .

2.2.3 Prepare a Glycerol Stock of the Transformed Cells

1. Microcentrifuge tubes (1.5 mL).
2. Overnight culture from LB/carb agar plate.
3. Sterile 60% glycerol/LB solution.

2.2.4 Preparing Cells for SDS-PAGE Analysis

1. Benchtop microcentrifuge (e.g., Eppendorf 5424, ThermoFisher AccuSpin Micro 17).
2. $2\times$ SDS-Loading Buffer.
3. Heat block.
4. SDS-PAGE gel running apparatus (e.g., Bio-Rad PROTEAN system).
5. Precast SDS-PAGE gel (e.g., Bio-Rad 4–15% Mini-PROTEAN TGX).
6. SDS-PAGE running buffers.
7. Gel stain (e.g., ThermoFisher PageBlue Stain).

2.2.5 Prepare a 5 mL Overnight Starter Culture

1. Round-bottom 14 mL culture tubes.
2. LB medium: dissolve 5 g NaCl, 10 g peptone, and 5 g yeast extract in 800 mL of water. Adjust the volume to 1 L and autoclave.
3. Ampicillin: stock solution prepared at 50 mg/mL in water and stored in 500 μ L aliquots at -20°C .
4. Expression-verified cells.
5. Temperature-controlled shaker (e.g., New Brunswick Innova 44R, ThermoFisher MaxQ 8000).

2.2.6 Large-Scale Expression

1. Overnight culture from LB/carb agar plate.
2. 2 L flasks.
3. TB medium: dissolve 20 g tryptone, 24 g yeast extract, 4 mL glycerol, and 100 mL of phosphate buffer (0.17 M, KH_2PO_4 , 0.72 K_2HPO_4) into 800 mL of water. Adjust the volume to 1 L and autoclave.
4. Ampicillin: stock solution prepared at 50 mg/mL in water and stored in 500 μ L aliquots at -20°C .
5. Temperature-controlled shaker (e.g., New Brunswick Innova 44R, ThermoFisher MaxQ 8000).

2.3 Purification

2.3.1 Preparation of Lysate

1. $1\times$ PBS (12 mM phosphate buffer, pH 7.4, 137 mM NaCl, 2.7 mM KCl).
2. DNase I: prepare a stock solution at 10 mg/ml in 10 mM NaCl and stored at -20°C .
Phenylmethylsulfonyl fluoride (PMSF): prepare a stock solution at 200 mM by dissolving solid PMSF in 100% ethanol and store in 0.5 mL aliquots at -20°C .
3. Apparatus for large scale cell lysis (e.g., an Avestin Emulsiflex C3 high-pressure homogenizer).

2.3.2 Solubilization from Whole Cell Lysate

1. High-speed centrifuge (e.g., ThermoFisher Lynx 6000).
2. *n*-Dodecyl β -D-maltoside (DDM): prepare a stock solution at 20% in water and store at 4°C .

2.3.3 Solubilization from Isolated Membranes

1. High-speed centrifuge (e.g., ThermoFisher Lynx 6000).
2. Ultracentrifuge (e.g., Beckman-Coulter XPN90).
3. 40 mL glass Dounce homogenizer (Wheaton).
4. *n*-Dodecyl β -D-maltoside (DDM): prepare a stock solution at 20% in water and store at 4°C .
5. Elugent: prepare a stock solution at 25% in water.

2.3.4 Isolation of Inclusion Bodies and Refolding

1. High-speed centrifuge (e.g., ThermoFisher Lynx 6000).
2. *n*-Dodecyl β -D-maltoside (DDM): prepare a stock solution at 20% in water and store at 4°C .
3. 40 mL dounce homogenizer (Wheaton).
4. Triton X-100: prepare a 20% stock solution in water.
5. 8 M urea: dissolve 480.5 g urea into 500 mL of hot water or $1\times$ PBS; adjust the volume to 1 L.

2.3.5 Purification of Solubilized Membrane Proteins

1. FPLC column (5 mL) packed with HisPur Ni-NTA resin (ThermoFisher).
2. FPLC automated purification system (e.g., GE Healthcare Pure 25 L).
3. Buffer A ($1\times$ PBS, 0.05% DDM).
4. Buffer B ($1\times$ PBS, 0.05% DDM, 1 M imidazole).
5. Dialysis tubing (Millipore Sigma).
6. Amicon Ultra Centrifugal filter units (30 and 100 MWCO) (Millipore Sigma).
7. Sephacryl S-300 HR 16/60 Hi-Prep column (GE Healthcare).
8. SEC buffer (25 mM Tris, pH 7.5, 150 mM NaCl, and 0.6% C_8E_4).

2.4 Crystallization**2.4.1 Aliquoting***Commercial Crystallization**Broad-Matrix Screens**(Hanging Drop)*

1. Commercial broad matrix screens in 96-well blocks (e.g., Shot Gun, Crystal Screen, MCSG Suite).
2. Plate dispensing/stamping instrument (e.g., Rainin Liquidator-96, ThermoFisher Versette, Beckman Biomek).
3. Automated microplate heat sealer (e.g., ThermoFisher ALPS 3000); or manual plate seals.
4. Round-bottom 96-well polypropylene or polystyrene plates (e.g., Greiner Bio-One #650201).
5. Microplate sealing film roller.

2.4.2 Preparing 96-Well*Optimization Screens*

1. Crystallization buffer components (from lead conditions in broad-matrix screening).
2. Automated liquid handling robot (e.g., SPT Dragonfly; Formulatrix Formulator).
3. Microplate mixer.

2.4.3 Performing*Broad-Matrix**and Optimization**Crystallization Screening**Using an Automated**Crystallization Robot**(Hanging**and Sandwich Drop)*

1. Broad-matrix crystallization screens in 96-well plates.
2. Optimization crystallization screens in 96-well plates.
3. Hydrophobic hanging drop crystallization sheets (e.g., Grace Bio-Labs ProCrystal #875238, SPT Labtech Viewdrop III).
4. LCP sandwich plates (e.g., Hampton Research LCP Sandwich Set, Molecular Dimensions Laminex Plates).
5. Bicelles (e.g., MemX Biosciences MemMagic Bicelle Screen kit).
6. Additives (e.g., Hampton Research Additive Screen).
7. Multichannel pipetter (12-channel; 0.5–10 μ L).
8. Automated crystallization robot (e.g., SPT Labtech mosquito LCP, Art Robbins Crystal Gryphon LCP).
9. V-bottom 96 well plates.
10. Centrifuge with microplate holders (e.g., ThermoFisher Lynx 4000, Beckman Allegra 15R).

2.4.4 Performing*Optimization Screening**Manually*

1. Large format 24-well plates with sealant (e.g., VDXm, VDX, EasyXtal).
2. LCP sandwich plates (e.g., Hampton Research LCP Sandwich Set, Molecular Dimensions Laminex Plates).
3. Siliconized crystallization coverslips (18 mm round and square).
4. Repeating dispenser for Hamilton syringes.
5. Crystallization buffer components (from lead conditions in broad-matrix screening).
6. Additives.

*2.4.5 Visualizing
and Assessing
Crystallization Leads*

1. Automated crystallization imaging robot (e.g., Formulatrix Rock Imager, Art Robbins CrysCam, JanSi UVEX).
2. UV imaging microscope (e.g., Korima PRS-1000).
3. Stereomicroscope (e.g., Zeiss SteREO Discovery.V8).
4. Micro tools (e.g., Hampton Research Micro Tools, Mitegen MicroTools Kit).
5. Scalpel or razor blade.
6. Crystal staining dye (e.g., Hampton Research IZIT Crystal, Jena Bioscience Black Light).

**2.5 Data Collection
and Structure
Determination**

2.5.1 Harvesting Crystals

1. Crystallization trays containing crystals.
2. Stereomicroscope (e.g., Zeiss SteREO Discovery.V8).
3. Cryoprotectant solution(s).
4. Magnetic wand (e.g., Hampton Research CrystalWand, Mitegen Magnetic Wand Straight).
5. Scalpel or razor blade.
6. Reverse (self-closing) tweezers (e.g., Dumont style N2a).
7. Petri dish.
8. Mounted crystallization pins/assemblies (loops/bases).
9. Uni-pucks (e.g., Mitegen Universal V1-Puck).
10. Puck tools and tongs.
11. Foam dewar (e.g., Spearlab Foam dewars, Mitegen Double Puck Loading Dewar).
12. Liquid nitrogen.
13. Puck log sheets.

2.5.2 Storing Crystals

1. Uni-pucks containing crystals.
2. Puck cane (e.g., Mitegen Shelved Puck Shipping Cane).
3. Liquid nitrogen.
4. Storage Dewar (e.g., Taylor-Wharton VHC35 Dewar or CX100 Cryo Dry Shipper).

*2.5.3 Screening and Data
Collection*

1. X-ray source (synchrotron or home source including X-ray generator, optics, cryo system, and detector).
2. Frozen crystals in uni-pucks.
3. Puck tools and tongs.
4. Magnetic wand (e.g., Hampton Research CrystalWand, Mitegen Magnetic Wand Straight).
5. Cryo pin tongs (18 mm).

6. Foam dewar (e.g., Spearlab Foam dewars, Mitegen Double Puck Loading Dewar).
7. Workstation.
8. Data acquisition software (e.g., JBluIce, BluIce, SERGUI).

2.5.4 Diffraction Analysis and Structure Determination

1. Diffraction images.
2. Workstation.
3. Diffraction data processing and reduction software (e.g., HKL2000, MOSFLM, XDS, DIALS, Xia2).
4. Diffraction data analysis software (e.g., phenix.xtriage (PHENIX), Cell Content Analysis (CCP4)).
5. Structure determination software packages (e.g., PHENIX, CCP4, SHELX).

3 Methods

3.1 Cloning

3.1.1 Polymerase Chain Reaction (PCR)

1. Prepare 138 μL Phusion PCR master mix by combining 100.5 μL dH₂O, 30 μL 5 $\times$ HF Phusion buffer, 4.5 μL 100% DMSO, and 3 μL dNTP mix (*see Note 1*).
2. For each PCR reaction (each construct you plan to clone), add 23 μL master mix, 1 μL template DNA, and 0.5 μL each forward and reverse primers to a thin-walled PCR tube. Lastly, add 0.5 μL HF Phusion polymerase to each reaction.
3. Place the tubes containing PCR reactions into the thermocycler and use according to manufacturer instructions. General PCR conditions include an initial denaturation step of 98 °C for 10–30 s, 25–40 cycles of [98 °C for 10 s, 55–60 °C for 30 s, and 72 °C for 15–30 s per kb of amplicon (for Phusion HF polymerase)], and a final extension of 72 °C for 5–10 min before holding at 4 °C until the reactions can be removed from the thermocycler (*see Note 2*).
4. Remove the PCR reaction from the thermocycler and prepare for gel purification as noted in the next section, or store at 4 °C for short term or –20 °C for long term.

3.1.2 Agarose Gel Electrophoresis and DNA Gel Extraction/PCR Cleanup

1. Prepare a 1% agarose gel by mixing 0.25 g agarose with 25 mL 1 $\times$ TBE buffer into a conical tube or Erlenmeyer flask.
2. Microwave for 20–30 s or until fully dissolved.
3. Once dissolved, allow to cool until the container can be handled, then add 2.5 μL of ethidium bromide to the solution and mix gently by swirling the solution (*see Note 3*).

4. Carefully pour the agarose solution into the gel assembly, checking that the teeth of the comb are mostly covered for adequate well formation. Allow to cool for 15–20 min or until solidified (*see Note 4*).
5. Place the gel into a DNA gel electrophoresis apparatus. Add 1× TBE running buffer to the “max” line of the of gel apparatus or until the buffer is just above the gel surface.
6. Add 5 μL DNA ladder to the first well.
7. Mix 5 μL 6× DNA loading dye with each 25 μL PCR reaction.
8. Carefully pipet each into separate wells of the gel and record.
9. Run the gel at 100 V for 25–40 min or until the dye front is at the bottom of the gel.
10. Remove the gel from the apparatus and, with proper protection, use a UV transilluminator to visualize the bands in the gel.
11. Use a scalpel to carefully extract the slice of the gel containing your PCR insert, ideally removing a 300–800 mg slice of agarose, being sure to exclude any excess gel not containing your sample.
12. Following manufacturer instructions, use a DNA gel extraction kit to dissolve the gel slice and purify the DNA, eluting with 50–100 μL of ultrapure microbiology grade water (*see Note 5*).

3.1.3 Restriction Enzyme Digestion

1. Measure the concentration of the purified PCR insert using standard methods for DNA, preferably using a nanoliter volume instrument such as a NanoDrop or Epoch2 (*see Note 6*).
2. Combine the gel purified PCR insert(s) (100 μL) with 1 μL XhoI, 1 μL NcoI, and 11.3 μL of 10× Cutsmart buffer in a 1.5 mL microcentrifuge tube.
3. Repeat **steps 1** and **2** with the vector DNA, adjusting the volumes of the components according to the concentration and volume being used.
4. Incubate for at least 1 h at 37 °C for adequate digestion; however, longer incubation times even up to overnight may be used to ensure complete digestion (*see Note 7*).
5. Once the digest incubation period is complete, perform a DNA cleanup of the PCR insert using a DNA extraction/cleanup kit, eluting with 50–75 μL of ultrapure microbiology grade water.
6. For the vector DNA digest reaction(s), first run on a 1% agarose gel as instructed in Subheading 3.1.2. Then excise the digested vector band and use a DNA gel extraction kit to dissolve the gel slice and purify the DNA, eluting with 50–75 μL of ultrapure microbiology grade water.

3.1.4 Ligation of Insert and Vector

1. Measure the concentration of the digested PCR insert and vector as indicated in Subheading 3.1.3, step 1.
2. Mix the digested PCR insert and vector at a molar ratio of 5–10:1 (insert:vector); then add T4 DNA ligase (1 μ L per 10 μ L of ligation reaction volume), and 10 $\times$ T4 ligase buffer to a final 1 $\times$ concentration.
3. Incubate at room temperature for at least 1 h; however, longer incubation times up to overnight will ensure maximal ligation yields (*see* **Note 8**).

3.1.5 Transformation of Ligation Reaction into Chemically Competent DH5 α

1. Thaw 50 μ L of chemically competent DH5 α cells on ice. Once thawed, add 1–5 μ L of the ligation reaction to the cells, mix by gently tapping the tube, and incubate on ice for 20 min (*see* **Note 9**).
2. During the incubation, place an LB-agar plate containing the appropriate antibiotic in a 37 °C incubator to pre-equilibrate.
3. Heat shock the cells in a 42 °C water bath for 45 s and immediately place back on ice for 2 min.
4. Add 200 μ L of SOC outgrowth medium and shake at 37 °C for 30–60 min at ~1000 RPM using a thermomixer.
5. Using a flame and sterile cell spreader, plate all the cells onto the pre-equilibrated agar plate containing the appropriate antibiotic(s) (e.g., For a pHIS2-parallel vector, use an LB-agar/carbenicillin or LB-agar/ampicillin plate).
6. Incubate the agar plate(s) overnight (~20 h) at 37 °C inverted.
7. Check the plate(s) and count the number of colonies present for the ligation reaction and compare to the control plate with vector only (if this was also made). While the colonies are best when fresh, the plates may be sealed with parafilm and stored at 4 °C for up to two weeks.

3.1.6 Overnight Cultures and DNA Plasmid Minipreps

1. Select 2–5 colonies to prepare 5 mL cultures of LB with an appropriate antibiotic for each in separate 14-mL culture tubes.
2. Use a flame and a sterile loop each time to carefully transfer each colony into the premade, prelabeled culture tube, ensuring the cap is loose to allow proper aeration.
3. Shake the culture tubes at ~200 RPM overnight at 37 °C or until stationary phase.
4. Centrifuge the cultures at 4500 RPM using a benchtop centrifuge for 15 min at 4 °C.
5. Pour off the medium, being sure to remove as much of the liquid as possible.

6. The plasmid DNA may be purified using a commercially available DNA plasmid isolation/miniprep kit according to the manufacturer instructions.
7. Measure the concentration of each of the miniprep DNA samples using standard methods for DNA as described in Subheading 3.1.3, step 1.

3.1.7 DNA Sequencing Analysis

1. For each miniprep DNA sample, adjust the concentration and volume required by the sequencing facility being used. Do the same for each sequencing primer being used.
2. Submit samples for sequencing analysis.
3. Use a vector analysis program or a pairwise BLAST program to compare the sequencing results to the known sequence to determine if the cloning was successful.

3.2 Expression

3.2.1 Transformation into BL21 (DE3) Cells

1. Place an LB-carbenicillin agar plate into a 37 °C incubator to preequilibrate.
2. Thaw a 50 mL tube of chemically competent BL21(DE3) cells on ice (*see Note 10*).
3. Pipet 1 µL of your sequence verified plasmid into the cells, close the cap, and mix by gently tapping the end of the tube.
4. Place back on ice for 20 min.
5. Heat shock the cells in a 42 °C water bath for 45 s and immediately place back on ice for 2 min.
6. Add 200 µL of SOC medium to the tube.
7. Place into a thermomixer and shake at 1000 RPM at 37 °C for at least 30 min (*see Note 11*).
8. Pipet 30 µL of the transformation reaction onto one side of the prewarmed LB-carb agar plate and use an inoculation loop to spread the cells in a streaking fashion across the plate, rotating by 90° and repeating until back at the starting position.
9. Allow the plate to dry completely and then incubate at 37 °C inverted overnight (16–24 h).

3.2.2 Small-Scale Expression Test

1. Inspect your plate for well grown colonies. Select one or more colonies, circle, and label them on the bottom of the plate (*see Note 12*).
2. Prepare a 14 mL culture tube with 3–5 mL of LB medium supplemented with ampicillin (50 µg/mL).
3. Use a sterile loop to gently pick up your colony and transfer into your medium with rapid stirring for 2–5 s.
4. Repeat **steps 1–3** for each colony you have chosen to test for expression.

5. Shake (~200 RPM) the cells at 37 °C until they have reached an OD₆₀₀ between 0.6 and 1.0.
6. For each culture in **step 5**, transfer 1 mL of the cells into a 1.5 mL microcentrifuge tube and label as preinduction control (Pre).
7. Prepare the preinduction controls (Pre) for SDS-PAGE analysis as described in Subheading 3.2.3.
8. Transfer 1 mL of the cells from **step 5** into a fresh 14 mL culture tube and label as postinduction sample (Pre).
9. Store the remaining cultures from **step 5** at 4 °C.
10. Induce the postinduction cultures from **step 8** by adding IPTG to a final concentration of 1 mM (5 µL of stock).
11. Allow the cells to continue to grow an additional 1–2 h.
12. Prepare the postinduction samples (Post) for SDS-PAGE analysis as described in Subheading 3.2.3.

3.2.3 Preparing Cells for SDS-PAGE Analysis

1. Adjust the OD₆₀₀ of the cells to 0.6 in a final volume of 1 mL in a new 1.5 mL microcentrifuge tube.
2. Spin down the cells using a microcentrifuge at 16,000 × *g* (14,000 RPM) for 1 min.
3. Remove and discard of the medium supernatant.
4. Add 200 µL of 2× SDS-Loading Buffer and gently resuspend the cells by pipetting.
5. Heat the samples for 5 min at 95 °C.
6. Remove and centrifuge the samples for 5 min at full speed using a microcentrifuge (*see Note 13*).
7. Remove the samples and be careful not to disturb or shake them.
8. Analyze the pre- and postinduction samples, including a suitable protein marker, using SDS-PAGE using manufacturer instructions and stain/destain the gel to visualize the results.
9. Compare the pre- versus postinduction samples to determine if an increase in the levels of protein expression is observed in the presence of IPTG. Make a note if there were differences between the different colonies, if more than one was analyzed (*see Note 14*).

3.2.4 Prepare a Glycerol Stock of the Transformed Cells

1. Collect and label a 1.5 mL microcentrifuge tube for each sample in Subheading 3.2.3, **step 9** with confirmed expression (*see Note 15*).
2. Pipet 250 µL of your overnight culture from Subheading 3.2.2, **step 9**.

3. Pipet 250 μL of 60% glycerol–LB solution and mix by pipetting.
4. Place your glycerol stock into a freezer at -80°C until future use; flash freezing in liquid nitrogen is preferred when possible.

3.2.5 Prepare a 5 mL Overnight Starter Culture for Large-Scale Expression

1. Prepare a 14 mL culture tube with 6 mL of LB medium supplemented with ampicillin (50 $\mu\text{g}/\text{mL}$) (*see* **Note 16**).
2. Use a sterile loop to inoculate the culture using the glycerol stock prepared in Subheading 3.2.4. Do this by scraping across the top of the frozen glycerol stock with the sterile loop and transferring to the culture medium with rapid stirring for 2–5 s. Alternatively, fresh colonies may be used if no difference in expression was observed across multiple colonies in Subheading 3.2.3, step 9.
3. Shake (~ 200 RPM) the culture at 37°C until the OD_{600} is ~ 0.5 , do not overgrow (*see* **Note 17**).

3.2.6 Large-Scale Expression

1. Prepare five 2 L baffled flasks with 1 L of TB medium in each, autoclave using manufacturer instructions (*see* **Note 18**).
2. Let the flasks cool and add carbenicillin to a final concentration of 100 $\mu\text{g}/\text{mL}$.
3. Collect the overnight culture from Subheading 3.2.5, step 3.
4. Pipet 1 mL of the starter culture into each of the 1 L TB/carb baffled flasks.
5. Shake (~ 200 RPM) the flasks for 72 h at 20°C .
6. Harvest the cells by centrifugation at $8,000 \times g$ (5,000 RPM) for 15 min at 4°C .

3.3 Purification

3.3.1 Preparation of Lysate

1. Weigh the cell pellet (either fresh or frozen) from Subheading 3.2.6, step 6 and add 10 mL of $1\times$ PBS per gram of cells.
2. Stir the cells either on ice or in a cold room rapidly until fully resuspended (*see* **Note 19**).
3. Add DNase to a final concentration 10 $\mu\text{g}/\text{mL}$, mix well, and incubate for 10 min.
4. Add PMSF to a final concentration of 200 μM and immediately lyse the cells by three passages through an Emulsiflex C3 high-pressure homogenizer (Avestin) using manufacturer instructions maintaining a pressure of $\sim 15,000$ PSI (*see* **Note 20**).
5. The lysate is now ready for further purification (*see* **Note 21**).

3.3.2 Solubilization from Whole Cell Lysate

1. Centrifuge the lysate at $12,000 \times g$ ($\sim 8,000$ RPM) for 10 min to remove large insoluble cell debris.
2. Pour the supernatant into a fresh beaker and add DDM to a final concentration of 0.5% and stir overnight at 4°C .

3. Centrifuge the solution at $40,000 \times g$ (~18,000 RPM) to remove large insoluble debris.
4. Collect the supernatant and filter (0.45 μm); the solubilized protein is in the supernatant.
5. Proceed with further purification as described in Subheading 3.3.5 (*see* **Note 21**).

3.3.3 Solubilization from Isolated Membranes

1. Centrifuge the lysate at $12,000 \times g$ (~8,000 RPM) for 10 min to remove large insoluble cell debris.
2. Pour the supernatant into round-bottom ultracentrifuge tubes (e.g., for a Type 45 Ti rotor which has a maximum volume of 70 mL) (*see* **Note 22**).
3. Centrifuge at $235,000 \times g$ (~45,000 RPM) for 90 min using an ultracentrifuge. The resulting pellet contains the cellular membranes.
4. Pour off the supernatant and transfer the membrane pellet into a 40 mL dounce homogenizer.
5. Homogenize the membranes in ~60 mL of $1 \times \text{PBS}$.
6. Bring the volume to 80 mL with $1 \times \text{PBS}$, then add Elugent to a final concentration of 5% with rapid stirring, and stir overnight at 4 °C.
7. Centrifuge the solubilized sample at $235,000 \times g$ (~45,000 RPM) for 60 min.
8. Collect the supernatant and filter (0.45 μm); the solubilized protein is in the supernatant.
9. Proceed with further purification as described in Subheading 3.3.5 (*see* **Note 21**).

3.3.4 Isolation of Inclusion Bodies and Refolding

1. After cell lysis, harvest the inclusion bodies by centrifuging at $12,000 \times g$ (~8,000 RPM) for 10 min.
2. Transfer the inclusion bodies into a dounce homogenizer and wash by homogenizing in 40 mL of $1 \times \text{PBS} + 2\% \text{ Triton X-100}$.
3. Centrifuge at $12,000 \times g$ (~8,000 RPM) for 10 min to isolate the inclusion bodies, discarding the supernatant.
4. Repeat the washes in **steps 2** and **3** two more times. The inclusion body pellet should ideally become whiter with each wash (*see* **Note 23**).
5. Repeat the washes in **steps 2** and **3** three more times with $1 \times \text{PBS}$ only to remove as much Triton X-100 as possible.
6. After the final wash step, dissolve the inclusion bodies in 40 mL of freshly prepared 8 M urea.

7. Add DDM to a final concentration of 0.1%.
8. Allow the urea-solubilized sample to rock at room temperature for 30 min.
9. Centrifuge the sample at $12,000 \times g$ ($\sim 8,000$ RPM) for 10 min and collect the supernatant.
10. Transfer the supernatant into preequilibrated dialysis tubing and dialyze against 1 L of $1 \times$ PBS overnight at 4°C (*see Note 24*).
11. Collect the dialyzed sample and centrifuge at $40,000 \times g$ ($\sim 18,000$ RPM) for 30 min to remove insoluble debris.
12. Collect the supernatant and filter (0.45 μm); the refolded protein is in the supernatant.
13. Proceed with further purification as described in Subheading 3.3.5 (*see Note 21*).

3.3.5 Purification of Solubilized Membrane Proteins

1. Attach the 5 mL HisPur Ni-NTA column to the FPLC system according to manufacturer instructions.
2. Write a standard method using a method wizard (or manually) to perform affinity chromatography as follows monitoring at 280 nm: (a) equilibrate the column with 5 column volumes (CV) of Buffer A, (b) apply the solubilized protein sample using a sample pump connected to an air sensor, (c) wash out unbound sample using 5 CV of Buffer A, (d) perform elution with fractionation (2 mL per fraction) using a linear gradient from 0 to 500 mM imidazole over the course of 25 CV; or by using a step gradient with 25, 50, 100, 250, and 500 mM imidazole steps with 5 CV each, and lastly (v) reequilibrate the column with 5 CV of Buffer A.
3. Use SDS-PAGE analysis of the peak fractions to determine which contain the target protein.
4. Combine the fractions containing the protein of interest (*see Note 25*).
5. Concentrate the sample to 5 mL or less and run over a Sephacryl S-300 HR16/60 Hi-Prep size-exclusion column (GE Healthcare) in SEC buffer using the FPLC system with a flow rate of 0.5 mL/min with fraction volumes of 1 mL each (*see Note 26*).
6. Use SDS-PAGE analysis of the peak fractions to determine which contain the target protein.
7. Combine the purest fractions containing the protein of interest.
8. Concentrate the sample to 10 mg/mL (*see Note 27*).

3.4 Crystallization

3.4.1 Aliquoting

Commercial Crystallization

Broad-Matrix Screens

(Hanging Drop)

1. Centrifuge the commercial screen deep well block at $250 \times g$ ($\sim 1,500$ RPM) for 2 min.
2. Label the needed number of round-bottom plates (*see Note 28*).
3. Carefully remove the seal of the deep well block and place into the pipetting robot according to manufacturer instructions.
4. Add the labeled round-bottom plates into the available positions on the robot, according to manufacturer instructions.
5. Pipet 50 μL of solution from the deep well block into each of the round-bottom plates. If the instrument has more than a single dispensing position, a single aspiration may allow multiple dispenses; follow manufacturer instructions.
6. Replace the filled plates in the dispensing positions with empty ones and repeat **step 5**.
7. Repeat **steps 5** and **6** until all the solution in the deep well block has been dispensed (*see Note 29*).
8. Seal the round-bottom plates using an automated microplate heat sealer or manually using single plate seals (*see Note 30*).
9. Repeat **steps 1–8** for each commercial broad-matrix screen that will be used. Alternatively, **steps 1–8** can be performed in parallel for all the screens to improve efficiency.
10. Store the aliquoted plates between 4 and 10 $^{\circ}\text{C}$ or freeze at -20°C .

3.4.2 Preparing 96-Well

Optimization Screens

1. Gather the information for the lead crystallization condition that will be optimized.
2. Prepare sufficient quantities of the necessary stock solutions for each of the components (*see Note 31*).
3. Using the screen design software of the liquid handling robot, design an appropriate screen around the lead condition that will pipet 50–100 μL into each well. Ideally, precipitants ($\pm 10\%$) are varied along the *X*-axis and salts are varied along the *Y*-axis (± 25 –100 mM, depending on the starting concentration), while buffers are kept constant across each single plate, yet, can be varied across multiple plates. Alternatively, buffer pH can also be varied along one of the axes if desired (*see Note 32*).
4. Insert/load the stock solutions onto the automated liquid handling robot and prime the system according to the manufacturer instructions.
5. Load the round-bottom 96-well plate onto the instrument and run the program to dispense the optimization screen. If desired, larger volumes can be designed and dispensed into 0.5 or 1 mL deep well blocks.

6. Mix the optimization plate using a microplate mixer or vortexer fitted with a microplate adapter. If making a deep well block, seal and then mix by inverting continuously until homogeneous. Then centrifuge at $250 \times g$ (1,500 RPM) for 2 min prior to using.
7. Repeat **steps 1–6** for each lead crystallization condition that will be optimized.
8. Additionally, an additive optimization screen can be prepared by preparing 5 mL of each lead crystallization condition and dispensing 50 μ L into each well of a 96-well round-bottom plate. Then, 5.6 μ L of each additive from a 96-well additive screen is pipetted into each well of the round-bottom plate using a 12-channel multipipetter, and follow **step 6**.
9. Store the optimization plates between 4 and 10 °C or freeze at -20 °C.

*3.4.3 Performing
Broad-Matrix
and Optimization
Crystallization Screening
Using an Automated
Crystallization Robot
(Hanging
and Sandwich Drop)*

1. Design a hanging drop method that dispenses 200 nL of protein solution and 200 nL of well solution onto the hanging drop crystallization sheet using an automated crystallization robot, according to manufacturer instructions. Where possible, utilize a humidification chamber or alternatively, a humidity flow from a standard humidifier may directed over the plate deck (*see* **Note 33**).
2. Load a V-bottom 96 well polypropylene plate (for a SPT Labtech mosquito LCP, but may vary for other crystallization robots) onto the plate deck according to the designed method and pipet a sufficient volume of the protein sample for setting 1–5 plates into a full column.
3. Centrifuge the broad-matrix or optimization 96-well crystallization screen at $250 \times g$ (1,500 RPM) for 2 min and load onto the robot plate deck according to the designed method.
4. Add a hanging drop crystallization sheet onto the plate deck, according to robot manufacturer instructions (*see* **Note 34**).
5. Run the designed method to dispense the hanging drop crystallization tray.
6. When finished dispensing, immediately invert the crystallization sheet from **step 4** and carefully affix to the top of the crystallization plate from **step 3** such that the drops are facing toward the well solution, being sure that the crystallization drops are properly aligned to the corresponding wells. Use a plate roller to ensure proper sealing across the entire plate.
7. For crystallization in bicelles, mix 40 μ L of protein solution with 10 μ L of bicelles and incubate on ice for at least 30 min. Then repeat **steps 2–6** for each crystallization screen being used.

8. For crystallization using LCP, first prepare the LCP–protein mix according to published protocols. Then attach the syringe containing the LCP–protein mix according to instrument manufacturer instructions. Next, prepare a new method for LCP crystallization with sandwich plates consisting of 50–100 nL of LCP–protein mix and 1 μ L of well solution. Run the method according to manufacturer instructions for each crystallization screen being used (*see* **Note 35**).
9. Transfer the crystallization trays to an automated crystallization imaging robot with an attached hotel according to manufacturer instructions at an appropriate temperature, if available. Alternatively, crystallization trays may be stored in standalone temperature-controlled incubators.

3.4.4 Performing Optimization Screening Manually

1. Gather the information for the lead crystallization condition that will be optimized.
2. Prepare sufficient quantities of the necessary stock solutions for each of the components (*see* **Note 31**).
3. Design the optimization screen for either a 15 or 24-well large format optimization tray using an online tool such as Hampton Research's Make Tray Tool (<https://hamptonresearch.com/make-tray.php>) (*see* **Note 36**). Print the summary which lists the pipetting volumes of each component for each well of the tray.
4. Pipet the components into each well of the tray as indicated and mix the tray thoroughly using a plate mixer or vortexer with a plate adapter.
5. For detergent and bicelle crystallization, pipet 1–2 μ L of sample onto an 18 mm round coverslip, followed by an equal volume of well solution. Then carefully invert the coverslip and place over the corresponding well and gently press slightly into the grease until a seal is formed; it is not necessary to press fully down to the plate rim. Repeat for each well of the optimization plate.
6. For LCP crystallization, prepare the LCP–protein mix in a 10 μ L Hamilton syringe according to instrument manufacturer instructions. Install the syringe containing the LCP–protein mix into a repeating dispenser. Pipet 200 nL (a single dispense) of LCP–protein mix and 1 μ L of well solution into each well of a 2×2 section of a 96-well sandwich plate. Seal with an 18 mm square coverslip. Repeat for each of the remaining 2×2 sections of the optimization plate.
7. Store the optimization trays at an appropriate temperature ranging 4–21 $^{\circ}$ C in standalone temperature-controlled incubators (*see* **Note 35**).

3.4.5 *Visualizing
and Assessing
Crystallization Leads*

1. If using an automated crystallization imaging system, image the trays on a routine schedule at least once per week for up to 3 months. Images are typically visualized remotely and interactively scored and annotated (*see Note 37*).
2. If using a stereomicroscope to manually visualize the crystallization trays, monitor the trays at least one time per week for up to 3 months. Keep a log of the most promising conditions and those conditions producing crystals.
3. For lead conditions that produce crystals, image the drop using UV imaging either on the automated imaging robot or manually using a standalone UV imaging microscope (*see Note 38*).
4. If UV imaging is inconclusive, use a stereomicroscope and a cryoloop or microtool to crush one of the smaller crystals in the drop. Protein crystals are fragile and should crush relatively easily, while salt crystals would require significant force and would typically break off in chunks. Alternatively, small molecule dyes may be used to stain protein crystals (*see Note 39*).

**3.5 Data Collection
and Structure
Determination**

3.5.1 *Harvesting
and Storing Crystals*

1. Use the puck tools to open a uni-puck, placing the base into an empty foam dewar.
2. Add liquid nitrogen to the dewar until it just covers the uni-puck base. Allow to cool for a few minutes until the liquid nitrogen is no longer boiling.
3. Fill the dewar to the rim and place the cover/cap on to allow the dewar and uni-puck to equilibrate. Ensure that the dewar stays topped off the entire time you are harvesting crystals and avoid frost buildup and ice in the liquid nitrogen.
4. To harvest crystals from 96-well plates, use a razor blade to carefully cut around the drop of interest by scoring a box, being sure to cut completely through the crystallization sheet.
5. Gently lift the edge of the excised square, cut the underlying glue, and quickly invert the drop.
6. Use reverse action tweezers to hold the drop or place it onto a petri dish to allow easy movement on the microscope. Align the drop to the focus region of the microscope (*see Note 40*).
7. Place a crystal loop/base onto a magnetic crystal wand and quickly, but carefully, bring the loop into proximity of the drop and gently submerge the loop in the drop, capture a crystal, and then rapidly freeze the crystal in the liquid nitrogen, removing the dewar lid immediately before plunging the loop (*see Note 41*).
8. Once frozen, carefully place the loop in the appropriate position within the puck, being sure not to reexpose it to air.

9. Repeat **steps 7 and 8** for each crystal in the drop that you wish to harvest. Record the necessary crystal information in the puck log sheet for each position.
10. Repeat **steps 4–9** for each drop in the tray you wish to harvest crystals from.
11. If cryoprotectants are being used, they can be pipetted in close proximity to the drop with the crystals being directly transferred and incubated for the desired time period prior to looping and freezing as outlined in **steps 7 and 8**. Alternatively, the cryoprotectant can be directly added to the drop. If longer incubation periods are desired, the drop can be inverted and placed back over the well solution and taped to reduce evaporation.
12. Once a puck is filled, use the puck tools to reinstall the puck top and place the puck into a precooled puck cane according to manufacturer instructions and place into a storage dewar until X-ray diffraction screening and data collection. Repeat for other pucks that are filled while harvesting crystals.

3.5.2 Screening and Data Collection

1. If screening at a home source, transfer the puck to a precooled dewar filled with liquid nitrogen.
2. Use the puck tools to open the top of the puck to allow access to the crystal pins.
3. Using a magnetic wand and cryo pin tongs, transfer the crystal pin of interest from the puck to the goniometer as instructed by the facility scientist.
4. Align the crystal to the center position as instructed, rotating the crystal 90° and realigning to the center position. Repeat until the crystal is properly aligned along the axis of rotation and always in the beam.
5. Move the detector to where the edge would produce ~3 Å resolution diffraction.
6. Set the parameters for collecting initial images at 0° and 90°. For example, for a Rigaku system equipped with a MicroMax-007 HF X-ray generator and a CCD detector, one would screen with an initial exposure of 30–300 s over 0.5–1° oscillation. The goal here is to determine if the crystal diffracts and what an approximate resolution may be. All final data collection should be performed at a synchrotron as described below.
7. If screening at a synchrotron, coordinate with the beamline staff to load the pucks into the robotic dewar (*see Note 42*).
8. Using the puck log sheet, use the data acquisition software to mount the desired crystal by selecting the puck and position to be loaded onto the goniometer.

9. Align the crystal to the center position as instructed by the beamline staff, rotate the crystal 90° and realign to the center position. Repeat until the crystal is properly aligned along the axis of rotation and always in the beam.
10. Move the detector to where the edge would produce ~3 Å resolution diffraction.
11. Setup for initial screening by attenuating the X-ray beam for only a short exposure. For example, at the GM/CA ID-B beamline with a 20 µm beam size and an Eiger 16M detector, we routinely screen with 80% attenuation for 0.25 s over 0.25–0.5° oscillation, collecting at 0°, 45°, and 90°.
12. Once the set of initial images have been collected, inspect them to determine if there are diffraction spots, the shape and intensities of the diffraction spots, the resolution, and whether there is ice present or not. From these initial observations, decide if the crystal is suitable to continue with data collection or if you should move on to screen other crystals. Keep in mind that now, it is often faster to just collect a dataset rather than spending time discussing it with a colleague or carefully inspecting the diffraction images.
13. To set up for data collection, refer to the strategy program within the data acquisition software for parameters to load and adjust as necessary, including detector distance. When in doubt, just collect a full 180° or 360° dataset if using molecular replacement to solve the structure. For experimental phasing using an anomalous signal, coordinate with the beamline staff to perform an edge scan and collect at 1–3 wavelengths (peak, edge, remote) being sure to collect as highly redundant data as possible, yet avoiding radiation damage (*see Note 43*).

3.5.3 Diffraction Analysis and Structure Determination

1. Most synchrotron beamlines now employ automated data processing and analysis tasks once a dataset is collected. These are convenient for quickly analyzing the data to assist in deciding how to proceed further.
2. Other programs such as phenix.xtriage (PHENIX) and Cell Content Analysis (CCP4) may also be used to analyze the data to determine other important information including the number of molecules per asymmetric unit, space group options, strength of anomalous signal, overall data quality, if twinning is present, and if pseudo translational symmetry is present.
3. If solving the structure using molecular replacement, use the information gathered in **step 2** to set up a molecular replacement job within PHENIX or CCP4 using standard protocols. For example, if using PHASER, select and load your sequence file, your processed data file, and your cleaned search model.

Complete the cell composition details and set up for a search procedure, being sure to turn on the option to try “All possible in same pointgroup.” Once a correct solution has been found, proceed to iterative cycles of refinement and model building until the model matches the data as best as possible with suitable R/R_{free} values.

4. If solving the structure using experimental phasing, use the information gathered in **step 2** to set up an experimental phasing job within PHENIX, CCP4, SHELX or other suitable program using standard protocols. For example, if using AutoSol (PHENIX), define the phasing method and select and load your sequence file, your processed data file(s) and corresponding values for wavelength, f' , f'' , the anomalous scattering atom, and the estimated number of anomalous atoms present. Adjust the parameters for the substructure search as desired, including whether or not a model will be built and the number of processors to use. Once a correct substructure solution has been found, a density-modified map will be calculated which you can open and view using COOT or PyMOL. A structure is deemed “solved” if secondary structure elements and side chains are observed in the electron density. From here, perform an autobuilding job that should build much of the structure (with good density) and then proceed to iterative cycles of refinement and model building until the model matches the data as best as possible with suitable R/R_{free} values.

4 Notes

1. We often prepare our master mix in bulk and freeze in aliquots at -20 . While it is usually stable at 4°C for short time periods, we try to keep frozen for longer time periods. If problems arise with our PCR reactions, we try a new tube of frozen stock, or remake fresh. As an alternative to $5\times$ Phusion HF Buffer, a GC-rich buffer may be used for GC-rich regions/primers or troublesome reactions.
2. Variations of the PCR protocol can vary depending on if the template is purified plasmid or genomic DNA, the percentage of GC-content in the template/primers, size of the amplicon, and the polymerase being used.
3. While proper personal protective equipment should be used at all times, it is critical that safety gloves be worn when working with ethidium bromide which is a strong mutagen and carcinogen. Other safer options are available if preferred.

4. Carefully transferring to a fridge at 4 °C can speed solidification. We routinely cast a large gel and store at 4 °C with a damp paper towel in an airtight container. Then cut into the number of lanes needed for each use.
5. To prevent buffer effects in the reaction, we prefer to elute with water rather than elution buffer during cloning.
6. To determine DNA concentration, measure the absorbance at 260 nm (A_{260}) without dilution. Then multiply the absorbance by 50; the resulting number will be your final DNA concentration in $\mu\text{g}/\text{mL}$. To check for purity, measure the absorbance at 280 nm (A_{280}) also and then divide the A_{260} by the A_{280} to yield a A_{260}/A_{280} ratio; values above 1.8 indicate good quality pure DNA.
7. Optimal incubation times will depend on the amount of DNA being digested, temperature, and efficiency of the enzyme(s) being used. We recommend using high-fidelity enzymes when possible.
8. While room temperature incubators work well for our work, other temperatures may be tried, with many protocols suggesting 16 °C.
9. If desired, a control can be done in parallel with an equivalent volume of vector only. However, we often skip the control here once we have become adept in cloning to save on resources.
10. We use BL21(DE3) cells here since they are known to have “leaky” expression, which allows very slow induction at low temperatures to provide sufficient time for the protein to fold into the outer membrane, which is specific to β -barrel membrane proteins. This may not be necessary for other membrane proteins including those with an α -helical membrane domain. Further, other cell types have also been used with success for membrane proteins including T7 Express, C41(DE3), C43(DE3), and Rosetta(DE3).
11. This step may be skipped if using vectors with amp/carb resistance; however, efficiency may be reduced. For other antibiotics, the outgrowth up to 1 h is necessary and should not be skipped.
12. As expression can vary significantly between colonies, especially if the protein is somewhat toxic, we recommend performing colony screening by choosing 5–10 colonies to test in a small-scale expression assay. Then proceed to large-scale expression with the colonies that produced the highest level of expression by SDS-PAGE analysis.
13. Centrifuging the sample here sediments the genomic DNA which will aid in loading the samples onto the gel. We

recommend pipetting from the very top of the solution. If the samples are still viscous, they can be reheated and then quickly loaded, avoiding the use of gel loading tips.

14. Expression may not always be observable and may require Western blot analysis to determine if expression was successful.
15. By preparing glycerol stocks, we can skip the transformation step each time and prepare cultures from this frozen stock which saves time and money.
16. Prepare enough starter culture such that you can dispense 1 mL per flask and have some extra if needed. If available, we would recommend the use of large-volume fermenters which have many advantages over shaker flasks.
17. While likely protein dependent, we have found that inoculating with cultures that are lower in initial cell density yields better results for some of our β -barrel membrane proteins; an observation we cannot explain.
18. We prefer to use baffled flasks here which allow for better aeration and growth.
19. Once the cells have been resuspended, all subsequent steps should be done at 4 °C or on ice.
20. PMSF has a short half-life in water of ~1 h, so be sure to add just before lysis. If you find that PMSF alone is insufficient, additional inhibitors may be used with several varieties available commercially such as the cOmplete Inhibitor Cocktail (Millipore Sigma), SIGMAFAST Protease Inhibitor (Millipore Sigma), and HALT Protease Inhibitor Cocktail (ThermoFisher).
21. Do not stop here. Once you lyse the cells, you must continue immediately to purification, else risk sample degradation and/or precipitation.
22. Follow manufacturer instructions here and be sure to fill the centrifuge bottles completely to prevent their damage and them getting stuck within the rotor.
23. If the inclusion bodies do not become visibly purer after each wash step, Sarkosyl (<1%) and/or urea (<2 M) can be added at low concentrations to aid the removal of impurities.
24. If using DDM, it is not necessary to add detergent in the dialysis buffer, as long as a smaller MWCO (<10 kDa) tubing is used. If using other detergents, supplementing the dialysis buffer may be necessary.
25. If the protein is expressed with a solubility fusion partner or affinity tag which is preferred to be removed, the cleavage process can also be performed during this step. This is usually done at 4 °C overnight, often with dialysis to remove the

imidazole. The solution can be reapplied to the Ni-NTA column (reverse IMAC) with the cleaved protein being collected in the flow through.

26. DDM forms large micelles which can often prevent crystal nucleation, especially with β -barrel membrane proteins. While DDM has been used with success for the crystallization of α -helical membrane proteins, it may be necessary to exchange into other detergents with smaller micelles such as *n*-octyl- β -D-glucopyranoside (OG, 1%), lauryl maltose neopentyl glycol (LMNG, 0.01%), or N,N-dimethyldodecylamine-N-oxide (LDAO, 0.05%).
27. The standard starting concentration for initial crystallization screening is ~ 10 mg/mL; however, this may vary depending on the sample properties and other factors including solubility, size, charge, isoelectric point, and conformational stability.
28. By prealiquoting the trays in advance, we use one box of tips for the entire deep well block, greatly reducing tip waste. For example, if using a 1.5 mL deep well block with 50 μ L in each tray, we label and make 30 trays. Another advantage is that the trays are ready to be used at all times, rather than having to aliquot them each time a sample is ready.
29. It sometimes occurs that there is insufficient solution left in the deep well block to prepare a full final plate; therefore we often just discard the last tray or skip it (e.g., we get 29 trays rather than 30 out of a 1.5 mL deep well block).
30. While we use a microplate roller to seal the manual plates, we also use the roller to roll the plates sealed with the automated heat sealer, to ensure a proper airtight seal.
31. Stock solutions are often described by the commercial screen manufacturers; therefore, this is a good place to find details about how the screens were prepared to improve reproducibility. Further, most companies also sell the individual components and even optimization screens.
32. Many liquid handling robots come with software that will enable simple to complex formulations across the entire plate such as setting up different conditions on each quadrant of a plate and pH gradients.
33. Other options that can be typically customized in the method include drop volume, drop ratio, the number of drops per well, mixing options, transfer sequences, tip change options, and humidification options.
34. With membrane proteins that require detergent-based buffers, it is very important to use hydrophobic crystallization sheets, being sure to remove the thin protective film prior to setting up the crystallization tray.

35. The LCP matrix is not easy to work with as it has a consistency of epoxy and requires patience and practice. We recommend starting with monoolein-based LCP crystallization which requires the trays to be stored at 17 °C or above; however, other lipids have been developed for lower temperatures if needed.
36. One important variable that can be assayed and optimized here is the well volume. For VDXm plates, we typically use 300–500 μ L.
37. Modern crystallization imaging systems offer many advanced features such as automated drop scoring, automated crystal detection, Z-slice imaging, UV imaging, and focused regions of interest imaging. In addition, there is a log of the drop history which can be viewed instantly.
38. Be aware that UV imaging will not be useful for proteins lacking tryptophan residues, therefore, other methods will need to be used to verify if the crystal is a protein. To circumvent this, during cloning, a tryptophan residue can be engineered at one of the termini to allow UV imaging during crystallization, but also for absorbance monitoring at 280 nm during purification.
39. Note that there are exceptions to many of these tests and therefore the ultimate test for an ambiguous crystal is to conduct an X-ray diffraction experiment.
40. The use of a petri dish can allow for reduced evaporation during harvesting. Additionally, one can also use a humidifier to provide a gentle stream of humidity to allow more time to harvest crystals from quickly evaporating drops.
41. Care must be taken here to prevent plunging too deep and hitting the bottom or side of the dewar. Harvesting crystals should be performed quickly and with care to not reexpose a frozen sample back into the air while transferring it into the desired position within the puck.
42. All beamlines are different and have their own standard operating protocols. We strongly recommend checking in advance with the beamline staff to determine their requirements for types of pins and lengths, base type, puck type, shipping procedures, and other relevant information.
43. For experimental phasing, it is a good idea to take sufficient time to carefully set up the data collection experiment, possibly selecting only the best crystals for this. Then, once the data has been collected, we recommend taking some time to analyze the data to determine if suitable phases can be calculated with an interpretable electron density map, rather than collecting all your data using the same parameters only to find out later that one parameter was not optimal or incorrect.

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References

- Slusky JS (2017) Outer membrane protein design. *Curr Opin Struct Biol* 45:45–52
- Krogh A, Larsson B, von Heijne G et al (2001) Predicting transmembrane protein topology with a hidden Markov model: application to complete genomes. *J Mol Biol* 305 (3):567–580
- Fairman JW, Noinaj N, Buchanan SK (2011) The structural biology of beta-barrel membrane proteins: a summary of recent reports. *Curr Opin Struct Biol* 21(4):523–531
- Tsirigos KD, Govindarajan S, Bassot C et al (2018) Topology of membrane proteins: predictions, limitations and variations. *Curr Opin Struct Biol* 50:9–17
- Chaturvedi D, Mahalakshmi R (2017) Transmembrane beta-barrels: evolution, folding and energetics. *Biochim Biophys Acta Biomembr* 1859(12):2467–2482
- Hedin LE, Illergard K, Elofsson A (2011) An introduction to membrane proteins. *J Proteome Res* 10(8):3324–3331
- von Heijne G (2006) Membrane-protein topology. *Nat Rev Mol Cell Biol* 7 (12):909–918
- Walther DM, Rapaport D, Tommassen J (2009) Biogenesis of beta-barrel membrane proteins in bacteria and eukaryotes: evolutionary conservation and divergence. *Cell Mol Life Sci* 66(17):2789–2804
- Jeffery CJ (2016) Expression, solubilization, and purification of bacterial membrane proteins. *Curr Protoc Protein Sci* 83:29.15.1–29.15.15
- Ki MR, Pack SP (2020) Fusion tags to enhance heterologous protein expression. *Appl Microbiol Biotechnol* 104(6):2411–2425
- Rosano GL, Morales ES, Ceccarelli EA (2019) New tools for recombinant protein production in *Escherichia coli*: a 5-year update. *Protein Sci* 28(8):1412–1422
- Chun E, Thompson AA, Liu W et al (2012) Fusion partner toolchest for the stabilization and crystallization of G protein-coupled receptors. *Structure* 20(6):967–976
- Mancusso R, Karpowich NK, Czyzewski BK et al (2011) Simple screening method for improving membrane protein thermostability. *Methods* 55(4):324–329
- White JF, Noinaj N, Shibata Y et al (2012) Structure of the agonist-bound neurotensin receptor. *Nature* 490(7421):508–513
- Noinaj N, Kuszak AJ, Gumbart JC et al (2013) Structural insight into the biogenesis of beta-barrel membrane proteins. *Nature* 501 (7467):385–390
- Tropea JE, Cherry S, Waugh DS (2009) Expression and purification of soluble His(6)-tagged TEV protease. *Methods Mol Biol* 498:297–307
- Snijder HJ, Hakulinen J (2016) Membrane protein production in *E. coli* for applications in drug discovery. *Adv Exp Med Biol* 896:59–77
- Carpenter EP, Beis K, Cameron AD et al (2008) Overcoming the challenges of membrane protein crystallography. *Curr Opin Struct Biol* 18(5):581–586
- Zhang Z, Kuipers G, Niemiec L et al (2015) High-level production of membrane proteins in *E. coli* BL21(DE3) by omitting the inducer IPTG. *Microb Cell Factories* 14:142
- Deacon SE, Roach PC, Postis VL et al (2008) Reliable scale-up of membrane protein overexpression by bacterial auto-induction: from microwell plates to pilot scale fermentations. *Mol Membr Biol* 25(8):588–598
- Studier FW (2005) Protein production by auto-induction in high density shaking cultures. *Protein Expr Purif* 41(1):207–234
- Structural Genomics C, China Structural Genomics C, Northeast Structural Genomics C et al (2008) Protein production and purification. *Nat. Methods* 5(2):135–146
- Studier FW (2018) T7 expression systems for inducible production of proteins from cloned

- genes in *E. coli*. *Curr Protoc Mol Biol* 124(1):e63
24. Miroux B, Walker JE (1996) Over-production of proteins in *Escherichia coli*: mutant hosts that allow synthesis of some membrane proteins and globular proteins at high levels. *J Mol Biol* 260(3):289–298
 25. Dumon-Seignovert L, Cariot G, Vuillard L (2004) The toxicity of recombinant proteins in *Escherichia coli*: a comparison of overexpression in BL21(DE3), C41(DE3), and C43(DE3). *Protein Expr Purif* 37(1):203–206
 26. Wu R, Stephenson R, Gichaba A et al (2020) The big BAM theory: an open and closed case? *Biochim Biophys Acta Biomembr* 1862(1):183062
 27. Seddon AM, Curnow P, Booth PJ (2004) Membrane proteins, lipids and detergents: not just a soap opera. *Biochim Biophys Acta* 1666(1–2):105–117
 28. le Maire M, Champeil P, Moller JV (2000) Interaction of membrane proteins and lipids with solubilizing detergents. *Biochim Biophys Acta* 1508(1–2):86–111
 29. Garavito RM, Ferguson-Miller S (2001) Detergents as tools in membrane biochemistry. *J Biol Chem* 276(35):32403–32406
 30. Huang KS, Bayley H, Liao MJ et al (1981) Refolding of an integral membrane protein. Denaturation, renaturation, and reconstitution of intact bacteriorhodopsin and two proteolytic fragments. *J Biol Chem* 256(8):3802–3809
 31. Lilie H, Schwarz E, Rudolph R (1998) Advances in refolding of proteins produced in *E. coli*. *Curr Opin Biotechnol* 9(5):497–501
 32. Yamashita S, Lukacik P, Barnard TJ et al (2011) Structural insights into Ail-mediated adhesion in *Yersinia pestis*. *Structure* 19(11):1672–1682
 33. Cherezov V, Clogston J, Misquitta Y et al (2002) Membrane protein crystallization in meso: lipid type-tailoring of the cubic phase. *Biophys J* 83(6):3393–3407
 34. Heras B, Martin JL (2005) Post-crystallization treatments for improving diffraction quality of protein crystals. *Acta Crystallogr D Biol Crystallogr* 61(Pt 9):1173–1180
 35. Pflugrath JW (2015) Practical macromolecular cryocrystallography. *Acta Crystallogr F Struct Biol Commun* 71(Pt 6):622–642
 36. Li D, Boland C, Aragao D et al (2012) Harvesting and cryo-cooling crystals of membrane proteins grown in lipidic mesophases for structure determination by macromolecular crystallography. *J Vis Exp* 67:e4001
 37. Caffrey M (2015) A comprehensive review of the lipid cubic phase or in meso method for crystallizing membrane and soluble proteins and complexes. *Acta Crystallogr F Struct Biol Commun* 71(Pt 1):3–18
 38. Caffrey M (2003) Membrane protein crystallization. *J Struct Biol* 142(1):108–132
 39. Faham S, Bowie JU (2002) Bicelle crystallization: a new method for crystallizing membrane proteins yields a monomeric bacteriorhodopsin structure. *J Mol Biol* 316(1):1–6
 40. Faham S, Boulting GL, Massey EA et al (2005) Crystallization of bacteriorhodopsin from bicelle formulations at room temperature. *Protein Sci* 14(3):836–840
 41. Warren AJ, Axford D, Paterson NG et al (2016) Exploiting microbeams for membrane protein structure determination. *Adv Exp Med Biol* 922:105–117
 42. Birch J, Axford D, Foadi J et al (2018) The fine art of integral membrane protein crystallisation. *Methods* 147:150–162
 43. Casanas A, Warshamanage R, Finke AD et al (2016) EIGER detector: application in macromolecular crystallography. *Acta Crystallogr D Struct Biol* 72(Pt 9):1036–1048
 44. Cherezov V, Hanson MA, Griffith MT et al (2009) Rastering strategy for screening and centring of microcrystal samples of human membrane proteins with a sub-10 microm size X-ray synchrotron beam. *J R Soc Interface* 6(Suppl 5):S587–S597
 45. Hilgart MC, Sanishvili R, Ogata CM et al (2011) Automated sample-scanning methods for radiation damage mitigation and diffraction-based centering of macromolecular crystals. *J Synchrotron Radiat* 18(Pt 5):717–722
 46. Hauptert LM, Simpson GJ (2011) Screening of protein crystallization trials by second order nonlinear optical imaging of chiral crystals (SONICC). *Methods* 55(4):379–386
 47. Kissick DJ, Wanapun D, Simpson GJ (2011) Second-order nonlinear optical imaging of chiral crystals. *Annu Rev Anal Chem (Palo Alto, Calif)* 4:419–437
 48. Schulz GE (2011) Validation of the detergent micelle classification for membrane protein crystals and explanation of the Matthews Graph for soluble proteins. *Protein Sci* 20(10):1765–1770



Studying Membrane Protein Structures by MicroED

Michael W. Martynowycz and Tamir Gonen

Abstract

Microcrystal electron diffraction (MicroED) enables atomic resolution structures to be determined from vanishingly small crystals. Soluble proteins typically grow crystals that are tens to hundreds of microns in size for X-ray crystallography. But membrane protein crystals often grow crystals that are too small for X-ray diffraction and yet too large for MicroED. These crystals are often formed in thick, viscous media that challenge traditional cryoEM grid preparation. Here, we describe two approaches for preparing membrane protein crystals for MicroED data collection: application of a crystal slurry directly to EM grids, and focused ion beam milling in a Scanning Electron Microscope (FIB-SEM). We summarize the case of preparing an ion channel, NaK, and the workflow of focused ion-beam milling. By milling away the excess media and crystalline material, crystals of any size may be prepared for MicroED. Finally, an energy filter may be used to help minimize inelastic scattering leading to lower noise on recorded images.

Key words CryoEM, Microcrystal electron diffraction, MicroED, Membrane proteins, FIB milling, Energy filter

1 Introduction

Several membrane proteins have been studied by electron diffraction using 3D crystals. The very first example of a novel structure of a membrane protein determined solely by electron diffraction was that of the water channel aquaporin-0 which was determined from crystals that contained only two layers [1]. This study laid the foundations for using molecular replacement with electron diffraction data [2] and from which the field of MicroED grew a decade later. Shortly after AQP0, the gap junction protein connexin 26 was determined using crystals only three layers thick [3]. In 2015 the structure of calcium ATPase was solved from much thicker 3D crystals [4] and the structure of the ion channel NaK was solved in 2018 using crystals with only ~1000 diffracting units [5] by continuous rotation MicroED [6]. MicroED coupled with focused-ion beam milling allowed structure determination of difficult membrane proteins crystallized with lipids: the voltage

dependent anion channel (VDAC) in bicelles and the G-protein coupled receptor A2a in lipidic-cubic phase [7, 8]. These examples illustrate the utility of electron diffraction in determining structures of membrane proteins from vanishingly small crystals to atomic resolution.

Electrons interact with matter more strongly than X-rays [9]. This strong interaction means that adequate signal can be generated from smaller specimens. However, this also suggests that specimens can easily become too large for electrons to penetrate. Membrane protein crystals often grow too small for synchrotron X-ray diffraction studies, and therefore make ideal candidates for microcrystal electron diffraction (MicroED) [6, 10–18]. To investigate membrane protein crystals using MicroED, they first have to be transferred from their crystallization drop to an electron microscopy (EM) grid, have the excess solvent removed by blotting, and are vitrified to preserve their hydrated, crystalline state [19]. Under ideal conditions, a drop of crystal slurry can be prepared using procedures typical of cryoEM applications [16, 19, 20]. However, the liquid in the slurry can often be viscous and difficult to blot [7, 8]. Furthermore, the crystals may not survive the forces incurred during the blotting and plunging process [10, 22, 23]. Focused ion beam (FIB) milling under cryogenic conditions involves shaving large specimens down to sizes amenable for cryoTEM investigation [24–27]. Samples are thinned by ablating away material from the specimen using a focused beam of gallium or other ions [28]. CryoFIB milling has garnered attention as a method to prepare thin lamellae for electron cryo-tomography and MicroED investigation [25, 29–33]. MicroED is ideally suited for crystals smaller than 500 nm thick [34–37], whereas crystals for synchrotron X-ray diffraction are typically >20 μm thick [34, 38]. Many crystals grow in this range, including many membrane proteins. Even membrane protein crystals smaller than 500 nm thick may be difficult to investigate by MicroED using traditional blotting methods. Blotting away the viscous detergents present in these slurries is challenging. Here, we demonstrate how protein crystals that are too large for MicroED can be milled down to thin lamellae, and how protein crystals that are of appropriate size may be transferred onto EM grids for MicroED data collection. We summarize the case of the MicroED study of the ion channel, NaK [5], and discuss a general workflow for focused ion-beam milling of protein crystals [29–33].

These results demonstrate the applicability of both standard blotting and cryo-FIB milling to prepare membrane protein crystals that are challenging to handle for synchrotron X-ray diffraction. These approaches pave the way for the investigation of considerably more difficult systems to be amenable to MicroED data collection. Several groups are working in this area, and have all made contributions to this growing field [4, 31–33, 39–43].

2 Materials

A NaK construct with the first 19 residues removed was expressed and purified as described [5, 44, 45]. Purified NaK protein was concentrated to 10 mg/mL for crystallization. Crystals were grown as hanging drops in 96 well plates by mixing 0.2 μ L protein solution with 0.2 μ L 50–80% MPD, 100 mM HEPES pH 7.0. Small crystalline slurries appeared in the drops after 3–5 days at room temperature. Proteinase K (*E. Album*) was purchased from Sigma and used without further purification. Protein was dissolved in 100 mM Tris–HCl pH 7.5 to a concentration of 20 mg/mL. Crystals of proteinase K were grown in sitting drops by combining 1 μ L protein solution with 1 μ L reservoir solution containing 1.25 M ammonium sulfate, 100 mM Tris–HCl pH 7.5. Experiments were conducted using Quantifoil Cu200 R2/1 or Cu300 R2/2 EM grids, clipped using c-rings and autogrid clips purchased from Thermo-Fisher. Blotting was conducted using a Thermo-Fisher Vitrobot Mark IV. Milling experiments were conducted using a Thermo-Fisher Aquilos dual beam FIB-SEM. MicroED data and TEM images were collected using either a Thermo-Fisher Talos Arctica or TF-20 TEM at cryogenic temperatures, as described [6, 16, 46, 47].

List of critical supplies and equipment:

1. Membrane protein crystals.
2. TEM grids (any).
3. Plunge freezer for vitrification of samples (any).
4. Cryo-TEM with stage capable of continuous rotation along a single axis with a suitably fast camera for accurately reading out collected data.
5. (Optional) Cryo-FIB/SEM for milling protein crystals along with a suitable cryo-transfer stage to shuttle the grids safely between the FIB/SEM and the TEM
6. Personal computer for processing data (any).

3 Methods

3.1 Preparing the Grids

1. Wrap a glass cover slide in Parafilm.
2. Place EM grids with their carbon sides facing up on the paraffin covered glass cover slide (*see Note 1*).
3. Glow-discharge for 15 s using a current of 20 mA and negative polarity.
4. Remove glass cover slide and place near Vitrobot.

3.2 Blotting

3.2.1 Blotting Conditions for the Standard MicroED Workflow

1. Set Vitrobot to 4–8 °C and 80–100% humidity at least 20 min before you plan to begin blotting.
2. Load the Vitrobot with fresh filter papers on both sides.
3. Pick up a grid by its edge with mountable tweezers and lock them in place.
4. Load the tweezers into the Vitrobot using either the touch screen or the foot pedal.
5. Plan out a wide range of blot times and forces to screen for ideal conditions, for example 4 s at force 4–16 s at 16 force over 4 grids.
6. Set the number of blots to one and do not skip application.
7. Once grid is in the application position, open the side of the Vitrobot and pipette ~1–3 µL of crystal slurry onto the carbon side of the glow-discharged EM grid (*see Note 2*).
8. Allow liquid to incubate on the carbon side of the grid for approximately 30 s (Fig. 1).
9. Blot the grid.
10. Plunge the grid into liquid ethane.
11. Transfer the grid to a grid box in liquid nitrogen.
12. Store grids in a large LN2 dewar prior to use.

3.2.2 Blotting Conditions for the CryoFIB Workflow

1. Set Vitrobot to 4–8 °C and 80–100% humidity at least 20 min before you plan to begin blotting.
2. Cut long strips of filter paper ~10–20 mm wide from Whatman #1 discs.
3. Pick up a grid by its edge with mountable tweezers and lock them in place.
4. Place tweezers inside Vitrobot.
5. Select number of blots to zero, and do not skip application.
6. Once grid is in the application position, open the side of the Vitrobot and pipette ~3 µL of crystal slurry onto the carbon side of the glow-discharged EM grid.
7. Allow liquid to incubate on the carbon side of the grid for approximately 30 s.
8. Secure a long strip of filter paper into a bow shape at the end of a large pair of forceps (Fig. 2).
9. Gently touch the copper side of the EM grid with the filter paper.
10. Wait until the filter paper wets, trying not to drag the paper against the grid.
11. When the wet spot stops growing, remove the forceps and filter paper, closing the opening to the chamber.



Fig. 1 Vitrobot with loaded tweezers incubating with sample prior to plunge freezing

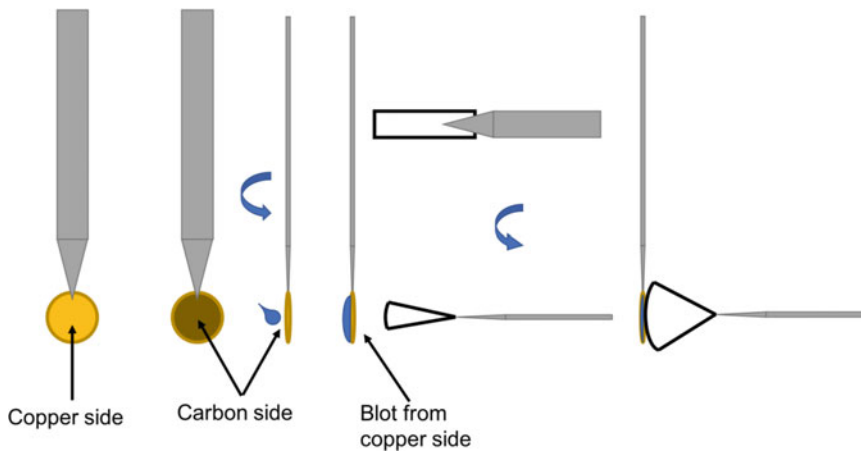


Fig. 2 Schematic diagram of blotting grids for FIB milling inside the Vitrobot. Blotting is done by gently touching the back (copper side) of the EM grid using strips of filter paper loaded from the side of the Vitrobot. The Vitrobot is kept at constant settings of 4 °C and ~100% humidity

12. Plunge freeze the grid into the liquid ethane.
13. Transfer the grid to a grid box in the liquid nitrogen.
14. Repeat for as many grids as needed.
15. Store grids in LN2 dewar.

3.3 (Optional) **Transfer to FIB/SEM**

1. Cool down FIB/SEM transfer station to LN2 temperature.
2. Remove grids from LN2 dewar, and place in transfer station.
3. Take out as many clips and c-rings as necessary for your experiments.

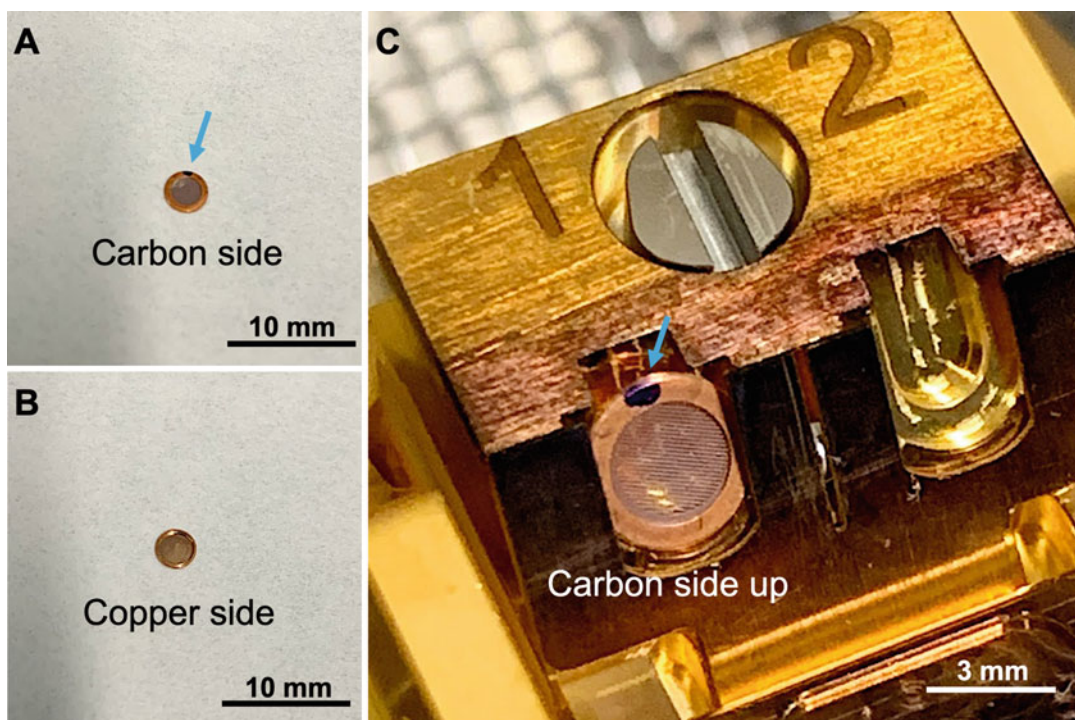


Fig. 3 (a) The correct clipping of a TEM grid for milling crystals, showing the carbon face flush with the flat face of the autogrid clip ring, and (b) the copper side of the grid facing the C-clip side of the autogrid clip. (c) Correct loading of the clipped TEM grid into the cryoFIB shuttle for a Thermo-Fisher Aquilos FIB-SEM dual beam instrument. The crystals are on the carbon face of the TEM grid, facing toward the observer. The blue dot is aligned with the milling direction of the gallium beam. The blue dot is indicated by a blue arrow in either panel

4. With a sharpie or permanent marker, place a small dot on the outside rim of the flat clip side (Fig. 3a) (*see Note 3*).
5. Clip the grids making sure the carbon side of the grid is on the flat side of the clip ring, with the copper side facing the C-ring (Fig. 3b).
6. Load the clipped EM grids into the cryo-FIB transfer shuttle with the carbon side of the grid facing up, and the marked dot aligned to the top or bottom (Fig. 3c).
7. Pump down the transfer rod, and load the cryo-FIB shuttle into the FIB/SEM.

3.4 (Optional) FIB Milling Crystals

1. In the mapping position, take a high-resolution image of each grid to check if they are of acceptable quality for milling (Fig. 4). This image should use a low magnification of 60–100 $\times$, an accelerating voltage of 5 kV or less, and a dwell time of 1 μ s or less. Resolutions of 1k–4k are acceptable.

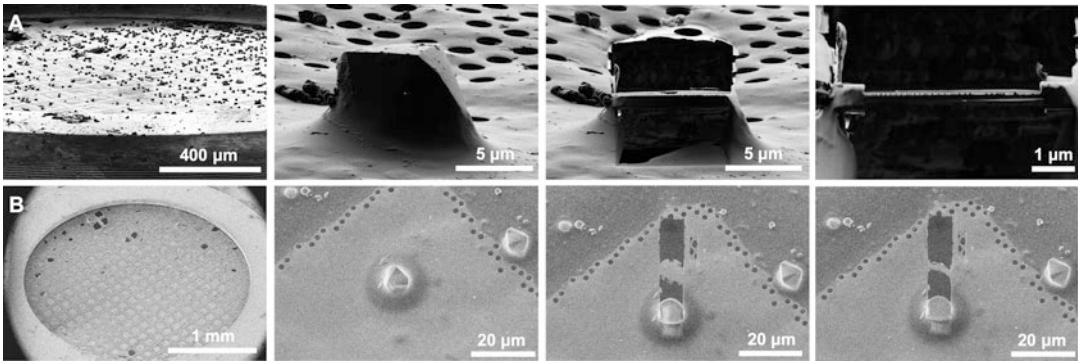


Fig. 4 The cryo-FIB/SEM workflow on microcrystals. (a) Images taken during the milling process by the ion beam, and (b) images taken by the SEM beam at similar stages. From left to right: overview of grid at low magnification, identification of protein crystal embedded in solvent, initial rough-milling of the crystal lamella, and final polished lamella after final milling step

2. Visually assess the quality of each grid from this initial image. Good grids will have a visible distinction between the carbon film and the grid bars with visible holes through the vitreous ice layer. Crystals should be on the carbon area, not within 5–10 μm of the grid bars. If grids are not badly damaged and visible crystals are in the grid square areas, move the FIB-shuttle to the sputtering position.
3. Sputter coat with 10–100 nm of fine platinum. Optionally, coat with an additional layer of carbon rich platinum using the gas injection system (GIS) for extra protection from the damaging gallium beam.
4. Recover the instrument from sputtering and move the grid into the milling position. Alternatively, the MAPS software (Thermo-Fisher) can be used to take a high magnification montage of the grid at mapping position. Crystals can be identified in this montage similar to viewing an all-grid atlas in standard TEM software. Here, sites can be added and aligned ahead of time, to increase the throughput of milling crystal lamellae.
5. Change the tilt angle from 18° to 0° .
6. Move about the grid looking for sharp edges of crystals, or dunes of solvent-covered crystals. A good crystal is near the center of the grid square as close to the center of the grid as possible.
7. Center the crystal in the SEM image. Tilt the sample to 18° and adjust the stage Z height to move the crystal back into the center of the SEM image. Link the SEM/FIB Z height.
8. Take images at increasing magnifications using the FIB to locate and center the crystal in the FIB image. Increase the

magnification until the crystal is just entirely visible in the FIB image. A well-calibrated stage will have the crystal centered in both the SEM and FIB images.

9. Draw a box in the FIB image of the crystal that covers from the crystal's center of mass to the top-most point of the crystal. Horizontally, the box should extend to cover all of the potential areas of the crystal. Change the mode from clearing to cleaning cross-section and change the direction of milling from bottom to top, to top to bottom. The goal is to slowly remove the crystalline material and have it leave this area. Exposing an area constantly rather than rastering away material can cause additional damage to the specimen.
10. Draw a second box that vertically extends from the carbon film below the crystal to $\sim 1\text{--}5\text{ }\mu\text{m}$ below the first box. This box should leave at least $1\text{--}5\text{ }\mu\text{m}$ on either side of the crystal or solvent. This extra space is necessary to hold the lamella and support the specimen after the material has been milled away. Change the milling mode from clearing to cleaning cross section, and assure this area is cleaning from bottom to top. This area will be removed and jettisoned below the crystal after the carbon film has been broken open.
11. For both boxes, roughly approximate the z thickness of the specimen using the SEM image. Ideally, the amount of milling time required for a specific thickness of each crystal would be calibrated for your instrument. This is difficult to ascertain for every possible sample and solvent type. A rough approximation would be to set the milling profile to Silicone, and divide the actual thickness of the specimen by 2.5. This is a rough approximation made by scaling the density of solid silicone to that of solid water ice (2.32 g/cm^3 to $0.9\text{--}1.0\text{ g/cm}^3$). This will give relatively accurate milling times for protein crystals. The time estimate the system gives should inform the ion beam current you use for the first few steps of milling. The first one or two steps are usually done using currents of $\sim 500\text{--}100\text{ pA}$, depending on the size of the crystals, and each set of milling takes approximately 10 min.
12. Mill these two boxes using the ion beam.
13. Switch the ion beam current back to $1\text{--}10\text{ pA}$ and take both an SEM and a FIB image (Fig. 4). Inspect the lamellae in both images to assess the success of the rough-milling (Fig. 4).
14. In the Aquilos software, change the distance between the two boxes from $1\text{--}1.5\text{ }\mu\text{m}$, to 500 nm . Adjust the size of both drawn boxes such that, vertically, they just encompass the crystal lamella. Horizontally, bring in the boxes by a small amount, perhaps 100 nm . Change the ion beam current to $10\text{--}100\text{ pA}$.

15. Mill these two boxes using the ion beam.
16. Repeat **step 13**, and then **step 14** changing the distance between boxes to 300 nm and the current to 10 pA.
17. These general steps can be broken down into smaller changes and additional steps down in ion beam current. However, we have found that the platinum coated crystals are more sensitive to the final stages of milling (polishing) than the rough first few stages.
18. Repeat **steps 6–18** for all the desired crystals on each grid. We recommend doing 4–6 per grid when operating manually. The contamination rate in the Aquilos is typically 10–30 nm/h of redeposited ice contamination. A diligent operator can mill six lamellae in 2 h. Alleviating the contamination can be accomplished by rough-milling all the lamellae on a grid, and then going back for the final low-current polishing step for each crystal after (*see Note 4*).
19. Remove the cryoFIB shuttle from the FIB/SEM.
20. Load the shuttle into a LN2 cooled sample loading station. Move the grids from the shuttle to a grid box.
21. Store the clipped grids with milled lamellae in a dewar until it is time to transfer them into a cryo-cooled TEM.

3.5 Transfer Grids into the TEM

1. Remove grids from dewar and load into autoloader cassette. For traditional grids, the rotational orientation of the grid does not matter. For milled grids, the blue dot should be rotated 90° in either direction from where it will be held by the grid transfer forceps. When the grid is loaded into the cassette, you should not be able to see the blue dot (Fig. 5).
2. Insert cassette into autoloader. Perform an inventory of the grids. Load the grid into the TEM column.
3. Identify crystals using low-dose search mode calibrated to either over-focused diffraction mode or low magnification imaging (~155×). Crystals typically appear as black, sharp-edged shapes in over-focused diffraction mode. In low-dose imaging, crystals will appear semitransparent with similar shape (Fig. 6). In milled grids, lamellae sites appear as long strips of empty space with material suspended between the edges (Fig. 4).
4. Using either over-focused diffraction or ~900× LM imaging mode, adjust the stage to the eucentric height using the wobbler. Image the crystal identified in **step 3** to inspect the specimen. In over-focused diffraction mode, the difference in contrast between the crystalline and amorphous area of the crystals is apparent for lamellae (Fig. 4).

5. Insert and center the SA aperture to assure data is collected only from the desired area of the crystal. Insert the beam stop.
6. Switch to parallel beam diffraction mode, assuring the beam is well aligned and the direct beam is behind the beam stop.
7. Take a single diffraction image to assess the resolution and quality of the crystal lamellae (Fig. 5).

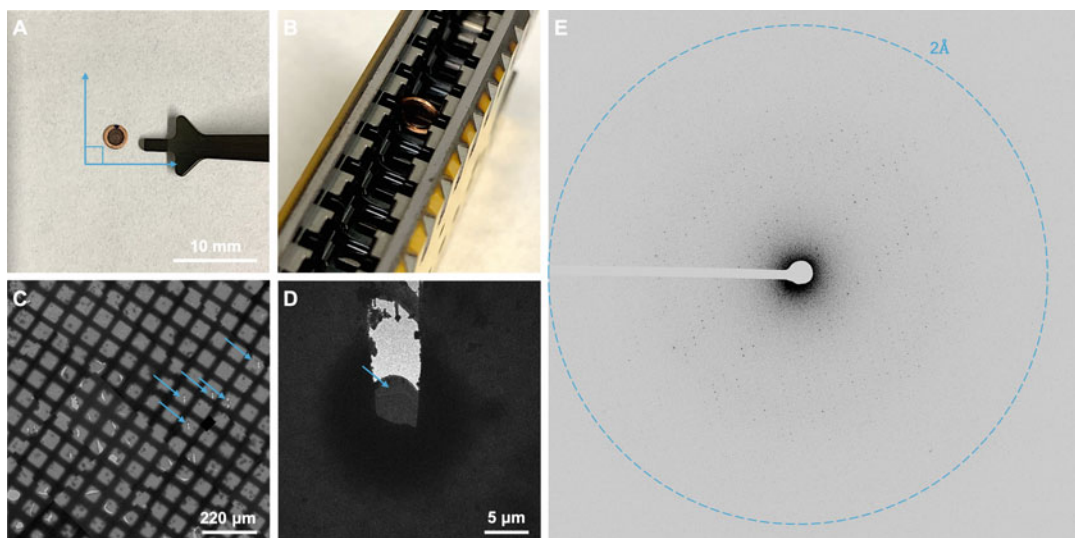


Fig. 5 Milled crystals in the TEM. (a) Loading of the clipped grid into the autoloader cassette at 90° relative to the loading of the cryoFIB shuttle. (b) Low-magnification grid atlas. Milled lamellae are visible as strips missing grid carbon and marked by blue arrows. (c) Over-focused diffraction image of the crystal milled in Fig. 2. (d) Diffraction pattern extending to ~2 Å resolution from this lamella

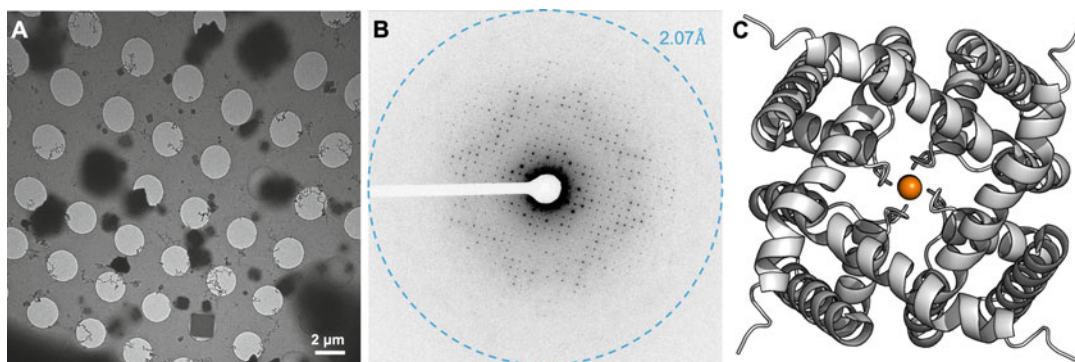


Fig. 6 Structure determination of the ion channel NaK. (a) Over-focused diffraction image of square NaK crystals with visible, sharp edges on a Quantifoil Cu300 R2/2 grid. Crystal slurries were applied directly to the grid and blotted in the Vitrobot. (b) Typical MicroED frame from a continuous rotation movie from a NaK microcrystal. (c) Final structure solution of NaK (PDB-6CPV) looking down the fourfold axis, with sodium atoms colored in orange and cartoon ribbons of the protein in gray

3.6 Collection of MicroED Data from Crystals

1. Data are collected as described. Briefly, a wedge between 30° and 120° is selected for collection and aligned to the eucentric height in either imaging or over-focused diffraction modes.
2. The camera and stage are set such that the stage begins rotating at a constant rate and the camera reads out frames at regular intervals of time. Typically, this is a rotation rate of $0.1\text{--}1.0^\circ/\text{s}$ with frames being read out every 1–5 s, aiming for 0.5° wedges per frame for most proteins (Fig. 6).
3. (Optional) MicroED data may be improved by using a zero-loss energy filter to minimize contributions from inelastic scattering (Fig. 7).
4. Data are converted using the MicroED tools available at cryoEM.ucla.edu from TVIPS, MRC, or SER to SMV format [20, 29].
5. Data processing from this stage follows the same procedures as X-ray crystallography data reduction pipelines. We recommend using XDS [48], iMOSFLM [49, 50], or DIALS [51, 52] to reduce the data, and then SHELXT [53] SHELXD [54], phaser [55], or MOLREP [56] to phase the data, depending on resolution. Structures can be solved using ab initio direct methods or molecular replacement depending on resolution [6, 57]. Recent MicroED studies have also demonstrated ab initio phasing by radiation damage [58].
6. Structure refinement and model building similarly follow X-ray crystallographic methods as described [14, 20].

4 Notes

1. The copper side of the grid appears visibly shiny with an orange hue, whereas the carbon side appears darker. Grids shipped in Quantifoil grid boxes typically have their carbon sides all facing the center of the box.
2. Large or visibly single crystals can also be used, but must be fragmented into smaller shards suitable for MicroED investigations [16, 20]. This is typically done by either pipetting the crystals repeatedly or sonicating the solution as described.
3. The blue dot is meant to identify the direction in which the grids were milled. There are also commercial autogrids available from Thermo-Fisher that include a dot, or thinned clipping areas that accomplish a similar goal. In our case, we rotate the grid to assure the TEM tilt axis is perpendicular to the milling direction. We do this because there are typically large areas of crystal remaining that occlude the view of the lamellae while tilting to high angle. Rotation to a position perpendicular to the milling direction allows us to achieve the largest possible tilt range for MicroED data collection.

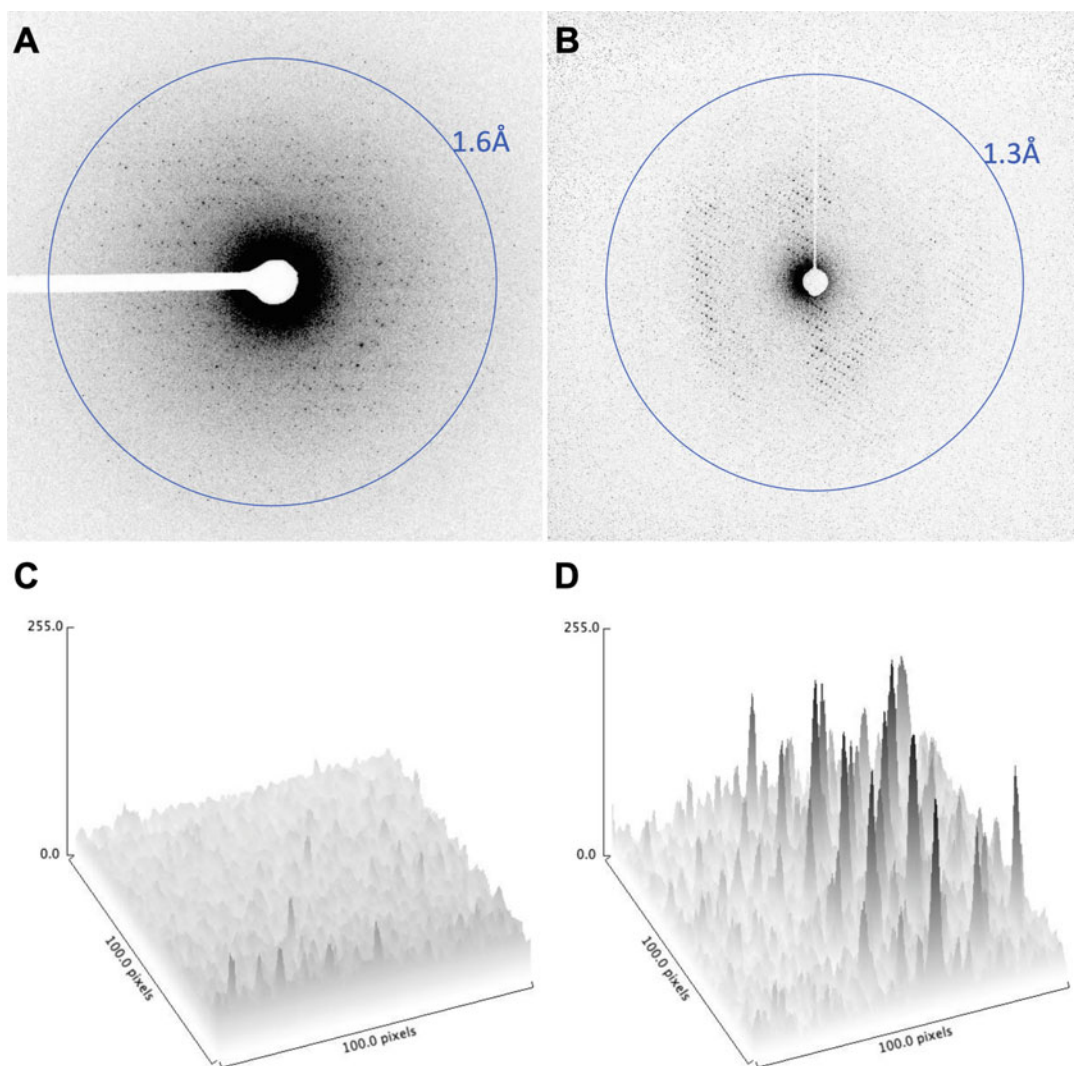


Fig. 7 Comparison of MicroED data collection with and without an energy filter. (a) Proteinase K MicroED data collected on a Thermo-Fisher F-20 at 200 keV without an energy filter, and (b) MicroED data from proteinase K collected on a JEOL 3200-FSC at 300 keV with an in-column omega energy filter. (c) Plot of 100×100 pixels around 1.6 Å resolution from (a), and a 100×100 pixel selection at 1.6 Å resolution from (d) showing the difference in measured intensities. Energy-filtered MicroED data were collected using an in-column Omega energy filter operating with a slit-width of 20 eV

4. We typically use software such as MAPS (Thermo-Fisher) to save the eucentric position of each lamella prior to milling. This allows milling in batch, where the rough-milling of all crystals is done in series, followed by fine-milling of each, and finally the polishing steps. This results in overall reduced ice contamination to the lamellae than milling each lamella completely before moving on to the next position.

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References

1. Gonen T, Cheng Y, Sliz P et al (2005) Lipid-protein interactions in double-layered two-dimensional AQP0 crystals. *Nature* 438:633–638
2. Wisedchaisri G, Gonen T (2013) Phasing electron diffraction data by molecular replacement: strategy for structure determination and refinement. *Methods Mol Biol* 955:243–272
3. Oshima A, Tani K, Hiroaki Y et al (2007) Three-dimensional structure of a human connexin26 gap junction channel reveals a plug in the vestibule. *Proc Natl Acad Sci* 104:10034–10039
4. Yonekura K, Kato K, Ogasawara M et al (2015) Electron crystallography of ultrathin 3D protein crystals: atomic model with charges. *Proc Natl Acad Sci* 112:3368–3373
5. Liu S, Gonen T (2018) MicroED structure of the NaK ion channel reveals a Na⁺ partition process into the selectivity filter. *Commun Biol* 1:38
6. Nannenga BL, Shi D, Leslie AGW et al (2014) High-resolution structure determination by continuous-rotation data collection in MicroED. *Nat Methods* 11:927–930
7. MicroED structure of lipid-embedded mammalian mitochondrial voltage-dependent anion channel <https://doi.org/10.1073/pnas.2020010117>
8. MicroED structure of the human adenosine receptor determined from a single nanocrystal in LCP <https://doi.org/10.1101/2020.09.27.316109>
9. Arnal F, Balladore JL, Soum G et al (1976) Calculation of the cross sections of electron interaction with matter. *Ultramicroscopy* 2:305–310
10. Shi D, Nannenga BL, Iadanza MG et al (2013) Three-dimensional electron crystallography of protein microcrystals. *elife* 2013:e01345
11. Nannenga BL, Shi D, Hattne J et al (2014) Structure of catalase determined by MicroED. *elife* 3:e03600
12. Hattne J, Shi D, Glynn C et al (2018) Analysis of global and site-specific radiation damage in cryo-EM. *Structure* 26:759–766.e4
13. Gallagher-Jones M, Glynn C, Boyer DR et al (2018) Sub-ångström cryo-EM structure of a prion protofibril reveals a polar clasp. *Nat Struct Mol Biol* 25(2):131–134
14. Jones CG, Martynowycz MW, Hattne J et al (2018) The cryoEM method microED as a powerful tool for small molecule structure determination. *ACS Cent Sci* 4:1587–1592
15. Vergara S, Lukes DA, Martynowycz MW et al (2017) MicroED structure of Au₁₄₆ (p-MBA)₅₇ at subatomic resolution reveals a twinned FCC cluster. *J Phys Chem Lett* 8:5523–5530
16. de l CMJ, Hattne J, Shi D et al (2017) Atomic-resolution structures from fragmented protein crystals with the cryoEM method MicroED. *Nat Methods* 14:399–402
17. Clabbers MTB, van GE, Wan W et al (2017) Protein structure determination by electron diffraction using a single three-dimensional nanocrystal. *Acta Crystallogr D Struct Biol* 73:738–748
18. Van GE, Clabbers MTB, Das PP et al (2016) Ab initio structure determination of nanocrystals of organic pharmaceutical compounds by electron diffraction at room temperature using a Timepix quantum area direct electron detector. *Acta Crystallogr A Found Adv* 72:236–242
19. Shi D, Nannenga BL, de l CMJ et al (2016) The collection of MicroED data for macromolecular crystallography. *Nat Protoc* 11:895–904
20. Hattne J, Reyes FE, Nannenga BL et al (2015) MicroED data collection and processing. *Acta Crystallogr A Found Adv* 71:353–360
21. Caffrey M (2003) Membrane protein crystallization. *J Struct Biol* 142:108–132
22. Petsko GA (1975) Protein crystallography at sub-zero temperatures: cryo-protective mother liquors for protein crystals. *J Mol Biol* 96:381–392
23. Kriminski S, Kazmierczak M, Thorne RE (2003) Heat transfer from protein crystals: implications for flash-cooling and X-ray beam heating. *Acta Crystallogr D Biol Crystallogr* 59:697–708

24. Marko M, Hsieh C, Schalek R et al (2007) Focused-ion-beam thinning of frozen-hydrated biological specimens for cryo-electron microscopy. *Nat Methods* 4:215–217
25. Rigort A, Bauerlein FJB, Villa E et al (2012) Focused ion beam micromachining of eukaryotic cells for cryoelectron tomography. *Proc Natl Acad Sci* 109:4449–4454
26. Schertel A, Snaidero N, Han H-M et al (2013) Cryo FIB-SEM: volume imaging of cellular ultrastructure in native frozen specimens. *J Struct Biol* 184:355–360
27. Giannuzzi LA, Stevie FA (1999) A review of focused ion beam milling techniques for TEM specimen preparation. *Micron* 30:197–204
28. Sergey G, Denis K, Ava H et al (2018) Oxygen plasma focused ion beam scanning electron microscopy for biological samples. *bioRxiv* 2018:457820. <https://doi.org/10.1101/457820>
29. Martynowycz MW, Zhao W, Hattne J et al (2019) Collection of continuous rotation microED data from ion beam-milled crystals of any size. *Structure* 27:545–548.e2
30. Martynowycz MW, Zhao W, Hattne J et al (2019) Qualitative analyses of polishing and precoating FIB milled crystals for microED. *Structure* 27:1594–1600.e2
31. Duyvesteyn HME, Kotecha A, Ginn HM et al (2018) Machining protein microcrystals for structure determination by electron diffraction. *Proc Natl Acad Sci* 115:9569–9573
32. Li X, Zhang S, Zhang J et al (2018) In situ protein micro-crystal fabrication by cryo-FIB for electron diffraction. *Biophys Rep* 4:339–347
33. Zhou H, Luo Z, Li X (2019) Using focus ion beam to prepare crystal lamella for electron diffraction. *J Struct Biol* 205(3):59–64
34. Martynowycz M, Glynn C, Miao J, et al (2017) MicroED structures from micrometer thick protein crystals. <https://doi.org/10.1101/152504>
35. Latychevskaia T, Abrahams JP (2019) Inelastic scattering and solvent scattering reduce dynamical diffraction in biological crystals. *Acta Crystallogr B Struct Sci Cryst Eng Mater* 75:523–531
36. Clabbers MTB, Abrahams JP (2018) Electron diffraction and three-dimensional crystallography for structural biology. *Crystallogr Rev* 24:176–204
37. Clabbers MTB, Gruene T, van Genderen E et al (2019) Reducing dynamical electron scattering reveals hydrogen atoms. *Acta Crystallogr A Found Adv* 75:82–93
38. Flot D, Mairs T, Giraud T et al (2010) The ID23-2 structural biology microfocus beamline at the ESRF. *J Synchrotron Radiat* 17:107–118
39. Palatinus L, Brázda P, Boullay P et al (2017) Hydrogen positions in single nanocrystals revealed by electron diffraction. *Science* 355:166–169
40. Brázda P, Palatinus L, Babor M (2019) Electron diffraction determines molecular absolute configuration in a pharmaceutical nanocrystal. *Science* 364:667–669
41. Gruene T, Wennmacher JTC, Zaubitzer C et al (2018) Rapid structure determination of microcrystalline molecular compounds using electron diffraction. *Angew Chem Int Ed* 57:16313–16317
42. Spurlino JC, Lu GY, Quirocho FA (1991) The 2.3-Å resolution structure of the maltose- or maltodextrin-binding protein, a primary receptor of bacterial active transport and chemotaxis. *J Biol Chem* 266:5202–5219
43. Yonekura K, Maki-Yonekura S (2016) Refinement of cryo-EM structures using scattering factors of charged atoms. *J Appl Crystallogr* 49:1517–1523
44. Alam A, Jiang Y (2009) Structural analysis of ion selectivity in the NaK channel. *Nat Struct Mol Biol* 16:35–41
45. Alam A, Jiang Y (2009) High-resolution structure of the open NaK channel. *Nat Struct Mol Biol* 16:30–34
46. Rodriguez JA, Gonen T (2016) High-resolution macromolecular structure determination by microED, a cryo-EM method. Elsevier, Inc.
47. Nannenga BL, Gonen T (2014) Protein structure determination by microED. *Curr Opin Struct Biol* 27:24–31
48. Kabsch W (2010) Xds. *Acta Crystallogr D Biol Crystallogr* 66:125–132
49. Leslie AGW, Powell HR (2007) Processing diffraction data with mosflm. https://doi.org/10.1007/978-1-4020-6316-9_4
50. Battye TGG, Kontogiannis L, Johnson O et al (2011) iMOSFLM: A new graphical interface for diffraction-image processing with MOSFLM. *Acta Crystallogr D Biol Crystallogr* 67:271–281
51. Winter G, Waterman DG, Parkhurst JM et al (2018) DIALS: implementation and evaluation of a new integration package. *Acta Crystallogr D Struct Biol* 74:85–97

52. Clabbers MTB, Gruene T, Parkhurst JM et al (2018) Electron diffraction data processing with DIALS. *Acta Crystallogr D Struct Biol* 74:506–518
53. Sheldrick GM (2015) SHELXT – integrated space-group and crystal-structure determination. *Acta Crystallogr A Found Crystallogr* 71:3–8
54. Schneider TR, Sheldrick GM (2002) Substructure solution with SHELXD. *Acta Crystallogr D Biol Crystallogr* 58:1772–1779
55. McCoy AJ, Grosse-Kunstleve RW, Adams PD et al (2007) Phaser crystallographic software. *J Appl Crystallogr* 40:658–674
56. Vagin A, Teplyakov A (1997) MOLREP: an automated program for molecular replacement. *J Appl Crystallogr* 30:1022–1025
57. Sawaya MR, Rodriguez J, Cascio D et al (2016) Ab initio structure determination from prion nanocrystals at atomic resolution by microED. *Proc Natl Acad Sci* 113:11232–11236
58. Martynowycz MW, Hattne J, Gonen T (2020) Experimental phasing of microED data using radiation damage. *Structure* 28:458–464.e2



Single-Particle Cryo-EM of Membrane Proteins

Dovile Janulienė and Arne Moeller

Abstract

In the recent years, the protein databank has been fueled by the exponential growth of high-resolution electron cryo-microscopy (cryo-EM) structures. This trend will be further accelerated through the continuous software and method developments and the increasing availability of imaging centers, which will open cryo-EM to a wide array of researchers with their diverse scientific goals and questions. Especially for structural biology of membrane proteins, cryo-EM offers significant advantages as it can overcome multiple limitations of classical methods. Most importantly, in cryo-EM, the sample is prepared as a vitrified suspension, which abolishes the need for crystallization, reduces the required sample amount and allows usage of a wide arsenal of hydrophobic environments. Despite recent improvements, high-resolution cryo-EM still poses some significant challenges, and standardized procedures, especially for the characterization of membrane proteins, are missing. While there can be no ultimate recipe toward a high-resolution cryo-EM structure for every membrane protein, certain factors seem to be universally relevant. Here, we share the protocols that have been successfully used in our laboratory. We hope that this may be a useful resource to other researchers in the field and may increase their chances of success.

Key words Cryo-EM, Negative-stain, Single-particle analysis, Membrane proteins, Microscopy, structural biology

1 Introduction

Single-particle cryo-EM has become the method of choice in many structural biology laboratories, especially, in the field of membrane proteins. This is highlighted by the appearance of many seminal structures of proteins that in the past had stubbornly resisted crystallization, as for example the TRP channels [1, 2]. Cryo-EM is also advantageous to tackle dynamic or flexible proteins and multimeric assemblies, like GPCR signaling complexes [3–5]. Adding to the versatility, the small sample amount, required for the experiment, further enables structural characterization of low-abundant targets or those that can only be purified from endogenous sources [6]. Even for membrane proteins, which are attainable by crystallography, for instance, several members of the

respiratory chain [7–13], cryo-EM seems to be advantageous, providing easy access to multimeric complexes and the possibility to characterize dynamics [14].

Compared to diffraction-based methods, sample preparation for cryo-EM is less demanding, as crystallization is not needed and overall sample amounts are significantly reduced. Furthermore, cryo-EM tolerates samples, embedded in many different hydrophobic environments, including various detergents [15], lipid nanodiscs [16], saposin-lipoprotein nanoparticles [17], peptidiscs [18], amphipols [19], beta-peptides [20] or styrene maleic acid lipid particles [21]. Accordingly, the solubilizing agent can be tailored to the individual specimen and the specific goal, which highly increases the chances of success, especially for challenging targets. From all the possible options, lipid nanodiscs seem to be a popular choice as they provide a lipid environment, high contrast and are flexible in their application. Finally, due to sophisticated algorithms [22–28], cryo-EM can tolerate some heterogeneity and certain degrees of flexibility, enabling analysis of conformational variability [29].

Due to the low signal-to-noise ratio in electron micrographs, one of the major limitations of cryo-EM is the size of the sample. While it has been long predicted that targets below 50 kDa can be analyzed in cryo-EM [30], only recently, structures in this MW range have been solved for soluble proteins [31, 32]. Complexities, inferred by the intrinsic properties of membrane proteins and their requirement for a hydrophobic environment, seem to currently additionally limit the attainable minimum molecular mass for membrane protein targets. However, utilizing antibodies [33] and nanobody technology [34] to increase mass and structural features, the structure of a 49 kDa membrane protein has recently been solved at a high resolution [35].

In cryo-EM, the current state of the art is structures at resolutions which allow de novo model building. For well-behaved soluble samples, streamlined workflows are often applicable and resolutions better than 2 Å have been achieved [36–41]. Future developments in sample preparation, imaging, and single-particle processing will further improve the attainable resolution and hold great promises for membrane proteins as well [42].

1.1 Sample Optimization in Negative-Stain EM

A typical cryo-EM experiment begins with the sample optimization (Fig. 1), which remains the major bottleneck toward high-resolution. While single-particle analysis can deal with some heterogeneity and sample flexibility, high-resolution studies require high quality samples. In conjunction with classical biophysical approaches like thermostability measurements [43–45], fluorescent size exclusion chromatography [46], blue native PAGE [47], and dynamic light scattering [48], high-throughput negative-stain EM offers a priceless direct view on the sample [20, 49]. Negative-stain

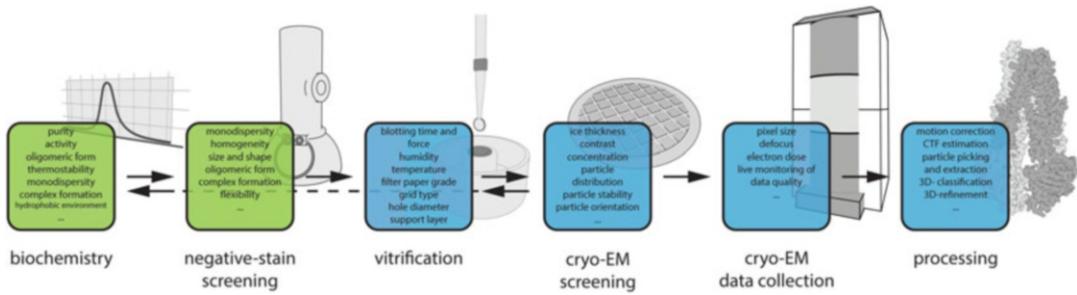


Fig. 1 Typical cryo-EM workflow. The experiment begins with sample preparation and optimization, reflected by interconnected and iterative steps of biochemistry and screening in negative-stain EM (marked in green). Once suitable quality is confirmed by negative-stain EM, the experiment continues with the actual cryo-EM steps (marked in blue). Screening in cryo-EM is another iterative process in conjunction with adjustment of the vitrification parameters. To counteract issues, like insufficient particle concentration, preferred orientation or dissociation, it is often necessary to revisit the initial biochemistry steps, as indicated by the dashed arrow. The main parameters, which require assessment and those that affect the final result, are listed in the colored boxes for each step

EM provides a quick, simple, and efficient way of screening for sample homogeneity, complex formation, effects of various buffer compositions and different hydrophobic environments [49]. However, when working with membrane proteins, empty detergent micelles can cause significant background in negative-stain and, when not carefully analyzed, can be mistaken for small detergent-embedded proteins [50]. Nevertheless, negative-stain EM provides a highly efficient way to identify and characterize low-abundance targets as it requires only minimal amounts of sample at low concentrations.

For negative-stain EM, the sample is absorbed onto a continuous support layer, typically, carbon, deposited on a copper object carrier (grid) (Fig. 2f). After blotting to a thin film, the adsorbed particles are covered with a layer of heavy metal solution. This provides high contrast between the protein particles and the surrounding electron-dense heavy metal stain, resulting in micrographs with white particles on a dark background. The quality of the result is heavily dependent on the grain size of the staining agent. Therefore, best images can be obtained using uranyl formate. In practice, negative-stain EM provides information about the overall shape of the particle and its assembly, possibly at domain resolution, and can deliver insight into conformational dynamics or reactions to specific ligands if the conformational changes are large enough [51].

1.2 Cryo-EM Single-Particle Analysis

In contrast to negative stain EM, cryo-EM allows for imaging of hydrated biological samples. Typically, the specimen is prepared by applying the protein solution onto a holey carbon or gold grid [52], followed by blotting to achieve a thin layer and then plunging

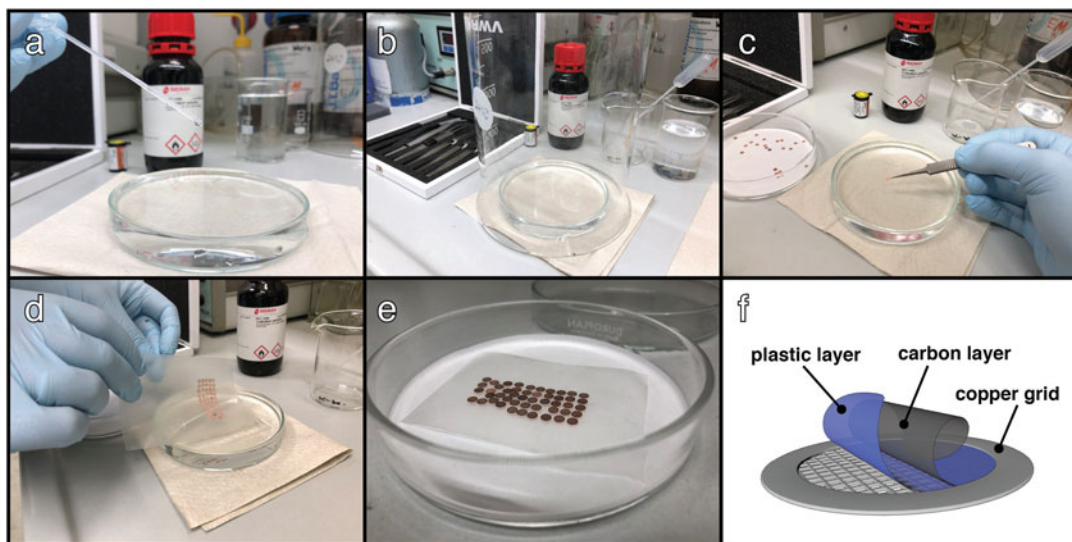


Fig. 2 Preparation of plastic-coated grids. (a) In the fume hood, apply the collodion solution on a flat water surface. (b) Cover the collodion surface with a large beaker to protect it from wrinkles, which may arise from air circulation. (c) Place the copper grids on top of the solidified collodion. (d) Transfer the grids with the collodion layer onto Parafilm. (e) Allow the plastic-coated grids to air-dry. (f) Anatomy of the plastic-coated grid after carbon deposition. For the detailed protocol, see Subheading 2.1, item 7

into liquid ethane for rapid vitrification. To increase the contrast between the target and the surrounding solution, the vitrified suspension should be as thin and inert as possible. Ideally, the embedded particles should exhibit random orientations to enable isotropic back-projection into 3D space. However, this is rarely the case and most of the time the investigated particles exhibit some preferred views. While this is not always problematic, pathological cases, where only a single or a few very similar views are represented, are frequently encountered and pose a major obstacle during sample optimization. Collecting tilted data can increase angular distribution to some extent [53], but, in general, treating the preferred orientation problem by shielding the sample from the air–water interface is the better option, though much harder to achieve. During the standard procedure of cryo-EM specimen preparation, typically involving blotting, the particles interact with the air–water interface [54], which can lead not only to preferred orientation but also to their denaturation [55]. Blotting is far from optimal in many aspects [56]. In addition to the air–water interface issue, shear forces may damage fragile samples and prolonged contact with the filter paper can cause sample aggregation or dissociation. As a countermeasure, new ways of sample preparation without the need of blotting are being developed [57–62]. They require much smaller volumes of the sample, but similar or even higher concentrations, that are not always easily obtainable for

fragile, low-abundance membrane protein complexes. Therefore, low-background support films, like graphene [55, 63–65], which enable imaging of low-concentration samples while protecting them from the air–water interface, offer an especially attractive remedy.

The potential of a cryo-EM dataset is dictated by the performance of the EM-setup and the quality of the cryo-EM specimen. Modern microscopes with direct electron detectors, automated data collection and microscope alignment software can typically ensure high-quality imaging, if curated well. Therefore, obtaining a high-quality cryo-EM specimen remains the biggest challenge and its preparation requires screening before high resolution data-collection. Screening is used to find optimal blotting conditions, as judged by ice thickness, and optimal sample concentration, as judged by particle distribution. Typically, it also involves collecting a test dataset to evaluate sample behavior, which for membrane proteins can be highly variable in different hydrophobic environments. As mentioned above, the visibility of individual particles strongly depends on the signal to noise ratio. Therefore, whenever possible, it is advisable to minimize the buffer components, ligands and detergents in a cryo-EM experiment to reduce the background noise. After the specimen is deemed suitable for high-resolution data collection, a large dataset is collected on a high-end electron microscope. Processing of the data of a membrane protein, generally, involves standard steps of single-particle analysis [66], starting with the alignment of the individual movie frames [67], followed by the estimation of the contrast transfer function (CTF) [68–70], particle picking [39, 71–75], and assembly into stacks. All these steps can be coupled with automated data collection and performed in real time [24, 76–79]. Such online monitoring provides a quick way to assess image quality and adjust the data collection strategy if needed.

Initial stacks usually contain large amounts of poor-quality particles and false positives, which have to be removed. Typically, 2D classification is used as the first round of particle cleaning; however, this step can often be skipped and cleaning can be performed directly in 3D. The level of heterogeneity within the sample determines the required size of the dataset, and can, therefore, severely impact the chances of success. Consequently, homogeneous samples are preferred for high-resolution structure determination. With the increasing speed of data acquisition and automation, in conjunction with software development, systems of higher complexity become more and more amenable to high-resolution studies and the analysis of conformational dynamics and heterogeneous samples is certainly going to be one of the main applications of cryo-EM in the near future.

Different single-particle analysis software [23–27, 80–86] are capable of achieving high-resolution structures and various cryo-EM sample preparation and data processing strategies have been shown to be successful. Here, we present our approach, which was used to obtain a series of high-resolution reconstructions of an ABC transporter [29], and is generally applied in our laboratory for structural studies of membrane proteins [87].

2 Materials

2.1

Negative-Stain EM

1. 120 keV electron microscope with CCD camera and automated data collection software.
2. Carbon evaporator.
3. Glow discharge system.
4. Regular and anti-capillary tweezers.
5. Filter paper (Whatman Grade 4).
6. 40 mm glass petri dish or microscope slide.
7. Carbon coated grids with plastic support (protocol adjusted from [88]): Fully fill a petri dish with double distilled water and ensure that the water surface is smooth and as flat as possible. In the fume hood, add a single drop of 2% collodion solution (in amyl acetate) on top of the water surface (Fig. 2a). Cover the petri dish with a large beaker to prevent surface disturbance due to air circulation and allow the collodion to solidify for around 10 min (Fig. 2b) until it forms an even film on top of the water. If it wrinkles, discard and repeat the procedure. Carefully place copper TEM grids (e.g., Gilder G-400) one by one on top of the plastic surface with the shiny, darker side of the grid facing down (Fig. 2c). With some practice it should be possible to deposit 100 single grids on the area of approximately a microscope slide. Using a piece of Parafilm, pick up the plastic foil with the grids (Fig. 2d) and air dry for at least 30 min (Fig. 2e). As soon as the grids are dry, coat them with a thin layer of carbon, using a carbon evaporator (the procedure depends on the specific instrument, so follow the instructions of the manufacturer). It is not advisable to delay the step of carbon coating and store uncoated grids as this may lead to formation of small holes in the final carbon surface. Carbon coated plastic grids can be stored in a desiccator cabinet for a long period of time, therefore, it is practical to make larger amounts.
8. 2% uranyl formate solution (protocol adjusted from [89]): Dissolve 0.1 g of uranyl formate in 5 ml degassed ultrapure H₂O (boiled and cooled to room temperature). Vortex for

10 min, add 10 μl of 10 N KOH and vortex again for 10 min. Spin the solution at maximum speed on a table top centrifuge for 10 min and filter the supernatant through a 0.2 μm filter. Make small, single-use aliquots (25–50 μl) of the uranyl formate stain, flash-freeze in liquid nitrogen and store at -80°C in the dark (*see Note 1*).

2.2 Cryo-EM

1. 200–300 keV electron microscope with direct electron detector and automated data collection software.
2. Grid clipping and cassette-loading station.
3. Tweezers and tools for grid clipping.
4. Oven for tool drying.
5. Storage boxes for cryo-EM grids.
6. Storage system for cryo-EM grid boxes.
7. Liquid nitrogen.
8. Liquid ethane.
9. Glow discharge system or plasma cleaner.
10. Plunge-freezing device (Vitrobot in this example).
11. Filter paper (Whatman Grade 595).
12. 40 mm glass petri dish or microscope slide.
13. Holey carbon or gold grids (C-flat or Quantifoil).

2.3 Single-Particle Analysis

1. Computer cluster or workstation.
2. Single-particle analysis software.

3 Methods

3.1 Sample Preparation for Cryo- and Negative-Stain EM

1. Purify the membrane protein in the detergent of choice or use a detergent free environment, for example, reconstitute the protein into nanodiscs. Confirm its purity and, if possible, activity (*see Note 2*).
2. Perform size-exclusion chromatography (SEC) to remove aggregates or to exchange the buffer. To reduce the background in cryo-EM, keep salt and detergent concentration in the buffer at minimum as long as it does not compromise sample stability. Avoid glycerol and other cryoprotectants in the sample.
3. Perform EM grid preparation immediately after SEC on the peak fraction. Whenever possible, avoid any further concentration of the sample (*see Note 3*).

3.2 Testing Sample Quality in Negative-Stain EM

1. Place a plastic-coated carbon grid (*see* Subheading 2.1, item 7) into a 40 mm glass petri dish (or on top of the microscope slide) with the carbon coated (darker) side of the grid facing up (Fig. 3a). Transfer the open petri dish to the glow discharge device and glow-discharge to make the grid hydrophilic (Fig. 3b, *see* Note 4).
2. Pick up the glow-discharged grid at its edge using anti-capillary tweezers and add 3 μ l of protein sample (Fig. 3c) at a concentration in the 100 nM range (*see* Note 5).
3. Blot the grid from one edge, using filter paper, and immediately add 3 μ l of uranyl formate solution (Fig. 3d). Repeat this step two more times (*see* Note 6), blotting always from the same side.
4. After the last blot (Fig. 3e), gently touch the grid with the edge of the filter paper at the point where it is held by the tweezers (Fig. 3f) to remove the last traces of the staining solution. Typically, no additional drying of the grid is required, but ensure that it is fully dry before inserting into the microscope.
5. Place the grid in the sample holder and insert it into the microscope.
6. At low-magnification ($\sim 500\times$) explore the grid for well-stained areas and examine particle distribution and appearance at a high magnification (Fig. 4, *see* Notes 6 and 7).

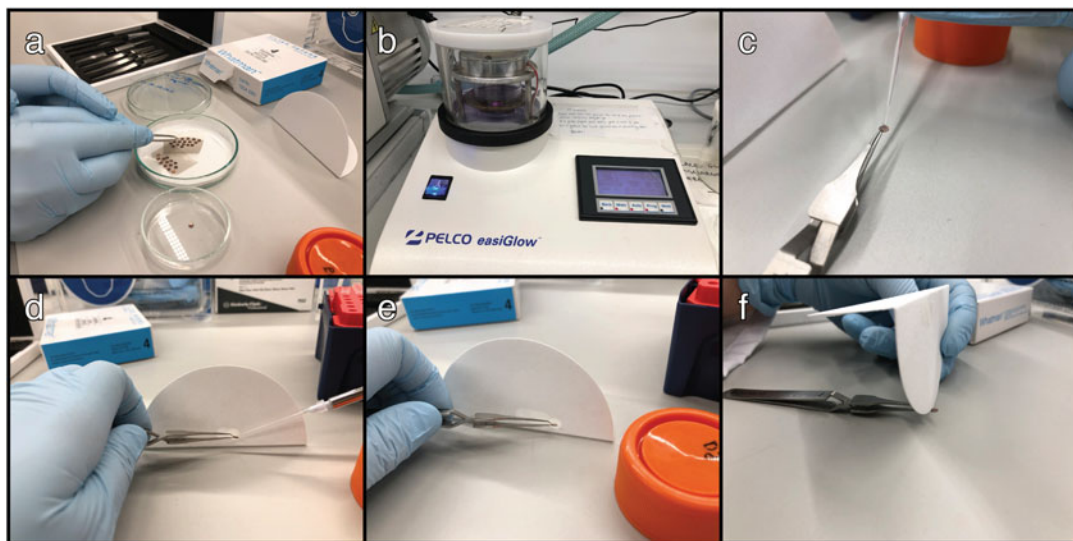


Fig. 3 Negative-stain sample preparation. (a) Transfer the carbon-coated plastic grids to a glass petri dish. (b) Glow-discharge the grids. (c) Apply the sample onto the grid. (d) Blot the sample and immediately apply the uranyl formate solution. (e) Blot the stain solution. (f) Use the edge of the filter to remove all of the remaining liquid. For details, *see* Subheading 3.2

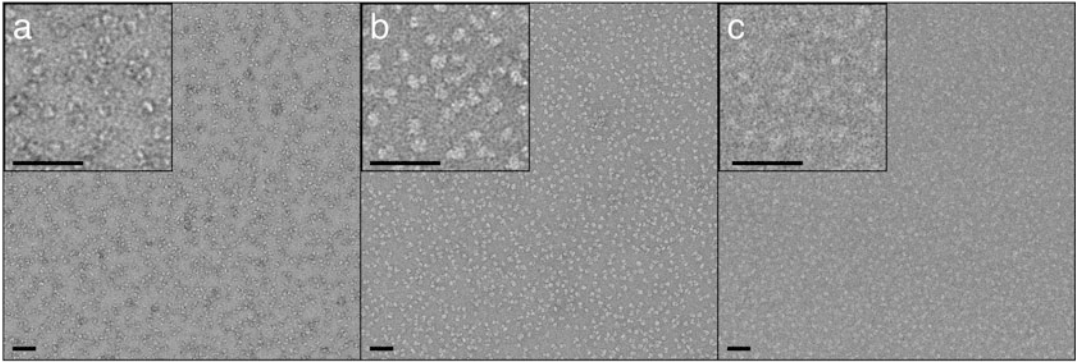


Fig. 4 Effect of the stain-thickness on visualizing structural features in negative-stain EM. (a) In too thin stain, the particles display dark edges, which hamper interpretation and particle detection. (b) Only properly stained particles look crisp and allow reliable assessment of sample quality. (c) In too thick stain, the particles appear fuzzy, which can be mistaken for poor quality of the sample. Scale bar corresponds to 30 nm

7. If a sufficient number of the target protein particles can be identified in a raw micrograph, collect 100–200 micrographs at a pixel size of around 3 Å and perform 2D classification on picked particles, using the software of choice (*see Note 8*).
8. Judge the sample quality, mainly the level of aggregation and contamination, based on the particle appearance in the micrographs and 2D-class averages, and either further optimize the sample preparation or proceed to cryo-EM.

3.3 Sample Vitrification, Using a Vitrobot

1. Turn on the Vitrobot, fill the humidifier with double distilled water, set the desired humidity and temperature (100% humidity and 4 °C typically works well), place the filter papers and allow them to equilibrate for 1 h (*see Note 9*).
2. Glow-discharge the holey carbon (or gold) grids to make them hydrophilic (*see Notes 4 and 10*).
3. Cool the liquid ethane container, its holder and a labeled cryo-EM grid storage box in liquid nitrogen (Fig. 5a).
4. Empty the ethane container (Fig. 5b) and keep it cooled in liquid nitrogen from the outside. Carefully fill the container with ethane (Fig. 5c) and allow it to cool until the walls are covered with ice (Fig. 5d) and the solution becomes milky (*see Note 11 and Fig. 5e*). Always wear hand and face protection, when working with ethane.
5. Place the container with liquid ethane in its holder and transfer to the Vitrobot.
6. Set the blot force and blotting time on the Vitrobot (*see Note 12*) and perform an empty run to test if everything is working properly.

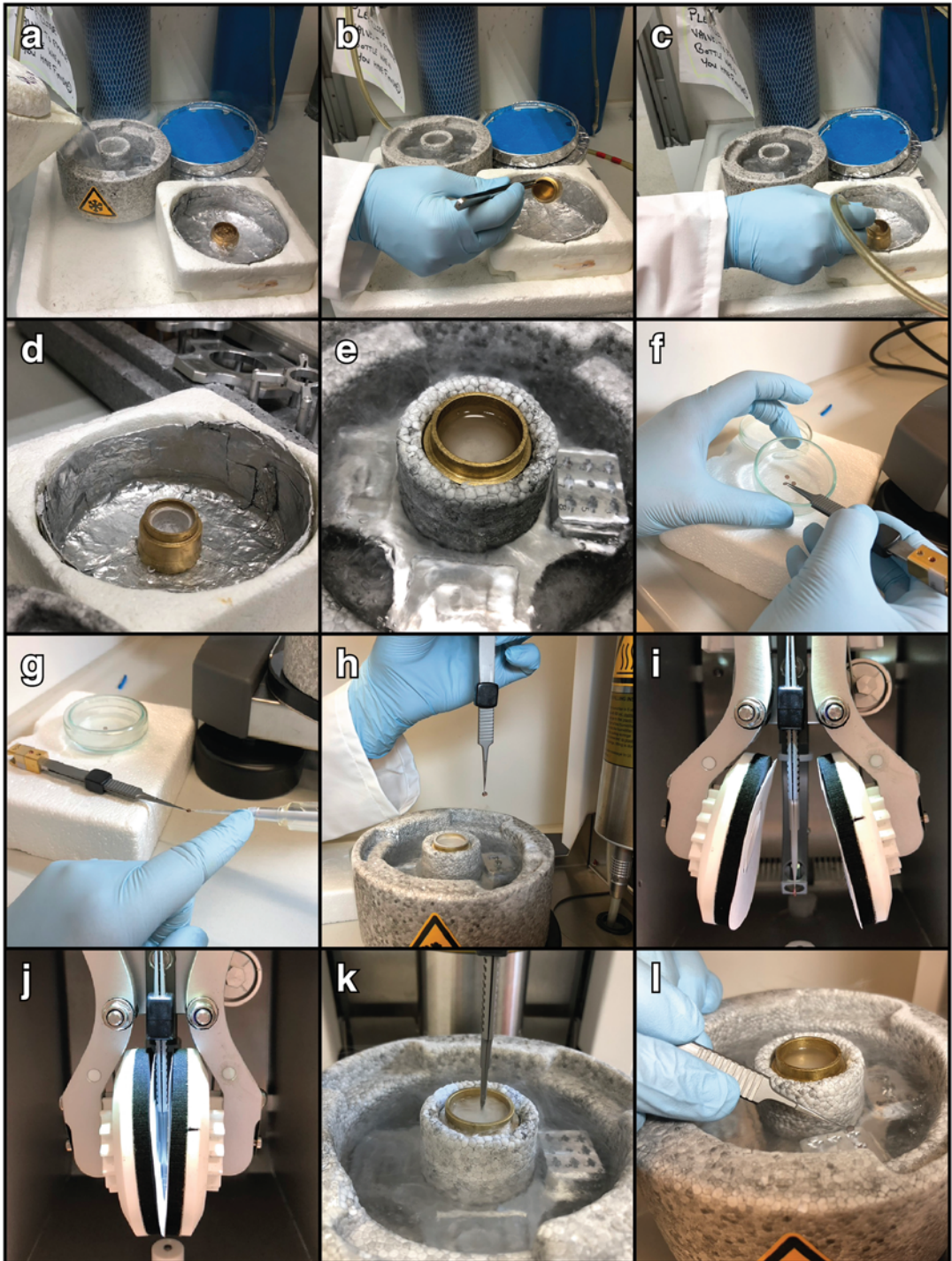


Fig. 5 Cryo-EM specimen preparation, using the Vitrobot (Thermo Fisher). (a) Cool the ethane cup and the grid storage box in liquid nitrogen. (b) Remove the liquid nitrogen from the ethane cup. (c) Fill the cooled cup with ethane. (d) Cool the ethane in the cup in a liquid nitrogen bath. (e) Wait for a milky appearance of the liquid ethane for optimal vitrification. (f) Pick up a glow-discharged holey carbon grid (C-flat was used in this photo) with the Vitrobot tweezers. (g) Apply the sample on the grid. (h) Mount the specimen to the Vitrobot (i–k) Autorun the Vitrobot to vitrify the grid in liquid ethane. (l) Transfer the vitrified grid to the storage box under liquid nitrogen

7. Pick up a glow-discharged grid with the Vitrobot tweezers (Fig. 5f) and add 3 μ l of protein sample to the grid (Fig. 5g), place the tweezers into the Vitrobot (Fig. 5h) and immediately operate it to blot the sample (Fig. 5i, j) and vitrify it in liquid ethane (Fig. 5k).
8. Quickly, but carefully, transfer the vitrified grid from the liquid ethane to the liquid nitrogen and place it in the precooled storage box (Fig. 5l).
9. Store the cryo-EM grids in the storage boxes in a liquid nitrogen dewar or load them directly in the cartridges for screening in the microscope.

3.4 Screening of the Cryo-EM Grids

1. Cool the grid clipping station with liquid nitrogen and transfer the grid box with cryo-EM specimen there. Precool the cartridge storage box and all the necessary tools.
2. Insert the C-clip into the insertion tool: using regular tweezers, grab the C-clip at the center, push it vertically into the C-clip insertion tool and then adjust it to a more horizontal position. Press the opening of the insertion tool with the C-clip against the clean flat surface and push the plunger down to properly align the C-clip around the rim of the tool. Cool the insertion tool with the C-clip in liquid nitrogen.
3. Assemble the cartridge (AutoGrid in this example): in the dedicated place of the grid clipping station, place the cryo-EM grid into the AutoGrid ring and fix it with the C-clip, by positioning the insertion tool on top of the grid and pushing the plunger of the tool all the way down to insert the C-clip into the AutoGrid ring. Using the AutoGrid tweezers, transfer the AutoGrid into the dedicated storage box. To reduce ice contamination, wear face mask throughout the procedure.
4. Fill the cassette-loading station and the autoloader capsule with liquid nitrogen. Transfer the cartridge box from the grid clipping to the cassette loading station. Carefully inspect each cartridge (AutoGrid) to ensure that the C-clip is in place and load the cartridges into the autoloader cassette. Discard any damaged or improperly clipped grids.
5. Transfer the cassette into the cooled autoloader capsule, check if the capsule pin moves freely and then load the microscope.
6. Perform the inventory of the autoloader after the temperatures have recovered to below 100 K.
7. Load the selected grid into the column and collect a grid atlas, using automated data collection software, to check the ice thickness. It may not always be necessary to collect a full atlas—the process can be stopped as soon as enough grid

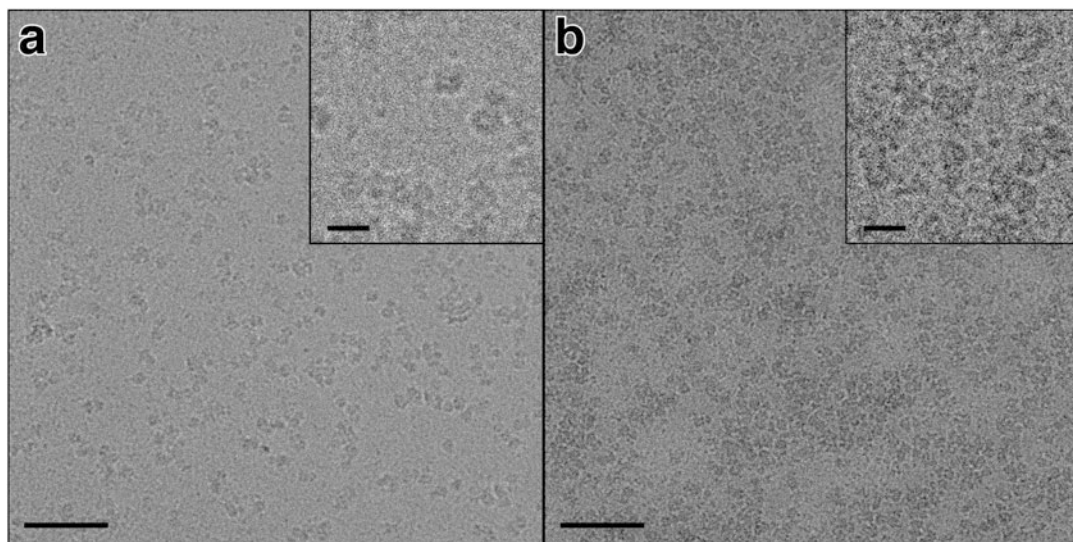


Fig. 6 Optimization of sample concentration for cryo-EM. The same sample at low (**a**) and high (**b**) concentration. While at first glance (**a**) seems to be vitrified optimally (particles are highly visible and nicely separated), in (**b**) approximately four times more particles can be retrieved per micrograph, even with better quality. Scale bars correspond to 50 and 10 nm in the inset

squares with appropriate ice thickness were obtained, especially during screening.

8. Select a square and adjust the eucentric height of the microscope stage. Then target several holes and collect test images to evaluate particle distribution and appearance (*see* **Note 13** and Fig. 6).
9. Based on the ice thickness and particle distribution, determine the optimal blotting conditions.
10. Depending on the screening results, either prepare new grids and repeat the screening until optimal conditions are found or collect a test dataset to evaluate the sample.
11. Process the data to check if the sample is intact and if it displays a sufficiently random angular distribution. If this is the case, continue with the collection of a large dataset. If not, optimize the biochemistry (*see* **Note 14**).

3.5 Cryo-EM Data Collection

1. Choose the pixel size for data collection (*see* **Note 15**) and optimal electron dose for the particular camera.
2. Set parallel illumination and ensure that the microscope is properly aligned at the specified magnification. Autofunctions in EPU or AutoCTF in Sherpa significantly simplify this procedure and optimize microscope performance.
3. Collect a gain reference, calibrate image shifts and choose your targets.

4. Check for the lowest defocus at which you can still identify the particles and set the defocus range for data collection around that value.
5. Start automated data collection and monitor the quality of the images (*see* **Note 16**).

3.6 Data Processing

There are many different software choices and strategies for data processing. It is always a good idea to test and possibly combine several of them (Fig. 7) to get the most out of the data. Often this choice highly depends on the sample as well as user preference and experience. Here, we share our strategy for single-particle analysis, which was optimized for ABC transporters, but appears to be applicable to most of the projects in our laboratory.

1. Before starting with your own project, get familiar with the processing software by executing the tutorial (RELION-3 [39] in this example).
2. Follow the suggested preprocessing steps for your own dataset, starting with the alignment of movie frames (*see* **Note 17**) and CTF estimation (*see* **Note 18**).
3. Select micrographs for further processing based on the estimated maximum resolution (*see* **Notes 16** and **19**).
4. To generate templates for auto-picking, perform 2D classification on a small subset of particles, that were picked during online data preprocessing (*see* **Note 20**), and select around five descriptive 2D class averages.
5. Run the RELION-3 template picker on your full dataset (*see* **Note 21**). Keep the picking threshold low to select all particles, even though this will include a lot of false positives.
6. Extract the picked particles binned 4–5 times and perform 3D classification to remove false positives (*see* **Note 22**).
7. Choose the good classes based on the map appearance and resolution (*see* **Note 23**), and reextract the recentered corresponding particles at full-pixel size.
8. Perform another round of 3D classification to sort out large conformational changes and residual false positives (*see* **Note 24**).
9. Refine each good class individually, create corresponding masks (*see* **Note 25**) and postprocess the obtained maps.
10. Perform particle polishing and CTF refinement, followed by 3D refinement after each step.
11. Further subclassify the particles in 3D, without additional alignment, to distinguish smaller conformational changes (*see* **Note 26**).
12. Refine and postprocess the resulting maps (*see* **Note 27**).

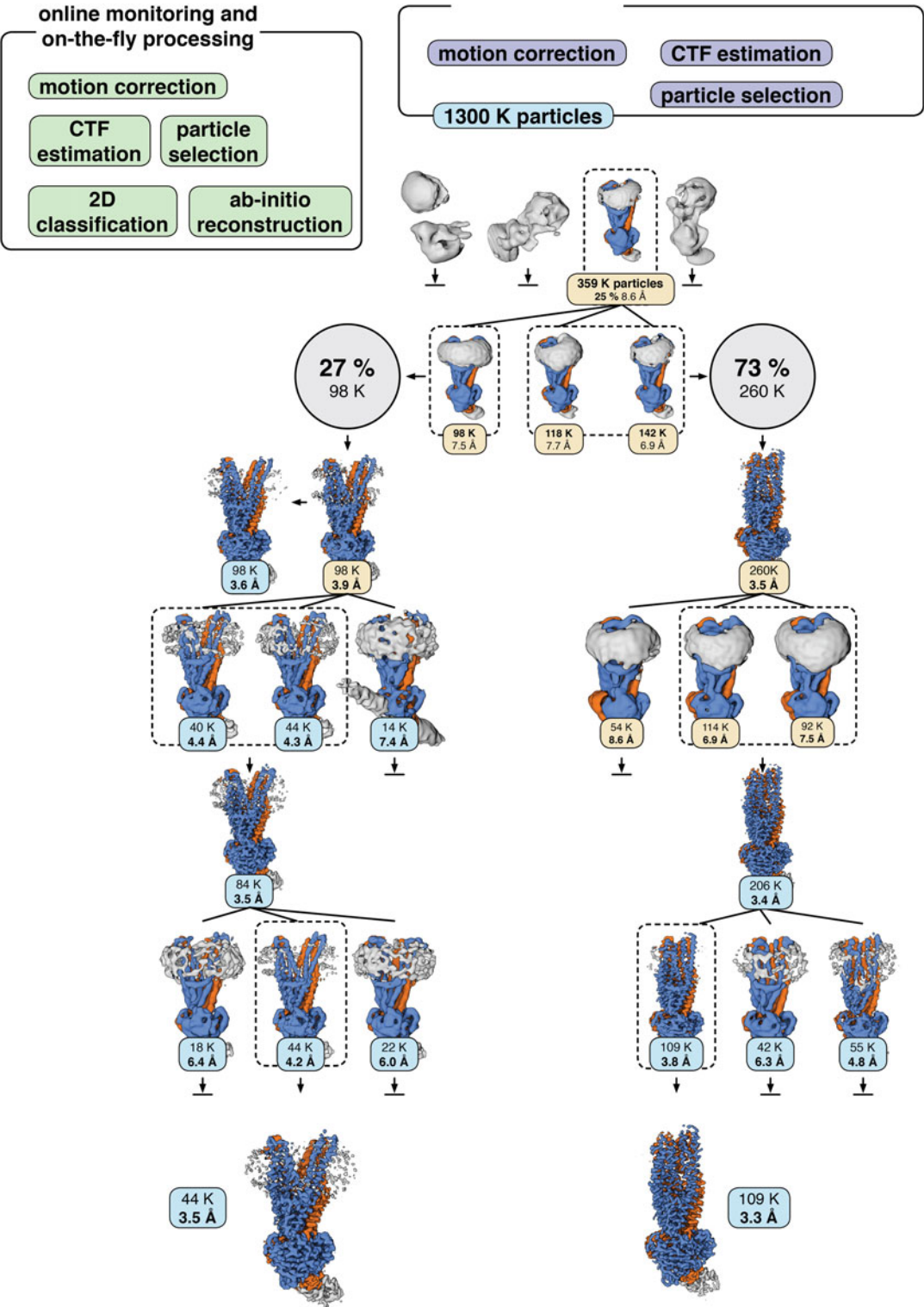


Fig. 7 Exemplary processing workflow. Here, we combined different software to obtain two high-resolution structures (EMD-4777 and EMD-4778) of the ABC transporter TmrAB from one dataset [29]. In this particular

13. Estimate the local resolution and, depending on the protein and the questions raised, perform multibody refinement [90] or apply a symmetry expansion approach [90], if applicable. Useful strategies for processing of heterogeneous data, are reviewed in [22, 91].

4 Notes

1. For preparing uranyl formate solution, only use fresh uranyl formate powder, which should be yellow. The stain solution is very unstable and light sensitive; therefore, it has to be kept protected from light and used within a day. In practice, we store small frozen aliquots (25 μ l) to avoid the wasteful and time-consuming process of preparation. Uranyl formate is radioactive and toxic, and hence must be treated accordingly.
2. As a minimal requirement, the sample purity needs to be confirmed by SDS-PAGE and SEC, but any further information about sample monodispersity, oligomeric form and thermostability increases the chances of success. In general, for cryo-EM, it is a good idea to use membrane proteins in a detergent-free system, for example, nanodiscs, as it not only represents a more native-like environment but also often increases contrast in cryo-EM micrographs. As different hydrophobic environments have a high impact on the sample behavior and distribution in the vitreous buffer, it is often beneficial to test several of them.

Fig. 7 (continued) example, the preprocessing steps were performed on the fly in Warp [75], followed by particle examination by 2D-classification in CryoSPARC [24], which was also used to generate an ab-initio model. Subsequently, the dataset was fully reprocessed in RELION-3 [39]. Here, the particles were template-picked (templates were derived from the initial picking in Warp), using a very generous threshold to minimize false negatives, even though this resulted in a large number of false positives. Accordingly, more than one million picked particles were extracted, binned four times, and applied to a first round of cleaning in 3D. Three hundred and fifty-nine thousands of all particles (class 3) formed a meaningful class and were recentered and reextracted at full-pixel size. The second round of 3D classification readily allowed to separate two major conformations in the dataset, which were subsequently treated independently. A final refinement after particle polishing and CTF-refinement resulted in reconstructions with resolutions of 3.6 and 3.5 Å, respectively. However, high B-factors and weaker densities at some regions suggested that further subclassification may be beneficial. Typically, at this stage 3D classification with no further alignment and applying masks can separate small conformational differences and particles of different quality. The best classification strategy depends on the sample and the particular problem, but it is always recommended to test different avenues and software to achieve the best result. In this example, the particles were transferred from RELION-3 (marked in yellow) to CryoSPARC (marked in blue) and underwent several rounds of heterogeneous refinement, which further improved the 3D densities. Nonuniform refinement was used to generate the final maps at 3.5 and 3.3 Å resolution

3. Current vitrification methods require high sample concentrations. While the typical concentration of a nanodisc-embedded membrane protein is in the range of 1 mg/ml for cryo-EM, vitrification in the presence of detergent may require up to ten times higher concentrations. It is not always trivial to reach such concentrations as many membrane proteins and their complexes tend to aggregate or fall apart during the sample concentration procedure. In such cases, gradient fixation Gra-Fix [92, 93] has been shown to be extremely beneficial. Another option is to freeze the sample on a continuous support film that is deposited onto conventional holey carbon film. While carbon typically adds a significant background, limiting imaging of small membrane proteins, graphene holds a great promise for such applications, as it is almost invisible to the electron beam. Like carbon, graphene as a substrate allows for lower protein concentrations and, additionally, protects the sample from the air–water interface. Despite these obvious advantages, graphene is still not routinely used due to technical difficulties associated with the contamination and hydrophobicity of the surface. The constantly increasing number of successful applications is, however, very promising [55, 63].
4. We use the same glow-discharging parameters for all types of grids. On our PELCO easiGlow™ system, glow-discharging at 15 mA for 45 s typically works well. The grids remain hydrophilic for several hours, so we always glow-discharge the required number of grids at once. The hydrophilicity of the grid can be assessed by the contact angle of the spread sample drop, but the ultimate answer about the suitability of the glow-discharging conditions is provided by examining the grid in electron microscope. If the grid is not hydrophilic enough, it typically displays stain or ice patches and uneven particle distribution.
5. The optimal protein concentration for negative-stain EM depends on the sample, but is typically in the range of 100 nM. To determine the optimal concentration, make one grid at the starting 100 nM concentration, examine it in electron microscope and make further dilutions or use the undiluted sample if necessary. Optimization of the sample concentration is a good time investment not only for cryo-EM but also for negative stain, as even doubling the concentration can significantly decrease the required imaging time. Normally, there is no need to incubate the sample on the grid before blotting, but if the concentration is very low, the protein can rest on the grid for a couple of minutes or even overnight (in a humidified environment) to increase the concentration.

6. Uranyl stains are most commonly used, as they are easy to make and provide excellent contrast. However, when working with phosphate containing buffers, or even high concentrations of ATP, the phosphate can cause stain precipitation. If the stain looks too thin, uranyl formate can be applied to a grid four or even five times instead of three. This can also provide a remedy if high levels of ATP cannot be avoided. Judgment of the stain thickness and quality is performed by examining the grid in the microscope. This step requires some practice. Therefore, especially for beginners, it is advisable to identify grid squares, which display a gradient or a wavy pattern to increase the chances of finding the right stain. When the stain is too thin, the particles display black edges (Fig. 4a), that can hamper automatic particle detection by SPA software. When the stain is too thick, particles become blurry and lose details (Fig. 4c).
7. In the case of small membrane proteins, which are fully embedded in the membrane, it may be difficult to distinguish them from the empty detergent micelles or nanodiscs. Whenever in doubt, the buffer should be tested alone and the sample should be examined after further dilution in the buffer with and without detergent. Further tips and example micrographs, demonstrating the background produced by various detergents, can be found in the recent study from our group [50].
8. While sample purity and homogeneity to some extent can be addressed on a single micrograph, in most cases, collecting only a few micrographs is insufficient to fully evaluate the quality of the sample. A statistical comparison is often imperative to check if the macromolecular complexes are intact or if individual particles are displaying different conformations. Thus, 2D-class averages with sufficient numbers of particles are necessary. The required number of micrographs can be easily obtained using automated data collection, for example, Legion [94]. For particle processing any SPA software can be used, but we typically use cis-TEM [23] to quickly get 2D-class averages of the stained particles. While 2D-class averages allow judging sample quality in most cases, for very small proteins, fully embedded in the membrane, the information that they can provide is limited.
9. Equilibrating filter papers prior to freezing grids increases reproducibility. Blotting parameters depend on the type of the filter paper, so always use the one, for which blotting conditions were optimized. We typically use Whatman grade 595 qualitative filter papers for preparing cryo-EM specimen.
10. For cryo-EM, we use the same glow-discharge parameters as for negative-stain (*see* **Note 4**), but repeat the procedure three

times instead of one. This provides sufficient reproducibility in most cases. Alternatively, a plasma cleaner can be used.

11. If the ethane warms up during the freezing procedure, place it back into the liquid nitrogen bath and cool it from the outside until the solution is milky again. Do not solidify it as this will damage the grid. Solidified ethane can be melted by dipping the back end of regular warm forceps into the ethane. Never use the Vitrobot tweezers for this, as this will damage and misalign them beyond repair.
12. Optimal blotting parameters, mainly blot time and blot force, depend on the instrument and its proper alignment. Periodical maintenance is advisable; however, blotting parameters may have to be readjusted afterwards. Additional parameters are the temperature and humidity, the type of blotting paper as well as the type of grids and presence or absence of detergent in the sample. If all of these parameters are being kept constant, an experienced user can achieve sufficient reproducibility in most of the cases. To identify good blotting conditions for your setup, use a test specimen, for example, apoferritin. As soon as starting conditions are established, often only slight adjustments are needed for each individual sample. We typically set the Vitrobot to 100% humidity and 4 °C, use a filter paper of grade 595, blot force -2 and blotting times between 11 and 13 s, which provides good results for most of our small membrane protein samples.
13. When screening for sample concentration, aim for the one that provides rather crowded images (Fig. 6) and, therefore, requires fewer micrographs to obtain a sufficient number of particles and increases the signal for better CTF estimation. As for negative stain, an increased particle concentration will result in significant reductions of precious data collection time. In addition, the crowded environment often improves particle stability. However, take care not to overcrowd the grids, especially when collecting tilted data. Keep in mind that the relationship between sample concentration and particle distribution on the grid is rarely linear. Furthermore, it is advisable to inspect different levels of ice thickness, and, also to check the carbon surface. When particles are not detected inside the holes, increase the concentration significantly, if possible. Alternatively, different hole sizes and types of grids as well as support films can be utilized to increase particle counts.
14. If particles are falling apart during the vitrification procedure, they can be crosslinked, for example, using GraFix [92], which should increase their stability. Furthermore, buffer conditions can be optimized based on thermostability measurements. Changing the hydrophobic environment, for instance

reconstituting into nanodiscs, also can increase particle stability. Most often, particles are damaged at the air-water interface. Therefore, methods that shield the particles from the air-water interface, like using support films or detergents, may be beneficial for such cases. The same strategy can also help in alleviating preferred orientation. In general, the hydrophobic environment is a very important parameter, and the same membrane protein often displays very different angular distributions in different detergents or nanodiscs. If available, FAB-fragments [33] or nanobody technology [34] can also be utilized to randomize the particle orientation. For pathological cases, tilted data collection can be an additional remedy [53].

15. Often, with membrane proteins, the resolution is limited by the sample rather than the pixel size. Therefore, it is typically not necessary to go to very high magnification. In our experience, 0.8–1 Å/pix is sufficient to reach sub-3-Å resolutions and provides a good compromise between field of view and high-resolution details.
16. Monitoring the data quality on the fly provides a fast readout, which allows for quick adjustment of the data collection strategy, and thus for optimal final results. An important parameter to monitor is the quality of the CTF-fit. To achieve high resolution, we typically only choose images with an estimated resolution better than 4 Å. For online monitoring, we like to use Warp [75] as it has a convenient interface to sort out poor images and outputs a particle star file, that can be directly piped to RELION [27] or CryoSPARC [24]. To quickly assess the spatial distribution of the particles and their overall quality while the data is being collected, we perform a 2D classification in CryoSPARC on the particles, picked in Warp. CryoSPARC also implements powerful algorithms for ab initio reconstruction, which allows the user to quickly generate a single model or multiple initial 3D models for subsequent 3D classification or refinement. CryoSPARC live can also be used for real-time data processing and performs the abovementioned steps in one go.
17. For motion correction, we typically use RELION's own implementation of MotionCor2 [95] with default parameters, only changing the number of patches to 5×5 . Sometimes it can be beneficial to save the non-dose-weighted images as well, but remember to delete them immediately after the CTF estimation to save storage space, which can quickly become a problem with large amounts of cryo-EM datasets. While beam-induced motion correction is often performed on the fly, to use the full functionality of a certain software, it may need to be repeated.

For example, currently, to enable Bayesian polishing [96], motion correction needs to be run within RELION-3.

18. For CTF estimation, either Gctf [70] or CTFFIND4 [69] can be used. We typically keep the default parameters and only adjust the FFT box size to 1024 pix and the maximum resolution parameter to 3 Å.
19. Micrograph selection can be performed in RELION; however, we typically do this in Warp [75] during online monitoring of the data, as we find its selection filters particularly convenient. We then perform the full processing in RELION-3 only on the micrographs preselected in Warp.
20. If no preprocessing of data was performed, particles, for generating 2D templates, can be picked using the reference-free Laplacian-of-Gaussian (LoG) auto-picker, as implemented in RELION-3 [39]. At this step, it is not necessary to invest a lot of time in optimizing picking parameters as these initial coordinates will only be used to generate templates. To speed up the procedure, this type of picking should be performed only on a small part of the dataset, and 2D classification can be performed on 4–5 times binned particles with a handful of classes.
21. Neuronal network based pickers, for example, Warp [75], crYOLO [74], or Topaz [73], provide a great alternative for particle picking and often result in better coordinates with a minimal number of false positives, which is especially beneficial for small, featureless and low contrast particles. However, for the very crowded micrographs that we like to use, the RELION-3 template picker currently seems to work best in our hands. With the current speed of data processing algorithms, removing large amounts of false positives is not a huge bottleneck anymore, especially, when done directly in 3D with highly binned particles on a GPU accelerated system. Particle sorting and selection based on the *Z* score can additionally be applied to quickly remove poor quality particles. Typically, manual picking can be skipped completely; however, in some very challenging cases careful manual inspection and curation can still be beneficial.
22. Particle cleaning, using 3D classification, requires an initial 3D model that can be used as a reference. We typically obtain it through the ab initio reconstruction in CryoSPARC [24] during the online data preprocessing, but an initial model can also be generated directly in RELION-3 [39]. Normally, for the initial 3D cleaning step, we use 3–4 classes and perform 30–50 iterations, as in most of the encountered cases involving small membrane proteins, the default 25 iterations were insufficient to reach convergence.

23. Good quality particles typically populate a single or two 3D classes, reaching the Nyquist resolution for the binned data (around 8 Å for 4–5 times binned particles). If the signal-to-noise ratio is very low, cleaning directly in 3D may fail, and in such cases additional sorting and selection based on the Zscore or conventional precleaning in 2D can help. However, based on our experience, most of those cases, in which direct cleaning in 3D fails, do not yield high-resolution maps even after cleaning in 2D or using alternative particle pickers.
24. For the 3D classification, choose the number of classes based on the expected number of different conformations in the dataset and add an extra class for junk particles. As a 3D reference, use the rescaled best map, obtained from the previous 3D-classification step, filtered to 25 Å. As mentioned in **Note 22**, often more than 25 iterations are required to reach class convergence. In some cases, the classification may need to be rerun with a different number of classes and with an increased sampling rate.
25. Mask creation is one of the steps requiring the most user intervention, especially when generating focused masks. Typically, the map is inspected in UCSF Chimera [97] to determine the initial binarization threshold and a soft mask is then created in RELION. The user needs to specify how far the binary map should be extended and how many pixels have to be added to the soft edge of the mask. This step typically requires several trials. It is strongly recommended to create rather loose masks at the initial processing steps and, as processing progresses and better alignment accuracy is achieved, to apply tighter masks. For membrane proteins, masking of the detergent micelle or nanodisc can significantly improve the alignment accuracy. For such masks, we combine different tools, present in UCSF Chimera [97], mainly, map segmentation and volume eraser. If atomic coordinates are available, the color zone command can be utilized as well. Signal subtraction can be additionally applied to remove the signal of the hydrophobic environment; however, in our hands, we rarely notice any benefit of signal subtraction over masking.
26. Applying masks and increasing the regularization parameter can be useful at this classification step. While turning off the alignment typically works better for small membrane proteins, 3D classification using only local searches can also be performed. A good resource on the processing of heterogeneous datasets in RELION can be found in [22].
27. In parallel to the refinement and postprocessing steps in RELION, which allow application of various masks, we typically also test nonuniform refinement [98], implemented in CryoSPARC [24]. While the resolution values estimated by both software are similar in most cases, nonuniform refinement sometimes improves the quality in certain areas of the map.

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References

1. Liao M, Cao E, Julius D, Cheng Y (2013) Structure of the TRPV1 ion channel determined by electron cryo-microscopy. *Nature* 504:107–112. <https://doi.org/10.1038/nature12822>
2. Madej MG, Ziegler CM (2018) Dawning of a new era in TRP channel structural biology by cryo-electron microscopy. *Pflügers Arch Eur J Physiol* 470:213–225
3. Liang YL, Khoshouei M, Radjainia M et al (2017) Phase-plate cryo-EM structure of a class B GPCR-G-protein complex. *Nature* 546:118–123. <https://doi.org/10.1038/nature22327>
4. Zhang Y, Sun B, Feng D et al (2017) Cryo-EM structure of the activated GLP-1 receptor in complex with a G protein. *Nature* 546:248–253. <https://doi.org/10.1038/nature22394>
5. García-Nafria J, Tate CG (2020) Cryo-electron microscopy: moving beyond X-ray crystal structures for drug receptors and drug development. *Annu Rev Pharmacol Toxicol* 60. <https://doi.org/10.1146/annurev-pharmtox-010919-023545>
6. Blees A, Janulienė D, Hofmann T et al (2017) Structure of the human MHC-I peptide-loading complex. *Nature* 551:525–528. <https://doi.org/10.1038/nature24627>
7. Vinothkumar KR, Zhu J, Hirst J (2014) Architecture of mammalian respiratory complex I. *Nature* 515:80–84. <https://doi.org/10.1038/nature13686>
8. Gu J, Wu M, Guo R et al (2016) The architecture of the mammalian respirasome. *Nature* 537:639–643. <https://doi.org/10.1038/nature19359>
9. Fiedorczuk K, Letts JA, Degliesposti G et al (2016) Atomic structure of the entire mammalian mitochondrial complex I. *Nature* 538:406–410. <https://doi.org/10.1038/nature19794>
10. Agip A-NA, Blaza JN, Fedor JG, Hirst J (2019) Mammalian respiratory complex I through the lens of cryo-EM. *Annu Rev Biophys* 48:165–184. <https://doi.org/10.1146/annurev-biophys-052118-115704>
11. Zhou A, Rohou A, Schep DG et al (2015) Structure and conformational states of the bovine mitochondrial ATP synthase by cryo-EM. *elife* 4. <https://doi.org/10.7554/eLife.10180>
12. Allegretti M, Klusch N, Mills DJ et al (2015) Horizontal membrane-intrinsic α -helices in the stator a-subunit of an F-type ATP synthase. *Nature* 521:237–240. <https://doi.org/10.1038/nature14185>
13. Kühlbrandt W (2019) Structure and mechanisms of F-type ATP synthases. *Annu Rev Biochem* 88:515–549. <https://doi.org/10.1146/annurev-biochem-013118-110903>
14. Murphy BJ, Klusch N, Langer J et al (2019) Rotary substates of mitochondrial ATP synthase reveal the basis of flexible F1-Fo coupling. *Science* 364:eaaw9128
15. Lyons JA, Shahsavari A, Paulsen PA et al (2016) Expression strategies for structural studies of eukaryotic membrane proteins. *Curr Opin Struct Biol* 38:137–144. <https://doi.org/10.1016/j.sbi.2016.06.011>
16. Bayburt TH, Grinkova YV, Sligar SG (2002) Self-assembly of discoidal phospholipid bilayer nanoparticles with membrane scaffold proteins. *Nano Lett* 2:853–856. <https://doi.org/10.1021/nl025623k>
17. Frauenfeld J, Löving R, Armache JP et al (2016) A saposin-lipoprotein nanoparticle system for membrane proteins. *Nat Methods* 13:345–351. <https://doi.org/10.1038/nmeth.3801>
18. Carlson ML, Young JW, Zhao Z et al (2018) The peptidisc, a simple method for stabilizing membrane proteins in detergent-free solution. *elife* 7. <https://doi.org/10.7554/eLife.34085>
19. Tribet C, Audebert R, Popot JL (1996) Amphipols: polymers that keep membrane proteins soluble in aqueous solutions. *Proc Natl*

- Acad Sci U S A 93:15047–15050. <https://doi.org/10.1073/pnas.93.26.15047>
20. Tao H, Lee SCC, Moeller A et al (2013) Engineered nanostructured β -sheet peptides protect membrane proteins. *Nat Methods* 10:759–761. <https://doi.org/10.1038/nmeth.2533>
 21. Lee SC, Knowles TJ, Postis VLG et al (2016) A method for detergent-free isolation of membrane proteins in their local lipid environment. *Nat Protoc* 11:1149–1162. <https://doi.org/10.1038/nprot.2016.070>
 22. Scheres SHW (2016) Processing of structurally heterogeneous cryo-EM data in RELION. In: *Methods in enzymology*. Academic, New York, NY, pp 125–157
 23. Grant T, Rohou A, Grigorieff N (2018) Cistem, user-friendly software for single-particle image processing. *elife* 7. <https://doi.org/10.7554/eLife.35383>
 24. Punjani A, Rubinstein JL, Fleet DJ, Brubaker MA (2017) CryoSPARC: algorithms for rapid unsupervised cryo-EM structure determination. *Nat Methods* 14:290–296. <https://doi.org/10.1038/nmeth.4169>
 25. Moriya T, Saur M, Stabrin M et al (2017) High-resolution single particle analysis from electron cryo-microscopy images using SPHIRE. *J Vis Exp* 2017. <https://doi.org/10.3791/55448>
 26. Bell JM, Chen M, Baldwin PR, Ludtke SJ (2016) High resolution single particle refinement in EMAN2.1. *Methods* 100:25–34. <https://doi.org/10.1016/j.ymeth.2016.02.018>
 27. Scheres SHW (2012) RELION: implementation of a Bayesian approach to cryo-EM structure determination. *J Struct Biol* 180:519–530. <https://doi.org/10.1016/j.jsb.2012.09.006>
 28. Nakane T, Kimanius D, Lindahl E, Scheres SHW (2018) Characterisation of molecular motions in cryo-EM single-particle data by multi-body refinement in RELION. *elife* 7. <https://doi.org/10.7554/eLife.36861>
 29. Hofmann S, Janulienė D, Mehdi pour AR et al (2019) Conformation space of a heterodimeric ABC exporter under turnover conditions. *Nature* 571:580–583
 30. Henderson R (1995) The potential and limitations of neutrons, electrons and X-rays for atomic resolution microscopy of unstained biological molecules. *Q Rev Biophys* 28:171–193. <https://doi.org/10.1017/S003358350000305X>
 31. Khoshouei M, Radjainia M, Baumeister W, Danev R (2017) Cryo-EM structure of haemoglobin at 3.2 Å determined with the Volta phase plate. *Nat Commun* 8. <https://doi.org/10.1038/ncomms16099>
 32. Fan X, Wang J, Zhang X et al (2019) Single particle cryo-EM reconstruction of 52 kDa streptavidin at 3.2 Angstrom resolution. *Nat Commun* 10. <https://doi.org/10.1038/s41467-019-10368-w>
 33. Wu S, Avila-Sakar A, Kim J et al (2012) Fabs enable single particle cryoEM studies of small proteins. *Structure* 20:582–592. <https://doi.org/10.1016/j.str.2012.02.017>
 34. Uchański T, Masiulis S, Fischer B, et al (2021) Megabodies expand the nanobody toolkit for protein structure determination by single-particle cryo-EM. *Nat Methods* 18:60–68. <https://doi.org/10.1038/s41592-020-01001-6>
 35. Kim J, Tan YZ, Wicht KJ et al (2019) Structure and drug resistance of the *Plasmodium falciparum* transporter PfCRT. *Nature*. <https://doi.org/10.1038/s41586-019-1795-x>
 36. Merk A, Bartesaghi A, Banerjee S et al (2016) Breaking cryo-EM resolution barriers to facilitate drug discovery. *Cell* 165:1698–1707. <https://doi.org/10.1016/j.cell.2016.05.040>
 37. Bartesaghi A, Aguerrebere C, Falconieri V et al (2018) Atomic resolution cryo-EM structure of β -galactosidase. *Structure* 26:848–856.e3. <https://doi.org/10.1016/j.str.2018.04.004>
 38. Danev R, Yanagisawa H, Kikkawa M (2019) Cryo-electron microscopy methodology: current aspects and future directions. *Trends Biochem Sci* 44:837–848
 39. Zivanov J, Nakane T, Forsberg BO et al (2018) New tools for automated high-resolution cryo-EM structure determination in RELION-3. *elife* 7. <https://doi.org/10.7554/eLife.42166>
 40. Tan YZ, Aiyer S, Mietzsch M et al (2018) Sub-2 Å Ewald curvature corrected structure of an AAV2 capsid variant. *Nat Commun* 9. <https://doi.org/10.1038/s41467-018-06076-6>
 41. Yip KM, Fischer N, Paknia E et al (2020) Atomic-resolution protein structure determination by cryo-EM. *Nature* 587:157–161. <https://doi.org/10.1038/s41586-020-2833-4>
 42. Nakane T, Kotecha A, Sente A et al (2020) Single-particle cryo-EM at atomic resolution. *Nature* 587:152–156. <https://doi.org/10.1038/s41586-020-2829-0>

43. Niesen FH, Berglund H, Vedadi M (2007) The use of differential scanning fluorimetry to detect ligand interactions that promote protein stability. *Nat Protoc* 2:2212–2221. <https://doi.org/10.1038/nprot.2007.321>
44. Mancusso R, Karpowich NK, Czyzewski BK, Wang DN (2011) Simple screening method for improving membrane protein thermostability. *Methods* 55:324–329. <https://doi.org/10.1016/j.ymeth.2011.07.008>
45. Chari A, Haselbach D, Kirves JM et al (2015) ProteoPlex: stability optimization of macromolecular complexes by sparse-matrix screening of chemical space. *Nat Methods* 12:859–865. <https://doi.org/10.1038/nmeth.3493>
46. Kawate T, Gouaux E (2006) Fluorescence-detection size-exclusion chromatography for precrystallization screening of integral membrane proteins. *Structure* 14:673–681. <https://doi.org/10.1016/j.str.2006.01.013>
47. Wittig I, Braun HP, Schägger H (2006) Blue native PAGE. *Nat Protoc* 1:418–428. <https://doi.org/10.1038/nprot.2006.62>
48. Stetefeld J, McKenna SA, Patel TR (2016) Dynamic light scattering: a practical guide and applications in biomedical sciences. *Biophys Rev* 8:409–427
49. Kang Y, Zhou XE, Gao X et al (2015) Crystal structure of rhodopsin bound to arrestin by femtosecond X-ray laser. *Nature* 523:561–567. <https://doi.org/10.1038/nature14656>
50. Gewering T, Janulienė D, Ries AB, Moeller A (2018) Know your detergents: a case study on detergent background in negative stain electron microscopy. *J Struct Biol* 203:242–246. <https://doi.org/10.1016/j.jsb.2018.05.008>
51. Moeller A, Lee SC, Tao H et al (2015) Distinct conformational spectrum of homologous multidrug ABC transporters. *Structure* 23:450–460. <https://doi.org/10.1016/j.str.2014.12.013>
52. Russo CJ, Passmore LA (2014) Ultrastable gold substrates for electron cryomicroscopy. *Science* 346:1377–1380. <https://doi.org/10.1126/science.1259530>
53. Zi Tan Y, Baldwin PR, Davis JH et al (2017) Addressing preferred specimen orientation in single-particle cryo-EM through tilting. *Nat Methods* 14:793–796. <https://doi.org/10.1038/nmeth.4347>
54. Noble AJ, Wei H, Dandey VP et al (2018) Reducing effects of particle adsorption to the air–water interface in cryo-EM. *Nat Methods* 15:793–795. <https://doi.org/10.1038/s41592-018-0139-3>
55. D’Imprima E, Floris D, Joppe M et al (2019) Protein denaturation at the air–water interface and how to prevent it. *elife* 8. <https://doi.org/10.7554/eLife.42747>
56. Armstrong M, Han B-G, Gomez S et al (2019) Micro-scale fluid behavior during cryo-EM sample blotting. *Biophys J*. <https://doi.org/10.1016/j.bpj.2019.12.017>
57. Razinkov I, Dandey VP, Wei H et al (2016) A new method for vitrifying samples for cryoEM. *J Struct Biol* 195:190–198. <https://doi.org/10.1016/j.jsb.2016.06.001>
58. Arnold SA, Albiez S, Bieri A et al (2017) Blotting-free and lossless cryo-electron microscopy grid preparation from nanoliter-sized protein samples and single-cell extracts. *J Struct Biol* 197:220–226. <https://doi.org/10.1016/j.jsb.2016.11.002>
59. Rubinstein JL, Guo H, Ripstein ZA et al (2019) Shake-it-off: a simple ultrasonic cryo-EM specimen-preparation device. *Acta Crystallogr D Struct Biol* 75:1063–1070. <https://doi.org/10.1107/S2059798319014372>
60. Tan YZ, Rubinstein JL (2020) Through-grid wicking enables high-speed cryoEM specimen preparation. *Acta Crystallogr Sect D Struct Biol* 76:1092–1103. <https://doi.org/10.1107/s2059798320012474>
61. Ravelli RBG, Nijpels FJT, Henderikx RJM, et al (2020) Cryo-EM structures from sub-nl volumes using pin-printing and jet vitrification. *Nat Commun* 11:1–9. <https://doi.org/10.1038/s41467-020-16392-5>
62. Kontziampasis D, Klebl DP, Iadanza MG, et al (2019) A cryo-EM grid preparation device for time-resolved structural studies. *IUCrJ* 6:1024–1031. <https://doi.org/10.1107/S2052252519011345>
63. Naydenova K, Peet MJ, Russo CJ (2019) Multifunctional graphene supports for electron cryomicroscopy. *Proc Natl Acad Sci U S A* 116:11718–11724. <https://doi.org/10.1073/pnas.1904766116>
64. Han Y, Fan X, Wang H et al (2019) High-yield monolayer graphene grids for near-atomic resolution cryoelectron microscopy. *Proc Natl Acad Sci* 2019:201919114. <https://doi.org/10.1073/pnas.1919114117>
65. Liu N, Zhang J, Chen Y, et al (2019) Bioactive Functionalized Monolayer Graphene for High-Resolution Cryo-Electron Microscopy. *J Am Chem Soc* 141:4016–4025. <https://doi.org/10.1021/jacs.8b13038>

66. Cheng Y, Grigorieff N, Penczek PA, Walz T (2015) A primer to single-particle cryo-electron microscopy. *Cell* 161:438–449
67. Campbell MG, Cheng A, Brilot AF et al (2012) Movies of ice-embedded particles enhance resolution in electron cryo-microscopy. *Structure* 20:1823–1828. <https://doi.org/10.1016/j.str.2012.08.026>
68. Huang Z, Baldwin PR, Mullapudi S, Penczek PA (2003) Automated determination of parameters describing power spectra of micrograph images in electron microscopy. *J Struct Biol* 144:79–94. <https://doi.org/10.1016/j.jsb.2003.10.011>
69. Rohou A, Grigorieff N (2015) CTFFIND4: fast and accurate defocus estimation from electron micrographs. *J Struct Biol* 192:216–221. <https://doi.org/10.1016/j.jsb.2015.08.008>
70. Zhang K (2016) Gctf: real-time CTF determination and correction. *J Struct Biol* 193:1–12. <https://doi.org/10.1016/j.jsb.2015.11.003>
71. Van Heel M (1982) Detection of objects in quantum-noise-limited images. *Ultramicroscopy* 7:331–341. [https://doi.org/10.1016/0304-3991\(82\)90258-3](https://doi.org/10.1016/0304-3991(82)90258-3)
72. Voss NR, Yoshioka CK, Radermacher M et al (2009) DoG Picker and TiltPicker: software tools to facilitate particle selection in single particle electron microscopy. *J Struct Biol* 166:205–213. <https://doi.org/10.1016/j.jsb.2009.01.004>
73. Bepler T, Morin A, Rapp M et al (2019) Positive-unlabeled convolutional neural networks for particle picking in cryo-electron micrographs. *Nat Methods* 16:1153–1160. <https://doi.org/10.1038/s41592-019-0575-8>
74. Wagner T, Merino F, Stabrin M et al (2019) SPHIRE-crYOLO is a fast and accurate fully automated particle picker for cryo-EM. *Commun Biol* 2. <https://doi.org/10.1038/s42003-019-0437-z>
75. Tegunov D, Cramer P (2019) Real-time cryo-electron microscopy data preprocessing with Warp. *Nat Methods* 16:1146–1152. <https://doi.org/10.1038/s41592-019-0580-y>
76. Lander GC, Stagg SM, Voss NR et al (2009) Appion: an integrated, database-driven pipeline to facilitate EM image processing. *J Struct Biol* 166:95–102. <https://doi.org/10.1016/j.jsb.2009.01.002>
77. Li X, Zheng S, Agard DA, Cheng Y (2015) Asynchronous data acquisition and on-the-fly analysis of dose fractionated cryoEM images by UCSFImage. *J Struct Biol* 192:174–178. <https://doi.org/10.1016/j.jsb.2015.09.003>
78. Fernandez-Leiro R, Scheres SHW (2017) A pipeline approach to single-particle processing in RELION. *Acta Crystallogr Sect D Struct Biol* 2017:496–502
79. Gómez-Blanco J, de la Rosa-Trevín JM, Marabini R et al (2018) Using Scipion for stream image processing at Cryo-EM facilities. *J Struct Biol* 204:457–463. <https://doi.org/10.1016/j.jsb.2018.10.001>
80. de la Rosa-Trevín JM, Otón J, Marabini R et al (2013) Xmipp 3.0: an improved software suite for image processing in electron microscopy. *J Struct Biol* 184:321–328. <https://doi.org/10.1016/j.jsb.2013.09.015>
81. Reboul CF, Kiesewetter S, Eager M et al (2018) Rapid near-atomic resolution single-particle 3D reconstruction with SIMPLE. *J Struct Biol* 204:172–181. <https://doi.org/10.1016/j.jsb.2018.08.005>
82. Van Heel M, Harauz G, Orlova EV et al (1996) A new generation of the IMAGIC image processing system. *J Struct Biol* 116:17–24. <https://doi.org/10.1006/jsbi.1996.0004>
83. Grigorieff N (2007) FREALIGN: high-resolution refinement of single particle structures. *J Struct Biol* 157:117–125. <https://doi.org/10.1016/j.jsb.2006.05.004>
84. Hohn M, Tang G, Goodyear G et al (2007) SPARX, a new environment for Cryo-EM image processing. *J Struct Biol* 157:47–55. <https://doi.org/10.1016/j.jsb.2006.07.003>
85. Baxter WT, Leith AD, Frank J (2007) SPIRE: the SPIDER reconstruction engine. *J Struct Biol* 157:56–63. <https://doi.org/10.1016/j.jsb.2006.07.019>
86. Heymann JB, Belnap DM (2007) Bsoft: Image processing and molecular modeling for electron microscopy. *J Struct Biol* 157:3–18. <https://doi.org/10.1016/j.jsb.2006.06.006>
87. Timcenko M, Lyons JA, Janulienė D et al (2019) Structure and autoregulation of a P4-ATPase lipid flippase. *Nature*. <https://doi.org/10.1038/s41586-019-1344-7>
88. Booth DS, Avila-Sakar A, Cheng Y (2011) Visualizing proteins and macromolecular complexes by negative stain EM: from grid preparation to image acquisition. *J Vis Exp*. <https://doi.org/10.3791/3227>
89. Ohi M, Li Y, Cheng Y, Walz T (2004) Negative staining and image classification – powerful tools in modern electron microscopy. *Biol Proced Online* 6:23–34. <https://doi.org/10.1251/bpo70>
90. Briggs JAG, Huiskonen JT, Fernando KV et al (2005) Classification and three-dimensional reconstruction of unevenly distributed or

- symmetry mismatched features of icosahedral particles. *J Struct Biol* 150:332–339. <https://doi.org/10.1016/j.jsb.2005.03.009>
91. Serna M (2019) Hands on methods for high resolution cryo-electron microscopy structures of heterogeneous macromolecular complexes. *Front Mol Biosci* 6. <https://doi.org/10.3389/fmolb.2019.00033>
 92. Kastner B, Fischer N, Golas MM et al (2008) GraFix: sample preparation for single-particle electron cryomicroscopy. *Nat Methods* 5:53–55. <https://doi.org/10.1038/nmeth1139>
 93. Stark H (2010) GraFix: Stabilization of fragile macromolecular complexes for single particle Cryo-EM. *Methods in Enzymology*. Academic Press Inc., In, pp 109–126
 94. Suloway C, Pulokas J, Fellmann D et al (2005) Automated molecular microscopy: the new Legion system. *J Struct Biol* 151:41–60. <https://doi.org/10.1016/j.jsb.2005.03.010>
 95. Zheng SQ, Palovcak E, Armache JP et al (2017) MotionCor2: anisotropic correction of beam-induced motion for improved cryo-electron microscopy. *Nat Methods* 14:331–332
 96. Zivanov J, Nakane T, Scheres SHW (2019) A Bayesian approach to beam-induced motion correction in cryo-EM single-particle analysis. *IUCrJ* 6:5–17. <https://doi.org/10.1107/S205225251801463X>
 97. Pettersen EF, Goddard TD, Huang CC, et al (2004) UCSF Chimera – A visualization system for exploratory research and analysis. *J Comput Chem* 25:1605–1612. <https://doi.org/10.1002/jcc.20084>
 98. Punjani A, Zhang H, Fleet DJ (2020) Non-uniform refinement: adaptive regularization improves single-particle cryo-EM reconstruction. *Nat Methods* 17:1214–1221. <https://doi.org/10.1038/s41592-020-00990-8>



Helical Membrane Protein Crystallization in the New Era of Electron Cryo-Microscopy

Mary D. Hernando, Joseph O. Primeau, and Howard S. Young

Abstract

Helical assemblies of proteins, which consist of a two-dimensional lattice of identical subunits arranged with helical symmetry, are a common structural motif in nature. For membrane proteins, crystallization protocols can induce helical arrangements and take advantage of the symmetry found in these assemblies for the structural determination of target proteins. Modern advances in the field of electron cryo-microscopy (cryo-EM), in particular the advent of direct electron detectors, have opened the potential for structure determination of membrane proteins in such assemblies at high resolution. The nature of the symmetry in helical crystals of membrane proteins means that a single image potentially contains enough information for three-dimensional structural determination. With the current direct electron detectors, we have never been closer to making this a reality. Here, we present a protocol detailing the preparation of helical crystals, with an emphasis on further cryo-EM analysis and structural determination of the sarco(endo)plasmic reticulum Ca^{2+} -ATPase in the presence of regulatory subunits such as phospholamban.

Key words Two-dimensional crystals, Helical crystals, Electron cryo-microscopy, Electron crystallography, SERCA, Phospholamban

1 Introduction

Starting in the 1970s, landmark advances in the field of electron microscopy established cryo-EM as a powerful tool in structural biology. A pioneering structural study of purple membranes from halobacteria in the absence of negative stain yielded a 7 Å three-dimensional structure of bacteriorhodopsin, a light-activated proton pump [1, 2]. Low dose imaging of raw, unstained bacterial membranes demonstrated that electron microscopy of crystalline objects was a viable alternative to X-ray crystallography, and advantageous in that the protein of interest could be probed in the native context of its endogenous membrane. Shortly after this landmark

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work, Jacques Dubochet and colleagues developed a method to reliably view biological samples by embedding them in water rapidly frozen to trap its liquid form and the native state of the biological sample [3, 4]. Imaging in the electron microscope no longer required staining, and the technique now known as electron cryo-microscopy, or cryo-EM for short, has gone on to revolutionize modern structural biology. With the recent surge in molecular structures determined from dispersions of biological macromolecules, it is important to consider that the foundations of cryo-EM were enabled by crystalline samples such as bacteriorhodopsin and catalase. The application of cryo-EM to two-dimensional crystals became known as electron crystallography, and it has remained a powerful research technique for the structural analysis of mainly integral membrane proteins. The application of cryo-EM to small three-dimensional crystals, the resurgence of a technique now referred to as micro-crystal electron diffraction or micro-ED for short, has emerged as a powerful alternative for protein and small molecule structure determination.

Prior to the commercial introduction of direct electron detectors in 2010, high-resolution structures of biological macromolecules were largely the purview of X-ray crystallography. For integral membrane proteins, electron crystallography was considered an ideal technique because of the compatibility of a membrane bilayer with the formation of planar two-dimensional crystals. Conversely, the requirement for a membrane bilayer was considered a major obstacle in the formation of three-dimensional crystals for X-ray crystallography. Of course, X-ray crystallography prevailed in the arena of membrane protein structures until the introduction of direct electron detectors catapulted single particle analysis into the realm of routine molecular resolution, even for membrane proteins. Nonetheless, electron crystallography continues to provide novel structures of membrane proteins in a lipid bilayer. Some recent examples include members of the aquaporin family [5] with AQP0 determined at the highest resolution to date of 1.9 Å [6], the gastric H⁺,K⁺-ATPase at a moderate resolution of 6.5 Å [7], and a cyclic-nucleotide modulated potassium channel at near-molecular resolution of 4.0 Å [8]. Besides these success stories, there have been numerous reports of membrane protein two-dimensional crystals with the majority being suitable for only low-resolution structure determination—in part, due to defects in the two-dimensional crystal lattice. Modern cryo-EM and direct electron detectors are just beginning to impact the field of electron crystallography, and it is likely that the combination of dose-fractionation and single particle analysis methods will improve poorly ordered two-dimensional crystals.

Since the 1980s, helical crystallization—a variation of two-dimensional crystallization—has been a viable alternative for the structure determination of membrane proteins. The

identification of helical crystals coincided with the development of cryo-EM, and the combined approaches provided the initial structural models for membrane transport proteins such as the nicotinic acetylcholine receptor [9, 10] and the sarcoplasmic reticulum calcium pump (a.k.a. Ca^{2+} -ATPase or SERCA [11, 12]). Helical assemblies are prevalent in nature, including actin filaments [13], tau filaments [14], bacterial flagella [15], amyloid fibrils [16], and tobacco mosaic virus [17], just to name a few. By virtue of their helical arrangement, a single image in an electron microscope contains enough information to calculate a three-dimensional structure; albeit the structure may be limited by factors that degrade image quality. Modern direct electron detectors restore image quality, and the averaging of many images can achieve high resolution. For membrane proteins, helical crystals offer an additional advantage of the presence of a membrane bilayer, often in the absence of detergents. The automation of modern cryo-EM and the dose-fractionation and electron counting capabilities of direct electron detectors have advanced the analysis of helical assemblies such as actin filaments, tau filaments, bacterial flagella, and amyloid fibrils. The helical filaments can be picked manually, and partially overlapping segments can be automatically selected along the length of the helix. These segments can be treated much the same as single particle images, including motion-correction, dose-weighting, and reference-free 2D classification. The segments or class averages can then be used in reconstruction algorithms that combine helical and single particle analyses (e.g., RELION 2.0 [18] or IHRSR [19]).

Helical assemblies of membrane proteins represent a crystal morphology that lies between the larger two-dimensional crystals (e.g., bacteriorhodopsin) and helical protein assemblies (e.g., actin). Helical crystals of the sarcoplasmic reticulum calcium pump SERCA are on the order of 70 nm in diameter [20], while large 2D crystals of SERCA are approximately 500 nm wide [21]. By comparison, helical protein assemblies such as actin filaments and microtubules are 6 and 25 nm in diameter, respectively. All three sample types are amenable to dose-fractionation (i.e., movie mode), motion-correction, and dose-weighting to restore information content. For helical membrane protein crystals, the alignment of frames for a region that encompasses the helical crystal can partially correct for specimen movement and improve resolution [22]. Such images typically contain a long stretch of the helical object that includes multiple repeats of the underlying lattice. Thus, the helical image can be rendered into a series of smaller, partially-overlapping segments, and motion correction of these segments can further correct for specimen movement and improve resolution (e.g., [13]).

The larger membrane helical crystals such as those formed by SERCA [20] or the boron transporter Bor1p [23] usually come in a range of tube diameters originating from a common 2D lattice rolled into different helical symmetries. Understanding and limiting the helical symmetries to one or a few classes is essential for 3D reconstruction. The larger helical crystals can also have defects such as helix flattening, variable twist, and the non-linear path of the helix through the vitreous ice. Extracting the maximum amount of information (i.e. resolution) means correcting for these distortions [24] or restricting acceptable segments to a narrow range of helical and orientation parameters (e.g., [25]). The implementation of these procedures should improve the general applicability of helical crystallization and reconstruction methods, even for the larger diameter tubes produced by membrane proteins. The helical crystals formed by SERCA are a challenging target for structure determination using these approaches. Because they are wider than many membrane protein helical crystals, the variability in helical symmetry and the potential for distortions are more pronounced. Nonetheless, the ability to co-crystallize SERCA with its regulatory subunits phospholamban and sarcolipin make it an attractive test of modern methods. This chapter revisits the protocol for cocrystallization of SERCA and its regulatory protein PLN as a basis for future cryo-EM structure determination combining helical and single particle analysis methods.

2 Materials

The materials required for this experimental procedure will be divided into three principal tasks required for helical crystallization and sample preparation for cryo-EM.

2.1 Co-reconstitution of SERCA and Phospholamban

2.1.1 Stock Solutions

1. *Ca²⁺-ATPase storage solution* (0.1% w/v C₁₂E₈, 20% v/v glycerol, 1 mmol/L CaCl₂, 20 mmol/L MOPS pH 7.0, 1 mmol/L DTT) (120 mL) (see Note 1).

To a glass beaker, add 70 mL npH₂O and a magnetic stir bar. Begin stirring and add the following volumes of stock solutions: 1.2 mL 10% w/v C₁₂E₈ detergent solution, 30 mL 80% v/v glycerol, 240 µL 0.5 mol/L CaCl₂, 4.8 mL 0.5 mol/L MOPS pH 7.0, 120 µL 1 mol/L DTT. Adjust the volume to 120 mL with npH₂O and aliquot 1.5 mL into microcentrifuge tubes. Store at −20 °C.

2. *Lipid stock solution* (see Note 2) (8 EYPC-1 EYPE-1 EYPA molar ratio).

To a 12 × 75 mm disposable glass culture tube, measure and add (using a gas-tight syringe) 65 µL 25 mg/mL phosphatidylcholine (egg yolk phosphatidylcholine, EYPC, PC),

10 μL 25 mg/mL phosphatidylethanolamine (egg yolk phosphatidylethanolamine, EYPE, PE), and 10 μL 25 mg/mL phosphatidic acid (egg yolk phosphatidic acid, EYPA, PA; *see Note 8*) (Avanti Polar Lipids, Alabaster, AL, USA). Prepare immediately before forming a thin-film of PLN and lipid.

3. *Phosphatidylcholine stock solution, 10 mg/mL (1 mL).*

Using a gas-tight syringe, add 400 μL of 25 mg/mL phosphatidylcholine to a 12×75 mm disposable glass culture tube (*see Note 3*). Dry down to a thin film by streaming $\text{N}_2(\text{g})$ onto the surface of the stock solution while vortexing. Once a thin film of lipid has formed, place in a desiccator under vacuum overnight. The following day, resuspend the thin film in 1 mL of npH_2O by vortexing. Aliquot 100 μL into microcentrifuge tubes and store at -20°C .

2.1.2 Co-reconstitution

1. *4 × 1 mm stir bars.*

2. *Purified Ca^{2+} -ATPase* Prepare purified Ca^{2+} -ATPase (SERCA1a) as described in [26].

3. *Purified phospholamban (lyophilized protein).*

Prepare purified phospholamban as described in [27].

4. *BioBeads*

Prepare SM-2 BioBeads in bulk (Bio-Rad, Hercules, CA, USA) before use. Place dry BioBeads in a 50 mL conical centrifuge tube and add methanol until all BioBeads are fully submerged. Place on a nutator for a minimum of 1 h. Decant methanol and add npH_2O until all BioBeads are fully submerged and nutate for a minimum of 1 h. Decant water and repeat. Once the final wash has been completed, store BioBeads in 40 mL of 0.02% w/v azide solution at 4°C .

5. *Reconstitution salt buffer (8 mmol/L imidazole pH 7.0, 20 mmol/L KCl) (65 μL).*

To a 1.5 mL microcentrifuge tube, add 40 μL 0.5 mol/L imidazole pH 7.0 and 25 μL 2 mol/L KCl. Vortex to mix. Prepare on the same day as reconstitution.

2.2 Crystallization

All solutions prepared for crystallization should be stored on ice at all times and used immediately (unless otherwise noted).

1. *Decavanadate solution, 50 mmol/L (10 mL).*

The preparation of this solution should be done on ice.

In a glass beaker, add ~ 7.75 mL npH_2O and a small magnetic stir bar. Add 0.092 g sodium orthovanadate (Na_3VO_4) (Sigma-Aldrich, St. Louis, MO, USA) and begin stirring (*see Note 4*). Lower the pH to ~ 1.8 by dropwise addition of 5 mol/L HCl (*see Note 5*) and stir for 30 min (*see Note 6*). Slowly raise the pH to between 4.0 and 5.0 using cold 1 mol/L KOH dropwise, while stirring. Raise the pH to between 6.7

and 7.0 using 0.1 mol/L KOH (*see* **Notes 7 and 8**). Adjust the volume to 10 mL with cold npH₂O and store on ice. The polyanion solution can remain stable for 3–5 days (*see* **Note 9**).

2. *Crystallization master mix (20 mmol/L imidazole pH 7.0, 100 mmol/L KCl, 0.5 mmol/L EGTA pH 8.0, 0.02% w/v azide) (9.4 mL).*

In a 15 mL conical centrifuge tube add: 8.36 mL npH₂O, 400 μ L 0.5 mol/L imidazole pH 7.0, 500 μ L 2 mol/L KCl, 100 μ L 50 mmol/L EGTA pH 8.0, and 40 μ L 5% w/v azide. Vortex to mix and store on ice.

3. *Crystallization buffer (20 mmol/L imidazole pH 7.0, 100 mmol/L KCl, 0.5 mmol/L EGTA pH 8.0, 0.02% w/v azide, 5–15 mmol/L MgCl₂, 0.25–0.50 mmol/L decavanadate) (1 mL).*

To a 1.5 mL microcentrifuge tube add 940 μ L of the crystallization master mix. Next, add the desired volume of 1 mol/L MgCl₂ (*see* **Note 10**) and the desired volume of 50 mmol/L decavanadate solution, with npH₂O composing the remaining volume (*see* **Note 11**).

4. *Thapsigargin solution, 1 mg/mL (1 mL).*

In a 2 mL glass vial, add 1 mg of thapsigargin to 1 mL of ice-cold 100% ethanol. Vortex to mix and store at -80°C .

2.3 Electron Cryo-Microscopy

1. *Whatman filter paper.*

2. *Carbon coated Cu/Rh grids, 400 mesh.*

400 mesh Maxtaform Rh-flashed Cu grids (Ted Pella, Redding, CA, USA) were carbon coated by floating a thin sheet of carbon, as adapted from [28], onto the grids.

3. *Uranyl acetate, 2% w/v (10 mL).*

To a 15 mL conical centrifuge tube, add ~9 mL of npH₂O, followed by 0.2 g of uranyl acetate (*see* **Note 12**) and vortex to mix. Filter through a 0.22 μ m filter and store in a 15 mL conical centrifuge tube covered in aluminum foil. Filter before each use.

4. *Lacey carbon grids, 300 mesh.*

Lacey carbon grids were home-made using 300-mesh Maxtaform Rh-flashed Cu grids (Ted Pella, Redding, CA, USA) as adapted from [29]. Holey plastic was floated onto the grid surface, followed by carbon coating and plastic dissolution using chloroform.

5. *Quantifoil grids, R2/2.*

R2/2 Quantifoil support-film coated, 300-mesh, Cu grids (Quantifoil Micro Tools, Großlöbichau, Germany) were purchased and lightly carbon coated prior to sample application.

6. *Vitrification.*

Vitrobot Mark IV (Thermo-Fisher, Hillsboro, OR, USA) is used for plunge freezing of sample coated grids. Liquid ethane cryogen is condensed in a brass cup thermo-coupled to liquid nitrogen (*see Note 13*).

7. *Electron microscope.*

Tecnai F20 200KV TEM (Thermo-Fisher, Hillsboro, OR, USA) equipped with a Falcon II direct electron detector and a Gatan 626 single-tilt cryo holder. Automated imaging under low dose conditions conducted with EPU software.

3 Methods

3.1 Reconstitution

3.1.1 Preparation of Reagents for Reconstitution

1. *CaCl₂, 0.5 mol/L (50 mL).*

Weigh out 2.77 g of CaCl₂ and add to ~40 mL of npH₂O in a glass beaker with a magnetic stir bar. Stir until mixed into a clear solution. Filter the solution using a 0.22 µm filter (Millipore) and adjust the volume with npH₂O to 50 mL using a volumetric flask. Store at room temperature in a 50 mL conical centrifuge tube. Can be stored for several weeks.

2. *EGTA, 50 mmol/L pH 8.0 (50 mL).*

Weigh out 0.95 g of EGTA (ethylene glycol tetraacetic acid, ethylene glycol-bis-(β-aminoethyl ether)-N,N,N',N'-tetraacetic acid, egtazic acid) (SIGMA ALDRICH) and add to ~40 mL of npH₂O in a glass beaker with a magnetic stir bar. EGTA will remain insoluble until a pH near 7.0 is reached. Raise the pH by the dropwise addition of 5 mol/L NaOH into the mixture (while stirring) until the desired pH of 7.0 is reached (solution should become clear). As the solution nears a pH of 7.0, avoid over-shooting by the dropwise addition of 1 mol/L NaOH. Adjust the volume with npH₂O to 50 mL and filter using a 0.22 µm filter (Millipore). Store at 4 °C in a 50 mL conical centrifuge tube. Can be stored for several weeks.

3. *MgCl₂, 1 mol/L (50 mL).*

Weigh out 4.76 g of MgCl₂ and add to ~40 mL of npH₂O in a glass beaker with a magnetic stir bar. Stir until mixed into a clear solution. Adjust the volume with npH₂O to 50 mL and filter using a 0.22 µm filter (Millipore). Store at 4 °C in a 50 mL conical centrifuge tube. Can be stored for several weeks.

4. *KCl, 2 mol/L (50 mL).*

Weigh out 7.46 g of KCl and add to ~40 mL of npH₂O in a glass beaker with a magnetic stir bar. Stir until mixed into a clear solution. Adjust the volume with npH₂O to 50 mL and filter using a 0.22 µm filter (Millipore). Store at 4 °C in a 50 mL conical centrifuge tube. Can be stored for several weeks.

5. *Imidazole, 0.5 mol/L pH 7.0 (50 mL).*

Weigh out 1.70 g of imidazole and add to ~40 mL npH₂O in a glass beaker with a magnetic stir bar. Stir until mixed and titrate with 5 mol/L HCl (*see Note 5*) dropwise until the desired pH of 7.0 is reached. Adjust the volume with npH₂O to 50 mL and filter using a 0.22 µm filter (Millipore). Store at 4 °C in a 50 mL conical centrifuge tube. Can be stored for several weeks.

6. *MOPS, 0.5 mol/L pH 7.0 (50 mL).*

Weigh out 5.23 g of MOPS (3-(*N*-morpholino)propane-sulfonic acid) and add to ~40 mL npH₂O in a glass beaker with a magnetic stir bar. Stir until mixed and titrate with 5 mol/L HCl dropwise until the solution reaches the desired pH of 7.0. Adjust the volume with npH₂O to 50 mL and filter using a 0.22 µm filter (Millipore). Store at 4 °C in a 50 mL conical centrifuge tube. Can be stored for several weeks.

7. *Azide, 5% w/v (50 mL).*

Weigh out 2.5 g of sodium azide (NaN₃) (*see Note 14*) and add to ~40 mL of npH₂O in a glass beaker with a magnetic stir bar. Stir until mixed into a clear solution. Adjust the volume with npH₂O to 50 mL. Store at room temperature in a 50 mL conical centrifuge tube. Can be stored for months.

8. *Glycerol, 80% v/v (200 mL).*

Measure out 160 mL of glycerol and add to a glass beaker with a magnetic stir bar. Adjust the volume to 200 mL with npH₂O and transfer to a storage vessel of appropriate size for autoclaving (*see Note 15*). Autoclave the vessel and store at room temperature. Can be stored for months.

9. *DTT, 1 mol/L (10 mL).*

Weigh out 1.54 g of DTT (dithiothreitol) and add to a 15 mL conical centrifuge tube with ~9 mL npH₂O. Vortex until solution is clear. Adjust the volume with npH₂O to 10 mL and aliquot 1 mL into 1.5 mL microcentrifuge tubes. Tubes are then stored at -20 °C. Can be stored for months.

10. *Sucrose storage solution, 50% w/v (20 mmol/L imidazole pH 7.0, 100 mmol/L NaCl, 0.02% azide w/v) (250 mL).*

Weigh out 125 g of sucrose and add to a glass beaker containing ~150 mL of npH₂O and a magnetic stir bar. Begin stirring the mixture until a clear solution is formed. Add 1.46 g of NaCl to the cleared solution and continue stirring. To this add 10 µL imidazole stock solution (*see Note 16*) and 1 mL azide stock solution and continue mixing until cleared. Adjust the volume with npH₂O to 250 mL and store at 4 °C. Can be stored for several weeks.

11. *Sucrose storage solution, 20% w/v (20 mmol/L imidazole pH 7.0, 100 mmol/L NaCl, 0.02% w/v azide) (250 mL).*

Weigh out 50 g of sucrose and add to a glass beaker containing ~150 mL of npH₂O and a magnetic stir bar. Begin stirring the mixture until a clear solution is formed. Add 1.46 g of NaCl to the solution and continue stirring. To this add 10 μ L imidazole stock solution (*see Note 16*) and 1 mL azide stock solution and continue mixing until clear. Adjust the volume with npH₂O to 250 mL and store at 4 °C. Can be stored for several weeks.

12. *C*₁₂E₈ detergent solution, 10% w/v (25 mL).

Weigh out 2.5 g of C₁₂E₈ (octaethylene glycol monododecyl ether; Nikkol Chemicals Co., Ltd., Japan) in a glass beaker containing 25 mL of npH₂O and a magnetic stir bar. Stir the mixture until a clear solution is formed (*see Note 17*).

3.1.2 *Preparation of Lipid-Peptide Thin-Films*

One day prior to proteoliposome reconstitution, prepare thin-films using lipid stock solution and purified PLN. This protocol assumes that four independent proteoliposome reconstitutions are occurring in parallel.

1. Resuspend 100 μ g of lyophilized PLN in 100 μ L of 80:20 tetrafluoroethane (TFE)–npH₂O (*see Note 18*) and equilibrate at room temperature for 5–10 min.
2. Using a gas-tight syringe, add 17 μ L of lipid stock solution to the tube containing resuspended PLN (*see Note 19*).
3. Dry down the mixture under a gentle stream of nitrogen while vortexing using the lowest setting. The goal is to generate a uniform thin-film within the bottom centimeter of the tube.
4. Store under a vacuum in a desiccator overnight (*see Note 20*).

3.1.3 *Coreconstitution of SERCA/Peptide into Proteoliposomes*

1. Resuspend the lipid-peptide thin-films in 140 μ L npH₂O by vortexing. The sample will appear cloudy.
2. Heat at 50 °C in a dry block for 30 min with intermittent vortexing (*see Note 21*). Allow the resuspended lipid-peptide mixture to cool to room temperature.
3. Add 7 μ L of 10% w/v C₁₂E₈.
4. Vortex at a high setting continuously for 3 min.
5. Add 13 μ L of reconstitution salt buffer to the lipid-peptide mixture.
6. Vortex thoroughly and remove any remaining thin-film from the sides of the tube using a microscopula.
7. Calculate the volume of SERCA required for 500 μ g of protein (*see Note 22*) or approximately 1.0 mg/mL concentration. If the required volume of SERCA is less than 300 μ L, determine the volume of Ca²⁺-ATPase storage buffer to bring the total volume added to 300 μ L.

8. If needed, add the calculated volume of Ca^{2+} -ATPase storage buffer and vortex.
9. Add the required volume of purified SERCA (calculated in **step 7** above). At this step, avoid aggressively mixing the sample (vortexing) as this may inactivate and denature the enzyme.
10. Add a 1 mm magnetic stir bar to the reconstitution mixture and stir at a low-medium setting at room temperature for 10–15 min.
11. Add 1.5 mg of BioBeads to the reconstitution mixture; BioBeads will be added slowly over 4 h at room temperature as follows: 1.5 mg of BioBeads are added after 30, 60, and 90 min; 7 mg of BioBeads are added after 120 min, and an excess amount of BioBeads (approximately 18 mg) are added after 180 min. Detergent removal is complete (remove from stirring) at 240 min (*see Note 23*).
12. In 11 × 34 mm polycarbonate centrifuge tubes, layer 800 μL of 20% w/v sucrose storage solution on top of 800 μL of 50% w/v sucrose storage solution to form a two-layer sucrose step-gradient (*see Note 24*).
13. Carefully layer the reconstitution mixture (approximately 500 μL) on top of the sucrose gradient (*see Note 25*).
14. Using a swinging-bucket rotor, centrifuge the coreconstitution containing sucrose gradient at $100,000 \times \text{rcf}$ and 4 °C for 1–1.5 h using gentle acceleration and deceleration (e.g., TLS 55 rotor in an Optima TLX Ultracentrifuge; Beckman Coulter, CA, USA).
15. A white, cloudy layer will form at the 50% w/v sucrose interface. Collect the layer using three 200 μL extractions with a micropipette and store the collected solution in a screw-top 1.8 mL microcentrifuge tube (*see Note 26*).
16. Flash-freeze the proteoliposomes in liquid nitrogen and store at –80 °C.

3.2 Crystallization

3.2.1 Preparation of 2D Crystals

1. Remove frozen proteoliposomes in microcentrifuge tubes from –80 °C and place on ice for 5–10 min.
2. Add 400 μL of crystallization master mix to the reconstitution and gently mix by pipetting up and down repeatedly. This is a rinsing step to remove sucrose from the stored proteoliposomes.
3. Centrifuge the proteoliposomes at $40,000 \times \text{rcf}$ and 4 °C for 35 min (e.g., F1202 rotor and Allegra 64R Centrifuge; Beckman Coulter, CA, USA). A pellet will form at the bottom of the tube.
4. Carefully aspirate the supernatant, taking care not to disturb the pellet.

5. Resuspend the pellet in 200 μL of crystallization buffer and centrifuge again at $40,000 \times \text{rcf}$ and 4°C for 35 min. The pellet will now appear to be a pale yellow.
6. Aspirate the supernatant and add 15–20 μL of crystallization buffer and 1 μL of thapsigargin solution to the pellet (*see Note 27*).
7. Without resuspending the pellet, gently mix the crystallization buffer and thapsigargin solution with a micropipette.
8. Perform two freeze–thaw cycles as follows: flash-freeze the reconstitutions by submersing the microcentrifuge tubes in liquid nitrogen, then gently thaw them between the thumb and forefinger. Tubes can be rolled between the fingers and gently tapped to ensure the mixture is just thawed. Do not allow samples to become warm. Once thawed, repeat.
9. After the second freeze–thaw cycle, gently resuspend the pellet with a micropipette avoiding the production of air bubbles.
10. Perform two additional freeze–thaw cycles.
11. After the final freeze–thaw cycle, incubate the sample at 4°C . Retain the crystallization master mix and crystallization buffer, storing at 4°C (*see Note 28*).

3.3 Cryo-EM

3.3.1 Sample

Preparation for Cryo-EM Negative Stain Screen

After allowing the crystals to form over 3–5 days at 4°C , the samples are screened by negative stain EM for the presence of crystals deemed suitable for further analysis by cryo-EM (*see Fig. 1*). **Steps 2–6** should be done at 4°C in a humid environment.

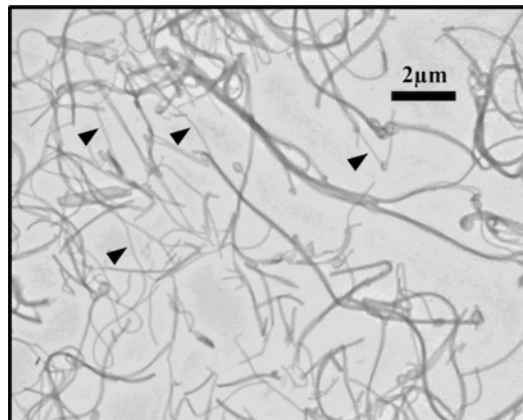


Fig. 1 An example micrograph of negatively stained SERCA and PLN helical membrane crystals (arrows) magnified at $2000\times$ on a portion of a grid square. The distribution and number of crystals would be suitable for further analysis by cryo-EM. The image was collected using a CCD camera and a Hitachi H-7650 TEM operated at 60 kV

1. Glow discharge carbon-coated 400 mesh Cu/Rh grids (*see Note 29*).
2. Place grids in self-closing tweezers (*see Note 30*) and apply 5 μL of the sample containing thin helical crystals. Allow sample to adsorb for 30–60 s.
3. Blot the grid by touching the edge of the grid with the edge of a torn strip of Whatman filter paper (grade No. 1) and quickly apply 5 μL of filtered 2% w/v uranyl acetate solution. Allow the negative stain to remain on the grid for 30–60 s.
4. Blot the grid dry by touching the edge of the grid with the edge of a torn strip of Whatman filter paper (grade No. 1). Bring the sample to a drier environment and allow to air-dry for 15 min.
5. Store negatively stained grids in a grid storage box for future use.
6. Image samples using a TEM to determine the quality of thin helical crystals generated. For a typical 300 or 400 mesh grid, there should be a large number of helical crystals per grid square (*see Fig. 2*) and the helical crystal lattice should be clearly visible.

3.3.2 Electron Cryo-Microscopy

There is some user preference in choosing the type of holey carbon grids for freezing and imaging helical crystals. The goal for cryo-EM imaging is to achieve a high frequency of single helical crystals spanning the center of holes in the carbon support film. Using a manufactured carbon support film with a regular array of holes, such as Quantifoil grids (Quantifoil Micro Tools GmbH) or C-flat grids (Protochips), can be useful for automated image acquisition. However, lacey carbon support films have a higher hole density and more heterogeneous hole size. This increases the probability that a helical crystal will be found spanning the center of a hole in the carbon support film and allows for optimal and more frequent image acquisition [30, 31] (*see Fig. 3*).

1. Lightly carbon-coat and glow discharge both sides of lacey/holey carbon grids.
2. Prepare liquid ethane cryogen (*see Notes 17 and 31*) as per [29]. Liquid ethane contained in a metal vessel and partially immersed in liquid N_2 achieves a temperature of -180°C , which is near the freezing point of ethane (-182.8°C). This temperature can be maintained and ensures that the vitrification of water can be achieved.
3. Dilute the helical crystal sample 1:10 or 1:20 with crystallization buffer (*see Note 32*).
4. Pipette 5 μL of the helical crystal dilution onto a lacey/holey carbon grid held by a pair of fine-tipped forceps. Because the grids have been lightly carbon coated and glow discharged on

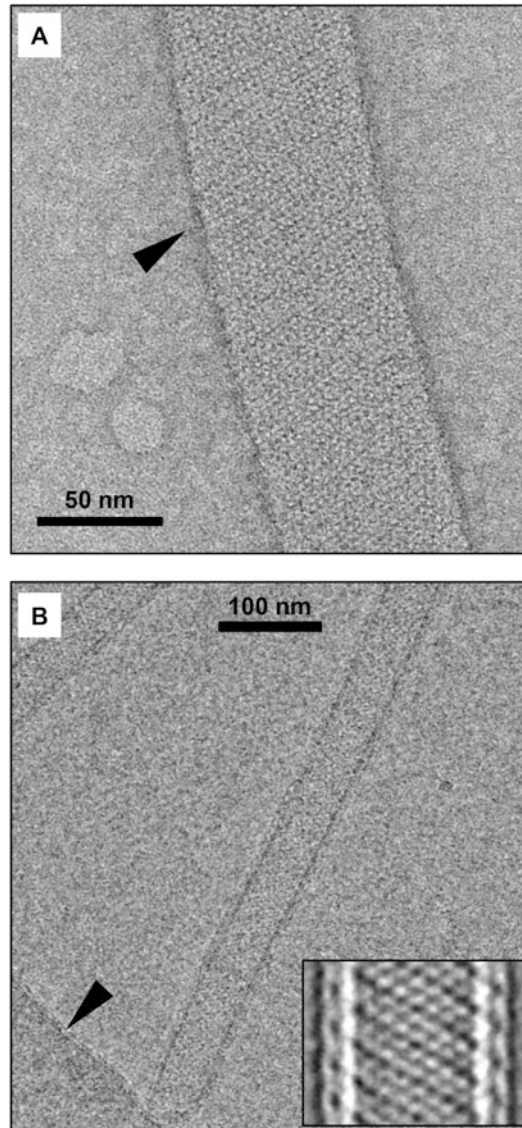


Fig. 2 Example image of helical crystals highlighting the lattice appearance at higher magnification. **(a)** A negatively stained helical crystal of SERCA and PLN showing the repeating array of SERCA dimer ribbons (arrow) running along the helical axis of the membranous tube. Note that though this image has high contrast, it lacks high-resolution information useful for structural determination ($50,000\times$ magnification). **(b)** The same helical crystal preparation imaged in the frozen-hydrated state. While the helical lattice is still apparent, the SERCA dimer ribbons are less visible because of the low inherent contrast ($29,000\times$ magnification). The arrow indicates the edge of a hole in the lacey carbon support film. Inset image is a class average for a limited set of segments for one helical class (motion correct has not been implemented). The density has been inverted such that higher densities are shaded white. Images were collected using a Tecnai F20 TEM equipped with a Falcon II direct electron detector using low-dose conditions

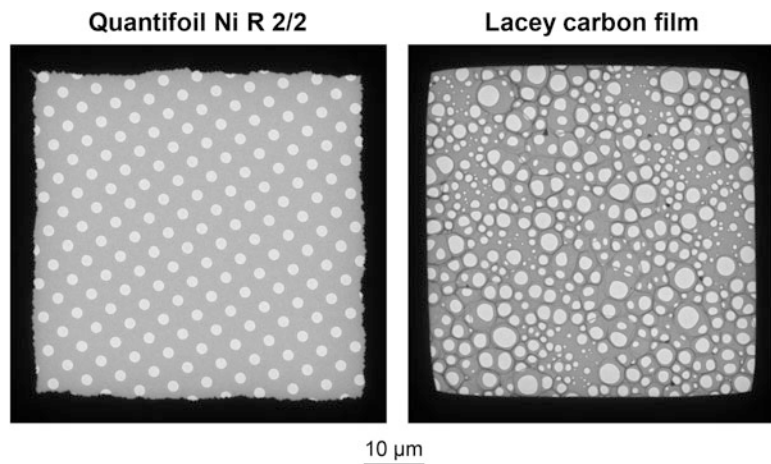


Fig. 3 Example images highlighting the differences between a Quantifoil holey carbon grid and a homemade lacey carbon grid. The Quantifoil grid is Cu R 2/2 and has a regular hole size and spacing in the carbon support film (2 μm hole size). The lacey carbon support film has a variable size and distribution of holes. Images of bare grids were collected using a Tecnai F20 TEM and an Eagle CCD camera under low-dose conditions at $97\times$ magnification

both sides, the solution will flow through the holes in the carbon support film and the 5 μL volume will distribute to both sides of the grid. This minimizes the blotting forces that can distort the helical crystals in the next few steps.

5. Mount the forceps in a plunge freezing apparatus of choice (*see* **Notes 33** and **34**).
6. Blot the solution on the grid to a very thin film using Whatman paper (grade No. 1; or Vitrobot blotting paper) and immediately plunge freeze into liquid ethane to vitrify the sample.
7. Once vitrified, store sample in a cryo grid storage box in liquid N_2 until needed for cryo-EM image acquisition. From this point on, the grid must be maintained at or near liquid nitrogen temperatures (-180°C or lower).
8. Once ready for cryo-EM, pre-cool the cryo-holder or autoloader in its respective transfer station using liquid N_2 ($\sim -185^\circ\text{C}$) (*see* **Note 35**). Load one or more cryo-grids as per the recommended procedure for the cryo-holder (typically one grid) or autoloader (e.g., 12 grids).
9. Carefully insert the sample-containing cryo-holder or autoloader into the microscope (*see* **Note 36**).
10. Allow 30–60 min for the temperature and vacuum to stabilize.
11. In a suitable 200 or 300 kV TEM, collect low-dose images ($\sim 20 \text{ e}^-/\text{\AA}^2$) of vitrified helical crystals suspended across the center of holes in the carbon support film (*see* **Note 37**). Image acquisition can be performed manually (pre-screening and

selection of suitable holes containing helical crystals) or automated (e.g., EPU and Quantifoil grids).

3.3.3 Image Processing Scheme

Images can be processed using a combination of helical reconstruction methods (e.g., [24, 32]) and single particle analysis methods (e.g., [33, 34]). This approach has not yet been applied to the large membranous helical crystals such as those described herein. Therefore, the following steps are based on a segmented helical image reconstruction approach (e.g., [35–37]).

1. Suitable images of helical crystals are evaluated for image quality by calculation of the Fourier transform and observation of the helical layer lines and Thon rings before and after whole-image motion correction. The highest resolution images are considered for further processing.
2. A region of the helical crystal is selected based on optimizing and maximizing the signal and resolution of the helical layer lines. This results in a rectangular boxed image composed of an integer number of helical repeats.
3. The images are sorted and grouped into helical classes, largely distinguishable by their diameters and helical layer line characteristics (positions, amplitudes, and phases). Helical symmetries are assigned for each class.
4. Each helical crystal is segmented into partially overlapping boxes along the length of the crystal. The boxes should be the same size within the helical class and related to the helical symmetry.
5. Low-resolution models and projections are generated for each helical symmetry and they are cross-correlated with the helical segments to assign a set of Euler angles Φ , Θ , and Ψ , and x , y translations.
6. With these parameters, the segments are reconstructed and averaged, and the helical symmetry is applied within each helical class. The resultant 3D volume is used for a new round of projection matching and the cycle is repeated until convergence.

The advances enabled by direct electron detectors, combined with helical reconstruction and single particle methods, should allow high resolution structures of large membranous helical crystals. For the sarcoplasmic reticulum calcium pump (SERCA), helical crystals provided the first structural insights [12] and they have provided a low resolution structure of the bound regulatory subunit phospholamban [38, 39]. The protocol described above can be used to generate helical crystals of SERCA in the presence of each of its regulatory subunits, including phospholamban, sarcolipin, and the “regulins” [40, 41]. In addition, we have a nearly complete understanding

of SERCA structure and function through X-ray crystallography (for a recent review, *see* [42]). Thus, high resolution structures of SERCA in complex with these regulatory subunits are the next horizon in SERCA structure and function. While high-resolution single particle methods may allow structure determination of SERCA complexes in detergents, bicelles, or nanodiscs, helical crystals maintain the complexes in a membrane environment with the additional benefit of repeating crystalline assemblies. A potential disadvantage is that the large membranous helical crystals are more prone to distortions such as bending and flattening. However, the automation and high-throughput capabilities of current generation electron microscopes will facilitate sufficient data collection to overcome these issues through the selection of well-preserved helical crystals. And one must recall that a single image of a helical crystal has the potential to provide complete 3D structural information. While this is likely out reach of current cryo-EM methodology and the high-resolution capabilities of direct electron detectors, it will be interesting to evaluate image datasets for these types of helical crystals.

4 Notes

1. Prior to the addition of Ca^{2+} -ATPase storage solution to the reconstitution mixture, phosphatidylcholine must be added to a final concentration of 0.25 mg/mL.
2. The final molar ratio of the reconstitution is 8 EYPC-1 EYPE-1 EYPA and 500 μg total lipid per reconstitution (final lipid-protein weight ratio of ~ 1.25). The lipid addition should take into account the lipid included in the Ca^{2+} -ATPase stock solution addition (75 μg).
3. Caution should be taken to avoid inhaling fumes, as the lipids are stored in chloroform. Additionally, take care to keep the solution cold, use immediately, and keep the container sealed when not in use. Chloroform evaporates rapidly and may change the concentration of the lipid stock solution.
4. The Na_3VO_4 will initially appear insoluble, but it will rapidly dissolve with the addition of acid to the solution.
5. Exercise caution while using concentrated acids. Wear the appropriate PPE and add 5 mol/L HCl dropwise with a Pasteur pipette. For increased accuracy and safety, when within ~ 0.5 pH units of the desired pH, begin adding 1 mol/L HCl dropwise until the solution is at the desired pH. Heat at 50 $^\circ\text{C}$ in a dry block for 30 min with intermittent vortexing (*see* **Note**

- 21). Allow the resuspended lipid-peptide mixture to cool to room temperature.
6. The preparation of the decavanadate polyanion follows several color-change cues (indicating the formation of the particular polyanion favored at the desired pH). At pH lower than 2.0, the solution will appear as a pale-yellow/straw color. At a pH between 2.0 and 7.0 the solution appears to be a gold-amber color.
 7. This is a critical step in the formation of decavanadate (as this solution is unbuffered and very sensitive to changes in pH). When nearing the desired pH, begin performing serial dilutions of 0.1 mol/L KOH (~1:2) and adding dropwise to carefully raise the pH.
 8. Should the desired pH of 7.0 be overshoot, or the pH rises too quickly, vanadate will precipitate in a shimmering, gold-like precipitate (quite pretty, but very disappointing to see). If this happens, the preparation needs to be discarded and begun again.
 9. Decavanadate can form an orange precipitate over time at a pH near 7.0.
 10. In the development of a reliable protocol for generating thin helical crystals of SERCA and PLN, our lab has found that a magnesium concentration of 5–15 mmol/L in this buffer is critical in promoting the helical crystal morphology [43]. It is necessary to screen a range of magnesium concentrations to optimize helical crystal formation.
 11. Previous experience in our lab has found that a concentration of 0.25–0.50 mmol/L decavanadate is optimal for helical crystal formation.
 12. Exercise caution when handling uranyl acetate as it is both radioactive and toxic. Laboratory grade uranyl acetate is generally made with depleted uranium and is mildly radioactive and exceedingly toxic if consumed. However, the radiation is not harmful unless ingested. Review relevant MSDS literature, wear required PPE, and dispose of as per the relevant nuclear commission.
 13. Exercise caution when using liquid ethane. Liquid ethane is explosive and can cause extreme cold burns with skin contact.
 14. Exercise caution while handling sodium azide as it is acutely toxic. Consult the MSDS for more information.
 15. Exercise caution while operating an autoclave. Autoclaved liquid must be stored in a vessel suitable for high pressures and temperatures. The volume of autoclaved liquid should not exceed 50% of the total volume of the vessel, and any screwtop lids must not be fully sealed (at the risk of boil-over or vessel

explosion). Loosening the screw-cap lid by a $\frac{1}{4}$ turn should be enough for safe autoclaving.

16. Substituting 0.5 mol/L MOPS pH 7.0 stock solution in place of 0.5 mol/L imidazole pH 7.0 is acceptable for this buffer.
17. Avoid vigorously mixing this detergent solution as it may foam.
18. Other mixtures of organics and aqueous solutions have been used, such as 2:1 chloroform–TFE or 4:1 2-propanol–TFE. The use of primarily organic solutions for resuspension is recommended as the hydrophobic peptide become more soluble.
19. Lipid stock solution has enough material for five independent reconstitutions.
20. These tubes can be placed in a high-vacuum lyophilizer for 4 h to expedite the reconstitution process, just ensure all organics have been removed prior to lyophilization.
21. This step may require additional time to form a dispersed suspension of peptide and lipid. Failure to do this will result in incomplete proteoliposome reconstitution.
22. The minimum concentration of purified SERCA is ~ 1.7 mg/mL for effective reconstitution. SERCA optimally reconstitutes at a final concentration of 1 mg/mL. Any lower than this will dilute the reconstitution and create difficulty in obtaining repeatable proteoliposome formation.
23. The final amount of BioBeads to be added is a 25:1 weight-to-weight ratio of BioBeads to detergent.
24. Carefully layer the 20% weight-to-volume sucrose layer onto the 50% weight-to-volume sucrose layer to avoid mixing and to maintain a defined interface.
25. The presence of a few BioBeads in the applied reconstitution layer will not negatively affect the preparation of the reconstitutions.
26. SERCA alone will produce a diffuse, cloudy band at the 50% weight-to-volume sucrose interface; the presence of PLN will cause the band to form a punctate disc. If two bands separated by a large distance are observed, the reconstitution has likely produced proteoliposomes with different lipid to protein ratios and should be discarded and repeated.
27. Thapsigargin is useful in helical crystallization, though crystallization is possible in its absence [39]. If crystallizing in the absence of thapsigargin, adding the same volume of 100% ethanol has been found to aid in crystal formation.
28. Crystallization buffer is used to dilute helical crystal samples for cryo-EM investigation.

29. Glow-discharged grids maintain their hydrophilic state for ~24 h if stored in a cool humid environment.
30. Ensure the fine-tipped forceps are grabbing the edge of the grid to avoid damaging the carbon support film.
31. Other cryogens such as liquid propane can be used. We prefer ethane because it is dispersive in its gaseous form.
32. Ideal screening condition is the appearance of approximately 25 helical crystals per grid square. For lacey carbon grids, samples should be diluted to about 5 helical crystals per grid square. A lower dilution, more helical crystals, may be desirable for Quantifoil or C-flat grids.
33. If using a Vitrobot, change the environmental parameters to fit your needs, including blotting pressure, humidity, and blotting time. Finding the correct parameters may take some trial and error, as the environment of your EM facility (humidity, temperature, etc.) can vary from facility to facility. In addition, the composition of the sample storage buffer (ion composition, viscosity, detergent) can affect how well liquid distributes on the grid surface. A good starting point to consider is ~80% humidity, blotting force of 4, and blotting time of 2.5 s.
34. Our lab uses a Vitrobot Mark IV vitrification system to plunge-freeze grids; however, we also use a traditional guillotine-style plunge-freezing system.
35. For cryo-EM screening, we use a single-tilt Gatan 626 cryo-holder cooled with liquid nitrogen until it reaches $\sim -185^{\circ}\text{C}$. The grid is then placed in its respective slot in the holder (this will change depending on the model of the cryoholder or microscope).
36. Always follow manufacturer instructions for sample insertion. Failure to do so will cause vacuum collapse in the column, or worse, damage to the stage/goniometer.
37. During image acquisition, the illuminated area of the sample includes a helical crystal and some or all of the carbon surrounding the hole. This helps to distribute the charge and reduce image drift during exposure of the sample to electrons [44].

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References

- Henderson R, Unwin PN (1975) Three-dimensional model of purple membrane obtained by electron microscopy. *Nature* 257 (5521):28–32. <https://doi.org/10.1038/257028a0>
- Henderson R, Unwin PN (1977) Structure of the purple membrane from *Halobacterium halobium*. *Biophys Struct Mech* 3(2):121. <https://doi.org/10.1007/bf00535804>
- Dubochet J, Adrian M, Chang JJ, Homo JC, Lepault J, McDowell AW, Schultz P (1988) Cryo-electron microscopy of vitrified specimens. *Q Rev Biophys* 21:129–228
- Lepault J, Booy FP, Dubochet J (1983) Electron microscopy of frozen biological specimens. *J Microsc* 129:89–102
- Tani K, Fujiyoshi Y (2014) Water channel structures analysed by electron crystallography. *Biochim Biophys Acta* 1840(5):1605–1613. <https://doi.org/10.1016/j.bbagen.2013.10.007>
- Gonen T, Cheng Y, Sliz P, Hiroaki Y, Fujiyoshi Y, Harrison SC, Walz T (2005) Lipid-protein interactions in double-layered two-dimensional AQP0 crystals. *Nature* 438 (7068):633–638. <https://doi.org/10.1038/nature04321>
- Abe K, Shimokawa J, Naito M, Munson K, Vagin O, Sachs G et al (2017) The cryo-EM structure of gastric H(+),K(+)-ATPase with bound BYK99, a high-affinity member of K(+) competitive, imidazo[1,2-a]pyridine inhibitors. *Sci Rep* 7(1):6632. <https://doi.org/10.1038/s41598-017-06698-8>
- Righetto RD, Biyani N, Kowal J, Chami M, Stahlberg H (2019) Retrieving high-resolution information from disordered 2D crystals by single-particle cryo-EM. *Nat Commun* 10 (1):1722. <https://doi.org/10.1038/s41467-019-09661-5>
- Brisson A, Unwin PNT (1984) Tubular crystals of acetylcholine receptor. *J Cell Biol* 99:1202–1211
- Toyoshima C, Unwin N (1990) Three-dimensional structure of the acetylcholine receptor by cryoelectron microscopy and helical image reconstruction. *J Cell Biol* 111:2623–2635
- Toyoshima C, Sasabe H, Stokes DL (1993) Three-dimensional cryo-electron microscopy of the calcium ion pump in the sarcoplasmic reticulum membrane. *Nature* 362:469–471
- Zhang P, Toyoshima C, Yonekura K, Green N, Stokes D (1998) Structure of the calcium pump from sarcoplasmic reticulum at 8 Å resolution. *Nature* 392:835–839
- Galkin VE, Orlova A, Vos MR, Schroder GF, Egelman EH (2015) Near-atomic resolution for one state of F-actin. *Structure* 23 (1):173–182. <https://doi.org/10.1016/j.str.2014.11.006>
- Fitzpatrick AWP, Falcon B, He S, Murzin AG, Murshudov G, Garringer HJ et al (2017) Cryo-EM structures of tau filaments from Alzheimer's disease. *Nature* 547(7662):185–190. <https://doi.org/10.1038/nature23002>
- Wang F, Burrage AM, Postel S, Clark RE, Orlova A, Sundberg EJ et al (2017) A structural model of flagellar filament switching across multiple bacterial species. *Nat Commun* 8(1):960. <https://doi.org/10.1038/s41467-017-01075-5>
- Kollmer M, Close W, Funk L, Rasmussen J, Bsoul A, Schierhorn A et al (2019) Cryo-EM structure and polymorphism of Aβ amyloid fibrils purified from Alzheimer's brain tissue. *Nat Commun* 10(1):4760. <https://doi.org/10.1038/s41467-019-12683-8>
- Fromm SA, Bharat TA, Jakobi AJ, Hagen WJ, Sachse C (2015) Seeing tobacco mosaic virus through direct electron detectors. *J Struct Biol* 189(2):87–97. <https://doi.org/10.1016/j.jsb.2014.12.002>
- He S, Scheres SHW (2017) Helical reconstruction in RELION. *J Struct Biol* 198 (3):163–176. <https://doi.org/10.1016/j.jsb.2017.02.003>
- Egelman E (2000) A robust algorithm for the reconstruction of helical filaments using single-particle methods. *Ultramicroscopy* 85:225–234
- Young HS, Rigaud JL, Lacapere JJ, Reddy LG, Stokes DL (1997) How to make tubular crystals by reconstitution of detergent-solubilized Ca²⁺-ATPase. *Biophys J* 72(6):2545–2558. [https://doi.org/10.1016/S0006-3495\(97\)78898-2](https://doi.org/10.1016/S0006-3495(97)78898-2)
- Stokes DL, Pomfret AJ, Rice WJ, Glaves JP, Young HS (2006) Interactions between Ca²⁺-ATPase and the pentameric form of phospholamban in two-dimensional co-crystals. *Biophys J* 90(11):4213–4223. <https://doi.org/10.1529/biophysj.105.079640>
- Coudray N, Lasala R, Zhang Z, Clark KM, Dumont ME, Stokes DL (2016) Deducing the symmetry of helical assemblies: applications to membrane proteins. *J Struct Biol* 195 (2):167–178. <https://doi.org/10.1016/j.jsb.2016.05.011>
- Coudray N, Lasala R, Zhang Z, Clark KM, Dumont ME et al (2017) Structure of the SLC4 transporter Bor1p in an inward-facing

- conformation. *Protein Sci* 26(1):130–145. <https://doi.org/10.1002/pro.3061>
24. Beroukhi R, Unwin N (1997) Distortion correction of tubular crystals: Improvements in the acetylcholine receptor structure. *Ultramicroscopy* 70:57–81
 25. Avery AW, Fealey ME, Wang F, Orlova A, Thompson AR, Thomas DD et al (2017) Structural basis for high-affinity actin binding revealed by a beta-III-spectrin SCA5 missense mutation. *Nat Commun* 8(1):1350. <https://doi.org/10.1038/s41467-017-01367-w>
 26. Stokes DL, Green NM (1990) Three-dimensional crystals of CaATPase from sarcoplasmic reticulum. Symmetry and molecular packing. *Biophys J* 57:1–14.
 27. Douglas JL, Trieber CA, Afara M, Young HS (2005) Rapid, high-yield expression and purification of Ca²⁺-ATPase regulatory proteins for high-resolution structural studies. *Protein Expr Purif* 40(1):118–125. <https://doi.org/10.1016/j.pep.2004.11.015>
 28. Booth, DS, Avila-Sakar, A, & Cheng, Y (2011) Visualizing proteins and macromolecular complexes by negative stain EM: from grid preparation to image acquisition. *JoVE* (58):3227
 29. Grassucci, RA, Taylor, DJ, & Frank, J (2007) Preparation of macromolecular complexes for cryo-electron microscopy. *Nature protocols* 2 (12): 3239–3246
 30. Nicholson WV, White H, Trinick J (2010) An approach to automated acquisition of cryoEM images from lacey carbon grids. *J Struct Biol* 172(3):395–399. <https://doi.org/10.1016/j.jsb.2010.08.014>
 31. Shi J, Williams DR, Stewart PL (2008) A Script-Assisted Microscopy (SAM) package to improve data acquisition rates on FEI Tecnai electron microscopes equipped with Gatan CCD cameras. *J Struct Biol* 164(1):166–169. <https://doi.org/10.1016/j.jsb.2008.05.011>
 32. Yonekura K, Toyoshima C (2000) Structure determination of tubular crystals of membrane proteins. II. Averaging of tubular crystals of different helical classes. *Ultramicroscopy* 84:15–28
 33. Fernandez-Leiro, R, & Scheres, SHW (2017) A pipeline approach to single-particle processing in RELION. *Acta Crystallogr D Struct Biol* 73(Pt 6):496–502. <https://doi.org/10.1107/S2059798316019276>
 34. Punjani A, Rubinstein JL, Fleet DJ, Brubaker MA (2017) cryoSPARC: algorithms for rapid unsupervised cryo-EM structure determination. *Nat Methods* 14(3):290–296. <https://doi.org/10.1038/nmeth.4169>
 35. Fromm SA, Sachse C (2016) Cryo-EM structure determination using segmented helical image reconstruction. *Methods Enzymol* 579:307–328. <https://doi.org/10.1016/b.smic.2016.05.034>
 36. Huber ST, Kuhm T, Sachse C (2018) Automated tracing of helical assemblies from electron cryo-micrographs. *J Struct Biol* 202 (1):1–12. <https://doi.org/10.1016/j.jsb.2017.11.013>
 37. Pothula KR, Smyrnova D, Schroder GF (2019) Clustering cryo-EM images of helical protein polymers for helical reconstructions. *Ultramicroscopy* 203:132–138. <https://doi.org/10.1016/j.ultramic.2018.12.009>
 38. Graves JP, Primeau JO, Espinoza-Fonseca LM, Lemieux MJ, Young HS (2019) The phospholamban pentamer alters function of the sarcoplasmic reticulum calcium pump SERCA. *Biophys J* 116(4):633–647. <https://doi.org/10.1016/j.bpj.2019.01.013>
 39. Young HS, Jones LR, Stokes DL (2001) Locating phospholamban in co-crystals with Ca²⁺-ATPase by cryoelectron microscopy. *Biophys J* 81(2):884–894. [https://doi.org/10.1016/S0006-3495\(01\)75748-7](https://doi.org/10.1016/S0006-3495(01)75748-7)
 40. Anderson DM, Makarewicz CA, Anderson KM, Shelton JM, Bezprozvannaya S, Bassel-Duby R, Olson EN (2016) Widespread control of calcium signaling by a family of SERCA-inhibiting micropeptides. *Sci Signal* 9(457):ra119. <https://doi.org/10.1126/scisignal.aaj1460>
 41. Primeau JO, Armanious GP, Fisher ME, Young HS (2017) The sarcoendoplasmic reticulum calcium ATPase. *Subcell Biochem*. <https://doi.org/10.3389/fmicb.2019.00209>
 42. Dyla M, Kjaergaard M, Poulsen H, Nissen P (2019) Structure and mechanism of P-type ATPase ion pumps. *Annu Rev Biochem*. <https://doi.org/10.1146/annurev-biochem-010611-112801>
 43. Graves JP, Fisher L, Ward A, Young HS (2010) Helical crystallization of two example membrane proteins MsbA and the Ca(2+)-ATPase. *Methods Enzymol* 483:143–159. [https://doi.org/10.1016/S0076-6879\(10\)83007-1](https://doi.org/10.1016/S0076-6879(10)83007-1)
 44. Russo CJ, Henderson R (2018) Charge accumulation in electron cryomicroscopy. *Ultramicroscopy* 187:43–49. <https://doi.org/10.1016/j.ultramic.2018.01.009>



NMR Spectroscopic Studies of Ion Channels in Lipid Bilayers: Sample Preparation Strategies Exemplified by the Voltage Dependent Anion Channel

Robert Silvers and Matthew T. Eddy

Abstract

We describe approaches for the preparation of membrane proteins in detergent micelles and lipid bilayers for solution and magic angle spinning NMR studies, respectively, as exemplified by the human voltage dependent anion channel 1 (hVDAC1). Here, we report protocols for the preparation of homogenous samples of recombinant hVDAC1 in detergent micelles and lipid two-dimensional crystals yielding high resolution NMR spectra. Procedures are described for the recombinant production of stable-isotope labeled hVDAC1 in *E. coli*, the isolation of hVDAC1 from inclusion bodies and the refolding into detergent micelles, as well as the reconstitution of hVDAC1 into lipids to form 2D crystals.

Key words Ion channels, Solution NMR spectroscopy, Solid-state NMR spectroscopy, Magic angle spinning, Membrane proteins, Lipids, 2D crystals, Voltage-dependent anion channel, Mitochondrial proteins

1 Introduction

Voltage-gated ion channels are integral membrane proteins activated by changes in membrane potential, which trigger conformational changes that regulate the passage of ions and other small molecules across lipid bilayers. The human voltage dependent anion channel isoform 1 (hVDAC1) is an exemplary voltage-gated channel that controls the transport of metabolites across the mitochondrial outer membrane. hVDAC1 is the most abundant protein in the mitochondrial outer membrane and has been implicated in multiple diseases, including various cancers, through its role in cellular apoptosis [1, 2]. Available crystal [3] and solution NMR structures [4] of the putative “open” state of recombinantly expressed hVDAC1 presented a β -barrel structure composed of 19 β strands and an inward-facing α -helical N-terminus reported to be critical for channel gating (Fig. 1a). Earlier structural models

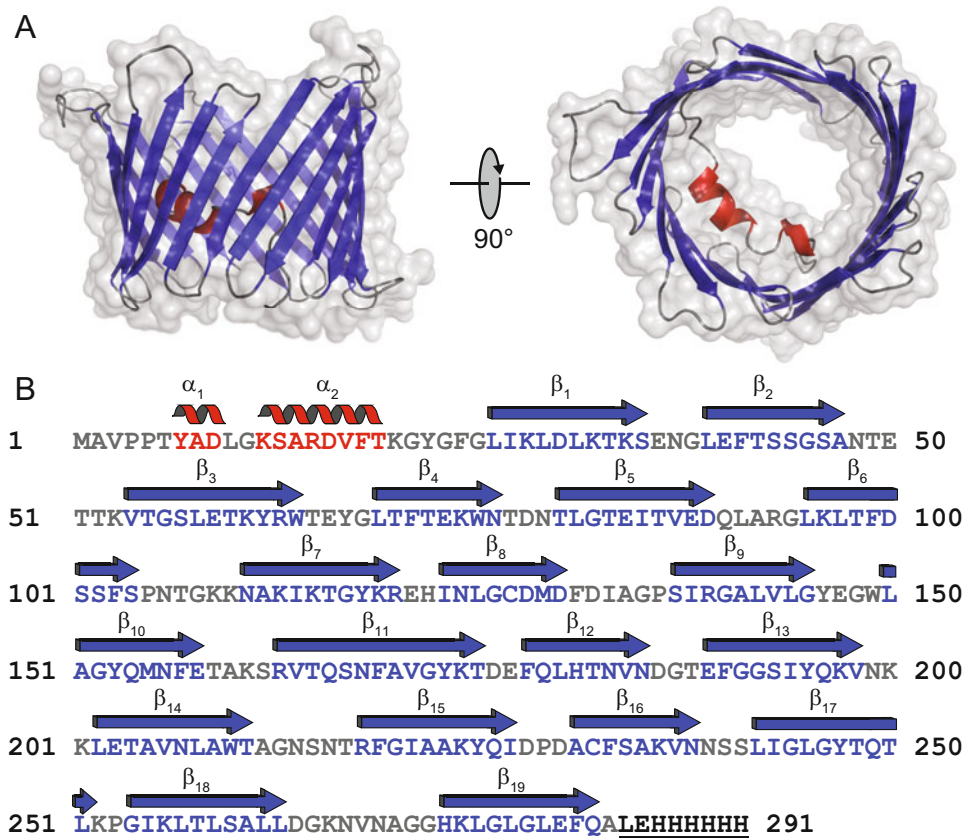


Fig. 1 (a) Crystal structure of hVDAC1 shown in space filling (transparent) and ribbon representations (PDB 3EMN [3]). (b) Amino acid sequence of hVDAC1 implemented in this protocol as encoded by the pET21a vector and as expressed in *E. coli*. Secondary structure elements are mapped onto the sequence. α -helices are shown in red, β -sheets in blue and loop regions in gray. The residues of the C-terminal tail carrying the His₆-tag are colored black and are underlined

developed from electrophysiological and biophysical data recorded with hVDAC1 in membrane lipids differed from the reported crystal and NMR structures [5–7], motivating studies of the structure of hVDAC1 in the same lipid environments used for activity measurements [8].

A unique advantage of magic angle spinning nuclear magnetic resonance (MAS NMR) studies of ion channels is the capability to investigate the conformation and structural plasticity of channels in the same lipid bilayers used for functional assays. For example, this enabled function-related structural data of hVDAC1 to be determined in diphytanyl phosphocholine lipids that were also used for electrophysiological experiments [9]. The general utility of this approach is documented also by MAS NMR studies of the functionally important hVDAC1 N-terminus [10] and exemplified in MAS NMR studies of the second isoform of human VDAC (hVDAC2) [11, 12].

Key to the success of MAS NMR studies of ion channels is the preparation of homogenous and properly folded protein samples that yield well resolved NMR spectra. A general scheme for the preparation of membrane protein samples for NMR studies in aqueous solutions is overexpression of the target protein into inclusion bodies, isolation and folding of the target protein using selected detergents, and further purification of properly refolded protein (Fig. 2). Samples for MAS NMR addend this procedure by mixing the purified, refolded, and detergent-solubilized protein with chosen lipids and subsequent removal of detergent (Fig. 2). For hVDAC1, mixing of purified detergent-solubilized protein with lipids and subsequent removal of the detergent forms 2-dimensional (2D) lipid crystals, which are multilamellar bilayers

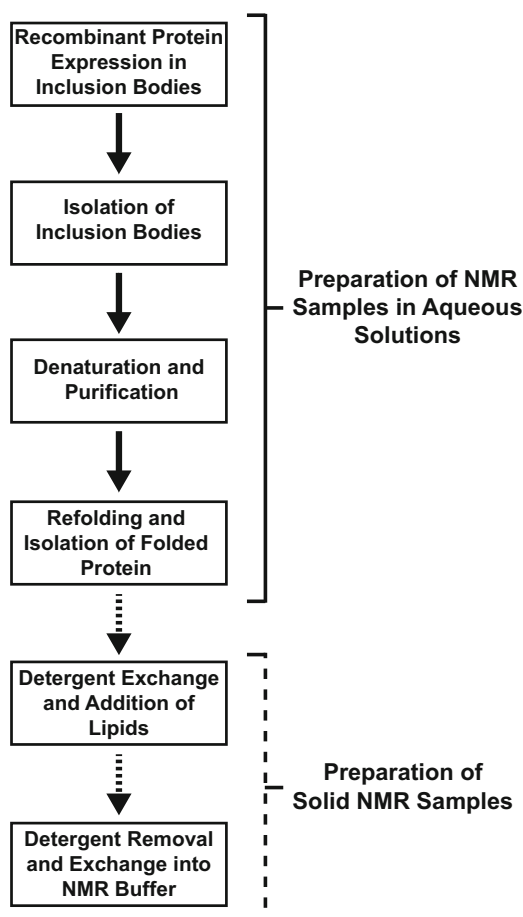


Fig. 2 Schematic diagram of the hVDAC1 sample preparation workflow. Steps for preparation of samples in solution are outlined in the bracket with solid lines, and in the bracket with dashed lines are additional steps for preparation of solid-state NMR samples

containing both lipids and proteins arranged in sheets that are approximately several to 10s of microns across (Fig. 5a, b) (vide infra). 2D crystals form with specific lipid-to-protein ratios for a number of different lipids and have also been observed to form with hVDAC2 [12]. MAS NMR spectra of hVDAC1 and hVDAC2 in 2D crystals are well resolved with typical ^{13}C and ^{15}N linewidths of <0.3 and <0.5 ppm, respectively [9, 15, 16]. The following description provides a detailed protocol for the preparation of hVDAC1 2D crystals for MAS NMR studies, as implemented in earlier MAS NMR studies (Figs. 5c, d, 6, and 7). Protocols for the production, purification, and refolding of stable-isotope labeled hVDAC1 are adapted from the Wagner laboratory [4] while protocols for preparation of 2D lipid crystals containing hVDAC1 were adapted from earlier cryogenic electron microscopy (cryo-EM) studies [17].

2 Materials

2.1 Reagents

1. The pET21a vector coding for the human voltage-dependent anion channel isoform 1 (hVDAC1) with a C-terminal His₆-tag was made available by the Wagner group (Harvard, Cambridge, MA) (*see Note 1*).
2. The *E. coli* strain BL21(DE3) was used for recombinant expression of hVDAC1 (*see Note 2*).
3. The stable-isotope labeled reagents $^{15}\text{NH}_4\text{Cl}$ and D-(+)- $^{13}\text{C}_6$ -glucose were obtained from Cambridge Isotope Laboratories (CIL, Andover, MA). The following reagents were obtained from Sigma-Aldrich unless otherwise specified. Ammonium chloride (NH_4Cl), β -mercaptoethanol, calcium chloride (CaCl_2), cOmplete™ EDTA-free protease inhibitor cocktail (Roche, Basel, Switzerland), D-(+)-glucose, 1,2-dimyristoyl-sn-glycero-3-phosphocholine (DMPC; Avanti Polar Lipids, Alabaster, AL), 1,4-dithiothreitol (DTT), ethylenediaminetetraacetic acid (EDTA), guanidine hydrochloride, hydrogen chloride (HCl), imidazole, isopropyl- β -D-thiogalactoside (IPTG), magnesium chloride (MgCl_2), magnesium sulfate (MgSO_4), N,N-Dimethyldodecylamine N-oxide (LDAO), 2-(N-morpholino)ethanesulfonic acid (MES), octylpolyoxyethylene (octyl-POE; Bachem, King of Prussia, PA), potassium phosphate monobasic (KH_2PO_4), sodium chloride (NaCl), sodium hydroxide (NaOH), sodium phosphate dibasic (Na_2HPO_4), sodium phosphate monobasic (NaH_2PO_4), thiamine hydrochloride, tris(hydroxymethyl)aminomethane (TRIS), Triton X-100, tryptone, urea, yeast extract.

2.2 Media Recipes

1. <i>Lysogeny Broth (LB)</i>	10 g/L tryptone 5 g/L yeast extract 10 g/L NaCl
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Mix all components until dissolved and either autoclave in an appropriately sized sterile flask or sterile filter into an appropriately sized, sterile culture flask. Add the appropriate antibiotic right before use.

2. <i>M9 minimal medium (M9)</i>	42 mM sodium phosphate dibasic (Na_2HPO_4) 22 mM potassium phosphate monobasic (KH_2PO_4) 0.5 g/L sodium chloride (NaCl)
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Combine all components above and add ddH₂O to ~90% of the final volume. Either autoclave in an appropriately sized culture flask or sterile filter into an appropriately sized, sterile culture flask. Before use add:

1 g/L	NH_4Cl
2 g/L	D-(+)-glucose
2 mL/L	1 M MgSO_4 (sterile filtered)
0.2 mL/L	1 M CaCl_2 (sterile filtered)
0.01% (w/v)	Thiamine hydrochloride (sterile filtered)

Add the appropriate antibiotic right before use. If necessary, adjust the pH to 7.2 by adding NaOH. If necessary, add sterile deionized water to bring the volume to 100% of the desired final volume. Ammonium chloride (NH_4Cl) and D-(+)-glucose are the only nitrogen and carbon sources in the medium, respectively. To achieve any combination of uniform [^{15}N]-, [^{13}C]-, or [^{13}C , ^{15}N]-labeling, the nitrogen and carbon sources in the medium can be exchanged by using [^{15}N]-labeled ammonium chloride ($^{15}\text{NH}_4\text{Cl}$) or [^{13}C]-labeled D-(+)-glucose (D-(+)- ^{13}C -glucose).

3. *Amino acid-specific M9 minimal medium:*

A modified M9 minimal media (see above) for the expression of hVDAC1 with specific [^{13}C , ^{15}N] amino acids can be used. Prepare the aforementioned M9 minimal medium using D-(+)- ^{12}C -glucose and $^{14}\text{NH}_4\text{Cl}$ and add [^{13}C , ^{15}N] amino acids that are desired to be labeled as well as [^{12}C , ^{14}N] amino acids for all that are desired to not be labeled to the medium at a final concentration of 1 mM each (see **Note 3**). Additionally, unlabeled amino acids can be added to a background of D-(+)- ^{13}C -glucose and $^{15}\text{NH}_4\text{Cl}$ to simplify the NMR spectra (Fig. 7).

2.3 Buffers

1. Lysis buffer (Buffer L):	50 mM 200 mM 20 mM 1 tab./ 50 mL	TRIS/HCl pH 7 NaCl β -Mercaptoethanol cOmplete protease inhibitor cocktail
2. Wash buffer (Buffer W):	50 mM 200 mM 20 mM 0.5% (w/v)	TRIS/HCl pH 7 NaCl β -Mercaptoethanol Triton X-100
3. HisTrap buffer A (Buffer HA):	50 mM 8 M 100 mM 10 mM 20 mM	TRIS/HCl pH 8 urea NaCl β -Mercaptoethanol imidazole
4. HisTrap buffer B (Buffer HB):	50 mM 8 M 100 mM 10 mM 250 mM	TRIS/HCl pH 8 urea NaCl β -Mercaptoethanol imidazole
5. Dialysis buffer 1 (Buffer D1):	50 mM 50 mM 1 mM 20 mM	TRIS/HCl pH 7.5 NaCl EDTA β -Mercaptoethanol
6. Size exclusion buffer (Buffer SEC):	100 mM 6 M 100 mM 1 mM 5 mM	$\text{Na}_2\text{HPO}_4/\text{NaH}_2\text{PO}_4$ pH 7.0 Guanidine hydrochloride NaCl EDTA DTT
7. Dialysis buffer 2 (Buffer D2):	25 mM 5 mM 1 mM	$\text{Na}_2\text{HPO}_4/\text{NaH}_2\text{PO}_4$ pH 7.0 DTT EDTA
8. Refolding buffer (Buffer R):	100 mM 100 mM 25 mM 10 mM 1% (w/v)	TRIS/HCl pH 8.0. NaCl Na_2HPO_4 DTT LDAO
9. Ion exchange buffer A (Buffer IA):	25 mM 0.1% 2 mM	$\text{Na}_2\text{HPO}_4/\text{NaH}_2\text{PO}_4$ pH 6.5 LDAO DTT

(continued)

10. Ion exchange buffer B (Buffer IB):	25 mM	Na ₂ HPO ₄ /NaH ₂ PO ₄
	1 M	pH 6.5
	0.1%	NaCl
	2 mM	LDAAO DTT
11. Dialysis buffer 3 (Buffer D3):	50 mM	TRIS/HCl pH 8.0
	0.6% (w/v)	octyl-POE
	5 mM	DTT
12. Octyl-POE buffer (Buffer O):	50 mM	TRIS/HCl pH 8.0
	1% (w/v)	octyl-POE
	5 mM	DTT
13. Crystallization buffer (Buffer X):	10 mM	MES pH 6.5
	150 mM	NaCl
	20 mM	MgCl ₂
	5 mM	DTT
14. Crystal wash buffer (Buffer XW):	25 mM	Na ₂ HPO ₄ /NaH ₂ PO ₄
		pH 7.0

3 Methods

The following methods for the expression and purification of hVDAC1 preparations are based on previously published protocols by the Wagner [4, 18] and Griffin [9, 15, 16] groups and include some relatively minor improvements developed through experience. Hence, the methods below represent a consensus protocol. Notes provided in Subheading 4 contain additional details.

3.1 Overexpression of hVDAC1 in *E. coli*

1. Transform the pET21a plasmid encoding hVDAC1 into the *E. coli* strain BL21(DE3) using standard heat shock transformation protocols. Plate the transformed cells onto LB agar plates containing 100 µg/mL ampicillin and incubate the plates at 37 °C overnight.
2. From an agar plate with freshly transformed colonies (or alternatively from a glycerol stock preserved at −80 °C) prepare overnight cultures in LB medium. For every 500 mL of main cell culture, prepare 5 mL overnight culture in an appropriately sized tube or flask containing LB medium supplemented with 100 µg/mL ampicillin. Place the overnight cultures in an incubated orbital shaker operating at 37 °C and around 250 RPM.
3. Prepare sterile LB or M9 minimal medium for main cultures and store at room temperature overnight in flasks covered to prevent the introduction of contaminants.
4. The next morning, preheat the flasks containing the culture medium (LB or M9 minimal medium) to 37 °C without

shaking. Meanwhile, remove the precultures and gently centrifuge the cells at $4000 \times g$ for 15 min. Carefully wash the cell pellet(s) by gently resuspending in ice-cold phosphate buffered saline (PBS) and centrifuging at $4000 \times g$ for 15 min. Afterward, resuspend the cells in an appropriate volume of PBS and use the suspension to inoculate the main cultures. Out of experience, reaching an initial OD₆₀₀ of around 0.1 after inoculation results in optimal growth conditions while ensuring efficient isotope labeling when using M9 minimal medium.

5. Monitor the cell optical density and when the OD₆₀₀ reaches 0.6–0.8, induce with adding IPTG (final concentration: 1 mM) from a 1 M stock solution and allow induction of protein expression to proceed for 3–5 h at 37 °C while shaking at around 250 RPM. Alternatively, you can use a lower amount of IPTG (final concentration: 200–400 μM) and immediately lower the temperature to 18–20 °C while shaking at around 250 RPM for overnight expression (~16 h).
6. Pellet the cells in a centrifuge, decant and discard the supernatant by appropriate means and either immediately proceed to protein purification (Subheading 3.2) or freeze the cell pellet in liquid nitrogen and store at –80 °C.

3.2 Isolation of Inclusion Bodies

1. Resuspend the fresh or frozen cell pellets (Subheading 3.1, step 6) in sufficient buffer **L**. Make sure all cells are resuspended well and the resulting solution is homogeneous.
2. Lyse the cells. In our experience, various options for cell lysis can be used and work equally well if done properly. The cells can be lysed by using a microfluidizer, French press, sonication, or freeze-thaw cycles. Lysis by multiple rounds of sonication or freeze-thaw cycles can be improved by adding 1% Triton X-100 and a spatula's tip worth of lysozyme directly to buffer **L** prior to lysis.
3. After cell lysis is complete, spin down the inclusion bodies by low-to-intermediate speed centrifugation ($\sim 10,000 \times g$, 20 min). Decant the supernatant, which typically appears translucent. Inclusion bodies will appear as solid precipitants in the bottom of the tubes.
4. Resuspend the inclusion body pellet in 30–50 mL of buffer **W**. If the inclusion bodies do not break up easily, additional sonication can be used to improve the resuspension of inclusion bodies. Spin down the resuspended inclusion bodies via centrifugation ($>20,000 \times g$, 20 min). Decant the supernatant, resuspend and spin down the inclusion bodies as described before. Repeat the wash a total of three times to remove impurities.

3.3 Purification of hVDAC1 from Inclusion Bodies

5. The inclusion bodies can be used directly for further purification (Subheading 3.3) or be frozen in liquid nitrogen and stored at -80°C .
1. Dissolve the inclusion body pellet (Subheading 3.2, step 5) in sufficient buffer **HA**, typically 50–100 mL. Once dissolved, clear the mixture of particles by centrifugation at the highest centrifugal force available. In our experience, centrifugation in an ultracentrifuge is beneficial, but not required.
 2. After centrifugation, pass the supernatant through a 0.4 μm filter to remove remaining particles that could clog the column during subsequent purification steps.
 3. Equilibrate your choice of resin for immobilized metal affinity chromatography (IMAC) in buffer **HA**. We routinely used prepacked 5 mL HisTrapTM HP columns (formerly GE Healthcare Life Sciences, now Cytiva Life Sciences). Multiple 5 mL columns can be connected for the purification of larger amounts of hVDAC1. Loose Ni-agarose resin (Qiagen) can also be used with a gravity flow column.
 4. Apply the filtered solution containing unfolded hVDAC1 to the affinity resin to bind the His₆-tagged hVDAC1. To monitor the removal of contaminating proteins and nucleic acids, observe the absorption of the flow through solution at a wavelength of 280 nm and at a wavelength of 260 nm, if available.
 5. Wash the resin with buffer **HA** until a baseline at a wavelength of 280 nm (and 260 nm) is reached, indicating that macromolecules that do not interact with the affinity resin have been removed sufficiently.
 6. Elute hVDAC1 by applying 100% buffer **HB** to the resin and combine all fractions containing hVDAC1. Elution fractions will contain both hVDAC1 and a hVDAC1 truncation product (see **Note 4** and Fig. 3).
 7. (optional) The combined eluate containing mainly hVDAC1 can be frozen and stored at -20°C for further purification at a later time.
 8. Dialyze hVDAC1 dissolved in a buffer containing 8 M urea over night at 4°C against buffer **D1** using a dialysis membrane with a MWCO of 10,000 Da. Purified hVDAC1 will form a white precipitate.
 9. Centrifuge precipitated hVDAC1, decant the supernatant, and dissolve the pellet in a small amount of buffer **SEC** (see **Note 5**).
 10. Equilibrate the size exclusion column with buffer **SEC** (see **Note 6**). We routinely used HiLoad[®] 16/600 Superdex[®] 75 pg or HiLoad[®] 26/600 Superdex[®] 75 pg columns (Cytiva

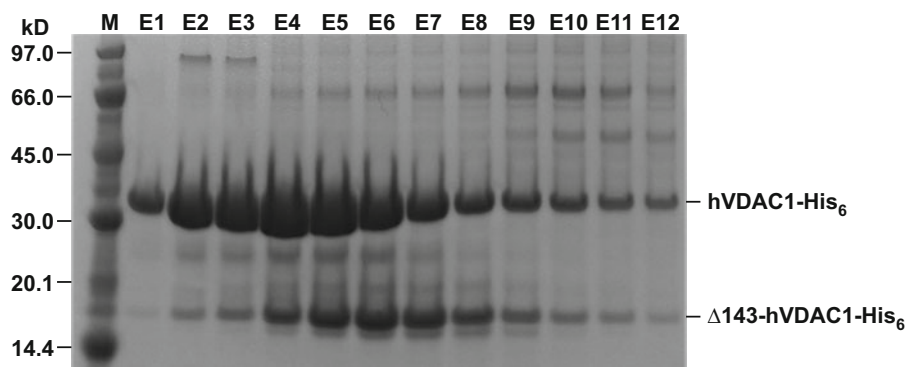


Fig. 3 SDS-PAGE (4–12% NuPAGE™ Bis-Tris, MES buffer, Thermo Fisher) of a representative IMAC hVDAC1 protein purification eluted using a gradient of increasing concentration of imidazole. Despite the use of this gradient, all proteins including hVDAC1 elute over a narrow range of the concentration of imidazole, essentially in one peak. Shown are the 12 fractions (E1-E12) of said peak. The protein marker (M) was the LMW-SDS Marker Kit from GE Healthcare Life Sciences (now Cytiva Life Sciences)

Life Sciences, formerly GE Healthcare Life Sciences) depending on the amount of hVDAC1 to be purified.

11. Filter the solution containing hVDAC1 through a 0.4 μ m filter and apply the sample to the equilibrated size exclusion chromatography column. hVDAC1 elutes closely to the Δ 143-hVDAC1-His₆ truncation that contains the C-terminal His tag and both peaks typically overlap to an extent (Fig. 4a).
12. Run samples on SDS-PAGE to confirm purity of individual fractions and combine only fraction containing hVDAC1, but no Δ 143-hVDAC1-His₆.
13. (Optional) The remaining fractions containing a mixture of hVDAC1 and its truncation can be combined, concentrated, and reapplied to the same size exclusion chromatography to improve yield (Fig. 4b).

3.4 Refolding of hVDAC1 into Detergent Micelles

1. Concentrate hVDAC1 using a spin concentrator to a concentration of 1 mg/mL.
2. Refold hVDAC1 at 4 °C by slow, dropwise dilution into buffer **R** (see Note 7). Adjust the amount of buffer **R** to achieve a ratio of 10 mL refolding buffer for every 1 mL hVDAC1 solution. After the dilution finished, continue to stir the refolding mixture over night at 4 °C.
3. For subsequent purification of refolded hVDAC1 by ion exchange chromatography, dialyze refolded hVDAC1 over night at 4 °C against buffer **D2** using a dialysis membrane with a MWCO of 10,000 Da. For every 1 mL hVDAC1 solution use 20 mL buffer **D2** (see Note 8).

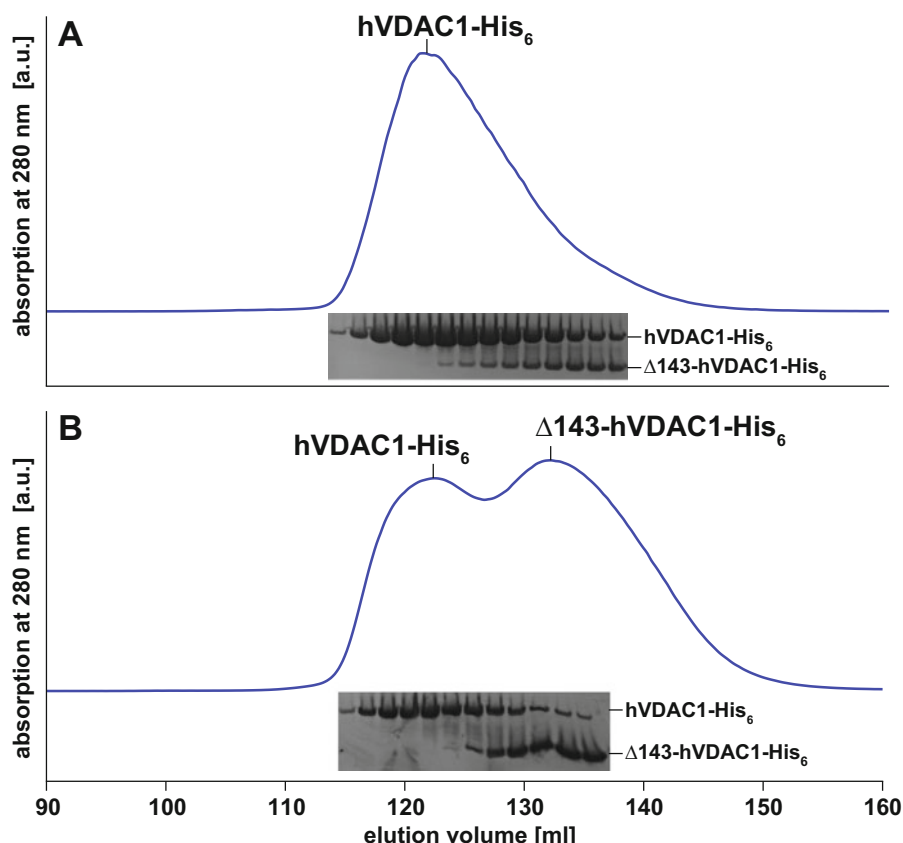


Fig. 4 The application of multiple sequential size exclusion chromatography steps separates hVDAC1 from an undesired hVDAC1 truncation ($\Delta 143$ -hVDAC-His₆) product. **(a)** Representative size exclusion chromatograms (SEC) with hVDAC1 performed under denaturing conditions are shown above SDS-PAGE gels that display the contents of the fraction above each lane. In an initial SEC step, the truncation product appears as a shoulder eluted at a later time than hVDAC1. Fractions containing predominantly hVDAC1 were pooled and set aside. The fractions containing a mixture of hVDAC1 and the truncation product were combined, concentrated, and reapplied multiple times. **(b)** On the third SEC run, hVDAC1-His₆ and $\Delta 143$ -hVDAC-His₆ appear at a 1:1 ratio

3.5 Purification of Natively Folded hVDAC1 in Detergent Micelles

1. Equilibrate the ion exchange chromatography column with buffer **IA** at 4 °C. We recommend the use of the HiLoad 26/10 SP Sepharose HP (Cytiva life sciences, formerly GE Healthcare Life Sciences).
2. Apply the solution containing refolded hVDAC1 to the IEX column at 4 °C. Running at a different temperature shifts the positions of peaks and is discouraged.
3. Elute properly folded hVDAC1 using a linear gradient from 25–50% buffer **IB** over a volume of 300 mL. Wash the column with 100 mL of 100% buffer **IB**. The correctly, natively folded hVDAC elutes around 30–35% buffer **IB**. The misfolded hVDAC1 elutes afterward around 40–45% buffer **IB**. Typically, the peak for folded hVDAC1 is significantly larger than the

peak for the misfolded hVDAC1, which is a good way to estimate folding efficiency (*see Note 6*).

4. Combine fractions containing correctly folded hVDAC1.
5. (Optional) For use in solution NMR spectroscopy or use in methods requiring hVDAC1 in LDAO micelles, concentrate fractions containing folded hVDAC1 using 10 kDa MWCO centrifugal concentrators.
6. For preparation of hVDAC1 samples in lipid bilayers for studies by solid-state NMR spectroscopy (or other methods requiring 2D crystalized hVDAC1) proceed to the next step (Subheading 3.6).

3.6 Formation of hVDAC1 2D Lipid Crystals

1. Dialyze the purified and refolded hVDAC1 in LDAO micelles for 24 h against 2–4 L of buffer **D3** containing octyl-POE at 4 °C using a dialysis cassette or tubing with a MWCO of 10 kDa. This step is necessary because octyl-POE has a lower critical micelle concentration than LDAO and can be more easily dialyzed during the formation of the 2D lipid crystals [17].
2. Concentrate hVDAC1 in octyl-POE to a final concentration of 1 mg/mL. Buffer **O** can be used to adjust the concentration of hVDAC1, if necessary.
3. Add dry DMPC to buffer **O** at a final DMPC concentration of 0.5 mg/mL. Solubilize by rotation or gentle stirring until the solution is clear.
4. Mix the 1 mg/mL hVDAC1 solution and the 0.5 mg/mL DMPC solution in a 1:1 ratio. This will result in a lipid–protein ratio of 1:2 (w/w) (*see Note 9*). Allow the solution to equilibrate for 30 minutes at room temperature.
5. Dialyze the mixture against copious amounts of buffer **X** for several days at room temperature using a 10 kDa MWCO dialysis membrane. Exchange buffer **X** twice per day (*see Note 10*).
6. Harvest crystals by centrifugation and wash the 2D crystals by resuspending the 2D crystals in buffer **XW**. Repeat 3–4 times.
7. Store 2D crystal samples at 4 °C (*see Note 11*).
8. For solid-state NMR spectroscopy (Figs. 5, 6, and 7), pack 2D crystals of hVDAC1 by centrifugation using the appropriate packing tool.

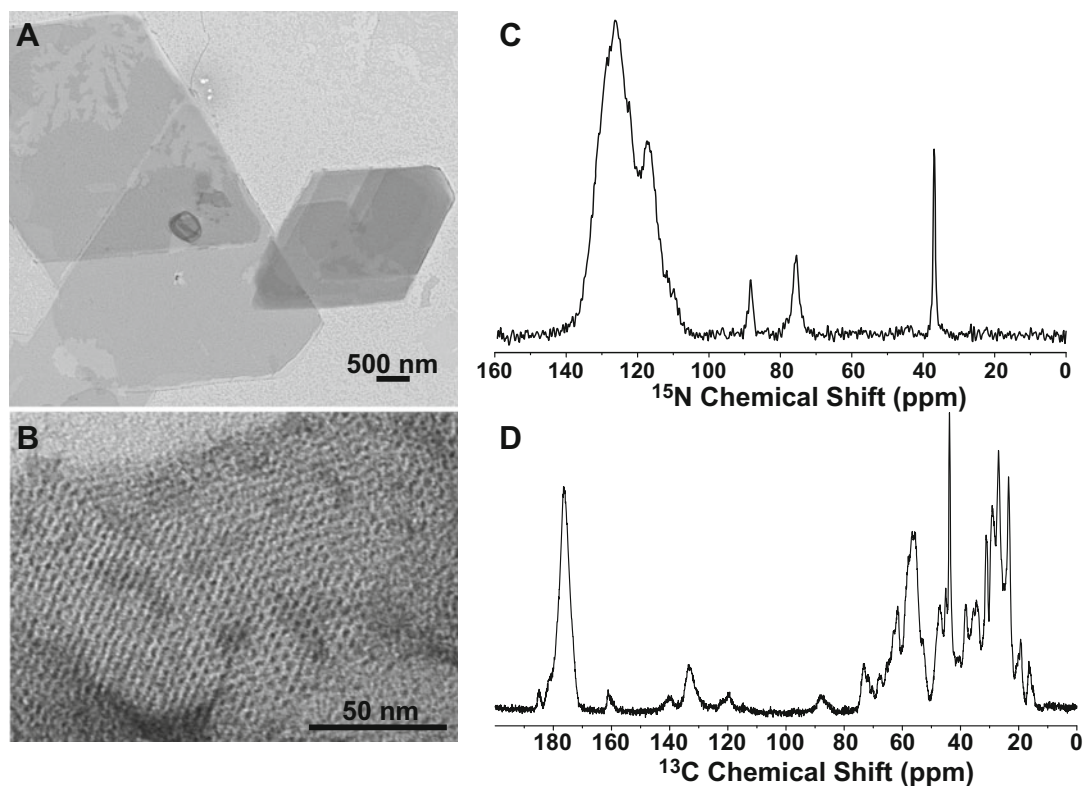


Fig. 5 Characterization of hVDAC1 in 2D lipid crystals by transmission electron microscopy and solid-state NMR spectroscopy. **(a, b)** Representative negative stain images of 2D lipid crystals formed with hVDAC1 and DMPC are shown. **(a)** Multiple 2D crystals can be observed with overlapping sections visible. **(b)** At higher magnification, hVDAC1 molecules appear as “indentations” in organized arrays. Representative **(c)** ^{15}N and **(d)** ^{13}C CPMAS spectra of the same sample measured at 750 MHz ^1H nutation frequency are shown

4 Notes

1. The plasmid coding for the hVDAC1 sequence (GenBank accession number BC008482) was generated by cloning the cDNA into the pET21a vector (Novagen) using the NdeI/XhoI restriction sites [18]. The generated construct added two amino acids (LE) as well as a His₆-tag to the C-terminus (Fig. 1b). This construct design was chosen to avoid disturbing the N-terminus that is critical for proper channel function.
2. For expression of other isoforms of hVDAC, such as hVDAC2, Rosetta (DE3) cells (Novagen) were found to be superior to BL21(DE3) [19].
3. Stable-isotope labeling of specific amino acid types in proteins can suffer from scrambling of stable-isotopes to undesired amino acids through transamination and other biochemical

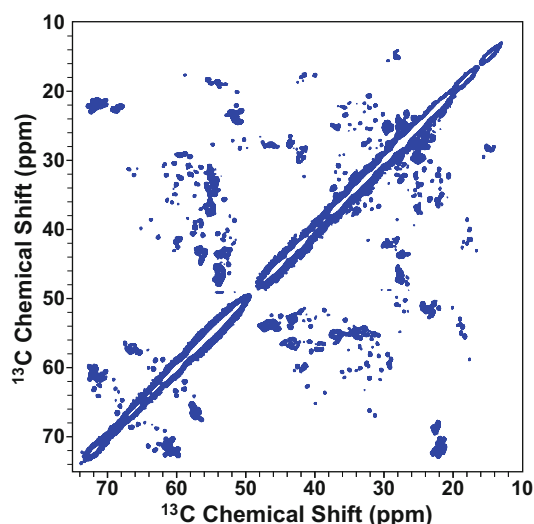


Fig. 6 A representative 2-dimensional MAS NMR correlation spectrum is shown of 2D lipid crystals containing hVDAC1. The 2D ^{13}C - ^{13}C correlation spectrum was recorded at a spinning frequency of 20 kHz and at a 900 MHz ^1H nutation frequency with RFDR [13] mixing and TPPM [14] decoupling during the direct and indirect acquisition periods using a 3.2 mm rotor. Figure adapted with permission from Eddy et al. [9]

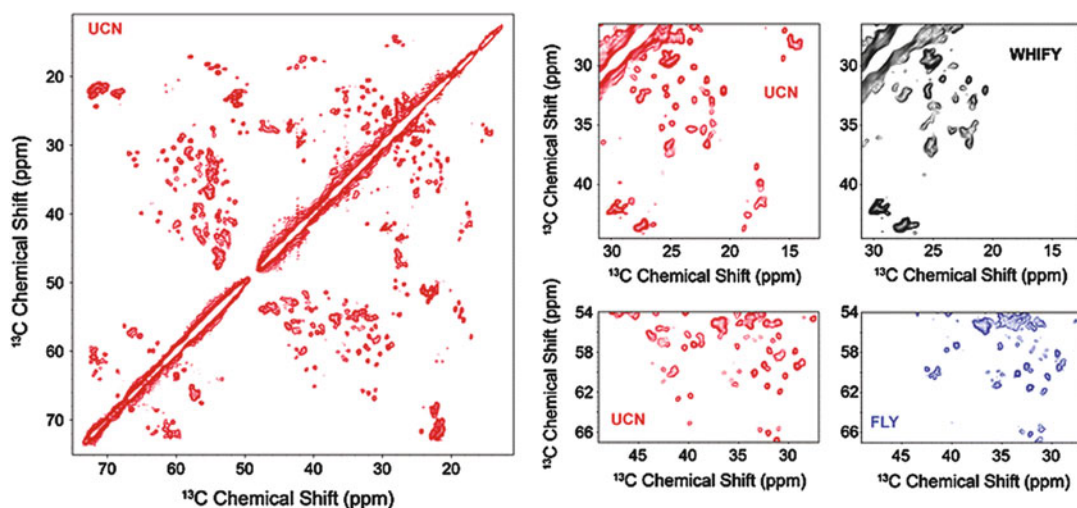


Fig. 7 The use of unlabeled amino acids in a background of uniform ^{13}C , ^{15}N labeling simplifies NMR spectra. 2D ^{13}C - ^{13}C RFDR correlation spectra are shown of 2D lipid crystals containing U- ^{13}C , ^{15}N hVDAC1 (red), and hVDAC1 samples prepared by adding the unlabeled amino acids F, L, and Y (blue) or W, H, I, F and Y (black) to the protein expression media. Spectra were recorded at 12.5 kHz MAS frequency and 750 MHz ^1H nutation frequency with 1.3 ms RFDR mixing. Middle and right panels show expanded sections of the full spectrum on the left. Figure used with permission from Eddy et al. [16]

pathways. In our experience, the amino acids Trp, His, Phe, Tyr, Ile, Leu, Val, and Pro showed only minor indications of the scrambling of ^{13}C labels [16]. Samples prepared with stable-isotope labeled amino acids from this group of eight amino acids have been reported in additional NMR studies of other membrane proteins [20].

4. hVDAC1 and an hVDAC1 truncation product containing the C-terminal His tag are coexpressed. The truncation product was determined to start with residue L144, hence, we chose the nomenclature $\Delta 143\text{-hVDAC1-His}_6$ for the His₆-tagged truncation. Separation of hVDAC1 from the truncation product prior to refolding and the formation of 2D lipid crystals is not strictly required; however, from our experience separation of the two components prior to refolding can improve the resolution of the hVDAC1 NMR spectra. Because the truncation product includes the His₆-tag, separation by application of an imidazole gradient during IMAC purification is not successful (Fig. 3) and must be done by application of multiple sequential denaturing size exclusion chromatograms. Furthermore, while Fig. 3 shows the result of an imidazole gradient, the purification does not require a gradient and everything bound to the resin can be eluted immediately.
5. Subheading 3.3, steps 8 and 9 are used to concentrate hVDAC1 for the next step. Alternatively, an appropriate spin-concentrator can be used and Subheading 3.3, steps 8 and 9 can be omitted.
6. The size exclusion chromatography can be performed with 8 M urea instead of 6 M guanidine hydrochloride, however, we found that refolding yields are typically higher when hVDAC1 is refolded from 6 M guanidine hydrochloride as compared to 8 M urea.
7. To improve the yield of properly refolded hVDAC1, the refolding step should be performed as slowly as possible and with all reagents at 4 °C. While there are various methods of performing slow refolding, a straightforward approach is the use of a peristaltic pump (e.g., Peristaltic Pump P1, Cytiva Life Sciences, formerly GE Healthcare Life Sciences) placed in a cold room and set to the slowest possible flow rate. Solution containing unfolded hVDAC1 is very slowly flowed into a beaker containing the refolding buffer placed on a magnetic stir plate with constant stirring. Allow the solution containing refolded protein to continue stirring for several hours to overnight after the last unfolded hVDAC1 is added.
8. This step is necessary to reduce the LDAO concentration to below 0.25% and above the CMC. We found, that hVDAC1 does not bind to the SP ion exchange chromatography resin at LDAO concentrations above 0.25%.

9. hVDAC1 forms 2D crystals at a lipid–protein ratio of 1:2 (weight/weight). The crystallization condition for hVDAC1 and DMPC is rather narrow at the lower end. Using less lipid will lead to the precipitation of hVDAC1 without the formation of 2D crystals while using more lipid than necessary can cause the formation of lipid vesicles.
10. Visible precipitation is usually formed after 12–24 h, and crystals appear fully formed after 36–48 h. Allow dialysis to continue for at least an additional 48–72 h to remove lingering detergent and ensure crystals are fully formed.
11. Freezing destroys hVDAC1 2D crystals by ice crystal formation.

Acknowledgments

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References

1. Mazure NM (2017) VDAC in cancer. *Biochim Biophys Acta* 1858:665–673. <https://doi.org/10.1016/j.bbabbio.2017.03.002>
2. Shoshan-Barmatz V, Keinan N, Zaid H (2008) Uncovering the role of VDAC in the regulation of cell life and death. *J Bioenerg Biomembr* 40:183–191. <https://doi.org/10.1007/s10863-008-9147-9>
3. Ujwal R et al (2008) The crystal structure of mouse VDAC1 at 2.3 Å resolution reveals mechanistic insights into metabolite gating. *Proc Natl Acad Sci U S A* 105:17742–17747. <https://doi.org/10.1073/pnas.0809634105>
4. Hiller S, Garces RG, Malia TJ, Orekhov VY, Colombini M, Wagner G (2008) Solution structure of the integral human membrane protein VDAC-1 in detergent micelles. *Science* 321:1206–1210. <https://doi.org/10.1126/science.1161302>
5. Colombini M (2009) The published 3D structure of the VDAC channel: native or not? *Trends Biochem Sci* 34:382–389. <https://doi.org/10.1016/j.tibs.2009.05.001>
6. Colombini M (2012) VDAC structure, selectivity, and dynamics. *BBA-Biomembranes* 1818:1457–1465. <https://doi.org/10.1016/j.bbammem.2011.12.026>
7. Mannella CA, Forte M, Colombini M (1992) Toward the molecular structure of the mitochondrial channel, VDAC. *J Bioenerg Biomembr* 24:7–19. <https://doi.org/10.1007/BF00769525>
8. Shanmugavadivu B, Apell H-J, Meins T, Zeth K, Kleinschmidt JH (2007) Correct folding of the β -barrel of the human membrane protein VDAC requires a lipid bilayer. *J Mol Biol* 368:66–78. <https://doi.org/10.1016/j.jmb.2007.01.066>
9. Eddy MT et al (2012) Lipid dynamics and protein-lipid interactions in 2D crystals formed with the β -barrel integral membrane protein VDAC1. *J Am Chem Soc* 134:6375–6387. <https://doi.org/10.1021/ja300347v>
10. Schneider R et al (2010) The native conformation of the human VDAC1 N terminus. *Angew Chem* 49:1882–1885. <https://doi.org/10.1002/anie.200906241>
11. Bauer AJ, Gieschler S, Lemberg KM, McDermott AE, Stockwell BR (2011) Functional model of metabolite gating by human voltage-dependent anion channel

2. *Biochemistry* 50:3408–3410. <https://doi.org/10.1021/bi2003247>
12. Eddy MT et al. (2019) Structural characterization of the human membrane protein VDAC2 in lipid bilayers by MAS NMR. *J Biomol NMR* 73:451–460. <https://doi.org/10.1007/s10858-019-00242-8>
13. Bennett AE, Griffin RG, Ok JH, Vega S (1992) Chemical shift correlation spectroscopy in rotating solids: radio frequency-driven dipolar recoupling and longitudinal exchange. *J Chem Phys* 96:8624–8627. <https://doi.org/10.1063/1.462267>
14. Bennett AE, Rienstra CM, Auger M, Lakshmi KV, Griffin RG (1995) Heteronuclear decoupling in rotating solids. *J Chem Phys* 103:6951–6958. <https://doi.org/10.1063/1.470372>
15. Eddy MT et al (2015a) Magic angle spinning nuclear magnetic resonance characterization of voltage-dependent anion channel gating in two-dimensional lipid crystalline bilayers. *Biochemistry* 54:994–1005. <https://doi.org/10.1021/bi501260r>
16. Eddy MT et al (2015b) Lipid bilayer-bound conformation of an integral membrane beta barrel protein by multidimensional MAS NMR. *J Biomol NMR* 61:299–310. <https://doi.org/10.1007/s10858-015-9903-1>
17. Dolder M, Zeth K, Tittmann P, Gross H, Welte W, Wallimann T (1999) Crystallization of the human, mitochondrial voltage-dependent anion-selective channel in the presence of phospholipids. *J Struct Biol* 127:64–71. <https://doi.org/10.1006/jsbi.1999.4141>
18. Malia TJ, Wagner G (2007) NMR structural investigation of the mitochondrial outer membrane protein VDAC and its interaction with antiapoptotic Bcl-xL. *Biochemistry* 46:514–525. <https://doi.org/10.1021/bi061577h>
19. Yu T-Y, Raschle T, Hiller S, Wagner G (2012) Solution NMR spectroscopic characterization of human VDAC-2 in detergent micelles and lipid bilayer nanodiscs. *BBA Biomembranes* 1818:1562–1569. <https://doi.org/10.1016/j.bbamem.2011.11.012>
20. Shi L, Lake EMR, Ahmed MAM, Brown LS, Ladizhansky V (2009) Solid-state NMR study of proteorhodopsin in the lipid environment: secondary structure and dynamics. *Biochim Biophys Acta* 1788:2563–2574. <https://doi.org/10.1016/j.bbamem.2009.09.011>



Chapter 12

Preparation of a Deuterated Membrane Protein for Small-Angle Neutron Scattering

Yuqi Wu, Kevin L. Weiss, and Raquel L. Lieberman

Abstract

This chapter outlines a protocol developed to prepare a purified deuterated membrane protein for a small-angle neutron scattering (SANS) experiment. SANS is a noninvasive technique well suited to studying membrane protein solution structures, and deuteration enhances the signal from the protein over the background (Breyton et al., *Eur Phys J E Soft Matter* 36 (7):71, 2013; Garg et al., *Biophys J* 101 (2):370–377, 2011). We present our workflow: transformation of our plasmid into *E. coli*, cell growth and expression of our deuterated protein, membrane isolation, detergent solubilization, protein purification, purity assessment, and final preparation for SANS.

Key words Membrane protein, α -Helix, Recombinant expression, Intramembrane proteolysis, Detergent, Contrast match point, Deuteration, Neutron scattering

1 Introduction

Membrane proteins are abundant and important players in many fundamental cell processes, including catalysis, cell adhesion, and the transport of molecules across membranes. As such, about half of the current drugs target membrane proteins [1]. Still, relatively few molecular structures of membrane proteins are available (<http://blanco.biomol.uci.edu/mpstruc/listAll/list>), especially in comparison to structures of soluble proteins deposited in the protein data bank (<http://www.rcsb.org/pdb>). There are numerous challenges associated with the study of the functions and structures of membrane proteins [2, 3]. Recombinant expression of human proteins is a common bottleneck, as the yield from a eukaryotic expression system is often low compared to levels needed for downstream applications for molecular characterization. Another problem is inherent in the nature of membrane proteins; detergent or other amphiphilic systems need to be used to maintain their active conformations [4]. Such reagents are not compatible with many

downstream analytical techniques. For example, in crystallography the phase behavior of the surfactant can inhibit crystal growth [5], and in small-angle X-ray scattering (SAXS) [6] and in size exclusion chromatography coupled to multiangle light scattering [7], the signal of the protein cannot always be readily distinguished from the signal from a similarly sized detergent micelle.

Among the biophysical techniques currently available for studying membrane proteins, neutron scattering stands out [8, 9]. Thermal or cold neutrons do not deposit energy into the samples, so unlike electrons or X-rays [10], neutrons are noninvasive and nondestructive to the sample [9]. Second, the method can be used to probe structure in lipid membranes or solution, as in small-angle neutron scattering (SANS). The signal from lipid or detergent can be suppressed using a method called “contrast matching”, which exploits the different scattering length densities (SLDs) of components such as proteins, detergents, and buffer, so that only the signal from the protein of interest remains [8, 11]. Traditionally, the SLDs of solvent and detergent are matched by adjusting the percentage of D₂O [12], but more sophisticated methods are emerging [13–16]. In addition, deuteration of the target macromolecule increases signal-to-noise, facilitating the data collection process [8, 12, 17].

Intramembrane aspartyl proteases (IAPs) are a subfamily of membrane-bound proteases that contain the conserved catalytic motif in YD...GXGD, where X is often a hydrophobic residue [18]. There are two main members of the IAP subfamily in humans: signal peptide peptidase (SPP) and presenilin. SPPs hydrolyze signal peptides and are involved, for example, in the production of the cell surface histocompatibility antigen-E epitope [19] and are exploited by hepatitis C for virus core protein maturation [20]. Presenilin is the catalytic subunit of the γ -secretase complex and is of interest due to its relevance to Alzheimer’s disease [21, 22]. Disease-causing mutations in presenilin or in one of its substrates, amyloid precursor protein (APP), alter the levels or atomic properties of amyloid- β , which ultimately deposit as plaques in the brain [21, 23]. Human presenilin cannot be studied structurally in solution by SANS because it is only active after the full γ -secretase complex is assembled, and after it is endoproteolyzed into N-terminal and C-terminal fragments [21, 22]. Human SPP and γ -secretase, which are expressed in eukaryotic cell lines, cannot be deuterated because eukaryotic cells cannot grow in media with more than 30–50% D₂O [24]. Therefore, to gain insight into the structural and biochemical properties of presenilin, we turned to an archaeal IAP ortholog from a methanogen *Methanoculleus marisnigri* JR1 (MCMJR1) as a model. MCMJR1 IAP (MCMIAP) and presenilin share sequence and structural similarity, including catalytic motifs within the same TM segments [25–27].

The protocol presented in this chapter describes the preparation of deuterated MCMIAP for SANS in order to study its structure and oligomeric state in detergent solution where it is enzymatically active [28, 29]. The MCMIAP gene was cloned into a commercially available vector flanked by an N-terminal *pelB* leader sequence for periplasmic insertion and a C-terminal hexahistidine tag for affinity chromatography (Subheading 3.1) for expression and purification in relatively high yield [30]. After transforming chemically competent *E. coli* (Subheading 3.2) with the MCMIAP-containing plasmid, cells were adapted to deuterium-labeled medium and then scaled up for overexpression (Subheadings 3.3–3.6). Following cell lysis and membrane isolation (Subheading 3.7), purification consisted of detergent solubilization of the membranes, affinity chromatography using nickel-nitrilotriacetic acid (Ni-NTA) resin, and further polishing using size exclusion chromatography (SEC). SEC was also used to exchange the protein into a buffer solution with the desired D₂O and deuterated detergent content before analysis by SDS-PAGE (Subheading 3.8) and SANS. This protocol was developed in close collaboration with the Center for Structural Molecular Biology Bio-Deuteration Laboratory at Oak Ridge National Laboratory and should be adaptable to other membrane proteins with transmembrane (TM) helices or with β -barrels expressed in *E. coli*.

2 Materials

2.1 Molecular Biology

1. The plasmid containing the gene of interest is the same as for the unlabeled protein [30].

2.2 Transformation of a Suitable *E. coli* Host

1. *E. coli* Rosetta™ 2(DE3) Singles™ competent cells (Novagen) (*see Note 1*).
2. Plasmid containing the gene of interest (*see* Subheading 3.1).
3. LB agar (e.g., LB Agar, Miller, Fisher Scientific) and petri dish.
4. Antibiotics for plasmid selection (ampicillin/chloramphenicol) (*see Note 2*).
5. SOC outgrowth medium (e.g., New England Biolabs).
6. Water bath set to 42 °C.
7. Incubator set to 37 °C.
8. Floor shaking incubator set to 37 °C.

2.3 Media and Solutions

1. LB agar plate with single colonies of transformed *E. coli* cells (*see* Subheading 3.2).
2. LB broth (e.g., Difco™ LB Broth, Miller, BD).
3. Minimal media salts (Table 1).

Table 1
Materials for minimal media preparation [31, 32]

Name of salt		Concentration
$(\text{NH}_4)_2\text{SO}_4$		7.0 g/L
Na_2HPO_4		5.25 g/L
KH_2PO_4		1.6 g/L
Diammonium hydrogen citrate		0.50 g/L
Glycerol		5.0 g/L
20% (w/v) $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$		1.0 mL/L
Holme trace metals (1.0 mL/L)	$\text{CaCl}_2 \cdot \text{H}_2\text{O}$	0.50 g/L
	CoCl_2	0.098 g/L
	CuSO_4	0.102 g/L
	$\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$	16.7 g/L
	$\text{MnSO}_4 \cdot \text{H}_2\text{O}$	0.114 g/L
	$\text{Na}_2\text{EDTA} \cdot 2\text{H}_2\text{O}$	22.3 g/L
	$\text{ZnSO}_4 \cdot \text{H}_2\text{O}$	0.112 g/L

4. D_2O (e.g., 99.8 atom % D deuterium oxide, Sigma-Aldrich).
5. Dry, sterile media bottles with GL45 threads.
6. Bottle top filters, 0.2 μm .
7. Appropriate antibiotics (e.g., carbenicillin and chloramphenicol) (*see* **Note 2**).
8. Isopropyl β -D-1-thiogalactopyranoside (IPTG) (e.g., Teknova).
9. Syringe or bottle filters, 0.2 μm .
10. 10% (w/v) NaOH in 90% D_2O (200 mL).
11. 30% (w/v) glycerol, 0.2% $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ in 90% D_2O with antibiotics (300 mL).
12. 70% (v/v) ethanol.

**2.4 Growth
of Deuterated
Inoculum**

1. Shaking incubator set to 37 °C.
2. Spectrophotometer (e.g., GE Ultrospec 10 cell density meter) for measuring optical density at 600 nm (OD_{600}) (*see* **Note 3**).
3. Baffled Fernbach shake flask.
4. Fixed angle centrifuge and rotor suitable for $\sim 6000 \times g$ set at 4 °C.

**2.5 Preparation
of the Bioreactor
Vessel
and Accessories**

1. Benchtop bioreactor system (e.g., Eppendorf BioFlo 310).
2. Bioreactor vessel (e.g., Eppendorf).
3. Inlet tubing 3/8" ID $\times$ 1/2" OD $\times$ 1/16" wall (e.g., VWR).

4. Exhaust tubing 5/16" ID $\times$ 9/16" OD $\times$ 1/8" wall (e.g., Versilic).
5. Inlet vent filter, 0.2 μ m (e.g., Pall Acro 37).
6. Exhaust vent filter, 0.2 μ m (e.g., Pall Acro 50).
7. Autoclave.
8. Electrode, pH (e.g., Mettler Toledo).
9. Calibration buffers (pH 4 and pH 7).
10. Dissolved oxygen (DO) sensor (e.g., Mettler Toledo).
11. DO electrolyte solution.
12. House cooling water or recirculating chiller.
13. House compressed air and regulator with a gas purifier (W. A. Hammond Drierite).
14. Nitrogen tank and regulator.
15. Stainless steel headpieces (Sartorius Stedim) with gaskets (Sartorius Stedim).
16. Vent filter, 0.2 μ m (e.g., GE Whatman 25 mm Polyvent filter).
17. PharMed peristaltic tubing 1/16" ID $\times$ 3/16" OD $\times$ 1/16" wall (e.g., Saint-Gobain).
18. Polypropylene glycol 2000 (Alfa Aesar).

2.6 Expression of Deuterated Membrane Protein

1. Deuterated inoculum (*see* Subheading 3.4).
2. Spectrophotometer (e.g., GE Ultrospec 10 cell density meter) for measuring optical density at 600 nm (OD₆₀₀) (*see* Note 3).
3. Semi-micro cuvettes, polystyrene.
4. 1 mM Isopropyl β -D-1-thiogalactopyranoside (IPTG) stock solution in 90% D₂O.

2.7 Membrane Isolation from Harvested Cell Paste

1. Cell paste containing the protein of interest (*see* Subheading 3.6).
2. Protease inhibitor (e.g., EDTA-Free SIGMAFAST Protease Inhibitor Cocktail Tablets, Sigma Aldrich).
3. Cell disruptor (e.g., Avestin EmulsiFlex C3) (*see* Note 4).
4. Ultracentrifuge and rotor suitable for $\sim 150,000 \times g$ set at 4 °C.
5. Dounce homogenizer, 40 mL, with loose and tight pestles (*see* Note 5).
6. Cell resuspension buffer: 50 mM HEPES (pH 7.5) and 200 mM NaCl.
7. Liquid N₂.

2.8 Purification of Deuterated Membrane Protein

1. Isolated membrane containing the protein of interest (*see* Subheading 3.7).
2. Dounce homogenizer, 40 mL, with loose and tight pestles (*see* **Note 5**).
3. Ni-NTA chromatography wash buffer (HisTrap buffer A): 50 mM HEPES (pH 7.5), 500 mM NaCl, and 20 mM imidazole.
4. Detergent for solubilizing membrane protein (e.g., Anagrade *n*-dodecyl β -D-maltoside (DDM), Anatrace), stored in desiccator at -80°C (*see* **Note 6**).
5. Ultracentrifuge and rotor suitable for $150,000 \times g$ set at 4°C .
6. Protein purification system (e.g., AKTA Pure). Depending on instrument, additional accessories may be required (e.g., Superloop 150 mL, GE Healthcare).
7. Ni-NTA column (e.g., HisTrap FF 1 or 5 mL, GE Healthcare) (*see* **Note 7**).
8. Wash buffer A for Ni-NTA chromatography: 50 mM HEPES (pH 7.5), 500 mM NaCl, 20 mM imidazole, and 0.1% DDM (*see* **Note 8**).
9. Elution buffer B for Ni-NTA chromatography: 50 mM HEPES (pH 7.5), 500 mM NaCl, 500 mM imidazole, and 0.1% DDM (*see* **Note 8**).
10. Gel filtration column (e.g., HiPrep 16/60 Sephacryl S-300, GE Healthcare).
11. Gel filtration buffer: 20 mM HEPES (pH 7.5), 250 mM NaCl, and 0.05% DDM (*see* **Note 8**).
12. Gel filtration column (e.g., Superose 12 10/300 GL, GE Healthcare).
13. Gel filtration buffer containing the correct D₂O percentage from contrast match point (CMP) calculation [11] (*see* Subheading 3.8, **step 10**) (*see* **Note 9**).
14. Amicon Ultra 15 mL centrifugal filter units (Millipore Sigma) with the appropriate molecular weight cut-off (MWCO) (*see* **Note 10**).
15. Swinging bucket benchtop centrifuge suited to $\sim 4500 \times g$ set at 4°C .
16. Spectrophotometer for measuring protein absorbance at 280 nm.
17. 12% polyacrylamide gel.
18. Electrophoresis unit and power supply.
19. Imager (e.g., Amersham Imager 600).

3 Methods

3.1 Molecular Biology

1. Clone the gene of the membrane protein of interest into the pET-22b(+) vector (Novagen) using the procedure and considerations described previously [30] (Fig. 1) (*see Note 11*).
2. Verify plasmid fidelity by DNA sequencing (e.g., MWG Operon or Genscript).

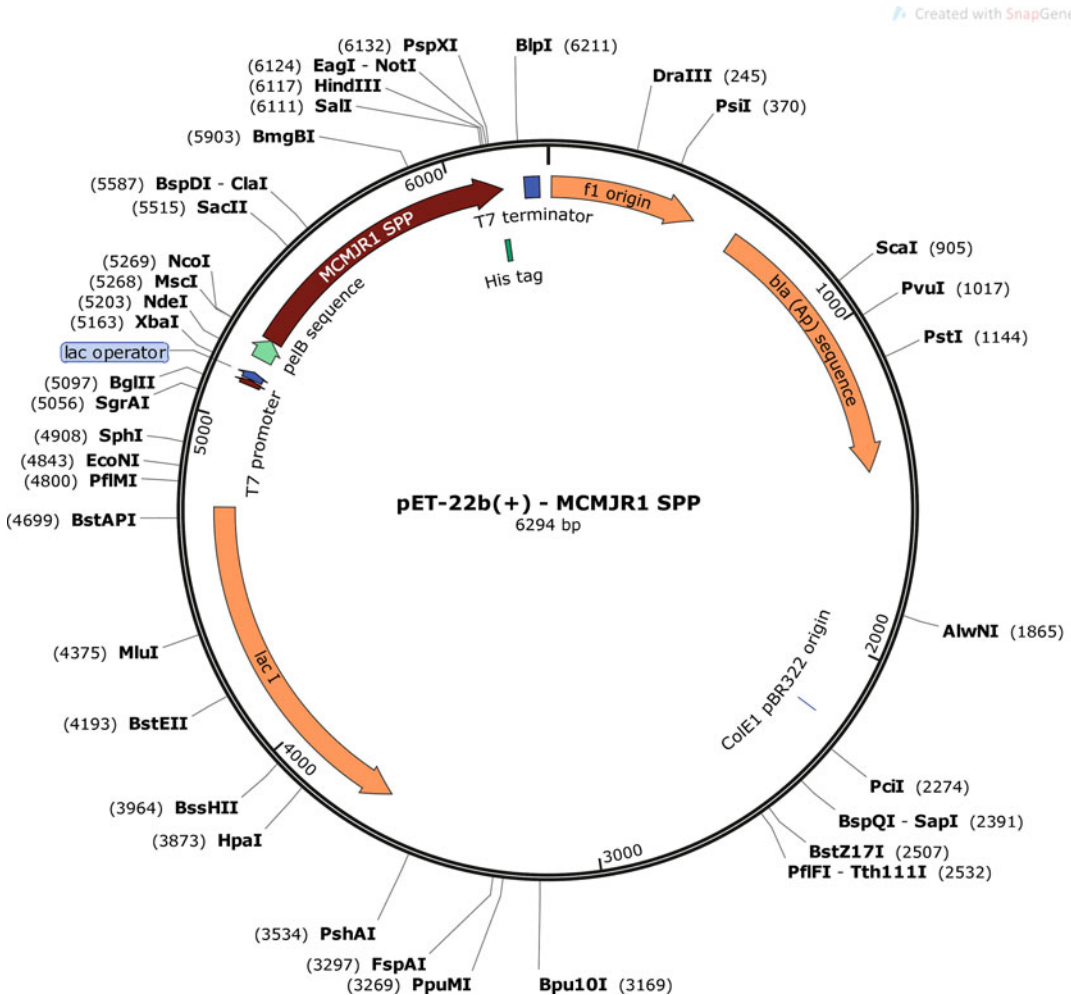


Fig. 1 Plasmid map of pET-22b(+) cloned with MCMIAP (MCMJR1 SPP). The *pepB* sequence precedes the protein of interest, MCMIAP in this case, to be inserted into the periplasmic membrane. The hexahistidine tag enables purification through Ni-NTA chromatography. The *bla* (Ap) sequence codes for beta-lactamase, providing ampicillin resistance for the plasmid, which is used as a selection marker in the protocol. Figure was produced using SnapGene Viewer

3.2 Transformation of a Suitable *E. coli* Host

1. Prepare an LB agar plate containing the appropriate amount of antibiotics (e.g., 60 µg/mL ampicillin and 34 µg/mL chloramphenicol) by following the manufacturer's protocol. Dry the plate overnight with lid slightly open at 37 °C in an incubator.
2. Thaw a ~50 µL aliquot of *E. coli* Rosetta 2 (DE3) cells on ice.
3. Add 30–50 ng plasmid DNA to the cells without mixing. Incubate the cells with DNA on ice for 30 min.
4. Heat shock the cells at 42 °C in the water bath for exactly 45 s. Return the cells to ice immediately afterward.
5. After 2 min, add ~250 µL prewarmed SOC outgrowth medium to the cells.
6. Shake the cells at 37 °C in a floor shaking incubator at 225 RPM for 1 h.
7. Using sterile technique, spread ~250 µL cells on the predried agar plate with a sterilized cell spreader.
8. Incubate the agar plate at 37 °C in an incubator overnight.
9. The next morning, confirm the presence of well-dispersed single colonies on the transformation plate and continue with growth of the deuterated inoculum (Subheading 3.4) or seal the plate with Parafilm and store it at 4 °C for later use.

3.3 Preparation of Media and Solutions

1. Prepare LB broth in H₂O according to manufacturer's instructions.
2. Prepare minimal medium in H₂O and D₂O according to Table 1. To guard against precipitation, add the magnesium and trace metals solutions only after the other components have been dissolved and diluted to at least a 2× concentration.
3. Prepare 1 mL of a 100 mg/mL stock solution of ampicillin/carbenicillin in H₂O and D₂O. Prepare 1 mL of a 34 mg/mL stock solution of chloramphenicol in ethanol.
4. Prepare 5 mL of a 1 M IPTG stock solution in H₂O and D₂O.
5. Using syringe or bottle top filters, filter all solutions into dry, autoclaved tubes or containers using sterile technique.

3.4 Growth of the Deuterated Inoculum

1. Using sterile technique, inoculate a single colony from the LB agar plate (*see* Subheading 3.2) into a clear polystyrene 17 × 100 mm culture tube containing 3 mL LB broth and antibiotics.
2. Incubate the cells at 37 °C in a shaking incubator at 250 RPM.
3. When the OD₆₀₀ of the cell culture reaches ~1, dilute the culture 1:20 (v/v) into 3 mL minimal medium (0% D₂O) in another polystyrene culture tube (*see* Note 12).

4. When the OD₆₀₀ of the minimal medium culture reaches ~1, dilute the culture 1:20 (v/v) into 3 mL minimal medium containing 50% D₂O.
5. To adapt the cells to deuterium-labeled medium, repeat **step 4** with minimal medium containing 75% and then 100% D₂O (or desired maximal percentage). To prepare fully deuterated (perdeuterated) protein, use 100% D₂O and a deuterated carbon source (e.g., glycerol-d8) (*see Note 12, 13*).
6. Once the cells are growing in the deuterated medium that will be used for protein expression, progressively scale up the adapted culture in fresh deuterated medium to a final inoculum that is one / tenth the initial volume to be used in the bioreactor. For the deuteration of MCMIAP, a 400 mL inoculum was prepared [11, 14].

3.5 Preparation of the Bioreactor Vessel and Accessories

1. Based on the protein expression level and the amount of material required, select a bioreactor vessel. For the deuteration of MCMIAP, a vessel with a maximum working volume of 5.5 L was used [11, 14].
2. Clean and assemble the bioreactor vessel per manufacturer's instructions. However, do not add the sensors (DO and pH) or deuterated medium.
3. Steam sterilize the empty bioreactor vessel, feed bottles, and peristaltic pump tubing in an autoclave.
4. Immediately after the autoclave cycle ends, carefully move the hot vessel to the bioreactor system and flow dry compressed air through the vessel via the inlet vent filter. Dry all other autoclaved labware in an oven set to 80 °C.
5. Prepare the DO sensor using manufacturer's instructions. Polarographic sensors must be polarized for several hours prior to calibration.
6. Attach the pH electrode to the bioreactor system and calibrate using pH standards.
7. Rinse the DO and pH sensors with deionized water and spray them down thoroughly with 70% ethanol.
8. After carefully blotting away the 70% ethanol, insert both sensors into the vessel and connect the corresponding cables. Insert the temperature probe into the thermowell within the vessel headplate.
9. Following manufacturer's instructions, connect the inlet and outlet water hoses to the corresponding ports on the vessel base and exhaust condenser.
10. Assemble two oven-dried feed bottles, add the filtered 10% (w/v) NaOH to one bottle and the 30% (w/v) glycerol feed solution to the other.

Table 2
Initial setpoints

Setpoint name	Setpoint value
Temperature	30 °C
Dissolved oxygen	30%
pH	6.9
Agitation	200 RPM
Gas flow	4 L/min
Glycerol feed	~12 mL/h

11. Attach peristaltic tubing to the bottles and pump heads then prime the lines with solution.
12. Pour sterile-filtered, deuterated medium into the vessel via an inoculation port.
13. Add 200 μ L polypropylene glycol 2000 (antifoam) per liter of medium.
14. Install the motor assembly, agitate the medium, and enter the temperature setpoint.
15. Once the temperature stabilizes, calibrate the DO sensor with nitrogen and compressed air per manufacturer's instructions.
16. Enter the remaining setpoints at the bioreactor system console. *See* Table 2 for the setpoint values that were used for the deuteration of MCMiAP [11, 14].
17. Enable cascaded control at the bioreactor system console to vary agitation (200–800 RPM) to maintain DO greater than 30%.
18. If available, initiate software for recording data and controlling the glycerol feed. The DO spike that occurs upon consumption of the initial 5 g/L of glycerol is typically used as the trigger that initiates glycerol feeding.

3.6 Expression of Deuterated Membrane Protein

1. Inoculate the vessel with deuterium-labeled culture prepared in Subheading 3.4.
2. Monitor the bioreactor system and collect periodic samples from the vessel for OD₆₀₀ measurements.
3. Once the deuterated cell mass increases (OD₆₀₀ ~ 8–10), set the desired induction temperature and add IPTG. The induction time, temperature, and post-induction feed rate varies with the protein being expressed.

4. Harvest the cell paste by centrifugation at $\sim 6000 \times g$ for 45 min at 4 °C and shut down the bioreactor system following manufacturer's instructions.
5. Flash-freeze the cell paste in liquid N₂ and store at –80 °C for later use or lyse directly as described in Subheading 3.7.

3.7 Membrane Isolation from Harvested Cell Paste

1. In centrifuge bottles or a separate glass media bottle, resuspend the deuterated cell paste in buffer (*see* Subheading 2.7, **item 6**) containing an appropriate amount of protease inhibitor on ice. For every 1 g of cell paste, use at least 5 mL of resuspension buffer. Resuspend the cell paste gently and thoroughly using a serological pipet.
2. Lyse the cells by three passages through a cell disruptor at 15,000 psi.
3. Centrifuge the cell lysate at $\sim 5000 \times g$ for 15 min at 4 °C to remove cell debris. Repeat 3–4 times to obtain clear supernatant.
4. Continue with membrane isolation (**step 5**) or aliquot the supernatant in 50 mL Falcon tubes or Nalgene bottles and store at –80 °C.
5. Ultracentrifuge the supernatant at $\sim 150,000 \times g$ for 30–45 min at 4 °C to isolate the membranes.
6. Pool the pelleted membranes into a 40 mL Dounce homogenizer. Resuspend the membrane in fresh cell resuspension buffer using 10–20 passes of the loose pestle and three passes of the tight pestle.
7. Ultracentrifuge again at $150,000 \times g$ for 30–45 min at 4 °C. Repeat **steps 6** and **7** to completely remove loosely bound proteins.
8. Remove the pelleted membrane from the ultracentrifuge tube with a spatula. Use in purification immediately or flash freeze in liquid N₂ for storage at –80 °C for later use (*see* **Note 14**).

3.8 Purification of Deuterated Membrane Protein

1. Add desired membrane mass to a 40 mL Dounce homogenizer using a spatula containing Ni-NTA wash buffer A (no DDM) on ice. For every 1 g of membrane, use 5 mL of buffer. If more than 40 mL is needed, divide the sample accordingly.
2. Resuspend the membrane fraction using 10–20 passes of the loose and tight pestles until membrane appears homogeneously resuspended in buffer.
3. After warming the DDM bottle to room temperature for 15–20 min, make a 2% DDM solution in Ni-NTA wash buffer A in a conical tube, matching the volume used for membrane resuspension. Place this solution on a 3D rocker for 10–15 min at 4 °C to fully dissolve the DDM.

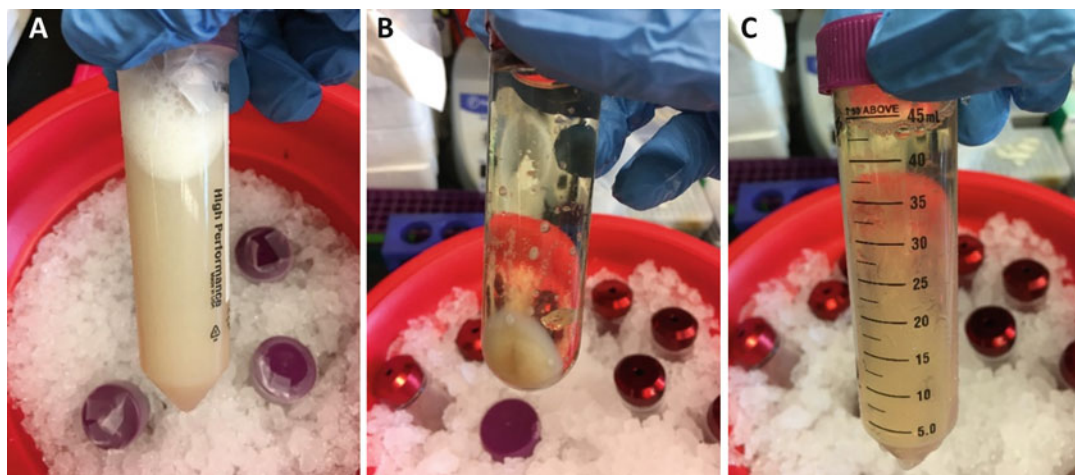


Fig. 2 Membrane solubilization. (a) MCMIAP resuspended membranes with 1% DDM solution added. Note that the solution has not become translucent after rocking even though membranes are predominantly solubilized as seen in **b**. (b) Insoluble material is pelleted after ultracentrifugation, fully separated from the solubilized membrane protein, which is in solution. (c) Solubilized membrane fraction prior to Ni-NTA chromatography

4. Gently add the resuspended membranes from **step 2** to the 2% DDM buffer solution from **step 3**; the final concentration of DDM will be 1%. Place the conical tube on the 3D rocker for 30–60 min at 4 °C to fully solubilize the membrane protein (Fig. 2a) (*see Note 15*).
5. Ultracentrifuge at $150,000 \times g$ for 30–45 min at 4 °C to remove any insoluble material (Fig. 2b).
6. Load the supernatant from **step 5** (Fig. 2c) onto the Ni-NTA column connected to a protein purification system and pre-equilibrated with Ni-NTA wash buffer A (containing 0.1% DDM). Wash with Ni-NTA wash buffer A until the absorbance returns to baseline, then wash with a mixture of 95% Ni-NTA wash buffer A and 5% Ni-NTA elution buffer B (containing 0.1% DDM) to remove unbound and loosely bound proteins. Elute the protein using a gradient from 5% to 60% Ni-NTA elution buffer B. Collect 1.5 mL fractions throughout the purification. For volumes higher than that of the Superloop, do consecutive purifications until all the supernatant has been processed (*see Note 7*).
7. As needed, concentrate each elution fraction to 500 μ L using Amicon Ultra concentration devices centrifugal filter units with appropriate molecular weight cutoff (MWCO) (*see Note 10*). Analyze the fractions by SDS-PAGE to confirm that the protein of interest is present and contains minimal impurities (*see Notes 7 and 16*).

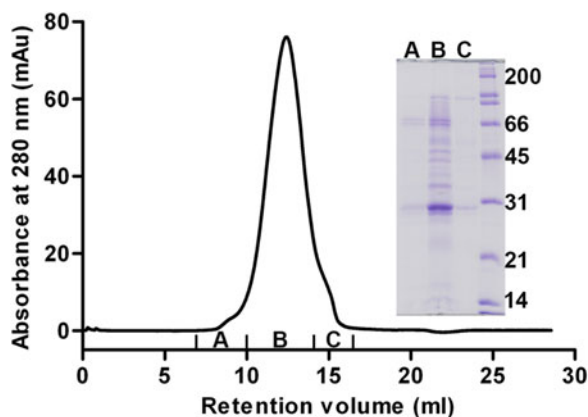


Fig. 3 Purification of deuterated MCMIAP for SANS. A Superose 12 column was used for final exchange into desired buffer (20 mM HEPES at pH 7.5, 250 mM NaCl, 48.5% D₂O, 0.05% DDM, 44% (w/v) of which is tail-deuterated). The SDS-PAGE gel shows relatively clean MCMIAP (~31 kDa) in fraction B, which was used for SANS experiment. Figure is reproduced with permission from Naing et al. [14]

8. Pool elution fractions and further concentrate to 1 mL as in **step 7** for polishing using SEC (e.g., HiPrep 16/60 Sephacryl S-300). Use gel filtration buffer (containing 0.05% DDM) for equilibration and elution.
9. To exchange purified protein into conditions for contrast matching in the SANS experiment, use the Superose 12 SEC column (Fig. 3) (*see Note 9*).
10. After SDS-PAGE analysis to confirm purity (Fig. 3), using concentration devices as needed, pool fractions and concentrate to the final volume for SANS (*see Note 17*).
11. Calculate the final concentration of the SANS sample by measuring the absorbance of the final sample at 280 nm and converting to concentration, for example, using the calculated extinction coefficient (e.g., using online ProtParam tool [33]) and molecular weight of the protein (*see Note 18*).

4 Notes

1. After optimization, *E. coli* Rosetta 2 (DE3) cells were selected to express MCMIAP because this line corrects for rare codons and active enzyme could be purified in sufficient yield for downstream applications. An option to avoid this cell line might be to conduct codon optimization on the plasmid [34].
2. Both antibiotics are used to select for only successfully transformed *E. coli* Rosetta 2 (DE3) cells. Ampicillin/carbenicillin

resistance comes from the pET22b(+) plasmid (Fig. 1). Chloramphenicol resistance comes from the plasmid which codes for rare codon tRNAs in *E. coli* Rosetta 2 (DE3) cells.

3. Since the measured OD₆₀₀ is due to light scattering, which is affected by the optical layout, OD₆₀₀ measurements made using different spectrophotometers can vary for the same culture [35].
4. Other cell disruption methods (sonication or French press) can be used depending on the scale required.
5. The Dounce homogenizer offers gentle homogenization and avoids disruption of the membrane.
6. The choice of detergent may vary for different proteins [3]. Previous detergent screening and protein stability tests in detergents suggested that DDM was optimal for the purification of MCMIAP [30].
7. It may be advantageous to use a larger (e.g., 5 mL or tandem 1 mL) HisTrap column for purification. In our experience, the column may not capture all of the protein of interest from the first round of column purification. Reloading the flow-through fractions onto a cleaned column may help rescue some of the protein of interest. Similarly, it is probably good practice to divide the samples into multiple batches and run each batch separately on a Ni-NTA column to prevent overloading the column.
8. 0.1% and 0.05% are much higher than the reported critical micelle concentration of DDM: 0.0087% [36]. Higher concentration ensures a high level of solubilization of membrane protein.
9. In our study of MCMIAP by SANS, 48.5% (w/v) tail-deuterated DDM (total with protonated DDM 0.05%) and 44% D₂O were used to extinguish the scattering signal from DDM [14]. A detailed protocol for determining the CMP can be found in Oliver et al. [11].
10. For MCMIAP, a 50 kDa MWCO membrane was used. To choose the optimal molecular mass cutoff for concentrating a membrane protein, both the protein loss and detergent loss need to be considered. For detailed procedures on choosing the best centrifugal filter, *see* other studies by us and others [30, 37].
11. Important features include an N-terminal *pelB* leader sequence for periplasmic membrane insertion and a C-terminal hexahistidine tag for Ni-NTA purification flanking the gene of interest. Depending on the membrane protein topology, *pelB* may not be appropriate. Other slight plasmid variations for MCMIAP have been reported [38, 39].

12. Dilutions other than 1:20 (v/v) may also be used. For small volume adaptation to media higher than 75% D₂O content, multiple dilutions are often grown in parallel to guard against failure.
13. Cell growth will slow as the D₂O content increases. The final D₂O percentage used in the deuteration of MCMiAP was 90% [14]. Due to the long growth times in minimal media, ampicillin is typically substituted with carbenicillin. For growth of the Rosetta 2(DE3) cells used in the expression of deuterated MCMiAP, the carbenicillin concentration used was 100 µg/mL. At 90% D₂O, the chloramphenicol concentration was decreased from 34 to 17 µg/mL [14].
14. Membranes can be stored at −80 °C for weeks. Start the purification close to the scheduled SANS beam time to ensure that the protein is as fresh as possible.
15. After rocking, the solution may turn from opaque to translucent, but if not, still proceed to the ultracentrifugation step. An additional resuspension and solubilization round can be employed to ensure complete membrane solubilization.
16. For MCMiAP, the *E. coli* outer membrane protein (OmpF, MW ~ 40 kDa monomer, ~100 kDa trimer) and multidrug efflux protein (AcrB, MW ~113 kDa monomer, ~300 kDa trimer) are common impurities identified during the purification process.
17. The final concentrated volume should consider the volume of the quartz cells used in the SANS experiment. Cylindrical quartz cells (banjo cells) with a 1 mm pathlength require ~300 µL each. Besides SDS-PAGE, mass spectrometry methods can be used to confirm the identity of the protein, although the deuteration can complicate analysis for typical peptide digestion experiments.
18. Protein expression and purification procedures require optimization for each different membrane protein for optimal yield.

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References

- Overington JP, Al-Lazikani B, Hopkins AL (2006) How many drug targets are there? *Nat Rev Drug Discov* 5(12):993–996. <https://doi.org/10.1038/nrd2199>
- Carpenter EP, Beis K, Cameron AD, Iwata S (2008) Overcoming the challenges of membrane protein crystallography. *Curr Opin Struct Biol* 18(5):581–586. <https://doi.org/10.1016/j.sbi.2008.07.001>
- Lin S-H, Guidotti G (2009) Purification of membrane proteins. In: Burgess RR, Deutscher MP (eds) *Methods in enzymology*, vol 463. Academic, New York, NY, pp 619–629. [https://doi.org/10.1016/S0076-6879\(09\)63035-4](https://doi.org/10.1016/S0076-6879(09)63035-4)
- Seddon AM, Curnow P, Booth PJ (2004) Membrane proteins, lipids and detergents: not just a soap opera. *Biochim Biophys Acta* 1666(1–2):105–117. <https://doi.org/10.1016/j.bbamem.2004.04.011>
- Wiener MC (2004) A pedestrian guide to membrane protein crystallization. *Methods* 34(3):364–372. <https://doi.org/10.1016/j.ymeth.2004.03.025>
- Hong X, Weng YX, Li M (2004) Determination of the topological shape of integral membrane protein light-harvesting complex LH2 from photosynthetic bacteria in the detergent solution by small-angle X-ray scattering. *Biophys J* 86(2):1082–1088. [https://doi.org/10.1016/s0006-3495\(04\)74183-1](https://doi.org/10.1016/s0006-3495(04)74183-1)
- Gimpl K, Klement J, Keller S (2016) Characterising protein/detergent complexes by triple-detection size-exclusion chromatography. *Biol Proced Online* 18:4. <https://doi.org/10.1186/s12575-015-0031-9>
- Breyton C, Gabel F, Lethier M, Flayhan A, Durand G, Jault JM, Juillan-Binard C, Imbert L, Moulin M, Ravaud S, Hartlein M, Ebel C (2013) Small angle neutron scattering for the study of solubilised membrane proteins. *Eur Phys J E Soft Matter* 36(7):71. <https://doi.org/10.1140/epje/i2013-13071-6>
- Garg S, Porcar L, Woodka AC, Butler PD, Perez-Salas U (2011) Noninvasive neutron scattering measurements reveal slower cholesterol transport in model lipid membranes. *Biophys J* 101(2):370–377. <https://doi.org/10.1016/j.bpj.2011.06.014>
- Shull CG, Wollan EO (1948) X-ray, electron, and neutron diffraction. *Science* 108(2795):69–75. <https://doi.org/10.1126/science.108.2795.69>
- Oliver RC, Naing S-H, Weiss KL, Pingali SV, Lieberman RL, Urban VS (2018) Contrast-matching detergent in small-angle neutron scattering experiments for membrane protein structural analysis and ab Initio modeling. *JoVE* 140:e57901
- Trewhella J (2006) Neutrons reveal how nature uses structural themes and variation in biological regulation. *Physica B* 385:825–830
- Oliver RC, Pingali SV, Urban VS (2017) Designing mixed detergent micelles for uniform neutron contrast. *J Phys Chem Lett* 8(20):5041–5046. <https://doi.org/10.1021/acs.jpclett.7b02149>
- Naing SH, Oliver RC, Weiss KL, Urban VS, Lieberman RL (2018) Solution structure of an intramembrane aspartyl protease via small angle neutron scattering. *Biophys J* 114(3):602–608. <https://doi.org/10.1016/j.bpj.2017.12.017>
- Midtgaard SR, Darwish TA, Pedersen MC, Huda P, Larsen AH, Jensen GV, Kynde SAR, Skar-Gislunge N, Nielsen AJZ, Olesen C, Blaise M, Dorosz JJ, Thorsen TS, Venskutonyte R, Krintel C, Moller JV, Frielinghaus H, Gilbert EP, Martel A, Kastrup JS, Jensen PE, Nissen P, Arleth L (2018) Invisible detergents for structure determination of membrane proteins by small-angle neutron scattering. *FEBS J* 285(2):357–371. <https://doi.org/10.1111/febs.14345>
- Ashkar R, Bilheux HZ, Bordallo H, Briber R, Callaway DJE, Cheng X, Chu XQ, Curtis JE, Dadmun M, Fenimore P, Fushman D, Gabel F, Gupta K, Herberle F, Heinrich F, Hong L, Katsaras J, Kelman Z, Kharlampieva E, Kneller GR, Kovalevsky A, Krueger S, Langan P, Lieberman R, Liu Y, Losche M, Lyman E, Mao Y, Marino J, Mattos C, Meilleur F, Moody P, Nickels JD, O'Dell WB, O'Neill H, Perez-Salas U, Peters J, Petridis L, Sokolov AP, Stanley C, Wagner N, Weinrich M, Weiss K, Wymore T, Zhang Y, Smith JC (2018) Neutron scattering in the biological sciences: progress and prospects. *Acta Crystallogr D Struct Biol* 74(Pt 12):1129–1168. <https://doi.org/10.1107/S2059798318017503>
- Meilleur F, Weiss KL, Myles DA (2009) Deuterium labeling for neutron structure-function-dynamics analysis. *Methods Mol Biol* 544:281–292. https://doi.org/10.1007/978-1-59745-483-4_18
- Erez E, Fass D, Bibi E (2009) How intramembrane proteases bury hydrolytic reactions in the membrane. *Nature* 459(7245):371
- Lemberg MK, Bland FA, Weihofen A, Braud VM, Martoglio B (2001) Intramembrane proteolysis of signal peptides: an essential step in the generation of HLA-E epitopes. *J Immunol* 167(11):6441–6446. <https://doi.org/10.4049/jimmunol.167.11.6441>

20. McLauchlan J, Lemberg MK, Hope G, Martoglio B (2002) Intramembrane proteolysis promotes trafficking of hepatitis C virus core protein to lipid droplets. *EMBO J* 21 (15):3980–3988. <https://doi.org/10.1093/emboj/cdf414>
21. Wolfe MS (2019) Structure and function of the gamma-secretase complex. *Biochemistry* 58 (27):2953–2966. <https://doi.org/10.1021/acs.biochem.9b00401>
22. Wolfe MS, Xia W, Ostaszewski BL, Diehl TS, Kimberly WT, Selkoe DJ (1999) Two transmembrane aspartates in presenilin-1 required for presenilin endoproteolysis and gamma-secretase activity. *Nature* 398 (6727):513–517. <https://doi.org/10.1038/19077>
23. Chow VW, Mattson MP, Wong PC, Gleichmann M (2010) An overview of APP processing enzymes and products. *NeuroMolecular Med* 12(1):1–12. <https://doi.org/10.1007/s12017-009-8104-z>
24. Katz JJ, Crespi HL (1966) Deuterated organisms: cultivation and uses. *Science* 151 (3715):1187–1194. <https://doi.org/10.1126/science.151.3715.1187>
25. Dang S, Wu S, Wang J, Li H, Huang M, He W, Li YM, Wong CC, Shi Y (2015) Cleavage of amyloid precursor protein by an archaeal presenilin homologue PSH. *Proc Natl Acad Sci U S A* 112(11):3344–3349. <https://doi.org/10.1073/pnas.1502150112>
26. Bai XC, Yan C, Yang G, Lu P, Ma D, Sun L, Zhou R, Scheres SHW, Shi Y (2015) An atomic structure of human gamma-secretase. *Nature* 525(7568):212–217. <https://doi.org/10.1038/nature14892>
27. Li X, Dang S, Yan C, Gong X, Wang J, Shi Y (2013) Structure of a presenilin family intramembrane aspartate protease. *Nature* 493(7430):56–61. <https://doi.org/10.1038/nature11801>
28. Naing SH, Kalyoncu S, Smalley DM, Kim H, Tao X, George JB, Jonke AP, Oliver RC, Urban VS, Torres MP, Lieberman RL (2018) Both positional and chemical variables control in vitro proteolytic cleavage of a presenilin ortholog. *J Biol Chem* 293(13):4653–4663. <https://doi.org/10.1074/jbc.RA117.001436>
29. Naing SH, Vukoti KM, Drury JE, Johnson JL, Kalyoncu S, Hill SE, Torres MP, Lieberman RL (2015) Catalytic properties of intramembrane aspartyl protease substrate hydrolysis evaluated using a FRET peptide cleavage assay. *ACS Chem Biol* 10(9):2166–2174. <https://doi.org/10.1021/acschembio.5b00305>
30. Johnson JL, Kalyoncu S, Lieberman RL (2016) Lessons from an alpha-helical membrane enzyme: expression, purification, and detergent optimization for biophysical and structural characterization. *Methods Mol Biol* 1432:281–301. https://doi.org/10.1007/978-1-4939-3637-3_18
31. Holme T (1970) Enzymes-laboratory scale production. *Process Biochem* 5:62–66
32. Törnkvist M, Larsson G, Enfors S-O (1996) Protein release and foaming in *Escherichia coli* cultures grown in minimal medium. *Bioprocess Eng* 15(5):231–237. <https://doi.org/10.1007/bf02391583>
33. Wilkins MR, Gasteiger E, Bairoch A, Sanchez JC, Williams KL, Appel RD, Hochstrasser DF (1999) Protein identification and analysis tools in the ExPASy server. *Methods Mol Biol* 112:531–552. <https://doi.org/10.1385/1-59259-584-7:531>
34. Norholm MH, Light S, Virkki MT, Elofsson A, von Heijne G, Daley DO (2012) Manipulating the genetic code for membrane protein production: what have we learnt so far? *Biochim Biophys Acta* 1818(4):1091–1096. <https://doi.org/10.1016/j.bbamem.2011.08.018>
35. Koch AL (1970) Turbidity measurements of bacterial cultures in some available commercial instruments. *Anal Biochem* 38(1):252–259. [https://doi.org/10.1016/0003-2697\(70\)90174-0](https://doi.org/10.1016/0003-2697(70)90174-0)
36. VanAken T, Foxall-VanAken S, Castleman S, Ferguson-Miller S (1986) Alkyl glycoside detergents: synthesis and applications to the study of membrane proteins. *Methods Enzymol* 125:27–35. [https://doi.org/10.1016/s0076-6879\(86\)25005-3](https://doi.org/10.1016/s0076-6879(86)25005-3)
37. Feroz H, Vandervelden C, Ikwuagwu B, Ferlez B, Baker CS, Lugar DJ, Grzelakowski M, Golbeck JH, Zydney AL, Kumar M (2016) Concentrating membrane proteins using ultrafiltration without concentrating detergents. *Biotechnol Bioeng* 113 (10):2122–2130. <https://doi.org/10.1002/bit.25973>
38. Zhou R, Shi Y, Yang G (2017) Expression, purification, and enzymatic characterization of intramembrane proteases. *Methods Enzymol* 584:127–155. <https://doi.org/10.1016/bs.mic.2016.09.046>
39. Torres-Arancivia C, Ross CM, Chavez J, Assur Z, Dolios G, Mancía F, Ubarretxena-Belandia I (2010) Identification of an archaeal presenilin-like intramembrane protease. *PLoS One* 5(9). <https://doi.org/10.1371/journal.pone.0013072>



Preparing Membrane Proteins for Simulation Using CHARMM-GUI

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Abstract

Molecular dynamics simulations of membrane proteins have grown dramatically in the last 20 years. Running these simulations first requires embedding the protein's three-dimensional structure in a lipid bilayer of a suitable composition, one that resembles its native environment. This step is far from trivial, especially for modeling heterogeneous mixtures of lipids. CHARMM-GUI, a webserver for simulation system preparation greatly simplifies this step, allowing for the construction of complex heterogeneous and/or asymmetric membranes. Here, we demonstrate how to use CHARMM-GUI to build the membrane for the outer-membrane protein BamA.

Key words Lipid bilayer, Molecular dynamics, Gram-negative bacteria, Outer-membrane protein

1 Introduction

Membrane proteins perform multiple functions and are vital to the survival of all organisms [1]. It is estimated that the genes coding for membrane proteins make up 20–30% of the genomes of organisms [2]. They serve as channels [3, 4], transporters [5–7], receptors [8], enzymes [9] and function in cell signaling [10, 11], translocation of substrates [12–14], energy transduction [15, 16] and cell–cell recognition [17–19]. Due to their vital significance, advanced technological methods such as NMR [20], cryogenic electron microscopy (cryo-EM) [21] and X-ray crystallography [22] have been developed to determine structures of membrane proteins. However, these experimental methods only provide a static state of proteins while molecular dynamics (MD) simulations are capable of probing their dynamic behavior [23]. Therefore, preparing a membrane-protein system at atomic resolution is a major concern of simulators.

Previous studies have emphasized the importance of building a native membrane: not only are the lipid–protein interactions responsible for regulating or stabilizing the conformation of membrane proteins [24–27], but the composition of the membrane will also influence their structure and function [28, 29]. Consequently, one should be especially careful to select the appropriate membrane for a given membrane protein.

To simplify and automate the building process for a native membrane-protein system for MD simulations, CHARMM-GUI (<http://www.charmm-gui.org>) [19, 30] provides a graphical user interface (GUI) of multiple modules for the biomolecular simulation program CHARMM [31]. Membrane Builder [19] is one of the modules in CHARMM-GUI, which offers users a relatively easy way to build complicated membranes with all types of lipids through both user-specified and automated steps, including PDB loading, protein orientation, system size determination, generation for lipids, pore water, bulk water as well as ions, and assembly of components [19].

In this chapter, we will go through the process of preparing a membrane-protein system step by step in CHARMM-GUI using BamA as an example (Fig. 1). BamA is the central component of the BAM complex [32–35]. It is an outer membrane protein (OMP) of Gram-negative bacteria, which is responsible for the folding and insertion of other OMPs [36]. It contains a transmembrane β -barrel of 16 strands along with five periplasmic polypeptide-transport-associated (POTRA) domains (Fig. 2). We are going to use the structure from *E. coli* (PDB ID: 5AYW [33]), which includes all five POTRA domains.

2 Materials

2.1 Programs

1. In order to access CHARMM-GUI, a web browser such as Chrome and Firefox is required.

2.2 Hardware

1. A computer or laptop with any modern operating system (Windows, Linux, Mac OS).

3 Methods

3.1 Read Protein Coordinates and Manipulate Structure

The process of building a membrane-protein system via CHARMM-GUI starts with the loading of protein coordinates, followed by several alternative manipulation options, and finally generating a Protein Structure File (PSF) and PDB file containing the entire system. Users can upload a preoriented protein structure or specify a Protein Data Bank (PDB) ID to download PDB files

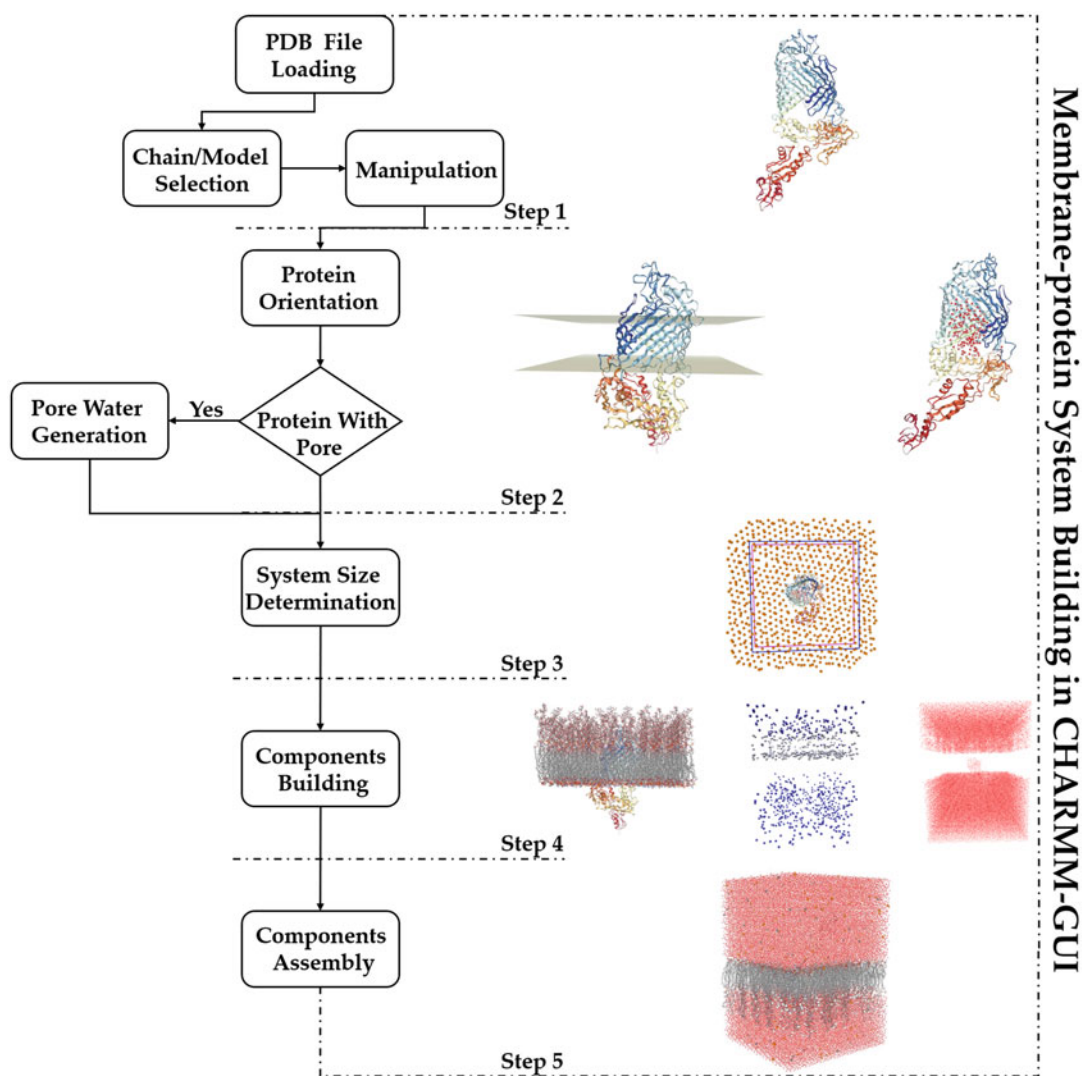


Fig. 1 Membrane-protein system building process in CHARMM-GUI

directly from either the Research Collaboratory for Structural Bioinformatics (RCSB) database [37, 38] or the Orientations of Proteins in Membranes (OPM) database [39] (Fig. 3).

3.1.1 Load PDB File

1. Open CHARMM-GUI (<http://www.charmm-gui.org>) in a web browser. Select the menu item **Input Generator** → **Membrane Builder** on the leftmost part of the website.
2. Drag the scroll bar to the middle. Two options will appear on the screen: **Protein/Membrane System** and **Membrane Only System**. Choose the former one.
3. Enter **5ayw** (PDB ID of one conformation of BamABCDE complex [33]) into the **Download PDB File** blank,

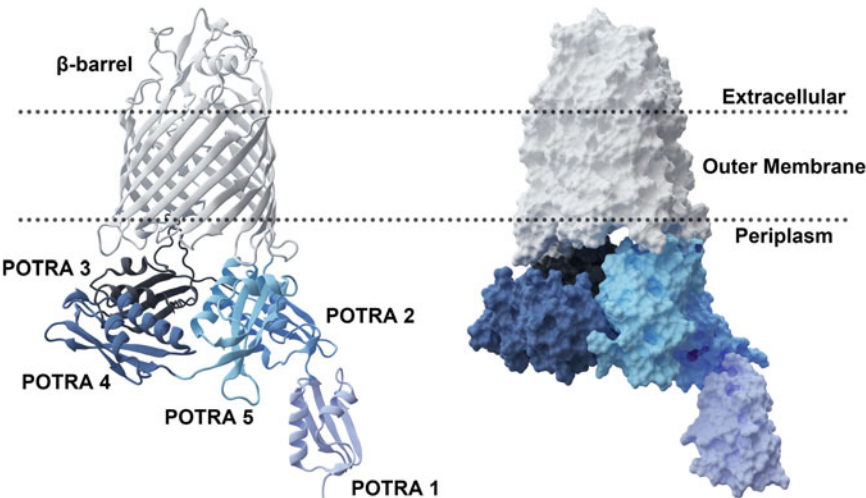


Fig. 2 Structure of *E. coli* BamA

```
SHEET 1 AA1 3 ASP A 28 GLU A 32 0
SHEET 2 AA1 3 THR A 83 GLU A 90 1 O VAL A 88 N GLU A 32
SHEET 3 AA1 3 PHE A 72 ARG A 79 -1 N ARG A 76 O GLN A 87
SHEET 1 AA2 3 THR A 93 SER A 100 0
SHEET 2 AA2 3 ARG A 162 VAL A 168 1 O VAL A 163 N ALA A 95
SHEET 3 AA2 3 LYS A 152 PRO A 157 -1 N LYS A 152 O VAL A 168
SHEET 1 AA3 4 GLU A 176 VAL A 183 0
SHEET 2 AA3 4 GLY A 253 THR A 261 1 O ILE A 254 N GLN A 178
SHEET 3 AA3 4 ASN A 239 LEU A 247 -1 N GLN A 244 O THR A 257
SHEET 4 AA3 4 SER B 193 LEU B 194 1 O SER B 193 N VAL A 245
SHEET 1 AA4 4 GLU A 294 LEU A 295 0
SHEET 2 AA4 4 LYS A 267 ASN A 276 -1 N LEU A 268 O GLU A 294
SHEET 3 AA4 4 THR A 334 ASP A 342 1 O VAL A 335 N LYS A 267
SHEET 4 AA4 4 ARG A 321 ASN A 329 -1 N MET A 325 O ARG A 338
SHEET 1 AA5 3 TYR A 348 GLU A 355 0
SHEET 2 AA5 3 GLN A 411 GLU A 420 1 O VAL A 412 N TYR A 348
SHEET 3 AA5 3 PHE A 395 ARG A 404 -1 N GLU A 396 O LYS A 419
```

Fig. 3 Part of the structural information in the original PDB file

meanwhile, selecting **OPM** as the **Download Source** (*see Note 1*). Then, click on the **Next Step: Select/Model Chain** button in the lower right corner.

4. View **Model/Chain Selection Option**. This PDB file contains five proteins. Information such as type, segID, PDB ID, first and last residue ID of chains and engineered residues are listed here as well.

Users are able to view the constitutive segments already present in the PDB file, which mainly include protein chains, substrates, crystallographic water molecules, ions, and crystallization

detergents. They can also select whatever segments they want to use as well.

5. Here, we will focus on BamA alone. Check on the box of **PROA** only. Then, click on the **Next Step: Manipulate PDB** button in the lower right corner.

Beyond deciding which segments to include in their system, users also have the ability to select a subset of residues of a chain, rename every segment, and remove engineered residues. These operations will not be used in this chapter.

3.1.2 Manipulate PDB File

In order to generate a system suitable for simulation, additional manipulation is required. CHARMM-GUI provides users with diverse options for manipulation to meet multiple demands, including terminal group patching, modeling missing residues, mutation, protonation, disulfide bonds, add lipidation, etc. Here, we will focus on terminal group patching and disulfide bonds manipulation options only. Readers can explore other options on their own.

1. Check the box labeled **Terminal group patching**. Select **NTER** for **First** and **CTER** for **Last**.
2. Check the box labeled **Disulfide bonds**. Set Pair 1 Residue ID to **690** while set Pair 2 Residue ID to **700** (*see Note 2*).
3. Click on the **Next Step: Generate PDB and Orient Molecule** button in the lower right corner. Users can view the loading structure (Fig. 4) in the next step by clicking on the **view structure** button on the top of the website.

You may notice another option called **Symmetry Operation Options** when you scroll down to the end of the page. This option is only supported when the PDB file contains the information about oligomerization, in which the protein oligomer is composed of two or more associating monomers with different or identical structure [40, 41].

3.2 Orient the Protein

After PDB loading and manipulation, the protein needs to be oriented and positioned properly relative to the membrane bilayer. This step consists of two subsections, that is, orient and position protein and generate pore water.

3.2.1 Orient and Position Protein

CHARMM-GUI's Membrane Builder defines the Z axis as the membrane normal and $Z = 0$ Å as the center of membrane bilayer [19, 42]. Therefore, to build a system with the proper protein orientation and position, it must be aligned with the Z axis and its hydrophobic region centered on $Z = 0$ Å. Since we use a pre-oriented protein from OPM, orientation and positioning are not necessary here.

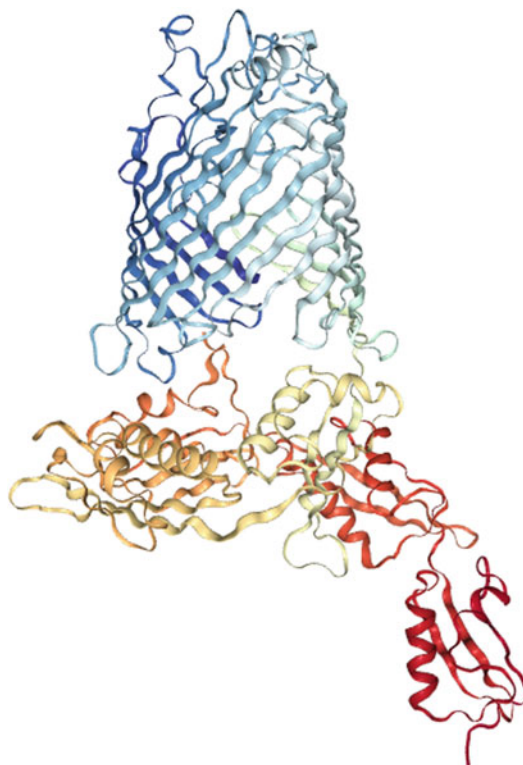


Fig. 4 Structure of *E. coli* BamA as seen in CHARMM-GUI

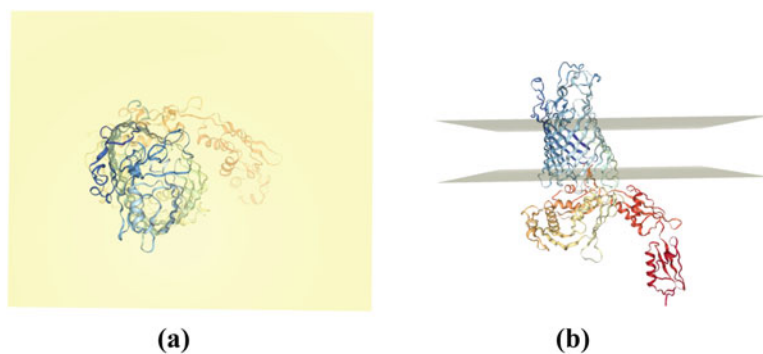


Fig. 5 Protein orientation. The yellow sheets are the *XY*-planes of membrane. (a) Top view. (b) Side view

1. Locate **Orientation Options**. Four options are provided here. Each option is labeled with the situation it is intended for (*see Note 3*).
2. Subsequently, select **Use PDB Orientation**. Users can see the orientation file (Fig. 5) in the next step by clicking on the **view structure** button on the top of the website.

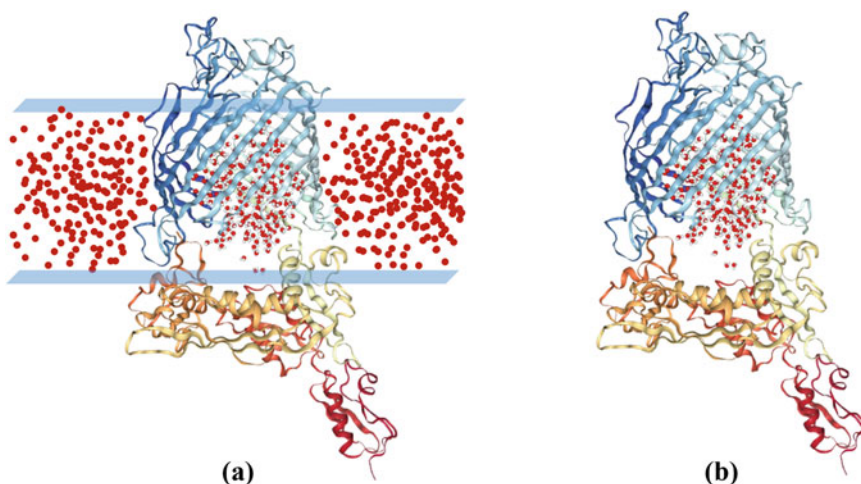


Fig. 6 Pore water generation. **(a)** Solvating the transmembrane region with water. **(b)** Pore water remains after high-temperature dynamics

Usually, proper orientation information is not available for PDB files from the RCSB database, such that proteins most likely need to be reoriented and repositioned in this step. Users can select **Use PDB Orientation** if they use PDB files from the OPM database (*see Note 4*).

3.2.2 Generate Pore Water

In general, proteins with pores, such as ion channels, transporters and porins, can accommodate water molecules inside their internal cavity. CHARMM-GUI provides a general approach for pore water generation (Figs. 6 and 7).

1. Locate the **Area Calculation Options**.
2. Click on the box of **Generate Pore Water** (*see Note 5*) and **Measure Pore Size**.
3. Select **Using protein geometry**.
4. Click on the **Next Step: Calculate Cross-Sectional Area** button in the lower right corner.

Note that the cross-sectional area of the protein will be calculated in this subsection to help determine the system size in the next step.

3.3 Determine the System Size

According to the cross-sectional area of the protein calculated in the previous step and lipid surface areas from experiments, the system size in the XY -plane and along the Z axis can be determined by multiple user-specified parameters in **System Size Determination Options**, including lipid types, system shape, water thickness along the Z axis on the top and bottom of the membrane, and numbers or ratios of lipid components.

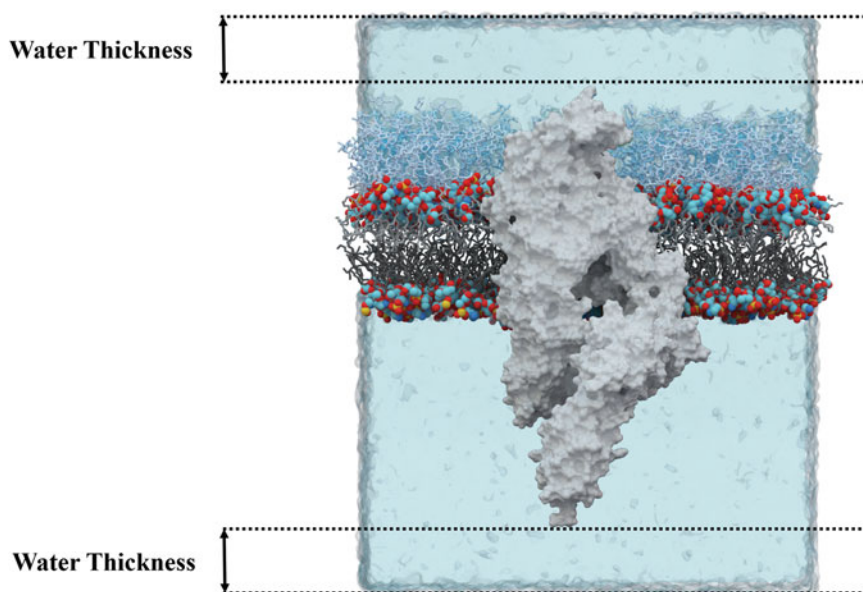


Fig. 7 Water thickness above and below the protein

Since we are building the membrane for BamA in *E. coli*, we will use an *E. coli* membrane. *E. coli* is a Gram-negative bacteria enveloped by two membranes, an inner membrane (IM) and outer membrane (OM). BamA resides in the OM. In Gram-negative bacteria, there is a special outer membrane component, lipopolysaccharide (LPS), that consists of a lipid A and a polysaccharide on the upper leaflet. The lower leaflet of the OM is a mixture of phospholipids. Here, we will use LPS for the upper leaflet while using PVCL2, PMPE, PMPG, PVPE, and PVPG for the lower leaflet, with a ratio of 2:8:1:8:2 [43, 44] (see Note 6).

1. Locate **System Size Determination Options**.
2. Select the **Heterogeneous Lipid** option.

Presently, the **Homogeneous Lipid** option is not supported, but users can select one type of lipid when using the **Heterogeneous Lipid** option to generate a homogeneous lipid bilayer.

3. Select **Rectangular** as the **Box Type**.
4. In the **Length of Z based on** option, select **Water thickness**. Change its initial parameter from 22.5 to 30 Å (see Note 7).
5. In the **Length of XY based on** option, select **Ratios of lipid components**.

Membrane Builder gives users two options to determine the system size in the *XY*-plane: **Ratio of lipid components**, which corresponds to the **Length of X and Y**, and **Numbers of lipid components**, which corresponds to the *XY* dimension ratio.

LPS Sequence:

Species: Escherichia coli

Lipid A: Type1 Image! Phosphate Charge: Phos.A: -2 Phos.B: -2

Core: User
 $\rightarrow 1) \text{bDGalNAc}(1-6) \text{aDGlc}(1-2) \text{aDGlc}(1-3) \text{aDGal}(1-6) \text{aDGlc}(1-3) \text{aLDHep}(1-7) \text{aLDHep}(1-3) \text{aLDHep}(1-5) \text{aDKdo}(2-4) \text{aDKdo}(2-4) \text{aDKdo}(2-4)$

O-units: 0

O-antigen: O1 $\rightarrow 3) \text{bDMannNAc}(1-2) \text{aLRha}(1-2) \text{aLRha}(1-2) \text{aDGal}(1-3) \text{bDGlcNAc}(1-3)$

available O-antigens

O1	O2	O3	O4	O5	O6	O7	O8	O9	O10	O11	O12	O13	O15	O16	O17	O18	O19	O20	O21
O22	O23	O24	O25	O26	O28	O29	O30	O32	O35	O36	O37	O38	O39	O40	O41	O42	O43	O44	O45
O46	O48	O49	O52	O53	O55	O56	O58	O59	O61	O62	O64	O65	O66	O69	O70	O71	O73	O74	O75
O76	O77	O78	O79	O82	O83	O85	O86	O87	O88	O90	O91	O96	O97	O98	O99	O100	O101	O102	O103
O104	O105	O107	O108	O109	O110	O111	O112	O113	O114	O115	O116	O117	O118	O119	O120	O121	O123	O124	O125
O126	O127	O128	O129	O130	O131	O132	O133	O135	O136	O137	O138	O139	O140	O141	O142	O143	O145	O146	O147
O148	O149	O150	O151	O152	O153	O154	O155	O156	O157	O158	O159	O160	O161	O164	O165	O166	O167	O168	O169
O170	O171	O172	O173	O174	O175	O176	O177	O178	O180	O181	O182	O183	O184	O185	O186	O187			

Core Sequence:

α	$\downarrow$	3-deoxy-D-manno-oct-2-ulosonic acid	$\downarrow$	-	+		
4	$\leftarrow$	α	$\downarrow$	3-deoxy-D-manno-oct-2-ulosonic acid	$\downarrow$	-	+
4	$\leftarrow$	α	$\downarrow$	3-deoxy-D-manno-oct-2-ulosonic acid	$\downarrow$	-	+
5	$\leftarrow$	α	$\downarrow$	LD-mannoheptose	$\downarrow$	-	+
3	$\leftarrow$	α	$\downarrow$	LD-mannoheptose	$\downarrow$	-	+
7	$\leftarrow$	α	$\downarrow$	LD-mannoheptose	$\downarrow$	-	+
3	$\leftarrow$	α	$\downarrow$	D-glucose	$\downarrow$	-	+
6	$\leftarrow$	α	$\downarrow$	D-galactose	$\downarrow$	-	+
3	$\leftarrow$	α	$\downarrow$	D-glucose	$\downarrow$	-	+
2	$\leftarrow$	α	$\downarrow$	D-glucose	$\downarrow$	-	+
6	$\leftarrow$	β	$\downarrow$	N-acetyl-D-galactosamine	$\downarrow$	-	+

☐ Chemical modification:

Next Step:
Update LPS 

Fig. 8 LPS type and core sequence

- Go to **Lipid Type** column. In **CL (cardiolipin) Lipids**, set **PVCL2's Lowerleaflet Ratio** as **2** and **Upperleaflet Ratio** as **0**. In **Bacterial Lipids**, set the **Lowerleaflet Ratio** of **PMPE**, **PMPG**, **PVPE** and **PVPG** as **8**, **1**, **8** and **2**, respectively, while keeping the **Upperleaflet Ratio** of all of them as **0**.
- Locate **LPS (lipopolysaccharides)**. Set the **Upperleaflet Ratio** as **1** and the **Lowerleaflet Ratio** as **0**.
- Click on **LPSA** button. In the pop-up, set all the parameters to match those shown in Fig. 8. Then click on the **Next Step: Update LPS** button in the lower right corner.
- Returning to the **Length of $X\Upsilon$ based on** option, enter **135** in the **Length of X and Υ** box as an initial guess. Then click on the **Show the system info** button and you should see the information shown in Fig. 9a.

Calculated Number of Lipids:

Lipid Type	Upperleaflet Number	Lowerleaflet Number
PVCL2	0	24
PMPE	0	96
PMPG	0	12
PVPE	0	96
PVPG	0	24
LPSA	92	0

Calculated XY System Size:

	Upperleaflet	Lowerleaflet
Protein Area	1655.20977	1960.7824
Lipid Area	17480	17306.4
# of Lipids	92	252
Total Area	19135.20977	19267.1824
Protein X Extent	35.19	
Protein Y Extent	52.57	
Average Area	19201.20	
A	138.57	
B	138.57	

Calculated XY System Size:

	Upperleaflet	Lowerleaflet
Protein Area	1655.20977	1960.7824
Lipid Area	17670	17306.4
# of Lipids	93	252
Total Area	19325.20977	19267.1824
Protein X Extent	35.19	
Protein Y Extent	52.57	
Average Area	19296.20	
A	138.91	
B	138.91	

The upperleaflet can have more lipids

(a)

(b)

Fig. 9 Feedback information for determining the membrane size. **(a)** Only using ratio to determine the membrane size may lead to one leaflet having too few lipids. **(b)** Adjusting the lipid numbers slightly will eliminate this problem

This situation is caused by the difference in areas between the upper leaflet and the lower leaflet of membrane. Generally, in order to solve it, we will use the **Ratio of lipid components** option first to determine the numbers of every membrane component under a certain initial guess. Then, use the **Numbers of lipid components** option to fine tune the number of lipids according to the feedback.

10. Select the **Numbers of lipid components** option. Change the upperleaflet lipid number of LPS from 92 to 93. Click on **Show the system info** button and you will *see* the information in Fig. 9b.
11. Click on the **Next Step: Determine the System Size** button.

3.4 Build the Components

On the basis of the system size, the generation of individual components for the system, including the membrane, bulk water, and counter ions will be completed in this step.

1. Locate the **System Building Options**. Then select **Replacement method** (*see Note 8*).
2. Move on to **Component Building Options**. Check the **Include Ions** box.
3. Use **KCl** as neutralizing species with a concentration of **0.15 M**. Choose **Mg²⁺** as the counter ions for both lipid A and core. Keep the **Ion Placing Method** as **Distance** (*see Note 9*).
4. Go to **Pore Water Options**. Inappropriately placed water molecules can be removed here (*see Note 10*). Usually, there are no extra water molecules that need to be removed and this step can be skipped.
5. Click on the **Next Step: Build Components** button in the lower right corner. The lipid bilayer will be generated first in this step.
6. To generate water molecules and ions, click on the **Next Step: Assemble Components** button in the lower right corner.

3.5 Assemble the Components

Components generated in the previous steps will be assembled in this step.

1. Check carefully to ensure the system is built as intended. If no problem exists, then click on the **Next Step: Assemble Components** button in the lower right corner to complete the assembly. Otherwise, go back to rebuild the system.
2. Download all the output files by click on **download.tgz**.

At this point, the entire system containing protein, lipid bilayer, bulk water, and ions has been generated through users-specified parameters and options in CHARMM-GUI. Users can load the system into a molecular visualization program such as VMD for further inspection. If desired, simulation input files can also be generated by CHARMM-GUI.

4 Notes

1. Users can also select **RCSB** or upload their own preoriented PDB file. Options for **PDB Format** need to be chosen when using your own PDB file. Note that PDB files obtained from the OPM database have already been preoriented with respect to the membrane normal (*Z* axis by definition) while those

from RCSB database need to be oriented manually by users themselves using **VMD** or in the subsequent step through CHARMM-GUI.

2. Generally, CHARMM-GUI detects structural information automatically, such as missing residues, disulfide bonds and others if indicated by remarks in PDB files (Fig. 3). However, depending on the source of the PDB, these remarks may have been written inadequately or even lost altogether. If that occurs, CHARMM-GUI cannot load those kinds of structure information, requiring users to add them manually in this step.
3. In CHARMM-GUI, the protein can be placed appropriately in the membrane by reorienting it via the alignment of its principal axis or a vector between two residues with the *Z* axis in **Orientation Options**, and repositioning it by means of the rotation with respect to the *X* or *Y* axis, or translation along the *Z* axis in **Positioning Options**. Users can also just utilize the original orientation and position information contained in the PDB file.
4. Users can also use move and rotate commands in the software program VMD to write a preoriented PDB file and then upload it to CHARMM-GUI.
5. During the pore water generation process, CHARMM-GUI solvates the transmembrane region of protein with a water box and runs high temperature dynamics with the protein fixed and water restrained in the transmembrane region. Water molecules inside the pore will remain while water molecules outside the pore will evaporate [19] (Fig. 6). Water staying close to the protein exterior due to strong interactions, can be removed by a refinement step in Subheading 3.4.
6. Ideally, the types and numbers of lipids are chosen to match the native membrane. Users should search the literature to determine which species the protein is from as well as the composition of its membrane in advance.
7. The scale of the entire system along the *Z* axis is determined by the height of the protein in *Z* and the thickness of the added water slabs (Fig. 7). In general, the default water thickness of 22.5 Å, is sufficient. For a membrane-only system, users can select the **Hydration number** (number of water molecules per one lipid molecule) option to define the total number of water molecules [19].
8. Replacement method (Fig. 10) distributes lipid-like pseudo atoms around the protein first, and then replaces them with lipid molecules selected randomly from a lipid molecule library, which contains 2000 different conformations of lipids from

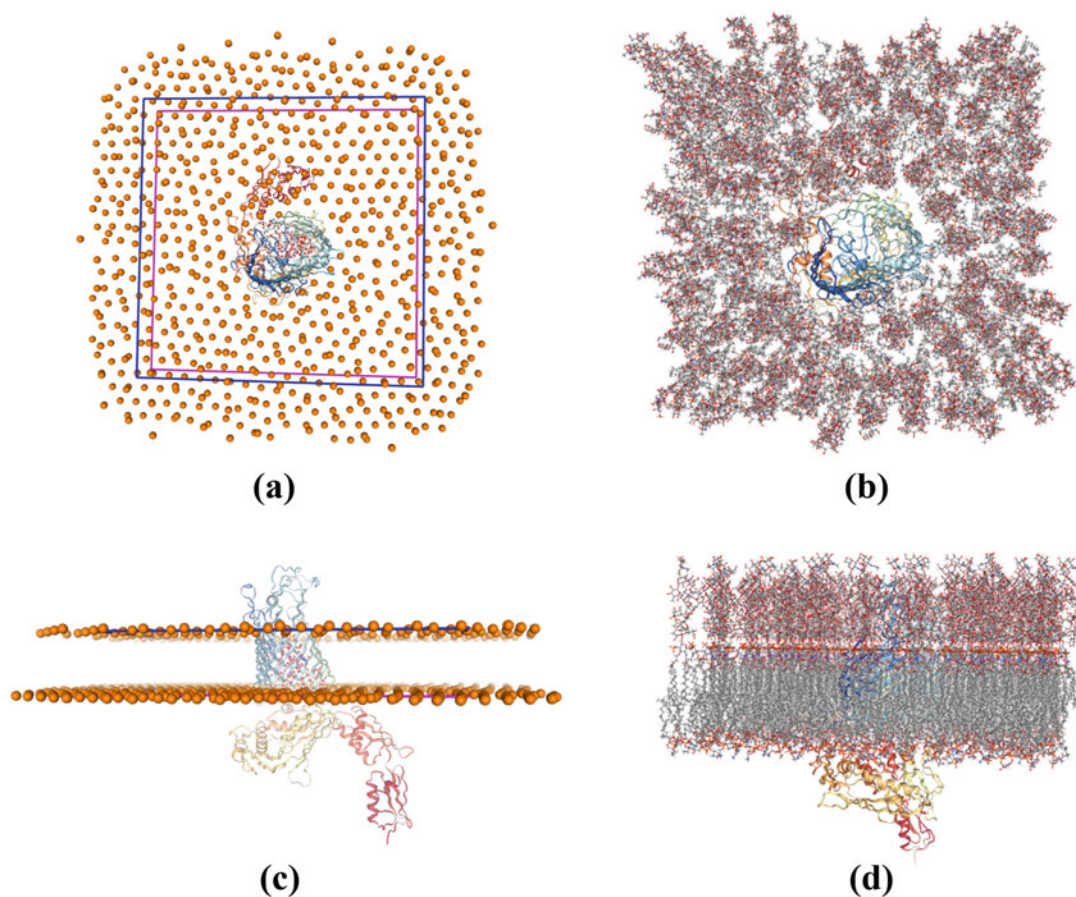


Fig. 10 The replacement method uses lipid-like pseudo-atoms to build lipids around the protein

MD simulations of pure bilayers [19]. Note that **Insertion method** is no longer supported in CHARMM-GUI.

9. In order to neutralize the system, Membrane Builder creates an appropriate number of ions based on the user-specified ion concentration and type. The initial configuration of ions is then determined through Monte Carlo simulations using a simplified model, that is, van der Waals and scaled Coulombic interactions [19].
10. Pore water generated in Subheading 3.2.2 can be refined in this step, to ensure that no water molecules are left outside of the protein in the membrane hydrophobic core region. Users can download the structure file to verify whether those water molecules are removed and select the residue numbers of water molecules needing to be removed on the website.

Acknowledgments

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References

- Almen MS, Nordstrom KJ, Fredriksson R, Schioth HB (2009) Mapping the human membrane proteome: a majority of the human membrane proteins can be classified according to function and evolutionary origin. *BMC Biol* 7:50
- Krogh A, Larsson B, von Heijne G, Sonnhammer EL (2001) Predicting transmembrane protein topology with a hidden Markov model: application to complete genomes. *J Mol Biol* 305(3):567–580
- Hidalgo P, MacKinnon R (1995) Revealing the architecture of a K⁺ channel pore through mutant cycles with a peptide inhibitor. *Science* 268:307–310
- Agre P, King LS, Yasui M, Guggino WB, Ottersen OP, Fujiyoshi Y, Engel A, Nielsen S (2002) Aquaporin water channels—from atomic structure to clinical medicine. *J Physiol* 542(Pt 1):3–16
- Rahman KS, Cui G, Harvey SC, McCarty NA (2013) Modeling the conformational changes underlying channel opening in CFTR. *PLoS One* 8(9):e74574
- Fairman JW, Noinaj N, Buchanan SK (2011) The structural biology of beta-barrel membrane proteins: a summary of recent reports. *Curr Opin Struct Biol* 21(4):523–531
- Gadsby DC (2007) Ion pumps made crystal clear. *Nature* 450:957–959
- Shukla AK, Manglik A, Kruse AC, Xiao K, Reis RI, Tseng WC, Staus DP, Hilger D, Uysal S, Huang LY, Paduch M, Tripathi-Shukla P, Koide A, Koide S, Weis WI, Kossiakoff AA, Kobilka BK, Lefkowitz RJ (2013) Structure of active beta-arrestin-1 bound to a G-protein-coupled receptor phosphopeptide. *Nature* 497(7447):137–141
- Cojocaru V, Balali-Mood K, Sansom MS, Wade RC (2011) Structure and dynamics of the membrane-bound cytochrome P450 2C9. *PLoS Comput Biol* 7(8):e1002152
- Jordan JD, Landau EM, Iyengar R (2000) Signaling networks: the origins of cellular multitasking. *Cell* 103:193–200
- Hunter T (2000) Signaling—2000 and beyond. *Cell* 100:113–127
- Fu D, Libson A, Miercke LJ, Weitzman C, Nollert P, Krucinski J, Stroud RM (2000) Structure of a glycerol-conducting channel and the basis for its selectivity. *Science* 290(5491):481–486
- Khademi S, O’Connell J 3rd, Remis J, Robles-Colmenares Y, Miercke LJ, Stroud RM (2004) Mechanism of ammonia transport by Amt/MEP/Rh: structure of AmtB at 1.35 Å. *Science* 305(5690):1587–1594
- Yellen G (2002) The voltage-gated potassium channels and their relatives. *Nature* 419:35–42
- Dong J, Yang G, McHaourab HS (2005) Structural basis of energy transduction in the transport cycle of MsbA. *Science* 308:1023–1028
- Elston T, Wang H, Oster G (1998) Energy transduction in ATP synthase. *Nature* 391:510–513
- Medley QG, Kedersha N, O’Brien S, Tian Q, Schlossman SF, Streuli M, Anderson P (1996) Characterization of GMP-17, a granule membrane protein that moves to the plasma membrane of natural killer cells following target cell recognition. *Proc Natl Acad Sci* 93:685–689
- Brooks JM, Lee SP, Leese AM, Thomas WA, Rowe M, Rickinson AB (2009) Cyclical expression of EBV latent membrane protein 1 in EBV-transformed B cells underpins heterogeneity of epitope presentation and CD8⁺ T cell recognition. *J Immunol* 182(4):1919–1928
- Jo S, Kim T, Im W (2007) Automated builder and database of protein/membrane complexes for molecular dynamics simulations. *PLoS One* 2(9):e880
- Hong M, Zhang Y, Hu F (2012) Membrane protein structure and dynamics from NMR spectroscopy. *Annu Rev Phys Chem* 63:1–24
- Earl LA, Falconieri V, Milne JL, Subramaniam S (2017) Cryo-EM: beyond the microscope. *Curr Opin Struct Biol* 46:71–78

22. Garavito RM, Picot D, Loll PJ (1996) Strategies for crystallizing membrane proteins. *J Bioenerg Biomembr* 28:13–27
23. Gumbart J, Wang Y, Aksimentiev A, Tajkhorshid E, Schulten K (2005) Molecular dynamics simulations of proteins in lipid bilayers. *Curr Opin Struct Biol* 15(4):423–431
24. Lee AG (2011) Biological membranes: the importance of molecular detail. *Trends Biochem Sci* 36(9):493–500
25. Landreh M, Marklund EG, Uzdavinyas P, Degiacomi MT, Coincon M, Gault J, Gupta K, Liko I, Benesch JL, Drew D, Robinson CV (2017) Integrating mass spectrometry with MD simulations reveals the role of lipids in Na(+)/H(+) antiporters. *Nat Commun* 8:13993
26. Dawaliby R, Trubbia C, Delporte C, Masurel M, Antwerpen PV, Kobilka BK, Govaerts C (2016) Allosteric regulation of G protein-coupled receptor activity by phospholipids. *Nat Chem Biol* 12:35–39
27. Dowhan W, Bogdanov M (2011) Lipid-protein interactions as determinants of membrane protein structure and function. *Biochem Soc Trans* 39(3):767–774
28. Vigh L, Escriba PV, Sonnleitner A, Sonnleitner M, Piotto S, Maresca B, Horvath I, Harwood JL (2005) The significance of lipid composition for membrane activity: new concepts and ways of assessing function. *Prog Lipid Res* 44(5):303–344
29. Saeedimasine M, Montanino A, Kleiven S, Villa A (2019) Role of lipid composition on the structural and mechanical features of axonal membranes: a molecular simulation study. *Sci Rep* 9(1):8000
30. Jo S, Kim T, Iyer VG, Im W (2008) CHARMM-GUI: a web-based graphical user interface for CHARMM. *J Comput Chem* 29(11):1859–1865
31. Brooks BR, REB BDO, States DJ, Swaminathan S, Karplus M (1983) CHARMM: a program for macromolecular energy, minimization, and dynamics calculations. *J Comput Chem* 4(2):187–217
32. Gu Y, Li H, Dong H, Zeng Y, Zhang Z, Paterson NG, Stansfeld PJ, Wang Z, Zhang Y, Wang W, Dong C (2016) Structural basis of outer membrane protein insertion by the BAM complex. *Nature* 531(7592):64–69
33. Han L, Zheng J, Wang Y, Yang X, Liu Y, Sun C, Cao B, Zhou H, Ni D, Lou J, Zhao Y, Huang Y (2016) Structure of the BAM complex and its implications for biogenesis of outer-membrane proteins. *Nat Struct Mol Biol* 23(3):192–196
34. Iadanza MG, Higgins AJ, Schiffrin B, Calabrese AN, Brockwell DJ, Ashcroft AE, Radford SE, Ranson NA (2016) Lateral opening in the intact beta-barrel assembly machinery captured by cryo-EM. *Nat Commun* 7:12865
35. Bakelar J, Buchanan SK, Noinaj N (2016) The structure of the β -barrel assembly machinery complex. *Science* 351(6269):180
36. Noinaj N, Kuszak AJ, Gumbart JC, Lukacik P, Chang H, Easley NC, Lithgow T, Buchanan SK (2013) Structural insight into the biogenesis of beta-barrel membrane proteins. *Nature* 501(7467):385–390
37. Berman HM, Westbrook J, Feng Z, Gilliland G, Bhat TN, Weissig H, Shindyalov IN, Bourne PE (2000) The Protein Data Bank. *Nucleic Acids Res* 28:235–242
38. Rose PW, Prlic A, Bi C, Bluhm WF, Christie CH, Dutta S, Green RK, Goodsell DS, Westbrook JD, Woo J, Young J, Zardecki C, Berman HM, Bourne PE, Burley SK (2015) The RCSB Protein Data Bank: views of structural biology for basic and applied research and education. *Nucleic Acids Res* 43(Database issue):D345–D356
39. Lomize MA, Lomize AL, Pogozheva ID, Mosberg HI (2006) OPM: orientations of proteins in membranes database. *Bioinformatics* 22(5):623–625
40. Ali MH, Imperiali B (2005) Protein oligomerization: how and why. *Bioorg Med Chem* 13(17):5013–5020
41. Protein complex oligomerization. <https://www.ebi.ac.uk/QuickGO/term/GO:0051259>. Accessed 14 Oct 2004
42. Jo S, Lim JB, Klauda JB, Im W (2009) CHARMM-GUI membrane builder for mixed bilayers and its application to yeast membranes. *Biophys J* 97(1):50–58
43. Pandit KR, Klauda JB (2012) Membrane models of *E. coli* containing cyclic moieties in the aliphatic lipid chain. *Biochim Biophys Acta* 1818(5):1205–1210
44. Lind TK, Wacklin H, Schiller J, Moulin M, Haertlein M, Pomorski TG, Cardenas M (2015) Formation and characterization of supported lipid bilayers composed of hydrogenated and deuterated *Escherichia coli* lipids. *PLoS One* 10(12):e0144671



Chapter 14

Coarse-Grained Molecular Dynamics Simulations of Membrane Proteins: A Practical Guide

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Abstract

Current computer architectures, coupled with state-of-the-art molecular dynamics simulation software, facilitate the in-depth study of large biomolecular systems at high levels of detail. However, biological phenomena take place at various time and length scales and as a result a multiscale approach must be adopted. One such approach is coarse-graining, where biochemical accuracy is sacrificed for computational efficiency. Here, we present a practical guide to setting up and carrying out coarse-grained molecular dynamics simulations.

Key words Molecular dynamics, Coarse-grained, Membranes, Membrane protein, Molecular modelling

1 Introduction

Biological processes of the cell span an enormous timescale, from femto-second through to months and longer. Molecular simulation provides a complementary toolkit for many experimental methods and is most commonly employed to provide dynamic information on underlying physical processes that drive emergent biologically relevant behavior. Thanks to developments in both the algorithms and the underpinning hardware, the timescales accessible to atomistic molecular dynamics (MD) simulations of membrane proteins are currently around tens of microseconds for systems of ~200,000–400,000 atoms with resources that are widely available. Examples of longer simulations exist [1], but these are extraordinary cases.

Currently, atomistic simulations are largely limited to systems of between 1 and 3 proteins within the membrane, and timescales that do not enable study of self-assembly of complete organelles nor

multiple protein–protein interaction events. Consequently, there is a tremendous drive to develop models and methods that enable simulation of longer timescales and bigger systems. A popular approach is to use coarse-grain (CG) models. Here, instead of representing each atom (or in the case of united-atom models, each atom excluding nonpolar hydrogen atoms) explicitly, small groups of atoms are grouped together into particles; typically, four or five heavy atoms are grouped into one spherical particle. The interaction properties of the particles are modelled as closely as possible to those of the atoms they represent. The speed-up in simulation arises as the dimensionality of the system has been reduced, requiring fewer calculations at each integration timestep. The reduction in degrees of freedom results in smoother energy landscapes leading to faster dynamics, which provides a further source of speed-up. Furthermore, given the particles are much heavier than individual atoms, the associated vibrational frequencies are slower, enabling the use of longer timesteps. Thus, overall atomistic resolution is sacrificed for the ability to simulate large systems for extended timescales. The most popular CG force-field for membrane proteins is MARTINI [2–8] (currently version 2, with version 3 imminent) developed by Siewert-Jan Marrink and coworkers, largely due to its versatility, ease of use, free availability, and support within popular MD codes. Table 1 provides some performance benchmarks to demonstrate the timescales achievable with MARTINI.

Alternative CG force-fields exist which are suitable for membrane protein simulations, such as SIRAH [9, 10] and ELBA [11] (*see Note 1*). However, for this guide we focus on MARTINI, with some indications of points of departure in other models. Although this guide serves as a practical approach to the setup and running of CG simulations, we assume the reader is familiar with the basics of atomistic MD simulations. Additionally, since it is most familiar to the authors, we refer primarily to the GROMACS [12–17] MD engine throughout this guide. It is not our intention to provide a thorough review of the capabilities of the MARTINI force-field, but rather to enable the reader to set up and run equilibrium MD simulations of membrane protein systems using MARTINI within GROMACS, for a showcase of capabilities and many other useful tools, tutorials and details of new developments we highly recommend the MARTINI website (<http://cgmartini.nl>). In this guide, firstly, we describe common ways to begin a CG simulation setup from atomistic coordinates and some commonly used tools. Secondly, we outline a number of ways to embed a membrane protein and some common parameters to consider when preparing the system for simulation. Finally, we highlight some important areas to consider when using CG models such as conversion to atomistic representations and choice of water model.

Table 1

Some benchmarks of MARTINI membrane protein simulations with GROMACS2018.1. Details of the hardware are as follows

ns/day		Notes
CPU ^a cores		
24	443	One MPI thread and one OpenMP thread per core
48	804	
96	1435	
144	1478	
192	1934	
288	2207	
384	3628	
480	3999	
576	4219	
GPU ^b		
1	197	One MPI thread and ten OpenMP threads per GPU
4	377	
8	445	

^aCPU: Each of the ARCHER standard compute nodes consist of two Intel Xeon 2.7GHz 12 core E5-2697 v2 (Ivy Bridge) processors, each core can support two hardware threads. On node the two processors are connected via two QuickPath Interconnect (QPI) links. Each node has 64 GB of memory arranged into two NUMA regions, one for each of the two processors. Communications between the compute nodes is provided by Cray Aries interconnect

^bGPU: Each of the JADE standard compute nodes consists of an Nvidia DGX-1 deep learning system. Each of the DGX-1 nodes contains two Intel Xeon 2.2GHz 20 core E5-2698 v4 (Broadwell) processors, each core can support two hardware threads. Each node has 512 GB of system memory and eight Nvidia Tesla V100 GPUs. Each Nvidia Tesla V100 GPU has 5120 CUDA cores and 16 GB of high speed memory bringing the on node totals up to 40,960 CUDA cores and 256 GB GPU memory. Inter-GPU communications on node are handled via Nvidia's high speed NVLink with up to 300 GB/s bandwidth. Internode communications are handled by a quad EDR infiniband interface resulting in up to 800 GB/s of internode bandwidth

2 Materials

In this section, we give a brief overview of the typical resources and computer architectures needed to perform CG MD simulations.

2.1 Obtaining the Protein Structure

The first stage in any membrane protein MD simulation is to ensure the protein structure is suitable and complete. This is also the case for CG simulations. One starting option is to extract the protein-lipid configuration from an atomistic/united-atom simulation and use the MARTINI utility code *martinize.py* (<http://www.cgmartini.nl/images/tools/martinize/martinize-2.6/martinize.py>) to convert this to CG resolution. An alternative route is to obtain the structure directly from X-ray, NMR, cryo-EM or another source of structural information. In either case the usual care must be taken to ensure the starting structure is complete, the

side chains are in their expected tautomeric states, ionization states are correctly assigned and that there are no unexpected mutations (these are sometimes required to, e.g., crystallize the protein). Often in the case of atomistic structures, loop regions of proteins may be missing from acquired structures due to their inherent flexibility and as a result can often be difficult to resolve. To model these regions homology modelling can be used and is offered through a wide variety of software packages such as MODELLER [18] as well as online servers including SWISS-MODEL [19]. When using any method of modelling in missing regions, it is advisable to create a number of homology models. Each model can then be evaluated using statistical scoring functions such as Discrete Optimized Protein Energy (DOPE) [20] and/or Qualitative Model Energy Analysis (QMEAN) [21]. Visual inspection of the final model is highly recommended, especially in regions of the protein that are known to be potentially critical to its function.

2.2 Software

A large variety of simulation packages are available for carrying out MD simulations. A number of these support both atomistic and CG force fields that support ease of use and production of efficient workflows. The most popular and freely available of these include GROMACS [12–17], AMBER [22, 23], NAMD [24], and LAMMPS [25].

2.3 Hardware

The specific hardware requirements for an MD simulation will be dictated by the underlying hardware specification, simulation code, and the timeframe in which the user is comfortable waiting for the simulation to finish. Since the use of a CG approach reduces the total number of particles in an MD simulation the underlying physics (and necessary calculations needing to be performed therein) remains unchanged when compared to the system's atomistic analogue. This permits the construction of large and often complex biomolecular systems that can be simulated on a modern desktop machine. To take advantage of the longer timescales afforded by a CG approach, a computing cluster is often utilized consisting of both CPUs and GPUs. Although modern simulation code can run on both CPUs and GPUs, at the time of writing there is a continued trend in the use of GPUs and GPU-only clusters.

3 Methods

3.1 Conversion to a Coarse Grained Representation

Once an atomistic model has been generated (as described in Subheading 2.1), it can either be directly converted to CG resolution using the *martinize.py* script or simulated first and subsequently converted. Conversion of an atomistic model to a MARTINI CG model requires consideration of the conservation of protein structure (this is not applicable to some CG force-fields

such as SIRAH (*see Note 1*). Simulations of membrane proteins usually require preservation of secondary structure, whereas tertiary structure is allowed more flexibility as the rearrangements are usually important to the study, and quaternary structure is rarely restrained at all. Differing levels of restraining the protein structure is achieved through an elastic network approach. In MARTINI there are two options that can be implemented in the *martinize.py* script, the first of which uses the in-built MARTINI elastic network methodology. Here, harmonic distance restraints are introduced between nonbonded backbone particles based on recommended upper and lower cutoff values of 0.5 and 0.9 nm, respectively. Additionally, the recommended elastic bond force constant should be set to 500 kJ/mol/nm². An alternative approach is the use of the ElNeDyn [26] network that places more emphasis on retaining atomistic structure. In this method, backbone particles are centred on the C α atoms as opposed to the center of mass of the backbone N, C α , C, and O atoms in standard MARTINI. To solve this difference between backbone particle locations the ElNeDyn model adapts the MARTINI bonded parameters (as well as mapping for aromatic residues). Both the upper cutoff and elastic bond strength values in the ElNeDyn network approach have been finely tuned and are typically set automatically at 0.9 nm and 500 kJ/mol/nm².

The standard MARTINI elastic network and the ELNeDyn network are both easily implemented in the *martinize.py* script and parameters such as cutoff distances and force constants can be changed by the user. It is important to note here that there is no “perfect combination” of elastic network parameters. The user must consider the type of protein dynamics that are being probed and thus the appropriate level of flexibility to achieve this when parameterizing their CG model.

3.2 Embedding the Protein into a Membrane

The next step is to embed the protein into a suitable model membrane. Several tools are available to achieve this. Perhaps the most useful and powerful is the Martini-Maker module of CHARMM-GUI developed by Im and coworkers (<http://www.charmm-gui.org/?doc=input/martini.bilayer>) [27, 28]. This online utility enables the user to easily set up simulations of proteins within spherical vesicles, nanodiscs, and flat lipid bilayers. A range of lipid types from simple phospholipids through to the complex lipopolysaccharides found in gram-negative bacterial membranes, are supported. An additional tool developed for use with the MARTINI model by Wassenaar et al. is the INSert membrANE code (*insane.py* available at <http://www.cgmartini.nl/images/tools/insane/insane.py>) [29]. This python script can be used to easily embed a CG protein in a user defined membrane composition, and it also has the option to include water and appropriate ions. Another added benefit of the latter approach is its potential

utilization in automated workflow approaches to biomolecular simulations. The use of such tools is well suited to embedding single protein systems but can also be utilized in the construction of systems containing multiple proteins. Here, the supplied coordinate file must contain all proteins to be studied in a suitable starting configuration. Such tools, online or otherwise, are indispensable for the computational study of membrane proteins, primarily due to their ease of use, simplicity, and free availability.

The aforementioned methods of protein embedding can also be used as a starting point for constructing large CG systems for applications such as protein clustering within membranes. After construction of the initial membrane and embedded protein, it is often advisable to perform energy minimization and equilibration to ensure the setup is stable. The user can then use their equilibrated system as a “unit cell” from which to construct a larger one. This approach is particularly well suited for the construction of flat bilayer systems. In the GROMACS MD engine the *genconf* tool can be used to easily achieve this. Here, the supplied system is copied a number of times (defined by the user) typically in directions in the same plane as the membrane (Fig. 1), although the tool can be used for any direction. In order to decrease the resulting periodicity of the system it is advisable to randomize the starting orientation of each protein copy as much as possible. This can be achieved by position restraining a CG particle (e.g., a backbone particle) within the protein whilst keeping all other protein particles unrestrained. When simulated (e.g., as an extension to a standard equilibration) each protein copy can randomly orientate itself on the membrane surface. Several studies have utilized this approach in protein clustering [30–32]. Construction of double membrane systems is achievable by employing a similar approach, by using *genconf* to replicate the system in the plane perpendicular to the membrane. If the compositions of the two membranes differ then they should be constructed individually using the approach described above, ensuring they both have the same x and y dimensions. The water and ions should be removed. Then the *editconf* tool should be used to translate one of the membranes in the z dimension to the required separation between the two membranes and the resulting files should then be concatenated. The system should then be re-solvated and the required ions added. Spherical vesicles with diameter ~20 nm can be constructed using the CHARMM-GUI online server [28], but it tends to struggle with larger vesicles. Vesicles of size ~65 nm diameter have been simulated using MARTINI [33]. The procedure to construct the vesicles involved placing the center of mass of each lipid on the surface of an ideal sphere, with the lipids spaced equally according to their average area per lipid in fluid phase, assuming a surface area of an ideal sphere located at the center of mass of the lipid.

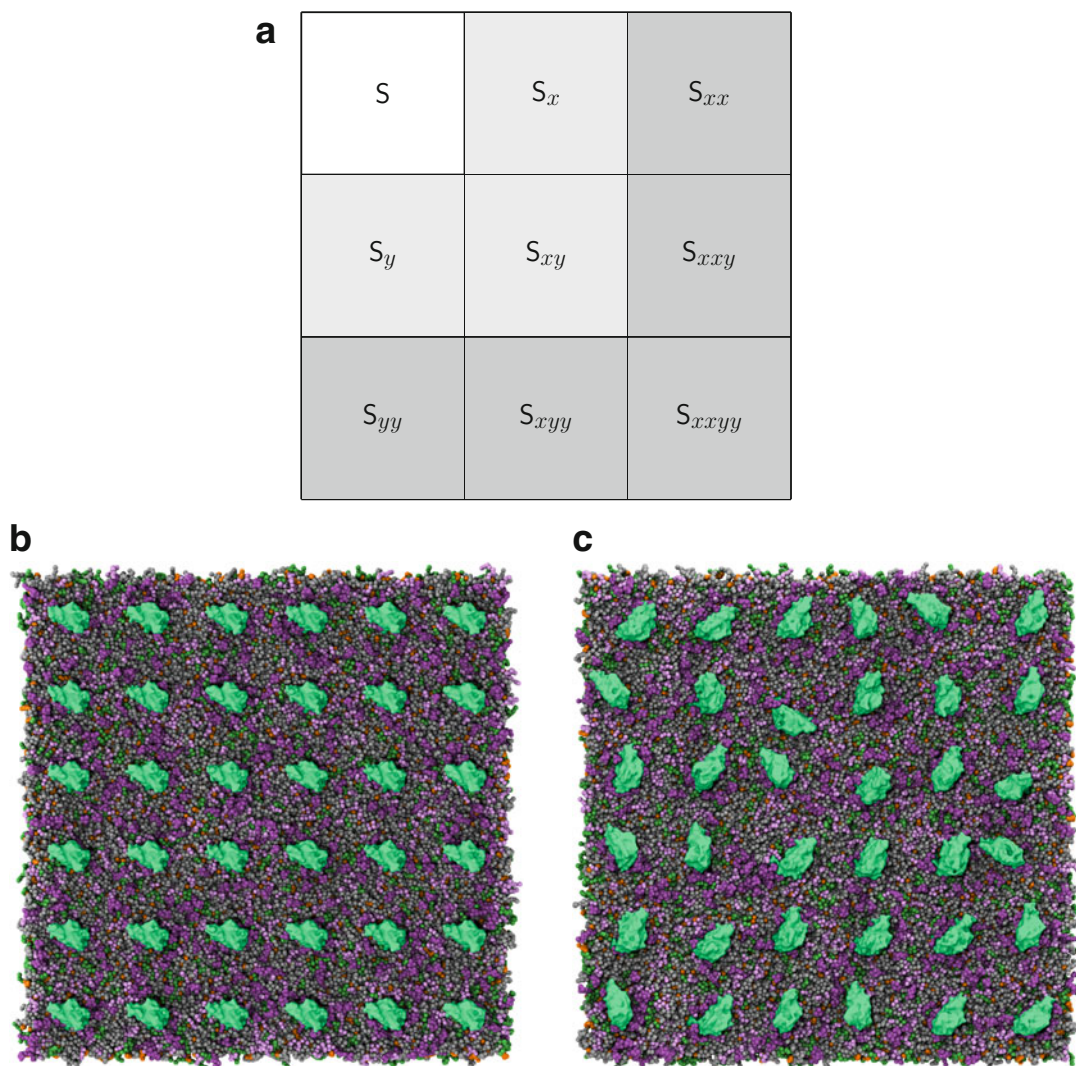


Fig. 1 Tiling of a CG system. **(a)** A generated CG system (S) can be tiled in the x and y plane (defined as parallel to the membrane plane) to create larger bilayer systems with multiple protein copies using the *genconf* command in the GROMACS MD engine. **(b)** An extracellular view of a *genconf* generated CG system **(c)** the same system as in **(b)** but with protein orientations randomized through position restraining one or a few backbone particle(s). Protein copies are of the voltage-gated sodium channel $\beta 3$ subunit (green) adapted with permission from Glass et al. [30]. Each protein is embedded in a mixed lipid bilayer where the upper leaflet (shown) consists of 1-palmitoyl-2-oleoyl-glycero-3-phosphatidylcholine (grey), palmitoyl-2-oleoyl-glycero-3-phosphatidylethanolamine (dark green), sphingomyelin (pink), monosialodihexosylganglioside (purple), and cholesterol (orange). The full lipid composition can be found in [30]

Once the protein is embedded in the membrane (either a simple construct or large system) it is advisable to perform a series of energy minimization steps. This ensures that any high-energy interactions between CG particles are removed. Typically, a steepest descent algorithm is employed to adjust system coordinates to reduce the overall net forces acting upon them. This method is

repeated until some user-defined energy cutoff or maximum step number is reached. Such minimization protocols are standard in atomistic simulations and are easily extended to CG systems. Once minimized, the system is ready to undergo simulation. First, a set of equilibration steps need to be performed where the protein is restrained whilst the bilayer and solvent are free to move. Once equilibrated and the system is stable, the protein restraints can be released and a production run performed. Checks for equilibration include, but are not restricted to, checking system volume and the area per lipid as a function of time; equilibration is achieved when the former only shows small fluctuations rather than a systematic drift, and the latter matches expected values for the lipids being simulated. Furthermore, any errant water molecules should have either moved out of the membrane during these stages or should be manually removed. A general protocol for setting up a CG simulation is shown in Fig. 2. Setting up simulations within the CHARMM-GUI server has the advantage of providing the user with all the files required for a multistep minimization and equilibration procedure (*see Note 2*). These comprise two energy minimization steps, the first using soft-core steepest descents minimization and the second standard steepest descents. This is followed by five equilibration steps, the first of which employs

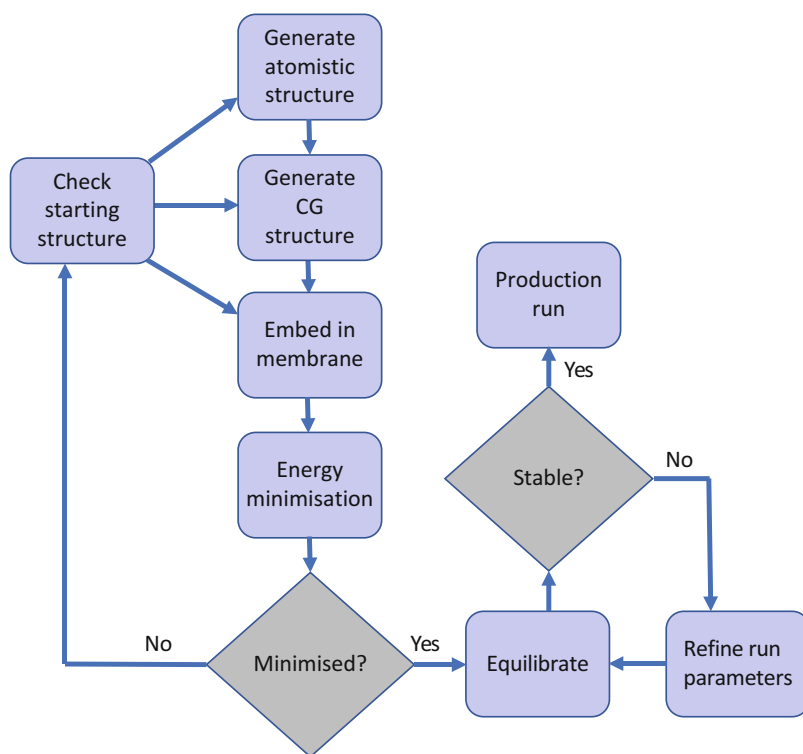


Fig. 2 General protocol for setting up a CG simulation of a membrane protein system

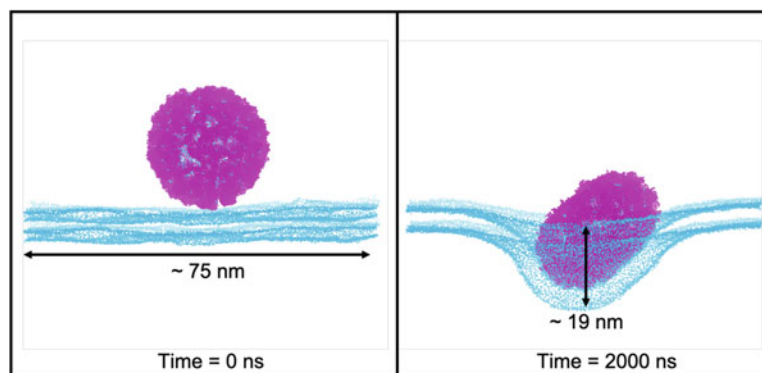


Fig. 3 Demonstration of undulations forming in large membranes. A vesicle with a 20-nm diameter composed of rough lipopolysaccharide molecules (purple) in the outer leaflet and phospholipids (cyan) in the inner leaflet and an initially flat bilayer composed of phospholipids. The bilayer deforms considerably to wrap around the vesicles within 2000 ns of simulation. Care was taken here to ensure the vesicle did not interact with itself through periodic boundaries [34]

positional restraints on the lipid head-groups to prevent movement in the z dimension (force constant of 200 kJ/mol) and an integration timestep of 2 fs. In the subsequent four steps the restraints are gradually reduced and the timestep is increased after which the system is ready for the production run. CHARMM-GUI provides all of the parameters files and also instruction for running each step and thus is recommended for nonexperts in MD simulations.

We note here that simulations of large membranes often lead to significant undulations in the membrane [34] (Fig. 3), and we advise the reader to check such simulations regularly to ensure any undulations do not become so large that the molecules interact with their own periodic images. And finally, as with all simulations, the user is advised to visualize the system (*see Note 3*) after each step of the systems setup, as numerical stability alone is not a sign of (bio)chemical accuracy.

3.3 Simulation Parameters

As with atomistic simulations the choice of simulation parameters is important and will ultimately depend on the specific system being studied. An example GROMACS input file is described in **Note 4**. Below is a general guide for the most common parameters encountered in simulations with a specific focus on settings used in CG simulations with the MARTINI force-field. Thus far we have excluded the description of parametrization of new molecules other than those in the standard MARTINI force field, for a brief description of tools available for this *see Note 5*.

3.3.1 Thermostat

Control of temperature can be achieved through the coupling of the system to a heat bath. One of the simplest ways to achieve this is through direct rescaling of the particle velocities to a reference temperature as the simulation progresses (Eq. 1).

$$\frac{dT(t)}{dt} = \tau_t^{-1}(T_{\text{ref}} - T(t)) \quad (1)$$

where $T(t)$ is the instantaneous temperature at a particular time step, T_{ref} is the reference temperature, and τ_t determines the strength of the coupling of the system to the bath. This method is motivated by the direct relationship of the kinetic energy of the system to its instantaneous temperature and is the algorithm used by the Berendsen thermostat [35]. Although such a thermostat can be used in both atomistic and CG simulations it is advised that it not be used since it has been shown to not reproduce the correct canonical ensemble, in addition to the introduction of artefacts (<https://doi.org/10.1021/acs.jctc.8b00446>). For these reasons it is recommended to use the closely related velocity-rescale (v-rescale) thermostat [36] that does reproduce the correct canonical ensemble by the introduction of a stochastic term. Instead of using a strict reference, the temperature is periodically drawn from the canonical equilibrium distribution from the kinetic energy of the system. Although other stochastic thermostats exist, the v-rescale thermostat is used extensively in MARTINI simulations and is the recommended choice when simulating CG MARTINI systems.

As with atomistic simulations, it is also advised to join individual components of the system into appropriate groups. This is to ensure that all parts of the simulation will be at or close to the target temperature without risking spurious “hot” or “cold” regions. In membrane protein simulations these groups are typically the bilayer, protein, and solvent (including ions). Typically, the τ_t is set to 1 ps for a MARTINI CG simulation (values approaching zero represent tighter coupling than higher values).

3.3.2 Barostat

Simulations performed within the NPT ensemble employ barostats to control pressure. In a similar fashion to the Berendsen thermostat, the Berendsen barostat [35] rescales the pressure of the system in accordance with a reference pressure bath. Here, the system coordinates and volume are rescaled with reference to a coupling matrix. Although useful for reaching a target pressure, this approach does not reproduce the correct NPT ensemble (see Ensembles described in the next section). It is therefore recommended to use the Parrinello–Rahman barostat [37, 38] that does reproduce the correct ensemble during production run simulations. Thus, a typical simulation protocol would be to use the Berendsen barostat to achieve the average target pressure during system equilibration and then switch to the Parrinello–Rahman

barostat for the production run, from which thermodynamic properties can be more reliably calculated.

It is important to note here that in membrane protein simulations the x and y box vectors are often coupled and scaled by the same amount whilst the z vector (defined as normal to the membrane plane) is scaled independently. In contrast to protein + solvent systems, for which it is usual to employ isotropic pressure coupling, membrane systems should be treated with semi-isotropic coupling, with the membrane plane (typically x and y dimensions) coupled separately from the plane that is perpendicular to the membrane plane (typically z dimension). As with temperature coupling, individual groups present within the system should be coupled together. For membrane protein simulations it is standard (and recommended) to perform MD simulations in the NPT ensemble. If performed in the NVT ensemble artefacts may arise as lipid groups will be unable to alter their conformation correctly to give rise to changes in lateral area of the bilayer. Typical values for the Parrinello–Rahman pressure coupling are $t_p = 12$ ps, compressibility of $3 \times 10^{-4} \text{ bar}^{-1}$ and a reference pressure of 1 bar.

3.3.3 Nonbonded Interactions

On average, the most computationally demanding part of any MD simulation is calculation of the nonbonded interactions. The methods employed to evaluate these interactions are the focus of much ongoing development and improvements are often reported in the literature. The methods chosen for particular force-fields or simulation types are arrived at by balancing accuracy and computational feasibility through rigorous testing. Here we provide a summary of methods and protocols used for MARTINI simulations and note that these are likely to differ slightly from those used for atomistic simulations and also other CG force-fields. In any simulation, atomistic or CG, the nonbonded interactions can be split into contributions from both van der Waals (vdW) and permanent electrostatic interactions.

3.3.4 Van der Waals Interactions

The vdW forces are described by the Lennard–Jones (LJ) potential [39, 40] in the MARTINI force-field. The LJ potential contains both an r^{-12} term to model short-range repulsion and an r^{-6} term to model longer-range dispersion forces. The latter notably decay rapidly toward zero as the interparticle distance increases. Intuitively, several cutoff schemes have been introduced to only consider particles within a particular distance for LJ calculations (Fig. 4). The most commonly used method (and the one we recommend) for simulations performed using the MARTINI force-field is the shifted-force potential, in which the potential is shifted toward zero at a defined distance. The alternative is to use a simple cutoff, but this has the disadvantage of introducing discontinuities in the function.

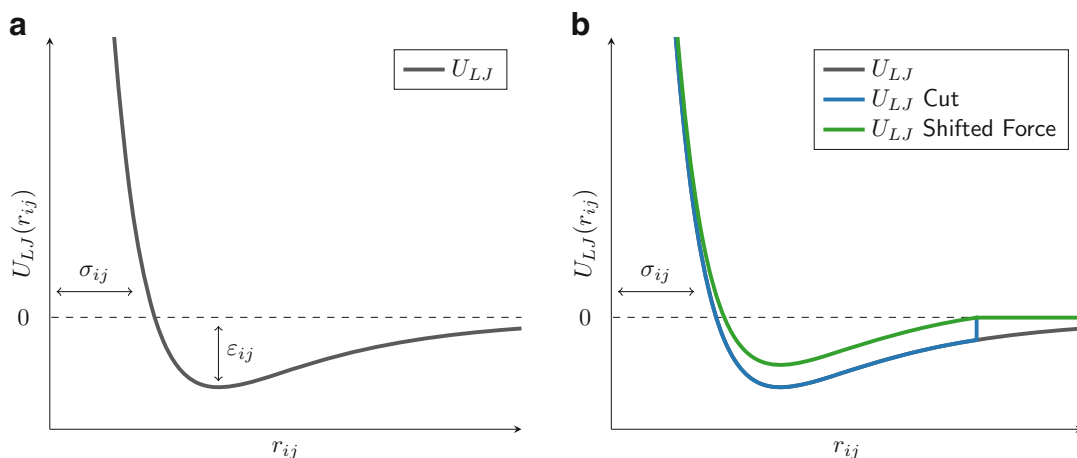


Fig. 4 The Lennard–Jones potential and cutoff schemes used in MD simulations. **(a)** The standard 12–6 Lennard–Jones potential (U_{LJ}) between two particles at distance r_{ij} . σ_{ij} and ϵ_{ij} represent the vdW radius and well-depth respectively. **(b)** An overlay of both the simple cut off (U_{LJ} Cut) and shifted-force (U_{LJ} Shifted-Force) potentials (ϵ_{ij} removed for clarity)

3.3.5 Electrostatics Interactions

The permanent electrostatic interactions are longer in range (r^{-1}) than the dispersion forces (r^{-6}) modelled using the LJ term. Consequently, a lot of simulation time is spent calculating the contribution from electrostatic interactions. There are a number of ways to improve the efficiency of calculating these contributions and many of these are implemented within all popular MD codes. MARTINI simulations using the standard water models should employ the reaction-field approach, given this was used for parametrization of the force-field. This method treats the Coulombic contributions inside a spherical region of the system explicitly, but once past a certain cutoff there is an assumed continuum of uniform dielectric constant. It is possible to use a particle mesh Ewald (PME) [41] approach with the MARTINI force-field; however, it should be used with caution as deviations from expected system behavior may occur.

3.3.6 Polarizable Water and Electrostatics

There are usually more water molecules in a membrane protein simulation than any other single species, and therefore the consideration of water model is significant. In the standard MARTINI water model, four atomistic water molecules are represented by a single particle. This has two important consequences; firstly, the size of the particles precludes this model being used to study very narrow protein channels which are known to be occupied by single-file water molecules. We note here that implicit screening of charges is achieved in MARTINI by using a fixed dielectric constant of 15. Secondly, the lack of a dipole means that electroporation processes cannot be studied. To circumvent the latter limitation, a polarizable water model is available for MARTINI. In this model, additional charged particles are bound via a harmonic potential to the central

water particle. Their rotation is controlled by a harmonic angle potential that allows the dipole moment to vary. Although useful, the cost of simulating extra interactions that arise from such an approach is often high in terms of computational power. Since the philosophy of any CG technique is to simplify the interactions, it is often unnecessary to use such a model, unless one is specifically studying phenomena that rely heavily on the water dipole, such as electroporation.

3.3.7 Time-Step

One of the major benefits of a CG approach is the access it gives to longer time scales. Once the system is constructed, and all appropriate thermodynamic parameters have been set, a decision must be made regarding the simulation length. Clearly this will depend heavily on the system and phenomena being studied. Any CG method enables the use of a longer time step in MD simulations due to the reduction in the number of degrees of freedom and subsequent smoothing of the energetic landscape. In the MARTINI force-field a time-step between 20 and 40 fs is achievable (*see Note 6*). Numerical instabilities will occur if the timestep is too long for the system (this is dependent on the bond vibrations within the system, we refer the reader to a review by Krieger et al. [42] for details of choosing appropriate simulation timesteps). If this is the case, GROMACS will warn the user at the stage of creating the input file for the simulation (i.e., running the *grompp* command) and the simulation will stop running at some point (during the *mdrun* phase). Here, the user is recommended to decrease the time-step. However, if instabilities persist it may be a deeper underlying problem with the system and further investigation would be needed.

3.4 Conversion Between Coarse-Grained and Atomistic Representations

Conversion of CG systems to atomistic (commonly referred to as back mapping) are less straightforward than the reverse transformation as there are multiple atomistic configurations that correspond to the same CG system. Although certainly achievable, an atomistic representation of a simulated CG system gives more of a “global” picture of protein conformation and any specific residue conformations/interactions must be interpreted tentatively. Additionally, the use of an elastic network model in the MARTINI force-field will influence the conformation that any secondary structure can adopt. With this in mind, it is highly recommended to perform MD simulations of any resulting atomistic model derived from a CG system to ensure that the system is stable.

A major benefit of using the MARTINI force-field is the useful *backward* back mapping tool [43] developed by Wassenaar and coworkers. Here, a predefined library of mapping definitions is utilized with predefined geometric parameters to produce united-atom or all-atom models from CG systems. Additional molecules can easily be incorporated by the user expanding the library to include molecules of interest. The CHARMM-GUI server can also be used for reverse-mapping to atomistic detail although this is currently limited in the types of molecules that are supported.

However, for a beginner studying a membrane protein in lipids supported by CHARMM-GUI, this would be an ideal option due to ease of use.

4 Notes

1. *Other popular CG force-fields.*

There are a number of other CG force-fields that are popular for membrane protein simulations, and while it is not our intention to provide a comprehensive list here, below we provide brief practical details for running simulations using SIRAH and ELBA.

The SIRAH Force-field.

SIRAH background.

The SIRAH force-field [9, 10] has an approach to coarse-graining that lies between the atomistic and the MARTINI coarse-grained resolutions. The force-fields has 66 particle types including those for water, proteins, nucleic acids, and lipids. One specific feature of SIRAH is that protein backbone is described by three separate beads. This enables modelling of secondary structures without adopting additional restraints. Four water molecules are represented as four particles at the extremes of a tetrahedron. Two vertices bear a positive charge (+0.41 e) and two the opposite one, ensuring a dipole moment. Moreover, the geometry resembles the network of interactions created by water in its liquid state.

SIRAH is fully compatible with AMBER and GROMACS. Similarly to MARTINI, it includes a utility tool to convert atomistic structures into their corresponding coarse-grained topology. The *cgconv* script works for both proteins and a variety of lipid types. It is compatible with several atomistic force-fields (AMBER [22, 23], CHARMM [44–46], OPLA-AA [47], OPLS-UA [48], GROMOS suite [49–51]), and supports the conversion from both residue-based lipid topologies (e.g., lipids identified by their residue name) or fragment-based ones (lipids identified by the head group and tails separately). As such, protein embedding and lipid packing can be obtained with CHARMGUI [52], MemBuilder [53], the membrane builder extension of VMD [54] or the HTMD database [55]. The output file can be parsed to the SIRAH conversion script, *cgconv* to obtain the coarse-grained coordinates.

SIRAH simulation parameters.

The SIRAH force-field was parameterized to work with a direct cutoff of 1.2 nm for the nonbonded interactions. A plain cutoff scheme is used for van der Waals interactions, and a PME

long-range treatment is used for the electrostatics. Temperature coupling is performed with a v-rescale thermostat and coupling time of 1 ps, or a Langevin thermostat [56] and coupling time of 5 ps (GROMACS and AMBER respectively). The pressure is maintained at 1 bar using either the Parrinello–Rahman or Berendsen barostat (GROMACS and AMBER) and 8 ps coupling time. The time step is 20 fs; the neighbor list is updated every ten steps.

Simulations are typically performed with neutralizing counter ions (Na^+ or Cl^-), plus additional ions to give the desired ionic concentration. The standard equilibration protocol includes a steepest descent energy minimization (5000 steps) and an equilibration in the NPT ensemble (for at least 5–15 ns). For a system in which a protein is embedded in the membrane, it is recommended to first perform an equilibration step in the NPT ensemble with positional restraints on the protein (of the order of 1000 kJ/mol/nm^2) on the vertical position of the phosphate head group particles (of the order of 100 kJ/mol/nm^2). This allows the lipids to repack around the protein without distorting the membrane thickness. A second equilibration step restraining lipid phosphate and protein backbone (100 kJ/mol/nm^2) particles, which serves to relax the interactions between protein sidechains and lipid tails, is also recommended. Subsequently, one can proceed with unrestrained production runs.

The ELBA Force-field.

ELBA background.

Hybrid molecular-mechanics/coarse-grain (MM/CG) simulations can also be carried out using the ELBA force-field to simulate the membrane and solvent [11], and an atomistic model for the protein or an associated small molecule [57, 58]. The applicability of the ELBA model to hybrid MM/CG simulations has been demonstrated when using an atomistic solute in CG water, by reproducing hydration free energy data of amino acid side chains, observing appropriate globular protein dynamics, and by the correct folding of short peptides. In the context of hybrid ELBA-based MM/CG simulations in CG-membranes, potentials of mean force for the permeation of atomistic amino-acid chains were used to parameterize the balance between the CG and atomistic force-fields, with the resulting model being validated against experimental partition coefficient data for small molecules in 1,2-Dimyristoyl-sn-glycero-3-phosphocholine (DMPC) lipids, and on the structure and dynamics of membrane bound peptides and the glycophorin A dimer [58].

ELBA setup.

The setup for the hybrid protein simulations involves placing the protein, which is aligned with the membrane normal,

alongside a preequilibrated membrane/water slab, and then subjecting the system to a short, restrained, high pressure (~1000 atm) equilibration phase, followed by longer equilibration at atmospheric pressure. The heavy atoms of the protein are initially restrained during this process, so that its structure is not perturbed by the pressure. This protocol exploits the ability of ELBA membrane models to rapidly self-assemble and reorganize, thereby generating an equilibrated, membrane-bound, protein structure. A number of example scripts for the insertion of a simple peptide into a membrane, along with scripts for the mapping of atomistic membranes to ELBA membranes, can be found on GitHub [59].

2. *Notes on CHARMM-GUI.*

We note here that the number of steps involved in the energy minimization and equilibration stages that are provided with CHARMM-GUI can usually be safely reduced. For example, the soft-core energy minimization is set to 5000 steps by default; this can be reduced to 500 or so steps.

3. *Visualization of simulations performed with MARTINI.*

Visualization of CG structures/trajectories is not straightforward as molecular visualization packages typically assign bonding using distance criteria and libraries of possible bond lengths based on atomistic structures. Given CG particles are larger than atoms and the distances and therefore “bonds” between them are longer than their atomistic counterparts, they are not identified by most visualization packages, leading to representation of structures as unconnected spheres. A script that can be used with the popular molecular visualization package VMD is available to read in CG model connectivities. Instructions on usage are provided at <http://cgmartini.nl/index.php/tools2/visualization>. Bendix [60] is a useful tool for analysis of CG helices and comes with a comprehensive tutorial (<http://sbc.bioch.ox.ac.uk/Bendix/tutorial.html>). The UnityMol visualization package [61] can also be used in a fairly straightforward manner to visualize MARTINI simulations. UnityMol attempts to facilitate the visualization of coarse-grained molecular simulations [62] by reading a variety of topology formats to assign correct bonds and interpret elastic networks, draw simplified cartoon representations [63] for such models and enable interactive simulations.

4. *A sample input file for MARTINI simulations in GROMACS.*

The .mdp file in GROMACS provides the simulation parameters such as integration timestep, methods for evaluating short- and long-range interactions. Below (Fig. 5a, b) is an example of an .mdp file for a CG MD simulation using the MARTINI forcefield. Descriptions and annotations are provided for guidance. We note that MARTINI 3 beta version

a

Parameter	Value	Description	Notes
Integrator	= md	Do molecular dynamics	
tinit	= 0.0	Time at start of simulation	
dt	= 0.020	Integration timestep (ps)	Total simulation length is dt x nsteps
nsteps	= 1500000	Number of steps	
nstxout	= 5000	Interval (in steps) for writing coordinates, velocities and forces to the trr file	Reduce these values for larger files which incorporate more frames
nstfout	= 5000		
nstlog	= 5000	Interval for writing to the log and ener files	
nstenergy	= 5000		
nstxout-compressed	= 5000	Interval (in steps) for writing to the xtc file and its (real) precision	
compressed-x-precision	= 100		
cutoff-scheme	= Verlet	Method for generating the neighbour list	
nstlist	= 20	Frequency of updating the neighbour list	
ns_type	= grid	Method of checking neighbour list	
pbc	= xyz	Directions in which to apply PBC	
verlet-buffer-tolerance	= 0.005	Max error tolerated for pair interactions per particle caused by Verlet cut-off	
epsilon_r	= 15	Dielectric constant	Use PME instead of reaction field
coulombtype	= reaction-field	Method of evaluating electrostatics	for electrostatics with polarisable
rcoulomb	= 1.1	Cut-off distance for long-range interactions	MARTINI water
vdw_type	= cutoff	Method for evaluating van der Waals interactions	
vdw-modifier	= Potential-shift-verlet	Method for modifying van der Waals potential	
rvdw	= 1.1	Cut-off for short-range interactions	

b

Parameter	Value	Description
tcoupl	= v-rescale	Method for implementing Temp coupling
tc-grps	= membrane solute	Groups to couple together
tau_t	= 1.0 1.0	Strength of coupling
ref_t	= 310 310	Required Temp (K)
Pcoupl	= Parrinello-Rahman	Method of implementing Pressure coupling
Pcoupltype	= semiisotropic	Type of coupling
tau_p	= 12.0	Strength of coupling
compressibility	= 3e-4 3e-4	Compressibility (bar ⁻¹)
ref_p	= 1.0 1.0	Required pressure (bar)
refcoord_scaling	= all	Whether and how reference coordinates are scaled
gen_vel	= no	Whether velocities are generated or set to zero, if none are provided with the input files.

Fig. 5 An example of typical parameters set in a GROMACS .mdp file for a MARTINI CG simulation. Both (a) and (b) outline various .mdp file parameters that need to be taken into consideration when performing a CG simulation using MARTINI. Both a description and notes are given to explain various parameter names

is available from the MARTINI website and will be in general release soon. The .mdp file provided here has been tested with MARTINI2.2, so recommended values for some parameters may differ for MARTINI3.

5. Parameterization of new molecules.

Parameterizing new molecules involves two considerations, (a) the mapping of atoms to MARTINI bead types and (b) assigning nonbonded parameters. Currently there are a number of options to facilitate these processes. Automartini

[64] provides mapping and nonbonded parameters based on static atomistic structures. PyCGTOOL [65] extracts bonded parameters from atomistic trajectories but requires mapping information from the user. As with all parameterization of new molecules, the user is advised to carefully check and validate the resultant parameters. A custom version of *martinize* is available on github (<https://github.com/raziel81/martinize.py/>) to convert atomistic protein structures containing palmitoylated cysteines to CG resolution [66].

6. *Benchmarking of MARTINI membrane protein simulations.*

Benchmark results for typical MARTINI membrane protein simulations are shown in Table 1. Gromacs 2020.1 was used for all simulations, with the mdp file provided in **Note 4** of this guide. The integration timestep was increased to 30 fs. The system size was 306,617 particles and consisted of membrane proteins, lipids, water, and Na⁺ ions.

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References

1. Shaw DE, Maragakis P, Lindorff-Larsen K et al (2010) Atomic-level characterization of the structural dynamics of proteins. *Science* 330:341–346. <https://doi.org/10.1126/SCIENCE.1187409>
2. Marrink SJ, Risselada HJ, Yefimov S et al (2007) The MARTINI force field: coarse grained model for biomolecular simulations. *J Phys Chem B* 111:7812–7824. <https://doi.org/10.1021/jp071097f>
3. Monticelli L, Kandasamy SK, Periole X et al (2008) The MARTINI coarse-grained force field: extension to proteins. *J Chem Theory Comput* 4:819–834. <https://doi.org/10.1021/ct700324x>
4. de Jong DH, Singh G, Bennett WFD et al (2013) Improved parameters for the Martini coarse-grained protein force field. *J Chem Theory Comput* 9:687–697. <https://doi.org/10.1021/ct300646g>
5. Marrink SJ, de Vries AH, Alan EM (2003) Coarse grained model for semiquantitative lipid simulations. <https://doi.org/10.1021/JP036508G>
6. Arnarez C, Uusitalo JJ, Masman MF et al (2015) Dry Martini, a coarse-grained force field for lipid membrane simulations with implicit solvent. *J Chem Theory Comput* 11:260–275. <https://doi.org/10.1021/ct500477k>
7. Uusitalo JJ, Ingólfsson HI, Akhshi P et al (2015) Martini coarse-grained force field: extension to DNA. *J Chem Theory Comput* 11:3932–3945. <https://doi.org/10.1021/acs.jctc.5b00286>
8. Uusitalo JJ, Ingólfsson HI, Marrink SJ, Faustino I (2017) Martini coarse-grained force field: extension to RNA. *Biophys J* 113:246–256. <https://doi.org/10.1016/J.BPJ.2017.05.043>
9. Machado MR, Barrera EE, Klein F et al (2019) The SIRAH 2.0 force field: altius, fortius, citius. *J Chem Theory Comput* 15:2719–2733. <https://doi.org/10.1021/acs.jctc.9b00006>
10. Barrera EE, Machado MR, Pantano S (2019) Fat SIRAH: coarse-grained phospholipids to explore membrane–protein dynamics. *J Chem Theory Comput* 15:5674–5688. <https://doi.org/10.1021/acs.jctc.9b00435>
11. Orsi M, Essex JW (2011) The ELBA force field for coarse-grain modeling of lipid membranes. *PLoS One* 6:e28637. <https://doi.org/10.1371/journal.pone.0028637>

12. Berendsen HJC, van der Spoel D, van Drunen R (1995) GROMACS: a message-passing parallel molecular dynamics implementation. *Comput Phys Commun* 91:43–56. [https://doi.org/10.1016/0010-4655\(95\)00042-E](https://doi.org/10.1016/0010-4655(95)00042-E)
13. Abraham MJ, Murtola T, Schulz R et al (2015) GROMACS: high performance molecular simulations through multi-level parallelism from laptops to supercomputers. *SoftwareX* 1–2:19–25. <https://doi.org/10.1016/j.softx.2015.06.001>
14. Van Der Spoel D, Lindahl E, Hess B et al (2005) GROMACS: fast, flexible, and free. *J Comput Chem* 26:1701–1718. <https://doi.org/10.1002/jcc.20291>
15. Lindahl E, Hess B, van der Spoel D (2001) GROMACS 3.0: a package for molecular simulation and trajectory analysis. *J Mol Model* 7:306–317. <https://doi.org/10.1007/s008940100045>
16. Pronk S, Páll S, Schulz R et al (2013) GROMACS 4.5: a high-throughput and highly parallel open source molecular simulation toolkit. *Bioinformatics* 29:845–854. <https://doi.org/10.1093/bioinformatics/btt055>
17. Hess B, Kutzner C, van der Spoel D, Lindahl E (2008) GROMACS 4: algorithms for highly efficient, load-balanced, and scalable molecular simulation. *J Chem Theory Comput* 4:435–447. <https://doi.org/10.1021/ct700301q>
18. Fiser A, Sali A (2003) Modeller: generation and refinement of homology-based protein structure models. *Methods Enzymol* 374:461–491. [https://doi.org/10.1016/S0076-6879\(03\)74020-8](https://doi.org/10.1016/S0076-6879(03)74020-8)
19. Waterhouse A, Bertoni M, Bienert S et al (2018) SWISS-MODEL: homology modelling of protein structures and complexes. *Nucleic Acids Res* 46:W296–W303. <https://doi.org/10.1093/nar/gky427>
20. Shen M-Y, Sali A (2006) Statistical potential for assessment and prediction of protein structures. *Protein Sci* 15:2507–2524. <https://doi.org/10.1110/ps.062416606>
21. Benkert P, Tosatto SCE, Schomburg D (2008) QMEAN: a comprehensive scoring function for model quality assessment. *Proteins* 71:261–277. <https://doi.org/10.1002/prot.21715>
22. Cornell WD, Cieplak P, Bayly CI et al (1995) A second generation force field for the simulation of proteins, nucleic acids, and organic molecules. *J Am Chem Soc* 117:5179–5197. <https://doi.org/10.1021/ja00124a002>
23. Maier JA, Martinez C, Kasavajhala K et al (2015) ff14SB: improving the accuracy of protein side chain and backbone parameters from ff99SB. *J Chem Theory Comput* 11. <https://doi.org/10.1021/acs.jctc.5b00255>
24. Phillips JC, Braun R, Wang W et al (2005) Scalable molecular dynamics with NAMD. *J Comput Chem* 26:1781–1802. <https://doi.org/10.1002/jcc.20289>
25. Plimpton S (1995) Fast parallel algorithms for short-range molecular dynamics. *J Comput Phys* 117:1–19. <https://doi.org/10.1006/JCPH.1995.1039>
26. Periole X, Cavalli M, Marrink S-J, Ceruso MA (2009) Combining an elastic network with a coarse-grained molecular force field: structure, dynamics, and intermolecular recognition. *J Chem Theory Comput* 5:2531–2543. <https://doi.org/10.1021/ct9002114>
27. Qi Y, Ingólfsson HI, Cheng X et al (2015) CHARMM-GUI Martini maker for coarse-grained simulations with the Martini force field. *J Chem Theory Comput*. <https://doi.org/10.1021/acs.jctc.5b00513>
28. Hsu P-C, Bruininks BMH, Jefferies D et al (2017) CHARMM-GUI Martini maker for modeling and simulation of complex bacterial membranes with lipopolysaccharides. *J Comput Chem* 38:2354–2363. <https://doi.org/10.1002/jcc.24895>
29. Wassenaar TA, Ingólfsson HI, Böckmann RA et al (2015) Computational lipidomics with insane: a versatile tool for generating custom membranes for molecular simulations. *J Chem Theory Comput* 11:2144–2155. <https://doi.org/10.1021/acs.jctc.5b00209>
30. Glass WG, Duncan AL, Biggin PC (2020) Computational investigation of voltage-gated sodium channel $\beta 3$ subunit dynamics. *Front Mol Biosci* 7:40. <https://doi.org/10.3389/fmolb.2020.00040>
31. Duncan AL, Reddy T, Koldsø H et al (2017) Protein crowding and lipid complexity influence the nanoscale dynamic organization of ion channels in cell membranes. *Sci Rep* 7:16647. <https://doi.org/10.1038/s41598-017-16865-6>
32. Chavent M, Duncan AL, Rassam P et al (2018) How nanoscale protein interactions determine the mesoscale dynamic organisation of bacterial outer membrane proteins. *Nat Commun* 9:2846. <https://doi.org/10.1038/s41467-018-05255-9>
33. Holdbrook DA, Huber RG, Piggot TJ et al (2016) Dynamics of crowded vesicles: local and global responses to membrane composition. *PLoS One* 11:e0156963. <https://doi.org/10.1371/journal.pone.0156963>

34. Jefferies D, Khalid S (2020) To infect or not to infect: molecular determinants of bacterial outer membrane vesicle internalization by host membranes. *J Mol Biol* 432:1251–1264. <https://doi.org/10.1016/J.JMB.2020.01.008>
35. Berendsen HJC, Postma JPM, van Gunsteren WF et al (1984) Molecular dynamics with coupling to an external bath. *J Chem Phys* 81:3684–3690. <https://doi.org/10.1063/1.448118>
36. Bussi G, Donadio D, Parrinello M (2007) Canonical sampling through velocity rescaling. *J Chem Phys* 126:014101. <https://doi.org/10.1063/1.2408420>
37. Parrinello M, Rahman A (1980) Crystal structure and pair potentials: a molecular-dynamics study. *Phys Rev Lett* 45:1196–1199. <https://doi.org/10.1103/PhysRevLett.45.1196>
38. Parrinello M, Rahman A (1981) Polymorphic transitions in single crystals: a new molecular dynamics method. *J Appl Phys* 52:7182–7190. <https://doi.org/10.1063/1.328693>
39. Jones JE (1924) On the determination of molecular fields. I. From the variation of the viscosity of a gas with temperature. *Proc R Soc A Math Phys Eng Sci* 106:441–462. <https://doi.org/10.1098/rspa.1924.0081>
40. Jones JE (1924) On the determination of molecular fields. II. From the equation of state of a gas. *Proc R Soc London A* 106:463–477. <https://doi.org/10.1098/rspa.1924.0082>
41. Darden T, York D, Pedersen L (1993) Particle mesh Ewald: an Nlog(N) method for Ewald sums in large systems. *J Chem Phys* 98:10089–10092. <https://doi.org/10.1063/1.464397>
42. Krieger E, Vriend G (2015) New ways to boost molecular dynamics simulations. *J Comput Chem* 36:996–1007. <https://doi.org/10.1002/jcc.23899>
43. Wassenaar TA, Pluhackova K, Böckmann RA et al (2014) Going backward: a flexible geometric approach to reverse transformation from coarse grained to atomistic models. *J Chem Theory Comput* 10:676–690. <https://doi.org/10.1021/ct400617g>
44. MacKerell AD, Banavali N, Foloppe N (2000) Development and current status of the CHARMM force field for nucleic acids. *Biopolymers* 56:257–265. [https://doi.org/10.1002/1097-0282\(2000\)56:4<257::AID-BIP10029>3.0.CO;2-W](https://doi.org/10.1002/1097-0282(2000)56:4<257::AID-BIP10029>3.0.CO;2-W)
45. Huang J, MacKerell AD (2013) CHARMM36 all-atom additive protein force field: validation based on comparison to NMR data. *J Comput Chem* 34:2135–2145. <https://doi.org/10.1002/jcc.23354>
46. MacKerell AD, Bashford D, Bellott M et al (1998) All-atom empirical potential for molecular modeling and dynamics studies of proteins. *J Phys Chem B* 102:3586–3616. <https://doi.org/10.1021/jp973084f>
47. Maciejewski A, Pasenkiewicz-Gierula M, Cramariuc O et al (2014) Refined OPLS all-atom force field for saturated phosphatidylcholine bilayers at full hydration. *J Phys Chem B* 118:4571–4581. <https://doi.org/10.1021/jp5016627>
48. Ulmschneider JP, Ulmschneider MB (2009) United atom lipid parameters for combination with the optimized potentials for liquid simulations all-atom force field. *J Chem Theory Comput* 5:1803–1813. <https://doi.org/10.1021/ct900086b>
49. van Gunsteren WF, Billeter SR, Eising AA et al (1996) Biomolecular simulation: the GROMOS96 manual and user guide. Vdf Hochschulverlag AG an der ETH Zürich, Zürich
50. Schuler LD, Daura X, van Gunsteren WF (2001) An improved GROMOS96 force field for aliphatic hydrocarbons in the condensed phase. *J Comput Chem* 22:1205–1218. <https://doi.org/10.1002/jcc.1078>
51. van Gunsteren WF, Berendsen HJC (1987) Groningen molecular simulation (GROMOS) library manual. Biomos, Groningen
52. Jo S, Cheng X, Lee J et al (2017) CHARMM-GUI 10 years for biomolecular modeling and simulation. *J Comput Chem* 38:1114–1124. <https://doi.org/10.1002/jcc.24660>
53. Ghahremanpour MM, Arab SS, Aghazadeh SB et al (2014) MemBuilder: a web-based graphical interface to build heterogeneously mixed membrane bilayers for the GROMACS biomolecular simulation program. *Bioinformatics* 30:439–441. <https://doi.org/10.1093/bioinformatics/btt680>
54. Balabin I VMD membrane plugin. <https://www.ks.uiuc.edu/Research/vmd/plugins/membrane>
55. Doerr S, Giorgino T, Martínez-Rosell G et al (2017) High-throughput automated preparation and simulation of membrane proteins with HTMD. *J Chem Theory Comput* 13:4003–4011. <https://doi.org/10.1021/acs.jctc.7b00480>
56. Wu X, Brooks BR (2003) Self-guided Langevin dynamics simulation method. *Chem Phys Lett* 381:512–518. <https://doi.org/10.1016/J.CPLETT.2003.10.013>

57. Orsi M, Ding W, Palaiokostas M (2014) Direct mixing of atomistic solutes and coarse-grained water. *J Chem Theory Comput* 10:4684–4693. <https://doi.org/10.1021/ct500065k>
58. Genheden S, Essex JW (2015) A simple and transferable all-atom/coarse-grained hybrid model to study membrane processes. *J Chem Theory Comput* 11:4749–4759. <https://doi.org/10.1021/acs.jctc.5b00469>
59. Genheden S (2018) ELBA. <https://github.com/SGenheden/Elba>
60. Dahl ACE, Chavent M, Sansom MSP (2012) Bendix: intuitive helix geometry analysis and abstraction. *Bioinformatics* 28:2193. <https://doi.org/10.1093/BIOINFORMATICS/BTS357>
61. Lv Z, Tek A, Da Silva F et al (2013) Game on, science – how video game technology may help biologists tackle visualization challenges. *PLoS One* 8:e57990. <https://doi.org/10.1371/journal.pone.0057990>
62. Doutreligne S, Gageat C, Cragolini T et al (2015) UnityMol: interactive and ludic visual manipulation of coarse-grained RNA and other biomolecules. In: 2015 IEEE 1st international workshop on virtual and augmented reality for molecular science (VARMS@IEEEVR). IEEE, New York, NY, pp 1–6
63. Doutreligne S, Cragolini T, Pasquali S et al (2014) UnityMol: interactive scientific visualization for integrative biology. In: 2014 IEEE 4th symposium on large data analysis and visualization (LDAV). IEEE, New York, NY, pp 109–110
64. Bereau T, Kremer K (2015) Automated parametrization of the coarse-grained Martini force field for small organic molecules. *J Chem Theory Comput* 11:2783–2791. <https://doi.org/10.1021/acs.jctc.5b00056>
65. Graham JA, Essex JW, Khalid S (2017) PyCG-TOOL: automated generation of coarse-grained molecular dynamics models from atomistic trajectories. *J Chem Inf Model* 57:650–656. <https://doi.org/10.1021/acs.jcim.7b00096>
66. Atsmon-Raz Y, Tieleman DP (2017) Parameterization of palmitoylated cysteine, farnesylated cysteine, geranylgeranylated cysteine, and myristoylated glycine for the Martini force field. *J Phys Chem B* 121:11132–11143. <https://doi.org/10.1021/acs.jpcb.7b10175>



Continuous Constant pH Molecular Dynamics Simulations of Transmembrane Proteins

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Abstract

Many membrane channels, transporters, and receptors utilize a pH gradient or proton coupling to drive functionally relevant conformational transitions. Conventional molecular dynamics simulations employ fixed protonation states, thus neglecting the coupling between protonation and conformational equilibria. Here we describe the membrane-enabled hybrid-solvent continuous constant pH molecular dynamics method for capturing atomic details of proton-coupled conformational dynamics of transmembrane proteins. Example protocols from our recent application studies of proton channels and ion/substrate transporters are discussed.

Key words Molecular dynamics, Electrostatics, pK_a , Proton channels, Secondary active transporters, G-protein-coupled receptors

1 Introduction

1.1 pH-Dependent Conformational Dynamics of Membrane Proteins

Solution pH plays a critical role in the functions of many transmembrane proteins, including secondary active transporters, proton channels, and G-protein-coupled receptors (GPCRs). For example, a pH gradient across the membrane drives the transport of sodium ions through the sodium-proton antiporter NhaA, which is essential for cellular sodium and proton homeostasis in *Escherichia coli* [1, 2]. Proton channels, such as the voltage-gated proton channel Hv1 [3], conduct protons across the membrane in response to a change in cellular pH [4, 5]. Solution pH also modulates the functions of membrane receptors. For example, the ovarian cancer GPCR1 and the closely related GPR4 regulate cell-mediated responses to acidosis in bone [6].

pH-dependent transmembrane proteins often couple proton sensing with conformational changes [7]. For example, the inward-

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and outward-facing crystal structures of the sodium–proton antiporter NapA from *Thermus thermophilus* [8, 9] reveal an elevator-like alternating-access model, although the detailed mechanism of the pH-dependent conformational transition remains elusive. In many other cases, however, the conformational change is unknown due to the lack of an active-state structure. For example, only inactive crystal structures have been resolved for the aforementioned antiporter NhaA [10, 11] and proton channel Hv1 [12]. Apart from high-resolution structure models, pK_a 's of active-site residues are important for mechanistic elucidation, as they inform protonation states at a certain pH, thereby unveiling residues involved in proton binding/release coupled to the conformational transition. However, pK_a 's of membrane proteins are extremely challenging if possible at all to experimentally measure. Taking NhaA as an example, electrophysiology and biochemical experiments established that it transports two protons in exchange for one sodium above pH 6.5 [13, 14]; however, in the absence of an outward-facing crystal structure and experimental pK_a 's, the identities of proton-binding active-site residues remain controversial.

1.2 Continuous Constant pH Molecular Dynamics Methods

Constant pH molecular dynamics (pHMD) simulations [15–17] offer a unique means to describe proton-coupled conformational dynamics and calculate pK_a values [16, 18, 19]. Current pHMD methods can be grouped into two classes. In discrete pHMD (DpHMD) methods [20, 21], protonation states of titratable groups are periodically updated using Monte-Carlo sampling following molecular dynamics steps. Rooted in the λ -dynamics method for free energy calculations [22], continuous constant-pH molecular dynamics (CpHMD) methods [23–33] make use of fictitious λ particles to sample protonation states simultaneously with atomic coordinates. A λ particle carries a mass of a regular heavy atom (e.g., ten atomic mass units) and the λ coordinate continuously evolves between 0 and 1, where 0 and 1 represent protonated and deprotonated states, respectively. The total Hamiltonian includes the kinetic energy of the fictitious particles and the nonbonded term is modified by linearly scaling electrostatic and van der Waals energies according to λ coordinates. Thus, CpHMD directly couples conformational dynamics of the protein with protonation equilibria of all ionizable sites at a specified pH. Following a CpHMD simulation at a single pH or a set of replica-exchange CpHMD simulations at multiple pH conditions, the trajectories of λ coordinates are extracted, allowing the calculation of pK_a 's for all titratable sites.

The first two CpHMD methods [23, 24], implemented in the CHARMM molecular dynamics package [34], make use of the generalized Born (GB) implicit-solvent models for both proton titration and conformational dynamics. Since then, several

CpHMD methods have been developed and implemented in CHARMM. The hybrid-solvent CpHMD [25] was developed, which takes advantage of accurate description of conformational states in explicit solvent and rapid convergence of protonation state sampling in GB solvent. Additionally, the pH replica-exchange Scheme [25] was developed to accelerate convergence of protonation and conformational state sampling. To overcome the limitations of implicit-solvent models, all-atom CpHMD^{MSAD} with force switching was developed that uses explicit solvent to sample both titration and conformational dynamics [32, 33]. To avoid the potential artifacts due to electrostatic cutoff and to further improve accuracy, particle-mesh Ewald (PME) all-atom CpHMD [29] was developed. CpHMD methods have also been implemented in the popular Amber [35] and GROMACS [36] packages, including GBNeck2-CpHMD [31, 37] in Amber and the all-atom CpHMD [26] in GROMACS [36]. Here we focus on the extensively validated [38–46] pH replica-exchange hybrid-solvent CpHMD method in CHARMM [25], which has been further developed to enable mechanistic studies of membrane proteins [41].

1.3

Membrane-Enabled Hybrid-Solvent CpHMD with pH Replica-Exchange

In the membrane-enabled hybrid-solvent CpHMD method [41], conformational dynamics of a transmembrane protein are sampled in explicit lipids and water, while protonation states are sampled using the membrane Generalized Born with a simple SWitching (GBSW) model [47]. Combined with the pH replica-exchange protocol [25], this method allows detailed description of proton-coupled conformational dynamics of transmembrane proteins (Fig. 1). For example, in a simulation study of NhaA [41], a mechanism emerged that explains the pH-dependent activation of NhaA, reconciling a long standing controversy. According to the calculated pK_a values and conformational dynamics [41], Asp164 and Lys300 were identified as proton binding residues, consistent

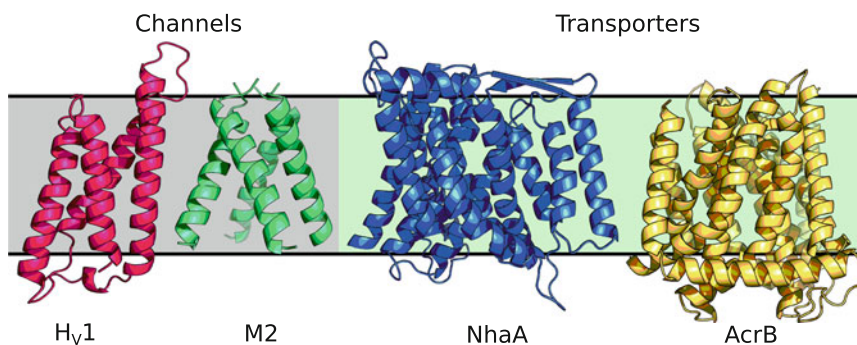


Fig. 1 Transmembrane proteins that have been studied by membrane-enabled hybrid-solvent CpHMD simulations with pH replica-exchange [41]: M2 proton channel [40] (PDB: 3LBW [48]), voltage-gated Hv1 proton channel (Henderson and Shen, unpublished data, PDB: 3WKV [12]), sodium–proton antiporter NhaA [41] (PDB: 4AU5 [11]), and multidrug efflux pump AcrB [42] (PDB: 4DX5 [49])

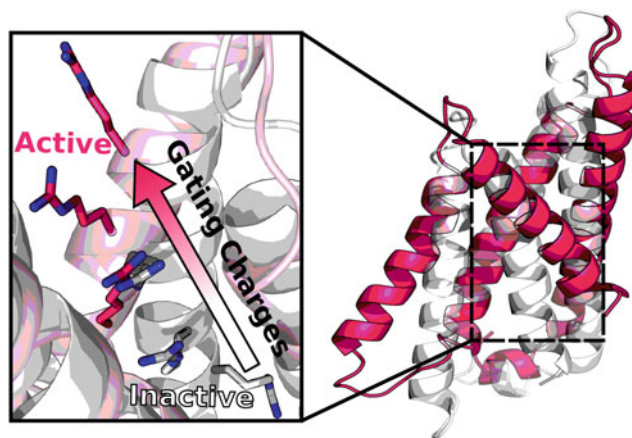


Fig. 2 Membrane-enabled hybrid-solvent CpHMD simulations with pH replica-exchange revealed conformational activation of the voltage-gated proton channel Hv1, as indicated by the upward movement of gating charges (Henderson and Shen, unpublished data)

with the recent crystal structures of NhaA [11] and the homologous protein NapA [9]. When NhaA is exposed to the cytoplasmic pH, Asp164 releases the first proton and triggers the hydrophobic gate on the cytoplasm side to open, allowing the entrance of a sodium ion [41]. The sodium ion then binds to three active-site residues, which induces the release of a second proton from Lys300 [41]. The same protocol used to study NhaA has also been applied to investigate the proton-coupled conformational dynamics of the proton channel M2 from influenza virus [40] and the *E. coli* multi-drug efflux pump AcrB [42]. Most recently, a uniform electric field feature has been added to the membrane-enabled hybrid-solvent CpHMD simulations, allowing us to investigate the activation mechanism of a voltage-gated proton channel Hv1 (Fig. 2).

Despite the recent successes of membrane-enabled hybrid-solvent CpHMD simulations (Fig. 1), several important limitations remain. For example, the implicit representation of solvent and bilayer employed in the titration dynamics limits the accuracy of pK_a estimation, particularly for buried active-site residues and those interacting with lipids or small-molecule ligands. In general, pK_a shifts relative to the solution (model) pK_a are underestimated for interior residues and overestimated for lipid-interacting ones [40–42]. Further, explicit interactions with lipids and ions, which may become important for protonation state changes, cannot be accounted for. While PME all-atom CpHMD [29] has the potential to overcome these limitations, our preliminary study of the M2 channel indicated that the calculated pK_a 's from all-atom CpHMD have large systematic errors, likely a result of underpolarization with additive force fields. Thus, the hybrid-solvent CpHMD method remains a method of choice until a solution is found to remedy

the force field issue. A future direction is to incorporate polarizable force fields such as the Drude model [50] in PME all-atom CpHMD. Due to the high computational cost of simulations with polarizable force fields, a GPU implementation is desirable.

2 Methods

2.1 Setting Up a CpHMD Simulation for Transmembrane Proteins

Here we discuss the protocol of membrane-enabled hybrid-solvent CpHMD simulations in CHARMM [34], which can be divided into three stages (Fig. 3). In Stage 1, the protein–bilayer complex is constructed and undergoes an initial equilibration. In Stage 2, with the protein structure restrained the lipids are equilibrated to the appropriate structural and phase behavior. In Stage 3, protein side-chains are made titratable and CpHMD is invoked to equilibrate the system at a specified pH condition followed by production replica-exchange CpHMD at multiple pH conditions.

2.1.1 Stage 1: Prepare Protein–Bilayer Complex

First, we retrieve the protein coordinates from the database, Orientations of Proteins in Membranes [51], which provides coordinates

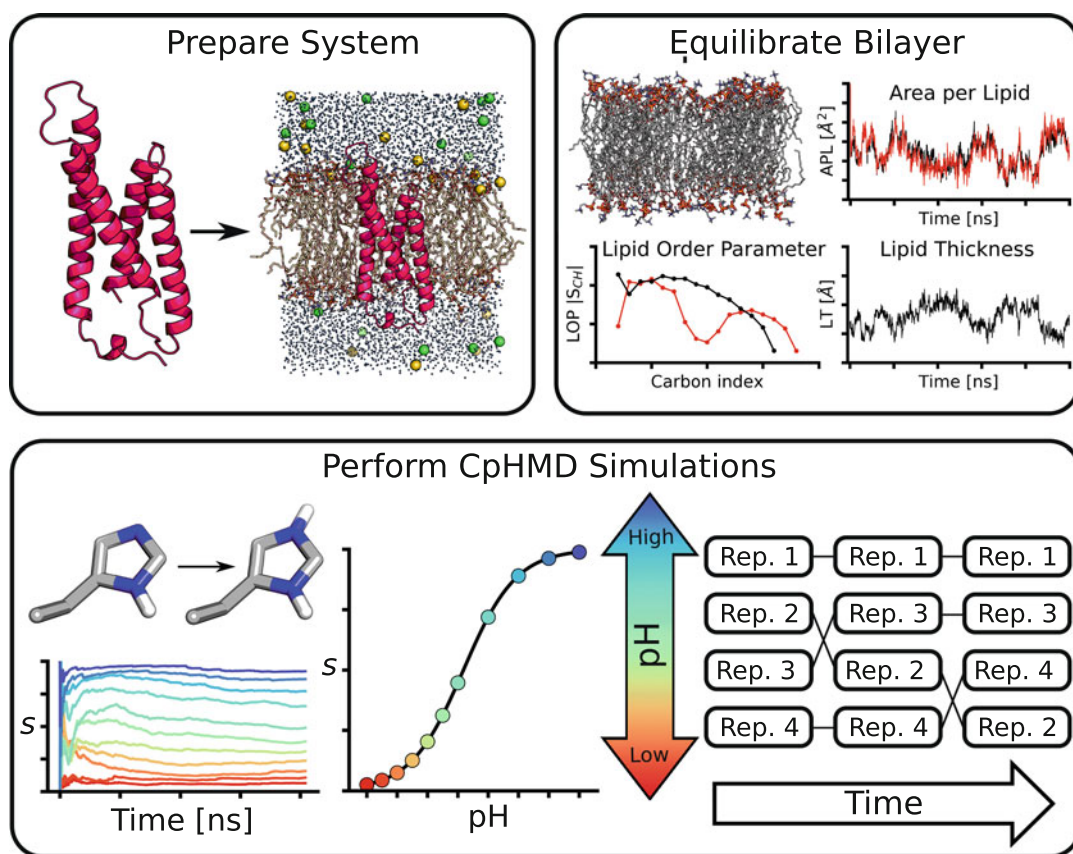


Fig. 3 Overview of the three-stage protocol for membrane-enabled hybrid-solvent CpHMD simulations with pH replica-exchange

of membrane proteins by orienting the structure with respect to the membrane normal using theoretical calculations and comparisons to experimental data [51]. We then use MMTSB toolset [52] or other tools such as SWISS-MODEL [53] to add or replace missing residues, heavy atoms, or mutation sites. The HBUILD facility [54] in CHARMM is used to add missing hydrogens. Note, in this stage, all titratable sites are assumed to be in the “anticipated” protonation states based on the experimental pH condition and hydrogen bond/salt bridge patterns (for His). If no experimental pH can be found, default protonation states corresponding to physiological pH (7.4) should be used: Asp(-)/Glu(-)/Lys(+)/Arg(+)/Cys(0). Unless His is in a salt bridge, HSE or HSD (i.e., protonation on the ϵ or δ nitrogen, respectively, based on the hydrogen bonding pattern) should be used.

Since amino acid sequences given in a PDB file are often truncated, we suggest capping the terminal groups of each protein chain: for N-terminus use CH_3CO (patch ACE), and for C-terminus use NH_2 (patch CT2) or NHCH_3 (patch CT3). However, if a terminal group interacts with a part of the protein, the free ionized form, NH_3^+ and COO^- , should be used. Since CpHMD simulation will be conducted at multiple pHs in Stage 3, we should consider possible protonation state changes. This consideration is especially relevant for the N-amino group, since its solution $\text{p}K_a$ is 8.5 [55], only one unit above the physiological pH. If the N-amino group is allowed to titrate, make sure the model parameters are included in the PHMD parameter file.

Once the protein structure is complete, a very short minimization in implicit solvent (e.g., with the GBSW model [56]) may be conducted with the steepest descent (SD) and adopted basis Newton–Raphson (ABNR) algorithms to relax unfavorable positions of hydrogens. It is important that during the minimization, protein heavy atoms should be fixed or restrained. For example, two rounds of 50 steps of SD followed by ten steps of ABNR are performed, first with heavy atoms fixed and then with a harmonic force constant of $5 \text{ kcal/mol/\AA}^2$. If an extensive number of atoms have been added we also suggest that these sites be energy minimized, but the overall structure should be kept as close as possible to the crystal structure. If crystal waters are resolved in the structure, these should be retained.

The rest of Stage 1 follows the CHARMM-GUI protocol [57] for preparing a protein–membrane complex for MD simulations. The protein is placed in a lipid bilayer with water above and below the bilayer. Counterions for neutralizing the system and additional ions to reach the specified ionic strength are added. The assembled protein–membrane complex undergoes a series of initial equilibration steps (Table 1, Stage 1) to relax the lipid positions before subjecting to a lengthy membrane equilibration step in Stage 2.

Table 1
Example equilibration steps in all three stages

Step	Ensemble	Timestep (fs)	Length (ns)	Restraint force constant (kcal/mol/Å ²)			
<i>Stage 1. Initial equilibration</i>				BB ^a	SC + XWat ^a	Lipid ^b	Wat ^c
1	NVT	1	0.05	10.0	5.0	2.5	2.5
2	NVT	2	0.05	5.0	5.0	2.5	2.5
3	NPT	2	0.1	5.0	5.0	1.0	1.0
4	NPT	2	0.1	5.0	2.5	0.5	0.5
5	NPT	2	0.1	2.5	2.5	0.1	0.1
6	NPT	2	0.1	2.5	2.5	0.0	0.0
<i>Stage 2. Membrane equilibration</i>				BB + SC			
1	NVT	2	0.2	2.5			
2	NPT	2	100	2.5			
<i>Stage 3. CpHMD equilibration</i>				BB + SC			
1	NVT	2	0.1	1.0			
2	NPT	2	0.1	1.0			
3	NPT	2	0.8	0.0			

^aBB and SC refer to protein backbone and side chain heavy atoms, respectively. XWat and BWat refer to crystallographic and bulk water oxygen atoms, respectively

^bLipid heavy atoms are restrained with a planar harmonic potential to ensure that lipid tails are close to the bilayer center ($-5 \text{ Å} < Z < 5 \text{ Å}$), and lipid head groups are close to the bilayer surface ($Z = \pm 19 \text{ Å}$ for POPE)

^cA planar harmonic restraint is applied to the bulk water oxygen atoms to keep them away from the membrane hydrophobic region

2.1.2 Stage 2: Equilibrate Bilayer

Once a protein–membrane complex is assembled and relaxed, the lipid bilayer needs to be fully equilibrated before protein conformational states can be sampled. Simulation in this stage can be carried out using any scalable simulation engine, such as GRO-MACS [58], NAMD [59], or OpenMM [60]. Typically, a 0.2-ns restrained MD is performed in the NVT ensemble followed by a 100-ns restrained MD in the NPT ensemble (*see* Table 1, Stage 2). During the equilibration, protein backbone and side chain heavy atoms are restrained with a harmonic force constant of 1.0 kcal/mol/Å².

Surface area per lipid, bilayer thickness, and lipid order parameters are quantities commonly used to assess the phase behavior and fine structure of the simulated bilayer [61]. A lipid bilayer is considered equilibrated once all three quantities are converged to experimental values for the specified lipid type or those obtained in the force field validation study [61]. The program VTMC [62] can be used to calculate surface area per lipid; it excludes lipids within 5 Å from the protein, as protein–lipid interactions can artificially

decrease the value. Bilayer thickness here refers to the hydrophobic thickness, which is the distance between the average Z-positions of the lipid C2 atoms from the top and bottom leaflets. Lipid order parameters S_{CH} indicate the rigidity of the lipid tails [63] and are defined as

$$S_{CH} = \frac{1}{2} \langle 3 \cos^2 \Theta - 1 \rangle, \quad (1)$$

where Θ is the angle between the C-H bond of the lipid tail methylene and the membrane normal (z), and the angular brackets denote ensemble average. S_{CH} can be calculated using, for example, a Tcl script for VMD [64].

2.1.3 Stage 3: Perform CpHMD Titration Simulations

Following equilibration of bilayer, the system can be further equilibrated using membrane-enabled [41] hybrid-solvent CpHMD simulation at a single pH. From the final snap shot of Stage 2, dummy protons need to be added to all Asp, Glu, and His sidechains. For Asp and Glu, dummy protons are placed in the *syn* position. Note, the dihedral energy barrier to the rotation around the carboxylic bond is raised (to 3 kcal/mol) in the force field [24] to prevent a dummy hydrogen from moving to the less favorable *anti* position where it becomes unlikely to protonate (“ghost” proton) [21, 24]. For His, a proton is added to either δ or ϵ nitrogen to form a doubly protonated HSP. Addition of dummy hydrogens requires a brief restrained energy minimization, for example, two rounds of 50 steps of SD followed by ten steps of ABNR. In the first round all heavy atoms’ positions are fixed, and in the second round a harmonic restraint of 5.0 kcal/mol/Å² is applied to them. Next, three short restrained CpHMD simulations are performed at crystallization or physiological pH (see Table 1, Stage 3). The first and second CpHMD equilibration steps are run for 0.1 ns in the NVT and NPT ensembles, respectively, with protein heavy atoms restrained with a harmonic force constant of 1.0 kcal/mol/Å². The third unrestrained equilibration step is run in the NPT ensemble for 0.8 ns. To prevent lateral drift of the protein, a cylindrical restraint with a harmonic force constant of 1.0 kcal/mol/Å² is applied using the MMFP facility in CHARMM [34].

Following equilibration at a single pH, production CpHMD simulations can begin using the pH replica-exchange protocol [25]. First, a simulation pH range is selected by considering the active pH range of the protein or the one that represents experimental or physiological pH conditions. The pH replicas need to be placed close enough to obtain sufficient exchange probabilities (e.g., >20%), which requires overlap between the potential energy distributions of neighboring replicas. Typically, a 0.5 pH unit increment provides the desired exchange probabilities. Each replica

is simulated in the NPT ensemble, with exchange attempts occurring every 500 or 1000 MD steps between adjacent replicas. We typically observe an average exchange rate of at least 40%. More replicas at smaller intervals (e.g., 0.25 pH units) may be added to increase the exchange rate. The effectiveness of sampling enhancement can be tracked by monitoring the movement of replicas through pH space as the simulation proceeds. At least several replicas should move across the entire pH ladder to achieve enhanced sampling.

2.1.4 Settings in Molecular Dynamics

The protein is represented by the CHARMM22/CMAP force field [65, 66], lipids and ions are represented by the CHARMM36 force field [61], and water is represented by the CHARMM modified TIP3P model. Van der Waals energies are calculated using a switching function from 10 to 12 Å with a cutoff at 14 Å, consistent with the protocol used in the development of the CHARMM36 lipid force field [61]. Simulations are conducted under periodic boundary conditions with the PME method [67] for long-range electrostatic energies and forces. The real space calculation uses a 12-Å cutoff and the reciprocal space calculation uses a 1-Å grid spacing and sixth-order spline interpolation. The nonbonded neighbor list is heuristically updated in CHARMM [34] or every 10 MD steps in GROMACS [36].

To allow for a 2-fs integration time step, all bonds involving hydrogens are constrained using SHAKE [68] in CHARMM [34], LINCS [69] in GROMACS [36], or rigidBonds in NAMD [59]. In NPT simulations, the temperature is maintained at 310 K by Nosé–Hoover thermostat [70, 71] and pressure is maintained at 1 atm by the Langevin piston pressure-coupling algorithm in CHARMM [72] or the Parrinello–Rahman barostat [73] in GROMACS.

2.1.5 CpHMD Specific Settings

Unless otherwise noted, default CpHMD settings should be used. In membrane-enabled hybrid-solvent CpHMD simulations [41], the protein is centered in the lipid bilayer and an exclusion cylinder, for example, with a radius of 25 Å, is placed along the membrane normal to exclude the low dielectric slab used in the implicit membrane model [47] from influencing the titration of interior residues in the protein. The cylinder should encompass the protein with no or minimal overlap with lipids. A center of mass restraint is applied to the protein with a harmonic force constant of 1.0 kcal/mol/Å² via the MMFP facility in CHARMM [34]. The thickness of the low-dielectric slab is set to the hydrophobic thickness of the lipid bilayer, defined as the difference between the average *z* positions of lipid C2 atoms from the top and bottom leaflets, for example, about 35 Å for POPE. The half-membrane switching length [47] for the dielectric transition region between the low-dielectric slab and bulk water is set to 2.5 Å. The default

GBSW atomic input radii [74, 75] are used for the protein. For small molecules, the generic atomic radii in the GBSW radii file or van der Waals radii in the force field [76] should be used. Titration (λ) and tautomeric state (x) coordinates are updated every 5 MD steps (default for hybrid-solvent CpHMD), allowing for water and lipid relaxation [25]. Langevin dynamics with a friction coefficient of 5 ps^{-1} is used (default for CpHMD). The mass of λ and x particles are set to ten atomic mass units (default for CpHMD).

2.2 pK_a Calculations

To calculate the pK_a of a titratable site, the deprotonated fraction at each pH, $S(\text{pH})$, is obtained and fit to the generalized Henderson-Hasselbalch equation (Hill equation),

$$S(\text{pH}) = \frac{1}{1 + 10^n(\text{p}K_a - \text{pH})}, \quad (2)$$

where n is the Hill coefficient representing the slope of the transition region in the titration curve. Deviation of n from 1 indicates the extent of cooperative ($n > 1$) or anticooperative ($n < 1$) interactions between two or more titrating sites [16]. The deprotonated fraction of a titratable site is calculated as the fraction of time λ samples the deprotonated state,

$$S = \frac{N_{\text{deprot}}}{N_{\text{deprot}} + N_{\text{prot}}}, \quad (3)$$

where deprotonated and protonated states are defined as $\lambda > 0.8$ and $\lambda < 0.2$, respectively.

For coupled titrating sites, that is, titration of one site is affected by the protonation state of the other, apparent or macroscopic sequential pK_a s are of interest [37]. In the statistical mechanics formulation, the two sequential pK_a 's, pK_1 and pK_2 , are obtained by fitting the total number of protons bound to the two sites, N_{prot} , to the coupled titration model below [27, 77].

$$N_{\text{prot}} = \frac{10^{\text{p}K_2 - \text{pH}} + 2 \times 10^{\text{p}K_1 + \text{p}K_2 - 2\text{pH}}}{1 + 10^{\text{p}K_2 - \text{pH}} + 10^{\text{p}K_1 + \text{p}K_2 - 2\text{pH}}}. \quad (4)$$

Alternatively, the total deprotonated fraction of the two sites can be fit to the following uncoupled model [33, 78],

$$S_1 + S_2 = \frac{1}{1 + 10^{\text{p}K_1 - \text{pH}}} + \frac{1}{1 + 10^{\text{p}K_2 - \text{pH}}}, \quad (5)$$

where S_1 and S_2 are the deprotonated fractions of the two sites. pK_1 and pK_2 are the two uncoupled pK_a 's.

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References

1. Mager T, Rimón A, Padan E et al (2011) Transport mechanism and pH regulation of the Na^+/H^+ antiporter NhaA from *Escherichia coli*. *J Biol Chem* 286:23570–23581
2. Padan E (2014) Functional and structural dynamics of NhaA, a prototype for Na^+ and H^+ antiporters, which are responsible for Na^+ and H^+ homeostasis in cells. *Biochim Biophys Acta* 1837:1047–1062
3. Ramsey IS, Moran MM, Chong JA et al (2006) A voltage-gated proton-selective channel lacking the pore domain. *Nature* 440:1213–1216
4. DeCoursey TE, Hosler J (2014) Philosophy of voltage-gated proton channels. *J R Soc Interface* 11:20130799
5. DeCoursey TE (2018) Voltage and pH sensing by the voltage-gated proton channel, Hv1. *J R Soc Interface* 15:20180108
6. Ludwig MG, Vanek M, Guerini D et al (2003) Proton-sensing G-protein-coupled receptors. *Nature* 425:93–98
7. Drew D, Boudker O (2016) Shared molecular mechanisms of membrane transporters. *Annu Rev Biochem* 85:543–572
8. Lee C, Kang HJ, von Ballmoos C et al (2013) A two-domain elevator mechanism for sodium/proton antiport. *Nature* 501:573–577
9. Coincon M, Uzdavinyas P, Nji E et al (2016) Crystal structures reveal the molecular basis of ion translocation in sodium/proton antiporters. *Nat Struct Mol Biol* 23:248–255
10. Hunte C, Screpanti E, Venturi M et al (2005) Structure of a Na^+/H^+ antiporter and insights into mechanism of action and regulation by pH. *Nature* 435:1197–1202
11. Lee C, Yashiro S, Dotson DL et al (2014) Crystal structure of the sodium-proton antiporter NhaA dimer and new mechanistic insights. *J Gen Physiol* 144:529–544
12. Takeshita K, Sakata S, Yamashita E et al (2014) X-ray crystal structure of voltage-gated proton channel. *Nat Struct Mol Biol* 21:352–357
13. Taglicht D, Padan E, Schuldiner S (1991) Overproduction and purification of a functional Na^+/H^+ antiporter coded by *nhaA* (*ant*) from *Escherichia coli*. *J Biol Chem* 266:11289–11294
14. Padan E, Danieli T, Keren Y et al (2015) NhaA antiporter functions using 10 helices, and an additional 2 contribute to assembly/stability. *Proc Natl Acad Sci U S A* 112:E5575–E5582
15. Chen J, Brooks CL III, Khandogin J (2008) Recent advances in implicit solvent-based methods for biomolecular simulations. *Curr Opin Struct Biol* 18:140–148
16. Wallace JA, Shen JK (2009) Predicting pK_a values with continuous constant pH molecular dynamics. *Methods Enzymol* 466:455–475
17. Chen W, Morrow BH, Shi C et al (2014) Recent development and application of constant pH molecular dynamics. *Mol Simul* 40:830–838
18. Wallace JA, Wang Y, Shi C et al (2011) Toward accurate prediction of pK_a values for internal protein residues: the importance of conformational relaxation and desolvation energy. *Proteins* 79:3364–3373
19. Alexov E, Mehler EL, Baker N et al (2011) Progress in the prediction of pK_a values in proteins. *Proteins* 79:3260–3275
20. Baptista AM, Teixeira VH, Soares CM (2002) Constant-pH molecular dynamics using stochastic titration. *J Chem Phys* 117:4184–4200
21. Mongan J, Case DA, McCammon JA (2004) Constant pH molecular dynamics in generalized born implicit solvent. *J Comput Chem* 25:2038–2048
22. Kong X, Brooks CL III (1996) λ -Dynamics: a new approach to free energy calculations. *J Chem Phys* 105:2414–2423
23. Lee MS, Salsbury FR Jr, Brooks CL III (2004) Constant-pH molecular dynamics using continuous titration coordinates. *Proteins* 56:738–752
24. Khandogin J, Brooks CL III (2005) Constant pH molecular dynamics with proton tautomerism. *Biophys J* 89:141–157
25. Wallace JA, Shen JK (2011) Continuous constant pH molecular dynamics in explicit solvent with pH-based replica exchange. *J Chem Theory Comput* 7:2617–2629

26. Donnini S, Tegeler F, Groenhof G et al (2011) Constant pH molecular dynamics in explicit solvent with λ -dynamics. *J Chem Theory Comput* 7:1962–1978
27. Wallace JA, Shen JK (2012a) Charge-leveling and proper treatment of long-range electrostatics in all-atom molecular dynamics at constant pH. *J Chem Phys* 137:184105
28. Chen W, Wallace J, Yue Z et al (2013) Introducing titratable water to all-atom molecular dynamics at constant pH. *Biophys J* 105: L15–L17
29. Huang Y, Chen W, Wallace JA et al (2016b) All-atom continuous constant pH molecular dynamics with particle mesh ewald and titratable water. *J Chem Theory Comput* 12:5411–5421
30. Arthur EJ, Brooks III CL (2016) Efficient implementation of constant pH molecular dynamics on modern graphics processors. *J Comput Chem* 37:2171–2180
31. Huang Y, Harris RC, Shen J (2018) Generalized born based continuous constant pH molecular dynamics in amber: implementation, benchmarking and analysis. *J Chem Inf Model* 58:1372–1383
32. Goh GB, Knight JL, Brooks CL III (2012) Constant pH molecular dynamics simulations of nucleic acids in explicit solvent. *J Chem Theory Comput* 8:36–46
33. Goh GB, Hulbert BS, Zhou H et al (2014) Constant pH molecular dynamics of proteins in explicit solvent with proton tautomerism. *Proteins* 82:1319–1331
34. Brooks BR, Brooks III CL, Mackerell AD Jr et al (2009) CHARMM: the biomolecular simulation program. *J Comput Chem* 30:1545–1614
35. Case DA, Ben-Shalom IY, Brozell SR et al (2018) AMBER 2018. University of California, San Francisco, CA
36. Pronk S, Páll S, Schulz R et al (2013) GRO-MACS 4.5: a high-throughput and highly parallel open source molecular simulation toolkit. *Bioinformatics* 29:845–854
37. Harris RC, Shen J (2019) GPU-accelerated implementation of continuous constant pH molecular dynamics in amber: predictions with single-pH simulations. *J Chem Inf Model* 59:4821–4832
38. Wallace JA, Shen JK (2012b) Unraveling a trap-and-trigger mechanism in the pH-sensitive self-assembly of spider silk proteins. *J Phys Chem Lett* 3:658–662
39. Ellis CR, Shen J (2015) pH-dependent population shift regulates BACE1 activity and inhibition. *J Am Chem Soc* 137:9543–9546
40. Chen W, Huang Y, Shen J (2016) Conformational activation of a transmembrane proton channel from constant pH molecular dynamics. *J Phys Chem Lett* 7:3961–3966
41. Huang Y, Chen W, Dotson DL et al (2016a) Mechanism of pH-dependent activation of the sodium-proton antiporter NhaA. *Nat Commun* 7:12940
42. Yue Z, Chen W, Zgurskaya HI et al (2017) Constant pH molecular dynamics reveals how proton release drives the conformational transition of a transmembrane efflux pump. *J Chem Theory Comput* 13:6405–6414
43. Ellis CR, Tsai CC, Hou X et al (2016) Constant pH molecular dynamics reveals pH-modulated binding of two small-molecule BACE1 inhibitors. *J Phys Chem Lett* 7:944–949
44. Harris RC, Tsai CC, Ellis CR et al (2017) Proton-coupled conformational allostery modulates the inhibitor selectivity for β -secretase. *J Phys Chem Lett* 8:4832–4837
45. Henderson JA, Harris RC, Tsai CC et al (2018) How ligand protonation state controls water in protein-ligand binding. *J Phys Chem Lett* 9:5440–5444
46. Tsai CC, Yue Z, Shen J (2019) How electrostatic coupling enables conformational plasticity in a tyrosine kinase. *J Am Chem Soc* 141:15092–15101
47. Im W, Feig M, Brooks CL III (2003a) An implicit membrane generalized Born theory for the study of structure, stability, and interactions of membrane proteins. *Biophys J* 85:2900–2918
48. Acharya R, Carnevale V, Fiorin G et al (2010) Structure and mechanism of proton transport through the transmembrane tetrameric M2 protein bundle of the influenza A virus. *Proc Natl Acad Sci U S A* 107:15075–15080
49. Eicher T, Cha HJ, Seeger MA et al (2012) Transport of drugs by the multidrug transporter AcrB involves an access and a deep binding pocket that are separated by a switch-loop. *Proc Natl Acad Sci U S A* 109:5687–5692
50. Lemkul JA, Huang J, Roux B et al (2016) An empirical polarizable force field based on the classical drude oscillator model: development history and recent applications. *Chem Rev* 116:4983–5013
51. Lomize MA, Lomize AL, Pogozheva ID et al (2006) OPM: orientations of proteins in membranes database. *Bioinformatics* 22:623–625
52. Feig M, Karanicolas J, Brooks CL III (2004) MMTSB tool set: enhanced sampling and multiscale modeling methods for applications in

- structural biology. *J Mol Graph Model* 22:377–395
53. Waterhouse A, Bertoni M, Bienert S et al (2018) SWISS-MODEL: homology modelling of protein structures and complexes. *Nucleic Acids Res* 46:296–303
 54. Brünger AT, Karplus M (1988) Polar hydrogen positions in proteins: empirical energy placement and neutron diffraction comparison. *Proteins* 4:148–156
 55. Nozaki Y, Tanford C (1967) Examination of titration behavior. *Methods Enzymol* 11:715–734
 56. Im W, Lee MS, Brooks CL III (2003b) Generalized Born model with a simple smoothing function. *J Comput Chem* 24:1691–1702
 57. Jo S, Kim T, Im W (2007) Automated builder and database of protein/membrane complexes for molecular dynamics simulations. *PLoS One* 2:e880
 58. Abraham MJ, Murtola T, Schulz R et al (2015) GROMACS: high performance molecular simulations through multi-level parallelism from laptops to supercomputers. *SoftwareX* 1-2:19–25
 59. Phillips JC, Braun R, Wang W et al (2005) Scalable molecular dynamics with NAMD. *J Comput Chem* 26:1781–1802
 60. Eastman P, Swails J, Chodera JD et al (2017) OpenMM 7: rapid development of high performance algorithms for molecular dynamics. *PLoS Comput Biol* 13:e1005659
 61. Klauda JB, Venable RM, Freites JA et al (2010) Update of the CHARMM all-atom additive force field for lipids: validation on six lipid types. *J Phys Chem B* 114:7830–7843
 62. Mori T, Ogushi F, Sugita Y (2012) Analysis of lipid surface area in protein-membrane systems combining voronoi tessellation and monte carlo integration methods. *J Comput Chem* 33:286–293
 63. Vermeer LS, de Groot BL, Réat V et al (2007) Acyl chain order parameter profiles in phospholipid bilayers: computation from molecular dynamics simulations and comparison with ^2H NMR experiments. *Eur Biophys J* 36:919–931
 64. Humphrey W, Dalke A, Schulten K (1996) VMD: visual molecular dynamics. *J Mol Graph* 14:33–38
 65. MacKerell AD Jr, Bashford D, Bellott M et al (1998) All-atom empirical potential for molecular modeling and dynamics studies of proteins. *J Phys Chem B* 102:3586–3616
 66. MacKerell AD Jr, Feig M, Brooks CL III (2004) Extending the treatment of backbone energetics in protein force fields: limitations of gas-phase quantum mechanics in reproducing protein conformational distributions in molecular dynamics simulations. *J Comput Chem* 25:1400–1415
 67. Darden T, York D, Pedersen L (1993) Particle mesh Ewald: an $N \log(N)$ method for Ewald sums in large systems. *J Chem Phys* 98:10089–10092
 68. Ryckaert JP, Ciccotti G, Berendsen HJC (1977) Numerical integration of the cartesian equations of motion of a system with constraints: molecular dynamics of *n*-alkanes. *J Comput Phys* 23:327–341
 69. Hess B, Bekker H, Berendsen HJC et al (1997) LINCS: a linear constraint solver for molecular simulations. *J Comput Chem* 18:1463–1472
 70. Nosé S (1984) A molecular dynamics method for simulations in the canonical ensemble. *Mol Phys* 52:255–268
 71. Hoover WG (1985) Canonical dynamics: equilibration phase-space distributions. *Phys Rev A* 31:1695–1697
 72. Feller SE, Zhang Y, Pastor RW et al (1995) Constant pressure molecular dynamics simulation: the Langevin piston method. *J Chem Phys* 103:4613–4621
 73. Parrinello M, Rahman A (1981) Polymorphic transitions in single crystals: a new molecular dynamics method. *J Appl Phys* 52:7182–7190
 74. Nina M, Beglov D, Roux B (1997) Atomic radii for continuum electrostatics calculations based on molecular dynamics free energy simulations. *J Phys Chem B* 101:5239–5248
 75. Chen J, Im W, Brooks CL III (2006) Balancing solvation and intramolecular interactions: toward a consistent generalized born force field. *J Am Chem Soc* 128:3728–3736
 76. Knight JL, Brooks CL III (2011) Surveying implicit solvent models for estimating small molecule absolute hydration free energies. *J Comput Chem* 32:2909–2923
 77. Ullmann GM (2003) Relations between protonation constants and titration curves in polyprotic acids: a critical view. *J Phys Chem B* 107:1263–1271
 78. Webb H, Tynan-Connolly BM, Lee GM et al (2011) Remeasuring HEWL pK_a values by NMR spectroscopy: methods, analysis, accuracy, and implications for theoretical pK_a calculations. *Proteins* 79:685–702



Molecular Dynamics–Based Thermodynamic and Kinetic Characterization of Membrane Protein Conformational Transitions

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Abstract

Molecular dynamics (MD) simulations are routinely used to study structural dynamics of membrane proteins. However, conventional MD is often unable to sample functionally important conformational transitions of membrane proteins such as those involved in active membrane transport or channel activation process. Here we describe a combination of multiple MD based techniques that allows for a rigorous characterization of energetics and kinetics of large-scale conformational changes in membrane proteins. The methodology is based on biased, nonequilibrium, collective-variable based simulations including nonequilibrium pulling, string method with swarms of trajectories, bias-exchange umbrella sampling, and rate estimation techniques.

Key words String Method, Umbrella Sampling, Nonequilibrium Pulling, Orientation Quaternion, Membrane Protein, Conformational Landscape, Transition Rate Estimation

1 Introduction

With advances in supercomputing technology, continued improvement of all-atom force fields, and increasing number of available structures of membrane proteins, molecular dynamics (MD) simulation technique [1–3] has emerged as a prominent computational method for determining the structural dynamics of membrane proteins in their membrane environment. MD is a technique that is routinely used to study the local fluctuations of membrane proteins around given functional states, often determined by X-ray crystallography, cryogenic electron microscopy (cryoEM), or homology modeling. The timescale gap between local fluctuations and large-scale conformational changes, however, has hindered the use of MD to study the functionally important conformational changes such as those involved in the state transition of transporters or activation of channels and receptors.

Conformational dynamics of proteins can be modeled as a diffusion in the protein conformational landscape [4–6]. The conformational free energy landscape has various basins and saddle points that represent the stable and transition states, respectively. A membrane protein, whether it is a channel, transporter, receptor, etc., is associated with many free energy minima, most of which are clustered around a few major free energy basins, representing the functional states of protein. For instance, a channel may have various closely related free energy minima around a large free energy basin that represents its active state, and one or a few free energy basins that represent its inactive state(s). Similarly, a membrane transporter has at least three free energy basins: one for the outward-facing (OF) state, one for the inward-facing (IF) state, and one for the occluded (Occ) state. Although the conformational landscape of a protein is generally very vast, most of this landscape is associated with large free energies and can be ignored. This is due to the presence of intra- and intermolecular forces that restrict the movement of atoms and molecular domains. These forces allow fluctuations around free energy basins and “rarely” allow the system to jump between these free energy basins by crossing the free energy barriers.

MD practitioners often use the available structures to study the local fluctuations of proteins around given free energy basins employing MD simulations of tens to hundreds of nanoseconds, and more recently up to several microseconds. Jumping between major free energy basins, however, rarely happens. In order to induce such jumps, one often needs to employ biased and/or nonequilibrium MD simulations. Many enhanced sampling techniques have been developed that can be employed to facilitate the sampling of rare events [7–18]. A handful of these methods are used routinely for the study of functionally relevant transitions; however, applying many of these methods to complex biological systems such as membrane proteins is challenging. Here we outline a number of methods to be used in finding and characterizing the membrane protein conformational transition pathways through the use of path-finding algorithms and enhanced sampling techniques. The specific techniques include: nonequilibrium pulling simulations (such as targeted MD, steered MD, and similar methods), path optimization algorithms (such as string method with swarms of trajectories [19, 20]), along-the-path free-energy calculations (such as bias-exchange umbrella sampling [20]), and transition rate estimation methods. The following section outlines the theory behind some of the techniques employed in this protocol.

2 Theory

2.1 Introduction

Dimensionality reduction is a necessary part of computational studies of protein dynamics due to the large number of degrees of freedom in the atomic coordinate space of proteins. Collective variable (colvar) based enhanced sampling techniques such as umbrella sampling (US) [7, 8] and its nonequilibrium counterparts [9–11, 13, 21, 22] effectively work within a reduced space. Collective variables can be defined intuitively to describe the slow degrees of freedom associated with functionally important protein conformational changes [23], for example, an interdomain molecular distance as in steered MD (SMD) [11] or the root-mean-square deviation (RMSD) from a reference structure as in targeted MD [9]. Several collective variable suites/modules [24–27] have recently been developed to allow for the system-specific design of collective variables such as path colvars [28, 29] and orientation colvars [26, 30].

An ideal collective variable could represent the “reaction coordinate” [31] as often described in transition state theory. However, even if a one-dimensional reaction coordinate exists, it is not known a priori. This has led to the development of several path-finding algorithms, which implicitly or explicitly approximate the reaction coordinate by the arc-length of a curve in the multidimensional space of atomic coordinates [32, 33] or colvars [4, 19]. Many of the colvar based enhanced sampling techniques implicitly or explicitly use a diffusion model to describe the effective dynamics in the colvar space [4, 5]. We have recently developed a Riemannian diffusion model for protein conformational dynamics that provides a robust framework for conformational free energy calculation methods and path-finding algorithms [34]. Unlike their Euclidean counterparts, the Riemannian potential of mean force (PMF) and minimum free energy path (MFEP) are *invariant under coordinate transformations* [34]. However, the protocol discussed here can be used with or without the Riemannian treatment of the colvar space.

Suppose that the dynamics of a high-dimensional atomic system ($\mathbf{x}$) can be simplified as effective dynamics in a reduced but generally multidimensional colvar space, $\boldsymbol{\xi}$. The effective dynamics can be described by a Brownian motion in the $\boldsymbol{\xi}$ space with an effective potential energy $G(\boldsymbol{\xi})$, that is the PMF of the atomic system in the $\boldsymbol{\xi}$ space:

$$G(\boldsymbol{\xi}_0) = -k_B T \log \langle \delta(\boldsymbol{\xi}(\mathbf{x}) - \boldsymbol{\xi}_0) \rangle \quad (1)$$

is an ensemble average, k_B and T are the Boltzmann constant and temperature, respectively, and δ is the Dirac delta function.

One may sample the regions around a given point $\boldsymbol{\xi}_0$ in the colvar space by adding a biasing term to the potential of the atomic system such as $U_0(\boldsymbol{\xi}) = \frac{k}{2} (\boldsymbol{\xi} - \boldsymbol{\xi}_0)^2$, in which k is the force constant.

The free energy of the biased system (or the perturbed free energy) is:

$$F(\xi_0) = -k_B T \log \int d\xi \exp(-\beta(G(\xi) + U_0(\xi))) \quad (2)$$

where $\beta = (k_B T)^{-1}$. For large force constants, the PMF can be approximated using the perturbed free energy $F(\xi)$. Otherwise, other methods can be used to estimate the PMF as briefly described below. If the biasing center is different in different simulations (or windows/images), we have $U_i(\xi) = \frac{k}{2} (\xi - \xi_i)^2$, where i is the window/image index.

Methods such as US rely on calculating the relative free energy of different points by biasing the system around those points in different simulations. Alternatively, the biasing center could change by time, for example, replacing ξ_i by $\eta(t)$. Examples of such simulations are SMD and targeted MD. Here, we refer to such methods as nonequilibrium pulling simulations, which may use any colvar(s) with any schedule of change (i.e., $\eta(t)$ may or may not be linear in time). The biasing potential is described by: $U(\xi, t) = \frac{k}{2} (\xi - \eta(t))^2$. The accumulated nonequilibrium work at any given time is: $W(t) = \int_0^t dt' \frac{\partial}{\partial t'} U(\xi, t')$. Nonequilibrium work can be used to estimate the perturbed free energy using the Jarzynski relation [35] or other nonequilibrium work relations [36, 37]. In this protocol, we use pulling simulations only to generate initial pathways for other simulation protocols. The nonequilibrium pulling simulations may use multidimensional colvars; however, a specific 1D pathway needs to be selected in order to perform the simulations.

Ideally, one may use a 1D collective variable for defining the effective dynamics as well as the biasing protocol. In practice, however, this may only be possible for extremely simple systems. A practical solution to this problem is to keep the collective variable space multidimensional, while sampling only around a particular pathway, represented by a 1D curve $\xi(\xi)$, parametrized by ξ . The choice of the pathway is obviously crucial here and determines the relevance of the free energy results to the transition of interest. Now $\xi(\mathbf{x})$ can be treated as a 1D colvar defined as a function of atomic coordinates $\mathbf{x}$, and $G(\xi_0)$ is the PMF associated with ξ_0 ,

$$\exp(-\beta G(\xi_0)) = \langle \delta(\xi(\mathbf{x}) - \xi_0) \rangle. \quad (3)$$

Assuming ξ dynamics can be effectively described by a diffusive model, we have,

$$d\xi = \left(-\beta D(\xi) \frac{d}{d\xi} G(\xi) + \frac{d}{d\xi} D(\xi) \right) dt + \sqrt{2D(\xi)} dB, \quad (4)$$

in which $D(\xi)$ is a position-dependent diffusion constant, and $B(t)$ is a Wiener process such that $\langle B(t) \rangle = 0$ and $\langle B^2(t) \rangle = t$. Fokker-Planck (or Smoluchowski) equation associated with this process is

$$\frac{\partial}{\partial t} p(\xi, t | \xi_0, 0) = \frac{\partial}{\partial \xi} \left(D \exp(-\beta G(\xi)) \frac{\partial}{\partial \xi} (\exp(\beta G(\xi)) p(\xi, t | \xi_0, 0)) \right), \quad (5)$$

in which $p(\xi, t | \xi_0, 0)$ is the likelihood of finding the system at ξ after time t , given it was at ξ_0 at time 0. If two major free energy minima exist at points A and B , with no other basins outside the region spanning from A to B , the mean-first-passage time (MFPT) from A to B ($\bar{\tau}_{FP}$) can be estimated using the following relation [38]:

$$\bar{\tau}_{FP} = \int_{\xi_A}^{\xi_B} d\xi \frac{\int_{\xi_A}^{\xi} d\xi' \exp(-\beta G(\xi'))}{D(\xi) \exp(-\beta G(\xi))}. \quad (6)$$

The aim of the protocols described here is to find the MFEP in a multidimensional colvar space, representing the most probable transition path between two free energy basins associated with two functional states of a protein. We start by generating an approximate path using nonequilibrium pulling simulations [23] (path generation), followed by path optimization in the multidimensional colvar space using string method [19, 20], followed by along-the-path free energy calculations using bias-exchange US [20], and finally followed by estimating the transition rate between the two states using the estimated free energies and diffusion constants.

2.2 String Method with Swarms of Trajectories (SMwST)

The SMwST algorithm [19, 20] starts from an initial string, defined by N points/images $\{\mathbf{x}_i\}$, where i is any integer from 1 to N . Colvar ξ primarily defines the biasing potential, which is $U_i(\xi) = \frac{k}{2} (\xi - \xi_0)^2$ for M copies of image i . The initial values for the image centers are determined from the initial string: $\xi_i = \xi(\mathbf{x}_i)$. The SMwST algorithm consists of three iterative steps as follows. (Step I) Restraining: Each system is restrained for τ_R (restraining time) using the harmonic potential described above centered at the current image ξ_i . (Step II) Drifting: The simulations are continued after being released from restraints for τ_D (drifting time). (Step III) Reparameterization: The new center for each image i is determined by averaging over all observed $\xi(\mathbf{x})$ values of M systems associated with image i at time $\tau_R + \tau_D$ and using a linear interpolation algorithm to keep the image centers equidistant. By iterating over these steps, the string will converge to the zero-drift path, around which the string centers oscillate (upon convergence). The zero-drift path is an approximation of the MFEP [6, 34].

2.3 Bias Exchange Umbrella Sampling (BEUS)

Once the MFEP (parametrized by ξ) is known, $F(\xi)$ can be estimated using a generalization of US [39], termed BEUS [20]. Similar to the SMwST method, ξ is discretized and N umbrella

windows/images are defined with biasing potentials $U_i(\xi) = \frac{k}{2} (\xi - \xi_0)^2$ for $i = 1, \dots, N$. This scheme can be thought of as a 1D US along the reaction coordinate ξ with an additional restraint on the (shortest) distance from the $\xi(\xi)$ curve. Perturbed free energies $F_i = F(\xi_i)$ can be estimated (up to an additive constant) by self-consistently solving the equations [40–42]:

$$e^{-\beta F_i} = \sum_t \frac{e^{-\beta U_i(\xi^t)}}{\sum_j T_j e^{-\beta(U_j(\xi^t) - F_j)}} \quad (7)$$

in which Σ_t sums over all collected samples (irrespective of which replica or image they belong to) and T_j is the number of samples collected for image j . With appropriate reweighting, PMF can be reconstructed in any arbitrary collective variable space, given sufficient sampling in that space. w^t , the unnormalized weight of configuration $\mathbf{x}^t$ can be estimated via [41]:

$$w^t = \left(\sum_i T_i e^{-\beta(U_i(\xi^t) - F_i)} \right)^{-1} \quad (8)$$

in which $\{F_i\}$ are estimated via Eq. (7). Alternatively [41], one may estimate $\{w^t\}$ and

$\{F_i\}$ by iteratively solving Eq. (8) and:

$$e^{-\beta F_i} = \sum_t w^t e^{-\beta U_i(\xi^t)} \quad (9)$$

The PMF in terms of $\boldsymbol{\eta}(\mathbf{x})$, an arbitrary collective variable, is estimated (up to an additive constant) as:

$$G(\boldsymbol{\eta}_0) = -\beta^{-1} \log \left(\sum_t w^t K(\boldsymbol{\eta}(\mathbf{x}^t) - \boldsymbol{\eta}_0) \right) \quad (10)$$

in which K is a kernel function. The above estimator is not accurate if the sampling in $\boldsymbol{\eta}(\mathbf{x})$ is not converged which is the case if $\boldsymbol{\eta}(\mathbf{x})$ is associated with slow dynamics and is not strongly correlated with ξ . For the special case of $\boldsymbol{\eta} = \xi$, the perturbed free energies $\{F_i\}$ can be used directly to estimate the PMF only within the stiff-spring approximation.

Finally, for averaging an arbitrary quantity $A(\mathbf{x})$ along the pathway $\xi(\xi)$, one may use the weighted average $\bar{A}(t) = \sum_t w^t A(\mathbf{x}^t) \delta(\xi^t - \xi(\xi))$. However the unweighted estimator $\bar{A}_i = \langle A(\mathbf{x}^t) \rangle_i$ is more efficient. $\bar{\sigma}_i^2 = \frac{\langle A^2(\mathbf{x}^t) - \bar{A}^2 \rangle_i}{g}$ provides an estimate for the variance, given $g = 1 + 2 \frac{\tau_{ac}^A}{\tau_{lag}}$ is the statistical inefficiency in which τ_{ac}^A is the autocorrelation time associated with quantity A and τ_{lag} is the lag time between the data points used in the analysis [23].

2.4 Transition Rate Estimation

Discretizing Relation (5) results in [43]:

$$\mathbf{P}(\delta t) = (\mathbf{I} + \mathbf{R}\delta t)\mathbf{P}(0), \quad (11)$$

where $\mathbf{P}(t)$ is a vector with elements $P_i = p(\xi, t | \xi_0, 0)$, δt is a small-time step, and $\mathbf{R}$ is a tridiagonal matrix with elements $R_{i \pm 1} = -R_{i+1} - R_{i-1}$, and:

$$R_{i \pm 1} = \delta \xi^2 D(\xi_{i \pm \frac{1}{2}}) \exp(-\beta(G(\xi_i) - G(\xi_{i \pm \frac{1}{2}}))), \quad (12)$$

where $\delta \xi = \xi_{i+1} - \xi_i$ for any i . More generally, for any lag time Δt and any time t , we have

$$\mathbf{P}(t + \Delta t) = \exp(\mathbf{R}\Delta t)\mathbf{P}(t), \quad (13)$$

which implies that the likelihood of finding a system at bin j at time $t + \Delta t$, given it was at bin i at time t , is proportional to $\exp(\mathbf{R}\Delta t)_{ij}$. Therefore, assuming neither $G(\xi)$ nor $D(\xi)$ is known, one may find both, as in Ref. [43], by maximizing the likelihood $L = \Pi_\alpha (\exp(\mathbf{R}\Delta t))_{i_\alpha j_\alpha}$ (Π_α runs over all observations of trajectories starting at the bin i_α at a given time t and being found at the bin j_α at time $t + \delta t$). Assuming $G(\xi)$ is known, one may find $D(\xi)$ using a similar maximum likelihood approach [44]. For any given $D(\xi)$, $\mathbf{R}$ can be evaluated, resulting in the log-likelihood,

$$l = \sum_\alpha \log((\exp(\mathbf{R}\Delta t))_{i_\alpha j_\alpha}), \quad (14)$$

which can be maximized using a Metropolis Monte Carlo algorithm. We first estimate the factors $\exp(-\beta(G(\xi_i) - G(\xi_{i \pm 1/2})))$ in $R_{i \pm 1}$, where $G(\xi_i)$ is determined for $i = 1, 2, \dots, N$ from the BEUS simulations and $G(\xi_{i+1/2})$ is estimated by interpolation. An arbitrary series $D_{i+1/2}, 1, \dots, N-1$ can be used as an initial guess for $D(\xi_{i+1/2})$. $R_{i \pm 1}$ and R_{ii} values are then calculated to estimate the log-likelihood l . For a faster convergence, one may start with the estimates of $\mathbf{R}$ associated with the $\Delta t \rightarrow 0$ limit of Relation (12) (i.e., Relation (11)) to maximize the log-likelihood in (14). It is easy to show that the following values for $R_{i \pm 1}$ maximize the log-likelihood in (14) at the $\Delta t \rightarrow 0$ limit:

$$R_{i \pm 1} = \frac{1}{\Delta t} \frac{N_{i \pm 1} + N_{i \pm 1 i}}{N_{ii} \exp(-\beta(G(\xi_{i \pm 1}) - G(\xi_i))) + N_{i \pm 1 i \pm 1}}, \quad (15)$$

in which N_{ij} is the number of observed jumps from bin i to j with lag time Δt . Diagonal values of $\mathbf{R}$ can be also estimated using $R_{ii} = -R_{i+1} - R_{i-1}$, while the other elements are zero. For an arbitrary lag time Δt , the log-likelihood in Relation (14) can be evaluated using the values of $\mathbf{N}$ matrix as

$$l = \sum_\alpha N_{ij} \log((\exp(\mathbf{R}\Delta t))_{ij}). \quad (16)$$

Starting from $\Delta t \rightarrow 0$ limit of $\mathbf{R}$, one can use a Metropolis Monte Carlo algorithm to maximize the log-likelihood l in Relation (16). $D_{i+1/2}$ can be estimated using

$$D_{i+1/2} = \delta \xi^2 R_{i \pm 1} \exp(\beta(G(\xi_i) - G(\xi_{i \pm 1/2}))). \quad (17)$$

$D(\xi_i)$ can be estimated by interpolation $(D_{i+1/2} + D_{i-1/2})/2$. Finally, the MPFT ($\bar{\tau}_{FP}$) can be estimated numerically using

$$\bar{\tau}_{FP} = \sum_{i=1}^N \frac{\sum_{j=1}^i \exp(-\beta G(\xi_j))}{D(\xi_i) \exp(-\beta G(\xi_i))}. \quad (18)$$

3 Methods

3.1 Initial Preparation

1. Begin by preparing a membrane-embedded, water-solvated model of protein using one of its available structures. The suggested protocol here may use information from multiple structures in the next steps, but only one initial model needs to be prepared for all MD simulations (*see* **Notes 1** and **2**).
2. Before employing any biased or nonequilibrium simulations, it is important to run an equilibrium, unbiased simulation of the protein as in a conventional MD simulation. The next steps of the protocol will suffer particularly in terms of convergence if they are initiated from unequilibrated structures.
3. The length of the initial equilibration simulation can vary on how quickly a stable conformation can be reached. This is typically examined by monitoring the RMSD of the protein (*see* **Note 3**).
4. The last snapshot of the equilibrium MD simulation can be used as the initial conformation for the pulling simulations. If longer simulations have been performed, multiple snapshots may be used to examine the reproducibility as long as the selected structures resides in the equilibrated region.

3.2 Path Generation: Nonequilibrium Pulling Simulations

1. The choice of colvars should be specific to the protein of interest. A set of colvars used successfully to induce the transition of interest in one protein may not be applicable to another protein. Some common examples include the RMSD with respect to a target structure (as in targeted MD) and the distance between the mass centers of two specific molecules or molecular domains (as in SMD). The orientation based colvars have particularly proven to be very effective in describing the orientations of transmembrane helices or helical bundles of transmembrane proteins and are highly recommended as an alternative to RMSD and distance (*see* **Note 4**).

2. When determining what colvar to use, it is important to be familiar with the proposed conformational transition of the protein, in particular the target state. It is always useful to have a model of the target state even if the model is not complete or not accurate. The target model will not be used to run an actual MD simulation, so it can typically only contain the C_α atoms of the protein or the domains that will be steered (e.g., transmembrane helices).
3. Once a target model is available, one may run a targeted MD simulation with the target model to generate a transition path. The targeted MD based transition pathways are not often reliable since they typically generate pathways that are not close to the MFEP. However, they could provide a reference to compare other protocols that are based on other colvars. Multistep targeted MD simulations also provide an alternative method for generating initial pathways, if intermediate target models are available (*see Note 5*).
4. The number of colvars used is completely dependent on how simple or complex the transition pathway may be. Multiple colvars may be used in a single nonequilibrium pulling protocol; however, a specific schedule needs to be provided for changing the center of bias for each colvar. In other words, the system is steered along a specific path in the colvar space, if multiple colvars are used.
5. The choice of force constant is also dependent on both the colvar type—the unit of force constant is that of energy divided by the square of the colvar unit—and the particular transition of interest—the barriers that need to be crossed as the system is driven along the predetermined pathway is particularly important. The force constant should be large enough to induce the conformational transition of interest. If the force constant is too large, however, the simulation could become unstable, and the molecular system could undergo deformation or distortion. One may need to start with an educated guess and perform a short simulation to determine whether or not the colvars change according to their schedule. If not, the force constant can be increased. If there is very little deviation from the schedule, the force constant can be lowered to allow some deviation without introducing a delay in the schedule that increases significantly over time (*see Note 6*).
6. The simulation time is also dependent on the choice of the colvars, the desired transition, and even the force constant chosen. It is reasonable to start with relatively short simulations (a few nanoseconds) to roughly determine the quality of the protocol and fine tune the parameters; however, the final simulation that will seed the next step of our protocol (i.e., SMwST)

should be long enough (at least 100 ns) to allow for relaxation of orthogonal degrees of freedom not involved in the colvars used at least to some extent.

7. Since pulling simulations are relatively inexpensive, it is advantageous to repeat and try many different protocols to identify one or more that may lead to a reasonable transition pathway for the given protein of interest (*see* Fig. 1).
8. A reasonable protocol must satisfy the following four criteria:
 - (a) The desired transition must occur (*see* **Note 7**). To monitor this, one may use various measures depending on the particular transition of interest. For instance, for the activation of a channel, one may monitor the number of water molecules within the transmembrane region of protein (*see* Fig. 1) or measure the pore radius using programs such as HOLE [45].
 - (b) The protocol must not introduce undesired structural distortions such as major secondary structural changes (unless it is part of the transition mechanism).
 - (c) The protocol should not require large amounts of work (e.g., over 500 kcal/mol). Large work indicates that the generated transition pathway is too far from equilibrium and it may not easily relax to a converged pathway in the next step (i.e., SMwST).
 - (d) Finally, it is important to follow up the pulling simulations with an equilibrium simulation to release the system of all restraints and to ensure that the system will maintain a relatively stable conformation following the transition. This is to be done by slowly releasing all restraints and then allowing the system to converge toward a new equilibrium state. A protocol may satisfy all three criteria above but the final conformation may not maintain the desired functional state (e.g., the channel may not stay open). In this case, the protocol is not considered a successful protocol (*see* **Note 8**).

3.3 Path Optimization: String Method with Swarms of Trajectories

1. After performing the pulling simulations and the post equilibrium simulations, an initial set of conformations along the generated transition pathway can be extracted from the non-equilibrium trajectories to initiate the SMwST algorithm (*see* **Note 9**).
2. The number of conformations extracted as snapshots from the trajectory files will be the number of images used in the algorithm. The number of images to be used in the algorithm may depend on the complexity of the pathway that is to be refined (*see* **Note 10**).

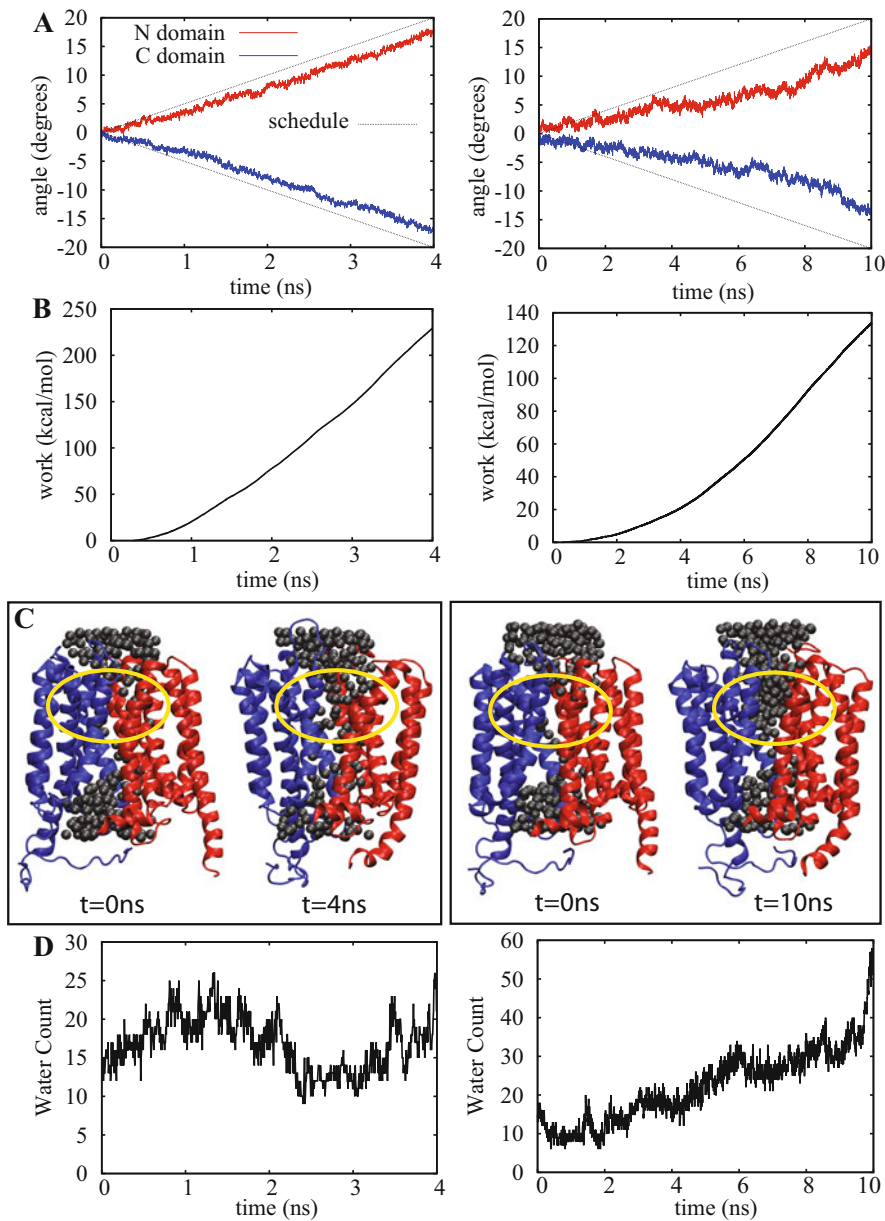


Fig. 1 Comparing two nonequilibrium pulling protocols to induce an IF→OF transition in membrane transporter GlpT [20]. Both protocols induce rotational changes on the N- and C-bundle domains of GlpT using spin angles (left) or orientations (right) of the two domains. Both protocols use a force constant of 3 kcal/mol deg^2 . (a) Time series of spin (left) and orientation (right) angles. The black line shows the schedule of the time-dependent center of harmonic potentials. (b) Time series of nonequilibrium work. (c) Snapshots of protein at the beginning and end of simulations. The water molecules within the transmembrane region are shown. (d) The number of water molecules in the periplasmic side of the transmembrane region as a function of time

3. SMwST may be performed as a series of simulations, as originally implemented or it may be run in parallel as a single job as currently implemented in NAMD (using a TCL script) and Amber (using the NFE suite). In the parallel version, each image is represented by a number of independent copies that are first restrained to stay around the current image center (the restraining step) and then released (drifting stage). The total number of replicas is determined by the number of copies of each image multiplied by the number of images (*see Note 11*).
4. The collective variables to be used in the string method simulations may or may not be similar to those used in the pulling simulations. Obviously, there needs to be more than one colvar to have a meaningful path optimization. There is no particular limitation on the number of colvars used as long as the colvar space represents a smooth space (*see Note 12*).
5. The force constants to be used for the restraining step may be at least as large as those used in previous pulling simulations if the same colvars are used. However, it is recommended to use larger force constants at this stage to ensure that the restrained conformations reach and stay around the desired image center during short restraining simulations. Distortion is unlikely in these short simulations and thus larger force constants can be employed (*see Note 13*).
6. The number of iterations to be employed in the string method simulations will be dependent on the initial pathway generated. The closer to the MFEP the initial pathway is, the faster the SMwST simulations converge. It is thus important to ensure a reasonable pulling protocol is used to generate the initial pathway before employing computationally costly SMwST simulations.
7. The convergence of the SMwST can be monitored using measures such as string RMSD from a reference structure such as initial string or final string. The string RMSD between two strings (of N images) is defined as the root mean square distance of the individual colvars. For instance, if n colvars are used, the string RMSD between strings i and j would be $\sqrt{\frac{1}{nN} \sum_{k=1}^N \sum_{l=1}^n d(\xi_{k,l,i}, \xi_{k,l,j})^2}$, where $d(a, b)$ is the distance between colvars a and b , and $\xi_{k,l,i}$ is the l th colvar of the k 'th image of string i (*see Fig. 2*).

3.4 Free Energy Calculations: Bias-Exchange Umbrella Sampling

1. The converged SMwST string provides an approximation for the MFEP. To estimate the free energy along this pathway, the US simulations can be carried out using the converged SMwST image centers as the center of US windows with the same colvars used for the SMwST simulations and the last snapshots of SMwST simulations as the initial conformations of the US simulations.

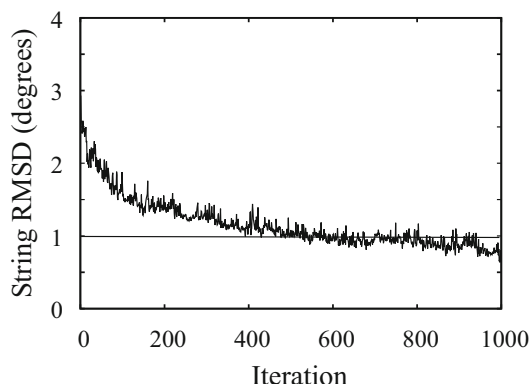


Fig. 2 Monitoring the convergence of the SMwST algorithm. String RMSD at each iteration with respect to the last string from SMwST simulations of IF→OF GlpT transition using 12 transmembrane orientations. Comparing the horizontal line with measured RMSD string shows that after about 400 iterations, the SMwST does not significantly change

2. The BEUS scheme is recommended over the conventional US scheme since it allows for the diffusion of the individual replicas along the pathway through the exchange procedure of the BEUS scheme. The faster and more reliable convergence is expected as a result.
3. Rather than using one copy per image/window, one may use multiple copies of each image in the BEUS scheme. This is particularly convenient since the SMwST algorithm already provides us with multiple copies of each image if the fully parallel version is used (*see Note 14*).
4. BEUS uses a similar restraining bias as that used by the SMwST simulations, but unlike the restraining used in SMwST, the force constant cannot be too large. The restraining is used to keep the protein from drifting away from the current image centers as in the SMwST method; however, an additional feature of BEUS is that the biasing potentials are also used to determine the exchange criteria. Therefore, the force constant cannot be too high, otherwise, there will not be adequate exchange between the neighboring windows. The force constant can be selected such that all replicas have 10 to 50% of successful exchange at every attempt (*see Note 15*).
5. By employing a nonparametric reweighting scheme as discussed above in Subheading 2, the perturbed free energies can be estimated to reconstruct the free energy profile (*see Fig. 3*) and various other ensemble averages along the MFEP. The perturbed free energies are only a good estimate of the PMF along the path, if the force constant is large enough (*see Note 16*).

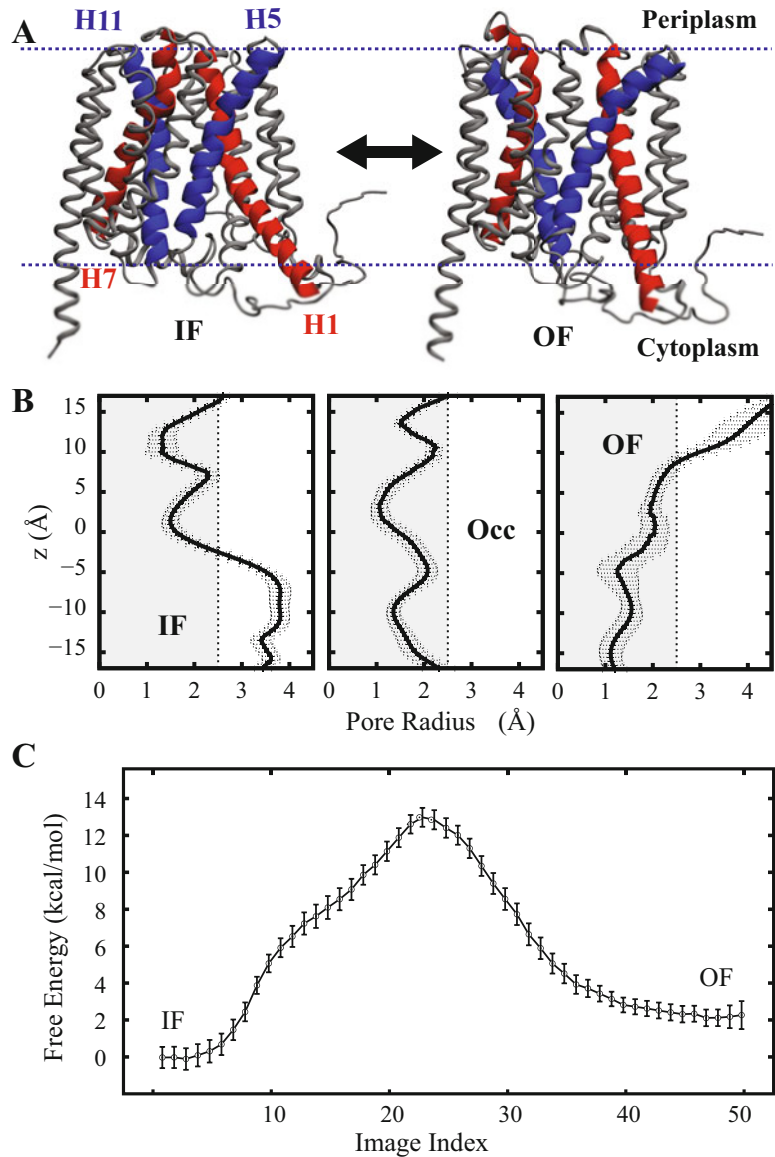


Fig. 3 BEUS simulation results of IF→OF GlpT transition. (a) Snapshots of the first ($i = 1$) and last ($i = 50$) images, representing the IF and OF state, respectively. (b) The pore radius along the pore based on the snapshots of the first (IF) and last (OF) as well as an intermediate (Occ) image. The latter represents an occluded state. (c) The perturbed free energies estimated from the BEUS simulations [20]

6. An alternative method to estimate the PMF is to use the weighting factors (w^i 's) to construct a PMF along a given 1D collective variable such as the first principal component obtained from the principal component analysis of C_α atoms of proteins. The latter method has an advantage over the former method in that it does not require the large force constant condition.

3.5 Transition Rate Estimation

1. Assuming that we have identified the MFEP relatively accurately in a relevant colvar space such that the effect dynamics of the system can be assumed to be diffusive along the identified MFEP, we can use a 1D diffusion model to describe the effective kinetics of the system as described in Subheading 2 above.
2. To determine the transition rate, the free energy profile (or the PMF) along the MFEP is needed, which is already obtained from BEUS simulations. In addition, the position-dependent diffusion constant along the MFEP is also needed to accurately estimate the transition rate using Relation (6). The diffusion constant estimation can be carried out by estimating inter-image transition rates measured using unbiased simulations along the MFEP.
3. Multiple copies of conformations per image/window will be extracted from BEUS trajectories to initiate these unbiased simulations. If multiple copies of images were used in BEUS simulations as recommended, one may simply use the last snapshots of all BEUS trajectories as the starting point for these unbiased simulations.
4. Although these are unbiased simulations, it is recommended to collect the colvar values during the simulations to monitor jumps between the images. These are the same colvars used in the BEUS simulations and the colvar values can be used to first assign an image to each sampled conformation at any given time and then count the number of transitions between different images. One may then build an empirical transition matrix based on these counts. The empirical transition matrix will be dependent on the lag time used to collect the data (or to count the jumps). It is recommended to use multiple lag times to determine the behavior of the estimated transition rates as a function of the lag time (*see Note 17*).
5. Once an empirical transition rate is constructed for a given lag time, a Metropolis Monte Carlo algorithm will be used as described in Subheading 2, to estimate diffusion constants $D(\xi_i)$ from the empirical transition matrix and the BEUS estimate free energies $G(\xi_i)$ and eventually estimate the MFPT and the overall transition rate (*see Fig. 4 and Note 18*).

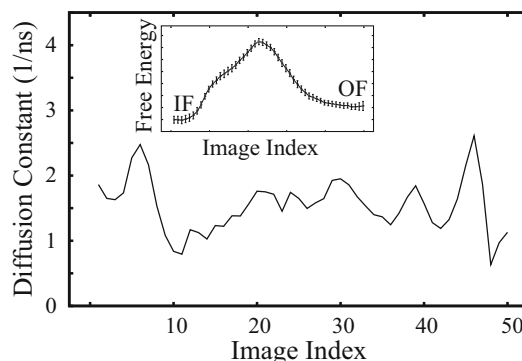


Fig. 4 The diffusion constant as a function of image index estimated based on GlpT BEUS simulations (for free energies) and follow-up equilibrium simulations using a lag time of 0.5 ns, which was determined to be the optimum lag time. The MFPT estimated based on these calculations for the IF→OF GlpT transition is approximately 6 s

4 Notes

Initial Preparation:

1. In the case of membrane transporters, a typical crystal structure, cryo-EM structure, or homology model will be in an inward-facing (IF), outward-facing (OF), or occluded (Occ) state. The choice of the starting point is often based on the quality and reliability of the structure. For example, a homology model is less reliable than a crystal structure; a mutant or engineered crystal structure is less reliable than a wild-type one.
2. The choice of lipid composition, salt type and concentration, protonation states of titratable amino acids, force field parameters, temperature, box size, and other MD simulation parameters is determined at this stage. Care must be taken in making these choices as any changes of these parameters in the next steps may complicate the interpretability of the results.
3. Although it is common to monitor the RMSD of the protein with respect to initial frame (or preferably initial model that usually represents the known (e.g., crystal) structure), it is recommended to also monitor the RMSD with respect to the last frame. If the RMSD with respect to the last frame stays small for a long enough period, it is a much stronger evidence for the stability of the final conformation than if the RMSD with respect to the initial frame stays constant.

Nonequilibrium Pulling:

4. The orientation based colvars as implemented in NAMD, LAMPS, and Amber are based on the orientation quaternion formalism that measure the rotation of a semirigid-body

molecule or molecular domain with respect to the same molecule or molecular domain in a reference conformation. The colvar may fully describe all three rotational degrees of freedom (that is in the form of a unit orientation quaternion) or only describe the angle of rotation (orientation angle) or the angle of rotation with respect to a specific axis (tilt angle or spin angle). A 1D orientation based colvar such as an orientation, spin, or tilt angle is typically easier to use particularly at this stage of protocol. In SMwST/BEUS simulations, the orientation colvar that contains all three degrees of freedom is more appropriate to use.

5. Another method to employ the pulling simulations is the definition of individual initial, final, and hypothetical intermediate state along the transition pathway. This will be carried out in many individual targeted simulations until the final state has been reached. Targeting of the intermediates allows to target local minima which would otherwise not be sampled by employing a single target. The intermediate may be based on available crystal structures (e.g., the Occ state of a transporter) or they may be generated using other modeling techniques (e.g., coarse-graining or isotropic network models).
6. Although a short simulation could be quite informative with regards to the choice of the force constant, one needs to also examine whether the force constant is large enough for the other stages of the transition. Due to the presence of many metastable states and barriers, it is quite possible for the system to get trapped in one of the metastable states along the pathway and may never reach the final desired state. Increasing the force constant may help overcoming such issues; however, often-times, changing the choice of the colvars or their schedule is more effective.
7. The first step to monitor whether the desired transition has occurred is to monitor the colvars and how they follow their imposed schedule. However, it is important to note that the final targeted value may not be always reached, which admittedly, such is the case for TMD simulations. As long as the final conformation has the desired functional features (e.g., it is an open channel), the desired transition is considered to have been induced.
8. The postpulling equilibrium MD simulations must be long enough to allow for the relaxation of the system. If the system relaxes to a conformation that is at the desired functional state but significantly differs from the initial target used for pulling simulations, one may use the equilibrated conformation as a new target to generate an alternative transition path.

Path Optimization:

9. The extracted snapshots maybe extracted from the nonequilibrium pulling simulations in an equitemporal manner. In addition, one may include a few snapshots from the post-pulling equilibrium simulation trajectory as well. This often helps speeding up the convergence of the SMwST simulations.
10. If the number of images is small, the reparameterization step may introduce large changes to the image centers that cannot be easily achieved during the restraining step. Using too many images, on the other hand, introduces unnecessary curvatures. The ideal number of images is highly dependent on the nature and number of the colvars. A typical number of images would be between 50 and 200.
11. Depending on the computational resources available, one may use a hybrid version of serial and parallel SMwST. This is again implemented in both NAMD (smwst script) and Amber (NFE suite). In this version, fewer number of copies (or only one copy) per image is used, but each copy generates more than one sample before averaging the drift and updating the image centers. One may choose to use one copy per image and 20 samples per copy, or 20 copies per image and only one sample per copy, or anything in between, for example, 5 copies per image and 4 samples per copy. The recommended number of copies $\times$ number of samples is at least 20.
12. One may even use the atomic coordinate space (of select atoms, e.g., C_α 's). The orientation based colvars, however, provide a smoother space and are expected to provide a faster and more reliable convergence.
13. It is important to monitor the progress of the SMwST simulations, particularly in the first few iterations, to ensure the appropriateness of the parameters chosen. For instance, one should check whether all copies of each image end up around the desired image center during the restraining stage before they are released. For instance, one may plot both the image centers and the actual colvar values of the sampled conformations in different 2D colvar spaces to ensure the sampled colvar values are closely distributed around each image center at the end of each restraining stage.

Free Energy Calculations:

14. Using multiple copies of an image/window in a BEUS simulation, given the availability of supercomputers that allow executing large jobs, not only allows for faster sampling but more importantly also allows for a better uncertainty estimation. The multiple copies of images are effectively independent simulations and provide uncorrelated data points for unbiased uncertainty estimation.

15. Having adequate sampling overlap between the neighboring windows is important in US simulations. Similarly, having adequate exchange rate is important in BEUS simulations. Care must be taken in choosing the number of images and the force constants to ensure each image is close enough to its neighboring image, allowing some exchange between the neighboring replicas. This can be remedied by either lowering the force constant between neighboring images that may be experiencing lower exchange rates or by adding more images allowing for more evenly spaced neighboring images and promoting a much greater exchange rate between neighboring images. The above parameters can be optimized iteratively using short runs with the goal of achieving similar rates of exchange between neighboring replicas.
16. Within the stiff-spring approximation, that is, the large force constant assumption, the perturbed free energy and the PMF are equal. However, the first-order correction of the approximation can be estimated by *some posteriori* from the estimated perturbed free energy $F(\xi)$ as: $\frac{1}{2\beta k} \left(\beta \left(\frac{d}{d\xi} F(\xi) \right)^2 - \frac{d^2}{d\xi^2} F(\xi) \right)$.

Transition Rate Estimation:

17. If the lag time is too short, the data will be too correlated and the diffusion constants and transition rates will be overestimated. If the lag time is too long, given the limited simulation time, few transitions will be observed and the estimated diffusion constants and transition rates will be associated with large errors. Using multiple lag times allows for identifying the optimum lag time.
18. The simulation time is again system dependent; however, since the estimated free energies already provide relative inter-image transition rates $\frac{R_{i \rightarrow i+1}}{R_{i+1 \rightarrow i}} = \exp(-\beta(G(\xi_i) - G(\xi_{i+1})))$, the only information needed to fully construct the transition rate matrix is the downhill interimage transition rates. Without the BEUS free energy estimates, both uphill and downhill inter-image transition rates need to be estimated, which requires considerably more time.

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References

- Hansson T, Oostenbrink C, van Gunsteren WF (2002) Molecular dynamics simulations. *Curr Opin Struct Biol* 12:190–196
- Karplus M, McCammon JA (2002) Molecular dynamics simulations of biomolecules. *Nat Struct Biol* 265:654–652
- Karplus M, Kuriyan J (2005) Molecular dynamics and protein function. *Proc Natl Acad Sci U S A* 102:6679–6685
- Maragliano L, Fischer A, Vanden-Eijnden E et al (2006) String method in collective variables: minimum free energy paths and isocommittor surfaces. *J Chem Phys* 125:24106
- E W, Vanden-Eijnden E (2010) Transition-path theory and path-finding algorithms for the study of rare events. *Annu Rev Phys Chem* 61:391
- Johnson ME, Hummer G (2012) Characterization of a dynamic string method for the construction of transition pathways in molecular reactions. *J Phys Chem B* 116:8573–8583
- Torrie GM, Valleau JP (1977) Nonphysical sampling distributions in Monte Carlo free-energy estimation: umbrella sampling. *J Comp Phys* 23:187–199
- Northrup SH, Pear MR, Lee CY et al (1982) Dynamical theory of activated processes in globular proteins. *Proc Natl Acad Sci U S A* 79:4035–4039
- Schlitter J, Engels M, Krüger P et al (1993) Targeted molecular dynamics simulation of conformational change—application to the T-R transition in insulin. *Mol Simulation* 10:291–308
- Huber T, Torda AE, van Gunsteren WF (1994) Local elevation: a method for improving the searching properties of molecular dynamics simulation. *J Comput Aided Mol Des* 8:695
- Izrailev S, Stepaniants S, Balsera M et al (1997) Molecular dynamics study of unbinding of the avidin-biotin complex. *Biophys J* 72:1568–1581
- Sugita Y, Okamoto Y (1999) Replica-exchange molecular dynamics method for protein folding. *Chem Phys Lett* 314:141–151
- Laio A, Parrinello M (2002) Escaping free energy minima. *Proc Natl Acad Sci U S A* 99:12562–12566
- Darve E, Rodríguez-Gómez D, Pohorille A (2008) Adaptive biasing force method for scalar and vector free energy calculations. *J Chem Phys* 128:144120
- Abrams CF, Vanden-Eijnden E (2010) Large-scale conformational sampling of proteins using temperature-accelerated molecular dynamics. *Proc Natl Acad Sci U S A* 107:4961–4966
- Templeton C, Chen SH, Fathizadeh A et al (2017) Rock climbing: a local-global algorithm to compute minimum energy and minimum free energy pathways. *J Chem Phys* 147:152718
- Chong LT, Saglam AS, Zuckerman DM (2017) Path-sampling strategies for simulating rare events in biomolecular systems. *Curr Opin Struct Biol* 43:88–94
- Laio A, Panagiotopoulos AZ, Zuckerman DM (2018) Preface: special topic on enhanced sampling for molecular systems. *J Chem Phys* 149:072001
- Pan AC, Sezer D, Roux B (2008) Finding transition pathways using the string method with swarms of trajectories. *J Phys Chem B* 112:3432–3440
- Moradi M, Enkavi G, Tajkhorshid E (2015) Atomic-level characterization of transport cycle thermodynamics in the glycerol-3-phosphate:phosphate antiporter. *Nat Commun* 6:8393
- Hummer G, Kevrekidis IG (2003) Coarse molecular dynamics of a peptide fragment: free energy, kinetics, and long-time dynamics computations. *J Chem Phys* 118:10762
- Moradi M, Tajkhorshid E (2013) Driven metadynamics: reconstructing equilibrium free energies from driven adaptive-bias simulations. *J Phys Chem Lett* 4:1882–1887
- Moradi M, Tajkhorshid E (2014) Computational recipe for efficient description of large-scale conformational changes in biomolecular systems. *J Chem Theory Comp* 10:2866–2880

24. Bonomi M, Branduardi D, Bussi G et al (2009) PLUMED: a portable plugin for free energy calculations with molecular dynamics. *Comput Phys Commun* 180:1961
25. Babin V, Karpusenko V, Moradi M et al (2009) Adaptively biased molecular dynamics: an umbrella sampling method with a time-dependent potential. *Int J Quantum Chem* 109:3666–3678
26. Fiorin G, Klein ML, Hénin J (2013) Using collective variables to drive molecular dynamics simulations. *Mol Phys* 111:3345
27. Sidky H, Colón YJ, Helfferich J et al (2018) Ssages: software suite for advanced general ensemble simulations. *J Chem Phys* 148:044104
28. Branduardi D, Gervasio FL, Parrinello M (2007) From a to b in free energy space. *J Chem Phys* 126:054103
29. Berteotti A, Cavalli A, Branduardi D et al (2009) Protein conformational transitions: the closure mechanism of a kinase explored by atomistic simulations. *J Am Chem Soc* 131:244–250
30. Moradi M, Tajkhorshid E (2013) Mechanistic picture for conformational transition of a membrane transporter at atomic resolution. *Proc Natl Acad Sci U S A* 110:18916–18921
31. Legoll F, Lelièvre T (2010) Effective dynamics using conditional expectations. *Nonlinearity* 23:2131
32. Czerminski R, Elber R (1989) Reaction path study of conformational transitions and helix formation in a tetrapeptide. *Proc Natl Acad Sci U S A* 86:6963–6967
33. Mills G, Jónsson H (1994) Quantum and thermal effects in dissociative adsorption: evaluation of free energy barriers in multidimensional quantum systems. *Phys Rev Lett* 72:1124–1127X
34. Fakharzadeh A, Moradi M (2016) Effective Riemannian diffusion model for conformational dynamics of biomolecular systems. *J Phys Chem Lett* 7:4980–4987
35. Jarzynski C (1997) Nonequilibrium equality for free energy differences. *Phys Rev Lett* 78:2690–2693
36. Crooks GE (2000) Path-ensemble averages in systems driven far from equilibrium. *Phys Rev E* 61:2361–2366
37. Hummer G, Szabo A (2001) Free energy reconstruction from nonequilibrium single-molecule pulling experiments. *Proc Natl Acad Sci U S A* 98:3658–3661
38. Lifson S, Jackson JL (1962) On the self-diffusion of ions in a polyelectrolyte solution. *J Chem Phys* 36:2410–2414
39. Torrie GM, Valleau JP (1977) Nonphysical sampling distributions in Monte Carlo free-energy estimation: umbrella sampling. *J Comput Phys* 23:187
40. Kumar S, Bouzida D, Swendsen RH et al (1992) The weighted histogram analysis method for free-energy calculations on biomolecules. I. the method. *J Comp Chem* 13:1011–1021
41. Bartels C (2000) Analyzing biased Monte Carlo and molecular dynamics simulations. *Chem Phys Lett* 331:446
42. Shirts MR, Chodera JD (2008) Statistically optimal analysis of samples from multiple equilibrium states. *J Chem Phys* 129:124105
43. Hummer G (2005) Position-dependent diffusion coefficients and free energies from bayesian analysis of equilibrium and replica molecular dynamics simulations. *New J Phys* 7:34
44. Singharoy A, Chipot C, Moradi M et al (2017) Chemomechanical coupling in hexameric protein-protein interfaces harness energy within V-type ATPases. *J Am Chem Soc* 139:293–310
45. Smart OS, Neduvelil JG, Wang X et al (1996) HOLE: a program for the analysis of the pore dimensions of ion channel structural models. *J Mol Graph* 14:354–360



Chapter 17

Concepts, Practices, and Interactive Tutorial for Allosteric Network Analysis of Molecular Dynamics Simulations

Wesley M. Botello-Smith and Yun Lyna Luo

Abstract

Over the past decade, concepts of network theory in combination with dynamical information from conformational ensembles have been widely applied to gain insights in understanding allosteric regulation in biomolecules. In this chapter, we introduce the basic theories and protocols used in protein dynamics network analysis through a series of interactive python Jupyter notebook scripts. While various network analysis methods exist in the literature, here we focus on the two popular methods based on correlated atomic motions and pairwise interaction energies. While the tutorial is based on a small prototypic protein, the workflow and protocol introduced here are optimized to handle large membrane proteins.

Key words Molecular Dynamics, Allostery, Network Analysis, Pairwise interaction energy, Pairwise correlation, Current flow betweenness

1 Introduction

Proteins are molecular machines that are capable of sensing environmental change and undergo orchestrated motions to propagate biological signals. The result of an allosteric mechanism is a change in the free energy, which results from changes in enthalpy and/or entropy. While the overall concept of allostery is generally well accepted in biology, the application of allostery has its own dilemma. For instance, the outcome of an allosteric perturbation may be estimated from the shift in the probabilities, that is, relative free energy measurements, to map out the signaling propagation pathway, a large number of mutagenesis studies coupled with functional analysis will be required, which often becomes infeasible for large membrane proteins. Therefore, computational approaches, especially atomistic simulations, are indispensable tools in investigating allosteric mechanisms.

The methodology for applying graph theory and network analysis to structural modeling and simulation of proteins has been

developed relatively recently (within the last two decades). It is sometimes difficult to choose among a wide range of techniques that have been developed. Indeed, even which interaction property should be used to generate a network topology is still a topic of some debate [1–5]. In this chapter, we aim to delineate the basic theories and protocols used in protein dynamics network analysis by providing a flexible and interactive notebook style script (https://github.com/LynaLuo-Lab/network_analysis_scripts). While reading this book chapter, users are encouraged to download this script and execute from Google Colab on Google’s cloud servers or Jupyter Notebook on a desktop. A small prototypical allosteric enzyme is used as the tutorial example; however, the streamlined protocol is specially optimized here for handling large proteins containing thousands of amino acids. All figures (except Fig. 1) can be regenerated using the scripts under the `network_analysis_example` folder.

In the interest of brevity, we will present two common and relatively straightforward methods for generating a network topology and edge weights. The first utilizes minimum atomic distances between residue pairs to define the network topology and employs correlation/correspondence of residue motions as edge weights. The other method is based upon the mean interaction energy (e.g., over electrostatic and van der Waals nonbonded interactions), which may serve both as a network topology (via a high pass filter) and edge weights.

On a related note, there are a number of computational tools, packages, toolkits, etc. available that can be used to construct the needed interaction networks. We present here only a small subset that were used in our own previous studies and have been shown to be stable, readily accessible, and (relatively) straightforward to install and use. This is by no means an exhaustive listing, and, for a given particular choice of network generation scheme, there may be more efficient or accurate alternatives.

Finally, all of the network generation methods below require an appropriate molecular model and corresponding ensemble of structures, that is, simulation(s) of the protein being studied. The utility and quality of results from these network analysis methods are contingent upon the accuracy and appropriateness of the modeling and simulation(s) used. The details of how to generate or attain these structural models and conduct appropriate simulations are separate topics in their own right, and detailed discussion of those topics is beyond the scope of this chapter.

2 Methodological Background

2.1 Generating Interaction Networks

A network can be conceptualized as a mathematical object known as a graph, which consists of a set of points (nodes) along with a set of edges connecting pairs of those points together. Each node and edge may also have an associated weight. In general graphs may have multiple edges connecting a pair of points. Here we will only consider the simpler case where a single edge connects any given pair of points. Additionally, a network may be either directed (i.e., edges run from one particular node in a pair to the other) or undirected. Again, we will limit our discussion here and consider only the undirected case.

This model of an interaction network also lends itself well to a matrix description. We will use the two descriptions interchangeably throughout this chapter. To translate from a graph into a matrix representation, one would assign each node to a particular column/row index; for example, if a graph has N nodes, an N by N matrix would be required to describe it. So for a graph of N nodes with M edges denoted i_j (where i and j are distinct nodes in the graph) can be modeled as a matrix A such that $A_{i,j}$ is equal to the weight of the edge i_j . If the edges have no associated weights, or all weights are equivalent, $A_{i,j}$ is set to 1 if edge i_j is in the graph and 0 if not. This matrix is known as an adjacency matrix for the graph or network and is particularly useful as it is a representation of the underlying connectivity of the network (network topology). Figure 1 below illustrates these two representations for a simple toy example.

To apply this model to our investigation of allostery in proteins from a molecular modeling and simulation perspective, we may represent each amino acid in the protein as a node in a graph or network, then compute relevant interaction scores between each pair of amino acids and represent these interactions as (possibly weighted) edges in the network. These interaction scores could be

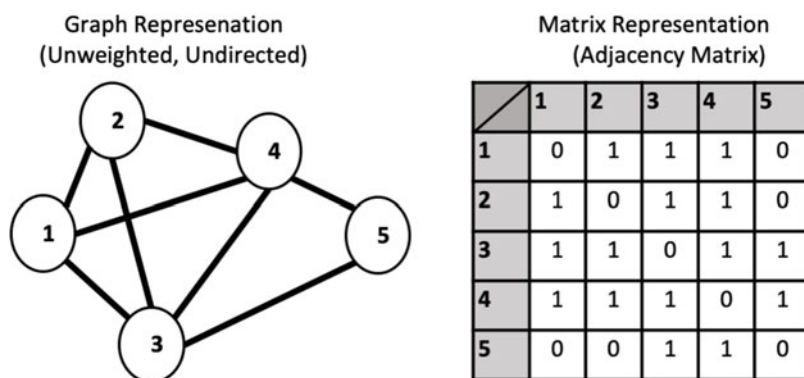


Fig. 1 Two equivalent representations of a simple undirected, unweighted, 5 node network

derived from almost any pairwise interaction, for example, distance and interaction energy are commonly used to generate the network topology, often after applying an appropriate cutoff or reweighting function.

Depending on the approach being taken, weights of the corresponding edges can be assigned using the same interaction or property that was used to generate the network topology, or they can be based upon a different property. In the case of simulations of large proteins structures, that is, proteins containing thousands of amino acids or more and simulations yielding tens of thousands of structures, it is often advantageous to compute the network topology first, based on quantities that can be computed quickly and efficiently. This can reduce the computational workload when generating edge weights based upon more computationally complex methods.

2.2 Important Graph Theory Terminology

1. **Network (Graph)**—A set of objects (nodes) along with a set of edges connecting the nodes together.
2. **Node**—an object in a graph (may also be referred to as a vertex or a point).
3. **Edge**—a connection between two nodes in a graph.
4. **Weight**—a value assigned to an edge or node in a graph, generally used in computing “path length.”
 - (a) Often the weight of a node is defined to be equal to the sum of its connected edges. This will be the case here, although it is not a requirement.
5. **Path**—a sequence of unique nodes and/or edges leading from one node to another in a network.
6. **Source Node**—the node at the beginning of a path.
7. **target (Destination) node**—the node at the end of a path,
8. **Path Length**—the sum of the weights of edges in a path.
9. **Betweenness**—a metric that defines how important a node or edge is with respect to the connectivity of a graph (or to connecting subset(s) thereof).
10. **Optimal (Shortest) Path**—the path leading between a pair of nodes which has the smallest (shortest) length over all possible paths.
11. **Suboptimal Path**—a path leading between a pair of nodes that is longer than the optimal (shortest) path.
12. **Betweenness Centrality**—one of the earliest and simplest betweenness metrics, defined as the number of times a node or edge appears over all shortest paths between pairs of nodes in a graph.

13. **Usage Frequency Betweenness**—A variation of betweenness centrality in which a specific set of paths is used, that is, the number of times a node or edge is used in a given set of paths divided by the total number of paths in that set.
 - (a) **Suboptimal Path Betweenness**—a betweenness metric defined as the usage frequency betweenness wherein the paths are chosen to be the top N paths between some subset of source and target (destination) nodes.
14. **Graph Topology**—A representation of the connectivity of a graph wherein edges are given a weight on the unit interval $[0,1]$. Often edges are only assigned either a 1 if they are present, or 0 if not. In some cases, this may be extended to include fractional weights (as when generating a “smoothed” topology).
15. **Adjacency Matrix**—A matrix representation of the connectivity (topology) of a graph. This will be a square matrix of size N (where N is the number of nodes in a graph). For each edge in the graph, the corresponding entry in the matrix will be assigned the weight/value of that edge. Note that the diagonal of the adjacency matrix is generally equal to zero (in our discussion here we do not consider graphs with self-loops, that is, where edges connecting nodes to themselves).
16. **Graph Laplacian**—Also known as the admittance matrix, Kirchhoff matrix, and discrete Laplacian. The graph Laplacian is a matrix representation of a graph. The diagonal terms are set equal to the row sums of the adjacency matrix and the off diagonal terms are set equal to the inverse of the adjacency matrix off diagonal entries. That is, $L = D - A$ where D is the diagonal matrix whose entries are the row sums of A .
17. **Current Flow Betweenness**—A betweenness metric in which an edge’s betweenness is equal to the weighted sum of its usage frequency, the “count” of its usage is weighted by the reciprocal of the path length in which the count is observed. In practice, this is not directly computed by summing over paths, but rather by using the pseudoinverse of the graph Laplacian.

2.3 Network Construction

2.3.1 Distance Based Topology Using a Collision Padding Radius

One of the simplest methods (at least relative to most other methods available) of generating a network topology is to consider the distance between each pair of residues. In this particular instance, the “distance” in question is taken to be the minimum distance between any atom in one amino acid (residue) to any atom in the other residue, for example,

$$D_{i,j} \text{ min} = (|r_{i,a} - r_{j,b}|) \quad (1)$$

where $D_{i,j}$ is the distance score for residues i and j , $r_{i,a}$ is the position of the a 'th atom of residue i , and $r_{j,b}$ is the b 'th atom of residue j . This equation is relatively trivial to apply to a single protein structure configuration (conformation), however, it is often desirable to generate a network topology as an agglomeration over an ensemble of structures, for example as one would obtain from a molecular dynamics simulation (i.e., a trajectory). This can be particularly useful if one seeks to use a computationally taxing weighting scheme to generate edge weights and therefore seeks to use the network topology to reduce the number of times such computations must be run. To that end, it is common practice to subsequently compute a mean value for each $D_{i,j}$ over the entire trajectory and retain only those pairs for which the mean falls below a particular cutoff. Alternatively, one could count the number of times the distance was observed to be below the cutoff distance and only keep the pair if this occurs for at least some specified percentage of times (e.g., an occurrence frequency cutoff).

Common choices for distance cutoff range from about 4 to 5 Å. This range corresponds roughly to the distance between heavy atoms in a hydrogen bond or salt bridge between neighboring residues. Indeed, as some papers have shown [3] this corresponds to a common peak in the distribution of minimum inter-residue (i.e., amino acid) distance between residue pairs over a wide range of proteins. On the other hand, choosing an occurrence frequency cutoff seems to be more or less arbitrary.

If one is comfortable with the somewhat generic choice of using 4.5 Å for the distance cutoff and a 75% occurrence frequency cutoff, the molecular visualization software VMD [5] comes with the Network View plugin³ which can be used to compute a network topology for a given simulation trajectory. It will also provide edge weights⁴ by computing a Pearson correlation of motion for each pair of residues' center (apparently taken as the alpha carbon for each residue). The resulting network topology and edge weights are stored in matrix representation form. The Network View package also provides a number of tools to conduct various other analyses of the resulting network and provides facilities for visualizing the results from within VMD, overlaying the network onto the three dimensional structure of the protein for a given configuration. Besides Network View plugin for generating a distance based network topology using a preset distance or occurrence frequency cutoffs, there are also more general purpose analysis tools that allow to generate network topology, such as ccpptraj and pytraj [4] (a python based wrapper of ccpptraj), MDAnalysis [6, 7], GRO-MACS [8–13], and the R-Bio3D package [14], to name a few. The second example below will show how this can be done using pytraj.

The examples below show steps for computing a distance based cutoff network for a 50-ns simulation of the imidazole glycerol phosphate synthase (IGPS) protein along with depictions of the

results. For a full description of the simulation protocol see the reference paper “Robust Determination of Protein Allosteric Signaling Pathways” [15], and its previous papers on dynamical network models [16, 17]. There are a number of software packages available that can accomplish this task; however, they are not always fully transparent or flexible in how their distance topology is generated. Here we make use of the pytraj package from AMBER [18] to perform the needed calculations and provide within an interactive notebook style format (via Jupyter Notebook). The notebook and script file should provide a flexible and transparent method. It should be noted that the online version of the code provided herein may be subject to further optimization to speed up the streamlined approach. Whenever possible, this “in-house” approach will be taken in this chapter, with the one notable exception being the computation of pairwise Generalized Born (GB) interaction energies. In that case, we make use of the MMPBSA.py [19] utility from the AmberTools [18] suite to generate the needed pairwise interaction terms.

The model system presented here (IGPS) is relatively small (~450 amino acids), and our trajectories are likewise relatively short (only 50 frames each), so iterating over all possible residue pairs is quite feasible. For larger proteins over thousands of amino acids, however, this can quickly become intractable. Since computing the minimum atom-atom inter-residue distance can be relatively slow over large trajectories, we will perform a first pass to filtering step using “padded bounding box collisions” to illustrate what can be done for larger systems. To do so, one needs to compute the minimum and maximum atomic coordinates of each residue at each frame. A “collision radius” is then added to the coordinate bound maxima and subtracted from the minima. Using these “padded bounding box” trajectories, we then count the total number of “collisions” between every residue pair (i.e., count how many times the bounding boxes of the two residues overlapped). This, as it turns out, can be completed very quickly and vectorizes quite readily, allowing for this calculation to be computed much faster than computing the pairwise minimum distances. Any residue pairs which did not exhibit at least one collision in at least one of the trajectories being considered are then discarded. Even with relatively large collision radii, this can greatly reduce the number of residue pairs to consider. After that, pairwise minimum distances of each remaining residue pair are calculated.

For a simple contact topology, we then count how many times a residue pair had a minimum distance below a specified threshold and divide by the total number of frames to get a contact frequency for each residue pair. Finally, similar to the NetworkView and WISP methods, we remove any residue pairs for which the contact frequency was below 75% to yield the contact topology. There are other options here of course; for instance one could use the contact

frequencies directly, or linearly interpolate between 1 and 0 over a frequency range¹⁶. More advanced smoothing can be applied by using a smoothing function to the pairwise distances. In either case, the “smoothed” topology can then be applied by multiplying it with the corresponding value matrix to obtain the final weighted network matrix. Figure 2 below shows the results for the construction of contact topology matrices for the apo and holo states IGPS simulations using a collision padding radius and contact distance cutoff of 8 Å, and a minimum contact frequency cutoff of 75%.

The choice of collision padding radius is somewhat arbitrary, but in general it should be equal to or larger than the contact cutoff for compute contact frequency from pairwise minimum distances. Here we chose a value of 8 Å, which corresponded to the first long range plateau in the pairwise residue distance distribution Fig. 3. The peak at about 4 Å and plateau shortly after 8 Å seem to be consistently present over many proteins. For the WISP and NetworkView packages, a value of 4.5 is typically used (which likely corresponds to maximum hydrogen bond distances). We opted for a larger value here to ensure inclusion of slightly longer range interactions.

2.3.2 Correlated Atomic Motion Weights

Once the distance based topology is completed, edge weights must be assigned. Here we will consider those based on correlation/correspondence between atomic motion of the two residues. Particularly, we will consider motion of the center of mass. While alpha carbon motion is commonly used, it may miss relevant information contained in the side chain motion. As with distance, in house python based methods are provided to calculate Pearson correlation and the generalized linear Mutual Information coefficient. While full Mutual Information will, in fact capture a wider range of interactions (specifically it can handle both linear and nonlinear interactions) it is much more computationally demanding and requires a larger sample size (number frames per trajectory) to converge, making it less feasible for large membrane proteins. Those interested are referred to the `g_correlation` utility [2].

In each case, “atomic motion” refers more specifically to the displacement of each residue’s center of mass from its mean position over all frames. It should be noted here that the trajectory should be aligned by performing a root-mean square (RMS) structural alignment to a chosen frame. Choice of alignment structure can significantly impact results, which is a known shortcoming. This is particularly true for the basic Pearson correlation, but choice of alignment pose will still affect both linear and full Mutual Information coefficients as well. The Pearson correlation method is the most basic of the three and will be discussed first (in fact, it forms an underpinning for the normalization of the other two methods).

To start we consider a pair of residues, i and j , and the corresponding mean centered center of mass displacements X_i

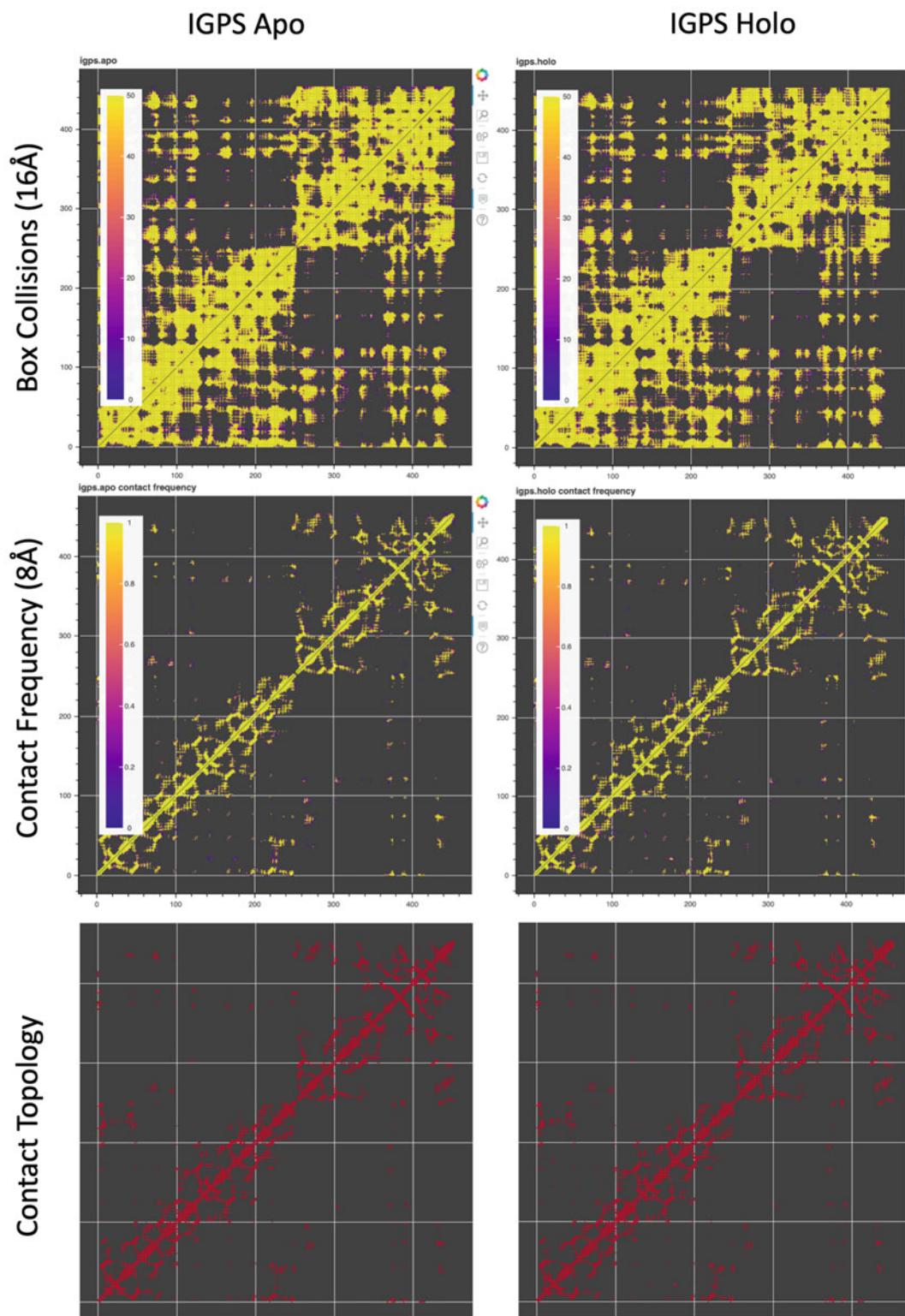


Fig. 2 Results for the construction of contact topology matrices for the apo and holo states IGPS simulations using a collision padding radius and contact distance cutoff of 8 Å, and a minimum contact frequency cutoff of 75%. (This figure was generated using `/network_analysis_examples/Distance_Topology_Generation.ipynb`)

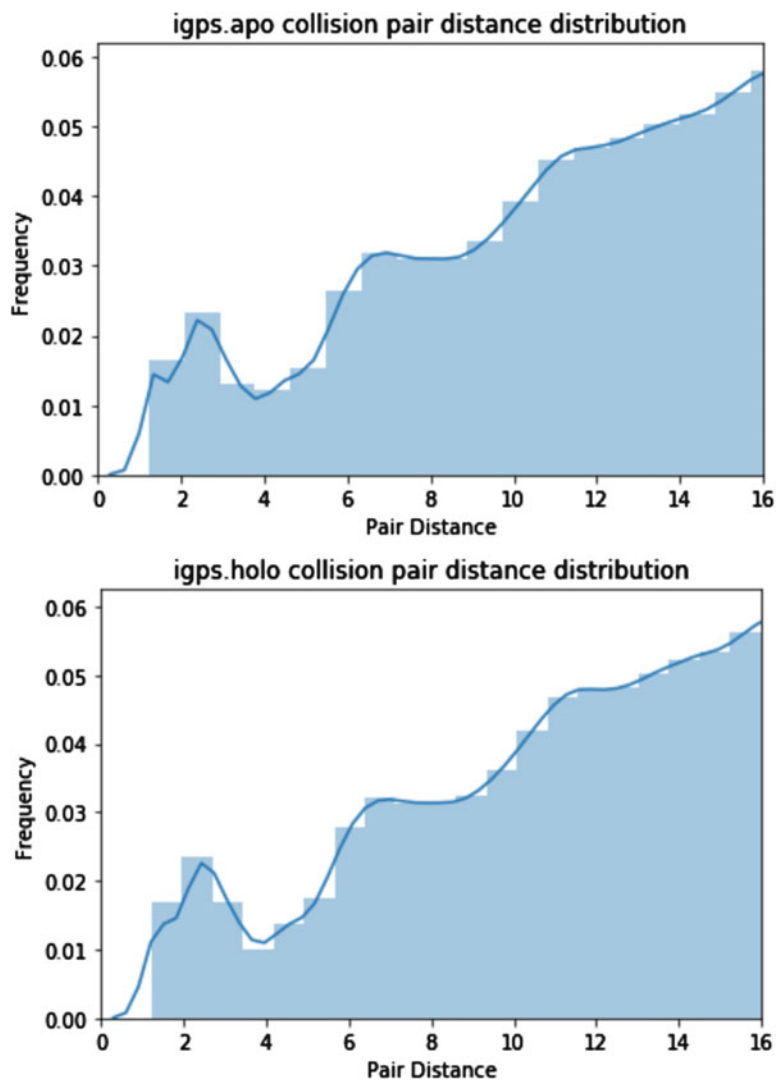


Fig. 3 Pairwise residue distance distribution calculated from apo and holo states of IGPS simulations. (This figure was generated using `/network_analysis_examples/Distance_Topology_Generation.ipynb`)

and X_j respectively (i.e., $\langle X_i \rangle = 0$ and $\langle X_j \rangle = 0$). Here, X_i and X_j are matrices where each row contains the displacement of the center of mass from its mean value along x, y, and z (i.e., columns are the x, y, and z coordinates). We then compute the Pearson correlation coefficient, r , as:

$$r = \frac{\langle X_i \cdot X_j \rangle}{\sqrt{\langle X_i \rangle^2 \langle X_j \rangle^2}} \quad (2)$$

As noted above, this form of correlation is perhaps the most ubiquitous metric and its computation is provided under WISP,

NetworkView (CARMA [2]), and g_correlation packages (among likely a large list of others). Unfortunately, it suffers from two distinct drawbacks. First, as one may note by inspection of the numerator, it is based upon the dot product of the centered motion vectors. This means that, like the dot product, any correlated motion along orthogonal directions will be ignored. And second, it is accurate only for linear correspondences.

The second correspondence metric to be discussed is the linearized Mutual Information approximation presented in “Generalized Correlation for Biomolecular Dynamics” [20]. In brief, the full Mutual Information coefficient models the difference in informational entropy between the margin coordinate distributions (C_i and C_j) and the joint distribution (C_{ij}). The interested reader is referred to the above mentioned reference for an in depth discussion of the full Mutual Information formulation. The linearized Mutual Information then approximates C_i , C_j , and C_{ij} from the corresponding covariance matrices such that:

$$\begin{aligned} C_i &= \langle X_i^T X_i \rangle \\ C_{ij} &= \langle (X_i, X_j)^T (X_i, X_j) \rangle \end{aligned} \quad (3)$$

Using C_i , C_j , and C_{ij} , we then compute the linearized Mutual Information by approximating the entropy of the marginal distributions (C_i , C_j) and the joint distribution, C_{ij} using the inverse log of their determinants, that is,

$$I_{\text{linear}} = \ln(\det(C_i)) + \ln(\det(C_j)) - \ln(\det(C_{ij})) \quad (4)$$

Finally, this linearized Mutual Information, which would normally be in the range from 0 to $+\infty$ is “renormalized” to the range from 0 to 1. This is accomplished by computing the Pearson correlation coefficient for a corresponding pair of colinear, multidimensional gaussian distributions (and their corresponding joint distribution) which would have an equivalent Mutual Information. This yields the analytical expression for our linearized Mutual Information correlation coefficient, r_{MI} :

$$r_{\text{MI}} = \sqrt{1 - e^{-\frac{2}{3}I_{\text{linear}}}} \quad (5)$$

The full matrix of Pearson and linear Mutual Information correlation coefficients for IGPS apo systems are shown in Fig. 4. They can be easily combined with the contact topology matrix via pairwise multiplication (Fig. 5). Alternatively, it may be beneficial to only compute correlations for those residue pairs present in the contact matrix. This can be useful for very large systems, particularly if the trajectory is also very large and/or a computationally expensive correspondence metric is chosen.

One difference that is immediately noticeable between Pearson and Linear Mutual Information metrics is that the latter seems to

IGPS-Apo State Simulation
Residue-pair Atomic Motion
Correlation Coefficient Matrices

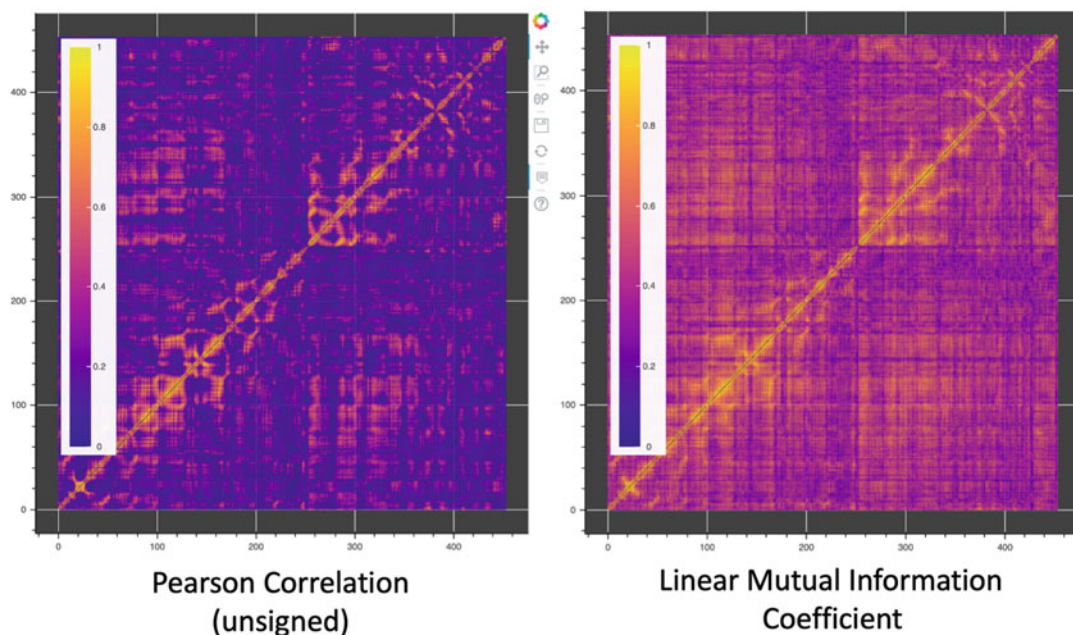


Fig. 4 Residue-pair atomic motion correlation coefficient matrices, calculated from IGPS apo state simulation. (This figure was generated using `/network_analysis_examples/Generate_ContactCorrelation_Matrix.ipynb`)

yield higher correspondence. This is because Pearson Correlation will only detect the portion of pairwise correlated motion that occurs along parallel directions. Linear Mutual Information, on the other hand, will pick up concurrent displacement/motion even when the directions of displacement are orthogonal. Thus, the Pearson Correlation metric would actually be a lower bound for linear Mutual Information. It should be noted, however, that both metrics will still have difficulty in cases where the correspondence/correlation is significantly nonlinear. In such cases, it may be necessary to employ more advanced metrics, such as the full Mutual Information score, provided via the `g_correlation` utility.

2.3.3 Energy Based Topology/Weights

While using distance based methods to generate network topology provides an intuitive and relatively straightforward method for generating network topologies, it may be argued that proximity between a pair of residues does not guarantee a strong or relevant interaction. Furthermore, it has been suggested that such methods may suffer from instability with respect to trajectory length/sampling, particularly when coupled with edge weighting based upon correlated motion (such as the networks produced by the Network View plugin [1]). As an alternative, interaction energy has been

Contact Correlation Matrices

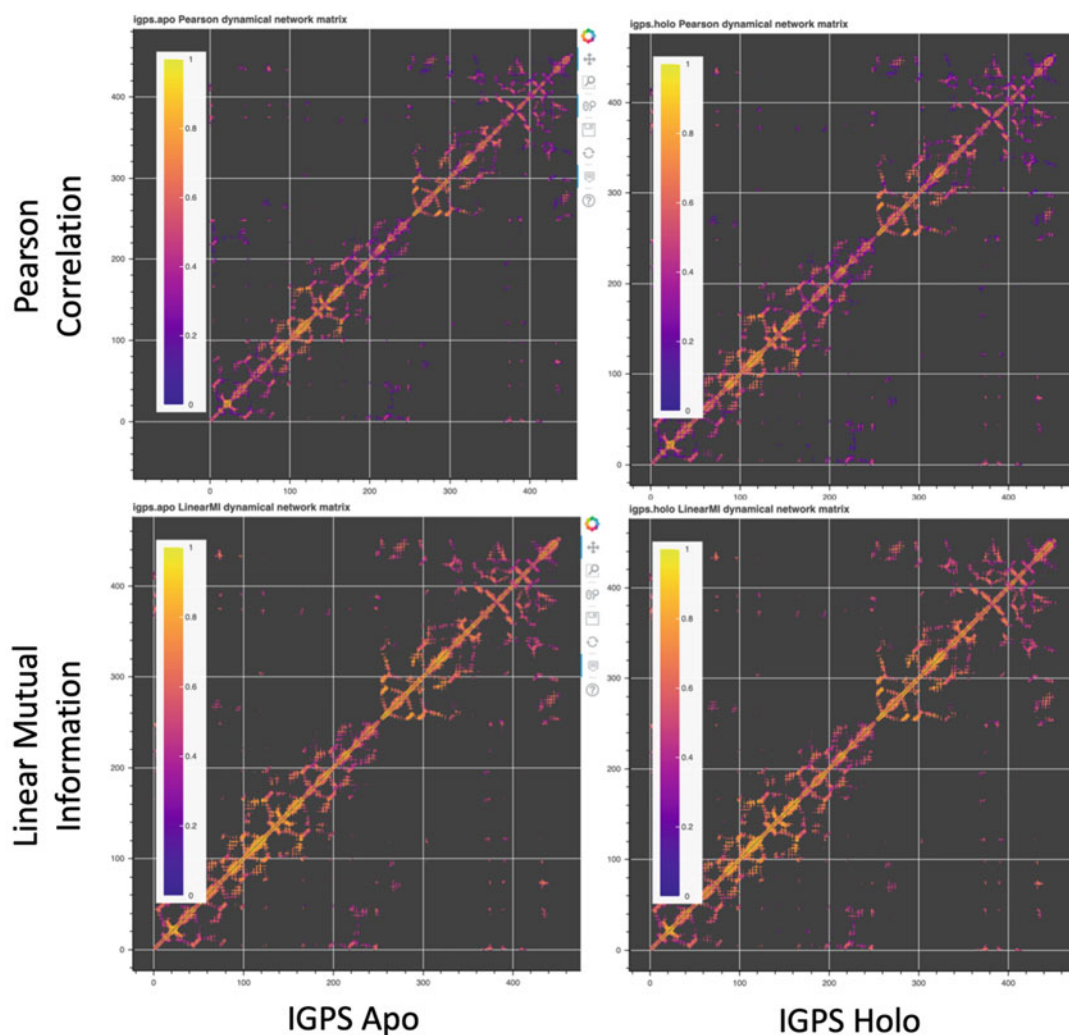


Fig. 5 Residue-pair atomic motion correlation coefficient matrices for IGPS apo and holo states, filtered by contact topology in Fig. 2. (This figure was generated using `/network_analysis_examples/Generate_Contact-Correlation_Matrix.ipynb`)

suggested to be a better metric for construction of network topology. On a similar note, interaction energy can also serve as an edge weight, whereas the previous method required either applying two different calculations, one to determine network topology via pairwise distance and the other to compute edge weights via a suitable correlation/correspondence metric.

While computing the interaction energy between a pair of residues is certainly more involved than a simple distance computation, there are plenty of tools available to facilitate this endeavor. Indeed, every simulation engine must, in fact, compute this for

every simulation step. Unfortunately, writing this data to disk while the simulation is running is not generally supported and could potentially slow the simulation down significantly. Fortunately, this information can be generated from the output trajectory. Here, we will describe how to do so using MMPBSA.py [19] from the AMBER molecular modeling and simulation suite [18]. While this tool is actually geared at computing ligand binding free energies by post processing an appropriate molecular dynamics simulation, it has an option to print out full pairwise interaction energy information as well. For the interested reader, there is an excellent online tutorial is available at: <http://ambermd.org/tutorials/advanced/tutorial3/> which provides an in-depth discussion and example of how to use this tool from start to finish for the purpose of computing ligand binding free energies (including setting up and running an appropriate simulation).

For the purpose of constructing a network topology and edge weights, we need to make use of one particular set of output options which causes MMPBSA to return a pairwise energy decomposition of the system. This, as it turns out, is exactly the information that is needed. This will yield a breakdown of each term that contributed to that energy for each pair of amino acids. Additionally, this information can be returned both on a frame by frame basis and/or as a summary (i.e., average, standard deviation, and standard error) over the entire trajectory.

In the case of a globular protein one could potentially use the full solvation free energy directly. For the membrane proteins, however, this could be inaccurate since the underlying solvation calculations do not provide support for including the effects of a membrane environment (technically, the Poisson-Boltzmann mode does provide support for membrane environments, but is prohibitively slow when a full pairwise decomposition is needed). Nevertheless, as was just mentioned above, the contributions from all relevant energy terms can be returned. Namely, the coulombic term (electrostatic) term and van der Waals term are provided. In the case of vacuum like approach (i.e., no consideration of solvent interactions) approach, the sum of these two terms yields the noncovalent energy used to construct our interaction network.

$$\begin{aligned}
 E_{i,j} &= E_{coul}(i,j) + E_{vdw}(i,j) \\
 E_{coul}(i,j) &= \sum_{a \in i, b \in j} k_e \frac{q_{i,a} q_{j,b}}{|r_{i,a} - r_{j,b}|} \\
 E_{vdw}(i,j) &= \sum_{a \in i, b \in j} \epsilon \left[\left(\frac{R_{(i,a)} + R_{(j,b)}}{r_{i,a} - r_{j,b}} \right)^{12} - 2 \left(\frac{R_{(i,a)} + R_{(j,b)}}{r_{i,a} - r_{j,b}} \right)^6 \right]
 \end{aligned}
 \tag{6}$$

where i and j are a given pair of residues and a and b are atoms in i and j respectively, k_e is coulomb's constant, $q_{i,a}$ and $q_{j,b}$ are the

partial charges of atoms a and b , $r_{i,a}$ and $r_{j,b}$ are the positions of atoms a and b , $R_{(i,a)}$ and $R_{(j,b)}$ are the van der Waals radii of atoms a and b , and ϵ is well depth of the van der Waals interaction potential for atoms a and b . Note that the r_m terms are constants and are defined for each individual atom (or more specifically “atom type”). Also worth noting is that ϵ is a constant that must be defined for each *pair* of atoms. While in many cases this can be decomposed into contributions from individual atoms, some simulations (or more accurately, the force field used in the simulations) will include “correction” terms for specific atom type pairs. Similarly, while the van der Waals term presented here is the most commonly used form, the r^{12} is in fact somewhat arbitrary and thus other alternative van der Waals forms have been proposed and see common use. Therefore, one should take care whether coding these calculations from scratch or selecting an appropriate tool. For edge weights, one simple approach is to use either reciprocal or inverse log of the absolute value of the interaction energies. Another approach, as outlined the paper “Determination of Signaling Pathways in Proteins Through Network Theory: The Importance of the Topology” [21], is to compute an appropriate scaling coefficient which sets bonded interaction pairs equal to a value of 0.99 and scales nonbonded interaction energies to values between 0 and 0.99 based upon their interaction energies relative to the average interaction energy of nonbonded pairs. We will employ the simpler form here, however, interested readers are encouraged to read the abovementioned paper (the required scaling equation is Discussed in section 3.4 of [21]). Again, for purposes of path finding or calculation of betweenness scores, the additive inverse of the log of these scaling weights would be employed.

One shortcoming of using a vacuum approach is that it may over-favor strong electrostatic interactions between residues. In particular, salt bridges will exhibit very high interaction energies, even over very long ranges. This will over emphasize such interactions which are often mitigated by solvent-solute interactions. One possible approach to compensate for this is to use an implicit solvent model, such as Generalized Born (GB). Such methods are well studied and are commonly used in other areas such as prediction of binding free energy and binding affinity, docking, and even simulations themselves. A detailed review of these methods is far beyond the scope of this chapter. In brief, these methods modify the electrostatic component by treating the solvent as a region of high dielectric constant and the solute as region of low dielectric constant. Additionally, they add a term to include hydrophobic-like effects, often based on some function of the molecular surface/volume, that is, solvent exposed or solvent excluded surface area/volume (GBSA). One major advantage of the GBSA solvent model is that it can be computed in a pairwise manner over individual

atomic interactions (which can then be summed to provide pairwise residue interactions). This makes it ideal for use here.

These methods are well suited to globular proteins; however, their direct application to membrane proteins is more complicated. Accurate modeling of such systems would require inclusion of an additional region in the model. While application/extension of GBSA to membrane bound systems has been studied for several decades, and such models are available under a number of modeling and simulation packages, their application for this particular purpose is cumbersome. Currently the automated MMPBSA.py tool available under the AMBER modeling and simulation package, which was selected for this paper because it allows a (relatively) easy to use interface to generate pairwise residue interaction energies in an automated manner, does not support GBSA models with inclusion of implicit membranes. The MMPBSA.py tool does allow for inclusion of membrane effects under the Poisson–Boltzmann Surface Area (PBSA) model, however, this method may be extremely slow when pairwise decompositions are requested.

While the GBSA method presented here can, potentially, be applied to membrane proteins, it may yield biased results because it will not include contributions of the membrane in its calculation. This may result in an underestimation of electrostatic contributions as well as an overestimation of hydrophobic contributions in regions of the protein that are in contact with the membrane.

In practice, both the coulombic term and van der Waals term are only calculated using the above direct pair sum equations for pairs of atoms that are relatively close (common cutoffs range from 8 to 12 Å). In the case of the van der Waals term, the interaction energy falls off so quickly as to rapidly become negligible. On the other hand, coulombic interactions can act over very large ranges. In the case of periodic systems, these terms are often computed using grid and Fourier transform based approaches such as the particle mesh Ewald (PME) and particle–particle–particle mesh (P³M) methods. One major reason for such methods is to allow for rapid computation of energies for systems with very large numbers of particles, for example a molecular dynamics simulation where water molecules are explicitly included.

As a brief word of warning, the amount of space required to run this analysis will scale as the square of the number of residues times the length of the trajectory. Therefore, care should be taken when investigating very large proteins, especially if the simulation time is also very long. This can quickly become prohibitive in both memory and computation time. This is particularly true if one's edge weighting scheme requires the pairwise energy to be returned for each frame rather than simply working on a function of the mean over the trajectory. Fortunately, energy terms tend to fluctuate much more slowly than distance. Thus, one may use a subset of a

trajectory instead, for example by taking every tenth or even every hundredth frame depending on the simulation setup.

Below is an example of using the MMPBSA.py tool with pairwise decomposition to return the data needed to construct an interaction energy based network topology. Other programs, such as the PairInteraction function in NAMD package can also be used to produce the same pairwise decomposition. We will make use of python to process this data and construct the corresponding matrix representation. Figure 6 below shows a heatmap view of the interaction energy matrix using a cutoff of 1.0 kT. Figure 6 bottom shows the corresponding projection onto a three dimensional rendering of the protein structure.

2.4 Analyzing Interaction Network Paths

2.4.1 Computing Current Flow Betweenness and Interaction Paths

Now that we have means to describe relevant networks of pairwise interactions in our protein, one question we might want to ask is “How important is a given amino acid or interaction?” In graph theory, this is known as “centrality,” which is a measure of how important a given vertex is in a graph. There are many different metrics for measuring centrality. Here we will focus on “betweenness centrality” along with a specific variant “current flow betweenness.” The former is measure of how many times the vertex appears in a “shortest path” between all other pairs of vertices. A “path” in a network is defined as a sequence of connected nodes/vertices. The length of the path is defined as the sum of the “weights” of the edges. In the case of energy or correlation this could be computed as either the reciprocal or inverse log of the absolute value of the energy or correlation respectively.

While the betweenness centrality metric is straightforward to calculate, not all shortest paths will be relevant to transmission between regions of interest (e.g., an allosteric binding site and active site), so betweenness centrality may overestimate the value of a vertex that is not actually of interest. Moreover, as can be seen in the renderings of interaction networks above, there are potentially many paths that may be traversed to get from one vertex to another, which will be missed if one considers only a single shortest path. Thus betweenness centrality may also underestimate the importance of a vertex as well.

One approach to addressing these shortcomings is to consider “suboptimal” paths between a pair of disjoint subsets of vertices, for example the amino acids that make up the allosteric binding site and active site of a protein. A “suboptimal” path (between a pair of vertices) is a path that is longer than the shortest path between those vertices. One may then apply the betweenness centrality metric over the set of all suboptimal paths found. There are a number of packages available that can compute the needed suboptimal paths. For example, WISP [17] and Network View [1], which provide a means to compute suboptimal paths with path lengths that are no more than some specified additional distance.

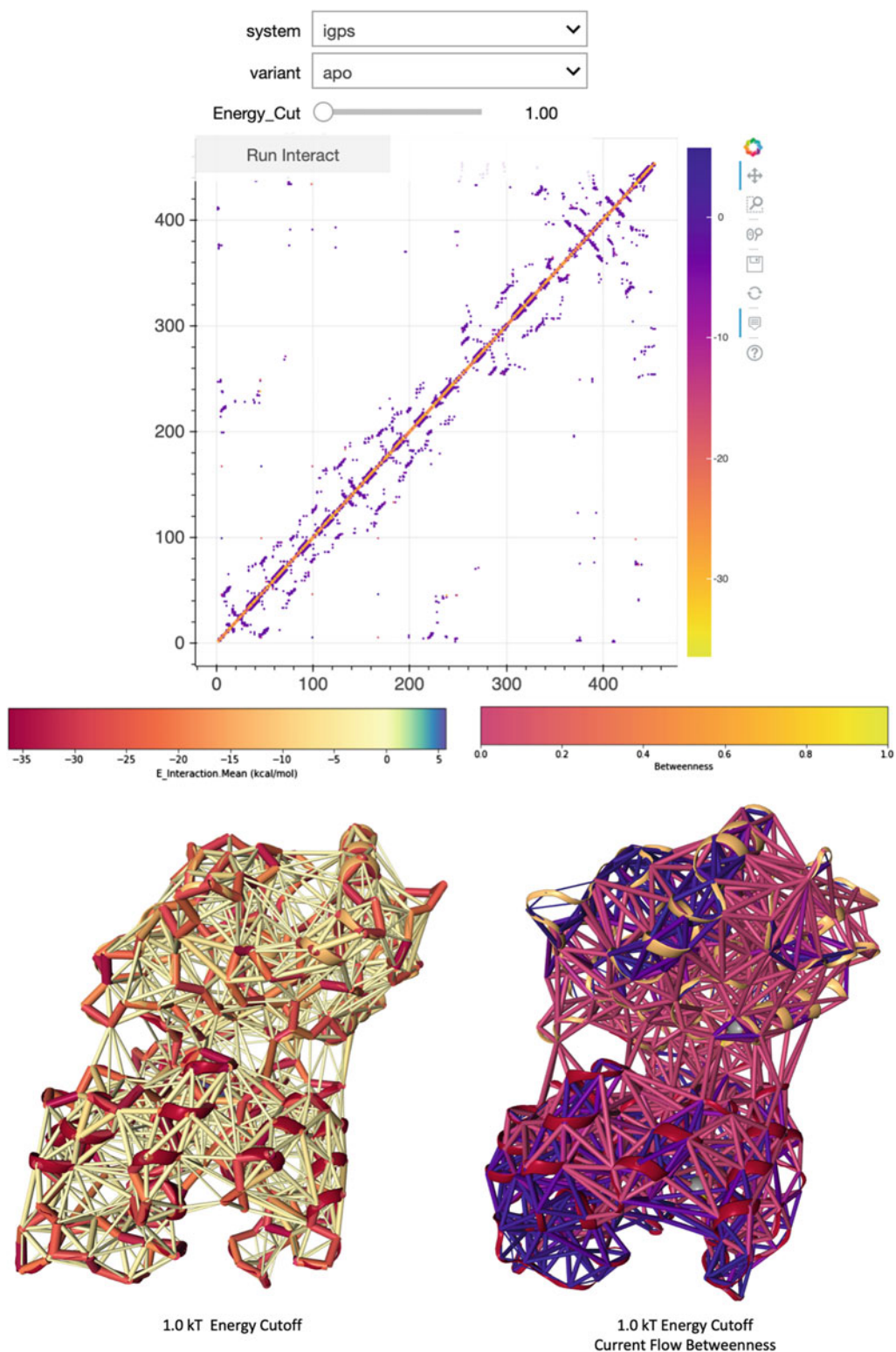


Fig. 6 Top: Interactive matrix visualization using the Bokeh package. This will let users set up the cutoff value, zoom, and query individual matrix elements by “hovering” the mouse over the data points in the matrix. Bottom: Structural heatmap view of the interaction energy matrix using a cutoff of 1.0 kT for IGPS apo system (left) and

Alternatively, if one is satisfied with computing the top K suboptimal paths, Yen's Algorithm [22] may be used. This is currently available using the command "shortest_simple_paths" found in the python package "networkx" [23]. For users of R, this could be accomplished via the package "igraph" [24]. There are resources that can provide the needed pseudo code/code for programming this oneself (e.g., Yen's algorithm in RStudio).

By using a suboptimal approach, both shortcomings of the original betweenness centrality metric are addressed. However, doing so raises the question of how many suboptimal paths (or what additional path length cutoff in the case of WISP or subopt) is needed. This is a relatively important question to consider because including too few suboptimal paths could yield poor estimates of a vertex's importance since the number of paths limits the lower bound of usage frequencies (i.e., the minimum observable betweenness score) and may miss contributions from important paths, particularly when there are many routes of similar path length. One solution to this would be to take increasingly more/higher-path-length paths until the betweenness scores observed converge. Unfortunately, the more paths that need to be generated, the longer the procedure will take. This is particularly true for large and/or highly connected networks, which is typical of the kind we are interested in here. Fortunately, as was mentioned at the beginning of this section, there are other centrality metrics available. One metric which builds on this concept of needing to include contributions from multiple paths is "current flow betweenness" (sometimes referred to as current-flow closeness centrality). Interestingly, this metric can be shown to be mathematically equivalent to information centrality [25, 26] which is derived from a somewhat more arcane approach conceptually, and both can be shown to be equivalent to random-walk betweenness centrality [25, 27]. The key advantage to this metric is that it essentially includes information from all possible paths implicitly, and thus is inherently more accurate and efficient than a random-walk approach.

Generalized computation of current-flow betweenness considers all possible pairs of source and target nodes. This again has the same shortcoming as in the original formulation of betweenness centrality. However, it can be adapted to consider only source and target nodes from a chosen subset of vertices corresponding to regions of interest. This has the added advantage that it reduces

Fig. 6 (continued) the corresponding current flow betweenness matrix generated using L50 as the source and E180 as the target. The corresponding interaction edges are projected onto a three dimensional rendering of the protein structure. (This figure was generated using `network_analysis_examples/Energy_Network_Analysis.ipynb` https://github.com/LynaLuo-Lab/network_analysis_scripts/network_analysis_examples/Energy_Network_Analysis.ipynb)

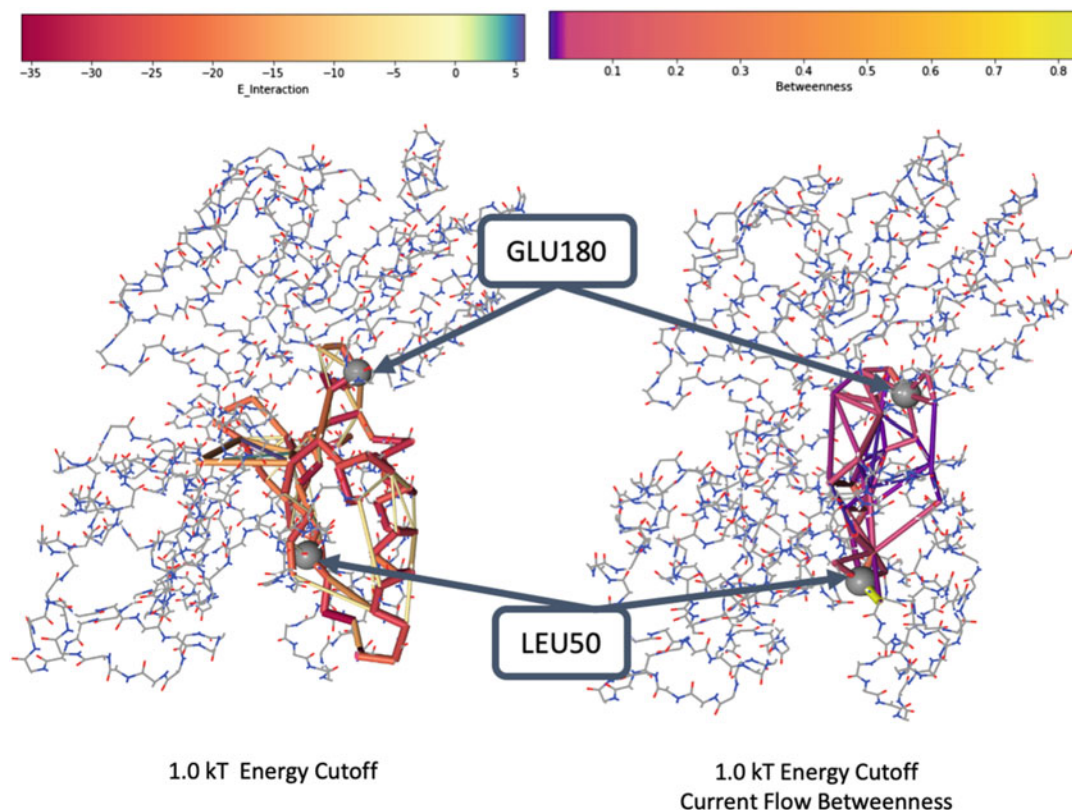


Fig. 7 Top 100 paths from Leu 50 to Glu 180 of the IGPS system generated using the reciprocal of interaction energy magnitude (energy cutoff 1.0 kT) as edge weights directly (left) versus the top 100 paths taken using the current flow betweenness of reciprocal interaction energy magnitude (right). (This figure was generated using `network_analysis_examples/Energy_Network_Analysis.ipynb`)

computation time as well since only contributions from paths between pairs of nodes in the chosen subsets need to be considered.

Lastly, it is often advantageous to combine suboptimal path computation with current flow betweenness. Specifically, this can yield a good way to visualize important transmission paths. This is quite useful in highly connected networks which are often quite cluttered. Moreover, one could apply the focused suboptimal path based betweenness centrality to the resulting current-flow betweenness weighted edges. While this once again leads to the need for determining a suitable cutoff, the convergence will often be markedly improved.

Figure 7 below shows an example of the top 100 paths from Leu 50 to Glu 180 of the IGPS system generated using the reciprocal of interaction energy magnitude as edge weights directly versus the top 100 paths taken using the current flow betweenness of reciprocal interaction energy magnitude. While this choice of total number of paths was somewhat arbitrary, the node usage

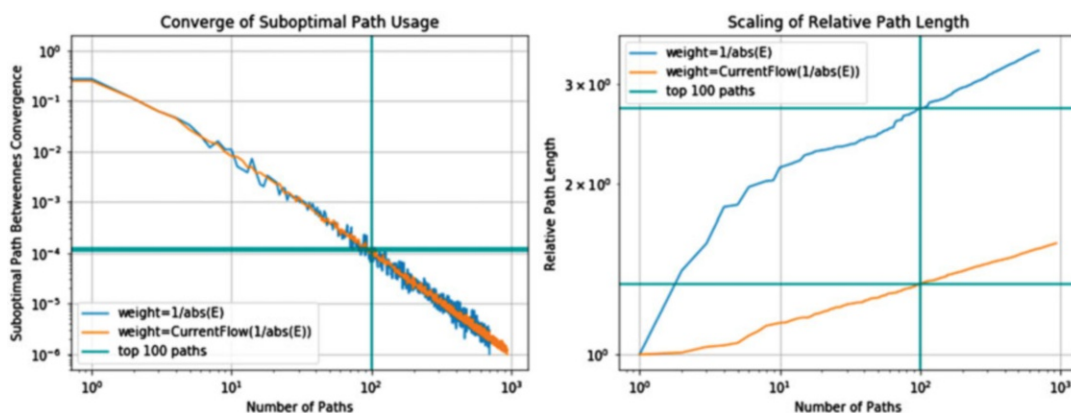


Fig. 8 Convergence as well as a comparison of relative path length scaling for using direct reciprocal energy magnitude (blue) and the corresponding flow betweenness as edge weights (orange). (This figure was generated using `network_analysis_examples/Energy_Network_Analysis.ipynb`)

frequency (suboptimal path betweenness) converges to nearly 10^{-4} in both cases. Figure 8 below shows a plot of this convergence as well as a comparison of relative path length scaling for using direct reciprocal energy magnitude and the corresponding flow betweenness as edge weights. This illustrates that while the suboptimal path betweenness converges at a similar rate, the relative path length of each additional path increases much more slowly under current flow betweenness. Current flow betweenness also yields more stable measurements of node and edge ranking as well [15].

One interesting feature worth noting is that interactions between bonded residues and between residues within the same secondary structure elements (e.g., 1–4 alpha helix backbone bonds) are prominent when using interaction energy directly as edge weights. While these entries are still present after corresponding current flow betweenness edge weighting (in fact all nonzero entries from the interaction energy matrix will also be present in the betweenness matrix) their values are in line with other nearby interactions. Perhaps more interesting is that more circuitous paths are present in the top 100 paths when directly using interaction energy for edge weighting while current flow betweenness yields a set of top 100 paths that lie more directly between the source and target residues.

2.4.2 Computing and Analyzing Node Betweenness

In addition to enabling more robust path finding and edge importance scoring, current flow betweenness can also be used to compute node importance scoring as well. Current flow betweenness scores for nodes may be computed by summing the current flow betweenness scores of all connected nodes. Again, with respect to the electrical network analogy, this is equivalent to computing the total magnitude of current flowing through a node. Since the current flow betweenness scores are unsigned, we have no way to

know which are flowing into and which are flowing out of the node. However, since this method is predicated on the assumption that the total current flowing into a node must equal the current flowing out of that node, we may simply divide by 2 for any nodes that are not either a source or target node.

It should be noted, that for source and target nodes, this sum does not hold. In the case of a single source, and a single target node, or in special cases where the network has a sufficiently tree-like structure, with sources and targets located on leaf nodes, one may simply omit this division since the sources and targets will exhibit only flow out of or into themselves respectively. This is not generally the case, however, as in a general graph some source nodes may exhibit flow into themselves and some targets may exhibit flow out of themselves if they lie on one or more paths between other source-target pairs. In short, one must take care when considering source or target node betweenness scores as it may not be possible to normalize these scores in the same manner as other nodes.

Despite this shortcoming, computing current flow betweenness for nodes can be useful in locating regions of importance in the protein structure. For instance, their values may be used to compute a structural heatmap, that is, a relevant rendering of the protein structure may be generated where the structure is “colored” based upon the current flow betweenness scores of the corresponding nodes. Additionally, if one is interested in comparing different states or variants of a protein, these scores may be subtracted and the resulting difference visualized in order to highlight regions where significant changes have occurred (Fig. 9).

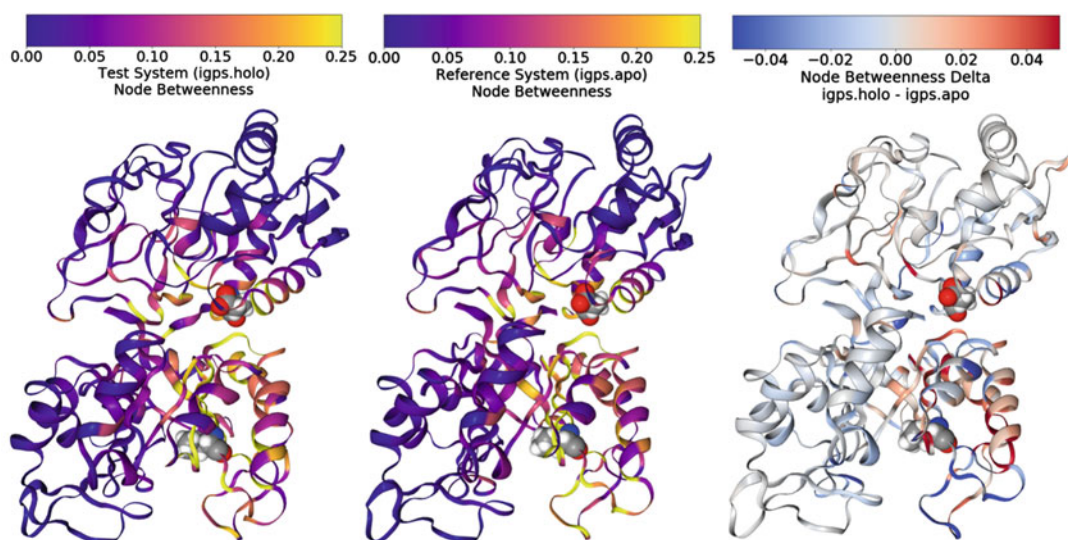


Fig. 9 Comparing the node betweenness between holo (left) and apo (middle) systems. The difference (holo–apo) is illustrated on the Delta-network (right). (This figure was generated using `network_analysis_examples/Energy_Network_Analysis.ipynb`)

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References

1. Stone J, Eargle J, Sethi A, Li L, Luthey-Schulten Z (2012) University of Illinois at Urbana-Champaign, Luthey-Schulten Group. NIH Resource for Macromolecular Modeling and Bioinformatics, Citeseer
2. Koukos PI, Glykos NM (2013) Grcarma: a fully automated task-oriented interface for the analysis of molecular dynamics trajectories. *J Comput Chem* 34(26):2310–2312
3. Vilorio JS, Allega MF, Lambrugh M, Papaleo E (2017) An optimal distance cutoff for contact-based protein structure networks using side-chain centers of mass. *Sci Rep* 7(1):1–11
4. Roe DR, Cheatham TE (2013) PTRAJ and CPPTRAJ: software for processing and analysis of molecular dynamics trajectory data. *J Chem Theory Comput* 9(7):3084–3095
5. Humphrey W, Dalke A, Schulten K (1996) VMD: visual molecular dynamics. *J Mol Graph* 14(1):33–38
6. Gowers R, Linke M, Barnoud J, Reddy T, Melo M, Seyler S et al (eds) (2016) MDAnalysis: a python package for the rapid analysis of molecular dynamics simulations. Python in science conference; 2016. Texas, Austin
7. Michaud-Agrawal N, Denning EJ, Woolf TB, Beckstein O (2011) MDAnalysis: a toolkit for the analysis of molecular dynamics simulations. *J Comput Chem* 32(10):2319–2327
8. Lindahl E, Hess B, van der Spoel D (2001) GROMACS 3.0: a package for molecular simulation and trajectory analysis. *J Mol Model* 7(8):306–317
9. Hess B, Kutzner C, van der Spoel D, Lindahl E (2008) GROMACS 4: algorithms for highly efficient, load-balanced, and scalable molecular simulation. *J Chem Theory Comput* 4(3):435–447
10. Pronk S, Páll S, Schulz R, Larsson P, Bjelkmar P, Apostolov R et al (2013) GROMACS 4.5: a high-throughput and highly parallel open source molecular simulation toolkit. *Bioinformatics* 29(7):845–854
11. Berendsen HJC, van der Spoel D, van Drunen R (1995) GROMACS: a message-passing parallel molecular dynamics implementation. *Comput Phys Commun* 91(1-3):43–56
12. Van Der Spoel D, Lindahl E, Hess B, Groenhof G, Mark AE, Berendsen HJC (2005) GROMACS: fast, flexible, and free. *J Comput Chem* 26(16):1701–1718
13. Abraham MJ, Murtola T, Schulz R, Páll S, Smith JC, Hess B et al (2015) GROMACS: high performance molecular simulations through multi-level parallelism from laptops to supercomputers. *SoftwareX* 1-2:19–25
14. Grant BJ, Rodrigues APC, ElSawy KM, McCammon JA, Caves LSD (2006) Bio3D: an R package for the comparative analysis of protein structures. *Bioinformatics* 22:2695–2696
15. Botello-Smith WM, Luo Y (2019) Robust determination of protein allosteric signaling pathways. *J Chem Theory Comput* 15(4):2116–2126
16. VanWart AT, Eargle J, Luthey-Schulten Z, Amaro RE (2012) Exploring residue component contributions to dynamical network models of Allostery. *J Chem Theory Comput* 8(8):2949–2961
17. Van Wart AT, Durrant J, Votapka L, Amaro RE (2014) Weighted implementation of suboptimal paths (WISP): an optimized algorithm and tool for dynamical network analysis. *J Chem Theory Comput* 10(2):511–517
18. Salomon-Ferrer R, Case DA, Walker RC (2013) An overview of the Amber biomolecular simulation package. *WIREs Comput Mol Sci* 3(2):198–210
19. Miller BR, TD MG, Swails JM, Homeyer N, Gohlke H, Roitberg AE (2012) *MMPBSA.py*: an efficient program for end-state free energy calculations. *J Chem Theory Comput* 8(9):3314–3321
20. Lange OF, Grubmüller H (2005) Generalized correlation for biomolecular dynamics. *Proteins* 62(4):1053–1061

21. Ribeiro AAST, Ortiz V (2014) Determination of signaling pathways in proteins through network theory: importance of the topology. *J Chem Theory Comput* 10(4):1762–1769
22. Yen JY (1970) An algorithm for finding shortest routes from all source nodes to a given destination in general networks. *Q Appl Math* 27(4):526–530
23. Hagberg AA, Schult DA, Swart PJ (eds) (2008) Exploring network structure, dynamics, and function using NetworkX2008. Pasadena, CA USA
24. Csardi G, Nepusz T (2005) The igraph software package for complex network research. *Int J Complex Syst* 1695:1–9
25. Brandes U, Fleischer D (2005) Centrality Measures Based on Current Flow. STACS; 2005. Springer, New York
26. Stephenson K, Zelen M (1989) Rethinking centrality: methods and examples. *Soc Networks* 11(1):1–37
27. Newman MEJ (2005) A measure of betweenness centrality based on random walks. *Soc Networks* 27(1):39–54



Chapter 18

Large-Scale Molecular Dynamics Simulations of Cellular Compartments

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Abstract

Molecular dynamics or MD simulation is gradually maturing into a tool for constructing in vivo models of living cells in atomistic details. The feasibility of such models is bolstered by integrating the simulations with data from microscopic, tomographic and spectroscopic experiments on exascale supercomputers, facilitated by the use of deep learning technologies. Over time, MD simulation has evolved from tens of thousands of atoms to over 100 million atoms comprising an entire cell organelle, a photosynthetic chromatophore vesicle from a purple bacterium. In this chapter, we present a step-by-step outline for preparing, executing and analyzing such large-scale MD simulations of biological systems that are essential to life processes. All scripts are provided via GitHub.

Key words Multiscale simulation, Molecular dynamics, Photosynthetic chromatophore, NAMD, VMD, Ensemble toolkit, High-performance computing

1 Introduction

Living cells are composed of hundreds of macromolecular complexes (proteins and nucleic acids) carrying out their assigned biological functions. These complexes are housed in key cellular compartments, such as the nucleus, mitochondrion, endoplasmic reticulum, and Golgi apparatus. These cell organelles form a detailed network to perform physicochemical reactions, which give rise to remarkable biological phenotypes such as growth, adaptation to environmental changes, and coaccommodation of competing functions [1, 2].

Eric Wilson and John Vant contributed equally to this work.

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Molecular simulations at atomistic resolution offer unprecedented detail of protein structure and functions [3]. Fulfilling this promise, the last decade has seen significant advancements in the development of multiphysics algorithms [4, 5] and their Graphical Processor Unit (GPU)-accelerated computational implementations [6]. Augmented by the deployment of peta- to exascale supercomputers application [7], these technologies are harnessed to estimate free-energy changes of dynamic biomolecules and ligands using “computational calorimetry” [8] in protein and ribonucleic acid (RNA) sequence analysis beginning the so-called big-data revolution in Biology and Chemistry [9], and to perform molecular simulations of living meso- to macroscopic systems (popularly called large systems) in chemical detail [10]. Here, we focus on how to perform molecular dynamics (MD) simulations of large biological systems to formulate a hands-on guide for modeling cell-scale objects (Fig. 1).

Notable large-scale all-atom simulations in recent years include the work on protein–protein interactions inside the bacterial cytoplasm characterized by a dense packing of many different macromolecules. This resulted in the generation of a cytoplasmic model comprising of 103-M atoms in a cubic box of 100 nm [11] and two different subsections of *Mycoplasma genitalium* (MG), MG_{m1} and MG_{m2}, with 12-M atoms were simulated [12]. Similar efforts have been taken to study viruses at unprecedented detail by Perilla and Schulten, simulating the Human immunodeficiency virus type

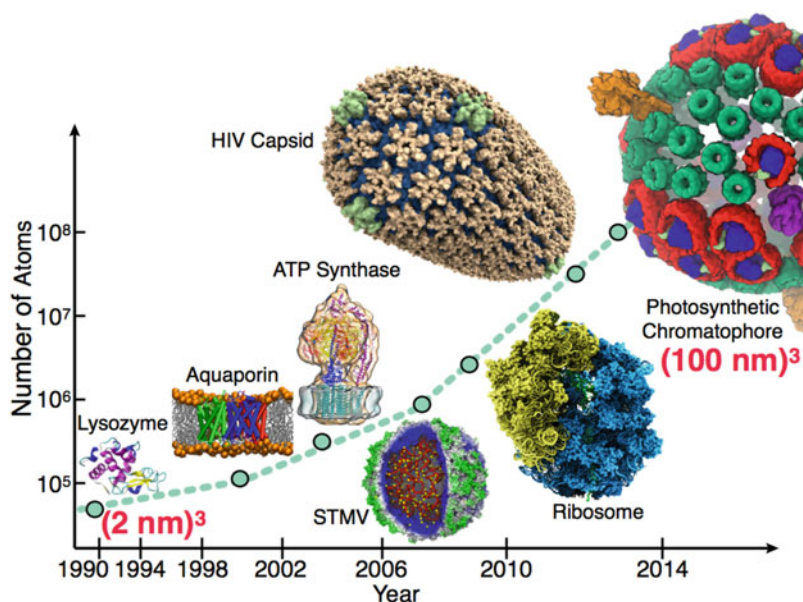


Fig. 1 Historical timeline of molecular dynamics simulations of biological systems. The system size increases from 1000 atoms in a lysozyme molecule ($\approx 2 \text{ nm}^3$ system volume) to over 100 M atoms in the photosynthetic chromatophore ($\approx 100 \text{ nm}^3$ system volume)

1 (HIV-1) capsid. The capsid is a large container, made of ≈ 1300 proteins with altogether four million atoms. Simulations of over 64-M atoms for over 1 μ s allowed for a detailed study of the chemical–physical properties of an empty HIV-1 capsid, including its electrostatics, vibrational and acoustic properties, and the effects of solvent (ions and water) on the capsid. The simulations reveal critical details about the capsid with implications to biological function [13]. Other exemplary targets of large-system simulation include the ribosomal translocation machinery [14, 15], the protein degradation pathway in proteasomes [16], enveloped viruses [17] and the periodic table of nonenveloped viruses [18], membrane-bending proteins [19, 20], and folding patterns of deoxyribonucleic acid (DNA) [21]. Most of these simulations are driven by ground-breaking advances in microscopic and imaging experiments [22].

Recently, we have reported a half microsecond simulation of a 136-M atom-scale model of an entire photosynthetic organelle, a chromatophore membrane vesicle from a purple bacterium, which revealed the rate-determining steps of membrane-mediated energy conversion [2]. Bioenergetic membranes are the key cellular structures responsible for coupled energy-conversion processes, which supply adenosine triphosphate (ATP) and important metabolites to the cell. A physical model of the emergence of phenotypic properties from detailed atomistic interactions is expected to offer direct insights on the rules of life [23]. Initial efforts to provide the structural and functional model of the bacterial photosynthetic membrane vesicle were undertaken by Sener et al. [24] and by Goh et al. [3]. Later, Singharoy et al. performed atomic-level investigations of cellular processes that have thus far been impeded by the sheer complexity of the network of interactions, the time-scales of a cell cycle, and the lack of essential experiment-inferred information. Molecular dynamics simulations of this bioenergetic organelle elucidate how the network of bioenergetic proteins influences membrane curvature and demonstrates the impact of thermal disorder on photosynthetic excitation transfer.

In subsequent sections, we provide a guide to large scale all-atom simulations, software required and description of input parameters. We then present a case study discussing how chromatophore converts sunlight into chemical energy, a process that is essential to life [25]. The necessary scripts and files to build the system and perform simulations are freely available to users on GitHub [26]. Note, while preparing this chapter it is assumed that the user feels comfortable using the UNIX command line and has basic experience with molecular dynamics using NAMD and scripting in VMD.

2 Materials

This section describes the input files and the software required to perform MD simulations on large systems (>100-M atoms). All simulations are performed using the popular MD tool called Nano-scale Molecular Dynamics or NAMD [27].

2.1 Input Files

Large-scale MD simulations with NAMD require the following four types of input files.

2.1.1 PDB

The starting structural coordinates of the given protein are provided by the Protein Data Bank (PDB) file. PDB files for the protein of interest can be accessed and downloaded from the PDB database <https://www.rcsb.org>.

2.1.2 JS Files

.js files are a binary file format which is more efficient than typical protein structure files (PSF) for large systems. Files of other formats can be converted to the .js format using the script found here: <https://www.ks.uiuc.edu/Research/vmd/minitutorials/largesystems/>

2.1.3 Force Field Parameters

Force field files contain numerical parameters (masses, charges and spring constants) needed to set up a potential energy function for evaluating how the atoms from different chemical constituents move due to bonded, angular, dihedral and nonbonded (electrostatic or van der Waals) interactions. These files typically consist of a topology file and a parameter file which has been generated either by spectroscopic data, quantum mechanical calculations, or through comparison of atoms with novel connectivity to known parameters. Topology files contain information about atom names, bond angles, and charges and end in the .inp file suffix for the CHARMM series of force fields. Parameter files contain constants required for an energy function, namely bond angle force constants and equilibrium angles bond and angle constants, and typically ends in the .par file suffix for the CHARMM simulations. More information on these files can be found at <https://www.ks.uiuc.edu/Training/Tutorials/science/forcefieldtutorial/forcefield-html/node6.html> or <https://www.ks.uiuc.edu/Training/Tutorials/namd/namd-tutorial-unix-html/node25.html>. Generation of a parameter file for a novel chemical constituent can be done with the CGenFF server (<https://cgenff.umaryland.edu>).

2.1.4 Gridforce Files

Since the construction of large molecular models involve iterative building and reequilibration of the simulation system, knowledge-based grid potentials will be employed as a constraint to maintain the stability of the intermediate models, while concomitantly maintaining the constant volume (NVT) or constant pressure (NPT)

conditions. The Gridforce simulations require a configuration file with the parameters of the potential represented as 3-dimensional volumetric data in .dx format, and also point to a PDB file which specifies which atoms will have forces applied to them. These grid potential or .dx files are also employed as input to Brownian Dynamics simulations of protein diffusion in coarse-grained computational tools such as ARBD [28]. A more in depth review of this method can be found at <https://www.ks.uiuc.edu/Training/Tutorials/science/forces/forces-tutorial-html/node4.html/>.

2.2 Software

The user should have access to the following software tools to run a large system simulation.

2.2.1 VMD

Visual Molecular Dynamics (VMD) is a computer program, designed for molecular modelling and visualization of biological systems [29]. It can be downloaded from <http://www.ks.uiuc.edu/Research/vmd/>.

2.2.2 NAMD

NAMD can be used to simulate large systems consisting of millions of atoms and is noted for its parallel efficiency, scaling to thousands of nodes (CPUs or GPUs). The software is widely used with CHARMM force fields but is also compatible with other popular force fields like AMBER and OPLS. It can be downloaded from <http://www.ks.uiuc.edu/Research/namd/>.

2.2.3 Charm++

Charm++ is a C++ based system for efficient parallelization of computational tasks. It was designed with efficient portability and latency tolerance in mind, being widely used by scientific and engineering applications for efficient computation across the entire range of hardware, from local computer clusters to large supercomputers. It can be downloaded as a binary and installed from <http://charm.cs.illinois.edu/software>.

2.2.4 APBS

Adaptive Poisson–Boltzmann Solver (APBS) numerically solves the Poisson–Boltzmann equation, which is a continuum model to describes the electrostatic interactions of molecular solutes in ionized solutions. APBS allows for the efficient evaluation of electrostatic properties of a wide range of length scales, including systems with millions of atoms. It can be downloaded from <https://sourceforge.net/projects/apbs/>. When working with large systems it can be useful to install an additional component called PDB2PQR. This software converts PDB files to PQR files which are read by the electrostatics software. PDB2PQR can be downloaded as a binary or compiled from source at <https://apbs-pdb2pqr.readthedocs.io/en/latest/pdb2pqr/index.html>.

2.2.5 ARBD

MD can be impractically expensive for monitoring the diffusive processes. ARBD or Atomic Resolution Brownian Dynamics simulations overcome this limitation of MD by approximating the proteins as rigid entities and capturing their diffusive dynamics within an implicit solvent environment assuming the so-called friction-dominated regime [30]. The potential of mean force required to drive Brownian dynamics is often derived from the average electrostatic potentials computed by APBS. ARBD supports point-like and grid-specified particles for representing the proteins. The performance is further enhanced by compatibility with GPUs.

All software tools used for performing large system simulations are free for academic use and binary executable are available for common operating systems and machine types.

2.3 Hardware

Many considerations need to be made when optimizing a simulation setup for a specific supercomputing architecture. Despite recent advancements in simulation software such as in NAMD 2.13, which redistributes computational workloads and reduces communication overhead, many bottlenecks remain a limiting factor. In today's biomolecular MD simulation literature, one typically reports production level runs of timescales on the order of microseconds. In order to achieve these timescales in a reasonable amount of time the software/hardware should produce 10–100 ns per day. The number of compute nodes needed to achieve the aforementioned performance varies with the computing architecture. Therefore, we provide a short description of the hardware with benchmarking from some example systems and our case study. It is important to note that before beginning production level runs for large system simulations, one should always thoroughly benchmark their system against the available computing architecture to ensure the most efficient use of resources.

2.3.1 Example of Applicable Hardware

Using the DOE's Summit supercomputer operated by the Oak Ridge Leadership Computing Facility (OLCF), a 21 M-atom system running on 64 compute nodes achieved a performance of 22 ns/day [31]. Each compute node of the IBM AC922 Summit system contains two 22 core-POWER9 CPUs and 6 NVIDIA Volta V100 GPUs. This benchmarking represents the state of the art for supercomputers.

2.3.2 Hardware Used for Case Study

The chromatophore model is comprised of 136 M-atoms. The MD software NAMD 2.12 was used for all equilibration and production runs. The production level runs of this model were conducted on the DOE's Titan supercomputer. Each compute node of the now decommissioned Cray XK7 Titan system had a 16 core AMD Opteron processor and an NVIDIA Kepler K20 GPU. Using a 1 femtosecond timestep and 4096 nodes, Titan was able to produce 16 ns of MD trajectory per day. Simulation parameters can be found at [26].

3 Methods

This section provides a general workflow for performing large-scale MD simulations of protein–membrane systems. A number of major simulation checkpoints will be covered, which includes model building, equilibration, analysis, and automation. Tcl scripts pertaining to the execution of these checkpoints in NAMD and VMD are provided.

3.1 Model Building

The first step for performing an MD simulation is to compose the system by assembling the models of individual components derived from X-ray crystallography, cryo-EM, homology and ab initio computations. In what follows, we outline how to construct a large system using VMD and Tcl scripts provided at [26]. First, the proteins are arranged based on prior information of their quaternary interactions. Second, these proteins are embedded within a membrane bilayer of biologically relevant shape and composition.

Third, the protein–membrane system is solvated and ionized. For the large systems, the solvation box is often truncated to noncubic symmetries. This setup reduces the number of atoms without compromising the protein–protein separation need to impose periodic boundary conditions [32]. Finally, the solvated system is equilibrated in a series of iterative steps.

3.1.1 Arrange Proteins

Relative position and orientation of the individual protein models within larger assemblies are determined from low-resolution imaging or microscopic experiments. For example, inter-protein distances are determined from atomic force microscopy (AFM) data using the so-called inverse-Mollweide transformation [33]. Similarly, the overall shape of the assembly is enforced on the model by employing restraints from electron density maps via the molecular dynamics flexible fitting scheme in NAMD [3]. The relative stoichiometry and copy number of the proteins, as well as the protein–lipid ratio can be determined from mass spectrometry [34]. These constraints originating from the protein–lipid ratio are nonunique [10]. The underlying large scale structural data involves quantifiable uncertainties in terms of distance (interprotein separation in AFM is prone to error of 10–15 Å [35, 36]), geometry (cryo-EM and tomography data ranges in resolution between 3 and 20 Å [22]) or mass measures (The error bounds on quantitative mass spectrometry is 0.1–0.7 mole, implying that the mass of the proteins and lipids were determined to within a 10–20% uncertainty [34]). Consequently, an ensemble of models should be constructed to fit to the same data set. Although every problem requires customization, tools such as IMP [37], multiple instancing and X-MAS builder modules on VMD [29], and symmetry builder in ViperDB [38] are broadly applicable for assembling the proteins.

3.1.2 *Build Membrane and Generate Connectivity File*

The scripts in the GitHub repository [26] provide tools for building four different types of membrane systems. The script `General-Tools.tcl` defines four commands for producing a bilayer, vesicle, micelle or a curved membrane. These commands take the desired composition of the membrane as input and require a PDB file for each lipid type in your membrane. An example where the generate membrane commands gets called is in the wrapper `01-build-30 nm-vesicle.tcl`. Once the membrane is built, the structural and connectivity information needs to be stored in a memory optimized file format (i.e., the .js file format). This is accomplished by sourcing the following script `02-write-js-file.tcl`. The output of this step is a .js file which contains all the structural and connectivity information of the lipids in the membrane.

3.1.3 *Create and Crop a Solvation Box*

Once the membrane is built, it needs to be explicitly solvated. First, a large water box is created by patching together smaller equilibrated water boxes using the script `03-make-giant-waterbox.tcl`. It is generally necessary to crop large water boxes in order to minimize the number of water molecules needed to simulate the system. In the case of a vesicle, the large water box can be cropped into a hexagonal water box with the script `04-crop-waterbox-for-30 nm-vesicle.tcl`. The resulting water box needs to be renamed so that each water molecule has a unique identifier. The script `05-renumber-hexwaterbox.tcl` renames the hexagonal water box made in the previous step. In order to efficiently solvate the membrane, the water box needs to be rotated to maximize the distance between the membrane and the edge of the water box. This can be accomplished with the script `06-rotate-waterbox.tcl`. The resulting water box and lipids are then combined using `07-combine-waterbox-with-lipids.tcl`. Finally, waters which clash with lipids are removed with `08-remove-clashing-waters.tcl`, and the system is ionized with `09-ionize-example-part1.tcl`.

3.1.4 *Merge Membrane Water Box Assembly with Proteins*

Once the membrane system is ionized and solvated, proteins are embedded into the membrane. First, a fixed atom file is generated for subsequent MD simulations by sourcing the script `10-makeFixedAtomsFile.tcl`. The biomolecular scaffold .js file generated in the assemble proteins step is then merged with the solvated and ionized lipids .js file with the script `20-merge-with-TEMP-proteins.tcl`. Next, the lipids, water, and ions that clash with the inserted proteins are removed with `30-combine-lipidsphere-and-proteins-pack5.tcl` and `32-carvelipids-SASA-pack5.tcl`. A solvent accessible surface area calculation can be run as a quality check with the script `33-get-sasa-uncondensed.tcl`. Finally, `40a-place-some-quinones-outerleaflet.tcl` and `40a-place-some-quinones-innerleaflet.tcl` can be used to place quinones in the membrane. Once the final model is assembled, `99-load-everything.tcl` and `check-vesicle.tcl` can be used for quality assurance.

3.2 Equilibration

The model produced in the previous section is likely to be unstable due to the inaccurate number of lipids in the inner and outer leaflet and disproportionate number water molecules placed on either side of the membrane. Therefore, before moving onto production level runs, the model requires minimization and equilibration. This step is achieved through a combination of short MD simulations and Tcl scripts.

3.2.1 Water

First, an NPT MD simulation is run with lipid headgroups and proteins fixed so that the lipid tails are allowed to relax (0.5 ns). After the tails have been allowed to relax, another NPT MD simulation with no constraints is ran (10 ns). This will cause the membrane to explode and form holes. The holes are then used to allow the number of water molecules on each side of the membrane to converge. Once water has equilibrated, we use a combination of LipidWrapper [39] and grid forces capabilities of NAMD [40] to patch the holes in the membrane and equilibrate the rest of the model.

3.2.2 Protein & Lipids

First, we use the scripts in modeling/hole-fixing/Lipid-wrapper-scripts/ as described by [26].

Bad contacts are then removed by shrinking the relevant lipids with the scripts in modeling/hole-fixing/lipid-shrinking-scripts/. More detailed instructions are made available with the scripts. After patching the holes, a gridforce map is created by simulating the density of water in the system which effectively constrains the equilibrated water and prevents water from solvating the holes created in subsequent simulations. Lastly, another short NPT MD simulation is running, this time with gridforces on, to equilibrate the proteins and lipids as well as create more holes if the number of lipids is inadequate. This process is repeated iteratively until the membrane stabilizes and holes cease to form. A typical NAMD configuration script for equilibration `chromtest.namd` is provided. Assuming the successful completion of the previous steps, production level runs are now ready to be simulated.

3.3 Analysis

APBS and ARBD were both used to reveal the rate-limiting step of membrane-mediated energy conversion in the purple bacterium chromatophore. The following section will present a general outline for the implementation of each technique to large systems simulations. Links to sample inputs fills are provided.

3.3.1 APBS

Electrostatic properties of the system can be derived by solving the nonlinear Poisson–Boltzmann equation within each frame of the MD simulation trajectory and averaging over all frames. These computations can be repeated over a range of pH and salinity conditions to determine how robust the physical properties of the biological constructs are to environmental changes.

To begin, the .js file of the simulation system is converted to a .pdb file. Then, this .pdb for the system may be converted to a .pqr file which will be read by APBS. Notably a .pqr file resembles the PDB format with the exception of the occupancy and the beta columns being replaced by the charge and mass of every atom. Three additional commands can be utilized to allow APBS to read additional parameters, all of which have the format .dx, which is a flexible scalar data format known as OpenDX. First, the command `diel` tells APBS to read the dielectric function, and paths to the x-, y-, and z-shifted dielectric map files must be specified in the input file. Second, if the system has a nonzero ionic strength value, an additional command, `kappa`, must be implemented, allowing APBS to read the ion-accessibility function. Lastly, `charge` allows APBS to read the fixed molecular charge density function.

Five different types of electrostatics calculations can be performed on the system, however particular focus will be given to `mg-para`, the calculation used for the chromatophore. Automatic parallel focusing multigrid calculations (`mg-para`) perform single-point calculations on systems and evaluate the electrostatic potential on a large scale. All relevant keywords implemented in our input will be briefly covered below.

The amount of overlap between individual processors meshes is specified with `ofrac`, which is a value between 0 and 1. Generally, a value of 0.1 is sufficient to generate stable energies. `npbe` specifies that the nonlinear Poisson–Boltzmann equation will be solved. `bcfl` flag defines the type of boundary conditions, where flag can be zero, `sdh`, `mdh`, or `focus`. zero and `focus` are generally not used with `mg-para` calculations, while `sdh` and `mdh` refer to “Single-Debye-Huckel” and “MultipleDebye-Huckel” models, respectively. The former is used for larger systems, as it describes a model of a single sphere with a point charge. `Pdievalue` and `sdievalue` specify the dielectric constant of the molecule and solvent, respectively. These values range from 2 to 20 for molecules depending on the extent of polarization to be considered, while a value in the range of 78 - 80 is generally used for biological conditions. `Srfm` flag specifies the model used to construct the dielectric ion-accessibility coefficients, where flag is defined as `mol`, `smol`, or `spl2`. Here, we used `spl2` to define the coefficients by a cubic-spline surface that is very stable relative to grid parameters. `chgm` flag, the method by which point charges are mapped onto the grid accordingly used `spl2` as its discretization method. `sradius` value specifies the radius of solvent molecules, which is usually set to 1.4 for water. Lastly, `swin` value defines the size of the support for spline-based definitions, with a usual value of 0.3. Examples of APBS input files can be found at [26].

3.3.2 ARBD

Brownian Dynamics (BD) simulations are a coarse grain simulation method for simulating the dynamics of molecules in solution at time scales which are not achievable with typical MD simulations. Coarse graining is necessary to model the relatively slow micro-second diffusion timescales of molecules in solution. In this case, two separate BD simulations were used to simulate the dynamics of both quinone and cytochrome c_2 . These simulations make use of the electrostatic potential information created in the previous APBS calculations and diffusion coefficients created by the Hydropro software (<http://leonardo.inf.um.es/macromol/programs/hydropro/hydropro.htm>). The resulting diffusion coefficients and .dx files contain electrostatics information that is used by ARBD to calculate the forces on the various molecules in the simulation using a symplectic integrator [28].

Among other general parameters in the BD input file are the information collected from Hydropro, a defined dummy particle and rigid body, and various grid files. First, it is important to note that the timestep is described in nanoseconds. Next, a dummy atom is defined to act as a reference particle if a rigid body is to be added. Here, we defined cytochrome c_2 as our rigid body, and its mass, inertia, and damping coefficients were obtained from Hydropro. Lastly, two additional files must be generated: charge density and Van Der Waals force potential grid files. The former can be obtained by utilizing the volmap density command in VMD, then opt to weigh charge. The latter can also be generated in VMD, using the Implicit Ligand Sampling (ILS) plugin. Examples of ARBD implementation scripts can be found at [26].

3.4 Conventional Workflow

The recent hardware improvements and software efficiency provide continuing growth of the capacity for molecular biology, climate science, and chemistry, particularly for MD simulations and data analytics. A robust workflow management tool creates an automated environment to build, simulate and analyze large systems. The workflow management tool consists of RADICAL-Cybertools (RCT), including the following middleware building blocks: Ensemble-Toolkit (EnTK), RADICAL-Pilot (RP), and RADICAL SAGA (RS).

RADICAL Ensemble Toolkit (EnTK) [41] provides scalable workflow management capabilities on high performance computing (HPC). EnTK is a Python implementation of a workflow engine, specialized in supporting the programming and execution of applications with ensembles of tasks. EnTK executes tasks concurrently or sequentially, depending on their arbitrary priority relation. Tasks are scalar, MPI, OpenMP, multiprocess, and multithreaded programs that run as self-contained executables. Tasks are not functions, methods, threads, or subprocesses. EnTK uses the Pipeline, Stage, and Task model (PST) model to encode workflows. Tasks are grouped in stages, indicating that they have no

input–output relationships and can be executed concurrently, depending on resource availability. Stages are collected into pipelines, indicating that tasks of different stages have input/output dependencies and have to be executed sequentially.

RP is a portable, modular and extensible Pilot system [42], written in the Python programming language. The pilot abstraction enables the separation between resource acquisition and the scheduling of tasks on those resources. RP acquires resources by submitting a job to the HPC batch system. Once the job is scheduled, RP enables task scheduling, placement, and launching on the job's resources. Tasks can be scheduled concurrently and sequentially, depending on resource availability. In this way, RP enables scalable execution of multitask applications with high throughput on HPC resources. Note that RP is fully compliant with the policies of each HPC platform: resources are acquired via the platform's batch system and used until their walltime expires.

RS is a Python implementation of the Open Grid Forum SAGA standard GFD.90 [43], a high-level interface to distributed infrastructure components like job schedulers, file transfer, and resource provisioning services. RS enables interoperability across heterogeneous distributed infrastructures, improving on their usability and enhancing the sustainability of services and tools [44].

3.5 Iterative Workflow Using NAMD-EnTK

Large scale MD simulations presented in Subheading 3.1 require submitting numerous scripts for model building, simulation, and analysis. This makes managing the execution of these simulations difficult, especially when considering the need to scale these execution large HPC platforms in the world. In EnTK, the PST model allows to create and submit heterogeneous pipelines to support multiple iterations of model building, simulation, and analysis tasks. Data sharing among tasks, stages, and pipelines is simplified by enabling transferring intermediate files (i.e., simulated maps) while the workflow is being executed. EnTK interface lets users label each element of the workflow via unique identifiers to locate and relate data at each step of the workflow execution. Finally, users can code fine-grained specification of computational requirements, including a number of CPU/GPU, and number and type of process/thread for each task of the workflow.

Using EnTK we created a nine-stage pipeline on XSEDE Bridges supercomputer. The pipeline automates the procedure to iteratively assemble protein structures in membrane: the iterative model building, refinement and analysis step, refined model solvation and, finally, equilibration (*see* Subheadings 3.1 and 3.2). After the final system is prepared and equilibrated in an NPT ensemble, we setup another iterative pipeline between the production run in NVT ensemble and analysis (*see* Subheading 3.3). We used two nodes, utilizing 52 compute cores in total but EnTK is designed to seamlessly scale large scale iterative pipelines to up to 100,000 cores [41].

Multiple Gridforce based NAMD simulations, as implemented in molecular dynamics flexible fitting (MDFF) are encoded as tasks of the PST. The VMD preprocessing steps (i.e., generating gridforce file .dx, building NAMD configuration files .namd) with the initial structure are codified as the stages of pipelines. These EnTK pipelines are used to perform simulations and analysis described in Subheading 3.2. The current implementation of the workflow is available on a dedicated GitHub repository [45], offering a simple example of an iterative pipeline for the execution of an MD simulation and analysis workflow. The current workflow executes on OLCF Summit—which currently occupies the Number 1 position in the Top 500 Supercomputing list.

In the near future, we will extend existing capabilities to support the execution of up to 1024 concurrent pipeline. As illustrated in Fig. 2, the pipeline will support iterative execution of a cycle of simulations, analysis, and model refinement.

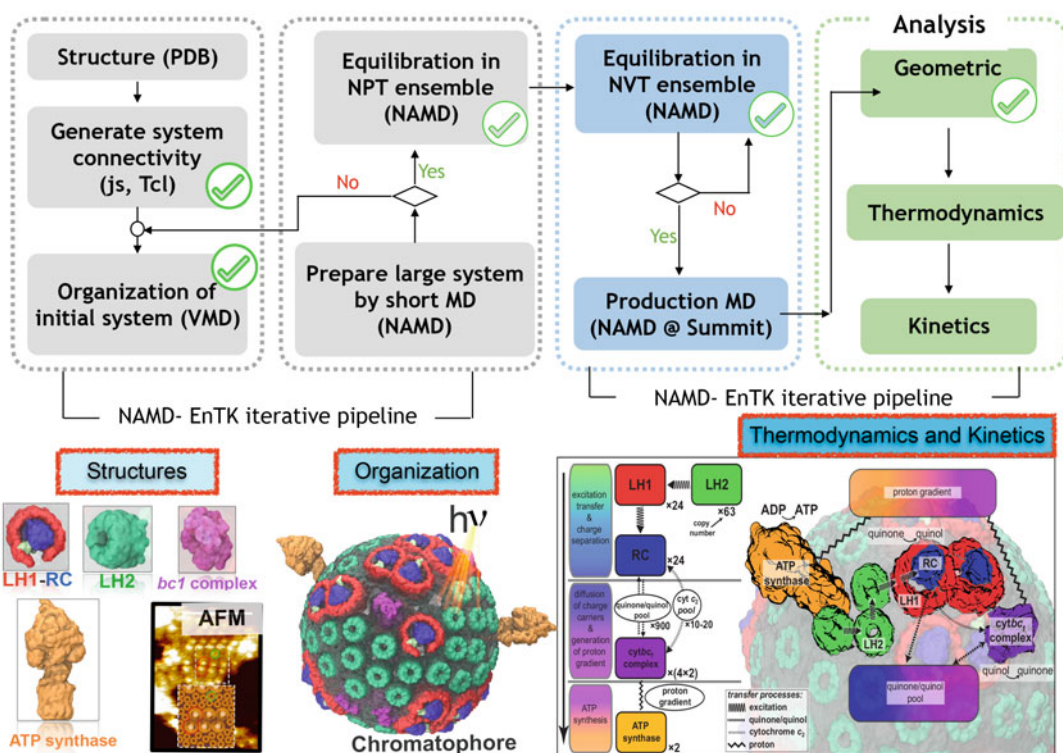


Fig. 2 NAMD-EnTK iterative pipeline on Summit. Flowchart illustrating the integration of molecular dynamics and visualization software (NAMD and VMD) in the ensemble toolkit (EnTK) iterative pipeline. Starting with initial structures of macromolecules, the structures are organized to assemble the photosynthetic chromatophore vesicle. Dynamics is performed at Summit, a GPU-accelerated petascale supercomputer. Checked boxes are steps of the algorithm that has been implemented using this pipeline [45]

4 Case Study

Here, we present an example of a large system simulation. This case study focuses on the simulation of a photosynthetic chromatophore [2], which absorbs solar energy to generate ATP, the so-called energy currency of life. The protein structure of the chromatophore vesicle was originally obtained by Cartron et al. [34].

4.1 Arrange Proteins

The first step is to arrange the proteins of the large complex in the shape of a sphere. Here, the vesicle patches imaged by AFM [34] are aligned relative to each other so that it is consistent with linear dichroism on intact membrane vesicles [2]. Overall, the model features 82 bioenergetic complexes (which includes 63 light-harvesting LH2 complexes, 11 dimeric and 2 monomeric RC-LH1 complexes, 4 bc_1 dimers, and 2 ATP synthases), together with 4011 light absorbing antenna molecules (comprised of 2469 bacteriochlorophyll and 1542 carotenoid). The spherical arrangement of these proteins are constructed by positioning the planar strips containing the RC-LH1 dimer arrays and their immediate LH2 neighbors along the north-south direction. The spaces between RC-LH1 strips are filled with LH2-rich regions. An area-preserving map, the inverse-Mollweide transformation [24] is used to map planar regions from the two-dimensional AFM images onto three-dimensional spherical ones. Structural conflicts are subsequently removed and gaps which arise between the pigment-protein complexes as a result of this projection process are removed manually by shifting the center of the proteins on the chromatophore sphere ensuring a tight packing of the structure. The “southern polar” region is left empty as a potential contact zone with the rest of the membrane.

4.2 Initial Equilibration

The chromatophore consists of a vesicle from AFM [34], and MD-equilibrated structures of individual POPC-embedded LH1-RC, LH2 [46], bc_1 [47], and ATP synthase models. Each protein within the chromatophore scaffold is overlaid with their membrane-embedded counterpart. The original protein inside the chromatophore is replaced by equilibrating the model with one lipid ring surrounding it. A POPC-only lipid vesicle of radius 30 nm is used to uniformly construct a lipid bilayer for the protein-excluded areas of the chromatophore surface. This lipid vesicle is then overlaid with the ring-encased proteins with a 2 Å exclusion radius, which is sufficient to avoid unfavorable steric interactions between the proteins and the lipids. The use of protein-lipid ring for membrane embedding instead of a protein-only embedding avoids the formation of ring-piercing artifacts. However, some unfavorable overlap remains due to the inter-twining of the lipid tails. Due to direct minimization being unable to remove

this artifact, the intercarbon distances in the lipid tail is shrunk (i.e., to a single carbon atom) until the intertwining was removed. Thereafter, brute-force energy minimization algorithm within NAMD resurrects the length of the lipid tails while avoiding the unphysical twining effect. After energy minimization, the vesicle is immersed in a water drop, followed by randomly mutating the lipid headgroups to obtain a lipid membrane composition containing 22% POPC, 22% POPG, and 56% POPE on the outer leaflet and 24% POPC, 10% POPG, and 66% POPE on the inner leaflet. The final step consisted of adding 900 quinone molecules resulting in the generation of a cubic simulation box of dimension 110 nm containing 136-M atoms.

4.3 Membrane Equilibration

A direct MD run starting with the 136-M atoms initial model leads to instabilities in the system within 1 ns of simulation (Fig. 3). These instabilities exhibit protein and/or lipid-excluded holes on the chromatophore surface. This is a result from a combination of (1) inaccurate number of lipid molecules and (2) water molecules on either side of the initial chromatophore membrane. The following strategy is employed to decouple the two issues and address them sequentially. First, a grid potential is defined at a resolution of 1 Å about the chromatophore surface, to softly repel the water away from the surface, while allowing the protein–lipid and lipid–lipid interactions to equilibrate during short 5 ns NVT simulations. Now, the present setup will only allow instabilities in the form of holes on the chromatophore surface due to insufficient number of lipid molecules. The imbalance in the number of water molecules on either side of the membrane will have minimal consequence on the chromatophore stability. The holes are allowed to form and equilibrate, but water passage is negated due to the presence of the soft grid-potential. The tool LipidWrapper [39] is employed to fix these holes through the insertion of filler lipids (Fig. 4). This script is available on GitHub [26]. Iteratively, after each round of hole formation, the holes are filled until LipidWrapper cannot identify any new holes. In the present case, a total of 4 iterations are performed for the number of lipids to converge.

4.4 Water Equilibration

In order to equilibrate the water on either side of the membrane, two LH1-monomers are removed from the chromatophore surface, creating few holes on the surface. Lipids within a 5 Å radius around these holes are constrained using harmonic restraints. Hole formation allows for passage of water molecules across the membrane, thereby letting the rest of the chromatophore relax without inducing any global instability. LH1 monomers were chosen for removal due to the observation that the area around these proteins were the most susceptible to instability. Thus, our choice of artificially creating holes in these areas and constraining the vicinity prevents any instability propagation through the membrane. It

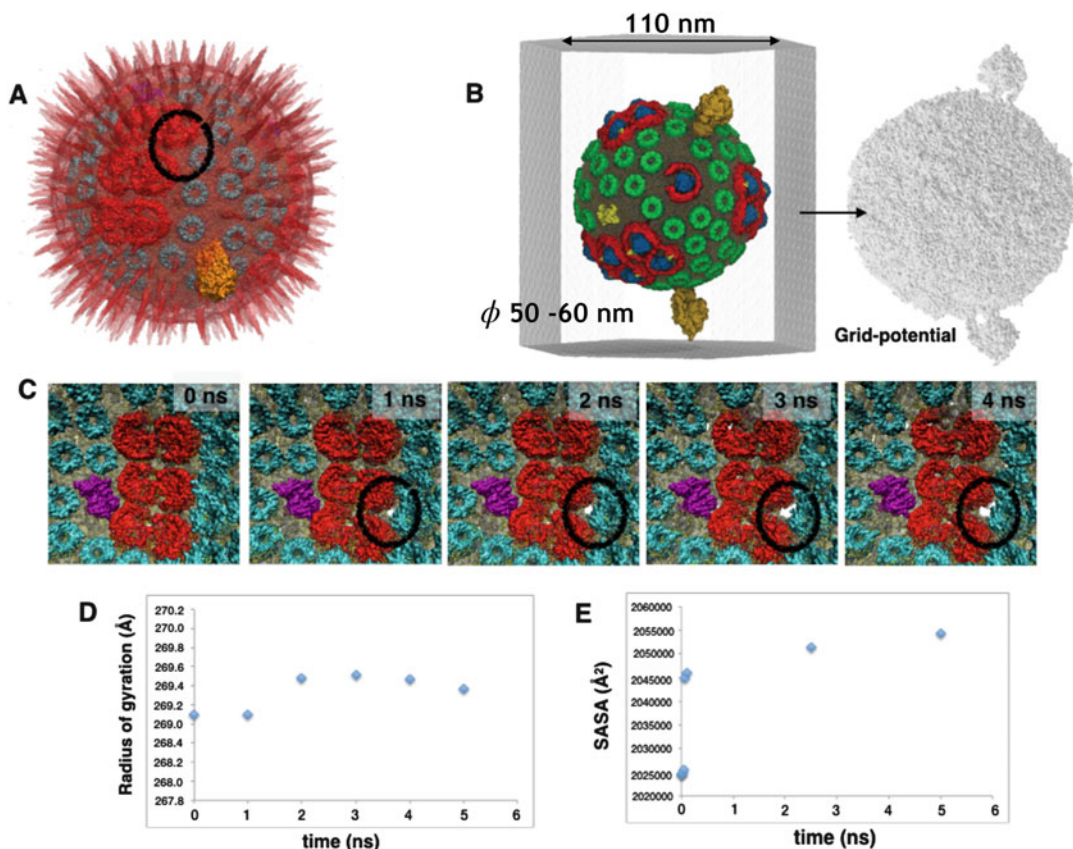


Fig. 3 Stepwise refinement of the initial chromatophore model. (a) Snapshot demonstrating instabilities in the 136 M-atom solvated and ionized chromatophore model after 1 ns of MD simulation. These instabilities, manifested as protein and/or lipid-excluded holes on the chromatophore surface, result from a combination of inaccurate number of lipid molecules and water molecules on either side of the initial chromatophore membrane. (b) A grid potential defined at the resolution of 1 Å about the chromatophore surface, which softly repels the water away from the surface, while allowing the protein-lipid and lipid-lipid interactions to equilibrate during short MD simulations for refining the initial model. (c) Snapshots illustrating equilibration of protein-lipid and lipid-lipid interactions during short 4 ns-MD simulations, which showcases hole formation (black ring) implying an inadequacy in the number of membrane lipids within the chromatophore model. The formation and propagation of holes due to inadequacy in the number of modeled lipids emerges as a surface artifact with minimal change in the overall radius of gyration (d), yet with significant alteration of the surface area (e)

takes approximately 25 ns for the solvent density to converge on either side of the membrane (Fig. 5). Subsequently, the missing LH1 monomers are reintroduced into the membrane by excluding some local water molecule.

The grid-constrained NVT simulations, for correcting the number of lipids on-the-fly, used a gscale of 0.3 [48]. After these short NVT simulations, the long NPT production runs are performed at 1 atm pressure using the Nose-Hoover Langevin piston with a period of 100 fs and damping timescale of 50 fs. Every model

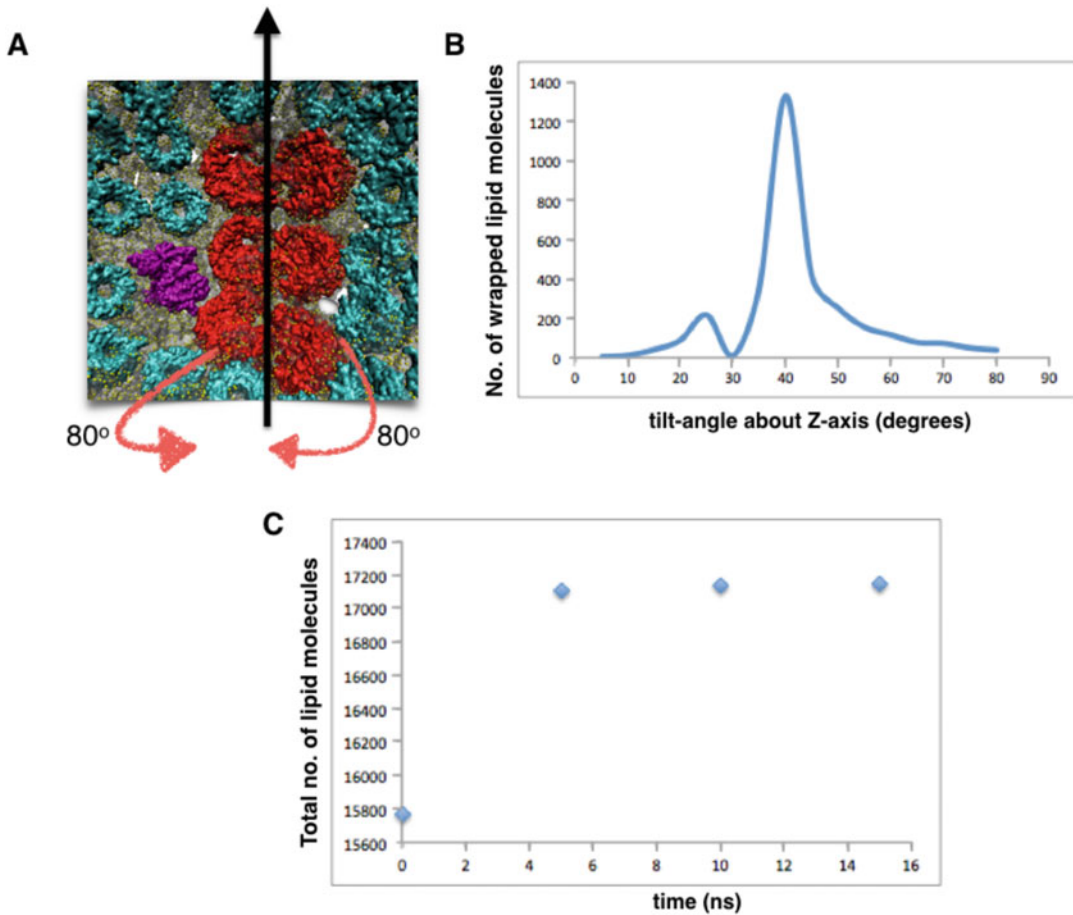


Fig. 4 Iterative refinement of the chromatophore membrane lipids with Lipid Wrapper. **(a)** schematic of lipid wrapper implementation [39], whereby a selected piece of membrane with holes is independently rotated about the X , Y , and Z axes to superimpose lipids from the rotated membrane with the holes from the original orientation. Lipid molecules are extracted from the rotated membrane and placed into holes of the original membrane to correct simultaneously the POPE, POPC, and POPE density of the chromatophore membrane. **(b)** Plot showing the number of extractable lipids at different rotation angles of a typical Lipid Wrapper scan. An angle of 40° is chosen about the X , Y , and Z axes for this particular instance as it provides the maximum number of extractable lipids for filling holes in the original membrane. **(c)** Four iterations of Lipid-wrapper-based lipid modeling each followed by a 4 ns of MD equilibration under grid forces were performed to converge to the final lipid count of 17,200

from an iteration of lipid updates (Fig. 5), is minimized for 10,000 steps prior to the adding more lipids in the next round; a total of $4 \times 10,000 = 40,000$ minimization steps are performed prior to solvent equilibration.

4.5 Production Run

Following the equilibration of water in the artificially water permeable chromatophore (created by removing two RC-LH1 monomers), these two proteins are reinserted into the equilibrated chromatophore.

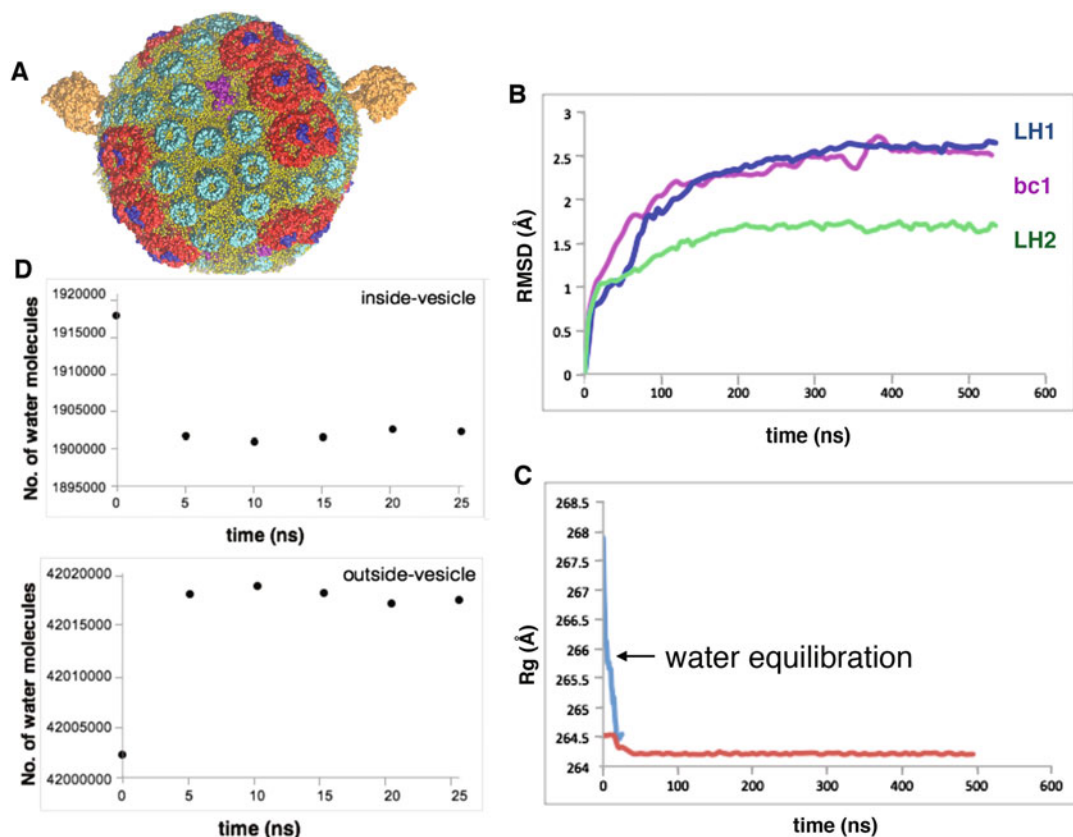


Fig. 5 Equilibration of a stable chromatophore model. (a) An all-atom model of the photosynthetic chromatophore vesicle in *Rhodobacter sphaeroides* derived from 50 ns of model-refinement and 0.5 ns of equilibrium MD simulations. The final simulation model features 82 bioenergetic complexes (67 Light-Harvesting LH2 complexes [green], 11 dimeric and 2 monomeric RC-LH1 complexes [LH1:red; RC:blue], 4 cytochrome bc1 complex dimers [magenta], and 2 ATP synthases [orange]), together with 4011 light absorbing antenna molecules (2469 bacteriochlorophylls and 1542 carotenoid) embedded in a membrane of 17,200 lipid molecules (Lipid phosphate indicated in yellow). (b) Root mean square deviation (RMSD) plotted as a function of simulated time showing that bioenergetic proteins have been sufficiently equilibrated over 0.5 ns. Consistent with simulations of a flat chromatophore patch [32], LH1 dimer is found to be the most flexible and LH2 the least flexible even when the chromatophore assembles as a vesicle. (c) Change in the radius of gyration of the vesicle showing a drop of 4 Å during the final 25 ns of model refinement stage, wherein the protein–lipid packing tightens and lipid composition reorganizes in the vicinity of the proteins, following which the value plateaus reinstating the achievement of a stable chromatophore model. (d) Number of water molecules inside and outside the chromatophore vesicle (defined by a radius of 280 Å from the center) showing the excess water molecules from inside are released through the hole outside the vesicle over 5 ns, after which the water molecules equilibrated

After 5000 more minimization steps are performed in the vicinity of the LH2 complex, restraining the rest of the chromatophore. Thereafter the long production run is performed for 500 ns.

All simulations are performed with the MD software NAMD 2.12 using the CHARMM36 force field for proteins and lipids

[27]. The force field parameters for the photosynthetic cofactors such as chlorophyll and quinone molecules have been obtained from our past studies of the individual proteins [32, 46]). A sample NAMD configuration file is provided on GitHub [26], including all the necessary input arguments employed by the simulation. Some key arguments are mentioned and requires special attention. It is assumed that the reader is familiar with basic NAMD configuration script and parameters employed by the simulation.

Simulations are performed with an integration time step of 1 fs where bonded interactions are computed every time step, short-range nonbonded interactions every two timesteps, and long-range electrostatic interactions every four timesteps. A cutoff of 12 Å is used for short-range electrostatic and van der Waals interactions: a switching function is started at 10 Å for van der Waals interactions to ensure a smooth cutoff. Periodic boundary conditions are used, with full-system, long-range electrostatics calculated by using the PME method with a grid point density of $1/\text{\AA}$. To avoid short-range interactions between adjacent copies of the system, it is ensured that the unit cell is large enough. Since the chromatophore dimensions vary between 50 and 60 nm, a water box of 110 nm (*see* Fig. 3b) maintains a padding of 25 nm on all sides. The systems were kept at constant temperature using Langevin dynamics for all nonhydrogen atoms with a Langevin damping coefficient of 5 ps^{-1} .

4.6 APBS Analysis

Cytochrome c_2 is responsible for shuttling electrons from the bc_1 to RC-LH1. Binding dynamics of Cytochrome c_2 to the integral membrane protein complexes is influenced by the charge state of the inside of the chromatophore [48]. Brownian dynamics simulations of cytochrome required APBS calculations at six different conditions: pH 4 with 0.02 M NaCl, pH 7 with 0.15 M NaCl, and pH 7 with 0.25 M NaCl, for both the reduced and oxidized forms of the chromatophore to represent microenvironments that single electron transfer proteins might experience while shuttling charges across a membrane. In the reduced chromatophore, bc_1 complexes were modeled as reduced, cytochrome c_2 was oxidized, and the RC-LH1 were neutral. In the oxidized chromatophore, the bc_1 complexes were set to neutral, cytochrome c_2 was reduced, and the RC-LH1 were oxidized. APBS simulations revealed that between the pH values of 6–7 and salt concentrations of 0.15 M to 0.95 M, the effective charge of the surface of the chromatophore was relatively unchanged. However, as salt was reduced below 0.15 M, the chromatophore charge dropped precipitously due to an absence of counter ions. Additionally, pH below 4 resulted in protonation of acidic side chains, where at higher pH, deprotonation enhances surface charge and protein–protein interactions with cytochrome c_2 .

4.7 ARBD Analysis

Each rigid body simulation began with a single cytochrome c_2 at the center of the chromatophore that was allowed to diffuse in solution and interact with the various surfaces on the inside of the chromatophore. For each condition, 500 independent 10 μ s simulations each with 100-fs timesteps were performed and cyt. c_2 coordinates were recorded every nanosecond. Interactions were tracked for each of the conditions, with an interaction defined as cytochrome c_2 being within 1 nm of a nonsolvent component of the chromatophore. Simulations in the low-salt conditions displayed higher surface binding overall; however, the specificity of binding was greatly reduced compared to physiological salt conditions. In simulations where proteins were modeled with oxidized conditions, cytochrome c_2 favored interactions with RC-LH1 complex as opposed to bc_1 . Conversely, in simulations under reducing conditions cytochrome c_2 was more likely to interact with bc_1 than RH. This follows the logic of the physiological role of cytochrome c_2 being to dock to the surface of a reduced bc_1 , accept an electron and become reduced, then unbind and interact with oxidized RC-LH1, reducing it and starting the cycle over again. The association times of cytochrome c_2 between bc_1 and RH-LH1 were 13 ns and 70 ns respectively when BD simulations were performed at pH 7 with 0.15 M NaCl. Reversible binding times (combined association and dissociation times) were 1 ms and 0.2 ms respectively. This binding time was found to be slower than the time it took cytochrome c_2 to shuffle between proteins.

Executing concurrent pipelines using represents an effective way of processing a series of simulations, model refinement, and analysis at supercomputers. Middleware solutions such as those provided by EnTK present the opportunity to utilize high-performance computers and large-scale computing. They provide the building blocks for iterative pipeline based workflows. In the near future, we plan to extend capabilities to support multiple pipelines of chromatophore simulations and analysis.

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References

1. Alberts B (2010) Cell biology: the endless frontier. *Mol Biol Cell* 21(22):3785
2. Singharoy A, Maffeo C, Delgado-Magnero K et al (2019) Atoms to phenotypes: molecular design principles of cellular energy metabolism. *Cell* 179(5):1098–1111.e23
3. Goh BC, Hadden JA, Bernardi RC et al (2016) Computational methodologies for real-space structural refinement of large macromolecular complexes. *Annu Rev Biophys* 45:253–278
4. Voth GA (2017) A multiscale description of biomolecular active matter: the chemistry underlying many life processes. *Acc Chem Res* 50(3):594–598
5. Davtyan A, Simunovic M, Voth GA (2016) Multiscale simulations of protein-facilitated membrane remodeling. *J Struct Biol* 196(1):57–63
6. Van Meel JA, Arnold A, Frenkel D et al (2008) Harvesting graphics power for md simulations. *Mol Simul* 34(3):259–266
7. Ananthraj V, De K, Jha S et al (2018) Towards exascale computing for high energy physics: The atlas experience at ornl. In: 2018 IEEE 14th international conference on e-science (e-science), pp 341–342
8. Kilburg D, Gallicchio E (2016) Recent advances in computational models for the study of protein–peptide interactions. *Adv Protein Chem Struct Biol* 105:27–57
9. Ourmazd A (2019) Cryo-em, xfels and the structure conundrum in structural biology. *Nat Methods* 16(10):941–944
10. Marrink SJ, Corradi V, Souza PC et al (2019) Computational modeling of realistic cell membranes. *Chem Rev* 119(9):6184–6226
11. Feig M, Harada R, Mori T et al (2015) Complete atomistic model of a bacterial cytoplasm for integrating physics, biochemistry, and systems biology. *J Mol Graph Model* 58:1–9
12. Yu I, Mori T, Ando T et al (2016) Biomolecular interactions modulate macromolecular structure and dynamics in atomistic model of a bacterial cytoplasm. *elife* 5:e19274
13. Perilla JR, Schulten K (2017) Physical properties of the hiv-1 capsid from all-atom molecular dynamics simulations. *Nat Commun* 8:15959
14. Wickles S, Singharoy A, Andreani J et al (2014) A structural model of the active ribosome-bound membrane protein insertase yidc. *elife* 3:e03035
15. Trabuco LG, Villa E, Mitra K et al (2008) Flexible fitting of atomic structures into electron microscopy maps using molecular dynamics. *Structure* 16(5):673–683
16. Schweitzer A, Aufderheide A, al Rudack T (2016) Structure of the human 26s proteasome at a resolution of 3.9 Å. *Proc Natl Acad Sci U S A* 113(28):7816–7821
17. Durrant JD, Bush RM, Amaro RE (2016) Microsecond molecular dynamics simulations of influenza neuraminidase suggest a mechanism for the increased virulence of stalk-deletion mutants. *J Phys Chem B* 120(33):8590–8599
18. Mannige RV, Brooks CL III (2010) Periodic table of virus capsids: implications for natural selection and design. *PLoS One* 5(3):e9423
19. Blood PD, Voth GA (2006) Direct observation of bin/amphiphysin/rvs (bar) domain-induced membrane curvature by means of molecular dynamics simulations. *Proc Natl Acad Sci U S A* 103(41):15068–15072
20. Arkhipov A, Yin Y, Schulten K (2008) Four-scale description of membrane sculpting by bar domains. *Biophys J* 95(6):2806–2821
21. Jung J, Nishima W, Daniels M et al (2019) Scaling molecular dynamics beyond 100,000 processor cores for large-scale biophysical simulations. *J Comput Chem* 40(21):1919–1930
22. Renaud J-P, Chari A, Ciferri C et al (2018) Cryo-EM in drug discovery: achievements, limitations and prospects. *Nat Rev Drug Discov* 17(7):471–492
23. Camargo C (2018) Physics makes rules, evolution rolls the dice. *Science* 361(6399):236–236
24. Şener MK, Olsen JD, Hunter CN et al (2007) Atomic-level structural and functional model of a bacterial photosynthetic membrane vesicle. *Proc Natl Acad Sci* 104(40):15723–15728
25. Blankenship RE (2014) Molecular mechanisms of photosynthesis. John Wiley & Sons, Hoboken, New Jersey
26. Vant, J. W. (2019). Chromatophore_large_system_simulation. https://github.com/jvant/Chromatophore_Large_System_Simulation GitHub
27. Phillips JC, Braun R, Wang W et al (2005) Scalable molecular dynamics with NAMD. *J Comput Chem* 26(16):1781–1802
28. Comer J, Aksimentiev A (2016) DNA sequence-dependent ionic currents in ultra-small solidstate nanopores. *Nanoscale* 8(18):9600–9613

29. Humphrey W, Dalke A, Schulten K (1996) VMD: Visual molecular dynamics. *J Mol Graph* 14(1):33–38
30. Singharoy A, Cheluvareja S, Ortoleva P (2011) Order parameters for macromolecules: application to multiscale simulation. *J Chem Phys* 134(4):044104
31. Acun B, Hardy DJ, Kale LV et al (2018) Scalable molecular dynamics with NAMD on the summit system. *IBM J Res Dev* 62(6):1–9
32. Chandler DE, Strümpfer J, Sener M et al (2014) Light harvesting by lamellar chromatophores in rhodospirillum photometricum. *Biophys J* 106(11):2503–2510
33. Şener M, Strümpfer J, Timney JA et al (2010) Photosynthetic vesicle architecture and constraints on efficient energy and harvesting. *Biophys J* 99(1):67–75
34. Cartron ML, Olsen JD, Sener M et al (2014) Integration of energy and electron transfer processes in the photosynthetic membrane of rhodobacter sphaeroides. *Biochim Biophys Acta* 1837(10):1769–1780
35. Kumar S, Cartron ML, Mullin N et al (2016) Direct imaging of protein organization in an intact bacterial organelle using high-resolution atomic force microscopy. *ACS Nano* 11(1):126–133
36. Scheuring S, Nevo R, Liu L-N et al (2014) The architecture of rhodobacter sphaeroides chromatophores. *Biochim Biophys Acta* 1837(8):1263–1270
37. Russel D, Lasker K, Webb B et al (2012) Putting the pieces together: integrative modeling platform software for structure determination of macromolecular assemblies. *PLoS Biol* 10(1):e1001244
38. Ho PT, Montiel-Garcia DJ, Wong JJ et al (2018) VIPERdb: a tool for virus research. *Annu Rev Virol* 5(1):477–488
39. Durrant JD, Amaro RE (2014) Lipidwrapper: an algorithm for generating large-scale membrane models of arbitrary geometry. *PLoS Comput Biol* 10(7):e1003720
40. Wells DB, Abramkina V, Aksimentiev A (2007) Exploring transmembrane transport through α -hemolysin with grid-steered molecular dynamics. *J Chem Phys* 127(12):09B619
41. Balasubramanian V, Turilli M, Hu W et al (2018) Harnessing the power of many: extensible toolkit for scalable ensemble applications. In: In 2018 IEEE international parallel and distributed processing symposium (ipdps). IEEE, New York, pp 536–545
42. Turilli M, Santcroos M, Jha S (2018) A comprehensive perspective on pilot-job systems. *ACM Comput Surv* 51(2):43:1–43:32
43. Goodale T, Jha S, Kaiser H et al (2006) SAGA: a simple API for grid applications, high-level application programming on the grid. *Comput Methods Sci Technol* 12(1):7–20
44. Merzky A, Weidner O, Jha S (2015) SAGA: a standardized access layer to heterogeneous distributed computing infrastructure. *Software-X* 1-2:3–8
45. MDFF Integration with EnTK on OLCF Summit. (2019). <https://github.com/radical-colaboration/MDFF-Error.GitHub>
46. Chandler DE, Hsin J, Harrison CB et al (2008) Intrinsic curvature properties of photosynthetic proteins in chromatophores. *Biophys J* 95(6):2822–2836
47. Singharoy A, Barragan AM, Thangapandian S et al (2016b) Binding site recognition and docking dynamics of a single electron transport protein: cytochrome c 2. *J Am Chem Soc* 138(37):12077–12089
48. Singharoy A, Teo I, McGreevy R et al (2016a) Molecular dynamics-based model refinement and validation for sub-5 angstrom cryo-electron microscopy maps. *eLife* 5:e16105

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